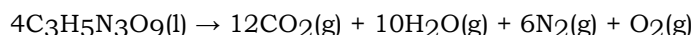


1.0 ATOMS, MOLECULES & STOICHIOMETRY

1. Nitroglycerine, $\text{C}_3\text{H}_5\text{N}_3\text{O}_9$, is an explosive which, on detonation, decomposes rapidly to form a large number of gaseous molecules. The equation for this decomposition is given below.



- (a) A sample of nitroglycerine was detonated and produced 0.350 g of oxygen gas.
- State what is meant by the term *one mole* of molecules.
 - Calculate the number of moles of oxygen gas produced in this reaction, and hence deduce the total number of moles of gas formed.
 - Calculate the number of moles, and the mass, of nitroglycerine detonated.

(7)

- (b) A second sample of nitroglycerine was placed in a strong sealed container and detonated. The volume of this container was $1.00 \times 10^{-3} \text{ m}^3$. The resulting decomposition produced a total of 0.873 mol of gaseous products at a temperature of 1100 K.

State the ideal gas equation and use it to calculate the pressure in the container after detonation.

(The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

(4)

(Total 11 marks)

2. Sodium chlorate(V), NaClO_3 , contains 21.6% by mass of sodium, 33.3% by mass of chlorine and 45.1% by mass of oxygen.

- (a) Use the above data to show that the empirical formula of sodium chlorate(V) is NaClO_3

- (b) Sodium chlorate(V) may be prepared by passing chlorine into hot aqueous sodium hydroxide. Balance the equation for this reaction below.



(3)

(Total 3 marks)

3. Potassium nitrate, KNO_3 , decomposes on strong heating, forming oxygen and solid **Y** as the only products.

- (a) A 1.00 g sample of KNO_3 ($M_r = 101.1$) was heated strongly until fully decomposed into **Y**.

- (i) Calculate the number of moles of KNO_3 in the 1.00 g sample.

- (ii) At 298 K and 100 kPa, the oxygen gas produced in this decomposition occupied a volume of $1.22 \times 10^{-4} \text{ m}^3$.

State the ideal gas equation and use it to calculate the number of moles of oxygen produced in this decomposition.

(The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

(5)

- (b) Compound **Y** contains 45.9% of potassium and 16.5% of nitrogen by mass, the remainder being oxygen.

- (i) State what is meant by the term *empirical formula*.

- (ii) Use the data above to calculate the empirical formula of **Y**.

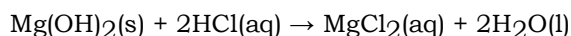
(4)

- (c) Deduce an equation for the decomposition of KNO_3 into **Y** and oxygen.

(1)

(Total 10 marks)

4. The equation for the reaction between magnesium hydroxide and hydrochloric acid is shown below.



Calculate the volume, in cm^3 , of 1.00 mol dm^{-3} hydrochloric acid required to react completely with 1.00 g of magnesium hydroxide.

(4)

(Total 4 marks)

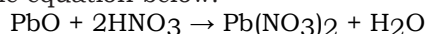
5. When aluminium is added to an aqueous solution of copper(II) chloride, CuCl_2 , copper

metal and aluminium chloride, AlCl_3 , are formed. Write an equation to represent this reaction.

(1)

(Total 1 mark)

6. (a) Lead(II) nitrate may be produced by the reaction between nitric acid and lead(II) oxide as shown by the equation below.



An excess of lead(II) oxide was allowed to react with 175 cm^3 of 1.50 mol dm^{-3} nitric acid. Calculate the maximum mass of lead(II) nitrate which could be obtained from this reaction.

(4)

- (b) An equation representing the thermal decomposition of lead(II) nitrate is shown below.



A sample of lead(II) nitrate was heated until the decomposition was complete. At a temperature of 500 K and a pressure of 100 kPa, the total volume of the gaseous mixture produced was found to be $1.50 \times 10^{-4} \text{ m}^3$.

- (i) State the ideal gas equation and use it to calculate the total number of moles of gas produced in this decomposition.

(The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

- (ii) Deduce the number of moles, and the mass, of NO_2 present in this gaseous mixture. (If you have been unable to calculate the total number of moles of gas in part (b)(i), you should assume this to be $2.23 \times 10^{-3} \text{ mol}$. This is not the correct answer.)

(7)

(Total 11 marks)

7. (a) Ammonia, NH_3 , reacts with sodium to form sodium amide, NaNH_2 , and hydrogen. Write an equation for the reaction between ammonia and sodium.

(1)

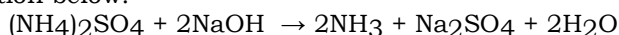
- (b) A salt, **X**, contains 16.2% by mass of magnesium, 18.9% by mass of nitrogen and 64.9% by mass of oxygen.

- (i) State what is meant by the term *empirical formula*.
(ii) Determine the empirical formula of **X**.

(3)

(Total 4 marks)

8. (a) Ammonium sulphate reacts with aqueous sodium hydroxide as shown by the equation below.



A sample of ammonium sulphate was heated with 100 cm^3 of $0.500 \text{ mol dm}^{-3}$ aqueous sodium hydroxide. To ensure that all the ammonium sulphate reacted, an excess of sodium hydroxide was used.

Heating was continued until all of the ammonia had been driven off as a gas.

The unreacted sodium hydroxide remaining in the solution required 27.3 cm^3 of $0.600 \text{ mol dm}^{-3}$ hydrochloric acid for neutralisation.

- (i) Calculate the original number of moles of NaOH in 100 cm^3 of $0.500 \text{ mol dm}^{-3}$ aqueous sodium hydroxide.
(ii) Calculate the number of moles of HCl in 27.3 cm^3 of $0.600 \text{ mol dm}^{-3}$ hydrochloric acid.
(iii) Deduce the number of moles of the unreacted NaOH neutralised by the hydrochloric acid.
(iv) Use your answers from parts (a) (i) and (a) (iii) to calculate the number of moles of NaOH which reacted with the ammonium sulphate.
(v) Use your answer in part (a) (iv) to calculate the number of moles and the mass of ammonium sulphate in the sample.
(If you have been unable to obtain an answer to part (a) (iv), you may assume that the number of moles of NaOH which reacted with ammonium sulphate

equals

2.78×10^{-2} mol. This is not the correct answer.)

(7)

- (b) A 0.143g gaseous sample of ammonia occupied a volume of $2.86 \times 10^{-4} \text{ m}^3$ at a temperature **T** and a pressure of 100 kPa.
State the ideal gas equation, calculate the number of moles of ammonia present and deduce the value of the temperature **T**.
(The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

(4)

(Total 11 marks)

9. Compound **A** is an oxide of sulphur. At 415 K, a gaseous sample of **A**, of mass 0.304 g, occupied a volume of 127 cm^3 at a pressure of 103 kPa.
State the ideal gas equation and use it to calculate the number of moles of **A** in the sample, and hence calculate the relative molecular mass of **A**.
(The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

(Total 5 marks)

10. A hydrocarbon, **W**, contains 92.3% carbon by mass. The relative molecular mass of **W** is 78.0

- (a) Calculate the empirical formula of **W**.
(b) Calculate the molecular formula of **W**.

(4)

(Total 4 marks)

11. (a) One isotope of sodium has a relative mass of 23.
Calculate the mass, in grams, of a single atom of this isotope of sodium.
(The Avogadro constant, L , is $6.023 \times 10^{23} \text{ mol}^{-1}$)

(2)

(Total 2 marks)

12. (a) State what is meant by the term *empirical formula*.
(b) A chromium compound contains 28.4% of sodium and 32.1% of chromium by mass, the remainder being oxygen.
Calculate the empirical formula of this compound.

(4)

(Total 4 marks)

13. (a) Titanium(IV) chloride reacted with water as shown in the following equation.
 $\text{TiCl}_4(\text{l}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 4\text{HCl}(\text{aq}) + \text{TiO}_2(\text{s})$

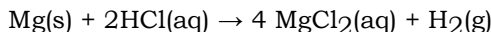
The reaction produced 200 cm^3 of a 1.20M solution of hydrochloric acid.
Calculate the number of moles of HCl in the solution and use your answer to find the original mass of TiCl_4 .

(4)

- (b) Calculate the volume of 1.10 M sodium hydroxide solution which would be required to neutralise a 100 cm^3 portion of the 1.20 M solution of hydrochloric acid.

(3)

- (c) An excess of magnesium metal was added to a 100 cm^3 portion of the 1.20 M solution of hydrochloric acid. Calculate the volume of hydrogen gas produced at 98 kPa and 20°C .



(4)

(Total 11 marks)

14. (a) Calculate the maximum mass of sodium sulphide that can be obtained from 10.0 g of sulphur.

(2)

- (b) Calculate the minimum volume of hydrogen, in cm^3 , at 298 K and 101.3 kPa, that is needed to form 5.00 g of hydrogen sulphide.

(3)

(Total 5 marks)

15. When a sample of liquid, **X**, of mass 0.406 g was vaporised, the vapour was found to

occupy a volume of $2.34 \times 10^{-4} \text{ m}^3$ at a pressure of 110 kPa and a temperature of 473 K.

- (a) Give the name of the equation $pV = nRT$. (1)
- (b) Use the equation $pV = nRT$ to calculate the number of moles of **X** in the sample and hence deduce the relative molecular mass of **X**. (4)
(The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)
- (c) Compound **X**, which contains carbon, hydrogen and oxygen only, has 38.7% carbon and 9.68% hydrogen by mass. Calculate the empirical formula of **X**. (3)
- (d) Using your answers to parts (b) and (c) above, deduce the molecular formula of **X**. (1)

(Total 9 marks)

16. (a) Calculate the concentration, in mol dm^{-3} , of the solution formed when 19.6 g of hydrogen chloride, HCl, are dissolved in water and the volume made up to 250 cm^3 . (3)
- (b) The carbonate of metal **M** has the formula M_2CO_3 . The equation for the reaction of this carbonate with hydrochloric acid is given below.

$$\text{M}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{MCl} + \text{CO}_2 + \text{H}_2\text{O}$$
A sample of M_2CO_3 , of mass 0.394 g, required the addition of 21.7 cm^3 of a $0.263 \text{ mol dm}^{-3}$ solution of hydrochloric acid for complete reaction.
- (i) Calculate the number of moles of hydrochloric acid used.
- (ii) Calculate the number of moles of M_2CO_3 in 0.394 g.
- (iii) Calculate the relative molecular mass of M_2CO_3
- (iv) Deduce the relative atomic mass of **M** and hence suggest its identity. (6)

(Total 9 marks)

17. (a) The mass of one mole of ^1H atoms is 1.0078 g and that of one ^1H atom is $1.6734 \times 10^{-24} \text{ g}$.
Use these data to calculate a value for the Avogadro constant accurate to five significant figures. Show your working. (2)
- (b) How does the number of atoms in one mole of argon compare with the number of molecules in one mole of ammonia? (1)
- (c) A sample of ammonia gas occupied a volume of 0.0352 m^3 at 298 K and 98.0 kPa. Calculate the number of moles of ammonia in the sample.
(The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) (3)
- (d) A solution containing 0.732 mol of ammonia was made up to 250 cm^3 in a volumetric flask by adding water. Calculate the concentration of ammonia in this final solution and state the appropriate units. (2)
- (e) A different solution of ammonia was reacted with sulphuric acid as shown in the equation below.

$$2\text{NH}_3(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow (\text{NH}_4)_2\text{SO}_4(\text{aq})$$
In a titration, 25.0 cm^3 of a 1.24 mol dm^{-3} solution of sulphuric acid required 30.8 cm^3 of this ammonia solution for complete reaction.
- (i) Calculate the concentration of ammonia in this solution.
- (ii) Calculate the mass of ammonium sulphate in the solution at the end of this titration. (6)
- (f) The reaction of magnesium nitride, Mg_3N_2 , with water produces ammonia and magnesium hydroxide. Write an equation for this reaction. (2)

(Total 16 marks)

18. (a) Define the term *relative molecular mass*. (2)
- (b) Give the meaning of the term *empirical formula*. (1)
- (c) Compound **X** contains 32.9% by mass of carbon and 1.40% by mass of hydrogen; the remainder is oxygen.
 (i) Calculate the empirical formula of **X**.
 (ii) The relative molecular mass of **X** is 146. Deduce its molecular formula. (4)
- (d) A 1.0 kg sample of methane was burned in air. It reacted as follows:

$$\text{CH}_4 (\text{g}) + 2\text{O}_2 (\text{g}) \rightarrow \text{CO}_2 (\text{g}) + 2\text{H}_2\text{O}(\text{g})$$

 (i) Calculate the number of moles in 1.0 kg of methane.
 (ii) Calculate the volume of oxygen gas, measured at 298 K and 100 kPa, which would be required for the complete combustion of 1.0 kg of methane. (6)
- (Total 13 marks)**
19. Give the meaning of the term *mole* as used in the phrase 'one mole of molecules'. (1)
- (Total 1 mark)**
20. (a) What is the name given to the number of molecules in one mole of carbon dioxide? (1)
- (b) (i) State the ideal gas equation.
 (ii) Calculate the volume of 1.00 mol of carbon dioxide gas at 298 K and 100 kPa. (The gas constant $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)
 (iii) Calculate the mass of carbon dioxide gas at 273 K and 500 kPa contained in a cylinder of volume 0.00500 m^3 . (7)
- (c) Hydrogen can be made by the reaction of hydrochloric acid with magnesium according to the equation

$$2\text{HCl} + \text{Mg} \rightarrow \text{MgCl}_2 + \text{H}_2$$

 What mass of hydrogen is formed when 100cm^3 of hydrochloric acid of concentration 5.0 mol dm^{-3} reacts with an excess of magnesium? (3)
- (d) A compound of iron contains 38.9% by mass of iron and 16.7% by mass of carbon, the remainder being oxygen.
 (i) Determine the empirical formula of the iron compound.
 (ii) When one mole of this iron compound is heated, it decomposes to give one mole of iron(II) oxide, FeO , one mole of carbon dioxide and one mole of another gas. Identify this other gas. (The molecular formula of the iron compound is the same as its empirical formula.) (4)
- (Total 15 marks)**
21. Ammonium nitrate can be prepared by the reaction between ammonia and nitric acid:

$$\text{NH}_3 + \text{HNO}_3 \rightarrow \text{NH}_4\text{NO}_3$$
- (a) The concentration of a nitric acid solution is 2.00 mol dm^{-3} . Calculate the volume of this solution which would be required to react with exactly 20.0 g of ammonia. (4)
- (b) A sample of ammonium nitrate decomposed on heating as shown in the equation below.

$$\text{NH}_4\text{NO}_3 \rightarrow 2\text{H}_2\text{O} + \text{N}_2 + \frac{1}{2}\text{O}_2$$

 On cooling the resulting gases to 298 K, the volume of nitrogen and oxygen together was found to be 0.0500 m^3 at a pressure of 95.0 kPa.
 (i) State the ideal gas equation and use it to calculate the total number of moles of nitrogen and oxygen formed. (The gas constant $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)
 (ii) Using your answer to part (b)(i), deduce the number of moles of ammonium nitrate decomposed and hence calculate the mass of ammonium nitrate in the sample.

- (6)
(Total 10 marks)
22. The mass of one atom of ^{12}C is 1.99×10^{-23} g. Use this information to calculate a value for the Avogadro constant. Show your working.
- (2)
(Total 2 marks)
23. The equation below represents the thermal decomposition of KClO_3 .
- $$2\text{KClO}_3(\text{s}) \rightarrow 2\text{KCl}(\text{s}) + 3\text{O}_2(\text{g})$$
- (a) Calculate the mass of oxygen which could be produced by the complete decomposition of 1.47 g of KClO_3 .
- (2)
- (b) Calculate the mass of KClO_3 required to produce 1.00 dm^3 (at 20°C and 101.3 kPa) of oxygen.
- (3)
(Total 5 marks)
24. (a) What experimental data are required in order to calculate the empirical formula of a compound?
- (1)
- (b) Give the meaning of the term *molecular formula*.
- (1)
- (c) When barium nitrate is heated it decomposes as follows:
- $$\text{Ba}(\text{NO}_3)_2(\text{s}) \rightarrow \text{BaO}(\text{s}) + 2\text{NO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g})$$
- (i) Calculate the total volume, measured at 298 K and 100 kPa , of gas which is produced by decomposing 5.00 g of barium nitrate.
- (ii) Calculate the volume of 1.20 M hydrochloric acid which is required to neutralise exactly the barium oxide formed by decomposition of 5.00 g of barium nitrate. Barium oxide reacts with hydrochloric acid as follows
- $$\text{BaO}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{BaCl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l})$$
- (7)
(Total 9 marks)
25. The mass spectrum of a compound has a molecular ion peak at $m/z = 168$. Elemental analysis shows it to contain 42.9% carbon, 2.4% hydrogen and 16.7% nitrogen by mass. The remainder is oxygen. Calculate the empirical and molecular formulae of this compound
- (4)
(Total 4 marks)
26. Compound **X** contains only boron and hydrogen. The percentage by mass of boron in **X** is 81.2% . In the mass spectrum of **X** the peak at the largest value of m/z occurs at 54 .
- (a) Use the percentage by mass data to calculate the empirical formula of **X**.
- (b) Deduce the molecular formula of **X**.
- (4)
(Total 4 marks)
27. (a) Define the term *relative molecular mass*.
- (2)
- (b) The mass of one atom of ^{12}C is 1.993×10^{-23} g. Use this mass to calculate a value for the Avogadro constant (L) showing your working.
- (1)
(Total 3 marks)
28. When iodine reacts directly with fluorine, a compound containing 57.2% by mass of iodine is formed.
- (a) Determine the empirical formula of this compound.
- (b) The empirical formula of this compound is the same as the molecular formula. Write a balanced equation for the formation of this compound.
- (4)
(Total 4 marks)
29. Compound **A** ($M_r = 215.8$) contains 22.24% carbon, 3.71% hydrogen and 74.05% bromine by mass. Show that the molecular formula of **A** is $\text{C}_4\text{H}_8\text{Br}_2$.
- (3)

(Total 3 marks)

30. (a) Define the term relative atomic mass. (2)
- (b) How would you calculate the mass of one mole of atoms from the mass of a single atom? (1)
- (c) Sodium hydride reacts with water according to the following equation.
- $$\text{NaH}_{(s)} + \text{H}_2\text{O}_{(l)} \rightarrow \text{NaOH}_{(aq)} + \text{H}_{2(g)}$$
- A 1.00 g sample of sodium hydride was added to water and the resulting solution was diluted to a volume of exactly 250 cm³
- Calculate the concentration in mol dm⁻³, of sodium hydroxide solution formed.
 - Calculate the volume of hydrogen gas evolved, measured at 293 K and 100 kPa.
 - Calculate the volume of 0.112 M hydrochloric acid which would react exactly with a 25.0 cm³ sample of sodium hydroxide solution.

(8)**[TOTAL 11 marks]**

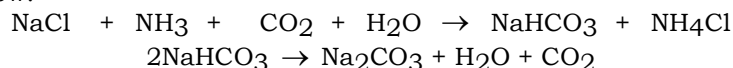
31. (a) Sodium carbonate forms a number of hydrates of general formula Na₂CO₃.xH₂O
A 3.01 g sample of one of these hydrates was dissolved in water and the solution made up to 250 cm³.
In a titration, a 25.0 cm³ portion of this solution required 24.3 cm³ of 0.200 mol⁻¹ dm⁻³ hydrochloric acid for complete reaction.
The equation for this reaction is shown below.
- $$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$$
- Calculate the number of moles of HCl in 24.3 cm³ of 0.200 mol dm⁻³ hydrochloric acid.
 - Deduce the number of moles of Na₂CO₃ in 25.0 cm³ of the Na₂CO₃ solution.
 - Hence deduce the number of moles of Na₂CO₃ in the original 250 cm³ of solution.
 - Calculate the *M_r* of the hydrated sodium carbonate.
- (b) In an experiment, the *M_r* of a different hydrated sodium carbonate was found to be 250.
Use this value to calculate the number of molecules of water of crystallisation, *x*, in this hydrated sodium carbonate, Na₂CO₃.xH₂O.
- (c) A gas cylinder, of volume 5.00 × 10⁻³ m³, contains 325 g of argon gas.
- Give the ideal gas equation.
 - Use the ideal gas equation to calculate the pressure of the argon gas in the cylinder at a temperature of 298 K.
(The gas constant *R* = 8.31 J K⁻¹ mol⁻¹)

(5)**(3)****(4)****[Total 12 marks]**

32. (a) A sample of ethanol vapour, C₂H₅OH (*M_r* = 46.0), was maintained at a pressure of 100 kPa and at a temperature of 366K.
- State the ideal gas equation.
 - Use the ideal gas equation to calculate the volume, in cm³, that 1.36 g of ethanol vapour would occupy under these conditions.
(The gas constant *R* = 8.31 J K⁻¹ mol⁻¹)
- (b) Magnesium nitride reacts with water to form magnesium hydroxide and ammonia.
- Balance the equation, given below, for the reaction between magnesium nitride and water.
$$\text{Mg}_3\text{N}_2 + \text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + \text{NH}_3$$
 - Calculate the number of moles, and hence the number of molecules, of NH₃ in 0.263 g of ammonia gas.
(The Avogadro constant *L* = 6.02 × 10²³ mol⁻¹)

(4)

- (c) Sodium carbonate is manufactured in a two-stage process as shown by the equations below.

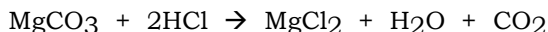


Calculate the maximum mass of sodium carbonate which could be obtained from 800 g of sodium chloride.

(4)

[Total 13 marks]

33. (a) The equation for the reaction between magnesium carbonate and hydrochloric acid is given below.



When 75.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ hydrochloric acid were added to 1.25 g of impure MgCO_3 some acid was left unreacted. This unreacted acid required 21.6 cm^3 of a $0.500 \text{ mol dm}^{-3}$ solution of sodium hydroxide for complete reaction.

- Calculate the number of moles of HCl in 75.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ hydrochloric acid.
- Calculate the number of moles of NaOH used to neutralise the unreacted HCl.
- Show that the number of moles of HCl which reacted with the MgCO_3 in the sample was 0.0267
- Calculate the number of moles and the mass of MgCO_3 in the sample, and hence deduce the percentage by mass of MgCO_3 in the sample.

(8)

- (b) A compound contains 36.5% of sodium and 25.5% of sulphur by mass, the rest being oxygen.

- Use this information to show that the empirical formula of the compound is Na_2SO_3
- When Na_2SO_3 is treated with an excess of hydrochloric acid, aqueous sodium chloride is formed and sulphur dioxide gas is evolved. Write an equation to represent this reaction.

(4)

[Total 12 marks]

34. (a) Give the meaning of the term empirical formula

(1)

- (b) Analysis of 3.150 g of compound X showed that it contained 0.769 g of calcium and 0.539 g of nitrogen; the remainder was oxygen. Calculate the empirical formula of X. (3)

- (c) What additional information is required in order to deduce the molecular formula of X. (1)

- (d) A sample of X when heated in alkaline solution with an aluminium-zinc alloy produced ammonia gas. After cooling to 293 K, the ammonia occupied a volume of $1.53 \times 10^{-3} \text{ m}^3$ at a pressure of 95.0 kPa. The ammonia was dissolved in water and made up to 250 cm^3 of aqueous solution. A 25.0 cm^3 sample of this was then titrated with a 0.150 M hydrochloric acid.

- Calculate the number of moles of ammonia gas in $1.53 \times 10^{-3} \text{ m}^3$, at a pressure of 95.0 kPa and a temperature of 293K.
- Calculate the concentration in mol dm^{-3} of ammonia in aqueous solution.
- Calculate the volume of 0.150 M hydrochloric acid required to neutralise the 25.0 cm^3 sample of ammonia solution.

(6)

35. In the sixteenth century, a large deposit of graphite was discovered in the Lake District. People at the time thought that the graphite was a form of lead. Nowadays, graphite is used in pencils but it is still referred to as 'pencil lead'. A student decided to investigate the number of carbon atoms in a 'pencil lead'. He found that the mass of the 'pencil lead' was 0.321 g.

- Calculate the amount, in mol, of carbon atoms in the student's pencil lead. Assume that the 'pencil lead' is pure graphite.

[1]

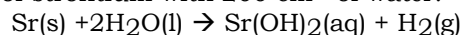
- Using the Avogadro constant, N_A , calculate the number of carbon atoms in the

student's 'pencil lead'.

[1]

[Total 2 marks]

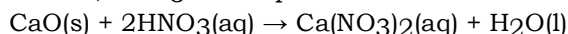
36. A student reacted 0.438 g of strontium with 200 cm³ of water.



- Calculate how many moles of Sr were reacted.
- Calculate the volume, in dm³, of H₂(g) produced.
- Calculate the concentration, in mol dm⁻³, of the Sr(OH)₂ produced.

[Total 3 marks]

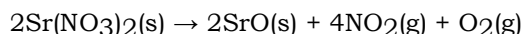
37. Calcium oxide neutralises acids such as nitric acid. A student neutralised 1.50 g of CaO with 2.50 mol dm⁻³ nitric acid, HNO₃. The equation for this reaction is shown below.



- How many moles of CaO were reacted?
- Calculate the volume of 2.50 mol dm⁻³ HNO₃ needed to exactly neutralise 1.50 g of CaO.

[Total 4 marks]

38. The element strontium forms a nitrate, Sr(NO₃)₂, which decomposes on heating as shown below.



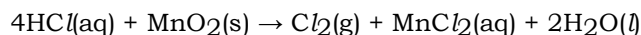
A student heats 5.29 g of Sr(NO₃)₂ and collects the gas at room temperature and pressure, RTP.

Calculate the volume of gas, in dm³, obtained by the student at RTP.

(Molar mass of Sr(NO₃)₂ = 211.6 g mol⁻¹)

[Total 3 marks]

39. Chlorine can be prepared by reacting concentrated hydrochloric acid with manganese(IV) oxide.



A student reacted 50.0 cm³ of 12.0 mol dm⁻³ hydrochloric acid with an excess of manganese(IV) oxide.

- Calculate how many moles of HCl were reacted.

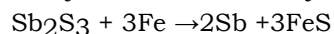
[1]

- Calculate the volume of Cl₂(g) produced, in dm³.

[2]

[Total 3 marks]

40. Antimony is found naturally in a number of minerals including stibnite. Stibnite typically contains 5% of Sb₂S₃. Antimony can be obtained by reducing Sb₂S₃ with scrap iron.



- How many moles of Sb₂S₃ are in 500 kg of a typical sample of stibnite containing 5% by mass of Sb₂S₃?

molar mass of Sb₂S₃ = 340 g mol⁻¹; relative atomic mass of Sb = 122

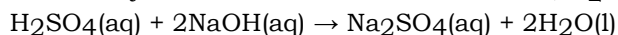
[2]

- Calculate the mass of antimony that could be obtained by processing 500 kg of stibnite.

[2]

[Total 4 marks]

41. (a) A student carries out a titration to find the concentration of some sulfuric acid. The student finds that 25.00 cm³ of 0.0880 mol dm⁻³ aqueous sodium hydroxide, NaOH, is neutralised by 17.60 cm³ of dilute sulfuric acid, H₂SO₄.



- Calculate the amount, in moles, of NaOH used.

[1]

- Determine the amount, in moles, of H₂SO₄ used.

[1]

- (iii) Calculate the concentration, in mol dm^{-3} , of the sulfuric acid. [1]
- (b) After carrying out the titration in (a), the student left the resulting solution to crystallise. White crystals were formed, with a formula of $\text{Na}_2\text{SO}_4 \cdot x \text{H}_2\text{O}$ and a molar mass of 322.1 g mol^{-1} . [1]
- (i) What term is given to the ' $x \text{H}_2\text{O}$ ' part of the formula? [1]
- (ii) Using the molar mass of the crystals, calculate the value of x . [2]
- [Total 6 marks]
42. Epsom salts can be used as bath salts to help relieve aches and pains. Epsom salts are crystals of hydrated magnesium sulfate, $\text{MgSO}_4 \cdot x \text{H}_2\text{O}$. A sample of Epsom salts was heated to remove the water. 1.57 g of water was removed leaving behind 1.51 g of anhydrous MgSO_4 . [1]
- (i) Calculate the amount, in mol, of anhydrous MgSO_4 formed. [2]
- (ii) Calculate the amount, in mol, of H_2O removed. [1]
- (iii) Calculate the value of x in $\text{MgSO}_4 \cdot x \text{H}_2\text{O}$. [1]
- [Total 4 marks]
43. Rubidium forms an ionic compound with silver and iodine. This compound has a potential use in miniaturised batteries because of its high electrical conductivity. The empirical formula of this ionic compound can be calculated from its percentage composition by mass: Rb, 7.42%; Ag, 37.48%; I, 55.10%. [1]
- (i) Define the term *empirical formula*. [1]
- (ii) Calculate the empirical formula of the compound. [2]
- [Total 3 marks]
44. A student reacted 1.44 g of titanium with chlorine to form 5.70 g of a chloride X . [1]
- (i) How many moles of Ti atoms were reacted? [1]
- (ii) How many moles of Cl atoms were reacted? [2]
- (iii) Determine the empirical formula of X . [1]
- (iv) Construct a balanced equation for the reaction between titanium and chlorine. [1]
- [Total 5 marks]
45. A small amount of solid magnesium oxide, MgO , was reacted with excess dilute hydrochloric acid. [1]
- (i) Define an acid. [1]
- (ii) Write a balanced equation for this reaction. [1]
- [Total 2 marks]
46. Both calcium carbonate, CaCO_3 , and calcium oxide, CaO , are white solids. Dilute hydrochloric acid, HCl , can be used to identify whether a sample of white solid is CaCO_3 or CaO . [1]
- (i) Write equations, including state symbols, for the reaction of HCl with CaCO_3 and the reaction of HCl with CaO . [3]
- (ii) How would observation of the reactions with hydrochloric acid allow the identification of the white solid? [1]
- [Total 4 marks]
47. A student carries out experiments using acids, bases and salts. Calcium nitrate, $\text{Ca}(\text{NO}_3)_2$, is an example of a salt.

The student prepares a solution of calcium nitrate by reacting dilute nitric acid, HNO_3 , with the base calcium hydroxide, $\text{Ca}(\text{OH})_2$.

- (i) Why is calcium nitrate an example of a salt? [1]
- (ii) Write the equation for the reaction between dilute nitric acid and calcium hydroxide. Include state symbols. [2]
- (iii) Explain how the hydroxide ion in aqueous calcium hydroxide acts as a base when it neutralises dilute nitric acid. [1]

[Total 4 marks]

48. Ammonium compounds such as ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$, can be used as fertilisers.

- (i) Write a balanced equation to show how ammonium sulfate could be formed by the reaction between aqueous ammonia and sulfuric acid. [1]
- (ii) Ammonium sulfate is an example of a salt formed when an acid is neutralised by a base. Explain what is meant by the term *salt*. [1]
- (iii) Why is ammonia acting as a base in this neutralisation? [1]
- (iv) What is the relative formula mass of $(\text{NH}_4)_2\text{SO}_4$? [1]
Give your answer to **one** decimal place.

[1]
[Total 4 marks]

Marking Scheme

1. (a) (i) Avogadro's number/constant of molecules/particles/species / 6×10^{23} 1
[Not 'atoms']
Or same number of particles as (there are atoms)
[Not molecules]
 in 12.(00)g of ^{12}C 1
- (ii) Moles $\text{O}_2 = \frac{0.350}{32} (= 1.09 \times 10^{-2} \text{ mol})$ 1
 $= 29 (\times 1.09 \times 10^{-2})$ 1
[Accept answers via 4 separate mole calculations]
 $= 0.316 - 0.317 \text{ mol}$ [answer to 3+ sf] 1
[Mark conseq on errors in M1/M2] (1)
- (iii) Moles of nitroglycerine $= 4 \times 1.09 \times 10^{-2} (= 0.0438 \text{ mol})$ 1
[Mark conseq on their moles of O_2]
 M_r of nitroglycerine $= 227$ or number string 1
 Moles of nitroglycerine $= 227 \times 0.0438 = 9.90 - 9.93(\underline{\text{g}})$
[answer to 3+ sf]
[If string OK but final answer wrong then allow M6 but AE for M7]
[Mark conseq on error in M_r] [Penalise wrong units]
[Penalise sig. fig. errors once only in whole question]
- (b) $pV = nRT$ **or** $pV = \frac{nRT}{V}$ **or** $p = \frac{nRT}{V}$ 1
 $p = \frac{nRT}{V} = \frac{0.873 \times 8.31 \times 1100}{1.00 \times 10^{-3}}$ 1
 $= 7980093$ **or** 7980 **or** 7.98 1
[ignore s.f.]
- units = Pa **or** kPa **or** MPa (as appropriate) 1
[If error in conversion from Pa, treat as a contradiction of the units mark]
[If transfer error, mark conseq but penalise M2]
[If data from outside of above used, penalise M2 and M3]
[If pV expression incorrectly rearranged, penalise M2 and M3]
[if $T = 1373 \text{ K}$ used, penalise M2]
- [11]
2. (a)
- | | | | |
|-------------------|---------------------|-------------------|---|
| <u>Na</u> | <u>Cl</u> | <u>O</u> | |
| $\frac{21.6}{23}$ | $\frac{33.3}{35.5}$ | $\frac{45.1}{16}$ | 1 |
| 0.9(39) | 0.9(38) | 2.8(2) | |
| 1 | 1 | 3 | 1 |
- Hence:
- Accept backwards calculation, i.e. from formula to % composition,
 and also accept route via M_r to 23; 35.5; 48, and then to 1:1:3
[If % values incorrectly copied, allow M1 only]
[If any wrong A_r values/atomic numbers used = CE = 0]
- (b) $3\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$ 1

[3]

3. (penalty for sig fig error = 1 mark per question)

- (a) (i) moles $\text{KNO}_3 = 1.00/101.1 = 9.89 \times 10^{-3}$ (mol) 1
 (ii) $pV = nRT$ or $n = pV/RT$ 1
 moles $\text{O}_2 = n = \frac{pV}{RT} = \textbf{(1)} \frac{100000 \times 1.22 \times 10^{-4}}{8.31 \times 298} \textbf{(1)}$ 2
 $= 4.93 \times 10^{-3}$ (mol) 1

(mark answer first – check back if wrong)

(transcription error lose M3, mark M4 conseq on error)

(if 'untraceable' figures used M3=M4=0)

(if wrong temp conversion – lose M3 – conseq M4)

(if $n = RT/pV$ CE, lose M3 and M4)

- (b) (i) simplest/lowest ratio of atoms of each / element/s in a compound / substance / species / entity / molecule 1
 (ii)

K	N	O
45.9	16.5	37.6
39.1	14	16
1.17	1.18	2.35
1	1	2

 KNO^2 (1) 3

(M3 tied to M2), (M3 can be transferred from equation if ratio correct but EF not given) (if calc inverted, lose M2 and M3), (if used At N₁ / wrong No for Ar then CE, lose M2 and M3) (if % of O missing, award M2 only)

- (c) $2\text{KNO}_3 \rightarrow 2\text{KNO}^2 + \text{O}^2$ or fractions/multiples 1
 (accept $2\text{KNO}_3 \rightarrow \text{K}^2\text{N}^2\text{O}^4 + \text{O}^2$)
 (do NOT accept 'Y' in equation)

[10]

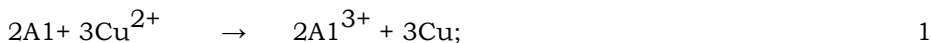
4. (penalty for sig fig error = 1 mark per question)

- $M_r(\text{Mg}(\text{NO}_3)_2) = 58(.3)$ (if At N° used, lose M1 and M2) 1
 moles $\text{Mg}(\text{OH})_2 = 0.0172$ (conseq on wrong M²) (answer to 3+ s.f.) 1
 moles $\text{HCl} = 2 \times 0.0172 = 0.0344$ or 0.0343 (mol) (process mark) 1
 $\text{vol HCl} = \frac{0.0343 \times 1000}{1} = 34.3 - 34.5$ (cm³) (unless wrong unit) 1
 (if candidate **used** 0.017 or 0.0171 lose M2)
 (just answer with no working, if in range = (4). if, say, 34 then =(2))
 (if not 2:1 ratio, lose M3 and M4)
 (if work on HCl, CE = 0/4)

[4]

5. $2\text{Al} + 3\text{CuCl}_2 \rightarrow 2\text{AlCl}_3 + 3\text{Cu}$;
 (accept multiples/fractions)

OR



[1]

6. (a) moles $\text{HNO}_3 = 175 \times 10^{-3} \times 1.5 = (0.2625 \text{ mol})$; 1
 moles $\text{Pb}(\text{NO}_3)_2 = \frac{1}{2} \times 0.2625 = (0.131 \text{ mol})$; 1
 $M_r \text{ Pb}(\text{NO}_3)_2 = 331(.2)$; 1
 mass $\text{Pb}(\text{NO}_3)_2 = 331.2 \times 0.131 = 43.5 \text{ g}$; 1

(accept 43.2 - 43.8)

(M1 & M2 are process marks. If error in M1, or in M2, do not mark M4)

consequently, i.e. do not award M4)

(if atomic numbers used in M3, do not award M4)

- (b) (i) $pV = nRT$; 1
 $n = \frac{pV}{RT} = \frac{100000 \times 1.5 \times 10^{-4}}{8.31 \times 500}$; 1
 $= 3.61 \times 10^{-3}$; 1
(If pressure not converted to Pa, max 2)
(If $n = \frac{RT}{pV}$ used = CE; M2 = M3 = 0)
- (ii) moles $\text{NO}_2 = 4/5 \times 3.61 \times 10^{-3}$; 1
[mark is for use of 4/5]
 $= 2.89 \times 10^{-3}$ OR 1.78×10^{-3} ; 1
 $M_r \text{ NO}_2 = 46$; 1
mass $\text{NO}_2 = 46 \times 2.89 \times 10^{-3} = 0.133(\text{g})$ OR $0.0821(\text{g})$; 1
(if atomic numbers used, M3 = M4 = 0)

[11]

7. (a) $2\text{Na} + 2\text{NH}_3 \rightarrow 2\text{NaNH}_2 + \text{H}_2$ (or multiples) 1
- (b) (i) Simplest ratio of atoms of each element in a compound / substance / species / entity / molecule 1
- (ii)
- | | | | | |
|---------|-------------|-----------|-----------|--------------------------|
| | Mg | N | O | |
| | 16.2 | 18.9 | 64.9 | |
| | <u>24.3</u> | <u>14</u> | <u>16</u> | |
| (24) | | | | |
| (0.675) | 0.667 | 1.37 | 4.06 | |
| | 1 | 2 | 6 | MgN_2O_6 |
- (Mark M1 first. If any wrong A_r used = CE = 0)*
(Accept $\text{Mg}(\text{NO}_3)_2$ for M3 if above working shown)

[4]

8. (a) (i) $100 \times 10^{-3} \times 0.500 = 5.00 \times 10^{-2} (\text{mol})$ accept $5 \times 10^{-2} / 0.051$
- (ii) $27.3 \times 10^{-3} \times 0.600 = 1.64 \times 10^{-2} / 1.638 \times 10^{-2} (\text{mol})$ only 1
- (iii) $1.64 \times 10^{-2} (\text{mol})$ 1
Mark conseq on (ii)
- (iv) $5.00 \times 10^{-2} - 1.64 \times 10^{-2} = 3.36 \times 10^{-2} (\text{mol})$ 1
Mark conseq on (i) & (iii)
- (v) $3.36 \times 10^{-2} \times \frac{1}{2} = 1.68 \times 10^{-2} (\text{mol})$ If 2.78×10^{-2} used 1.39×10^{-2} 1
Mark conseq on (iv)
- $1.68 \times 10^{-2} \times 132(.1)$ **or** $1.39 \times 10^{-2} \times 132(.1)$ 1
Mark for M_r
- $= 2.22 \text{ g}$ **or** 1.83 g 1
- (b) $pV = nRT$ 1
 $n = \frac{0.143}{17} = 8.4(1) \times 10^{-3} (\text{mol})$ 1
- $T = \frac{pV}{nR} = \frac{100000 \times 2.86 \times 10^{-4}}{8.31 \times 8.4 \times 10^{-3}} (1)$ 1
 $= 408.5 - 410.5 (\text{K})$ 1
Mark conseq on moles

Note Sig. fig. penalty - apply once if single sf given, unless calc works exactly

[11]

9. Ideal gas equation: $pV = nRT$ (1)

Calculation: $n = pV/RT = \frac{103000 \times 127 \times 10^{-6}}{(8.31 \times 415)}$ (1)

mark for volume conversion fully correct

$= 3.79 \times 10^{-3}$ (mol) (1)

range 3.79×10^{-3} to 3.8×10^{-3}

$M_r = m/n = .304 / 3.79 \times 10^{-3} = 80.1$ (1)

5

range 80 – 80.3

min 2 s.f. conseq

If 'V' wrong lose M2; 'p' wrong lose M3; 'inverted' lose M3 and M4

10.

(a)

M1 % by mass of H = 7.7(0)% (1)

[5]

M2 mol H = $7.70 / 1 = 7.70$

mol C = $92.3 / 12 = 7.69$ (1)

M3 (ratio 1:1 $\therefore$) CH

Credit variations for M2 $78 \times \frac{77}{100} = 6$ and $\frac{78}{12} \times \frac{92.3}{100} = 6$

Correct answer = 3 marks

(b) (CH has empirical mass of 13 and $\frac{78}{13} = 6 \therefore$) C_6H_6

Correct answer 1 mark

4

[4]

11.

$23 / 6.023 \times 10^{23}$ (1)

CE = 0 if inverted or multiplied

tied to M1 $3.8(2) \times 10^{-23}$ [2-5 sig figs] (1)

2

[2]

12.

(a) (simplest) ratio of atoms of each element in compound (1)

(b) % oxygen = 39.5% (1)

Na $28.4 / 23$

Cr $32.1 / 52$

O $39.5 / 16$ (1)

= 1.23

= 0.617

= 2.47

(2:1:4) so empirical formula = Na_2CrO_4 (1)

4

If % oxygen not calculated, only M2 available; if A_r values wrong, only M1 available

[4]

13. (a)

0.240 (1)

$M_r (TiCl_4) = 190$ (1)

moles $TiCl_4 = \frac{0.24}{4} = 0.06$ (1)

mass = $0.06 \times 190 = 11.4$ (1)

4

(b) moles NaOH = moles HCl = 0.12 (1)

vol NaOH = $\frac{\text{moles}}{\text{conc}}$ (1)

= $\frac{0.12}{1.1} = 109 \text{ cm}^3$ (1)

3

(c) moles $H_2 = \frac{\text{moles HCl}}{2} = 0.06$ (1)

(allow conseq)

$V = \frac{nRT}{p}$ (1) or $PV = nRT$

$$= \frac{0.06 \times 8.31 \times 293}{98000} \quad (1)$$

$$= 1.49 \times 10^{-3} \text{ m}^3 \quad (1) \text{ etc}$$

(if chemical error on moles H_2 , max2) **(2)**

4

[11]

- 14.** (a) moles of S = $\frac{10.0}{32.1}$ (= 0.3125) allow $\frac{10}{32}$ **(1)**
 mass of sodium sulphide = 0.312×78.1 i.e. 24.4 g or 24.37 g
 (accept 24.3 g or 24.33 g) (no sig.fig. penalty) (ignore units) **(1)** 2
- (b) moles of hydrogen sulphide = $\frac{5.00}{34.1}$ (i.e. 0.1466 or 0.147 mol) **(1)**
 moles of hydrogen gas needed = moles of H_2S **(1)**
 volume: 0.1466×24500 i.e. 3592 or 3590 cm^3
 (3602 or 3600 if 0.147 used) **(1)**
- answer must be in cm^3 i.e. $\times 24500$, not $\times 24.5$ / no s.f.
 penalty / ignore units
 allow $PV = nRT$ method with remembered value of R

3

[5]

- 15.** (a) Ideal gas equation law **(1)** 1

(b) Moles of **X**: $n = \frac{PV}{RT}$ **(1)** = $\frac{110000 \times 2.34 \times 10^{-4}}{8.31 \times 473}$
 = 6.55×10^{-3} **(1)**
6.5 to 6.6×10^{-3} , min 2 sig figs
If write $n = \frac{RT}{PV}$ zero here, but can score M_r

Relative molecular mass of **X**: $M_r = \frac{m}{n}$ **(1)**
 = 62 **(1)**

61.5 to 62.5

4

- (c) % oxygen = 51.6 **(2)**
 $\text{C} = 38.7 / 12$ $\text{H} = 9.68 / 1$ $\text{O} = 57.6(2) / 16$ **(1)**
 = 3.23 = 9.68 = 3.23
 1 : 3 : 1 $\therefore$ CH_3O **(1)**
If no % O or if wrong A_r used then max 1
Correct empirical formula earns all three marks

3

- (d) $(\frac{62}{31} \times \text{CH}_3\text{O}) = \text{C}_2\text{H}_6\text{O}_2$ **(1)** 1

[9]

- 16.** (a) Moles $\text{HCl} = \frac{\text{mass}}{M_r} = \frac{19.6}{36.5}$ **(1)** (= 0.537)
 Concentration = $\frac{0.537}{0.25}$ **(1)**
 = 2.15 (mol dm^{-3}) **(1)**

Conseq on $\frac{\text{mass}}{M_r}$ correct

min 2 d.p. 2.14 to 2.15

Ignore units

A.E. lose one mark

3

(b) (i) $\frac{21.7}{1000} \times 0.263 = 5.7(1) \times 10^{-3} \text{ (mol) (1)}$

5.7 to 5.71×10^{-3}

(ii) $\frac{5.71 \times 10^{-3}}{2} = 2.85 \times 10^{-3} \text{ (mol) (1)}$

Conseq

(iii) $\frac{0.394}{2.85 \times 10^{-3}} = 138 \text{ (1)}$

Conseq

(iv) *Relative atomic mass of M:* $138 - 60 = 78 \text{ (1)}$

$$\frac{78}{2} = 39 \text{ (1)}$$

Identify of M: Potassium or K or K^+ **(1)**

Conseq

If 78 = M_r then M = selenium

6

[9]

17. (a) $L = \frac{1.0078}{1.6734 \times 10^{-24}} \text{ (1) or } \frac{\text{mass of 1 mol}}{\text{mass of 1 atom}}$
must show working

$= 6.0225 \times 10^{23} \text{ (1)}$

Ignore wrong units

NB answer only scores 1

2

(b) equal **(1)**
Or same or 1:1

1

(c) $PV = nRT \text{ (or } n = \frac{PV}{RT} \text{) (1)}$

$= \frac{98000 \times 0.0352}{8.31 \times 298} \text{ (1)}$

$= 1.39 \text{ (1)}$

Allow 1.390 to 1.395

ignore units even if incorrect

answer = 1.4 loses last mark

3

(d) $0.732 \times \frac{1000}{250} = 2.93 \text{ (1) mol.dm}^{-3} \text{ (1)}$

OR M, mol/dm³, mol.l⁻¹

allow 2.928 to 2.93

Note unit mark tied to current answer but allow unit mark if answer = 2.9 or 3

2

(e) (i) $\text{moles H}_2\text{SO}_4 = \frac{25}{1000} \times 1.24 = 0.0310$

If use $m_1v_1 = m_2v_2$ scores 3 if answer is correct otherwise zero

$\text{moles NH}_3 \text{ in } 30.8 \text{ cm}^3 = 0.0310 \times 2 = 0.0620 \text{ (1)}$

Mark is for $\times 2$

CE if $\times 2$ not used

$\text{moles of NH}_3 \text{ in } 1 \text{ dm}^3 = 0.620 \times \frac{1000}{30.8} = 2.01 \text{ (1) (mol dm}^{-3}\text{)}$

Allow 2.010 to 2.015

No units OK, wrong units lose last mark

(ii) $\text{moles (NH}_4\text{)SO}_4 = \text{moles H}_2\text{SO}_4 = 0.310 \text{ (1)}$

Allow consequential wrong moles in part (i) if clear $\text{H}_2\text{SO}_4 = (\text{NH}_4)\text{SO}_4$

Wrong formula for $(\text{NH}_4)\text{SO}_4$ CE=0

$M_r (\text{NH}_4)\text{SO}_4 = 132.1 \text{ (1)}$

Allow (132)

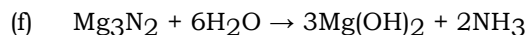
$\text{mass} = \text{moles} \times M_r = 0.0310 \times 132.1 = 4.10 \text{ (1)}$

if moles of $(\text{NH}_4)\text{SO}_4$ not clear CE

(g) wrong unit loses mark

Allow 4.09 – 4.1 – 4.11

6



Formulae (1)

Balanced equation (1)

2

[16]

18. (a) $\frac{\text{average mass of an entity (1)}}{\text{mass of 1 atom of } ^{12}\text{C}} \times 12 \text{ (1)}$

2

(b) simplest ratio of atoms of each element in a compound

1

(c) (i) $\%O = 65.7 \text{ (1)}$

$\text{C:H:O} = \frac{32.9}{12} : \frac{1.40}{1} : \frac{65.7}{16} \text{ (1)}$

$= 1.96 : 1 : 2.93$

$= \text{C}_2 \text{H}_2 \text{O}_3 \text{ (1)}$

(ii) $\text{C}_4 \text{H}_2 \text{O}_6 \text{ (1)}$

4

(d) (i) $\frac{\text{mass}}{M_r} = \frac{1000}{16} = 62.5 \text{ (1)}$

(1)

0.0625 scores one

(ii) $pV = nRT \text{ (1)}$

(1)

$V = \frac{nRT}{P} = \frac{2 \times 62.5 \times 8.31 \times 298}{100000} \text{ (1)}$

$= 3.10 \text{ m}^3 \text{ (1)}$

(allow consequential marking but if factor of 2 in missing max = 2)

6

[13]

19. Avogadro's number of molecules (1)

allow 6 to 6.1×10^{23} molecules OR No. of atoms in 12g of ^{12}C

1

[1]

20. (a) Avogadro's

1

(b) (i) $pV=nRT$ (1)

allow $V=\frac{nRT}{P}$ etc., $pV=\frac{m}{Mr}RT$

(ii) $V=\frac{nRT}{P}$

$$= \frac{1 \times 8.31 \times 298}{100000} \text{ (1)}$$

$$= 0.0248 \text{ m}^3 \text{ (1)}$$

(allow 0.0247 to 0.025)

allow use of 100 for P if ans given as 24.8 dm^3

(iii) $n = \frac{pV}{RT} = \frac{500000 \times 0.005}{8.31 \times 273} \text{ (1)}$

(treat 500 as AE-1 other values 1^s 2mar

0.005/1000 loses 1st two marks

$$= 1.10 \text{ mol (1)}$$

(allow 1.1 to 1.11)

$$M_r(\text{CO}_2) = 44 \text{ (1)}$$

treat wrong M_r as AE-1

$$\text{mass} = 44 \times 1.10 = 48.5 \text{ (g) (1)}$$

(allow 48-49)

(note can use calculation involving $\frac{P_1V_1}{T}$ and 22.4 dm^3 instead of 1st

mark - requires same accuracy for full marks)

7

(c) moles $\text{HCl} = \frac{100}{1000} \times 5.0 = 0.5(0) \text{ (mol) (1)}$

moles $\text{H}_2 = \frac{\text{moles HCl}}{2} \text{ (1)} = 0.25 \text{ (mol)}$

(if no factor of 2 CE = 0 from here)

mass $\text{H}_2 = 0.25 \times 2 = 0.5(0) \text{ (g) (1)}$

3

(d) (i) $\square\text{O} = 44.4 \text{ (1)}$

(if incorrect $\square\text{O}$, AE-1)

(if $\square\text{O}$ omitted can score max 1 for FeC_2)

$$\text{ratio Fe:C:O} = \frac{38.9}{55.8} : \frac{16.7}{12.0} : \frac{44.4}{16.0} \text{ (if use At, CE)}$$

$$= 1 : 2 : 4$$

Fe C_2 O_4 (1)

(ii) CO (1)

(mark independent of d (ii))

4

[15]

21. (a) moles of $\text{NH}_3 = \frac{20.0}{17.0} \text{ (1)}$

moles of $\text{HNO}_3 = \text{moles of } \text{NH}_3 = 1.18$

- volume $= \frac{1.18}{2}$ (1)
 $= 0.588$ (1) dm³ (1) 4
- (b) (i) $PV = nRT$ (1)
 $n = \frac{PV}{RT} = \frac{95000 \times 0.0500}{8.31 \times 298}$ (1)
 $= 1.92$ mol (1)
- (ii) Moles of ammonium nitrate $\frac{1.92}{1.5}$ (1) 1.28 mol
 Mass of ammonium nitrate mass = 1.28 × 80 (1) = 102 g (1) 6 [10]
22. $\frac{12}{1.99 \times 10^{-23} [[1]]} = 6.03 \times 10^{23}$ (1) 2
23. (a) mass ratio of O₂:KClO₃ = 96 / 245.2 (= 0.392)
 or moles of O₂ = 3/2 × 1.20 × 10⁻² or equivalent (1)
 mass of O₂ = 0.392 × 1.47 = 0.576g (1) 2
- (b) moles O₂ = 1.00/24.00 (= 0.0417 mol) (1)
 moles KClO₃ = 2/3 moles O₂ = 0.0417 × 2/3 = (0.0278 mol) (1)
 mass KClO₃ = 0.0278 × 122.6 = 3.41g (1) 3
 give one mark for $pV = nRT$ to get n for O₂,
 then second and third mark as above
 or 2/3 mole of KClO₃ → 1 Mole of O₂ (1)
 $81.7\text{g} \times 24 \text{ dm}^3 \text{ O}_2$ (1)
 $\frac{81.7}{24} \text{ g} = 3.41\text{g} \rightarrow 1 \text{ dm}^3 \text{ O}_2$ (1)
 penalise other than 2 – 5 sig. figs. once in (d)(i) and (ii)
 penalise missing or wrong units once in (d)(i) and (ii) [5]
24. (a) Mass of each element in the compound (1) 1
 (b) Number of atoms of each element in a molecule (1) 1
 (c) (i) Mr Ba(NO₃)₂ = 261 (1)
 moles Ba(NO₃)₂ = $\frac{5}{261}$ (1) = 0.0192
 moles gas = 2 $\frac{1}{2}$ × 0.0192 (1) = 0.0479
 $V = \frac{nRT}{p}$ (1) = $\frac{0.0479 \times 8.31 \times 298}{100000}$
 $= 1.19 \times 10^{-3} \text{ m}^3$ (1)
 (ii) moles HCl = 2 × 0.0192 (1) = 0.0384
 vol HCl = $\frac{0.0384}{1.2} = 0.032 \text{ dm}^3$ (1) (or 32 cm³) 7
25. percentage by mass of oxygen = 38.0% (1)
 ratio of elements = 42.9/12 : 2.4/1 : 16.7/14 : 38.0/16 (1)
 $= 3.6 : 2.4 : 1.2 : 2.4$
 empirical formula is C₃H₂NO₂ (1)
 molecular formula is C₆H₄N₂O₄ (1) 4 [4]
- 26.

- (a) %H = 18.8 (1)
 $B : H = \frac{81.2}{10.8} : \frac{18.8}{1}$ (1)
 = 2 : 5 : B₂H₅ (1)
- (b) B₄H₁₀ (1) 4 [4]
27. (a) mean mass of an entity (or molecule) (1) $\times 12$ (1)
 mass of 1 atom of ¹²C 2
- (b) $\frac{12}{1.993 \times 10^{-23}} = 6.02 \times 10^{23}$ (1)
 (allow 6.020 to 6.023) 1 [3]
28. (b) (i) %F = 42.8 (1)
 $I : F = \frac{57 \cdot 2}{127} = \frac{42 \cdot 8}{19}$ (1)
 = IF₅ (1)
- (ii) I₂ + 5F₂ → 2IF₅ (1) (or $\times \frac{1}{2}$) 4 [4]
29. C 22.24/12 = 1.85 H 3.71/1 = 3.71 Br 74.05/79.9 = 0.927 (1)
 ratio C:H:Br = 2:4:1 therefore C₂H₄Br (1)
 empirical mass = 107.9 ∴ mol formula = 215.8/107.9 $\times$ C₂H₄Br = C₄H₈Br₂ (1)
 must use % to justify answer
 or
 C (22.24/100) $\times$ 215.8 = 47.99 i.e. 48/12 = 4 carbon atoms (1)
 H (3.71/100) $\times$ 215.8 = 8.01 i.e. 8/1 = 8 hydrogen atoms (1)
 Br (74.05/100) $\times$ 215.8 = 159.8 i.e. 159.8/79.9 = 2 bromine atoms (1)
 or
 C (48/215.8) $\times$ 100 = 22.24% (1)
 H (8/215.8) $\times$ 100 = 3.71 % (1)
 Br (159.8/215.8) $\times$ 100 = 74.05% (1) 3 [3]
30. (a) mean(1) mass of an atom $\times 12$ (1)
 mass 1 atom of ¹²C 2
- (b) Multiply by Avogadro's number 1
- (c) (i) Moles NaH = $\frac{1}{24}$ (1)
 moles NaOH = Moles NaH = 0.0417 (1)
 = 0.167 mol dm⁻³ (1)
 (allow 0.166 – 0.168)
- (ii) pV = nRT (1)
 $= \frac{0.0417 \times 8.31 \times 293}{100000}$ (1) allow consequential
 = 0.00102 m³ (1)
 (allow 0.00101 – 0.00103) (allow in dm³ or cm³)
- (iii) vol HCl = $\frac{\text{moles}}{0.112} \times 1000 = \frac{0.00417}{0.112} \times 1000$ (1)
 = 37.3 cm³ (1)
 (allow 37.1 to 38.0 and conseq.) 8 [11]
31. (a) (i) 4.86×10^{-3} 1

- (ii) 2.43×10^{-3} 1
(mark conseq on (a)(i))
- (iii) 2.43×10^{-2} 1
(mark conseq on (a)(ii))
- (iv) $3.01/2.43 \times 10^{-2}$ 1
(mark conseq on (a)(iii))
- 124 1
(Do not allow 124 without evidence of appropriate calculation in (a)(iii))
- (b) $M_r(\text{Na}_2\text{CO}_3) = 106$ 1
 $M_r(x\text{H}_2\text{O}) = 250 - 106 = 144$ (mark conseq on M1) 1
 $x = 8$ (mark conseq on M2) 1
(Penalise sf errors once only)
- (c) (i) $PV = nRT$ 1
(ii) Moles Ar = $325/39.9 = 8.15$ 1
(accept $M_r = 40$)
- $P = nRT/V = (8.15 \times 8.31 \times 298)/5.00 \times 10^{-3}$ 1
 $= 4.03 \times 10^6 \text{ Pa}$ or $= 4.03 \times 10^3 \text{ kPa}$ 1
 $\text{Range} = 4.02 \times 10^6 \text{ Pa}$ to $4.04 \times 10^6 \text{ Pa}$
(If equation incorrectly rearranged, M3 & M4 = 0 If $n = 325$, lose M2)
(Allow M1 if gas law in (ii) if not given in (i))
- [12]**
- 32.** (a) (i) $pV = nRT$ **(1)**
(ii) Moles ethanol = $n = 1.36/46 (=0.0296 \text{ mol})$ **(1)**
 $V = nRT/p = \frac{0.0296 \times 8.31 \times 366}{100000}$ **(1)**
if $V = p/nRT$ lose M3 and M4
 $= 8.996 \times 10^{-4} \text{ (m}^3\text{)}$ **(1)**
 $= 899 \text{ (900) cm}^3$ **(1)** range = 895 – 905 5
If final answer = 0.899 award (2 + M1); if = 0.899 dm^3 or if = 912 award (3 + M1)
Note: If 1.36 or 46 or 46/1.36 used as number of moles (**n**) then M2 and M4 not available
Note: If pressure = 100 then, unless answer = 0.899 dm^3 , deduct M3 and mark consequentially
- (b) (i) $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Mg(OH)}_2 + 2\text{NH}_3$ **(1)**
(ii) Moles $\text{NH}_3 = \frac{0.263}{17} (=0.0155 \text{ mol})$ **(1)**
Number of molecules of $\text{NH}_3 = 0.0155 \times 6.02 \times 10^{23}$ **(1)** [mark conseq]
 $= 9.31 \times 10^{21}$ **(1)** [range 9.2×10^{21} to 9.4×10^{21}] 4
Conseq (**min** 2 sig fig)
- (c) Moles NaCl = $800/58.5$ (= 13.68) **(1)**
Moles of $\text{NaHCO}_3 = 13.68$ **(1)**
Moles of $\text{Na}_2\text{CO}_3 = 13.68/2 = 6.84$ **(1)**
Mass of $\text{Na}_2\text{CO}_3 = 6.84 \times 106 = 725 \text{ g}$ **(1)** [range = 724 – 727] 4
[1450 g (range 1448 – 1454) is worth 3 marks]
Accept valid calculation method, e.g. reacting masses or calculations via the mass of sodium present. Also, candidates may deduce a direct 2:1 ratio for $\text{NaCl}:\text{Na}_2\text{CO}_3$
- [13]**

33. (a) (i) $75.0 \times 10^{-3} \times 0.500 = 0.0375$ (mol) **(1)**
accept 0.037 or 0.038
- (ii) $21.6 \times 10^{-3} \times 0.500 = 0.0108$ (mol) **(1)**
accept 0.011
If both (i) and (ii) answers wrong, allow ONE process mark for both correct processes
- (iii) $\frac{0.0375 - 0.0108}{2} = 0.0267$ (mol) **(1)**
Not conseq – must use figures shown
- (iv) Moles of $\text{MgCO}_3 = 0.0267/2 = 0.01335$ (mol) **(1)**
allow 0.0134 - 0.0133
Mass of $\text{MgCO}_3 = 0.01335 \times 84.3$ **(1)**
allow 84
mark conseq on moles MgCO_3
 $= 1.125\text{g}$ **(1)**
accept 1.13g
mark conseq
Percentage $\text{MgCO}_3 = 1.125/1.25 \times 100$ **(1)**
mark conseq (check for inversion)
 $= 90\%$ **(1)**
mark conseq
range = 89.5 - 90.5%
If % expression inverted, lose M4 and M5
- (b) (i) % oxygen = 38.0 **(1)**
 $\text{Na} = 36.5/23$ $\text{S} = 25.5/32(.1)$ $\text{O} = 38.0/16$ **(1)**
 $= 1.587$ $= 0.794$ $= 2.375$
 $= 2:1:3$ **(1)**
If no % of oxygen Max 1 (allow M2 only)
If % for Na and S transposed, or atomic numbers used, M1 only available
- (ii) $\text{Na}_2\text{SO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{SO}_2$ **(1)**
allow multiples
allow $\text{SO}_3^{2-} + 2\text{H}^+ \rightarrow \text{H}_2\text{O} + \text{SO}_2$
- [12]**
34. (a) simplest ratio of atoms of each element in a compound **(1)**
- (b) mass of O = 1.842 g **(1)**
 $\text{Ca} : \text{N} : \text{O} = \frac{0.769}{40} : \frac{0.539}{14} : \frac{1.842}{16}$ **(1)**
 $= 1:2:6$
 CaN_2O_6 **(1)**
- (c) Mr **(1)**
- (d) (i) $n = \frac{PV}{RT}$ **(1)** (allow $PV = nRT$)
 $= \frac{95 \times 10^3 \times 1.53 \times 10^{-3}}{8.31 \times 293}$ **(1)**
 $= 0.0597$ **(1)** (allow 0.059 to 0.06)
- (ii) $0.0597 \times 4 = 0.239$ **(1)** (allow 0.24)
(allow conseq on (i))
- (iii) moles $\text{NH}_3 = 0.293 \times \frac{25}{1000} = 0.00597$ **(1)**

$$\text{volume} = 1000 \times \frac{0.00597}{0.15} = 39.8 \text{ (cm}^3\text{)} \text{ (1)}$$

(allow conseq
allow 0.0398 dm³)

6

35. (i) 0.0268 **OR** 0.027 **OR** 0.02675 mol ✓
NO OTHER ACCEPTABLE ANSWER

1

(ii) 1.61×10^{22} ✓

ALLOW 1.6×10^{22} up to calculator value

ALLOW

ECF answer to (i) $\times 6.02 \times 10^{23}$

ALLOW any value for N_A in the range:

$$6.0 \times 10^{23} - 6.1 \times 10^{23}$$

1

[2]

36. (i) $0.438/87.6 = 5.00 \times 10^{-3} / 0.00500 \text{ mol}$ ✓

1

(ii) $0.00500 \times 24.0 = 0.120 \text{ dm}^3$ ✓ (accept 120 cm^3)

1

(iii) $0.00500 \times 1000/200 = 0.0250 \text{ mol dm}^{-3}$ ✓

1

[3]

37. (i) Molar mass CaO = $56.1 \text{ (g mol}^{-1}\text{)}$ ✓ (anywhere)

2

$$\text{moles CaO} = \frac{1.50}{56.1} = 0.0267/0.027 \text{ ✓ calc: } 0.0267379$$

Allow 56 which gives 0.0268

(ii) moles HNO₃ = 2×0.0267

$$= 0.0534 \text{ or } 0.0535 / 0.053 \text{ mol ✓}$$

(i.e. answer to (i) $\times 2$)

$$\text{volume of HNO}_3 = \frac{0.0534(\text{or } 5) \times 1000}{2.50} = 21.4 \text{ cm}^3 \text{ ✓}$$

2

calc from value above = 21.3903743

If 0.053 mol, answer is 21 cm^3 but accept 21.2 cm^3

If 0.054 mol, answer is 22 cm^3 but accept 21.6 cm^3

[4]

38. Calculates correctly:

$$\text{Mol of Sr(NO}_3)_2 = \frac{5.29}{211.6} = 0.0250 \text{ ✓}$$

ALLOW 0.025

Calculates correctly:

$$\text{Mol of gas} = 5/2 \times 0.0250 = 0.0625 \text{ ✓}$$

ALLOW ECF for first answer $\times 2.5$ as calculator value or correct rounding to 2 significant figures or more but ignore trailing zeroes

Calculates correctly:

$$\text{Volume of gas} = 24.0 \times 0.0625 = 1.50 \text{ dm}^3 \text{ ✓}$$

ALLOW ECF for second answer $\times 24(.0)$ as calculator value or correct rounding to 2 significant figures or more but ignore trailing zeroes

DO NOT ALLOW ECF of first answer $\times 24(.0)$ (which gives $0.6(0) \text{ dm}^3$) as this has not measured the volume of any gas, simply 0.0250 mol of solid Sr(NO₃)₂ converted into a gas i.e. This answer would give **one** mark

ALLOW 1.5 dm^3

ALLOW ECF producing correct volume of NO_2 only

i.e. $1.2(0) \text{ dm}^3$ would give **two** marks

OR

ALLOW ECF producing correct volume of O_2 only

i.e. $0.3(0) \text{ dm}^3$ would give **two** marks

3

[3]

39. (i) $12 \times 50/1000 = 0.600 \text{ mol}$ ✓ 1
 (ii) $4 \text{ mol HCl} \rightarrow 1 \text{ mol Cl}_2$ / moles $\text{Cl}_2 = 0.15 \text{ mol}$ ✓
 $\text{vol of Cl}_2 = 0.15 \times 24 = 3.60 \text{ dm}^3$ ✓ 2
2nd mark is consequential on molar ratio given

[3]

40. (i) Mass Sb_2S_3 in stibnite = 5% of 500 kg = 25.0 kg ✓
 $\text{Moles Sb}_2\text{S}_3 = \frac{25.0 \times 10^3}{340} / 73.5 / 73.529 / 73.53 / 74 \text{ mol}$ ✓
 (calculator value: 73.52941176)
 If 5% is not used, 1471 mol; ecf for 2nd mark
 (calculator value: 1470.588235)
 If 5% is used 2nd, 73.6 mol: OK for both marks 2
 (ii) moles Sb = $2 \times 73.5 \text{ mol}$ ✓
 mass Sb = $2 \times 73.5 \times 122 \text{ g} = 17.9 \text{ kg}$ ✓
 If the 2 isn't used, answer = $73.5 \times 122 = 8.95$ ✓
ecf ans from (i) x 2
ecf ans above x 2

OR

% Sb = $244/340 = 71.7\%$ ✓

mass Sb = $25.0 \times 71.7/100 = 17.9 \text{ kg}$ ✓ (ecf as above)

2

[4]

41. (a) (i) Calculate correctly $\frac{0.0880 \times 25.0}{1000} = 2.20 \times 10^{-3} \text{ mol}$
OR 0.00220 mol ✓
ALLOW 0.0022 **OR** $2.2 \times 10^{-3} \text{ mol}$ 1
 (ii) Calculates correctly $\frac{0.00220}{2} = 1.10 \times 10^{-3} \text{ mol}$
OR 0.00110 mol ✓
ALLOW 0.0011 **OR** $1.1 \times 10^{-3} \text{ mol}$
ALLOW ECF for answer (i)/2 as calculator value or correct rounding to 2 significant figures or more but ignore trailing zeroes 1
 (iii) $\frac{0.00110 \times 1000}{17.60} = 0.0625 \text{ mol dm}^{-3}$
OR $6.25 \times 10^{-2} \text{ mol dm}^{-3}$ ✓
ALLOW 0.063 **OR** $6.3 \times 10^{-2} \text{ mol dm}^{-3}$
ALLOW ECF for answer (ii) $\times 1000/17.60$
OR
 ECF from (i) for answer (ii)/2 $\times 1000/17.60$ as calculator value or correct rounding to 2 significant figures or more but ignore trailing zeroes 1
 (b) (i) (The number of) Water(s) of crystallisation ✓
IGNORE hydrated **OR** hydrous

(ii) 142.1 ✓	1	[6]
ALLOW 142 ALLOW M_r expressed as a sum ALLOW ECF from incorrect M_r and x is calculated correctly	2	
$x = \frac{(322.1 - 142.1)}{18.0} = 10 \checkmark$ ALLOW ECF values of x from nearest whole number to calculator value ALLOW 2 marks if final answer is 10 without any working	2	
42. (i) $M(\text{MgSO}_4) = 120.4$ OR $120 \text{ (g mol}^{-1}\text{)} \checkmark$	2	
$\text{mol MgSO}_4 = \frac{1.51}{120.4} = 0.0125 \text{ mol} \checkmark$ ALLOW 0.013 up to calculator value of 0.012541528 correctly rounded (from $M = 120.4 \text{ g mol}^{-1}$) ALLOW 0.013 up to calculator value of 0.012583333 correctly rounded (from $M = 120 \text{ g mol}^{-1}$) ALLOW ecf from incorrect M i.e. $1.51 \div M$	2	
(ii) $\frac{1.57}{18.0} = 0.0872(2) \text{ (mol)} \checkmark$ ALLOW 0.09 up to calculator value of 0.08722222	1	
(iii) $\times = 7 \checkmark$ ALLOW ecf i.e. answer to (ii) $\div$ answer to (i) ALLOW correctly calculated answer from 1 significant figure up to calculator value, i.e. $\times$ does not have to be a whole number. Likely response = 6.95 ✓	1	[4]
43. (i) Simplest (whole number) ratio of atoms/moles/elements ✓ (ii) ratio Rb : Ag : I = $7.42/85.5 : 37.48/108 : 55.10/127$ or $0.0868 : 0.347 : 0.434$ or $1 : 4 : 5 \checkmark$ = $\text{RbAg}_4\text{I}_5 \checkmark$	1	
44. (i) moles Ti = 1.44/47.9 = 0.0301 mol/0.03 mol 1	2	[3]
(ii) mass of Cl = $5.70 - 1.44 = 4.26 \text{ g} \checkmark$ moles Cl = $4.26/35.5 = 0.120 \text{ mol} \checkmark$ $5.70/35.5 = 0.161 \text{ mol}$ gets 1 mark (iii) Ti:Cl = $0.0301 : 0.12 = 1:4$. Empirical formula = $\text{TiCl}_4 \checkmark$ $0.0301 : 0.161 \text{ mol}$ gives TiCl_5 for 1 mark (iv) $\text{Ti} + 2\text{Cl}_2 \rightarrow \text{TiCl}_4 \checkmark$ (ecf possible from (iii) covalent ✓	2	
45. (i) a proton donor ✓ (ii) $\text{MgO} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{O} \checkmark$	1	[5]
46. (i) $\text{CaCO}_3 (\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O} (\text{l})$ $\text{CaO}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{aq}) + \text{H}_2\text{O} (\text{l})$	1	
	1	[2]

- each balanced equation 1 mark (2)
all state symbols (1) 3
- (ii) CaCO_3 fizzes/ gas given off/ gas evolved / carbon dioxide evolved (1) 1
- [4]**
47. (i) The H^+ ion in an (nitric) acid has been replaced by a metal ion
OR by a Ca^{2+} ion ✓
DO NOT ALLOW it has been produced by the reaction of an acid and a base as this is stated in the question.
IGNORE references to replacement by NH_4^+ ions or positive ions.
ALLOW H **OR** Hydrogen for H^+ ;
DO NOT ALLOW Hydrogen atoms
ALLOW Ca **OR** Calcium for Ca^{2+} .
DO NOT ALLOW Calcium atoms
ALLOW 'metal' for 'metal ion' 1
- (ii) $2\text{HNO}_3(\text{aq}) + \text{Ca}(\text{OH})_2(\text{aq}) \rightarrow \text{Ca}(\text{NO}_3)_2(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$
Formulae ✓
Balance **AND** states ✓
ALLOW multiples
ALLOW (aq) **OR** (s) for $\text{Ca}(\text{OH})_2$ 2
- (iii) Accepts a **proton** **OR** accepts H^+ ✓
ALLOW $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$
ALLOW OH^- reacts with H^+ **OR** OH^- takes H^+
ALLOW OH^- 'attracts' H^+ if 'to form water' is seen
DO NOT ALLOW OH^- neutralises H^+ ('neutralises' is in the question) 1
- [4]**
48. (i) $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$ ✓
ALLOW $2\text{NH}_4\text{OH} + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4 + 2\text{H}_2\text{O}$
ALLOW $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$
ALLOW any correct multiple
IGNORE state symbols 1
- (ii) when the H^+ in an acid is replaced by a metal ion **OR** an ammonium ion **OR** a + ion ✓
ALLOW H for H^+ ;
ALLOW 'metal' for 'metal ion'
i.e.: H in an acid can be replaced by a metal 1
- (iii) accepts a proton **OR** accepts H^+ ✓
ALLOW donates a lone pair
ALLOW removes H^+
ALLOW forms OH^- ions 1
- (iv) 132.1 ✓
IGNORE units
NO OTHER ACCEPTABLE ANSWER 1
- [4]**

2.0 ATOMIC STRUCTURE

(Includes some questions from Chemical Periodicity)

1. (a) Copy and complete the following table.

	Relative mass	Relative charge
Neutron		
Electron		

(2)

- (b) An atom has twice as many protons as, and four more neutrons than, an atom of ${}^9\text{Be}$. Deduce the symbol, including the mass number, of this atom.

(2)

(Total 4 marks)

2. (a) Define the terms

- (i) *mass number* of an atom,
(ii) *relative molecular mass*.

(3)

- (b) (i) Write the electron arrangement for a copper atom.
(ii) Identify the block in the Periodic Table to which copper belongs.
(iii) Deduce the number of neutrons in one atom of ${}^{65}\text{Cu}$.

(3)

- (c) A sample of copper contains the two isotopes ${}^{63}\text{Cu}$ and ${}^{65}\text{Cu}$ only. It has a relative atomic mass, A_r , less than 64. The mass spectrum of this sample shows major peaks with m/e values of 63 and 65, respectively.

- (i) Explain why the A_r of this sample is less than 64.
(ii) Explain how Cu atoms are converted into Cu^+ ions in a mass spectrometer.
(iii) In addition to the major peaks at $m/e = 63$ and 65, much smaller peaks at $m/e = 31.5$ and 32.5 are also present in the mass spectrum. Identify the ion responsible for the peak at $m/e = 31.5$ in the mass spectrum. Explain why your chosen ion has this m/e value and suggest **one** reason why this peak is very small.

(6)

(Total 12 marks)

3. A sample of iron from a meteorite was found to contain the isotopes ${}^{54}\text{Fe}$, ${}^{56}\text{Fe}$ and ${}^{57}\text{Fe}$.

- (a) The relative abundances of these isotopes can be determined using a mass spectrometer. In the mass spectrometer, the sample is first vaporised and then ionised.
(i) State what is meant by the term *isotopes*.
(ii) Explain how, in a mass spectrometer, ions are detected and how their abundance is measured.

(5)

- (b) (i) Define the term *relative atomic mass* of an element.
(ii) The relative abundances of the isotopes in this sample of iron were found to be as follows.

m/e	54	56	57
Relative abundance (%)	5.8	91.6	2.6

Use the data above to calculate the relative atomic mass of iron in this sample. Give your answer to one decimal place.

(4)

- (c) (i) Give the electron arrangement of an Fe^{2+} ion.
 (ii) State why iron is placed in the d block of the Periodic Table.
 (iii) State the difference, if any, in the chemical properties of isotopes of the same element. Explain your answer.

(4)

(Total 13 marks)

4. (a) Copy and complete the following table.

Particle	Relative charge	Relative mass
Proton		
Neutron		
Electron		

(3)

- (b) An atom of element **Z** has two more protons and two more neutrons than an atom of ${}^{34}_{16}\text{S}$. Give the symbol, including mass number and atomic number, for this atom of **Z**.

(2)

- (c) Write the electronic configurations for the sulphur atom, S, and the sulphide ion, S^{2-} .

(2)

- (d) State the block in the Periodic Table in which sulphur is placed and explain your answer.

(2)

(Total 9 marks)

5. (a) Define the term *atomic number* of an element.

(1)

- (b) Give the symbol, including mass number and atomic number, for an atom of an element which contains 12 neutrons and 11 electrons.

(2)

- (c) In terms of s and p sub-levels, give the electronic configuration of an aluminium atom.

(1)

- (d) How many neutrons are there in one ${}^{27}\text{Al}$ atom?

(1)

- (e) Define the term *relative atomic mass* of an element.

(2)

- (f) Parts (i) to (iv) below refer to the operation of a mass spectrometer.

- (i) Name the device used to ionise atoms in a mass spectrometer.
 (ii) Why is it necessary to ionise atoms before acceleration?
 (iii) What deflects the ions?
 (iv) What is adjusted in order to direct ions of different mass to charge ratio onto the detector?

(4)

- (g) A meteorite was found to contain three isotopes of element **X**.
 A mass spectrometer gave the following information about these isotopes.

m/e	24.0	25.0	26.0
Relative abundance	64.2	20.3	15.5

- (i) Calculate the relative atomic mass of **X**.
 (ii) Using the Periodic Table, suggest the most likely identity of element **X**.
 (iii) Suggest one reason why the relative atomic mass of **X**, given in the Periodic Table, differs from your answer to part (g)(i).

(5)

(Total 16 marks)

6. (a) The diagram in **Figure 1** shows the behaviour of the three fundamental particles when passed through an electric field.

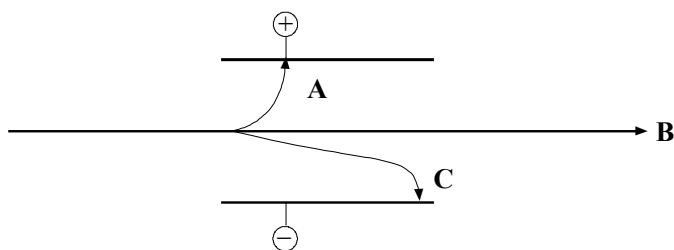


Figure 1

- (i) Identify the particles represented by **A**, **B**, and **C**. (1)
- (ii) Explain the shapes and directions of the paths traced by the fundamental particles as they pass through the electric field. (3)
- (Total 4 marks)**
7. (a) Define the term *atomic number* of an atom. (1)
- (b) Explain why atoms of the same element may have different mass numbers. (1)
- (c) The table below concerns a sample of krypton. (1)

Mass number	82	83	84	86
Relative abundance	12	12	50	26

- (i) Name an instrument which is used to measure the relative abundance of isotopes.
- (ii) Define the term *relative atomic mass* of an element.
- (iii) Calculate the relative atomic mass of this sample of krypton. (5)
- (d) Explain why the first ionisation energy of rubidium is less than the first ionisation energy of krypton. (2)
- (Total 9 marks)**

8. (a) Explain the terms:
- (i) *mass number*; (1)
- (ii) *relative atomic mass*. (2)
- (b) Sulphur consists of three isotopes. The table below shows the relative abundance of each isotope.

Mass number of isotope	32	33	34
Relative abundance/%	95.0	0.8	4.2

Figure 1

Using the data from **Figure 1**, calculate the relative atomic mass, A_r of sulphur, giving your final answer to 1 decimal place.

(2)
(Total 5 marks)

9. (a) Define the term *mass number* of an isotope.

(1)

- (b) Write the symbol, including mass number and atomic number, for the isotope which has eight electrons and nine neutrons in each atom.

(2)

- (c) The table below shows some data about fundamental particles.

Particle	Proton	neutron	Electron
Mass /g	1.6725×10^{-24}	1.6748×10^{-24}	0.0009×10^{-24}
Relative charge			

- (i) Complete the table by giving a value for the relative charge of each particle.
 (ii) Calculate the mass of an atom of hydrogen which is made from a proton and an electron.
 (iii) Calculate the mass of one mole of such hydrogen atoms giving your answer to four decimal places.
 (The Avogadro constant, $L = 6.0225 \times 10^{23} \text{ mol}^{-1}$)
 (iv) An accurate value for the mass of one mole of hydrogen atoms is 1.0080 g. Give one reason why this value is different from your answer to part (c)(iii).

(4)

(Total 7 marks)

10. (a) State, in terms of the fundamental particles present, the meaning of the term *isotopes*.

(1)

- (b) An atom contains one more proton than, but the same number of neutrons as, an atom of ^{36}S . Deduce the symbol, including the mass number and the atomic number, of this atom.

(2)

- (c) The table below gives the relative abundance of each isotope in a mass spectrum of a sample of germanium, Ge.

m/e	70	72	74
Relative abundance (%)	24.4	32.4	43.2

- (i) Give the electron arrangement of a Ge atom.
 (ii) Use the data above to calculate the relative atomic mass of this sample of germanium. Give your answer to one decimal place.
 (iii) State what is adjusted in a mass spectrometer in order to direct ions with different m/e values onto the detector. Explain your answer.
 (iv) One of the isotopes of Ge, given in the table in part (c), has an ion that forms a small peak in the mass spectrum which is indistinguishable from a peak produced by $^{36}\text{S}^+$ ions. Identify this Ge ion and explain your answer.

(8)

(Total 11 marks)

11. Aluminium, magnesium and vanadium are metals.

- (a) Give the electronic configurations for aluminium and vanadium.

(2)

- (b) State the block in the Periodic Table to which magnesium belongs.

(1)

(Total 3 marks)

12. (a) Give the symbol, including mass number and atomic number, for the isotope which has a mass number of 34 and which has 18 neutrons in each nucleus

(2)

- (b) Some data obtained from the mass spectrum of a sample of carbon are given below.

Ion	$^{12}\text{C}^+$	$^{13}\text{C}^+$
Absolute mass of one ion/g	1.993×10^{-23}	2.158×10^{-23}
Relative abundance/%	98.9	1.1

Use these data to calculate a value for the mass of one neutron, the relative atomic mass of ^{13}C and the relative atomic mass of carbon in the sample. You may neglect the mass of an electron.

(6)

(Total 8 marks)

13. (a) Describe the process by which particles are ionised in a mass spectrometer.

(2)

- (b) Give two reasons why particles must be ionised before being analysed in a mass spectrometer.

(2)

- (c) A sample of boron contains 20% by mass of ^{10}B and 80% by mass of ^{11}B . Calculate the relative atomic mass of boron in this sample.

(2)

(Total 6 marks)

14. (a) Give the following to show the electronic configuration of silicon.

(1)

- (b) Write chemical equations, including state symbols, for the following enthalpy changes:

(i) The first molar ionisation energy of silicon;

(ii) The second molar ionisation energy of silicon.

(3)

(Total 4 marks)

15. **Figure 1** contains data relating to the relative isotopic abundance of the element titanium, Ti.

Isotope	^{46}Ti	^{47}Ti	^{48}Ti	^{49}Ti	^{50}Ti
% abundance	8.02	7.31	73.81	5.54	5.32

Figure 1

- (a) Explain what is meant by the term *relative isotopic abundance*.

(2)

- (b) Using the data from **Figure 1**, calculate the relative atomic mass, A_r , of titanium.

(2)

(Total 4 marks)

16. (a) Write the electron arrangement for the Mg^{2+} ion.

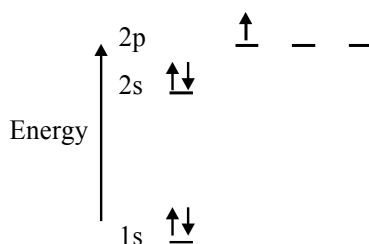
(1)

- (b) Identify the block in the Periodic Table to which magnesium belongs.

- (c) The Ne atom and the Mg^{2+} ion have the same number of electrons. Give **two** reasons why the first ionisation energy of neon is lower than the third ionisation energy of magnesium. (1)
- (d) There is a general trend in the first ionisation energies of the Period 3 elements, Na to Ar. (2)
- State and explain this general trend.
 - Explain why the first ionisation energy of sulphur is lower than would be predicted from the general trend.

(5)
(Total 9 marks)

17. The diagram below shows the electronic structure of boron.



- The electrons are represented by arrows. What property of the electrons do these 'up' and 'down' arrows represent? (1)
- Suggest why electrons which occupy the 2p sub-levels have a higher energy than electrons in the 2s sub-level. (1)
- Explain the meaning of the term *first ionisation energy*. (2)
- Explain why boron has a lower first ionisation energy than beryllium. (3)
- Explain why the first ionisation energy of helium is very large. (1)

(Total 8 marks)

18. (a) (i) State the general trend in the first ionisation energy of the Period 3 elements from Na to Ar. (4)
- State how, and explain why, the first ionisation energy of aluminium does not follow this general trend.

- Give the equation, including state symbols, for the process which represents the second ionisation energy of aluminium. (1)

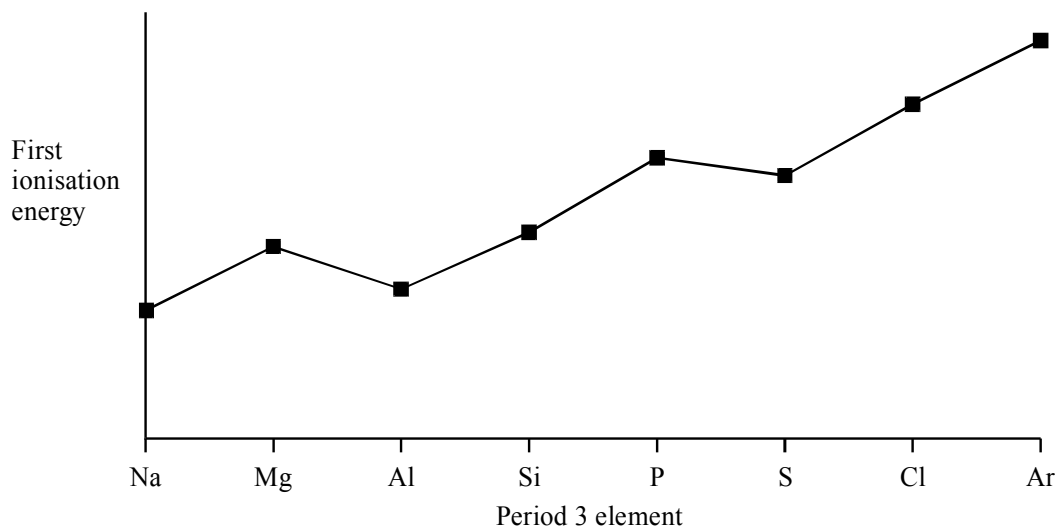
(Total 5 marks)

19. There is a general trend in the values of the first ionisation energies of the elements Na to Ar. The first ionisation energies of the elements Al and S deviate from this trend.

- Write an equation, including state symbols, to represent the process for which the energy change is the first ionisation energy of Na. (2)
- State and explain the general trend in the values of the first ionisation energies of the elements Na to Ar. (3)
- State how, and explain why, the values of the first ionisation energies of the elements Al and S deviate from the general trend. (5)

(Total 10 marks)

20. (a) What is meant by the term *first ionisation energy*? (2)
- (b) The diagram below shows the variation in first ionisation energy across Period 3.



- (i) What is the maximum number of electrons that can be accommodated in an s sub-level?
- (ii) What evidence from the diagram supports your answer to part (b(i))?
- (iii) What evidence from the diagram supports the fact that the 3p sub-level is higher in energy than the 3s?
- (iv) What evidence from the diagram supports the fact that no more than three unpaired electrons can be accommodated in the 3p sub-level?

(5)

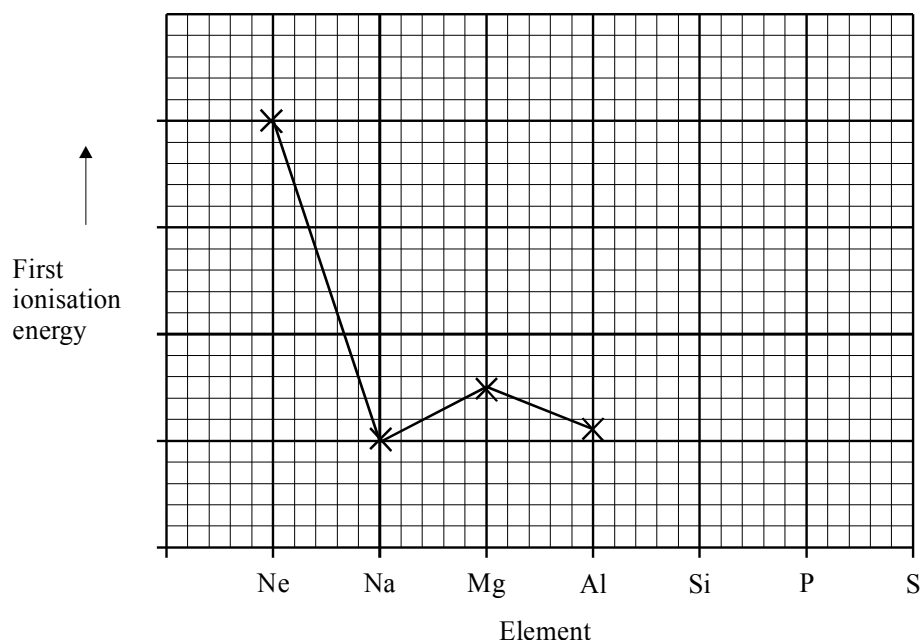
(Total 7 marks)

21. Explain why atomic radius decreases across Period 3 from sodium to chlorine.

(2)

(Total 2 marks)

22. The diagram below shows the trend in the first ionisation energies of the elements from neon to aluminium.



- (a) Draw crosses on the graph to show the first ionisation energies of silicon, phosphorus and sulphur.

- (b) Write an equation to illustrate the process which occurs during the first ionisation of neon. (3)
- (c) Explain why the first ionisation energy of neon and that of magnesium are both higher than that of sodium. (1)
- (d) Explain why the first ionisation energy of aluminium is lower than that of magnesium. (4)

(2)
(Total 10 marks)

23. Values for the covalent radii of the elements in Period 3 are given in the table below.

Elements	Na	Mg	Al	Si	P	S	Cl	Ar
Covalent radius/nm	0.157	0.136	0.125	0.117	0.110	0.104	0.099	-

Explain the decrease in the values shown in the table.

(Total 3 marks)

24. (a) State the relative charge and relative mass of a proton, of a neutron and of an electron.
In terms of particles, explain the relationship between two isotopes of the same element.
Explain why these isotopes have identical chemical properties. (7)
- (b) Define the term *relative atomic mass*. An element exists as a mixture of three isotopes.
Explain, in detail, how the relative atomic mass of this element can be calculated from data obtained from the mass spectrum of the element. (7)

(Total 14 marks)

25. (a) Ionisation is the first of the four main stages involved in obtaining the mass spectrum of a sample of gaseous titanium atoms. Explain how ionisation is achieved. Name the remaining three stages and, in each case, state how each stage is achieved. Explain why it would be difficult to distinguish between $^{48}\text{Ti}^{2+}$ and $^{24}\text{Mg}^+$ ions using a mass spectrometer. (10)
- (b) State any differences and similarities in the atomic structure of the isotopes of an element. State the difference, if any, in the chemistry of these isotopes. Explain your answer. (4)
- (c) The table below gives the percentage abundance of each isotope in the mass spectrum of a sample of titanium.

m/z	46	47	48	49	50
% abundance	8.02	7.31	73.81	5.54	5.32

Define the term *relative atomic mass* of an element. Use the above data to calculate the value of the relative atomic mass of titanium in this sample. Give your answer to two decimal places.

(4)
(Total 18 marks)

26. (a) Describe, in terms of charge and mass, the properties of protons, neutrons and electrons. Explain fully how these particles are arranged in an atom of ^{14}N . (6)
- (b) Account for the existence of isotopes. (2)
- (c) The mass spectrum of an element has peaks with relative intensity and m/z values shown in the table below.

m/z	80	82	83	84	86
Relative intensity	1	5	5	25	8

Identify this element and calculate its accurate relative atomic mass

(4)

(Total 12 marks)

27. A sample of element **Q** was extracted from a meteorite. The table below shows the relative abundance of each isotope in a mass spectrum of this sample of **Q**.

m/z	64	66	67	68
Relative abundance (%)	38.9	27.8	14.7	18.6

- (a) Define the term *relative atomic mass* of an element. (2)
- (b) Use the data above to calculate the relative atomic mass of this sample of **Q**. Give your answer to one decimal place. Suggest the identity of **Q**. (3)
- (c) In order to obtain a mass spectrum of **Q**, a gaseous sample is first ionised. Describe how ionisation is achieved in a mass spectrometer. Give **three** reasons why ionisation is necessary. (5)

(Total 10 marks)

28. Describe and explain the variation in first ionisation energy of the elements across Period 3 from sodium to argon.

(Total 9 marks)

Marking Scheme

1. (penalty for sig fig error = 1 mark per question)
- (a) neutron: relative mass = 1 relative charge = 0 1
(not 'neutral')
- electron: relative mass = $1/1800 \rightarrow 0$ /negligible or $5.56 \times 10^{-4} \rightarrow 0$ relative charge = -1 1
- (b) $^{17}\text{O}/\text{O}^{17}$ mass number (Do not accept 17.0) 1
- oxygen symbol 'O' 1
(if 'oxygen' + — 'mass number = 17'(1))
(if 'oxygen'+ — 'mass number = 17'(0))
(if at N^0 given but $\neq 8$, treat as 'con' for M2)
(if lp on Be, diagram = 0)
(ignore bond angles)
(not dot and cross diagrams)
- [4]**
2. (a) (i) p + n / number of nucleons (accept protons and neutrons) 1
(Incorrect reference to electrons = contradiction)
- (ii) Mean / average mass of a molecule/entity/formula 1
 $1/12^{\text{th}}$ mass of atom of ^{12}C 1
[Not $1/12^{\text{th}}$ mass of molecule of ^{12}C] (mark independently)
- OR** Mass of 1 mole of molecules/entities (1)
 $1/12^{\text{th}}$ mass of 1 mole of ^{12}C (1)
- OR** Average mass of a molecule/entity (1)
Relative to the mass of a ^{12}C atom taken as 12 / 12.000 (1)
(Mean/ average = stated or explained)
(mass = stated or explained)
(Penalise 'weight' once only)
(Ignore 'average ' mass of ^{12}C)
(Do not allow 'mass of average molecule)
- (b) (i) $2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$ (accept $3d^9 4s^2$) 1
(accept subscripts or caps)
[Penalise missing shell numbers]
- (ii) d / D [NOT 3d/ 'transition element'] 1
- (iii) 36 [NOT 36.0] 1
- (c) (i) More ^{63}Cu atoms than ^{65}Cu atoms 1
(idea of more abundant ^{63}Cu isotope - NOT just reference to peak heights)
- (ii) Electron from electron gun / high speed electron / high energy electron 1
(accept electron gun fired at)
[NOT 'bombarded with electrons']
- knock electron off (Cu atom) / idea of loss of e^- / appropriate equation 1
(Mark independently)

(iii)	$^{63}\text{Cu}^{2+}$ or equivalent	[NOT 63.0 - penalise this error once only]	1
	$m/z = 63/2 (=31.5)$ or equivalent		1
	More energy needed to remove second electron OR		1
	$^{63}\text{Cu}^{2+}$ statistically less likely to remove second electron		
	(Idea that not many $^{63}\text{Cu}^{2+}$ ions formed OR explains why few are formed e.g. more energy needed)		
	If ^{63}Cu not given, can only award M2 & M3		
Notes on	[If 65 used, lose M1 and M2]		
(c)	(iii)	[If mass number missing from identity but appears in explanation, penalise M1 but allow M2 if earned]	
[12]			
3.	(a)	(i)	(atoms with the) same number of protons / same atomic number / atoms of the same element; 1 (molecules = contradiction) But different number of neutrons / different mass number; 1 (not different atomic mass or A_r)
		(ii)	detected by: +ve ions collide with / are directed or deflected to / are collected at the detector; 1 causing current to flow / detected electrically / idea of electricity or voltage generated; 1 (not 'charge produced' or 'detected electronically') abundance measured: idea that current depends on abundance/number of ions hitting detector; 1
	(b)	(i)	<u>mean /average mass of an atom / all the isotopes;</u> $1/12^{\text{th}}$ mass of atom of ^{12}C ; (mark independently) OR <u>mass of 1 mole of atoms (of an element);</u> $1/12^{\text{th}}$ mass of 1 mole of ^{12}C ; OR average mass of a molecule/entity; relative to the mass of a ^{12}C atom taken as 12 / 12.000; 2 (penalise 'weight' once only) (ignore 'average' mass of ^{12}C) (do not allow 'mass of average atom')
		(ii)	$\frac{(54 \times 5.8) + (56 \times 91.6) + (57 \times 2.6)}{100};$ 1 $= 55.9;$ 1
	(c)	(i)	$1s^2 2s^2 2p^6 3s^2 3p^6;$ 1 (accept subscripts or caps; ignore 4s°) (penalise missing shell numbers)
		(ii)	highest energy level / last sub-shell to be filled / is (3)d; OR outermost electrons in the d sub-shell/orbital; 1 (not incomplete d sub-shell) (not valance electron in d sub-shell)
		(iii)	no difference; 1 OR same e^- arrangement / same number of e^- / same valence e^- .

OR same chemical properties;

chemical properties determined by electrons;
(M2 tied to correct answer for M1)

1

[13]

4. (a)

Particle	Relative charge	Relative mass	
Proton	+1 or 1+	1	(1)
Neutron	0 or no charge/neutral/zero	1 (<u>not</u> – 1)	(1)
Electron	–1 or 1–	1/1800 to 1/2000	(1)

or negligible

or zero

or 5.0×10^{-4} to 5.6×10^{-4}

**if 'g' in mass column - wrong
penalise once**

3

(b) ${}_{18}^{38}\text{Ar}$
(1)(1)

Allow numbers before or after Ar

2

(c) S: $1s^2 2s^2 2p^6 3s^2 3p^4$ **(1)**

Allow upper case letters

S^{2-} : $1s^2 2s^2 2p^6 3s^2 3p^6$ **(1)**

If use subscript penalise once

2

(d) Block: p **(1)**

Explanation: Highest energy or outer orbital is (3) p

OR outer electron, valency electron in (3) p

NOT 2p etc.

2

[9]

5. (a) number of protons in one atom or nucleus **(1)**

Allow protons & electrons

do not allow protons + electrons or electrons

1

(b) ${}_{11}^{23}\text{Na}$ **(1)**

OR Na^{23}_{11} **or** Na (1) + unambiguous statement of mass no. and
atomic no.

2

(c) $1s^2 2s^2 2p^6 3s^2 3p^1$ **(1)**

Allow Ne $3s^2 3p^1$

1

(d) 14 **(1)**

1

- (e) $\frac{\text{average mass of an atom (or isotope)}}{\text{mass of 1 atom of } ^{12}\text{C}}$ (1) $\times 12$ (1)
Reference to mass number not mass C.E. = 0
OR stated in moles
OR compared with 1/12 of a ^{12}C atom or relative to ^{12}C when taken as 12
2
- (f) (i) electron gun (1)
(ii) (particles must be charged) (ions) before attraction to a charged plate (or electric field) (1)
(or only ions can be attracted or accelerated by an electric field)
or converse; if not charge not attracted to electric field
(iii) magnetic field (or magnet) (1)
(iv) magnetic field (1) (accelerating potential or strength of magnet)
allow magnet
4
- (g) (i) $\frac{24.0 \times 64.2 + 25.0 \times 20.3 + 26.0 \times 15.5}{100}$ (1)
(1) mark for any $m/z \times \text{relative abundance}$
If numerator is correct but 100 has A.E. conseq A.E. -1
If A.E. on 100 allow conseq correct answer provided numerator is correct
= 24.5 (1)
Allow 24.5 to 24.52
ignore units
(ii) magnesium (1) (or Mg) (allow conseq on wrong A_r)
(iii) abundance of isotopes is different (1) **(or different isotopes)**
5
[16]
6. (a) (i) **A** = electron, **B** = neutron and **C** = proton **all** correctly identified (1)
do not give if any reference to cation / anion 1
(ii) neutron / n^0 / **B** has no charge and so is undeflected (1)
proton / p^+ / **C** and electron / e^- / **A** attracted to -ve and +ve plates respectively (1)
correct direction and shape for one particle gains one mark
mass $p > \text{mass } e^-$ so deflected less / mass **C** > mass **A** so deflected less (1)
allow this mark for valid difference in property and explanation of shape
allow reference to other particles here 3
[7]
7. (a) number of protons (1) 1
(b) different number of neutrons (1)
they are 'isotopes' – not enough for the mark 1
(c) (i) mass spectrometer (1)
allow *spectroscope*
(ii) (or average)

$$\frac{\text{mean mass of an atom (1)}}{(\text{mass}) 1 \text{ atom of } ^{12}\text{C}} \times 12 \text{ (1)}$$

$$\text{or } \frac{\text{mass of 1 mol of atoms (1)}}{\text{mass of 1 mol of } ^{12}\text{C atoms}} \times 12 \text{ (1)}$$

or mean mass of an atom (1)

or compared with {an atom of ^{12}C taken as 12

{ $\frac{1}{12}$ of a ^{12}C atom

If molecule, element or entity instead of 'atom' lose 1 mark

1

$$\text{(iii)} \quad \frac{(82 \times 12) + (83 \times 12) + (84 \times 50) + (86 \times 26)}{100} \quad \text{(1)} = 84.16 \quad \text{(1)}$$

allow 84.1 to 84.2 ignore units

5

- (d) outer electron in Rb is in 5th shell (or additional shell) (1)
further away (or more shielded) from nucleus (1)

mark independently but if there is contradiction – no marks

2

[9]

8. (a) (i) the number of protons and neutrons / number of nucleons
in the nucleus / atom
not in an element 1
- (ii) weighted average / mean mass of the isotopes or of
atoms **not** of an element / average mass in an isotopic mixture /
average mass of all naturally occurring isotopes (1)
measured on the ^{12}C scale / relative to $1/12$ th the mass of one
atom of the ^{12}C isotope (1)
independent of first mark 2

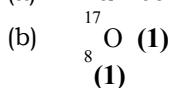
$$\text{(b)} \quad \frac{32 \times 95 + 33 \times 0.8 + 34 \times 4.2}{100} \quad \text{(1)}$$

32.1 (ignore units) – answer only gets both marks (1)

2

9. (a) Number of protons & number of neutrons (1)

1



2

- (c) (i)

Particle	proton	neutron	Electron
Mass /g	1.6725×10^{-24}	1.6748×10^{-24}	0.0009×10^{-24}
Relative charge	+1	0	-1

(ii) 1.6734×10^{-24} g (1)

(iii) 1.0078 g (1)

- (iv) other isotopes present (1)

4

[13]

10. (a) Atoms/isotopes/particles/species with the same (number of) protons and
different (number of) neutrons
[Not atomic number/mass number/molecules/same
element/diff electrons]

1

(b)	$^{37}_{17}\text{Cl}$	17 & Cl	1
		Mass number	1
		[Not 37.0] [Mark independently] [ignore charges]	
(c)	(i)	$2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^2$ [allow reversed $4s^2 3d^{10}$] [allow capitals/subscripts]	1
	(ii)	$A_r = \frac{(70 \times 24.4) + (72 \times 32.4) + (74 \times 43.2)}{100}$ [Wrong approach or not dividing by 100 = CE = 0] $= 72.4$ [Answer to 1 d.p.] [Mark conseq on transcription error]	1
	(iii)	Magnet/electromagnet/magnetic field / electric field/charge on negative/accelerator plate Correct link between deflection and m/z Correct link between deflection and field [Penalise 'reflected'/'diffracted' once only] [Ignore references to molecules/atoms/particles] [Consolation mark: allow correct link between mass and deflection for 1 mark out of the 2]	1 1 1
	(iv)	$^{72}\text{Ge}^{2+}$ <u>only</u> <u>Same</u> m/z as $^{36}\text{S}^+$ [Mark independently]	1 1
11.	(a)	aluminium: $1s^2 2s^2 2p^6 3s^2 3p^1$ (1) vanadium: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$ or $3d^3 4s^2$ (1)	2
	(b)	s-block	1
12.	(a)	$^{34}_{16}\text{S}$ (1) $^{16}_{16}$ (1)	2
	(b)	1.65×10^{-24} (1) $\frac{2.158 \times 10^{-23}}{1.993 \times 10^{-23}} \times 12$ (1) = 12.99 (1) or $6.023 \times 10^{23} \times 2.158 \times 10^{-23}$ = 12.998 (1) any use of ^{12}C & ^{13}C (1) $\frac{12 \times 98.9}{100} + \frac{13 \times 1.1}{100}$ (1) = 12.01 (1)	6
13.	(a)	Electron gun (1) knocks out electron(s) from the particle (1)	2

- (b) Reason 1 acceleration (1)
Reason 2 deflection (1) 2
- (c) $\frac{20}{100} \times 10 + \frac{80}{100} \times 11$ (1)
= 10.8 (1) 2
- [6]**
14. (a) $(1s^2 2s^2) 2p^6 3s^2 3p^2$ (1) 1
(b) $\text{Si(g)} \rightarrow \text{Si}^+(\text{g}) + \text{e}^-$ balanced (1)
 $\text{Si}^+(\text{g}) \rightarrow \text{Si}^{2+}(\text{g}) + \text{e}^-$ balanced (1)
correct state symbols in both equations (allow even if not balanced) (1) 3
- [4]**
15. (a) proportion / ratio / frequency / percentage / abundance of each isotope / different type of atom / specific atom
not amount unless relative amount compared to total amount (1)
present in the (natural) isotopic mixture / sample of the element / compound containing the element (1)
reference to relative atomic mass worth zero marks 2
(b) $(0.0802 \times 46) + (0.0731 \times 47) + (0.7381 \times 48)$
 $+ (0.0554 \times 49) + (0.0532 \times 50)$ (1)
= 47.9 (ignore units) (1) 2
- [4]**
16. (a) $1s^2 2s^2 2p^6$ (accept capitals and subscripts) 1
(b) 's' block (not a specific 's' orbital – e.g. 2s) 1
(c) Mg²⁺ ion smaller than Ne atom / $\text{Mg}^{2+} \text{e}^-$ closer to nucleus 1
(Not 'atomic' radius fo Mg^{2+})
Mg²⁺ has more protons than Ne / higher nuclear charge or 1
e⁻ is removed from a charged Mg^{2+} ion / neutral neon atom
(accept converse arguments)
(If used 'It' or Mg/magnesium/ Mg^{3+} etc. & 2 correct reasons, allow (1))
(d) (i) trend: increases (if 'decreases', CE = 0/3) 1
Explⁿ: more protons / increased proton number /
increased nuclear charge 1
(NOT increased atomic number)
same shell / same shielding / smaller size 1
(ii) **QoL** reference to the e^- pair in the 3p sub-level 1
(penalise if wrong shell, e.g. '2p', quoted)
repulsion between the e^- in this e^- pair 1
(if not stated, ' e^- pair' must be clearly implied)
(mark M4 and M5 separately)
- [9]**
17. (a) Spin (1) 1
(b) Further from nucleus (1)
(or more shielded) 1
(c) Energy to remove 1 electron (1) (or 1 mol electrons)
from a gaseous atom (1) (or molecule or 1 mol of atoms/molecules) 2

- (d) Be's outer electron is in an s (2s) orbital (1)
 B's outer electron is in a p (2p) orbital (1)
 B's outer electron is higher in energy (1) 3
 (e) Electron is not shielded from nucleus (1) 1 [8]
18. (a) (i) increases; 1
 (ii) lower than expected / lower than Mg / 1
 less energy needed to ionise; e^- removed from (3)p sub-level; 1
 (e^- removed' may be implied)
 of higher energy / further away from nucleus / shielded by 3s e^- s; 1
 (b) $Al^+(g) \rightarrow Al^{2+}(g) + e^-$; 1
19. (a) $Na(g) \rightarrow Na^+(g) + e^-$
 OR $Na(g) + e^- \rightarrow Na^+(g) + 2e^-$
 (-) on electron not essential
 equation (1)
 state symbols (1)
 Ignore state symbols on electrons
 2
 (b) Trend : Increases (1)
 Explanation : Increased nuclear charge or proton number (1)
 Stronger attraction (between nucleus and (outer) e^-) (1)
 Trend wrong
 Allow M2 only if M3 correct (con)
 3
 (c) How values deviate from trend: (both values) too low (1)
 Explanation for Al: e^- removed from (3) p (1)
 e^- or orbital is higher in energy level or better
 shielded than (3)s
 or p electron is shielded by 3s electrons (1)
 Allow e^- is further away
 Mark independently
 Explanation for S: e^- removed from (3)p electron pair (1)
 repulsion between paired e^- (reduces energy required) (1)
 Mark separately
 If deviation wrong allow M2 and M4
 If M3 and / or M5 right 0 (con)
 If used 'd' rather than 'p' orbital - lose M2 + M4 but may get M3, M5
 (explanation marks)
 5
 20. (a) Heat / enthalpy / energy for removal of one electron (1) from a gaseous atom (1)
 can score in an equation
 must have first mark to score the second
 2
 (b) (i) 2 (1)
 (ii) Two elements (or Na / Mg) before the drop (in energy) to Al (1)

(iii) ionisation energy of Al < that for Mg **(1)**

(iv) fall in energy from P to S **(1)**
or discontinuity in trend

From Al to P there are 3 additional electrons **(1)**
or three elements

5

For second mark idea of block of 3 elements

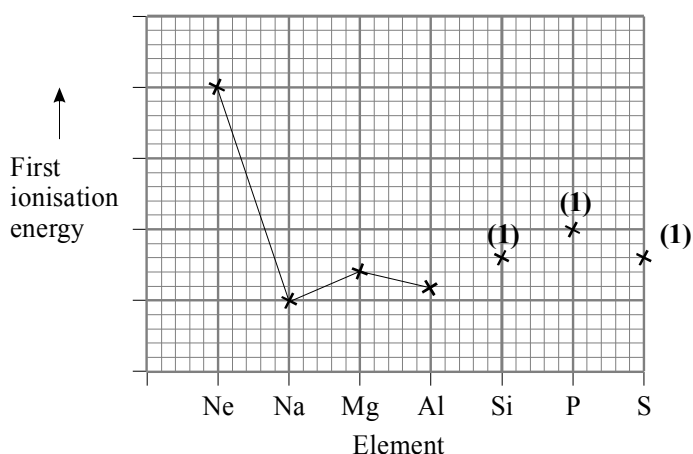
[7]

21. Number of protons increases **(1)**
 Electrons in same shell **(1)** (or same shielding)

2

[2]

22. (a)



3

(b) $\text{Ne} \rightarrow \text{Ne}^+ + \text{e}^-$ **(1)**

1

(c) *Explanation for neon* Neon's electron is in a lower (2p) shell **(1)**
 attracted more strongly to (or less shielded from) the nucleus **(1)**
Explanation for magnesium more protons **(1)**
 electrons in same shell
 or similar shielding **(1)**

4

(d) Al's outer electron is in a 3p sub-shell **(1)**

higher in energy than 3s in Magnesium **(1)**

2

[10]

23. increased nuclear charge / nuclear attraction number of protons **(1)**
 same shielding / electrons added to same or outer shell / increase in number **(1)**
 of electrons in outer shell
 therefore (outer) electrons attracted / pulled in more strongly or more closely **(1)**
 'increased effective nuclear charge' worth 2 marks

[3]

24. (a) Proton: mass 1, charge + 1 **(1)**
 Neutron: mass 1, charge 0 **(1)**
 Electron 1/1840, charge -1 **(1)**
Allow mass = 0, or negligible, or 1/1800 to 1/2000
 Isotopes have the same number of protons **(1)**
OR atomic number

different number of neutrons (1)

Isotopes have the same electronic configuration (1)

OR same number of electrons

Chemical properties depend on electrons

7

average(1) mass of an atom/isotopes

$$(b) \frac{\text{mass of 1 atom of } ^{12}\text{C}}{\text{average(1) mass of an atom/isotopes}} \times 12 \text{ (1)}$$

$$\text{OR } \frac{\text{mass of 1 mol of atoms}}{\text{mass of 1 atom of } ^{12}\text{C}} \times 12 \text{ or in words}$$

Spectrum gives (relative) abundance (1)

OR % or amount

And m/z (1)

Multiply m/z by relative abundance for each isotope (1)

Allow instead of m/z mass no, A_r or actual value from example

Sum these values (1)

Divide by the sum of the relative abundances (1)

only award this mark if previous 2 given

Max 2 if e.g. has only 2 isotopes

7

[14]

25. (a) Ionisation (1)

High speed **or** high energy electrons **or** electron gun (1)

NOT bombard

NOT beam or stream of electrons

Knocks out (outer) electron (1)

Forming positive ion - could be from $\text{Ti} \rightarrow \text{Ti}^+ + \text{e}^-$ (1)

Accept + ion later in question to clarify charge of ion

$\text{Ti} + \text{e}^- \rightarrow \text{Ti}^+ + 2 \text{e}^-$ worth 2 marks

Ignore state symbols

Acceleration (1)

By electric field **or** attraction to negative plate **or** electrostatic attraction (1)

NOT repelled by + plate

Allow passed through positive & negative plates / oppositely charged plates

Not just charged plates

Deflection (1)

By magnetic field or magnet **or** electromagnet (1)

Detection (1)

Idea that ions collected at detector and generate current (1)

Both ions have the same m/z value (of 24) **or** valid arguments

in terms of the doubled charge on $^{48}\text{Ti}^{2+}$ exactly counteracting its doubled mass (1)

Deflected equally (so detected together) **or** deflection dependent on m/e value (1)

Can't get this from previous section

10

(b) Differ in mass number **or** number of neutrons (1)

Same proton / atomic number (1)

Ignore reference to electrons here

Isotopes have the same chemical properties (1)

because all have the same electron configuration **or** number of electrons

or same number of valence electrons (so no chemical difference) (1)

This mark is tied to the above mark or near miss [similar etc] in M3

- 4
- (c) Mean mass of an atom or (isotope) **(1)** [NOT mass of average atom]
 Relative to $1/12$ mass of ^{12}C atom atc. **Or** to ^{12}C taken as 12.000 or exactly 12 **(1)**
Isotope can be accepted
OR $\frac{\text{mean (average) mass of an atom}}{\text{mass of an atom of } ^{12}\text{C}}$ **(1) × 12 (1)**
OR $\frac{\text{mass of 1 mol of atoms}}{\text{mass of 1 mol of } ^{12}\text{C}}$ **(1) × 12 (1)**
 $A_r = (46 \times .0802) + (47 \times .0731) + (48 \times .7381) + (49 \times .0554) + (50 \times .0532)$ **(1)**
 $= 47.93$ answer to 2 d.p **(1)**
47.92 is acceptable
Must be 5 sets of values
Ignore transcription errors BUT DON'T ignore missing 100 C.E.
If missing isotope C.E. **4**
- [18]**
26. (a) protons mass +1 charge + **(1)**
 neutrons mass +1 charge zero **(1)**
 electrons mass ~ zero (or <1/1800), charge - **(1)**
 nitrogen -14 has a nucleus with 7 protons and 7 neutrons **(1)**
 surrounded by electrons in shells (or orbitals) (or 2,5) **(1)**
 with configuration $1s^2 2s^2 2p^3$ (scores last 2 marks) **(1)** 6
- (b) isotopes have the same atomic number (or same number of protons or same element) **(1)**
 but different number of neutrons (or mass number) **(1)** 2
- (c) fraction of each component is rel. int./44 **(1)**
 [the mark is for 44, if 42 max 1]
 relative atomic mass
 $= 1 \times 80/44 + 5 \times 82/44 + 5 \times 83/44 + 25 \times 84/44 + 8 \times 86/44$ **(1)**
 $= 83.9$ **(1)**
 Krypton **(1)** 4
- [12]**
27. (a) Mean (average) mass of an atom / all the isotopes or 1
 $1/12^{\text{th}}$ mass of atom of ^{12}C 1
Mass of 1 mole of atoms of an element or
 $1/12^{\text{th}}$ mass of 1 mole of ^{12}C
 average mass of an atom / all the isotopes
 Relative to the mass of a ^{12}C atom taken as exactly 12 / 12.000
 (penalise 'weight' once only) (ignore 'average' mass of ^{12}C)
 (not 'mass of average atom')
- (b) $A_2 = (64 \times 0.389) + (66 \times 0.278) + (67 \times 0.147) + (68 \times 0.186)$ 1
 $= 65.7$
 (mark M2 conseq on transcription error or incorrect addition of %) 1
 identity: zinc / Zn 1
 (Conseq on A_2 but only if their A_2 is within range of Periodic Table)
- (c) electron gun (fires) electrons or high speed/energy electrons 1
 (not just 'bombarded by electrons' or 'bombarded by electron gun')
- knocks off e^- from **Q** 1
 (may be earned from a real or generic equation)

Reasons: to allow ions to be:	1
accelerated (by an electric field)	
deflected (by a magnet/magnetic field)	1
detected / description of current formed at the detector/sensor	1
<i>(accept in any order)</i>	
<i>(allow clear descriptions of 'accelerated', 'deflected', 'detected')</i>	

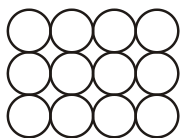
[10]**28. general increase across period (1)***can score this and points 4, 7 on graph*because number of protons (or nuclear charge) increases **(1)**but electrons in same shell (or similar shielding) **(1)**fall from Mg to Al **(1)**Al's outer electron is in a *p* orbital **(1)**higher in energy than *s* electron in Mg **(1)**
*if wrong period -1*fall from *P* to *S* **(1)**two of the *p* electrons in *S* are paired (or in same orbital) **(1)****[9]**

3.0 CHEMICAL BONDING & 4.0 STATES OF MATTER

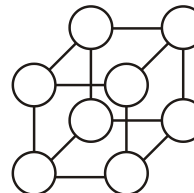
(Includes some questions from Chemical Periodicity)

1. At room temperature, both sodium metal and sodium chloride are crystalline solids which contain ions.

(a) On the diagrams for sodium metal and sodium chloride below, mark the charge for each ion.



Sodium metal

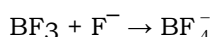


Sodium chloride

- (b) (i) Explain how the ions are held together in solid sodium metal. (2)
 (ii) Explain how the ions are held together in solid sodium chloride.
 (iii) The melting point of sodium chloride is much higher than that of sodium metal. What can be deduced from this information? (3)
- (c) Compare the electrical conductivity of solid sodium metal with that of solid sodium chloride. Explain your answer. (3)
- (d) Explain why sodium metal is malleable (can be hammered into shape). (1)

(Total 9 marks)

2. The equation below shows the reaction between boron trifluoride and a fluoride ion.



- (i) Draw diagrams to show the shape of the BF_3 molecule and the shape of the BF_4^- ion. In each case, name the shape. Account for the shape of the BF_4^- ion and state the bond angle present.
- (ii) In terms of the electrons involved, explain how the bond between the BF_3 molecule and the F^- ion is formed. Name the type of bond formed in this reaction.

(Total 9 marks)

3. Draw the shape of a molecule of BeCl_2 and the shape of a molecule of Cl_2O . Show any lone pairs of electrons on the central atom. Name the shape of each molecule.



(4)

(Total 4 marks)

4. Ammonia, NH_3 , reacts with sodium to form sodium amide, NaNH_2 , and hydrogen.

- (a) Draw the shape of an ammonia molecule and that of an amide ion, NH_2^- . In each case show any lone pairs of electrons.



- (b) State the bond angle found in an ammonia molecule.
 (c) Explain why the bond angle in an amide ion is smaller than that in an ammonia molecule.

(5)

(Total 5 marks)

5. (a) Describe the bonding that is present in metals. (3)
 (b) Explain how the bonding and structure lead to the typical metallic properties

of electrical conductivity and malleability.

(4)

- (c) Suggest a reason why aluminium is a better conductor of electricity than magnesium.

(2)

(Total 9 marks)

6. The table below shows the electronegativity values of some elements.

	Fluorine	Chlorine	Bromine	Iodine	Carbon	Hydrogen
Electronegativity	4.0	3.0	2.8	2.5	2.5	2.1

- (a) Define the term *electronegativity*.

(2)

- (b) The table below shows the boiling points of fluorine, fluoromethane (CH_3F) and hydrogen fluoride.

	F-F	$ \begin{array}{c} \text{F} \\ \\ \text{C} \\ / \quad \quad \backslash \\ \text{H} \quad \text{H} \quad \text{H} \end{array} $	H-F
Boiling point/K	85	194	293

- (i) Name the strongest type of intermolecular force present in:
 Liquid F_2
 Liquid CH_3F
 Liquid HF
 (ii) Explain how the strongest type of intermolecular force in liquid HF arises.

(6)

- (c) The table below shows the boiling points of some other hydrogen halides.

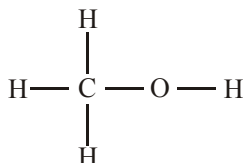
	HCl	HBr	HI
Boiling point / K	188	206	238

- (i) Explain the trend in the boiling points of the hydrogen halides from HCl to HI .
 (ii) Give **one** reason why the boiling point of HF is higher than that of all the other hydrogen halides.

(3)

(Total 11 marks)

7. (a) Methanol has the structure



Explain why the O-H bond in a methanol molecule is polar.

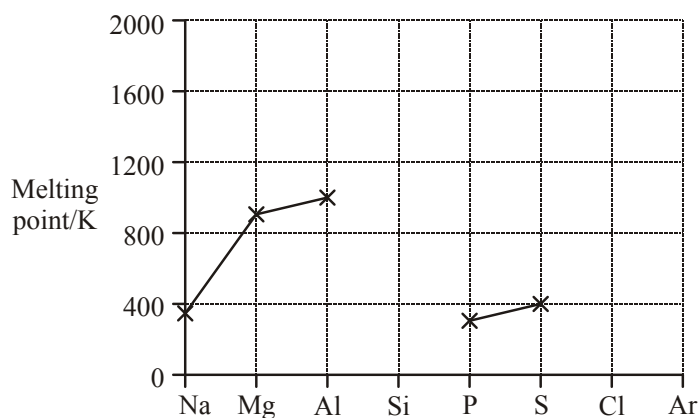
(2)

- (b) The boiling point of methanol is $+65^\circ\text{C}$; the boiling point of oxygen is -183°C . Methanol and oxygen each have an M_r value of 32. Explain, in terms of the intermolecular forces present in each case, why the boiling point of methanol is much higher than that of oxygen.

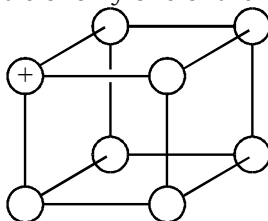
(3)

(Total 5 marks)

8. (a) The diagram below shows the melting points of some of the elements in Period 3.



- (i) On the diagram, use crosses to mark the approximate positions of the melting points for the elements silicon, chlorine and argon. Complete the diagram by joining the crosses.
- (ii) By referring to its structure and bonding, explain your choice of position for the melting point of silicon.
- (iii) Explain why the melting point of sulphur, S₈, is higher than that of phosphorus, P₄
- (b) State and explain the trend in melting point of the Group II elements Ca–Ba. (8)
- (3)
- (Total 11 marks)**
9. State and explain the trend in the melting points of the Period 3 metals Na, Mg and Al. (3)
- (Total 3 marks)**
10. (a) (i) Describe the bonding in a metal.
- (ii) Explain why magnesium has a higher melting point than sodium. (4)
- (b) Why do diamond and graphite both have high melting points? (3)
- (c) Why is graphite a good conductor of electricity? (1)
- (d) Why is graphite soft? (2)
- (Total 10 marks)**
11. Sodium sulphide, Na₂S, is a high melting point solid which conducts electricity when molten. Carbon disulphide, CS₂, is a liquid which does not conduct electricity.
- (a) Deduce the type of bonding present in Na₂S and that present in CS₂
- (b) By reference to all the atoms involved explain, in terms of electrons, how Na₂S is formed from its atoms.
- (c) Draw a diagram, including all the outer electrons, to represent the bonding present in CS₂
- (6)
- (Total 6 marks)**
12. (a) The diagram below represents a part of the structure of sodium chloride. The ionic charge is shown on the centre of only one of the ions.



- (i) On the diagram, mark the charges on the four negative ions.
- (ii) What change occurs to the motion of the ions in sodium chloride when it is

heated from room temperature to a temperature below its melting point?

(2)

- (b) Sodium chloride can be formed by reacting sodium with chlorine.
- (ii) A chloride ion has one more electron than a chlorine atom. In the formation of sodium chloride, from where does this electron come?
- (ii) What property of the atoms joined by a covalent bond causes the bond to be polar?

(2)

(Total 4 marks)

13. Phosphorus and nitrogen are in Group V of the Periodic Table and both elements form hydrides. Phosphine, PH_3 , reacts to form phosphonium ions, PH_4^+ , in a similar way to that by which ammonia, NH_3 , forms ammonium ions, NH_4^+

- (a) Give the name of the type of bond formed when phosphine reacts with an H^+ ion. Explain how this bond is formed.

(3)

- (b) Draw the shapes, including any lone pairs of electrons, of a phosphine molecule and of a phosphonium ion.
Give the name of the shape of the phosphine molecule and state the bond angle found in the phosphonium ion.



(4)

(Total 7 marks)

14. (a) State the meaning of the term *electronegativity*.

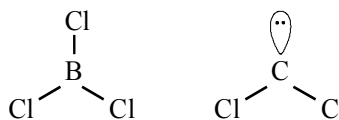
(2)

- (b) State and explain the trend in electronegativity values across Period 3 from sodium to chlorine.

(3)

(Total 5 marks)

15. (a) The shape of the molecule BCl_3 and that of the unstable molecule CCl_2 are shown below.



- (i) Why is each bond angle exactly 120° in BCl_3 ?
- (ii) Predict the bond angle in CCl_2 and explain why this angle is different from that in BCl_3

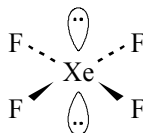
(5)

- (b) Give the name which describes the shape of molecules having bond angles of $109^\circ 28'$.

Give an example of one such molecule.

(2)

- (c) The shape of the XeF_4 molecule is shown below.



- (i) State the bond angle in XeF_4
- (ii) Suggest why the lone pairs of electrons are opposite each other in this molecule.
- (iii) Name the shape of this molecule, given that the shape describes the positions of the Xe and F atoms only.

(4)

- (d) Draw a sketch of the NF_3 molecule. Indicate in your sketch any lone pairs of electrons on nitrogen.

(2)

(Total 13 marks)

16. (a) Describe the motion of the particles in solid iodine and in iodine vapour. (3)
- (b) Explain why solid iodine vaporises when warmed gently. (2)
- (c) Silver and sodium chloride melt at similar temperatures. Give two physical properties of silver which are different from those of sodium chloride and, in each case, give one reason why the property of silver is different from that of sodium chloride. (4)
- (d) Draw the shapes of BeCl_2 , NCl_3 and BeCl_4^{2-} . In each case, show any lone-pair electrons on the central atom and state the value of the bond angle. (6)
- (Total 15 marks)**
17. Silicon dioxide has a macromolecular structure. Draw a diagram to show the arrangement of atoms around a silicon atom in silicon dioxide. Give the name of the shape of this arrangement of atoms and state the bond angle. (3)
- (Total 3 marks)**
18. (a) When considering electron pair repulsions in molecules, why does a lone pair of electrons repel more strongly than a bonding pair? (1)
- (b) The diagram below shows a hydrogen peroxide molecule.
-
- (i) On the diagram above, draw the lone pairs, in appropriate positions, on the oxygen atoms. (4)
- (ii) Indicate, on the diagram, the magnitude of one of the bond angles. (4)
- (iii) Name the strongest type of intermolecular force which exists between molecules of hydrogen peroxide in the pure liquid. (2)
- (c) Draw a diagram to illustrate the shape of a molecule of SF_4 and predict the bond angle(s). (4)
- (d) Name two types of intermolecular force which exist between molecules in liquid SF_4 . (2)
- (Total 11 marks)**
19. (a) Name the type of force that holds the particles together in an ionic crystal. (1)
- (b) What is a covalent bond? (1)
- (c) State how a co-ordinate bond is formed. (2)
- (d) Describe the bonding in a metal. (2)
- (e) A molecule of hydrogen chloride has a dipole and molecules of hydrogen chloride attract each other by permanent dipole-dipole forces. Molecules of chlorine are non-polar. (5)
- (i) What is a permanent dipole?
- (ii) Explain why a molecule of hydrogen chloride is polar.
- (iii) Name the type of force which exists between molecules of chlorine.
- (f) Show, by means of a diagram, how two molecules of hydrogen fluoride are attracted to each other by hydrogen bonding; include all lone-pair electrons and partial charges in your diagram. (3)
- (g) Why is there no hydrogen bonding between molecules of hydrogen bromide? (1)
- (Total 15 marks)**

20. The table below gives the boiling points, T_b , of some hydrogen halides.

Hydrogen halide	HF	HCl	HBr	HI
T_b /K	293	188	206	238

- (a) By referring to the types of intermolecular force involved, explain why energy must be supplied in order to boil liquid hydrogen chloride. (3)
- (b) Explain why the boiling point of hydrogen bromide lies between those of hydrogen chloride and hydrogen iodide. (2)
- (c) Explain why the boiling point of hydrogen fluoride is higher than that of hydrogen chloride. (2)
- (d) Draw a sketch to illustrate how two molecules of hydrogen fluoride interact in liquid hydrogen fluoride. (2)

(Total 9 marks)

21. Sulphur will combine separately with carbon, hydrogen and sodium to form carbon disulphide (CS_2), hydrogen sulphide (H_2S) and sodium sulphide (Na_2S) respectively. The bonding in these compounds is similar to that in CO_2 , H_2O and Na_2O .

- (a) Complete the table in **Figure 2** by classifying the compounds as either ionic or covalent.

	Melting point /K	Ionic or covalent
Carbon disulphide	162	
Hydrogen sulphide	187	
Sodium sulphide	1450	

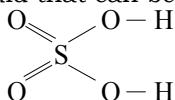
Figure 2

- (b) One of the compounds in **Figure 2** shows high electrical conductivity under appropriate conditions. Identify the compound, by name or formula, and state **one** condition under which it shows high electrical conductivity. (3)

(2)
(Total 5 marks)

22. (a) State what is meant by the term *polar bond*. (1)

- (b) Sulphuric acid is a liquid that can be represented by the formula drawn below. (1)



Given that the electronegativity values for hydrogen, sulphur and oxygen are 2.1, 2.5 and 3.5 respectively, clearly indicate the polarity of each bond present in the formula given. (2)

- (c) Suggest the strongest type of intermolecular force present in pure sulphuric acid. Briefly explain how this type of intermolecular force arises. (2)

(2)
(Total 5 marks)

23. (a) Sketch the shapes of each of the following molecules, showing any lone pairs of electrons. In each case, state the bond angle(s) present in the molecule and name the shape.

Molecule	Sketch of shape	Bond angle(s)	Name of shape
BF_3			
NF_3			
ClF_3			

- (b) State the types of intermolecular force which exist, in the liquid state, between pairs of BF_3 molecules and between pairs of NF_3 molecules. (9)
- (c) Name the type of bond which you would expect to be formed between a molecule of BF_3 and a molecule of NF_3 . Explain how this bond is able to form. (3)
- (3)
- (Total 15 marks)**
24. Sketch a diagram to show the shape of a molecule of NH_3 and indicate on your diagram how this molecule is attracted to another NH_3 molecule in liquid ammonia. (3)
- (Total 3 marks)**
25. (a) Define the term *electronegativity*. (2)
- (b) State and explain the trend in electronegativity of the elements across Period 3 from sodium to chlorine. (3)
- (c) State the bond type in sodium oxide and the bond type in sulphur dioxide. In each case, explain the link between the bond type and the electronegativity of the elements involved. (4)
- (Total 9 marks)**
26. (a) Co-ordinate bonding can be described as dative covalency. In this context, what is the meaning of each of the terms *covalency* and *dative*? (2)
- (b) Write an equation for a reaction in which a co-ordinate bond is formed. (2)
- (c) Why is sodium chloride ionic rather than covalent? (2)
- (d) Why is aluminium chloride covalent rather than ionic? (2)
- (e) Why is molten sodium chloride a good conductor of electricity? (1)
- (f) Explain, in terms of covalent bonding, why the element iodine exists as simple molecules whereas the element carbon does not. (3)
- (Total 12 marks)**
27. (a) Describe the nature and strength of the bonding in solid calcium oxide. (3)
- (b) Use the kinetic theory to describe the changes that take place as calcium oxide is heated from 25°C to a temperature above its melting point. (3)
- (c) State **two** properties of calcium oxide that depend on its bonding. (2)
- (Total 8 marks)**
28. What is a covalent bond? (1)
- (Total 1 mark)**
29. (a) **Figure 1** shows some data concerned with halogens.

Element	Electronegativity	Boiling point of hydride / K
Fluorine	4.0	293
Chlorine	3.0	188
Bromine	2.8	206
Iodine	2.5	238

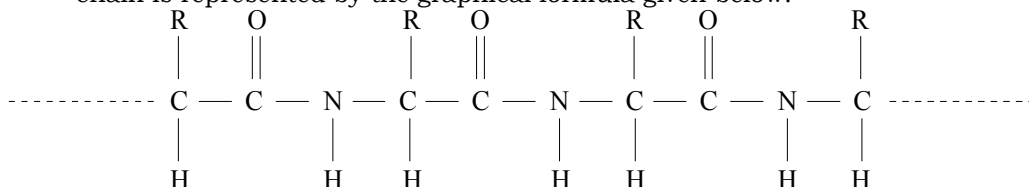
Figure 1

- (i) Define the term *electronegativity*. (2)

- (ii) Explain the trend in boiling points from hydrogen chloride to hydrogen iodide. (2)
- (iii) Explain why hydrogen fluoride does not fit this trend. (2)
- (b) The oxygen atoms in the sulphate ion surround the sulphur in a regular tetrahedral shape.
- (i) Write the formula of the ion. (1)
- (ii) State the O-S-O. bond angle. (1)
- (Total 8 marks)

30. (a) State the type of bonding in a crystal of potassium bromide. (1)
- (b) Sketch a diagram to show the shape of a BrF₃ molecule. Show on your sketch any lone pairs of electrons in the outermost shell of bromine and name the shape. (3)
- (Total 4 marks)

31. (a) (i) State **one** feature which molecules must have in order for hydrogen bonding to occur between them. (1)
- (ii) Give the name of the type of intermolecular bonding present in hydrogen sulphide, H₂S, and explain why hydrogen bonding does not occur. (2)
- (iii) Account for the much lower boiling point of hydrogen sulphide (-61 °C) compared with that of water(100 °C). (2)
- (b) Protein molecules are composed of sequences of amino acid molecules that have joined together, with the elimination of water, to form long chains. Part of a protein chain is represented by the graphical formula given below.



Explain the formation of hydrogen bonding between protein molecules.

- (4)
- (Total 9 marks)
32. (a) Describe the bonding found in metals. (3)
- (b) Use data from **table above** and your knowledge of the bonding in these metals to explain why the melting point of magnesium is higher than that of sodium. (3)
- (c) State and explain the similarities and differences in electrical conductivity of sodium, graphite and diamond. (4)

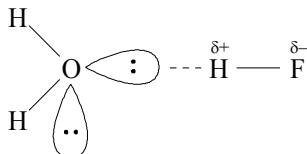
- (Total 10 marks)
33. The table below contains electronegativity values for the Period 3 elements, except chlorine.

Element	Na	Mg	Al	Si	P	S	Cl	Ar
Electronegativity	0.9	1.2	1.5	1.8	2.1	2.5		-

- (a) How can electronegativity values be used to predict whether a given chloride is likely to be ionic or covalent? (2)
- (b) State the type of bonding in sodium oxide. (1)

(Total 3 marks)

34. The diagram below shows how a water molecule interacts with a hydrogen fluoride molecule.



- What is the value of the bond angle in a single molecule of water? (1)
- Explain your answer to part (a) by using the concept of electron pair repulsion. (4)
- Name the type of interaction between a water molecule and a hydrogen fluoride molecule shown in the diagram above. (1)
- Explain the origin of the δ^+ charge shown on the hydrogen atom in the diagram. (2)
- When water interacts with hydrogen fluoride, the value of the bond angle in water changes slightly. Predict how the angle is different from that in a single molecule of water and explain your answer. (2)

(Total 10 marks)

35. (a) State which one of the elements neon, sodium, magnesium, aluminium and silicon has the lowest melting point and explain your answer in terms of the structure and bonding present in that element. (3)
- (b) State which one of the elements neon, sodium, magnesium, aluminium and silicon has the highest melting point and explain your answer in terms of the structure and bonding present in that element. (3)

(Total 6 marks)

36. Diamond and graphite are both forms of carbon. Diamond is able to scratch almost all other substances, whereas graphite may be used as a lubricant. Diamond and graphite both have high melting points. Explain each of these properties of diamond and graphite in terms of structure and bonding. Give **one** other difference in the properties of diamond and graphite. (Total 9 marks)

37. Iodine and diamond are both crystalline solids at room temperature. Identify one similarity in the bonding, and one difference in the structures, of these two solids. Explain why these two solids have very different melting points. (Total 6 marks)

38. Phosphorus exists in several different forms, two of which are white phosphorus and red phosphorus. White phosphorus consists of P_4 molecules, and melts at 44°C . Red phosphorus is macromolecular, and has a melting point above 550°C . Explain what is meant by the term *macromolecular*. By considering the structure and bonding present in these two forms of phosphorus, explain why their melting points are so different. (Total 5 marks)

39. (a) Predict the shapes of the SF_6 molecule and the AlCl_4^- ion. Draw diagrams of these species to show their three-dimensional shapes. Name the shapes and suggest values for the bond angles. Explain your reasoning. (8)
- (b) Perfume is a mixture of fragrant compounds dissolved in a volatile solvent. When applied to the skin the solvent evaporates, causing the skin to cool for a short time. After a while, the fragrance may be detected some distance away. Explain these observations. (4)

(Total 12 marks)

40. (a) Iodine and graphite crystals both contain covalent bonds and yet the physical properties of their crystals are very different.

For iodine and graphite, state and explain the differences in their melting points and in their electrical conductivities.

(9)

- (b) Draw the shape of the BeCl_2 molecule and explain why it has this shape.

(2)

(Total 11 marks)

41. (a) The table below gives the melting point for each of the Period 3 elements Na – Ar.

Element	Na	Mg	Al	Si	P	S	Cl	Ar
Melting point / K	371	923	933	1680	317	392	172	84

In terms of structure and bonding, explain why silicon has a high melting point, and why the melting point of sulphur is higher than that of phosphorus.

(7)

- (b) Draw a diagram to show the structure of sodium chloride. Explain, in terms of bonding, why sodium chloride has a high melting point.

(4)

(Total 11 marks)

42. Explain the meaning of the term *periodicity* as applied to the properties of rows of elements in the Periodic Table. Describe and explain the trends in atomic radius, in electronegativity and in conductivity for the elements sodium to argon.

(13)

(Total 13 marks)

43. (a) Describe the structure of, and bonding in, three different types of crystal. Illustrate your answer with a specific example of each type of crystal and sketch labelled diagrams of the structures. In each case, explain how the ability to conduct electricity is influenced by the type of bonding.

(18)

- (b) Explain how the concept of bonding and lone (non-bonding) pairs of electrons can be used to predict the shape of, and bond angles in, a molecule of sulphur tetrafluoride, SF_4 . Illustrate your answer with a sketch of the structure.

(8)

(Total 26 marks)

44. Sketch a graph to show how the melting points of the elements vary across Period 3 from sodium to argon. Account for the shape of the graph in terms of the structure of, and the bonding in, the elements.

(Total 21 marks)

45. (a) With the aid of diagrams, describe the structure of, and bonding in, crystals of sodium chloride, graphite and magnesium. In each case, explain how the melting point and the ability to conduct electricity of these substances can be understood by a consideration of the structure and bonding involved.

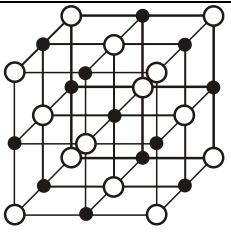
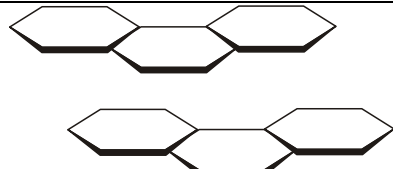
(23)

- (b) Explain how the electron-pair repulsion theory can be used to predict the shapes of the molecules H_2O and PF_5 . Illustrate your answer with diagrams of the molecules on which the bond angles are shown.

(7)

(Total 30 marks)

46. In this question, one mark is available for the quality of spelling, punctuation and grammar.
Many physical properties can be explained in terms of bonding and structure. The table below shows the structures and some properties of sodium chloride and graphite in the solid state.

substance	sodium chloride	graphite
structure		
electrical conductivity of solid	poor	good
melting and boiling point	high	high
solubility in water	good	insoluble

Explain these properties in terms of bonding and structure.

[7]
Quality of Written Communication [1]
[Total 8 marks]

47. Sodium reacts with excess oxygen to form sodium peroxide, Na_2O_2 .
 Na_2O_2 is used in laundry bleaches. When added to water a reaction takes place forming an alkaline solution and hydrogen peroxide, H_2O_2 .
- Construct a balanced equation for the formation of sodium peroxide from sodium. [1]
 - Construct a balanced equation for the reaction of sodium peroxide with water. [1]
 - Draw a 'dot-and-cross' diagram for a molecule of H_2O_2 . Show outer electrons only. [2]
- [Total 4 marks]**
48. In water treatment plants, care must be taken as chlorine can react with nitrogen compounds to form the highly explosive compound, nitrogen trichloride, NCl_3 . Molecules of NCl_3 have a bond angle of 107° .
- Name the shape of an NCl_3 molecule. [1]
 - Explain why a molecule of NCl_3 has this shape and a bond angle of 107° . [3]
- [Total 4 marks]**
49. Chemists have developed models for bonding and structure which are used to explain different properties.
 Ammonia, NH_3 , is a covalent compound.
- Explain what is meant by a *covalent bond*. [1]
 - Draw a 'dot-and-cross' diagram to show the bonding in NH_3 .
 Show **outer** electrons only. [1]
 - Name the shape of the ammonia molecule.
 Explain, using your 'dot-and-cross' diagram, why ammonia has this shape and has a bond angle of 107° . [3]
- [Total 5 marks]**
50. The shape of a water molecule is different from the shape of a carbon dioxide molecule.
- Draw the shapes of these molecules and state the bond angles.

water	carbon dioxide
bond angle in water =	bond angle in carbon dioxide =

- (ii) Explain why a water molecule has a different shape from a carbon dioxide molecule. [4]

[2]

[Total 6 marks]

51. Ammonia reacts with hydrogen chloride, HCl , to form ammonium chloride, NH_4Cl .

NH_4Cl is an ionic compound containing NH_4^+ and Cl^- ions.

- (i) Give the electron configuration of the Cl^- ion. [1]

- (ii) Draw a 'dot-and-cross' diagram to show the bonding in NH_4^+ .
Show **outer** electrons only. [1]

- (iii) State the shape of, and bond angle in, an NH_4^+ ion. [2]

- (iv) A student investigated the conductivity of ammonium chloride.
She noticed that when the ammonium chloride was solid it did **not** conduct electricity. However, when ammonium chloride was dissolved in water, the resulting solution did conduct electricity.
Explain these observations.

[2]

[Total 6 marks]

52. Compounds with covalent bonding often have polar bonds. Polarity can be explained in terms of electronegativity.

- (i) Explain the term *electronegativity*. [2]

- (ii) Use a suitable example to show how the presence of a polar bond can be explained in terms of electronegativity.
You may find it useful to draw a diagram in your answer.

[2]

[Total 4 marks]

53. Liquid ammonia, NH_3 , and water, H_2O , both show hydrogen bonding.

- (i) Draw a labelled diagram to show hydrogen bonding between two molecules of liquid **ammonia**. [3]

- (ii) Water has several anomalous properties as a result of its hydrogen bonding.
Describe and explain **one** anomalous property of water which results from hydrogen bonding.

[2]

[Total 5 marks]

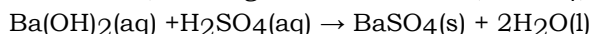
54. Some polar molecules are able to form hydrogen bonds.
Draw a diagram to show an example of hydrogen bonding.

[Total 2 marks]

55. At room temperature, **X** is a liquid which does **not** conduct electricity. What does this information suggest about the bonding and structure in **X**?

[Total 2 marks]

56. Sulphuric acid was added to aqueous barium hydroxide until the solution was just neutralised, forming the insoluble salt, BaSO_4 , and water.



The electrical conductivity of the solution steadily decreased as the sulphuric acid was added.

Explain why the electrical conductivity decreased.

[Total 2 marks]

57. The metal magnesium reacts with the non-metal chlorine to form a compound magnesium chloride, MgCl_2 , which has ionic bonding.

- (i) State what is meant by an *ionic bond*. [1]
- (ii) 'Dot-and-cross' diagrams are used to model which electrons are present in the ion. Draw a 'dot-and-cross' diagram, including outer electron shells only, to show the ions present in magnesium chloride, MgCl_2 . [2]
- (iii) A student finds that solid magnesium chloride and pure water do not conduct electricity. The student dissolved the magnesium chloride in the water and the resulting solution **does** conduct electricity. Explain these observations. [3]

[Total 6 marks]

58. In this question, one mark is available for the quality of use and organisation of scientific terms.

Nitrogen and oxygen are elements in Period 2 of the Periodic Table. The hydrogen compounds of oxygen and nitrogen, H_2O and NH_3 , both form hydrogen bonds.

- (i) Draw a diagram containing two H_2O molecules to show what is meant by *hydrogen bonding*. On your diagram, show any lone pairs present and relevant dipoles. [3]
- (ii) State and explain **two** anomalous properties of water resulting from hydrogen bonding. [4]

[Total 7 marks]

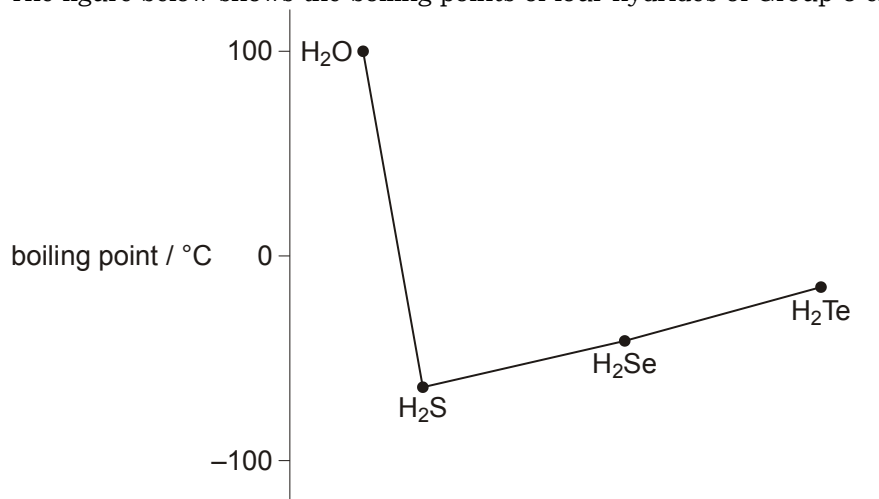
59. State and explain **two anomalous** properties of H_2O that depend on its intermolecular forces.

[4]

Quality of Written Communication [1]

[Total 5 marks]

60. The figure below shows the boiling points of four hydrides of Group 6 elements.



- (i) Explain, with the aid of a diagram, the intermolecular forces in H_2O that lead to the relatively high boiling point of H_2O . [3]
- (ii) Suggest why H_2S has a much lower boiling point than H_2O . [1]

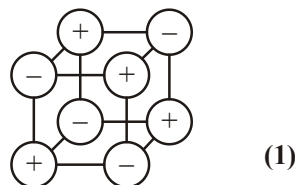
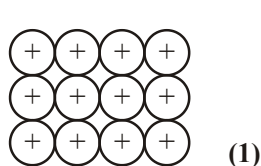
[Total 4 marks]

61. Water forms hydrogen bonds which influences its properties. Explain, with a diagram, what is meant by *hydrogen bonding* and explain **two** anomalous properties of water resulting from hydrogen bonding.

[Total 6 marks]

Marking Scheme

1. (a)



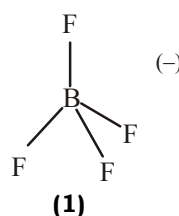
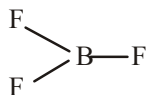
2

[Diagrams must be complete and accurate]

- (b) (i) Attraction / electrostatic forces/bonds/attractions between (positive) ions/lattice and delocalised/free electrons/sea of electrons. 1
[Not metallic bonding] [Not just 'forces']
- (ii) Electrostatic attractions/forces between ions or attractions between (oppositely charged) ions / Na^+ & Cl^- 1
[Not ionic bonding]
- (iii) (Here) the ionic bonding in NaCl is stronger/requires more energy to break than the metallic bonding in Na 1
QoL Accept 'bonding/forces of attraction in NaCl is stronger than in Na'
[If IMF/molecules/van der Waals'/dipole-dipole mentioned in parts(i) or (ii), then CE = 0 for parts (i) and/or(ii) and CE = 0 for part(iii)]
- (c) Comparison: 1
 Sodium conducts **and** sodium chloride does NOT conduct
Allow 'only Na conducts'
Accept 'Na conducts, NaCl only conducts when molten'
[Do not accept sodium conducts better than sodium chloride etc.]
- Explanation: 1
 (Delocalised) electrons flow through the metal
 Allow e^- move/carry current/are charge carriers/transfer charge.
[Not 'electrons carry electricity']
[Not 'NaCl has no free charged particles']
- Ions can't move in solid salt 1
- (d) Layers can slide over each other – idea that ions/atoms/particles move 1
[Not molecules] [Not layers separate]

[9]

2. (i)



2

[Do not allow shapes which show a lone pair]

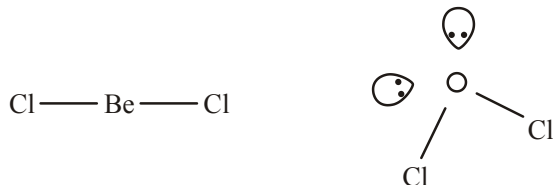
- BF_3 Trigonal planar/planar triangular 1
[Not plane triangle]
- BF_4^- Tetrahedral 1
[Not distorted tetrahedral]
- Equal repulsion between (4) bonding pairs/bonds/bonding electrons 1
 $109\frac{1}{2}^\circ$ 1

- (ii) Lone pair donated / both electrons supplied by one atom 1
 from F^- (to B) 1
[ignore missing charge or fluorine or 'atom']
 dative/dative covalent/coordinate bonding 1

[9]

3.

2

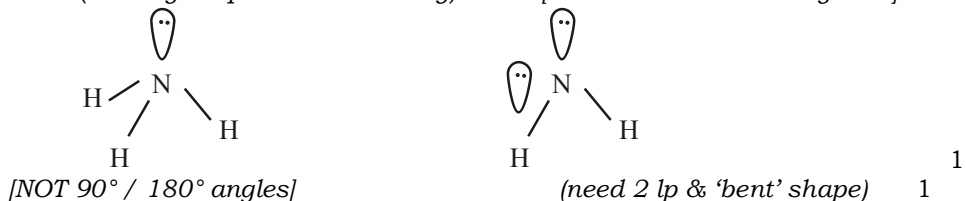


- QoL Linear **(1)** beat / V-shaped / angular **(1)** 2
(mark name and shape independently)
(accept (distorted) tetrahedral)
(if balls instead of symbols, lose M1 – can award M2)
(penalise missing 'Cl' once only)
(not 'non-linear')

[4]

4. (a)

(Missing 'H' penalise once only) [NOT dot-and-cross diagrams]



- (b) 107° 1

- (c) More lone pairs on NH_2^- , than on NH_3 1
 Lone pairs repel more than bonding pairs *Must be comparison* 1
(Mark separately)
[NOT repulsion between atoms or between bonds]

[5]

5.

- (a) (electrostatic forces of) attraction / held together by interactions between **(1)**
 positive ions / cations / nuclei **not** just metal ions **(1)**
 and delocalised or free (outer shell) electrons / 'sea' of electrons / cloud of electrons **(1)** 3
 allow 2 marks for 'electrons can flow'
- (b) *conductivity* clear indication that electrons are mobile / can move **(1)**
 electrons / carry charge / are charge carriers / move in same direction **not** 'so current can flow'
 allow 2 marks for 'electrons can flow'
malleability: cations / positive ions / atoms in the lattice are all identical / the same size / (planes of) ions (atoms) can slide easily over one another **(1)**
 attractive forces in the lattice are the same whichever ions (atoms) are adjacent / attractive forces remain the same throughout the lattice **(1)** 4
- (c) more electrons delocalised / more outer shell electrons **(1)**
 in aluminium compared with magnesium / reference to



2

[9]

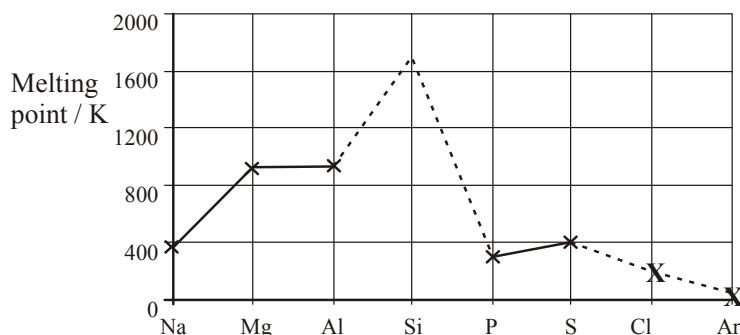
6. (a) tendency / strength / ability / power of an atom / element / nucleus to attract / pull / withdraw electrons / e⁻ density / bonding pair / shared pair 1
in a covalent bond 1
- (b) (i) F₂ = van der Waals' / induced/temporary dipole-dipole / dispersion / London forces 1
CH₃F dipole-dipole (*not just 'dipole'*) 1
HF = hydrogen bonding (*not just 'H' / 'hydrogen'*) 1
- (ii) large difference in electronegativity between H and F / F most/very/much more electronegative / values '4' & '2.1' quoted 1
(*not just 'higher'*)
 δ^+ H-F δ^- dipole created or dipole clearly implied 1
(*accept arguments such as 'uneven charge in bond' / 'polar bond'*
(F slightly negative/H slightly positive)
attraction/bond formed between δ^+ H and lone pair on F 1
(M2/M3 may be scored from a diagram)
(CE if full charges shown - lose M2 and M3)
- (c) (i) van der Waals' / induced/temporary dipole-dipole / dispersion / London forces / attractions (ignore references to dipole-dipole) 1
increase with the increasing M2 / size / mass / N1 of e⁻ / size of e⁻ cloud (in the hydrogen halides) 1
(*if ionic, or if 'covalent bonds broken' = CE = 0*)
(*mark M1 and M2 separately*)
- (ii) hydrogen bonding stronger than van der Waals' attraction/forces 1
(*accept hydrogen bonding is very strong / strongest*)
(*accept arguments such as 'HF has H-bonds, others only have van der Waals'*)
(*not just 'HF has H-bonding'*)

[11]

7. (a) Oxygen more/very/highly electronegative (than hydrogen)
OR oxygen has stronger attraction for bonding electrons / bonding electrons drawn towards oxygen; 1
causes higher e⁻ density round oxygen atom / causes H δ^+ O δ^- ; 1
- (b) van der Waals' forces between oxygen molecules; 1
Hydrogen bonding between methanol molecules; 1
H-B stronger than van der Waals' OR stronger IMF in methanol; 1
(*if dipole-dipole forces in O₂ or methanol, allow comparison, hence max 2*)
(*if ionic/covalent etc. max 1*)
(*mention of bond break = CE = 0*)

5]

8. (a) (i)



- M1 Si: cross $\square$ 1200 1
 M2 Cl: cross below S 1
 M3 Ar: cross below Cl 1
[allow, even if M2 wrong]
[If Cl cross missing and Ar below S, allow M3]

- (ii) Si is macromolecular/giant molecular/giant covalent/ giant atomic 1
 Covalent bonds need to be broken/accept 'overcome' 1
[Not loosened/weakened]
 Covalent bonds are strong / many covalent bonds involved/requires much energy/hard to break 1
[Tied to 'break' or near miss in M2] [Not 'structure' is broken]
[Must mention 'covalent' somewhere in part (a)(ii) to earn M2/M3]
 [If van der Waals'/IMF mentioned M2/M3 = CE = 0.
 [If ions mentioned M1/M2/M3 = CE = 0]

- (iii) Intermolecular force = van der Waals'/induced dipole-dipole/dispersion forces 1
QoL Sulphur has greater M_r / size / surface area/more electrons/more atoms **so** stronger intermolecular forces (comparison) 1
[Mark separately] [Not 'more shells']

- (b) Trend: Decreases 1
[If trend wrong = CE = 0]

Increase in size of ion/atom / more shells / decrease in charge density / decrease in charge size ratio 1

Weaker attraction for delocalised/free/sea of electrons / weaker metallic bonding 1
[Ignore shielding] [van der Waals' etc. = CE = 0 for M2 and M3]

[11]

9. trend: increases; 1
 more protons / higher charge on cation / more delocalised e^- / smaller atomic/ionic radius; 1
 stronger attraction between (cat)ions and delocalised/free/mobile e^-
 OR
 stronger metallic bonding; 1

[3]

10. (a) (i) positive ions (1)
 (attract) delocalised electrons (1) (or sea of or free or mobile) (1)
Confusion with -ve ions
or ionic lattice C.E. = 0

- (ii) more protons (1) (or Mg^{2+} more charge than Na^+)

attracts delocalised (or bonding) electrons more strongly (1)

Delocalised: can be brought forward from (a) (i)

OR more delocalised electrons (1)

Attacks positive ions more (1)

Metallic bonding is stronger scores one mark, only given if no other marks awarded

4

- (b) macromolecular (1) (or giant molecule etc)

covalent (1)

strong covalent bonds (1)

or bonds require much energy to break

3

- (c) delocalised (OR free or sea of or mobile) electrons (1)

1

- (d) Planes (1)

weak (bonds) forces between planes (1)

or vdw forces between planes

2

[10]

11. (a) Bonding in Na_2S : ionic (1)

Bonding in CS_2 : covalent (1)

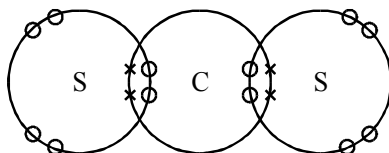
ignore other words such as dative / polar / co-ordinate

- (b) Clear indication of electron transfer from Na to S (1)

1 e^- from each (of 2) Na atoms or 2 e^- from 2 Na atoms (1)

QL correct English

- (c)



Correct covalent bonds (1)

All correct including lone pairs (1)

Allow all s or all x's

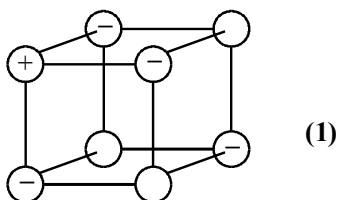
M2 tied to M1

NOT separate e^- s in S 2 1 p

6

[6]

12. (a) (i)



- (ii) vibrate faster (1)

Or bigger amplitude or more

NOT start to vibrate or other type of motion esp. Translation

Ignore rotation

2

- (b) (i) from sodium (1)

Allow from sodium ion, also from metal

1

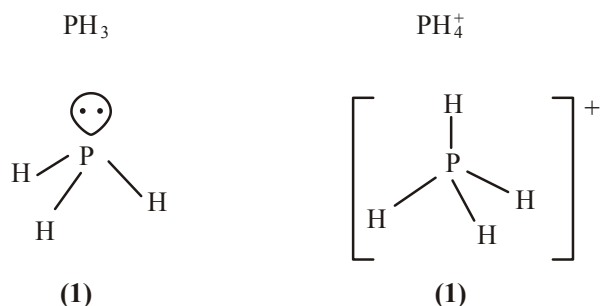
- (ii) difference in electronegativity (1)

1

[4]

13. (a) dative / coordinate (covalent) bond; 1
 Lone/non-bonding pair / both electrons; 1
 (donated) from P to H⁺; 1

(b)



pyramidal *OR* trigonal pyramid 109(1/2)°;
 (accept tetrahedral)

4

[7]

14. (a) Ability (or power) of an atom to attract electron density
 (or electrons or -ve charge) (1)
 in a covalent bond (1)
or shared pair
If remove an electron lose first mark 2

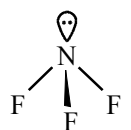
- (b) *Trend:* increases (1)
Explanation: nuclear charge (number of protons) increases (1)
 electrons in same shell (1)
OR similar shielding
OR atoms similar size or smaller 3

[5]

15. (a) (i) 3 (bonding) pairs of electrons (1)
allow 3 bonds
 repel equally (1) (or as much as possible)
Or get as far apart as possible
 (ii) *Predicted bond angle:* 118° (allow 117 - 119°) (1)
Explanation: lone pair (1)
 repels more than bonding pair (1)
Allow EXP if < 118°
but C.E. = 0 if more than 120° 5

- (b) *Name of shape:* Tetrahedral (1)
Example: CH₄ etc (1)
Allow correct ion 2

- (c) (i) 90° (1)
 (ii) lone pairs (or they) repel more than bonding pairs (or most) (1)
 (so are) as far apart as possible (1)
Mark independently
 (iii) square planar (1)
allow square 4

- (d)
- 

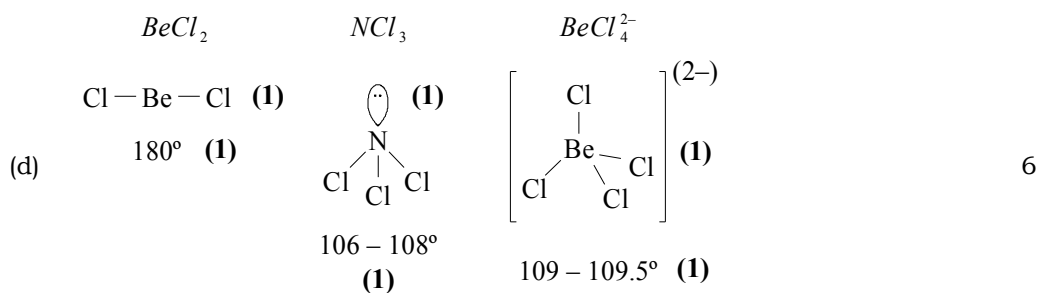
3 bonds + 1 lone pair (1)
 correct shape (1)
 (only give this mark if first mark also given

Penalise sticks (i.e. N-) once but N must be shown

- 2 [13]
16. (a) vibration (1)
about a fixed point (1)
random (1)
(or free to move) 3
- (b) weak intermolecular forces (1)
or v.d.w forces
easily broken (1)
or requires little energy to break 2
- (c) conducts electricity (1)
delocalised electrons (1)
(or converse argument for sodium chloride - no mobile charges)
malleable (1)
or ductile, softer
planes of atoms slide (1)
(or converse argument for NaCl - ionic lattice) 4
or Ag insol. in water (1)
not ionic (1)
or conducts heat (1)
deloc. electrons (1)
or shines (1)
metallic bonding (1)
or high density (1)
large A_r , close packed (1)
or sonorous (1)
metal (1)

Note 1. If property of NaCl given instead of Ag can score reason mark

2. for reasons can allow reason why NaCl is different



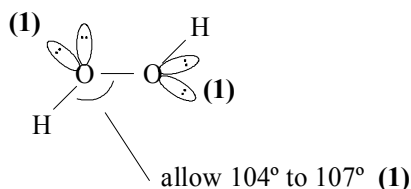
Note 1. Mark angles independently

BUT wrong number of electron pairs on diagram $CE=O$

2. penalise "sticks" (i.e no Cl) once only

3. accept 'dot-cross' diagram for $BeCl_2$ - only for NCl_3 $BeCl_4^{2-}$ if clear.

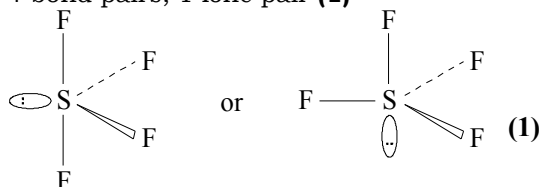
17. diagram showing a tetrahedral arrangement of four oxygen atoms around a central silicon atom (must include attempt at a 3-d representation) (1)
tetrahedral (1)
 $109.5^\circ / 109^\circ / 110^\circ$ (1) 3 [3]
18. (a) more compact (1) 1
- (b) (i) and (ii)



(iii) hydrogen bond (1)

4

(c) 4 bond pairs, 1 lone pair (1)



87° to 90° (1)

117° to 120° (1)

4

(d) Type 1 van der Waals' (1)
Type 2 dipole – dipole (1)

2

[11]

19. (a) electrostatic (1)

1

(b) shared electron pair (1)

1

(c) {both electrons, electron pair (1)

(donated) from {one, the same atom {or molecule, or species (1)

2

(d) (positive) ions (or atoms) (1)

delocalised electron (1) (or sea of, or mobile, or free)

(mention of ionic, covalent bonding or molecules $\text{CE}=\text{O}$)

2

(e) (i) Charge separation (or unequal sharing of electrons) (1)
leading to δ^+ & δ^- ends (1)
also gains first mark

(ii) chlorine more electronegative than H (1)

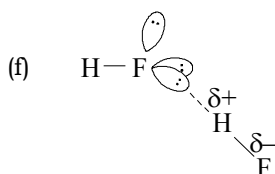
or {chlorine is electronegative, difference in electronegativity
attracts electrons towards Cl (1)

allow chloride instead of chlorine

(iii) van der Waals' (1)

or {v.d.w, temp dipole, include dipole, London

5



3 lone pairs on F (1)

δ^+ on H (1)

H (δ^+) adjacent to lone pair on F (1)

(note $\text{H}^+ - \text{F}^-$ CE = max 1)

3

(g) bromine not sufficiently electronegative (1)
{or lone pair too diffuse
{or bond not polar enough

{or little electronegativity difference
allow bromide

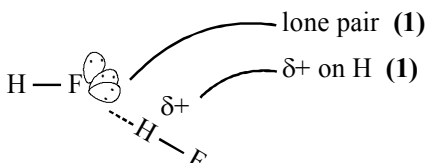
1

[15]

20. (a) van der Waals' (1)
 dipole - dipole (1)
 energy needed to overcome (*intermolecular or vdw or dipole-dipole*) forces (1) 3

- (b) HBr is intermediate in size (or polarisability) (1)
 van der Waals' forces depend on molecular size (1) 2

- (c) hydrogen bonding in HF (1)
 stronger than intermolecular forces in HCl (1)
 (or strongest intermolecular force) 2

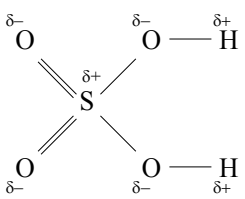
- (d)  2

21. (a) covalent (1)
 covalent (1)
 ionic (1) 3

- (b) sodium sulphide / Na₂S (1)
 liquid / molten / aqueous solution (independent of first mark) (1) 2

[9]

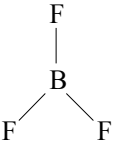
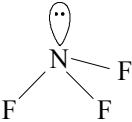
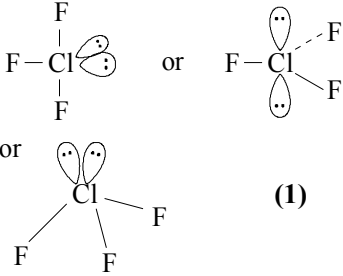
22. (a) partial charges on atoms of bond / centres of positive and negative charges do not coincide / bonding pair shared unequally (1)
not just bond between elements of different electronegativity
not just 'bond with dipole'
 allow clarification by diagram, but not just a diagram 1

- (b)  2
- at least one of each of the 2 polar bonds correctly labelled for each mark
 allow one mark for 2 arrows showing dipole on 2 different bonds

- (c) hydrogen bonding (1)
 attraction between (δ^+ of) H atoms and (δ^- of) O atoms in adjacent or different molecules (H of one molecule, O of another) (1)
 mark these points independently 2

[5]**[5]**

23. (a)

Molecule	Sketch of shape	Bond angle(s)	Name of shape
BF ₃	 (1)	120° (1)	Trigonal planar (1)
NF ₃	 (1)	107° ± 1° (1)	Tetrahedral or pyramidal (1)
ClF ₃	 (1)	87° to 90° and/or (1) 118° to 120°	Trigonal bipyramid or trigonal planar or T (1)

- (b) BF₃ van der Waals' **(1)**
- NF₃ van der Waals' **(1)** dipole-dipole **(1)**
- (c) Name of bond co-ordinate **(1)**
- Explanation lone pair on N **(1)**
- donated (or shared with) to B (or BF₃) **(1)**

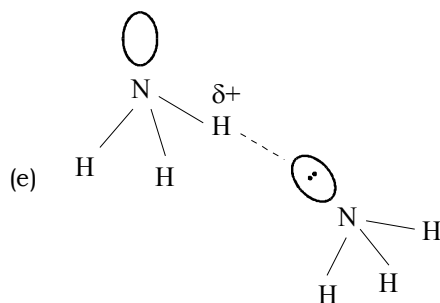
9

3

3

[15]

24.

shape **(1)**δ + on H **(1)**δ + attracted
to lone pair **(1)**

3

[3]

25. (a) power of an atom (allow element) to attract electron(s) **(1)**
- in a covalent bond **(1)**
- not ion or molecule

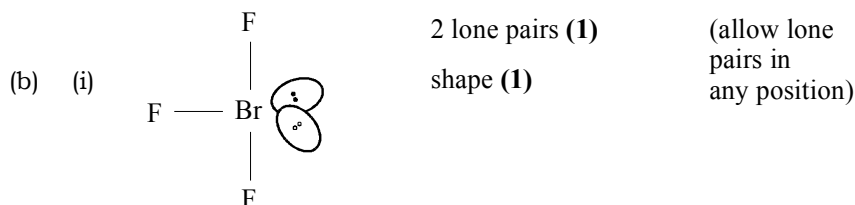
2

- (b) increases **(1)**

	nuclear charge increases (1) attracting electrons in same shell (1) (or similar shielding) <i>do not allow size arguments</i> <i>trend must be correct to score 'explanation' marks</i>	3	
	(c) ionic (1) big difference in electronegativity (1) <i>or idea of magnitude of difference</i> covalent (1) smaller electronegativity difference (1) <i>or similar electronegativity or $e.\text{neg}(S) > e.\text{neg}(Na)$</i>	4	
			[9]
26.	(a) shared electron pair (1) both electrons from one atom (1)	2	
	(b) $A + : B \rightarrow A : B$ (or $NH_3 + H^+ \rightarrow NH_4^+$ etc) equation (1) final species (1)	2	
	(c) chlorine is more electronegative than sodium (1) an electron transfers from sodium to chlorine (1)	2	
	(d) smaller electronegativity difference (1) so more equal sharing (1)	2	
	(e) mobile (free) ions (1)	1	
	(f) carbon forms 4 bonds (1) gives a macromolecule (1) iodine forms only one bond (1)	3	
			[12]
27.	(a) strong (accept if reasonably qualified) (1) electrostatic <u>attractions</u> or similar such as "held" (1) between ions <u>held in a lattice</u> (1)	3	
	(b) vibrations (1) increase (1) as temperature increases until the vibrations are so violent that the ions break free at the melting point. (some indication of translational movement of the particles) (1)	3	
	(c) high melting point/boiling point (1) electrical conductivity when molten (or in solution) (1)	2	
			[8]
28.	(a) Shared pair of electrons	1	
			[1]
29.	(a) (i) power / tendency / ability to attract electrons [must be plural] not gain (1) in a covalent bond / from a bonding pair / bonding electrons (1)	2	
	(ii) van der Waals' forces / induced dipole - induced dipole / intermolecular attractions increase (1) as M_r increases / number of electrons increases (1) size / radius / number of shells increase	2	
	(iii) higher due to hydrogen bonding / (strong) dipole - dipole attractions (1) caused by large (difference in) electronegativity of fluorine / clear indication (1) that intermolecular forces in HF are stronger than in (ii) (accept diagram)	2	
	(b) (i) SO_4^{2-} (1)	1	
	(ii) 109.5° (1)	1	
			[8]

30. (a) ionic (1)

1



trigonal bipyramid (1) (or T-shaped etc)

31. (a) (i) hydrogen atom bonded to highly electronegative atom / to N, O or F / hydrogen

containing molecule with a strong dipole (1)

condone mention of Cl if with other correct atoms

1

(ii) van der Waals' / (induced **or** temporary) dipole-dipole (1)

sulphur not sufficiently electronegative / too large / only small difference in

electronegativity (1)

2

(iii) intermolecular attractions in H₂S much weaker / in H₂O are stronger (1)
than the hydrogen bonding in water / so less energy needed to overcome forces (1)**not** more bonds in H₂O

2

(b) (i) both C=O and N-H bond polar / correct polarity of both bonds shown (1)

since electronegativity N>H and O>C (1)

H- bonding between -H^{δ+} and =O^{δ-} in different molecules(can allow H- bonding in **same** molecules if clear that

folding / higher order

structure is being referred to) (1)

using lone pair of electrons on O atom (could be on diagram) (1)

4

[9]

32. (a) (lattice of) metal ions / positive charges / cations **not** nuclei (1)with a 'sea' of/delocalised electrons **not** cloud (1)held together by (electrostatic) attraction / forces between ions / nuclei and electrons / opposite charges (**not** negative ions) (1)

3

(b) Mg / Mg²⁺ smaller than Na / Na⁺ metallic radius smaller / Mg close packed (1)Mg / Mg²⁺ higher charge than Na / Na⁺ (1)

magnesium contributes more electrons than Na (1)

therefore stronger (electrostatic) attractions / stronger metallic bond(ing) (1)max

3

(c) sodium and graphite conduct, diamond does not (1)

delocalised or free electrons in graphite (1)

single bonded structure of diamond / diamond has no free or delocalised e⁻ (1)

conduction due to delocalised electrons flowing (1)

4

[10]

33. (a) large difference in electronegativity → ionic (1)

small difference in electronegativity → covalent (1)

2

(b) ionic (1)

1

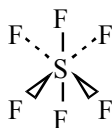
[3]

34. (a) 105° (allow 104–106°) (1) 1
 (b) 4 electron pairs round O (1)
 tetrahedral (or 109°) (1)
 lone pairs repel more than bonding pairs (1)
 therefore angle less than 109° (or less than tetrahedral angle) (1) 4
 (c) hydrogen bonding (1) 1
 (d) F more electronegative than H (1)
 pulls electrons from H (1) 2
 (e) *Prediction* more than 105° (or increases) (1)
Explanation lone pair more like bonding pair
 hence does not repel so strongly (1) 2
- [10]
35. (a) Element with lowest melting point neon (1)
Explanation free atoms (1)
 weak van der Waals forces (1) 3
 (b) *Element with highest melting point* silicon (1)
Explanation macromolecular (1)
 covalent bonds (1) 3
- [6]
36. Structure and hardness
- M1 Q of L both macromolecular/giant atomic/giant covalent/giant molecular; 1
 M2 C atoms in diamond joined to 4 other C atoms / diagram with min 5 C
 atoms i.e. shows tetrahedral shape / coordination number = 4; 1
 M3 C atoms in graphite joined to 3 other C atoms diagram with clear
 extended hexagonal plane/pattern i.e. shows trigonal planar shape /
 coordination number = 3; 1
 M4 diamond hard / crystal strong; 1
 (not diamond stronger than graphite)
 M5 because of 3-D structure / rigid structure / not layered; 1
 M6 graphite (soft) as layer can slide over each other; 1
 M7 Q of L as only (weak) van der Waals' forces between layers; 1
- Melting point (for either allotrope)
- M8 covalent bonds must be broken / overcome; 1
 M9 which are strong / many / hard to break; 1
 (M9 tied to M8)
- Other difference
- M10 diamond is non-conductor of electricity, graphite is conductor 1
 OR appropriate difference in appearance;
- [9]
37. QoL Bonding Both covalent (linked statement) 1
 Structure Iodine = molecular / I₂ (stated or in diagram) 1
 [treat incorrect diagram as contradiction]
- Diamond = giant molecular/macromolecular/giant
 covalent / giant atomic (stated only) 1
 Reference to van der Waals' / dipole-dipole = contradiction
- QoL Iodine Weak van der Waals' forces / induced dipole-induced 1
 dipole
- Diamond Covalent bonds would need to be broken 1
 Many / strong covalent bonds **OR** much energy needed 1
 Tied to M5 or near miss
 [If ionic/metallic structure suggested then CE for that
 substance] [If hydrogen bonding suggested, for I₂ lose M2 &
 M4; for diamond lose M3,M5&M6]
- [6]

38. giant lattice (1)
 of covalent bonds (1)
 need to break covalent bonds to melt red phosphorus (1)
 need to break Van der Waal's forces to melt white phosphorus (1)
 covalent bonds are stronger than Van der Waal's forces (1)

[5]

39. (a) SF₆ shown as octahedral / square based bipyramid (1)
 Bond angle: 90° **or** 180° and 90° (1)
 Shape = octahedral (1)
If lone pair shown then C.E. = 0 / 4



Wrong symbols - no diagram mark

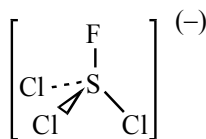
Equal repulsion between 6 bonding **or** shared electron pairs **QoL (1)**

AlCl₄⁻ shape shown as tetrahedral (1)

Bond angle = 109° to 109.5° (1)

Shape = tetrahedral (1)

If lone pair shown then C.E = 0/4



(Equal repulsion between) 4 bonding pairs **or** shared electron pairs (1)

QoL may be awarded here also

Mark all points independently

8

- (b) Solvent has low bp or weak intermolecular forces **or** evaporates quickly (1)
 (Solvent) needs energy to evaporate (**to overcome intermolecular forces**)
or valid reference to latent heat of vaporisation (**or evaporation is endothermic**) (1)

**OR higher energy or faster molecules more likely escape
 so mean energy (and hence temperature) falls**

Energy taken from the skin (and so it cools) (1)

Fragrance or perfume (molecule) slowly spreads (through the room) (1)

By random movement **or** diffusion (of the perfume / fragrance) (1)

4

[12]

40. (a) I₂ sublimates when heated / has low melting point AND graphite has (very) high melting point (1)
 I₂ is (simple) molecular / I₂ / I-I (1)
CE = 0 if 'ionic'
loses M2, M3, M4
Max 1 if I-I bond broken
 Van der Waals forces / induced or temporary dipole-dipole / London forces (1)
 Which are weak or easily overcome (so low melting point) (1)

Graphite is macromolecular / giant covalent / giant molecular / giant atomic / (1)

Diagram

min 3 correctly fused rings – ignore diagram unless no description offered

CE = 0 if 'ionic' or 'metallic' loses M5, M6, M7

(Many) covalent bonds need to be broken (1)
covalent may be inferred from structure

Which takes much energy / bonds are strong (1)

Only graphite conducts (1)

NOT just graphite is a better conductor

As it has delocalised / free / mobile electrons (1)

All e^- in iodine are used in bonding and lone pairs

OR as it has no delocalised / free / mobile e^- (1)

Max

9

M9 and M10 tied to correct statements about graphite = conductor and iodine = non-conductor

(b) Diagram Cl-Be-Cl (clearly linear) (1)

CE = 0 if lone pair(s) on Be, loses M1 and M2

(Equal) repulsion between 2 bonding pairs / bonds (1)

'2' may be inferred from a correct diag

NOT repulsion between Cl atoms

2

[11]

41. (a) Macromolecular **or** giant structure (1)

7

Accept diamond shaped lattice

Intermolecular forces / molecular lattice / comparison to graphite structure, = 'con'

Held together by covalent bonds (1)

'Giant covalent structure' earns both M1 and M2

(Much) energy needed to break bonds **Or many** bonds to be broken (1)

Mark tied to earning 'covalent' M2 If explanation is clearly of ionic bonding = CE

Vand der Waal / temporary induced dipole-dipole / London / disperse forces (1)

*Forces increase with size **or** with number of electrons **or** with surface area etc. (1)*

Description must be of the molecules of P and S

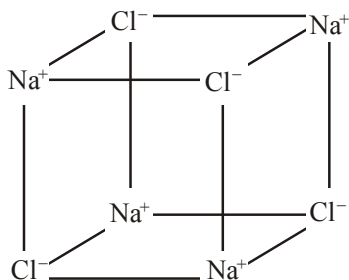
P₄ **or** M_r = 124 (1)

S₈ **or** M_r = 256 (1)

If M6 (i.e. P₄) and M7 (i.e. S₈) are not attempted, allow S molecule bigger / more surface area than P molecule for 1 mark

(b) Diagram NaCl = cubic (1)

Allow if 3 full faces shown correctly



Ions identified and placed properly (1)

If diagram shows '+' and '-' signs rather than symbols for ions, identification of the ions could be from the text

(Bonding) identified in writing as being ionic (1)

Not ionic molecule

Due to strong electrostatic attractions **or** similar description about attraction between oppositely charged ions. (1) **QoL**

4

Not just: 'ionic bonds are strong' / 'need much energy to

break bonding'

[11]

42. Pattern in the change in the properties of a row of elements (1)
OR Trend in the properties of elements across period
 Repeated in the next row (1)
OR element underneath (or in same group) has similar properties

atomic radius

decreases across the row (1)

CE if trend is wrong

number of protons increases (1) (or nuclear charge increases)

more attraction for electrons in the same shell (1)**electronegativity**

increases across the row (1)

number of protons increases (1) (or nuclear charge)

atomic radius decreases (1) (or shielding remains the same or electrons in the same shell) more attraction for bonding or shared electrons (1)**conductivity**

decreases row (1)

OR significant drop from Al to Si

Na-Al metals (1)

OR metallic bonding or description of metallic bonding

Two of Si - Ar non metals (1)

OR molecular or covalent

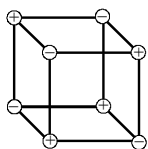
EITHER electrons free to move (delocalised) in metals

OR electrons unable to move in non-metals (1)

13

[13]

43. (a) Ionic (1)
 lattice (1)
 e.g. NaCl (1)
 conducts when molten (1)
 due to mobile ions (1)



sketch scores (2) { 4 ions placed correctly in a square (1)
 { extended to 3 dimensions (1)

max

6

macromolecular (1)

covalent (1)

e.g. diamond (or graphite) (1)

conductor (or conducts when solid) (1)

electrons localised in covalent bonds (or delocalised electrons in rings) (1)

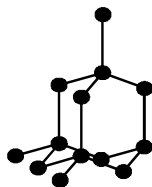
diamondgraphite

tetrahedral round a C (1)

2 'fused' hexagons (1)

extended from other Cs (1)

2 layers (planes) (1)



sketch scores (2)

max

molecular **(1)**

or molecules

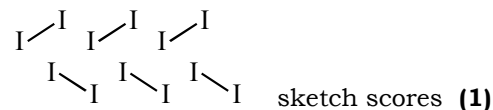
covalent **(1)**

e.g. iodine **(1)** etc

weak (van der Waals) forces between molecules **(1)**

non conductor **(1)**

electrons localised **(1)**



regular array of molecules

example of a metal (eg Mg) **(1)**

metallic bonding (or stated to be a metal) **(1)**

positive ions (or atoms) **(1)**

surrounded by delocalised electrons **(1)**

conducts when solid **(1)**

due to delocalised electrons **(1)**

sketch scores [1]

e.g.

Note if candidate deals with two crystals of the same type

eg diamond & graphite, award max 3 for worst of the two answers. **max 6**

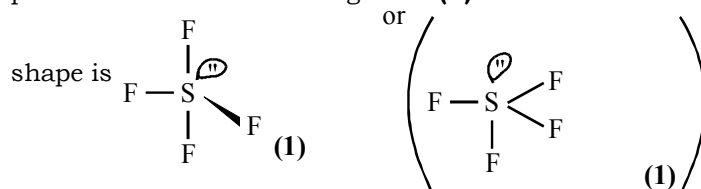
(b) 5 electron pairs **(1)**

including one lone pair **(1)**

repel as far apart as possible **(1)**

lone pair – bond pair repulsion > bp – bp **(1)**

pushes S-F bonds closer together **(1)**



angles <90 **(1)** and <120 **(1)** or (<90 **(1)** and 120 **(1)**)
 (85 – 89) (115 – 119) (85 – 89) (115 – 120)

N.B. if not 5 pairs of electrons can score only 3rd & 4th marks

max

[26]

44. From the graph:

Mg above Na **(1)**

Al similar to or above Mg **(1)**

Si above Na and Mg **(1)**

P lower than Mg, Al and Si **(1)**

S above P, lower than Si **(1)**

Ar lowest **(1)**

7

Sodium, magnesium and Al metallic **(1)**

Strength of metal-metal bonds increases from Na to Al **(1)**

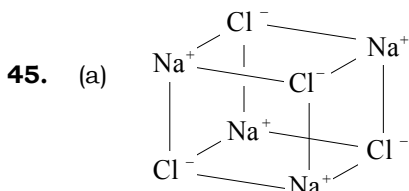
More protons (or bigger charge on ion or more delocalised electrons) **(1)**

Ions attract delocalised electrons more strongly **(1)**

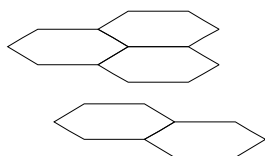
4

- Silicon has covalent bonds (1)
 Macromolecule (1)
 Many strong bonds to be broken (1) 3
- S₈ (1)
 P₄ (1)
 C₁₂ (1)
 weak intermolecular forces (1)
 van der Waals' (1)
 Stronger between larger molecules (1) 6
- Argon consists of free atoms with weak van der Waals' forces between atoms (1) 1

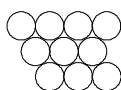
[21]



- from diag (cubic (1) ions (or Na, Cl or +,-) placed correctly (1)
 (minimum of 4 species)
 from diag or stated (ionic (1) lattice (1)
 high mpt (1) due to strong ionic bonds (1)
 conducts when molten (or in solution) (1) mobile ions (1)
 [OR] non conductor when solid (1) no mobile ions (or electrons) (1)] 8

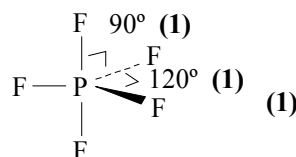
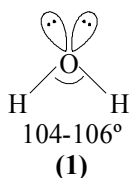


- from diag (3 hexagons in any one layer (1) planes (1))
 stated or from diag: macromolecular (1)
 must be stated: covalent (1)
 high mpt (1) due to strong covalent bonds (1)
 conducts when solid (1) delocalised electrons (1)
Note if free not delocalised mentioned penalise once only: for quality of language 8



- from diagram: spheres close together (1)
 stated or from diagram (hexagonal in one plane or lattice (1)
 ions (1) delocalised electrons (1)
 high mpt (1) strong metallic bonds (1)
 conducts (1) delocalised electrons (1) 8
 Max 23

- (b) electron pairs repel as far apart as possible (1)
 lone pairs repel more (1)



Note if angles not drawn on diagram penalise once only

7

[30]

46. **general**

NaCl: ionic/has ionic bonds ✓

Beware of contradictions for this mark, especially reference to intermolecular forces.**Ignore 'atoms'.**

graphite: covalent/giant molecular/macromolecular ✓

Ignore van der Waals', intermolecular, molecules**conductivity**NaCl: ions cannot move/
no free ions (or electrons) /
mobile ions only in solution or when molten ✓

graphite:

delocalised electrons/
free electrons (between layers)/
electrons conduct ✓**Ignore lone pair****melting point****both** graphite and NaCl:bonds are strong/
bonds difficult to break /
large amount of energy is needed to break bonds ✓**solubility**NaCl: Water is polar/water has a dipole/
ions interacts with water molecules ✓Graphite: no interaction with water/
no intermolecular forces with water/
graphite is non-polar ✓QWC: At least 2 complete sentences in which
the meaning is clear. ✓47. (i) $2\text{Na} + \text{O}_2 \rightarrow \text{Na}_2\text{O}_2$ ✓(ii) $\text{Na}_2\text{O}_2 + 2\text{H}_2\text{O} \rightarrow \text{H}_2\text{O}_2 + 2\text{NaOH}$ ✓

correct covalent bonds shown ✓

(iii) electron count (14) for rest of molecule correct ✓

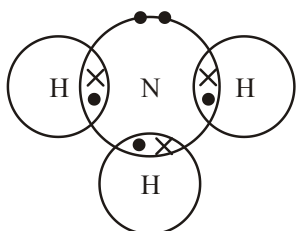
48. (i) (trigonal) pyramidal ✓

(ii) electron pairs repel/bonds repel
/electron pairs get as far apart as possible ✓
lone pairs repel more/forces 'them' closer ✓
4 electron pairs surround central atom or N
/diagram with 3 bonds and a lone pair ✓

49. (i) a shared pair of electrons ✓

ALLOW any response that communicates electron pair**ALLOW** shared pairs

(ii)

Must be 'dot-and-cross'
circles for outer shells **NOT** needed

IGNORE inner shells

Non-bonding electrons of N do not need to be shown as a pair

1

- (iii) Shape: pyramidal
- OR**
- (trigonal) pyramid ✓

Explanation:

There are 3 bonded pairs and 1 lone pair ✓

Lone pairs repel more than bonded pairs ✓

ALLOW 'bonds' for 'bonded pairs'**DO NOT ALLOW** 'atoms repel'**DO NOT ALLOW** electrons repel**ALLOW** LP for 'lone pair'**ALLOW** BP for bonded pair

3

[5]

50. (i) molecule shown as non-linear ✓

angle: 104 - 105° ✓

molecule shown as linear ✓

4

- (ii) angle: 180° ✓

shape of H₂Oshape of CO₂

Electron pairs repel/groups (or regions) of electrons

repel/electron pairs get as far apart as possible ✓

2

Oxygen in water surrounded by 4 areas of electron

density/2 bonds and 2 lone pairs

ANDCarbon in CO₂ surrounded by 2 regions of electron

density/2 double bonds ✓

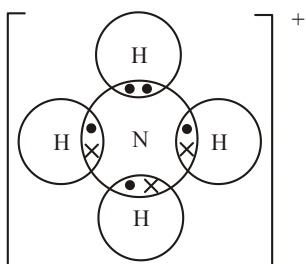
[6]

51. (i)
- $1s^2 2s^2 2p^6 3s^2 3p^6$
- ✓

ALLOW subscripts

1

- (ii)



'Dot-and-cross' diagram to show four shared pairs of electrons one of which is a dative covalent bond (which must consist of the same symbols) ✓

IGNORE inner shells**IGNORE** '+' sign **BUT DO NOT ALLOW** a '-' sign.

Brackets and circles not required

1

- (iii) tetrahedral ✓

109.5° ✓

ALLOW 109 - 110°

2

- (iv) ions
- OR**
- electrons cannot move in a solid ✓

ions can move **OR** are mobile in solution ✓

ALLOW ions can move in liquid

DO NOT ALLOW ions can move when molten

ALLOW 1 mark for:

'Ions can only move in solution'

2

[6]

52. (i) attraction of an **atom/element** for electrons ✓

in a (covalent) bond/bonded pair ✓

2

- (ii) one element attracts bonded pair more

/is more electronegative than other ✓

□ δ^- on more electronegative atom and δ^+ on less electronegative element in example ✓

2

May need to look for these marks below if not given here.

[4]

53. (i) (At least) two **NH₃** molecules with correct dipole shown with at

least one H with δ^+ and one N with δ^- ✓

DO NOT ALLOW first mark for ammonia molecules with incorrect lone pairs

DO NOT ALLOW first mark if H₂O, NH₂ or NH is shown

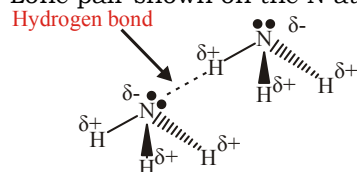
(Only) one hydrogen bond from N atom on one molecule to a

H atom on another molecule ✓

ALLOW hydrogen bond need not be labelled as long as it clear the bond type is different from the covalent N-H bond

ALLOW a line (i.e. looks like a covalent bond) as long as it is labelled 'hydrogen bond'

Lone pair shown on the N atom and hydrogen bond must hit the lone pair ✓



ALLOW 2-D diagrams

ALLOW two marks if water molecules are used. One awarded for a correct hydrogen bond and one for the involvement of lone pair

3

- (ii) Liquid H₂O is denser than solid ✓

In solid state H₂O molecules are held apart by hydrogen bonds

OR ice has an open lattice ✓

ORA

ALLOW ice floats for first mark

OR

H₂O has a relatively high boiling point **OR** melting point ✓

ALLOW higher melting **OR** boiling point than expected

DO NOT ALLOW H₂O has a high melting / boiling point

(relatively strong) hydrogen bonds need to be broken

OR a lot of energy is needed to overcome hydrogen bonds

OR hydrogen bonds are strong ✓

ALLOW other properties caused by hydrogen bonding not mentioned within the specification

E.g. high surface tension – strong hydrogen bonds on the surface

2

[5]

54. H-bond shown between H of one molecule and O, N or F of another ✓
H-bond shown going to a lone pair ✓ [2]
55. simple molecular ✓ 2 [2]
56. Original solution contains ions/there are mobile ions ✓
Charge carriers removed as reaction takes place
/as solid forms/ as BaSO₄ forms/as water forms ✓ 2 [2]
57. (i) attraction between oppositely charged ions 1
(ii)
-
- Mg and Cl both with 8 electrons in outer shell, (accept 0 electrons for Mg)
Cl must have one dot to seven crosses or vice versa (1)
correct charges on each ion (1) 2
- (iii) MgCl₂ does not conduct when solid because ions are fixed in lattice (1)
H₂O does not conduct as there are no free charge carriers/water molecules are uncharged (1)
MgCl₂ conducts when aqueous because ions are free to move (1) 3 [6]
58. (i) H bonding from O of 1 H₂O molecule to H of another ✓
dipoles shown ✓ with lone pair involved in bond ✓ 3
(ii) Two properties from:
Ice is lighter than water/ max density at 4°C ✓
explanation: H bonds hold H₂O molecules apart
/ open lattice in ice
/ H-bonds are longer ✓
Higher melting/boiling point than expected ✓
explanation:
strength of H bonds that need to be broken ✓
must imply that intermolecular bonds are broken
High surface tension/viscosity ✓
explanation strength of H bonds across surface ✓ 4 [7]
59. Two properties from:
Ice is less dense/lighter than water/floats on water/ max density at 4°C ✓
explanation: H bonds hold H₂O molecules apart
/ open lattice in ice
/ H-bonds are longer ✓ 2
Higher melting/boiling point than expected ✓
Not just high
Accept: 'unusually high/strangely high/relatively high'
explanation: H bonds need to be broken ✓ 2

must imply that intermolecular bonds are broken

High surface tension ✓

explanation strength of H bonds across surface ✓
mark 2 properties only □ 4 max

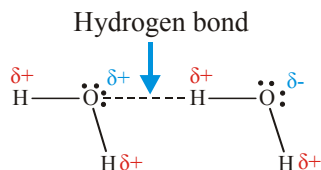
QoWC over whole question

□ legible text with accurate spelling, punctuation and grammar ✓

1

[5]

60. (i)



Shape of water with at least one H with $\delta+$ and at least one O with $\delta-$ ✓

H-bond between H in one water molecule and a lone pair of an O in another water molecule ✓

hydrogen bond labelled

OR H_2O has hydrogen bonding ✓

all marks can be awarded from a labelled diagram

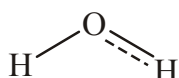
*If HO_2 shown then **DO NOT ALLOW** 1st mark*

*Dipole could be described in words so it does **not** need to be part of diagram.*

*At least one hydrogen bond **must** clearly hit a lone pair*

*Lone pair interaction could be described in words so it does **not** need to be part of diagram.*

DO NOT ALLOW hydrogen bonding if described in context of intramolecular bonding, ie



3

(ii) no hydrogen bonding

OR

weaker intermolecular forces ✓

DO NOT ALLOW 'weaker' / 'weak' hydrogen bonding

ALLOW weaker van der Waals' forces

ALLOW weaker dipole-dipole interactions

DO NOT ALLOW 'weak intermolecular forces'

(ie comparison essential here)

DO NOT ALLOW 'no intermolecular forces'

1

[4]

61. H bonding from lone pair on O of 1 H_2O molecule to H of another (1)

dipoles shown (1)

Two properties:

Ice is lighter than water / max density at 4°C (1)

explanation: H bonds hold H_2O molecules apart

/ open lattice in ice

/ H-bonds are longer (1)

Higher melting/boiling point than expected (1)

explanation: strength of H bonds that need to be broken **(1)**
must imply that intermolecular bonds are broken
High surface tension/viscosity **(1)**
explanation: strength of H bonds across surface **(1)**

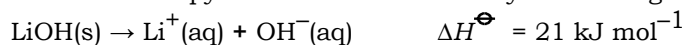
[6]

5.0 CHEMICAL ENERGETICS

1. A vessel and its contents of total heat capacity 120 J K^{-1} were heated using a methane burner. Calculate the maximum theoretical temperature rise when 0.10 g of methane was completely burned. The standard enthalpy of combustion of methane is -890 kJ mol^{-1} . (4)
(Total 4 marks)
2. When cyclohexanol, $\text{C}_6\text{H}_{11}\text{OH}$, is completely burned in oxygen, the products are carbon dioxide and water. Write an equation to represent the reaction that occurs during the measurement of the enthalpy change of combustion of cyclohexanol. (Total 1 mark)
3. The table below includes some values of standard enthalpies of formation ($\Delta H_f^\ominus$).

Substance	$\text{H}_2\text{O(l)}$	LiOH(s)	Li(s)
$\Delta H_f^\ominus / \text{kJ mol}^{-1}$	-286	-487	0

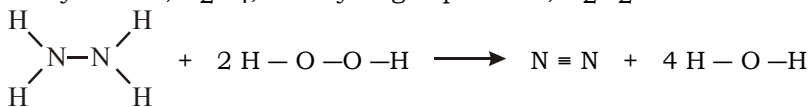
The standard enthalpy of solution of lithium hydroxide is given below.



- (a) State why the standard enthalpy of formation of lithium is quoted as zero. (1)
- (b) Write an equation for the chemical reaction which represents the formation of lithium hydroxide from its elements, in which the enthalpy change is equal to its standard enthalpy of formation. (2)
- (c) Write an equation, including state symbols, for the reaction of lithium with water in which lithium ions are formed. (2)
- (d) Use the data given above to calculate a value for the enthalpy change for the reaction of lithium with water. (3)
- (Total 8 marks)
4. (a) Explain the meaning of the terms *mean bond enthalpy* and *standard enthalpy of formation*. (5)
- (b) Some mean bond enthalpies are given below.

Bond	N-H	N-N	N≡N	H-O	O-O
Mean bond enthalpy/ kJ mol^{-1}	388	163	944	463	146

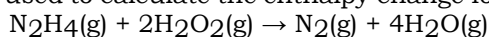
Use these data to calculate the enthalpy change for the following gas-phase reaction between hydrazine, N_2H_4 , and hydrogen peroxide, H_2O_2



- (c) Some standard enthalpies of formation are given below. (3)

	$\text{N}_2\text{H}_4(\text{g})$	$\text{H}_2\text{O}_2(\text{g})$	$\text{H}_2\text{O(g)}$
$\Delta H_f^\ominus / \text{kJ mol}^{-1}$	+75	-133	-242

These data can be used to calculate the enthalpy change for the reaction in part (b).

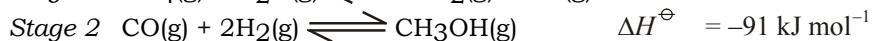
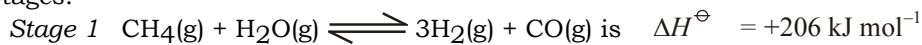


- (i) State the value of $\Delta H_f^\ominus$ for $\text{N}_2(\text{g})$.

- (ii) Use the $\Delta H_f^\ominus$ values from the table to calculate the enthalpy change for this reaction. (4)
- (d) Explain why the value obtained in part (b) is different from that obtained in part (c)(ii). (1)
- (Total 13 marks)**

5. Methanol, CH_3OH , is a convenient liquid fuel.

- (a) An experiment was conducted to determine the enthalpy of combustion of liquid methanol. The energy obtained from burning 2.12 g of methanol was used to heat 150 g of water. The temperature of the water rose from 298 K to 362 K. (The specific heat capacity of water is $4.18 \text{ J K}^{-1} \text{ g}^{-1}$)
- (i) Define the term *standard enthalpy of combustion*.
- (ii) Use the data above to calculate a value for the enthalpy of combustion of one mole of liquid methanol. (7)
- (b) Methanol can be synthesised from methane and steam by a process that occurs in two stages.



The standard enthalpies of combustion of carbon monoxide and of hydrogen are -283 kJ mol^{-1} and -286 kJ mol^{-1} , respectively. Use these data and the enthalpy change for *Stage 2* to calculate a value for the standard enthalpy of combustion of gaseous methanol. (3)

(Total 10 marks)

6. (a) Define the term *standard enthalpy of formation*. (3)
- (b) State Hess's Law and use it, together with the data given in the table below, to calculate the standard enthalpy change for the following reaction. (3)
- $$\text{MgO}(\text{s}) + 2\text{HCl}(\text{g}) \rightarrow \text{MgCl}_2(\text{s}) + \text{H}_2\text{O}(\text{l})$$

	MgO	HCl(g)	MgCl ₂	H ₂ O
$\Delta H_f^\ominus / \text{kJ mol}^{-1}$	-602	-92	-642	-286

(4)

- (c) In an experiment, an excess of solid magnesium oxide was added to 50 cm^3 of 3.0 mol dm^{-3} hydrochloric acid. The initial temperature of the solution was 21°C . After reaction, the temperature had risen to 53°C . (The specific heat capacity of water is $4.2 \text{ J K}^{-1} \text{ g}^{-1}$)
- Use this information to calculate the enthalpy change for the reaction of one mole of magnesium oxide with hydrochloric acid. For your calculation you should assume that all the heat from the reaction is used to raise the temperature of 50 g of water. (8)

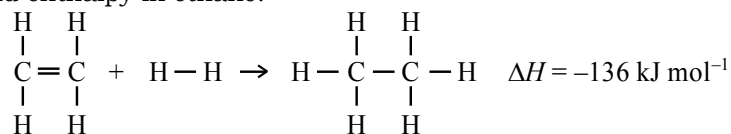
(Total 15 marks)

7. The table below contains some mean bond enthalpy data.

Bond	H—H	C—C	C=C	N≡N	N—H
Mean bond enthalpy / kJ mol^{-1}	436	348	612	944	388

- (a) Explain the term *mean bond enthalpy*. (2)
- (b) (i) Write an equation for the formation of one mole of ammonia, NH_3 , from its elements.
- (ii) Use data from the table above to calculate a value for the enthalpy of formation of ammonia.

- (c) Use the following equation and data from the table above to calculate a value for the C-H bond enthalpy in ethane. (4)



8. (a) Write an equation for the complete combustion of propanone, $\text{C}_3\text{H}_6\text{O}$, to form carbon dioxide and water. (3)
- (Total 9 marks)

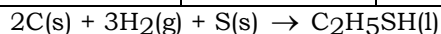
- (b) In a laboratory experiment, 1.45 g of propanone were burned completely in oxygen. The heat from this combustion was used to raise the temperature of 100 g of water from 293.1 K to 351.2 K. (1)
- (i) Calculate the number of moles of propanone in the 1.45 g.
- (ii) Calculate the heat energy required to raise the temperature of 100 g of water from 293.1 K to 351.2 K.

(The specific heat capacity of water is $4.18 \text{ J K}^{-1} \text{ g}^{-1}$)

- (iii) Hence, calculate a value, in kJ mol^{-1} , for the enthalpy of combustion of propanone. (5)
- (c) In a similar experiment, the enthalpy of combustion of butanone, $\text{C}_4\text{H}_8\text{O}$, was found to be $-1290 \text{ kJ mol}^{-1}$. A data book value for the same reaction is $\Delta H_{\text{C}}^{\ominus} = -2430 \text{ kJ mol}^{-1}$.
- (i) Suggest one reason why the experimental value is very different from the data book value.
- (ii) This data book value of $\Delta H_{\text{C}}^{\ominus}$ for butanone ($-2430 \text{ kJ mol}^{-1}$) refers to the formation of carbon dioxide gas and water in the gaseous state. How would this value differ if it referred to the formation of water in the liquid state? Explain your answer. (3)

- (d) Calculate a value for the standard enthalpy of formation for liquid ethanethiol, $\text{C}_2\text{H}_5\text{SH}$. Use the equation given below and enthalpy of combustion data from the following table. (3)

Substance	$\text{C}_2\text{H}_5\text{SH(l)}$	C(s)	$\text{H}_2\text{(g)}$	S(s)
$\Delta H_{\text{C}}^{\ominus} / \text{kJ mol}^{-1}$	-1170	-394	-286	-297

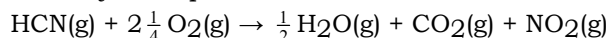


9. (a) Define the term *standard enthalpy of formation*. (3)

- (b) Write an equation, including state symbols, for a reaction for which the enthalpy change is the standard enthalpy of formation of liquid CH_3NO_2 (3)

- (c) Give the name of the principle or law which enables enthalpies of formation to be calculated from enthalpies of combustion. (2)

- (d) In the presence of a catalyst, gaseous hydrogen cyanide, HCN , burns in an excess of oxygen as shown by the equation below. (1)



Some standard enthalpies of combustion, $\Delta H_{\text{C}}^{\ominus}$ are given in the table below.

Substance	HCN(g)	H ₂ (g)	C(s)	N ₂ (g)
$\Delta H_f^\ominus / \text{kJ mol}^{-1}$	-611	-242	-394	+68

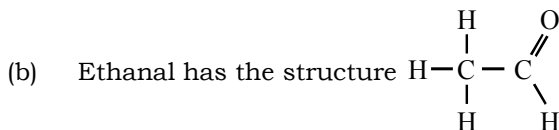
Use these data to calculate a value for the standard enthalpy of formation for gaseous hydrogen cyanide.

(4)

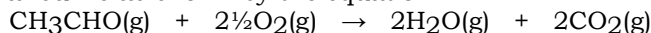
(Total 10 marks)

10. (a) State what is meant by the term *mean bond enthalpy*.

(2)



Gaseous ethanal burns as shown by the equation



Use the mean bond enthalpy data given below to answer the following questions.

Bond	Mean bond enthalpy/kJ mol ⁻¹
C—H	+413
C—C	+347
C=O	+736
O=O	+498
O—H	+464

- Calculate the enthalpy change which occurs when all the bonds in the reactants shown in the above equation are broken.
- Calculate the enthalpy change which occurs when all the bonds in the products shown in the above equation are formed.
- Hence, calculate the enthalpy change for the complete combustion of ethanal as shown in the equation above.

(5)

(Total 7 marks)

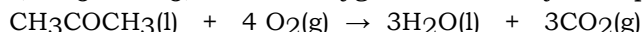
11. (a) Define the term *standard molar enthalpy of formation*, $\Delta H_f^\ominus$.

(3)

- (b) State Hess's law.

(1)

- (c) Propanone, CH₃COCH₃, burns in oxygen as shown by the equation



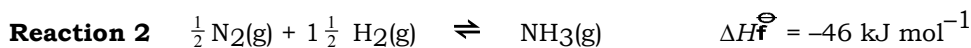
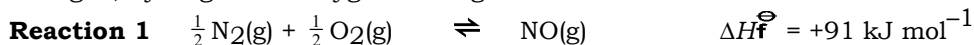
Use the data given below to calculate the standard enthalpy of combustion of propanone.

	CO ₂ (g)	H ₂ O(l)	CH ₃ COCH ₃ (l)
$\Delta H_f^\ominus / \text{kJ mol}^{-1}$	-394	-286	-248

(3)

(Total 7 marks)

12. Nitrogen, hydrogen and oxygen undergo the reactions shown below.



Use this information in answering the questions that follow.

- (a) What name is given to the symbol $\Delta H_f^\ominus$? Explain fully what it means.

(4)

- (b) (i) Write an equation for the reaction of nitrogen monoxide, NO, with hydrogen to form ammonia and steam.

- (ii) Calculate the value of $\Delta H^{\ominus}$ for this reaction, using the data given at the beginning of Question 3.

(4)

(Total 8 marks)

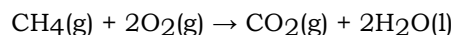
13. A student added 50.0 cm^3 of hydrochloric acid to 50.0 cm^3 of sodium hydroxide solution in a polystyrene cup. The temperature rose by 6.5°C . The initial concentration of each solution was 1.00 mol dm^{-3} .

- (a) Write an ionic equation for the reaction occurring. (1)
- (b) Calculate the number of moles of acid used in the reaction. (1)
- (c) Calculate the heat energy evolved in the reaction. (Assume that the final solution has a specific heat capacity of $4.18 \text{ J g}^{-1} \text{ K}^{-1}$ and a density of 1.00 g cm^3 .) (2)
- (d) Calculate the molar enthalpy change for the reaction. (2)

(2)

(Total 6 marks)

14. (a) Define the term *standard enthalpy of combustion*. (3)
- (b) Using the data given below, calculate the standard enthalpy change for the following reaction.



$$\Delta H_f^{\ominus} \text{CO}_2(\text{g}) = -394 \text{ kJ mol}^{-1}$$

$$\Delta H_f^{\ominus} \text{H}_2\text{O}(\text{l}) = -286 \text{ kJ mol}^{-1}$$

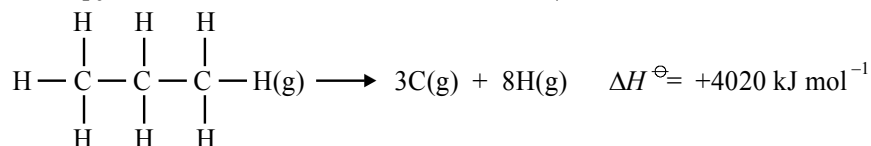
$$\Delta H_f^{\ominus} \text{CH}_4(\text{g}) = -75 \text{ kJ mol}^{-1}$$

(3)

- (c) (i) State what is meant by the term *mean bond enthalpy*.
- (ii) Using the standard enthalpy of formation of methane given in part (b) and the data given below, calculate the mean bond enthalpy of the **C-H** bond in methane.



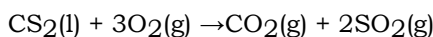
- (iii) Using the C-H bond enthalpy calculated in part (c)(ii) and the standard enthalpy change for the reaction given below, calculate the mean bond enthalpy of the C-C bond in propane.
- N.B. If you failed to complete part (c)(ii), you may assume that the mean bond enthalpy of the C-H bond is $+390 \text{ kJ mol}^{-1}$. (This is not the correct value.)



(7)

(Total 13 marks)

15. (a) Define the term *standard enthalpy of combustion*, $\Delta H_c^{\ominus}$. (3)
- (b) Methanethiol, CH_3SH , is added to natural gas in order to give it a smell. When methanethiol burns in air it forms carbon dioxide, water and sulphur dioxide. Write a balanced equation for this combustion reaction. (1)
- (c) A value for the enthalpy of formation of methanethiol can be determined using Hess's Law and enthalpies of combustion. In this calculation, which three enthalpy values would be needed in addition to the enthalpy of combustion of methanethiol? (2)
- (d) Carbon disulphide, CS_2 , burns in air as follows:



Some standard enthalpies of formation, $\Delta H_f^\ominus$ are given in the table below.

Substance	$\text{CS}_2(\text{l})$	$\text{CO}_2(\text{g})$	$\text{SO}_2(\text{g})$
$\Delta H_f^\ominus / \text{kJ mol}^{-1}$	+88	-394	-297

Use these data to calculate a value for the enthalpy of combustion of liquid carbon disulphide.

(3)

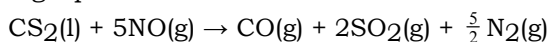
(Total 9 marks)

16. The tables below contain values of standard enthalpy of combustion, $\Delta H_c^\ominus$ and standard enthalpy of formation, $\Delta H_f^\ominus$.

	C(graphite)	S(s)
$\Delta H_c^\ominus / \text{kJ mol}^{-1}$	-394	-297

	$\text{CS}_2(\text{l})$	$\text{NO}(\text{g})$	$\text{CO}(\text{g})$	$\text{SO}_2(\text{g})$
$\Delta H_f^\ominus / \text{kJ mol}^{-1}$	88	90	-111	-297

- (a) State Hess's Law. (2)
- (b) Explain the meaning of the term *standard enthalpy of combustion* of a compound. (3)
- (c) (i) Write equations, with state symbols, for the formation of liquid carbon disulphide (CS_2) from its elements and for the combustion of carbon disulphide to form carbon dioxide and sulphur dioxide. (3)
- (ii) Calculate the enthalpy of combustion of carbon disulphide, using its enthalpy of formation and the enthalpies of combustion from the tables on above. (5)
- (d) Carbon disulphide reacts explosively with nitrogen monoxide (NO) according to the following equation.

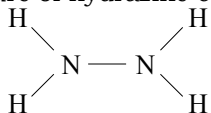


Use the enthalpies of formation given in the table to calculate the enthalpy change in this reaction.

(3)

(Total 13 marks)

17. The structure of hydrazine can be represented as shown below.



- (a) (i) Write an equation for the formation of hydrazine from its elements. (3)
- (ii) Steam and nitrogen gas are the combustion products of gaseous hydrazine. Write an equation for this reaction. (3)
- (iii) Use the data given below to calculate the standard enthalpy of formation of gaseous hydrazine. (4)

$$\Delta H_c^\ominus \text{N}_2\text{H}_4(\text{g}) = -568 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus \text{H}_2\text{O}(\text{g}) = -242 \text{ kJ mol}^{-1}$$

(4)

- (b) State what is meant by the term *mean bond enthalpy*. (2)
- (c) Use the enthalpy of formation of hydrazine calculated in part (a) and the mean bond enthalpies given below to calculate the enthalpy of the bond between the two nitrogen atoms in hydrazine. (2)

NB If you failed to complete part (a) you may assume that

$$\Delta H_f^\ominus \text{N}_2\text{H}_4(\text{g}) = +50 \text{ kJ mol}^{-1} \text{ (not the correct value).}$$

Bond	N-H	N≡N	H-H
Mean bond enthalpy/kJ mol ⁻¹	389	945	436

(3)

- (d) The measured N-N bond enthalpy in hydrazine is 158 kJ mol⁻¹. Explain why the bond enthalpy when calculated correctly in part (c) does not agree with the measured value.

(1)

(Total 10 marks)

18. The table below gives some *standard enthalpies of formation*, $\Delta H_f^\ominus$.

Substance	C ₂ H ₆ (g)	F ₂ (g)	CF ₄ (g)	HF(g)
$\Delta H_f^\ominus$ / kJ mol ⁻¹	-85	0	-680	-269

- (a) Define the term *standard enthalpy of formation*. (3)
- (b) Why is the value for the standard enthalpy of formation for fluorine equal to zero? (1)
- (c) Calculate the standard enthalpy change for the following reaction.
 $\text{C}_2\text{H}_6(\text{g}) + 7\text{F}_2(\text{g}) \rightarrow 2\text{CF}_4(\text{g}) + 6\text{HF}(\text{g})$ (3)
- (d) The table below shows some bond enthalpies.

Bond	C-C	C-H	F-F
Bond enthalpy / kJ mol ⁻¹	348	412	158

- (i) Why are the values for C-C and C-H described as **mean** bond enthalpies? (4)
- (ii) Suggest the most likely first step in the reaction shown in part (c) and give a reason for your answer.

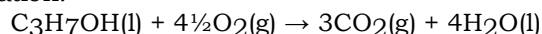
(Total 11 marks)

19. (a) Define the terms *standard enthalpy of formation* and *standard enthalpy of combustion*. (6)

- (b) Use the standard enthalpies of formation, $\Delta H_f^\ominus$, given below

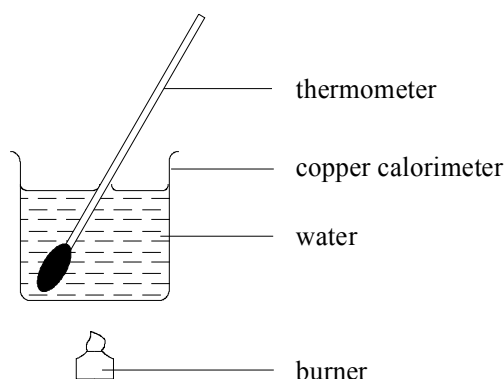
Compound	$\Delta H_f^\ominus$ / kJ mol ⁻¹
CO ₂ (g)	-394
C ₃ H ₇ OH(l)	-304
H ₂ O(l)	-286

to calculate the standard enthalpy of combustion of an alcohol C₃H₇OH, as shown by the equation:



(3)

- (c) A value for the enthalpy of combustion of the alcohol C₃H₇OH was determined in the laboratory using the apparatus shown below. The following results were obtained.



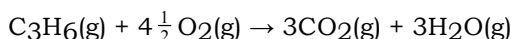
Mass of water in the calorimeter = 200 g
 Initial temperature of water = 15 °C
 Final temperature = 30 °C
 Mass of alcohol burned = 0.90 g

- Calculate the heat energy required to raise the temperature of the water from 15 °C to 30 °C. The specific heat capacity of water is $4.2 \text{ J g}^{-1} \text{ K}^{-1}$.
- Calculate the number of moles of the alcohol, $\text{C}_3\text{H}_7\text{OH}$, burned.
- Hence, calculate a value for the enthalpy of combustion of 1.0 mol of the alcohol.
- Give **two** reasons why you would expect your answer to part (c)(iii) to differ from that in part (b).

(8)

(Total 17 marks)

20. An equation for the combustion of cyclopropane, C_3H_6 , is shown below.



The standard enthalpy of combustion of cyclopropane can be calculated either from standard enthalpies of formation or by using mean bond enthalpies.

- (a) Use the standard enthalpies of formation given below to calculate the standard enthalpy of combustion of cyclopropane.

$$\Delta H_f^\ominus \text{C}_3\text{H}_6(\text{g}) = +53 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus \text{CO}_2(\text{g}) = -393 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus \text{H}_2\text{O}(\text{g}) = -242 \text{ kJ mol}^{-1}$$

(3)

- (b) State what is meant by the term *mean bond enthalpy*.

(2)

- (c) The following mean bond enthalpies have been obtained from a data book. Use these values to calculate the standard enthalpy of combustion of cyclopropane.

$$\text{C}-\text{C} \quad 347 \text{ kJ mol}^{-1}$$

$$\text{C}-\text{H} \quad 413 \text{ kJ mol}^{-1}$$

$$\text{C}=\text{O} \quad 805 \text{ kJ mol}^{-1}$$

$$\text{O}=\text{O} \quad 498 \text{ kJ mol}^{-1}$$

$$\text{O}-\text{H} \quad 464 \text{ kJ mol}^{-1}$$

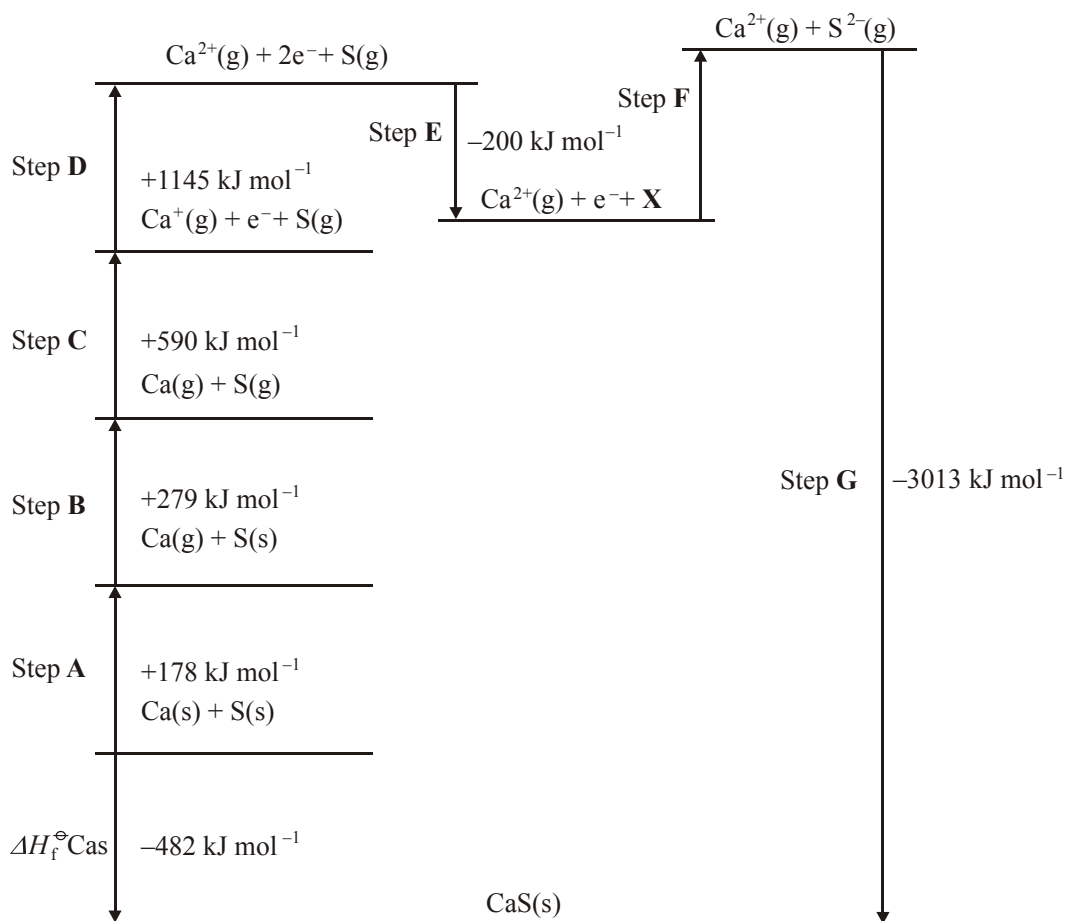
(3)

- (d) Explain, by reference to the structure of cyclopropane, why the enthalpy of combustion calculated in part (a) is more exothermic than that calculated in part (c).

(2)

(Total 10 marks)

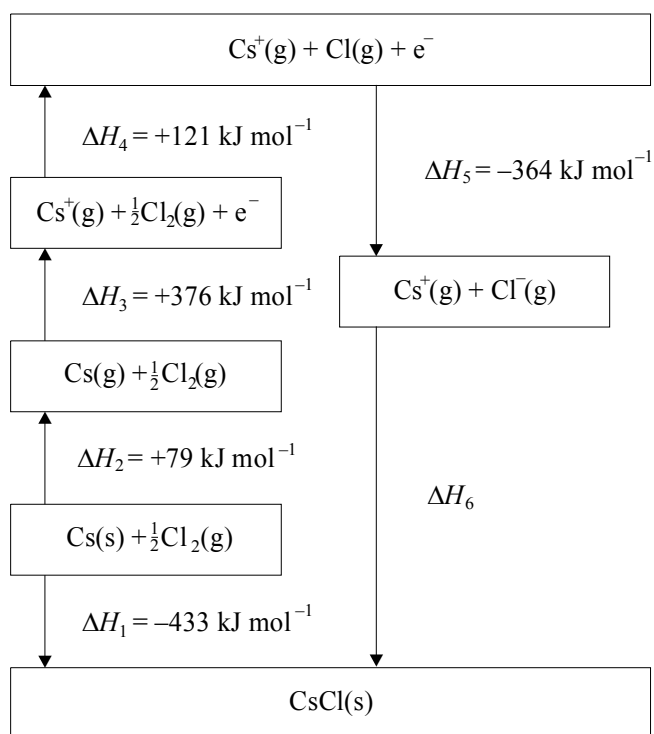
21. A Born-Haber cycle for the formation of calcium sulphide is shown below. The cycle includes enthalpy changes for all Steps except Step **F**. (The cycle is not drawn to scale.)



- Give the full electronic arrangement of the ion S^{2-} . (1)
- Identify the species **X** formed in Step **E**. (1)
- Suggest why Step **F** is an endothermic process. (2)
- Name the enthalpy change for each of the following steps. (3)
 - Step **B**
 - Step **D**
 - Step **F**
- Explain why the enthalpy change for Step **D** is larger than that for Step **C**. (2)
- Use the data shown in the cycle to calculate a value for the enthalpy change for Step **F**. (2)

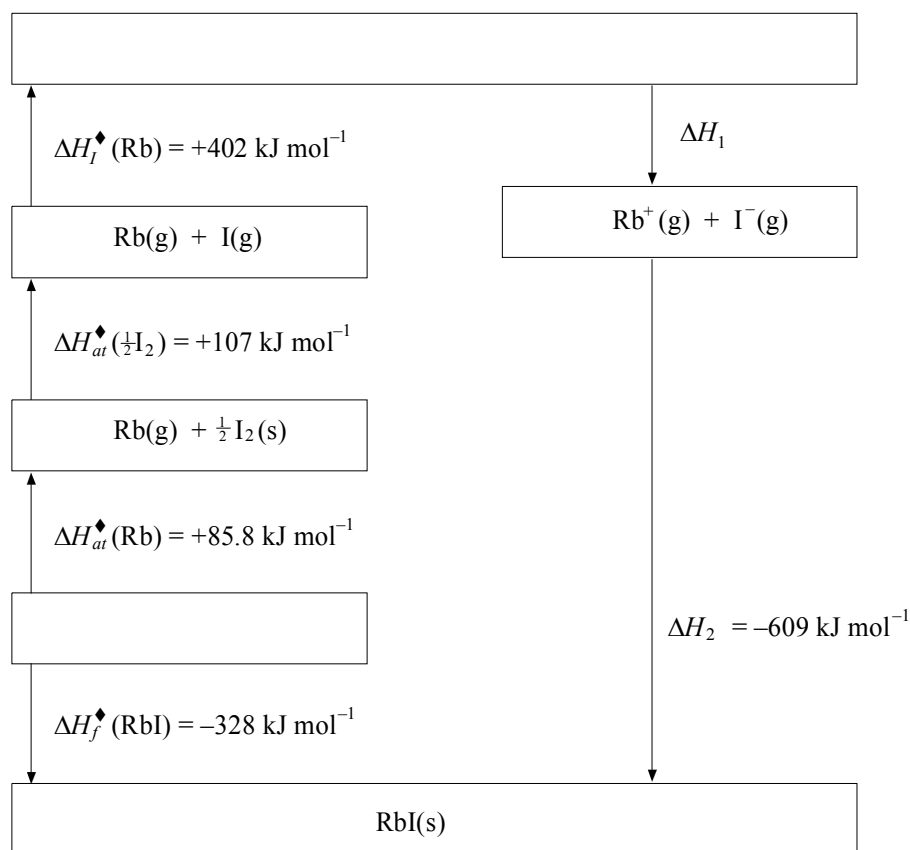
(Total 11 marks)

22. The energy level diagram (Born-Haber cycle) for caesium chloride is shown below.



- (a) Give the names of the enthalpy changes represented by ΔH_1 , ΔH_2 and ΔH_5 . (3)
- (b) Calculate the value of the lattice energy ΔH_6 . (2)
- (c) Explain why the enthalpy change represented by ΔH_3 has a lower magnitude for caesium than for sodium. (3)
- (Total 8 marks)**

23. **Figure 1** shows the energy level diagram (Born-Haber cycle) for the formation of rubidium iodide from its elements.

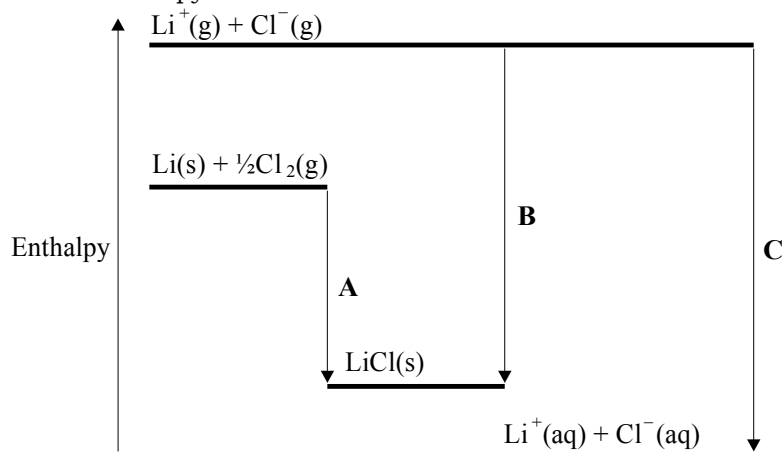
**Figure 1**

- Complete the diagram giving the identities of the missing species (2)
 - Give the names of the enthalpy changes represented by ΔH_1 and ΔH_2 . (2)
 - Calculate the value of the enthalpy change represented by ΔH_1 . (2)
- (Total 6 marks)**

24. The following data relate to lithium chloride.

Standard molar enthalpy change of solution is $-37.0 \text{ kJ mol}^{-1}$.

Lattice enthalpy is -846 kJ mol^{-1} .



- Give the name of each of the changes **A** and **B**. (2)
- Calculate the value of the enthalpy change represented by **C** and suggest the (2)

name(s) of the enthalpy change(s).

(3)

(Total 5 marks)

25. (a) (i) Construct a Born-Haber cycle for the formation of calcium chloride, CaCl_2 , from its elements. Name each step in the cycle and mark on the cycle each species involved together with its state symbol.

- (ii) When calcium reacts with chlorine under normal laboratory conditions CaCl_2 is formed not CaCl . Identify the stages in the Born-Haber cycle for the formation of CaCl which differ from those in the cycle for the formation of CaCl_2

(11)

- (b) (i) Use the data given below to calculate the standard enthalpy of solution of CaCl_2 .

Lattice dissociation enthalpy of CaCl_2 = $+2255 \text{ kJ mol}^{-1}$

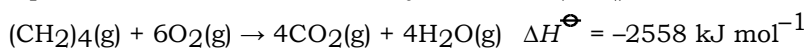
Hydration enthalpy of calcium ions = $-1650 \text{ kJ mol}^{-1}$

Hydration enthalpy of chloride ions = -384 kJ mol^{-1}

- (ii) Using your answer to part (b)(i), deduce how the solubility of CaCl_2 changes as temperature is increased. Explain your answer.

(6)

- (c) (i) Standard enthalpies of formation of hydrocarbons are difficult to measure directly but can be calculated indirectly using experimental data. The equation for the combustion of cyclobutane, $(\text{CH}_2)_4$, is



Calculate the enthalpy of formation of cyclobutane using the data given,

Standard enthalpy of formation of $\text{H}_2\text{O}(\text{g})$ = $-245.1 \text{ kJ mol}^{-1}$

Standard enthalpy of formation of $\text{CO}_2(\text{g})$ = $-393.5 \text{ kJ mol}^{-1}$

- (ii) Enthalpies of combustion can be calculated using bond enthalpies. Use the data given below to calculate the enthalpy of combustion of cyclobutane as shown by the equation above. Explain why the value obtained in this calculation is not $-2558 \text{ kJ mol}^{-1}$.

	C-H	C-C	O=O	O-H	C=O
Bond enthalpy/ kJ mol^{-1}	413	347	498	464	805

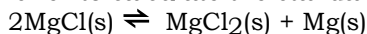
(13)

(Total 30 marks)

26. Construct a Born-Haber cycle for the formation of the hypothetical crystalline solid magnesium(I) chloride, $\text{MgCl}(\text{s})$.

The table below shows values of standard enthalpies for some processes involving magnesium and chlorine. Use these values to calculate the standard enthalpy of formation of the hypothetical $\text{MgCl}(\text{s})$.

Use your answer to calculate the standard enthalpy change for the reaction



given that the standard enthalpy of formation of $\text{MgCl}_2(\text{s})$, $\Delta H_f^\ominus$, is -653 kJ mol^{-1}

Explain why the standard entropy change in this reaction is likely to be negligibly small.

Comment on the stability of $\text{MgCl}(\text{s})$ relative to that of $\text{MgCl}_2(\text{s})$.

			$\Delta H^\ominus \text{ kJ mol}^{-1}$
$\text{Mg}^+(\text{g}) + \text{Cl}^-(\text{g})$	$\rightarrow$	$\text{MgCl}(\text{s})$	-364
$\text{Cl}(\text{g}) + \text{e}^-$	$\rightarrow$	$\text{Cl}^-(\text{g})$	+121
$\frac{1}{2} \text{Cl}_2(\text{g})$	$\rightarrow$	$\text{Cl}(\text{g})$	+146
$\text{Mg}(\text{s})$	$\rightarrow$	$\text{Mg}(\text{g})$	+736
$\text{Mg}(\text{g})$	$\rightarrow$	$\text{Mg}^+(\text{g}) + \text{e}^-$	-753

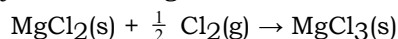
(Total 10 marks)

27. (a) State what is meant by the term *enthalpy change*. (2)
- (b) Construct a fully-labelled Born-Haber cycle for the formation of MgO and use the data given below to calculate a value for the enthalpy of lattice formation of this oxide.

Process	$\Delta H^\ominus / \text{kJ mol}^{-1}$
$\text{Mg(s)} + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{MgO(s)}$	-602
$\text{Mg(s)} \rightarrow \text{Mg(g)}$	+148
$\text{Mg(g)} \rightarrow \text{Mg}^+(\text{g}) + \text{e}^-$	+738
$\text{Mg}^+(\text{g}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{e}^-$	+1451
$\text{O}_2(\text{g}) \rightarrow 2\text{O(g)}$	+498
$\text{O(g)} + \text{e}^- \rightarrow \text{O}^-(\text{g})$	-141
$\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$	+798

(8)

- (c) By consideration of the entropy changes in the reaction and the data given below, explain why the following reaction is never feasible.



Compound	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
$\text{MgCl}_2(\text{s})$	-653
$\text{MgCl}_3(\text{s})$	+3904

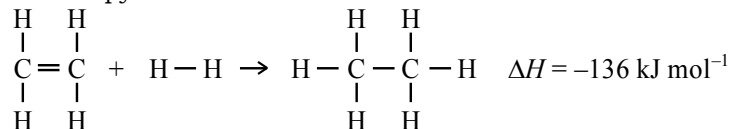
(5)

(Total 15 marks)

28. The table below contains some mean bond enthalpy data.

Bond	H-H	C-C	C=C	N≡N	N-H
Mean bond enthalpy / kJ mol ⁻¹	436	348	612	944	388

- (a) Explain the term *mean bond enthalpy*. (2)
- (b) (i) Write an equation for the formation of one mole of ammonia, NH₃, from its elements. (2)
- (ii) Use data from the table above to calculate a value for the enthalpy of formation of ammonia. (4)
- (c) Use the following equation and data from the table above to calculate a value for the C-H bond enthalpy in ethane. (4)



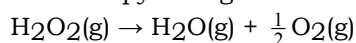
(3)

(Total 9 marks)

29. (a) The table below contains some mean bond enthalpy data.

Bond	H-O	O-O	O=O
Mean bond enthalpy/kJ mol ⁻¹	463	146	496

The bonding in hydrogen peroxide, H₂O₂, can be represented by H-O-O-H. Use these data to calculate the enthalpy change for the following reaction.



(3)

- (b) The standard enthalpy of formation, $\Delta H_f^\ominus$ for methane, is -74.9 kJ mol⁻¹. Write an

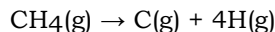
equation, including state symbols, for the reaction to which this enthalpy change applies.

(2)

- (c) The enthalpy changes for the formation of atomic hydrogen and atomic carbon from their respective elements in their standard states are as follows.



- (i) By reference to its structure, suggest why a large amount of heat energy is required to produce free carbon atoms from solid carbon.
- (ii) Parts (b) and (c) give enthalpy data for the formation of $\text{CH}_4(\text{g})$, $\text{H}(\text{g})$ and $\text{C}(\text{g})$. Use these data and Hess's Law to calculate the value of the enthalpy change for the following reaction.



- (iii) Use your answer from part (c)(ii) to calculate a value for the mean bond enthalpy of a C–H bond in methane.

(5)

(Total 10 marks)

30. (a) Define the term *standard enthalpy of combustion*.

(2)

- (b) (i) Write an equation for the complete combustion of ethane, C_2H_6 .
- (ii) Use the standard enthalpies of formation given below to calculate the standard enthalpy of combustion of ethane.

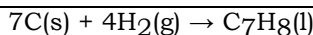
Formula and state of compound	$\text{C}_2\text{H}_6(\text{g})$	$\text{CO}_2(\text{g})$	$\text{H}_2\text{O}(\text{l})$
Standard enthalpy of formation (at 298 K) / kJ mol^{-1}	–85	–394	–286

(4)

(Total 6 marks)

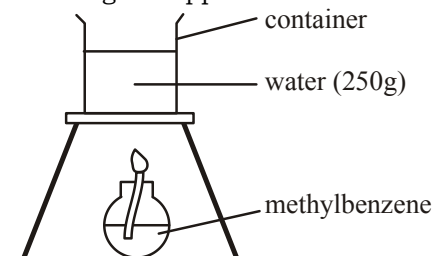
31. (a) Use the data in the table to calculate the standard enthalpy of formation of liquid methylbenzene, C_7H_8 .

Substance	$\text{C}(\text{s})$	$\text{H}_2(\text{g})$	$\text{C}_7\text{H}_8(\text{l})$
Standard enthalpy of combustion, ΔH_c° / kJ mol^{-1}	–394	–286	–3909



(3)

- (b) An experiment was carried out to determine a value for the enthalpy of combustion of liquid methylbenzene using the apparatus shown in the diagram.



Burning 2.5 g of methylbenzene caused the temperature of 250 g of water to rise by 60°C . Use this information to calculate a value for the enthalpy of combustion of methylbenzene, C_7H_8

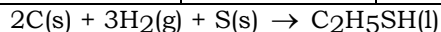
(The specific heat capacity of water is $4.18 \text{ J K}^{-1} \text{ g}^{-1}$. Ignore the heat capacity of the container.)

(3)

- (c) Calculate a value for the standard enthalpy of formation for liquid

ethanethiol, $\text{C}_2\text{H}_5\text{SH}$. Use the equation given below and enthalpy of combustion data from the following table.

Substance	$\text{C}_2\text{H}_5\text{SH}(\text{l})$	$\text{C}(\text{s})$	$\text{H}_2(\text{g})$	$\text{S}(\text{s})$
$\Delta H_c^\ominus / \text{kJ mol}^{-1}$	-1170	-394	-286	-297



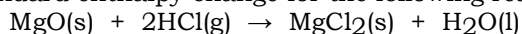
(4)

(Total 10 marks)

32. (a) Define the term *standard enthalpy of formation*.

(3)

- (b) State Hess's Law and use it, together with the data given in the table below, to calculate the standard enthalpy change for the following reaction.



	MgO	$\text{HCl}(\text{g})$	MgCl_2	H_2O
$\Delta H_f^\ominus / \text{kJ mol}^{-1}$	-602	-92	-642	-286

(4)

- (c) In an experiment, an excess of solid magnesium oxide was added to 50 cm^3 of 3.0 mol dm^{-3} hydrochloric acid. The initial temperature of the solution was 21°C . After reaction, the temperature had risen to 53°C . (The specific heat capacity of water is $4.2 \text{ J K}^{-1} \text{ g}^{-1}$)

Use this information to calculate the enthalpy change for the reaction of one mole of magnesium oxide with hydrochloric acid. For your calculation you should assume that all the heat from the reaction is used to raise the temperature of 50 g of water.

(8)

(Total 15 marks)

Marking Scheme

1. moles methane = $\frac{0.10}{16} = 6.25 \times 10^{-3}$ (1)
 kJ evolved = $6.25 \times 10^{-3} \times 890 = 5.56$ (1)
 5.56×10^3 joules = (mc) ΔT (1)
 $\Delta T = \frac{5.56 \times 10^3}{120} = 46.4$ K (1) 4
- [4]
2. $C_6H_{11}OH + 8\frac{1}{2}O_2 \rightarrow 6CO_2 + 6H_2O$ /double or multiple equation (1)
3. (a) by definition (1) 1
- (b) $Li(s) + \frac{1}{2}O_2(g) + \frac{1}{2}H_2(g) \rightarrow LiOH(s)$ equation (1)
 ss (1) 2
- (c) $Li(s) + H_2O(l) \rightarrow Li^+(aq) + OH^-(aq) + \frac{1}{2}H_2(g)$ equation (1)
 ss (1) 2
- (d) $\Delta H = \Delta H_f^\ominus (LiOH(aq)) - \Delta H_f^\ominus (H_2O(l))$ or cycle (1)
 $= -487 - 21 - (-286)$ (1)
 $= -222$ (kJ mol⁻¹) (1)
 (allow (1) for -201)
 reactivity increases (1)
Relationship between $\Delta H^\ominus$ and
 reactivity none (1)
Reason reactivity governed by rate (or E_a) (1) 3
- [8]
4. (a) enthalpy (or energy) to break (or dissociate) a bond; 1
 averaged over different molecules (environments); 1
 enthalpy (or heat energy) change when one mole of a compound; 1
 is formed from its elements; 1
 in their standard states; 1
- (b) enthalpy change = $\sum(\text{bonds broken}) - \sum(\text{bonds formed})$ or cycle; 1
 $= 4 \times 388 + 163 + 2 \times 146 + 4 \times 463 - (944 + 8 \times 463);$ 1
 (or similar)
 $= -789$; 1
 (+ 789 scores 1 only)
- (c) (i) zero; 1
 (ii) $\Delta H = \sum(\text{enthalpies of formation of products})$
 $- \sum(\text{enthalpies of formation of reactants})$ 1
 $= 4 \times -242 - (75 + 2 \times -133);$ 1
 $= -777;$ 1
 (+ 777 scores one only)
- (d) mean bond enthalpies are not exact 1
 (or indication that actual values are different from real values)
- [13]
5. (a) (i) enthalpy change when 1 mol of a substance (or compound) (QL mark) 1
 is (completely) burned in oxygen (or reacted in excess oxygen) 1
 at 298 K and 100 kPa (or under standard conditions) 1
- (ii) heat produced = mass of water $\times$ Sp heat capacity $\times \Delta T$ (or $mc\Delta T$) 1
 $= 150 \times 4.18 \times 64$ (note if mass = 2.12 lose first 2 marks then conseq) 1
 $= 40100$ J or $= 40.1$ kJ (allow 39.9-40.2 must have correct units) 1
 moles methanol = mass/ M_r = $2.12/32$ (1) 1

$$= 0.0663$$

$$\Delta H = -40.1/0.0663 = -605 \text{ kJ (mol}^{-1}\text{)} \quad 1$$

(allow -602 to -608 or answer in J)
(note allow conseq marking after all mistakes but note use of 2.12 g loses 2 marks)

$$(b) \quad \Delta H = \sum \Delta H_c^\ominus(\text{reactants}) - \sum \Delta H_c^\ominus(\text{products}) \text{ (or correct cycle)} \quad 1$$

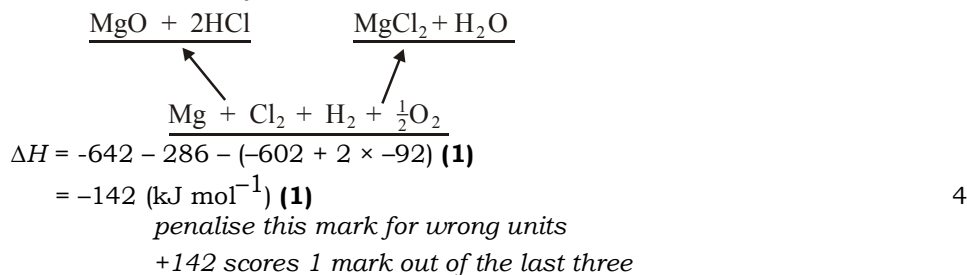
$$\begin{aligned} \Delta H_c^\ominus(\text{CH}_3\text{OH}) &= \Delta H_c^\ominus(\text{CO}) + 2 \times \Delta H_c^\ominus(\text{H}_2) - \Delta H \\ &= (-283) + (2 \times -286) - (-91) \text{ (mark for previous equation or this)} \\ &= -764 \text{ (kJ mol}^{-1}\text{)} \text{ (units not essential but lose mark if units wrong)} \\ &\text{(note + 764 scores 1/3)} \end{aligned} \quad 1$$

[10]

6. (a) (i) enthalpy (or heat or heat energy) change when 1 mol of a substance **(1)** (QL mark) is formed from its elements **(1)** all substances in their standard states **(1)** (or normal states at 298K, 100 kPa or std condits) 3

not STP, NTP

- (b) enthalpy change (or enthalpy of reaction) is independent of route **(1)**
 $\Delta H = \sum \Delta H_f^\ominus \text{ prods} - \sum \Delta H_f^\ominus \text{ reactants}$ (or cycle) **(1)**
 minimum correct cycle is:



- (c) $\Delta H = mcT$ **(1)** (or $mc\Delta T$)
 $= 50 \times 4.2 \times 32 = 6720 \text{ J} = 6.72 \text{ kJ}$ **(1)**
 mark is for 6720 J or 6.72 kJ

$$\begin{aligned} \text{moles HCl} &= \frac{\text{vol}}{1000} \times \text{conc} = \frac{50}{1000} \times 3 \text{ (1)} \\ &= 0.15 \text{ (1)} \\ &\text{if error here mark on conseq.} \end{aligned}$$

$$\begin{aligned} \text{Therefore moles of MgO reacted} &= \text{moles HCl}/2 \text{ (1) (mark is for /2, CE if not/2)} \\ &= 0.15/2 = 0.075 \end{aligned}$$

$$\begin{aligned} \text{Therefore } \Delta H &= 6.72/0.075 \text{ (1)} \\ &= -90 \text{ kJ (mol}^{-1}\text{)} \\ &\text{kJ must be given, allow 89 to 91} \\ &\text{value (1)} \end{aligned} \quad 8$$

sign **(1)**; this mark can be given despite CE for /2

Note various combinations of answers to part (c) score as follows:

$$\begin{aligned} -89 \text{ to } -91 \text{ kJ (8) (or } -89000 \text{ to } 91000\text{J)} \\ &\text{no units (7)} \\ +89 \text{ to } +91 \text{ kJ (7) (or } +89000 \text{ to } +91000\text{J)} \\ &\text{no units (6)} \\ -44 \text{ to } -46 \text{ kJ (5) (or } -44000 \text{ to } -46000\text{J)} \\ &\text{no units (4) if units after 6.72 or 6720 (5)} \\ +44 \text{ to } +46 \text{ kJ (4) (or } +44000 \text{ to } +46000\text{J)} \\ &\text{if no units and} \end{aligned}$$

if no units after 6.72 or 6720 (3)
otherwise check, could be (4)

[15]

7. (a) Enthalpy (Energy) to break a (covalent) bond (1) **OR dissociation energy**
Varies between compounds so average value used (1) **QL mark**
OR average of dissociation energies in a single molecule / e.g. CH₄
Do not allow mention of energy to form bonds
but with this case can allow second mark otherwise 2nd mark
consequential on first

2

- (b) (i) $1/2 \text{ N}_2 + 3/2 \text{ H}_2 \rightarrow \text{NH}_3$ (1)
Ignore s s
- (ii) $\Delta H = (\Sigma)\text{bonds broken} - (\Sigma)\text{bonds formed}$ (1)
 $= 1/2 \times 944 + 3/2 \times 436 - 3 \times 388$ (1)
 $= -38 \text{ kJ mol}^{-1}$ (1)
Ignore no units, penalise wrong units
Score 2/3 for -76
1/3 for +38
Allow 1/3 for +76

4

- (c) $4 (\text{C-H}) + (\text{C}=\text{C}) + (\text{H-H}) - (6 (\text{C-H}) + (\text{C-C})) = -136$ (1)
 $(\text{C}=\text{C}) + (\text{H-H}) - ((\text{C-C}) + 2 (\text{C-H})) = -136$
 $2 (\text{C-H}) = 836$ (1)
 $(\text{C-H}) = 418 (\text{kJ mol}^{-1})$ (1)
Note: allow (1) for -836
another (1) for -418

3

8. (a) $\text{C}_3\text{H}_6 + 4\text{O}_2 \rightarrow 3\text{CO}_2 + 3\text{H}_2\text{O}$ (1) (or multiple)

1

[9]

- (b) (i) $\frac{1.45}{58}$ (1) = 0.0250 (1)

allow 0.025

allow conseq on wrong M_r

$$1.45/100, \text{CE}; \frac{1.45}{58.1} \text{ C.E.}$$

- (ii) heat released = $mc\Delta T$
 $= 100 \times 4.18 \times 58.1$ (1)
if 1.45 used in place of 100 CE = 0
 $= 24300 \text{ J}$ (1) (or 24.3kJ)
allow 24200 to 24300
ignore decimal places
units tied to answer
If use $0.1 \times 4.18 \times 51.8$ allow $\frac{1}{2}$ for 24.3 with no units

5

- (iii) $\frac{24.3}{0.0250} = -972 (\text{kJ mol}^{-1})$ (1)

allow -968 to -973

allow +972

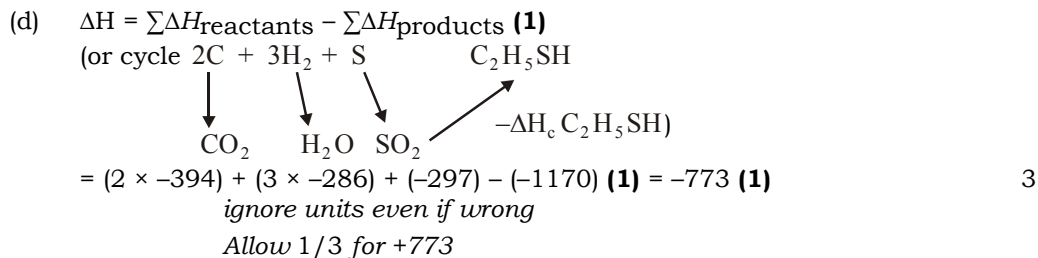
allow conseq

allow no units

penalise wrong units

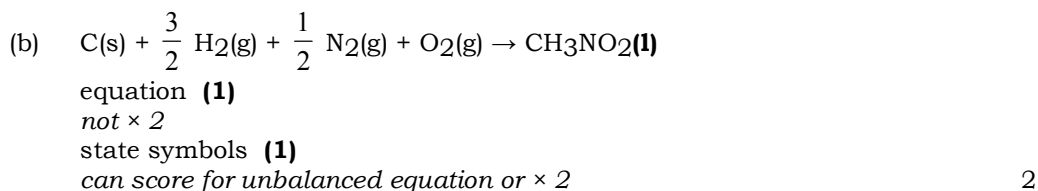
- (c) (i) Heat loss (1) or energy loss
do not allow incomplete combustion
- (ii) Difference: more negative (1) (or more exothermic)
QoL mark

Explanation: heat (or energy) released when water vapour condenses (1) 3
or heat/energy required to vaporise water
or water molecules have more energy in the gaseous state

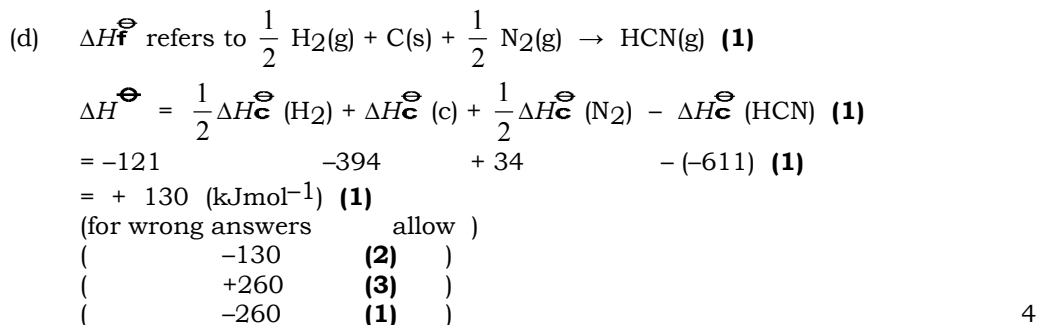


[12]

9. (a) {heat {molecule
{enthalpy change for formation of 1 mol of a {substance (1)
 {compound
from its elements (1)
in their standard states * (at 298 K, 100 kPa) (1)
(*) or natural/normal states at 298 K, 100 kPa
must have 2nd mark to score third 3

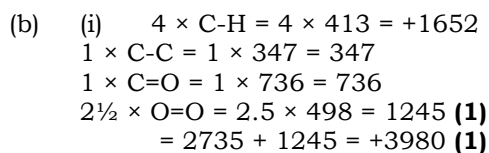


- (c) Hess's law (1)
or 1st Law or conservation of energy 1

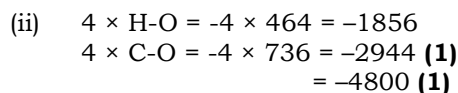


[10]

10. (a) (Energy required) to break a given covalent bond (1)
averaged over a range of compounds (1)
Penalise first mark if 'energy' / 'enthalpy' evolved 2



first mark for 4 : 1: 1 or 2735 ignore sign



First mark for 4 : 4

$$\begin{aligned} \text{(iii)} \quad \Delta H_R &= \sum \text{Bonds broken} - \sum \text{Bonds made} \\ &= +3980 - 4800 = -820 \text{ (1)} \end{aligned}$$

**Conseq Mark for incorrect answers in (i) and (ii) as
(i) Answer + (ii) Answer =**

5

[7]

11. (a) (Enthalpy change) when 1 mol (1) of a compound is formed from its constituent elements (1) in their standard states (1)

Allow energy or heat, Ignore evolved or absorbed

Mark each point independently

3

- (b) (The enthalpy change for a reaction is) independent of the route (1)

1

- (c) $\Delta H_R = \sum \Delta H_f \text{ products} - \sum \Delta H_f \text{ reactants}$ (1)

$$= [(3 \times -286) + (3 \times -394)] - (-248) \text{ (1)}$$

$$= -1792 \text{ (1) (kJ mol}^{-1}\text{)}$$

Deduct one mark for each error to zero

3

[7]

12. (a) standard enthalpy of formation (1)

allow enthalpy of formation under standard conditions

1 mol of a compound produced (1)

from elements under standard conditions (1)

reactants and products in standard states (1)

allow normal or most stable states under standard conditions

correct explanation not linked to correct name

4

- (b) (i) $\text{NO} + \frac{5}{2} \text{H}_2 \rightarrow \text{NH}_3 + \text{H}_2\text{O}$ or doubled (1)

allow two equations, single and doubled

mark on from wrong moles of H_2 (for which ΔH_f is zero)

$$\text{(ii)} \quad \text{NO} \rightleftharpoons \frac{1}{2} \text{N}_2 + \frac{1}{2} \text{O}_2 \quad \Delta H = -91 \text{ kJ mol}^{-1}$$

$$\frac{1}{2} \text{N}_2 + \frac{3}{2} \text{H}_2 \rightleftharpoons \text{NH}_3 \quad \Delta H = -46 \text{ kJ mol}^{-1}$$

$$\frac{1}{2} \text{O}_2 + \text{H}_2 \rightleftharpoons \text{H}_2\text{O} \quad \Delta H = -242 \text{ kJ mol}^{-1}$$

correct equations or cycle (1)

$$\sum \Delta H = -91 -46 -242 \text{ (1)} = -379 \text{ kJ mol}^{-1} \text{ (1)}$$

minus 1 for each wrong sign

if eqn in (i) and value in (ii) do not match, then MAX 2 ex 3

$\Delta H = \sum \Delta H_{\text{prod}} - \sum \Delta H_{\text{react}}$ (shown) gets 1 if zero otherwise

4

[8]

13. (a) $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$ or $\text{H}_3\text{O}^+ + \text{OH}^- \rightarrow 2\text{H}_2\text{O}$ (1)

Ignore state symbols; ignore 'spectator' ions

1

- (b) Mol H^+ = $50.0 \times 1.00 / 1000 = 5 \times 10^{-2}$ (1)

1

- (c) Heat energy evolved

=

MCAT/MS θ or in words (1)

$$= 100 \times 4.18 \times 6.5$$

=

$$2717 \text{ Allow } 2700 - 2717 \text{ (1)}$$

$$\text{or } = 0.1 \times 4.18 \times 6.5$$

=

$$2.717 \text{ Allow } 2.7 - 2.717 \text{ (1)}$$

$$\text{or } = 0.1 \times 4.2 \times 6.5 = 2.730 \text{ (or } 2730) \text{ if } 4.2 \text{ used for 'c'}$$

Ignore units in part (c)

2

- (d) Molar enthalpy change = $-2.717/5.00 \times 10^{-2}$ (1)
 = $-54.3 \text{ kJ mol}^{-1}$ (1)

Allow conseq to answers in (b) and (c)

Mark CE if this inverted

2

[6]

14. (a) (The enthalpy/heat change) when one mole (1) is completely burned/combusted in oxygen/air (1)
 under standard conditions or 100K Pa/lbar/atm/760mm Hg (1) and 298K or STP

Penalise first mark if heat adsorped. Penalise first mark if "energy change" stated 3

- (b) $\Delta H_R^\ominus = \sum \Delta H_f^\ominus \text{ products} - \Delta H_f^\ominus \text{ reactants}$ or cycle or
 $\Delta H_R^\ominus = \Delta H_f^\ominus \text{CO}_2 + 2\Delta H_f^\ominus \text{H}_2\text{O} - \Delta H_f^\ominus \text{CH}_4$ (1)
 = $(-394) + 2(-286) - (-75)$ (1)
 = $-891 \text{ (kJ mol}^{-1}\text{)}$

Allow +891 [max 1]

3

- (c) (i) Enthalpy (Do not allow energy) required to break a covalent bond (1)
 Allow second mark separately
 averaged over (many) compounds (1)
 (ii) $\Delta H_a = \sum \text{Bonds broken} - \Delta \text{Bonds made}$ or cycle (1)

	C(g) + 4H(g)		
Allow -415 to -416 Max (2)	C(g) + 2H ₂ (g)	$\uparrow 2\Delta H_{\text{BE}} \text{H}_2$	
	C(s) + 2H ₂ (g)	$\uparrow \Delta H_{\text{BE}} \text{C}$	$\uparrow 4\text{BE(C-H)}$
Allow 1662 Max (1)		$\downarrow \Delta H_f \text{CH}_4$	CH ₄ (g)

$$\text{BE (C-H)} = \frac{715 + 2(436) - (-75)}{4} \text{ (1)} = 415.5 \text{ (kJ mol}^{-1}\text{)} \text{ (1)}$$

Allow 415-416

- (iii) $4020 = 2\text{BE (C-C)} + 8\text{BE (C-H)}$ (1)
CE if 3BE(C-C) used

$$\text{BE (C-C)} + \frac{4020 - 8(415.5)}{2} = 348 \text{ (kJ mol}^{-1}\text{)} \text{ (1) Allow 346-350}$$

7

Mark conseq

Note: Using 390, the given answer, BE (C-C) = 450 [2]

The common wrong answer in C(ii) is 378 this gives BE(C-C) as 498 conseq [2]

[13]

15. (a) enthalpy change when 1 mol of a compound (1)
 is completely burned in oxygen (1)
 under standard conditions (1)

3

- (b) $\text{CH}_3\text{SH} + 3\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{SO}_2$

1

- (c) enthalpies of combustion of C, H₂, S

one correct (1)

three correct (2)

(or enthalpies of formation of CO₂, H₂O, SO₂)

one correct (1)

three correct (2)

2

- (d) $\Delta H^\ominus = -\Delta H_f^\ominus (\text{CS}_2) + \Delta H_f^\ominus (\text{CO}_2) + 2\Delta H_f^\ominus (\text{SO}_2)$ (1)
 or cycle with the same information

$$= -88 - 394 - 594 \text{ (1)}$$

$$= -1076 \text{ kJ mol}^{-1} \text{ (1)}$$

+1076 scores one, -900 scores 1 cycle mark

3

[9]

16. (a) enthalpy change (1)
independent of reaction route (1)

2

- (b) enthalpy change for 1 mol (1)
completely burned in oxygen (1)
under standard condition (1)

3

- (c) (i) Equation for formation of CS₂ C(s) + 2S(s) → CS₂(l) (1)
Equation for combustion of CS₂ CS₂(l) + 3O₂(g) → CO₂(g) + 2SO₂(g) (1)

(ii) $\Delta H_f^\ominus = \Delta H_c^\ominus(\text{C}) + 2\Delta H_c^\ominus(\text{S}) - \Delta H_f^\ominus(\text{CS}_2)$ (1)

(or correct cycle)

$$= -394 + 2(-297) - 88 \text{ (1)}$$

$$= -1076 \text{ kJ mol}^{-1} \text{ (1)}$$

5

- (d) $\Delta H^\ominus = \Delta H_f^\ominus(\text{CO}) + 2\Delta H_f^\ominus(\text{SO}_2) - (\Delta H_f^\ominus(\text{CS}_2) + 5\Delta H_f^\ominus(\text{NO}))$ (1)
(or correct cycle)

$$= -111 + 2(-297) - (88 + 5 \times 90) \text{ (1)}$$

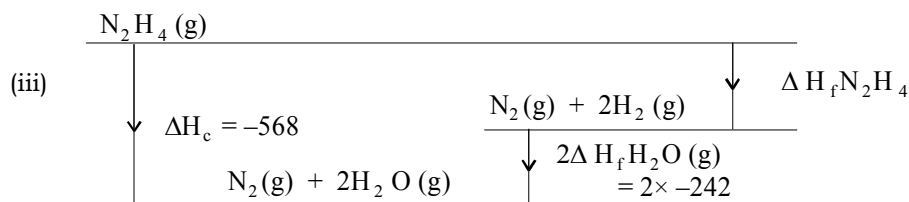
$$= -1243 \text{ kJ mol}^{-1} \text{ (1)}$$

3

[13]

17. (a) (i) N₂ + 2H₂ → N₂H₄ (1)

- (ii) N₂H₄ + O₂ → N₂ + 2H₂O (1)



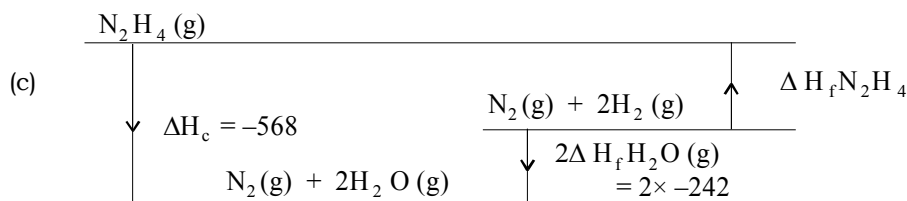
$$\Delta H_f = -484 + 568 = +84 \text{ kJ mol}^{-1} \text{ (1)}$$

$$[\text{OR } \Delta H_f = 2\Delta H_f \text{H}_2\text{O}(\text{g}) - \Delta H_c \text{N}_2\text{H}_4 \text{ (1)} = 84 \text{ kJ mol}^{-1} \text{ (1)}]$$

4

- (b) Enthalpy or heat required to break a covalent bond (1)
Average over a range of compounds

2



$$\text{Hence (N-N)} = 872 + 945 - 84 - 1556 \text{ (1)} = 177 \text{ kJ mol}^{-1} \text{ (1)}$$

Note:- Mark consequentially to $\Delta H_f \text{N}_2\text{H}_4$ from a(iii)

3

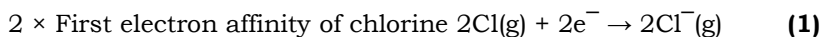
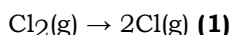
- (d) The N-H bond enthalpies used in the calculation were mean values or depend on environment (1)

1

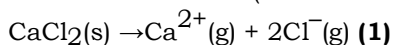
[10]

18. (a) enthalpy change for one mole of compound (1)
formed from its elements (1)
in their standard states (1) 3
- (b) by definition (1) 1
- (c) $\Delta H = 2 \times \Delta H_f^\ominus (\text{CF}_4) + 6 \times \Delta H_f^\ominus (\text{HF}) - \Delta H_f^\ominus (\text{C}_2\text{H}_6)$ (1)
 $= 2 \times -680 + 6 \times -269 - (-85)$ (1)
 $= -2889$ (1) (kJ mol^{-1}) 3
- (d) (i) bond energies (enthalpies) vary between compounds (1)
therefore an average value is used (1)
- (ii) First step in the reaction F-F bond breaks (1)
Reason weakest bond (1) 4
- [11]
19. (a) *Standard enthalpy of formation* The enthalpy change when one mole of a compound (1) is formed from its constituent element (1) in their normal or standard state (1) under standard conditions
Standard enthalpy of combustion The enthalpy change when one mole of a compound (1) is completely burnt in oxygen (1) under standard condition or at 298K and 100kPa (1)
- 6
- (b) $\Delta H = \sum \Delta H_f \text{ products} - \sum \Delta H_f \text{ reactants or cycle}$ (1)
 $= 3\Delta H_f \text{CO}_2 + 4\Delta H_f \text{H}_2\text{O} - \Delta H_f \text{C}_3\text{H}_7\text{OH}$ (1)
 $= -2022 \text{ kJ mol}^{-1}$ (1) 3
- (c) (i) Enthalpy = $200 \times 4.2 \times 15$ (1)
 $= 12.6 \text{ kJ}$ (1)
- (ii) Moles $\text{C}_3\text{H}_7\text{OH} = 0.90/60$ (1)
 12.6 kJ (1)
- (iii) Enthalpy of combustion =
 $-12.6/0.015$ (1)
 $= -840 \text{ kJ mol}^{-1}$ (1)
- (iv) Reason 1 Incomplete combustion
Heat lost to surroundings
Reason 2 Heat capacity of the apparatus
Any two (2) 8
- [17]
20. (a) $\Delta H = \sum \Delta H_f \text{ products} - \sum \Delta H_f \text{ reactants or cycle}$ (1)
 $= [(3 \times -393) + (3 \times -242)] - [+53]$ (1)
 $= -1905 - 53$
 $= -1958 \text{ kJ mol}^{-1}$ (1) 3
- (b) The enthalpy required to break a covalent bond (1)
average over a wide range of compounds (1) 2
- (c) In; $3 \times \text{C-C}; 3 \times 347 = +1041$
 $6 \times \text{C-H}; 6 \times 413 = +2478$
 $4.5 \times \text{O} = 0; 4.5 \times 498 = +2241$
Total in + 5760 (1)
- Out; $6 \times \text{C=O}; 6 \times -805 = -4830$
 $6 \times \text{O-H}; 6 \times -464 = -2784$
Total out -7614 (1)
- Enthalpy change = $5760 - 7614 = -1854 \text{ kJ mol}^{-1}$ (1) 3

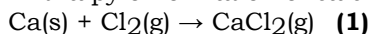
- (d) Cyclopropane has a strained ring structure **(1)**
 Bonds in cyclopropane are weaker OR
 Bond enthalpies depend on environment OR
 Mean bond enthalpies used in the calculation **(1)** 2 **[10]**
21. (a) $1s^2 2s^2 2p^6 3s^2 3p^6$ 1
- (b) $S^-(g)$ 1
- (c) The negative S^- ion
 repels the electron being added 1
 1
- (d) (i) Enthalpy of atomisation of sulphur 1
 (ii) Second ionisation enthalpy of calcium 1
 (iii) Second electron affinity of sulphur 1
- (e) Electron more strongly attracted 1
 nearer to the nucleus or attracted by Ca^+ ion 1
- (f) Correct cycle
 e.g. $+178 + 279 + 590 + 1145 - 200 + E - 3013 + 482 = 0$ 1
 $= 539$ 1
 Allow one mark for -539 **[11]**
22. (a) ΔH_1 formation **(1)**
 ΔH_2 atomisation / sublimation (of Cs not Cl) **(1)**
 ΔH_5 electron affinity (of Cl) **(1)** 3
- (b) $-433 = 79 + 376 + 121 - 364 + \Delta H_6$ **(1)**
 $\Delta H_6 = -645 \text{ (kJ mol}^{-1}\text{)}$ **(1)** 2
 both marks for correct answer alone
 allow 1 mark for +645 or 645
- (c) Cs has larger radius / larger atom (not ion) / more shells /
 more orbitals / more sub-shells **(1)**
 correct reference to shielding **(1)**
 less powerful attraction (of nucleus) for (outer) e^- in Cs **(1)** 3 **[8]**
23. (a) (i) diagram shows in top box $Rb^+(g) + e^- + I(g) / e^-$ must be there **(1)**
 diagram shows in lower box $Rb(s) + \frac{1}{2} I_2(s)$ **(1)** 2
- (ii) $\Delta H_1 =$ electron affinity (of iodine)/allow correct symbols
 including $I_2, I^-,$ **(1)**
 $\Delta H_2 =$ lattice enthalpy / energy (of RbI) **(1)** 2
- (iii) $\Delta H_1 = -(+402) - (+107) - (+85.8) + (-328) - (-609)$ **(1)**
 $= -314 \text{ (kJ mol}^{-1}\text{)}$ **(1)** 2
 give 1 mark for +313.8 or +314
 assume + if no sign given
 significant figure penalty for 310 **[6]**
24. (i) **A** is (enthalpy change) of formation **(1)**
B is lattice enthalpy **(1)** 2
- (ii) negative sign (only if a calculation) **(1)**
 883 kJ mol^{-1} **(1)**
 (only give both marks if working and answer correct)
 hydration / solvation enthalpies (of Li^+ and Cl^-) **(1)** 3 **[5]**
25. (a) (i) Steps
 Sublimation/atomisation enthalpy or
 $\Delta H_{fus} + \Delta H_{vap}$ $Ca(s) \rightarrow Ca(g)$ **(1)**
 First ionisation enthalpy of calcium $Ca(g) \rightarrow Ca^+(g) + e^-$ **(1)**
 Second ionisation enthalpy of calcium $Ca^+(g) \rightarrow Ca^{2+}(g) + e^-$ **(1)**
 Dissociation enthalpy/bond energy/2 $\times \Delta H_a$ chlorine



Lattice dissociation (or formation) enthalpy



Enthalpy of formation of calcium chloride



7

- (ii) 2nd IE needed when CaCl_2 formed or only 1st IE for CaCl (1)

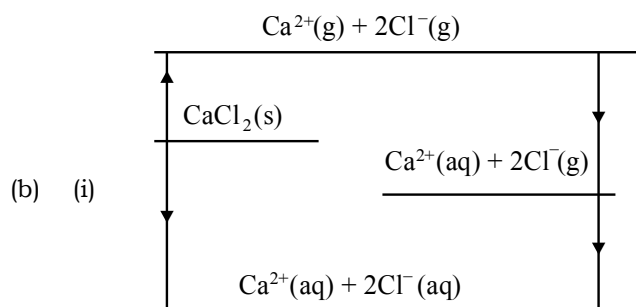
BE of Cl_2 needed when CaCl_2 formed but BE/2 for CaCl

(or equivalent) (1)

$2 \times$ electron affinity when CaCl_2 formed but $1 \times$ EA for CaCl (1)

LE of CaCl_2 greater (or different) than LE CaCl (1)

4



$+LE + (\Delta H_{\text{hyd}}\text{Ca}^{2+}) + 2(\Delta H_{\text{hyd}}\text{Cl}^-) - \Delta H_{\text{soln}} \text{CaCl}_2 = \text{CaCl}_2 = 0$ or correct cycle (1)

$$\Delta H_{\text{soln}} \text{CaCl}_2 = 2255 - 1650 - 768 \quad (1)$$

$$= -163 \text{ kJ mol}^{-1} \quad (1)$$

3

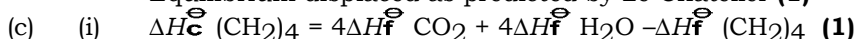
- (ii) Marked consequentially to answer in (c)(i)

Solubility decreases (1)

Enthalpy of solution negative or exothermic (1)

Equilibrium displaced as predicted by Le Chatelier (1)

3



$$-2558 = (4 \times -393.5) + (4 \times -245.1) - \Delta H_{\text{f}}^{\ominus} (\text{CH}_2)_4 \quad (1)$$

$$\Delta H_{\text{f}}^{\ominus} (\text{CH}_2)_4 = +3.6 \text{ kJ mol}^{-1} \quad (1)$$

3

- (ii) Enthalpy put in to break bonds

$$4 \times \text{C} - \text{C} = 4 \times 347 \quad (1) = 1388$$

$$8 \times \text{C} - \text{H} = 8 \times 413 \quad (1) = 3304$$

$$6 \times \text{O} = \text{O} = 6 \times 498 \quad (1) = 2988$$

$$\text{Total} = 7680$$

Enthalpy given out when bonds made

$$8 \times \text{C} = \text{O} = 8 \times -805 \quad (1) = -6440$$

$$8 \times \text{O} - \text{H} = 8 \times -464 \quad (1) = -3712$$

$$\text{Total} = -10152$$

Enthalpy change = Enthalpy in – enthalpy out

$$= 7680 - 10152 \quad (1)$$

$$= -2472 \text{ kJ mol}^{-1} \quad (1)$$

marked consequentially

Explanation

Mean bond enthalpies used or bond enthalpies depend on environment **(1)**

Cyclobutane less stable than suggested by mean bond enthalpies **(1)**

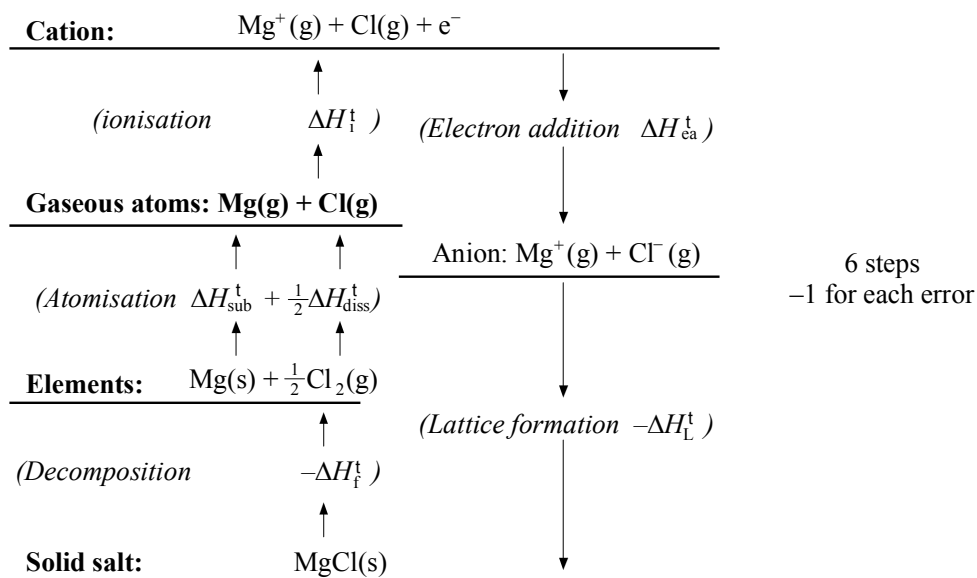
Cyclobutane a strained structure or bond angles 90° **(1)**

10

Stability of cyclobutane marked consequentially.

[30]

26

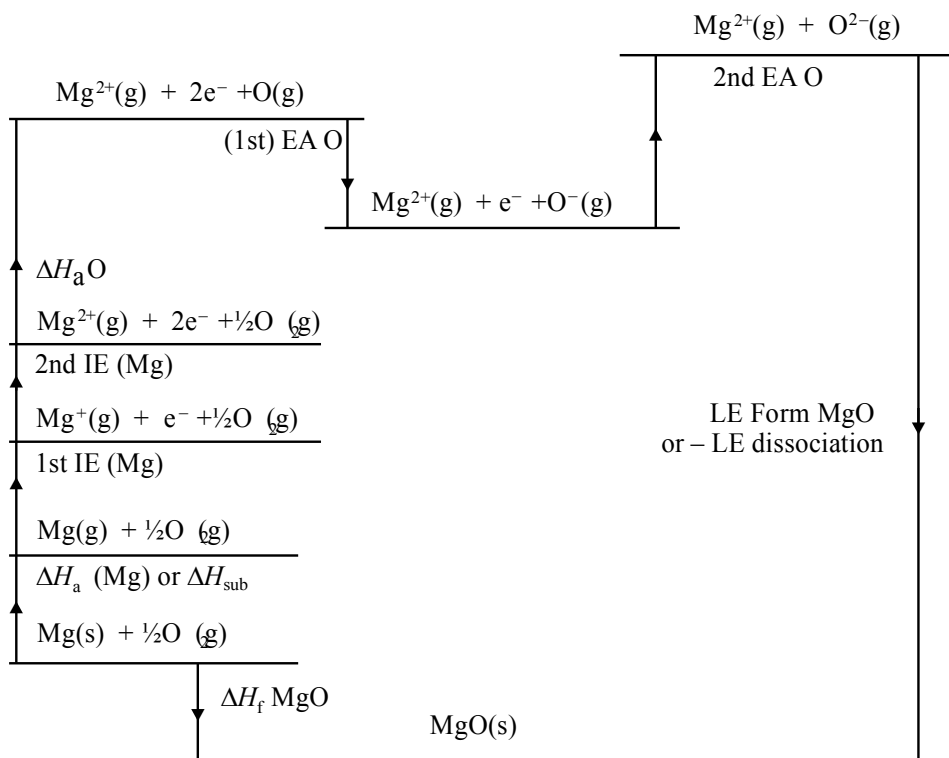


$$\begin{aligned}
 0 &= -\Delta H_{\text{f}}^{\ominus} + \Delta H_{\text{sub}}^{\ominus} + \Delta H_{\text{diss}}^{\ominus} + \Delta H_{\text{f}}^{\ominus} + \Delta H_{\text{ea}}^{\ominus} + \Delta H_{\text{f}}^{\ominus} \\
 \Delta H_{\text{f}}^{\ominus} &= \Delta H_{\text{sub}}^{\ominus} + \Delta H_{\text{diss}}^{\ominus} + \Delta H_{\text{f}}^{\ominus} + \Delta H_{\text{ea}}^{\ominus} + \Delta H_{\text{f}}^{\ominus} \\
 \Delta H_{\text{f}}^{\ominus} &= +146 + 121 + 736 - 364 - 753 \\
 \Delta H_{\text{f}}^{\ominus} &= -114 \text{ kJ mol}^{-1} \text{ (3)} \quad [-1 \text{ for each error}] \\
 \Delta H^{\ominus} &= \Delta H_{\text{f}}^{\ominus} (\text{MgCl}_2) - \Delta H_{\text{f}}^{\ominus} (\text{MgCl}) = -653 - 2(-114) = -425 \text{ kJ mol}^{-1} \\
 \text{MgCl}_2 &\text{ is always favoured over MgCl}
 \end{aligned}$$

- 27 (a) Heat (energy)/ internal energy change (not energy change) **(1)**
Measured at constant pressure **(1)**

(b)

[10]



8

Notes:- Ignore electrons in cycle

Penalise missing labels by max 2

Penalise missing state symbols in 'active step' by max 2

Penalise incorrect chemistry every time

Allow steps written horizontally

Calculation:

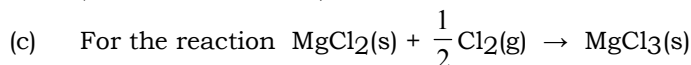
$$\Delta H_a \text{Mg} + 1^{\text{st}} \text{IE Mg} + 2^{\text{nd}} \text{IE Mg} + \Delta H_a \text{O} + 1^{\text{st}} \text{EA O} + 2^{\text{nd}} \text{EA O} + \text{LE Form -MgO} \Delta H_f \text{MgO} = 0 \quad (1)$$

$$\text{Hence; } 148 + 738 + 1451 + 249 - 141 + 798 + \text{LE} + 602 = 0 \quad (1)$$

(NB This scores 2)

$$\text{LE formation MgO} = -3845 \text{ (kJ mol}^{-1}\text{)} \quad (1)$$

(Allow +3845 Max 2)



$$\Delta H_r = \Delta H_f \text{MgCl}_3 - \Delta H_f \text{MgCl}_2$$

$$= +3904 - (-653) = 4557 \text{ (kJ mol}^{-1}\text{)} \quad (1)$$

ΔS is negative/ entropy decreases (as order increases) **(1)**

$$\Delta G = \Delta H - T\Delta S \quad (1)$$

ΔG must (always) be positive (since ΔH_r and $-T\Delta S$ are both positive) **(1)**

Reaction never/not feasible (or equivalent) **(1)**

5

[15]

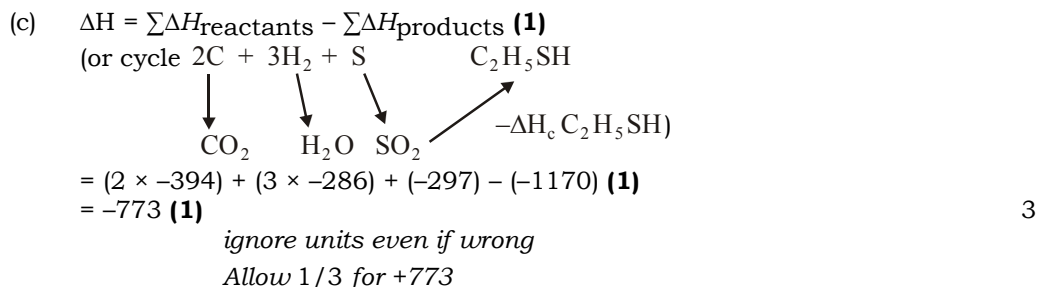
28. (a) Enthalpy (Energy) to break a (covalent) bond **(1)** OR **dissociation energy**
 Varies between compounds so average value used **(1)** **QL mark**
OR average of dissociation energies in a single molecule / e.g. CH₄
Do not allow mention of energy to form bonds
but with this case can allow second mark otherwise 2nd mark
consequential on first

2

- (b) (i) $\frac{1}{2} \text{N}_2 + \frac{3}{2} \text{H}_2 \rightarrow \text{NH}_3$ (1)
Ignore s s
- (ii) $\Delta H = (\sum)\text{bonds broken} - (\sum)\text{bonds formed}$ (1)
 $= \frac{1}{2} \times 944 + \frac{3}{2} \times 436 - 3 \times 388$ (1)
 $= -38 \text{ kJ mol}^{-1}$ (1)
Ignore no units, penalise wrong units
Score 2/3 for -76
1/3 for +38
Allow 1/3 for +76 4
- (c) $4 (\text{C-H}) + (\text{C}=\text{C}) + (\text{H-H}) - (6 (\text{C-H}) + (\text{C-C})) = -136$ (1)
 $(\text{C}=\text{C}) + (\text{H-H}) - ((\text{C-C}) + 2 (\text{C-H})) = -136$
 $2 (\text{C-H}) = 836$ (1)
 $(\text{C-H}) = 418 \text{ (kJ mol}^{-1}\text{)}$ (1)
Note: allow (1) for -836
another (1) for -418 3 [9]
29. (a) $\Delta H = \sum(\text{bonds broken}) - \sum(\text{bonds formed})$ (or cycle) 1
 $= +146 - 496/2$ (or $2 \times 463 + 146 - (2 \times 463 + 496/2)$) 1
 $= -102 \text{ (kJ mol}^{-1}\text{)}$ (1) 1
(accept no units, wrong units loses a mark; +102 scores (1) only)
- (b) $\text{C(s)} + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g})$ equation (1) Correct state symbols (1) 2
- (c) (i) Macromolecular 1
(accept giant molecule or carbon has many (4) bonds)
- (ii) $\Delta H = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$ (or cycle) 1
 $= 715 + 4 \times 218 - (-74.9)$ 1
 $= 1662 \text{ (kJ mol}^{-1}\text{)}$ 1
(accept no units, wrong units loses one mark, allow 1660 to 1663, -1662 scores one mark only)
- (iii) $1662/4 = 415.5$ 1
(mark is for divide by four, allow if answer to (c)(ii) is wrong) [10]
30. (a) The enthalpy change when 1 mol of a compound is completely burnt in oxygen under standard conditions, all reactants and products being in their standard states. 2
- (b) (i) $\text{C}_2\text{H}_6 + 3\frac{1}{2}\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$ (1)
- (ii) $\Delta H = 2 \times \Delta H_f^\ominus(\text{CO}_2) + 3 \times \Delta H_f^\ominus(\text{H}_2\text{O}) - \Delta H_f^\ominus(\text{C}_2\text{H}_6)$ (1)
 $= -788 - 858 - (-85)$
 $= -1561 \text{ kJ mol}^{-1}$
(1)(1) 4 [6]
31. (b) $\Delta H = \sum \Delta H_f^\ominus(\text{reactants}) - \sum \Delta H_f^\ominus(\text{products})$ 1
 $= (7x - 394) + (4x - 286) - (-3909)$ 1
 $= +7 \text{ kJ mol}^{-1}$ 1
- (b) $q = mc\Delta T = 250 \times 60 \times 4.18 = 62700 \text{ J} = 62.7 \text{ kJ}$ 1

$$\text{Number of moles} = 2.5/92 = 0.0272 \quad 1$$

$$\Delta H = -62.7/0.0272 = -2310 \text{ kJ mol}^{-1} \quad 1$$



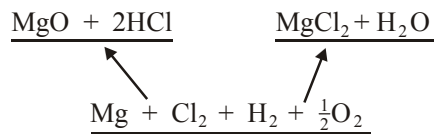
[9]

32. (a) (i) enthalpy (or heat or heat energy) change when 3
1 mol of a substance (1) (QL mark) is formed from its elements (1)
all substances in their standard states (1) (or normal states at 298K,
 100 kPa or std condits)
not STP, NTP

- (b) enthalpy change (or enthalpy of reaction) is independent of route (1)

$$\Delta H = \sum \Delta H_f^\ominus \text{ prods} - \sum \Delta H_f^\ominus \text{ reactants (or cycle) (1)}$$

minimum correct cycle is:



$$\Delta H = -642 - 286 - (-602 + 2 \times -92)$$
 (1)
 $= -142 \text{ (kJ mol}^{-1}\text{)}$ (1) 4
penalise this mark for wrong units
+142 scores 1 mark out of the last three

- (c) $\Delta H = mcT$ (1) (or $mc\Delta T$)
 $= 50 \times 4.2 \times 32 = 6720 \text{ J} = 6.72 \text{ kJ}$ (1)
mark is for 6720 J or 6.72 kJ
- moles HCl = $\frac{\text{vol}}{1000} \times \text{conc} = \frac{50}{1000} \times 3$ (1)
 $= 0.15$ (1)
if error here mark on consequ.
- Therefore moles of MgO reacted = moles HCl/2 (1) (mark is for /2, CE if not/2)
 $= 0.15/2 = 0.075$
- Therefore $\Delta H = 6.72/0.075$ (1) 8
 $= -90 \text{ kJ (mol}^{-1}\text{)}$
kJ must be given, allow 89 to 91
value (1)
sign (1); this mark can be given despite CE for /2

Note various combinations of answers to part (c) score as follows:

-89 to -91 kJ (8) (or -89000 to 91000J)
no units (7)

+89 to +91 kJ (7) (or + 89000 to +91000J)
no units (6)

-44 to -46 kJ (5) (or -44000 to -46000J)
no units (4) if units after 6.72 or 6720 (5)

+44 to +46 kJ **(4)** (or +44000 to + 46000)
if no units and
*if no units after 6.72 or 6720 **(3)***
*otherwise check, could be **(4)***

[15]

6.0 ELECTROCHEMISTRY

1. (a) Define the term *oxidising agent* in terms of electrons.

(1)

(Total 1 mark)

2. Use the data in the table below, where appropriate, to answer the questions which follow.

Standard electrode potentials	$E^\ominus/V$
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$	+1.36
$2\text{BrO}_3^-(\text{aq}) + 12\text{H}^+(\text{aq}) + 10\text{e}^- \rightarrow \text{Br}_2(\text{aq}) + 6\text{H}_2\text{O}(\text{l})$	+1.52
$\text{O}_3(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{O}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$	+2.08
$\text{F}_2\text{O}(\text{g}) + 2\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{F}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+2.15

Each of the above can be reversed under suitable conditions.

- (a) (i) Identify the most powerful reducing agent in the table.
 (ii) Identify the most powerful oxidising agent in the table.
 (iii) Identify **all** the species in the table which can be oxidised in acidic solution by $\text{BrO}_3^-(\text{aq})$.

(4)

- (b) The cell represented below was set up.



- (i) Deduce the e.m.f. of this cell.
 (ii) Write a half-equation for the reaction occurring at the negative electrode when current is taken from this cell.
 (iii) Deduce what change in the concentration of $\text{Fe}^{3+}(\text{aq})$ would cause an increase in the e.m.f. of the cell. Explain your answer.

(6)

(Total 10 marks)

3. Use the data below, where appropriate, to answer the following questions.

Standard electrode potentials	$E^\ominus/V$
$\text{S}_2\text{O}_8^{2-}(\text{aq}) + 2\text{e}^- \rightarrow 2\text{SO}_4^{2-}(\text{aq})$	+2.01
$\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$	+1.51
$\text{Cl}_2(\text{aq}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$	+1.36
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{e}^- \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$	+1.33
$\text{NO}_3^-(\text{aq}) + 3\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{HNO}_2(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+0.94
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$	+0.77

- (a) From the table above, select the species which is the most powerful reducing agent.

(1)

- (b) Deduce the oxidation state of

- (i) chromium in $\text{Cr}_2\text{O}_7^{2-}$
 (ii) nitrogen in HNO_2

(2)

- (c) (i) Calculate the e.m.f. of the cell represented by



- (ii) Deduce an equation for the reaction which occurs when an excess of $\text{S}_2\text{O}_8^{2-}(\text{aq})$ is added to an aqueous solution of $\text{Mn}^{2+}(\text{aq})$ ions.

(3)

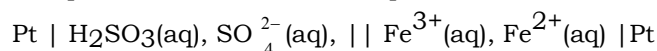
(Total 13 marks)

4. Use the standard electrode potential data given in the table below, where appropriate, to answer the questions which follow.

	$E^{\ominus}/V$
$V^{3+}(aq) + e^{-} \rightarrow V^{2+}(aq)$	-0.26
$SO_4^{2-}(aq) + 4H^{+}(aq) + 2e^{-} \rightarrow H_2SO_3(aq) + H_2O$	+0.17
$VO^{2+}(aq) + 2H^{+}(aq) + e^{-} \rightarrow V^{3+}(aq) + H_2O(l)$	+0.34
$O_2(g) + 2H^{+}(aq) + 2e^{-} \rightarrow H_2O_2(aq)$	+0.68
$Fe^{3+}(aq) + e^{-} \rightarrow Fe^{2+}(aq)$	+0.77
$VO_2^{+}(aq) + 2H^{+}(aq) + e^{-} \rightarrow VO^{2+}(aq) + H_2O(l)$	+1.00
$2IO_3^{-}(aq) + 12H^{+}(aq) + 10e^{-} \rightarrow I_2(aq) + 6H_2O(l)$	+1.19
$MnO_4^{-}(aq) + 8H^{+}(aq) + 5e^{-} \rightarrow Mn^{2+}(aq) + 4H_2O(l)$	+1.52

Each of the above can be reversed under suitable conditions.

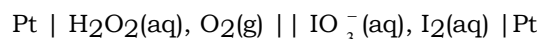
- (a) The cell represented below was set up under standard conditions.



- (i) Calculate the e.m.f. of this cell.
(ii) Write a half-equation for the oxidation process occurring at the negative electrode of this cell.

(2)

- (b) The cell represented below was set up under standard conditions.



- (i) Write an equation for the spontaneous cell reaction.
(ii) Give **one** reason why the e.m.f. of this cell changes when the electrodes are connected and a current flows.
(iii) State how, if at all, the e.m.f. of this standard cell will change if the surface area of each platinum electrode is doubled.
(iv) State how, if at all, the e.m.f. of this cell will change if the concentration of IO_3^{-} ions is increased. Explain your answer.

(7)

- (c) An excess of acidified potassium manganate(VII) was added to a solution containing $V^{2+}(aq)$ ions. Use the data given in the table to determine the vanadium species present in the solution at the end of this reaction. State the oxidation state of vanadium in this species and write a half-equation for its formation from $V^{2+}(aq)$.

(3)

(Total 12 marks)

5. Use the table of standard electrode potentials given below to answer the following questions.

	$E^{\ominus}/V$
$Cl_2(g) + 2e^{-} \rightarrow 2Cl^{-}(aq)$	+ 1.36
$Br_2(l) + 2e^{-} \rightarrow 2Br^{-}(aq)$	+1.07
$NO_3^{-}(aq) + 3H^{+}(aq) + 2e^{-} \rightarrow HNO_2(aq) + H_2O(l)$	+0.94
$Fe^{3+}(aq) + e^{-} \rightarrow Fe^{2+}(aq)$	+0.77
$I_2(aq) + 2e^{-} \rightarrow 2I^{-}(aq)$	+0.54
$VO^{2+}(aq) + 2H^{+}(aq) + e^{-} \rightarrow V^{3+}(aq) + H_2O(l)$	+0.34
$V^{3+}(aq) + e^{-} \rightarrow V^{2+}(aq)$	-0.26
$Fe^{2+}(aq) + 2e^{-} \rightarrow Fe(s)$	-0.44

- (a) In terms of electron transfer, define the term *oxidising agent*.

(1)

- (b) (i) Give the conditions under which the electrode potential for $Cl_2(g)/2Cl^{-}(aq)$ is +1.36 V.

- (ii) Give a change in one of these conditions which would result in the electrode potential becoming more positive. Explain your answer. (5)
- (c) (i) Which of the reducing agents in the table is the weakest? (5)
- (ii) Identify all the species in the table which could convert $\text{I}^-(\text{aq})$ into $\text{I}_2(\text{aq})$ but which could not convert $\text{Br}^-(\text{aq})$ into $\text{Br}_2(\text{l})$.
- (iii) Identify the metal ions which would be left in solution if an excess of powdered iron metal was added to an acidified solution containing $\text{VO}^{2+}(\text{aq})$ ions. (5)
- (Total 11 marks)**
6. Use the standard electrode potential data in the table below to answer the questions which follow.

	$E^\ominus / \text{V}$
$\text{Ce}^{4+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{Ce}^{3+}(\text{aq})$	+1.70
$\text{MnO}^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightleftharpoons \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$	+1.51
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-(\text{aq})$	+1.36
$\text{VO}_2^+(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{VO}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+1.00
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2\text{SO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+0.17

(a) Name the standard reference electrode against which all other electrode potentials are measured. (1)

(b) When the standard electrode potential for $\text{Fe}^{3+}(\text{aq}) / \text{Fe}^{2+}(\text{aq})$ is measured, a platinum electrode is required.

(i) What is the function of the platinum electrode?

(ii) What are the standard conditions which apply to $\text{Fe}^{3+}(\text{aq}) / \text{Fe}^{2+}(\text{aq})$ when measuring this potential? (3)

(c) The cell represented below was set up under standard conditions.

$$\text{Pt} | \text{H}_2\text{SO}_3(\text{aq}), \text{SO}_4^{2-}(\text{aq}) || \text{MnO}_4^-(\text{aq}), \text{Mn}^{2+}(\text{aq}) | \text{Pt}$$

Calculate the e.m.f. of this cell and write an equation for the spontaneous cell reaction. (3)

(d) (i) Which one of the species given in the table is the strongest oxidising agent? (3)

(ii) Which of the species in the table could convert $\text{Fe}^{2+}(\text{aq})$ into $\text{Fe}^{3+}(\text{aq})$ but could not convert $\text{Mn}^{2+}(\text{aq})$ into $\text{MnO}_4^-(\text{aq})$? (3)

(e) Use data from the table of standard electrode potentials to deduce the cell which would have a standard e.m.f. of 0.93 V. Represent this cell using the convention shown in part (c). (2)

(Total 12 marks)

7. Large blocks of magnesium are bolted onto the hulls of iron ships in an attempt to prevent the iron being converted into iron(II), one of the steps in the rusting process. Use the data below, where appropriate, to answer the questions which follow.

	$E^\ominus / \text{V}$
$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Mg}(\text{s})$	-2.37
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Fe}(\text{s})$	-0.44
$\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^- \rightleftharpoons 4\text{OH}^-(\text{aq})$	+0.40

(a) Calculate the e.m.f. of the cell represented by $\text{Mg}(\text{s}) | \text{Mg}^{2+}(\text{aq}) || \text{Fe}^{2+}(\text{aq}) | \text{Fe}(\text{s})$ under standard conditions. Write a half-equation for the reaction occurring at the

negative electrode of this cell when a current is drawn.

(2)

- (b) Deduce how the e.m.f. of the cell $\text{Mg(s)} | \text{Mg}^{2+}(\text{aq}) || \text{Fe}^{2+}(\text{aq}) | \text{Fe(s)}$ changes when the concentration of Mg^{2+} is decreased. Explain your answer.

(3)

- (c) Calculate a value for the e.m.f. of the cell represented by $\text{Pt(s)} | \text{OH}^-(\text{aq}) | \text{O}_2(\text{g}) || \text{Fe}^{2+}(\text{aq}) | \text{Fe(s)}$ and use it to explain why iron corrodes when in contact with water which contains dissolved oxygen.

(2)

(Total 7 marks)

8. The table below shows some values for standard electrode potentials.

Electrode	Electrode reaction	$E^\ominus / \text{V}$
A	$\text{Mn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Mn(s)}$	- 1.18
B	$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Fe(s)}$	- 0.44
C	$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Ni(s)}$	- 0.25
D	$\text{Sn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Sn(s)}$	- 0.14
E	$2\text{H}^+(\text{g}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$	?

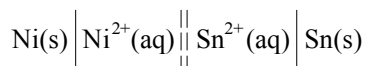
- (a) (i) Give the name of electrode **E** and indicate its role in the determination of standard electrode potentials.

(2)

- (ii) What is the value of the standard electrode potential for electrode **E**?

(1)

- (b) The electrochemical cell set up between electrodes **C** and **D** can be represented by the cell diagram:



- (i) Calculate the e.m.f. of this cell.

(1)

- (ii) State which would be the positive electrode.

(1)

- (iii) Write an equation to show the overall reaction in the cell.

(1)

- (c) Use the standard electrode potential data given in the table above:

- (i) to explain whether or not you would expect a reaction to occur if a piece of tin were to be added to a test tube containing aqueous iron(II) sulphate;

(2)

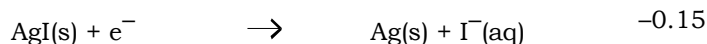
- (ii) to predict and explain two observations you would expect to make if a small piece of manganese were to be added to a test tube containing hydrochloric acid of concentration 1 mol dm^{-3} .

(4)

(Total 12 marks)

9. Use the data below to answer the questions that follow

Reaction at 298 K			$E^\ominus / \text{V}$
$\text{Ag}^+(\text{aq}) + \text{e}^-$	$\rightarrow$	Ag(s)	+0.08
$\text{AgF(s)} + \text{e}^-$	$\rightarrow$	$\text{Ag(s)} + \text{F}^-(\text{aq})$	+0.78
$\text{AgCl(s)} + \text{e}^-$	$\rightarrow$	$\text{Ag(s)} + \text{Cl}^-(\text{aq})$	+0.22
$\text{AgBr(s)} + \text{e}^-$	$\rightarrow$	$\text{Ag(s)} + \text{Br}^-(\text{aq})$	+0.07
$\text{H}^+(\text{aq}) + \text{e}^-$	$\rightarrow$	$\frac{1}{2} \text{H}_2(\text{g})$	0.00
$\text{D}^+(\text{aq}) + \text{e}^-$	$\rightarrow$	$\frac{1}{2} \text{D}_2(\text{g})$	-0.004



The symbol D denotes deuterium, which is heavy hydrogen, ${}^2_1\text{H}$.

(a) By considering electron transfer, state what is meant by the term *oxidising agent*. (1)

(b) State which of the two ions, $\text{H}^+(\text{aq})$ or $\text{D}^+(\text{aq})$, is the more powerful oxidising agent. Write an equation for the spontaneous reaction which occurs when a mixture of aqueous

H^+ ions and D^+ ions are in contact with a mixture of hydrogen and deuterium gas. Deduce the e.m.f. of the cell in which this reaction would occur spontaneously. (3)

(c) Write an equation for the spontaneous reaction which occurs when aqueous F^- ions and Cl^- ions are in contact with a mixture of solid AgF and solid AgCl . Deduce the e.m.f. of the cell in which this reaction would occur spontaneously. (2)

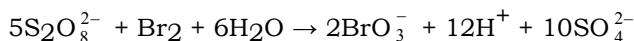
(d) Silver does not usually react with dilute solutions of strong acids to liberate hydrogen.

(i) State why this is so.

(ii) Suggest a hydrogen halide which might react with silver to liberate hydrogen in aqueous solution. Write an equation for the reaction and deduce the e.m.f. of the cell in which this reaction would occur spontaneously. (4)

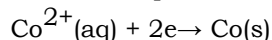
(Total 10 marks)

10. (a) The following reaction occurs in aqueous solution.



Identify the reducing agent in this reaction and write a half-equation for its action. (2)

(b) The electrode potential for the half-equation

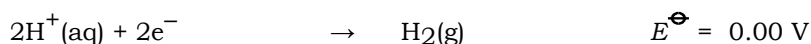
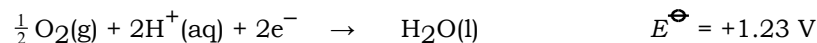
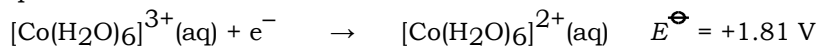


is measured by reference to a standard hydrogen electrode.

(i) State the temperature at which the standard electrode potential $E^\ominus$ is measured, and give the concentration of $\text{Co}^{2+}(\text{aq})$ that must be used.

(ii) Electrode potentials are usually measured by reference to a secondary standard electrode. Identify a secondary standard electrode and give a reason why it is used rather than a standard hydrogen electrode. (4)

(c) Cobalt in oxidation states +2 and +3 forms complex ions with water, ammonia and cyanide ligands. Use, where appropriate, the data given below to answer the questions which follow.



(i) Which of the six cobalt species shown above is the most powerful oxidising agent?

(ii) Identify a cobalt(II) species which cannot be oxidised by gaseous oxygen.

(iii) Hydrogen is evolved when a salt containing the cobalt species $[\text{Co}(\text{CN})_6]^{4-}(\text{aq})$ is reacted with a dilute acid. Use the electrode potentials given above to explain the formation of the hydrogen gas. (4)

(Total 10 marks)

11. The table below shows some values for standard electrode potentials. These data should be used, where appropriate, to answer the questions that follow concerning the chemistry

of copper and iron.

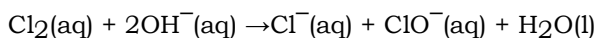
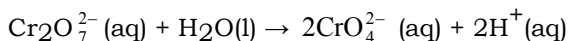
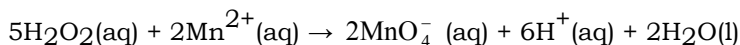
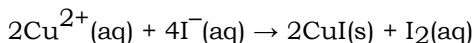
Electrode reaction	$E^\ominus / \text{V}$
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Fe}(\text{s})$	- 0.44
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$	0.00
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Cu}(\text{s})$	+ 0.34
$\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^- \rightleftharpoons 4\text{OH}^-(\text{aq})$	+ 0.40
$\text{NO}_3^-(\text{aq}) + 4\text{H}^+(\text{aq}) + 3\text{e}^- \rightleftharpoons \text{NO}(\text{g}) + 2\text{H}_2\text{O}(\text{l})$	+ 0.96

- (a) Write an equation to show the reaction that occurs when iron is added to a solution of a copper(II) salt. (1)
- (b) A similar overall reaction to that shown in (a) would occur if an electrochemical cell was set up between copper and iron electrodes.
- (i) Write down the cell diagram to represent the overall reaction in the cell. (2)
- (ii) Calculate the e.m.f. of the cell. (1)
- (c) (i) Use the standard electrode potential data given to explain why copper reacts with dilute nitric acid but has no reaction with dilute hydrochloric acid. (3)
- (ii) Write an equation for the reaction between copper and dilute nitric acid. (2)
- (d) Although iron is a widely used metal, it has a major disadvantage in that it readily corrodes in the presence of oxygen and water. The corrosion is an electrochemical process which occurs on the surface of the iron.
- (i) Use the standard electrode potential data given to write an equation for the overall reaction that occurs in the electrochemical cell set up between iron, oxygen and water. (1)
- (ii) State, with a reason, whether the iron acts as the anode or cathode of the cell. (2)
- (iii) Predict and explain whether or not you would expect a similar corrosion reaction to occur with copper in the presence of oxygen and water. (2)
- (Total 14 marks)**

12. (a) Name the standard reference electrode against which electrode potentials are measured and, for this electrode, state the conditions to which the term *standard* refers. (4)
- (b) The standard electrode potentials for two electrode reactions are given below.
- $\text{S}_2\text{O}_8^{2-}(\text{aq}) + 2\text{e}^- \rightarrow 2\text{SO}_4^{2-}(\text{aq}) \quad E^\ominus = + 2.01 \text{ V}$
- $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) \quad E^\ominus = + 0.80 \text{ V}$
- (i) A cell is produced when these two half-cells are connected. Deduce the cell potential, $E^\ominus$, for this cell and write an equation for the spontaneous reaction.
- (ii) State how, if at all, the electrode potential of the $\text{S}_2\text{O}_8^{2-} / \text{SO}_4^{2-}$ equilibrium would change if the concentration of SO_4^{2-} ions was increased. Explain your answer. (6)
- (Total 10 marks)**

13. For **each** of the reactions listed below
- (i) identify which species, if any, are acting as oxidising agents;
- (ii) determine the oxidation states before and after reaction of any species that are oxidised;

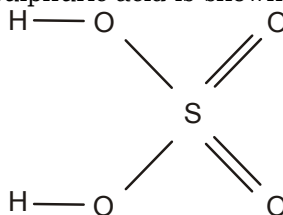
- (iii) write half-equations, including state symbols, for all redox reactions that occur.



(11)

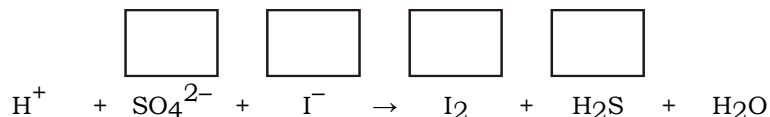
(Total 11 marks)

14. In order to obtain full marks in this question, you must show **all** your working clearly. In its reactions, sulphuric acid, H_2SO_4 , can behave as an acid, an oxidising agent and as a dehydrating agent. The displayed formula of pure sulphuric acid is shown below.



Concentrated sulphuric acid will readily oxidise halide ions to the halogen.

The equation below represents the unbalanced equation for the oxidation of iodide ions by sulphuric acid.



- (i) Write the oxidation numbers of sulphur and iodine in the boxes above the equation.

[2]

- (ii) Balance the equation above.

[1]

[Total 3 marks]

15. The carbonates and nitrates of Group 2 elements decompose when heated. Barium nitrate decomposes when heated to make barium oxide, nitrogen dioxide and oxygen.



Use oxidation states to explain why this decomposition reaction involves both oxidation **and** reduction.

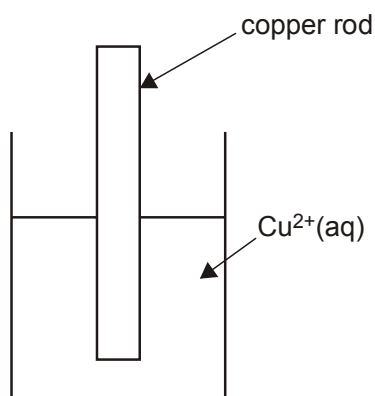
[3]

16. The standard electrode potential of $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightleftharpoons \text{Cu}(\text{s})$ is +0.34 V.

- (a) Define the term *standard electrode potential*.

[3]

- (b) Complete the diagram to show how the standard electrode potential of $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightleftharpoons \text{Cu}(\text{s})$ could be measured.



[3]

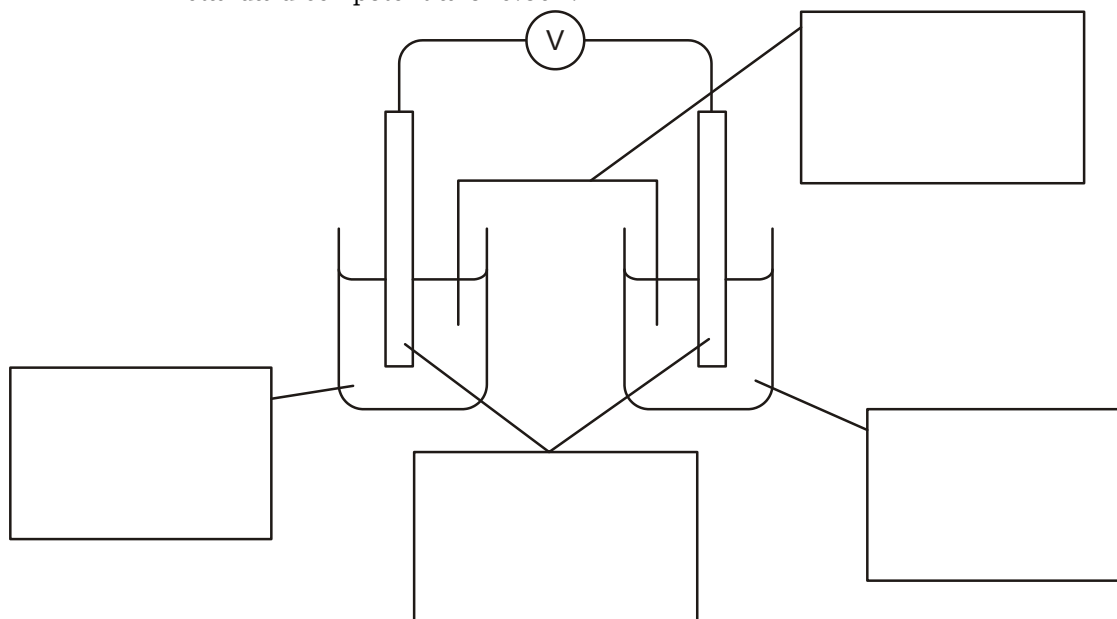
[Total 6 marks]

17. The standard electrode potentials for some redox systems involving vanadium are shown below.

These are labelled **A**, **B**, **C** and **D**.

		$E^\ominus / \text{V}$
A	$\text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightleftharpoons \text{VO}^{2+} + \text{H}_2\text{O}$	+1.00
B	$\text{V}^{3+} + \text{e}^- \rightleftharpoons \text{V}^{2+}$	-0.26
C	$\text{V}^{2+} + 2\text{e}^- \rightleftharpoons \text{V}$	-1.20
D	$\text{VO}^{2+} + 2\text{H}^+ + \text{e}^- \rightleftharpoons \text{V}^{3+} + \text{H}_2\text{O}$	+0.34

- (a) Which of the vanadium species shown in **A**, **B**, **C** and **D** is the most powerful oxidising agent? [1]
- (b) A student wishes to set up a cell with a standard cell potential of 0.60V. [1]
- (i) Which two of the redox systems, **A**, **B**, **C** or **D**, should he choose? [1]
- (ii) Complete the labelling of the following diagram which shows the cell with a standard cell potential of 0.60V.



- (iii) The emf of this cell is only 0.60 V under standard conditions. What do you understand by the expression *standard conditions*? [4]

[1]
[Total 7 marks]

18. Use the data below, where appropriate, to answer the questions which follow.

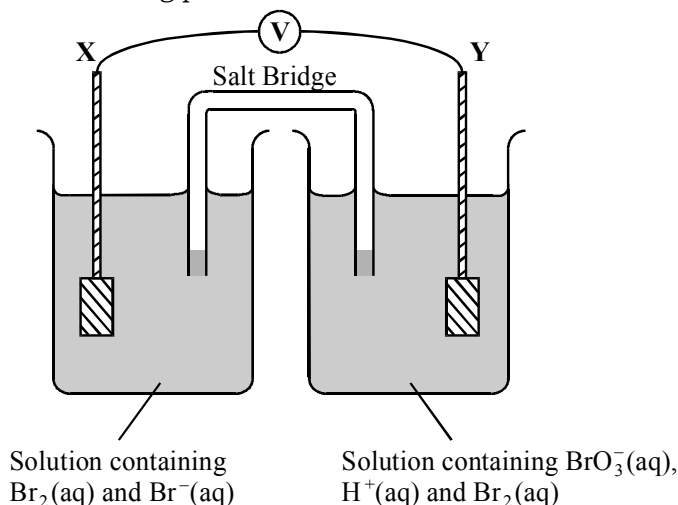
Standard electrode potentials	$E^\ominus / \text{V}$
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	0.00
$\text{Br}_2(\text{aq}) + 2\text{e}^- \rightarrow 2\text{Br}^-(\text{aq})$	+1.09
$2\text{BrO}_3^-(\text{aq}) + 12\text{H}^+(\text{aq}) + 10\text{e}^- \rightarrow \text{Br}_2(\text{aq}) + 6\text{H}_2\text{O}(\text{l})$	+1.52

Each of the above can be reversed under suitable conditions.

- (a) State the hydrogen ion concentration and the hydrogen gas pressure when, at 298 K, the potential of the hydrogen electrode is 0.00 V. (2)
- (b) The electrode potential of a hydrogen electrode changes when the hydrogen ion concentration is reduced. Explain, using Le Chatelier's principle, why this change occurs and state how the electrode potential of the hydrogen electrode changes.

(c) A diagram of a cell using platinum electrodes **X** and **Y** is shown below.

(3)



- (i) Use the data above to calculate the e.m.f. of the above cell under standard conditions.
- (ii) Write a half-equation for the reaction occurring at electrode **X** and an overall equation for the cell reaction which occurs when electrodes **X** and **Y** are connected.

(4)

(Total 9 marks)

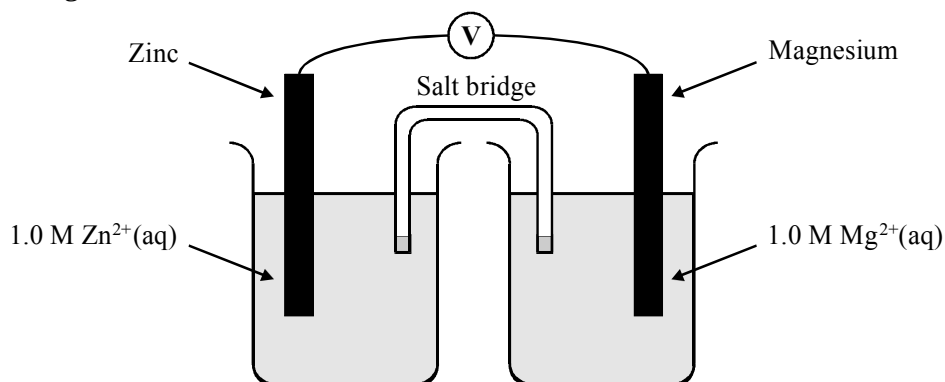
19. Use the data given below, where appropriate, to answer the questions which follow.

Standard electrode potentials in acid solution	$E^\ominus / \text{V}$
$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Mg}(\text{s})$	-2.37
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$\text{Sn}^{4+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn}^{2+}(\text{aq})$	+0.15
$\text{VO}^{2+}(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{V}^{3+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+0.34
$\text{VO}_2^+(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{VO}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+1.02
$\text{Ce}^{4+}(\text{aq}) + \text{e}^- \rightarrow \text{Ce}^{3+}(\text{aq})$	+1.70

- (a) Give the components of the standard reference electrode used in determining the standard electrode potentials above. State the conditions under which this standard reference electrode has a potential of 0.00 V.

(6)

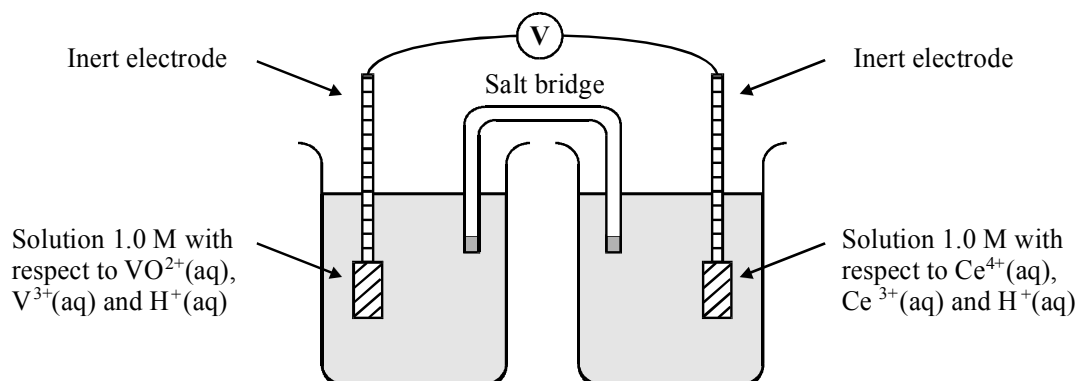
(b) A diagram of a cell is shown below.



- (i) Calculate the overall standard potential of this cell.
- (ii) State the polarity of the zinc electrode.

(2)

(c) A diagram of a cell is shown below.



- (i) Calculate the overall standard potential of the cell.
- (ii) Deduce the direction of electron flow in the external circuit when the inert electrodes are connected together. (2)
- (d) Using data from the table on page 6, derive an equation for the overall redox process which occurs when a solution containing $\text{Ce}^{4+}(\text{aq})$ is added to a solution containing $\text{V}^{3+}(\text{aq})$. (3)
- (e) Which, if any, of the four vanadium-containing species, $\text{V}^{2+}(\text{aq})$, $\text{V}^{3+}(\text{aq})$, $\text{VO}^{2+}(\text{aq})$ and $\text{VO}_2^+(\text{aq})$, will convert $\text{Sn}^{2+}(\text{aq})$ into $\text{Sn}^{4+}(\text{aq})$ in acid solution? (2)
- (Total 15 marks)**
20. Where appropriate, use the standard electrode potential data in the table below to answer the questions which follow.

	$E^\ominus/\text{V}$
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$\text{V}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{V}^{2+}(\text{aq})$	-0.26
$\text{SO}_4^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{SO}_3^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+0.17
$\text{VO}_2^+(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{V}^{3+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+0.34
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{VO}_2^+(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{VO}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	+1.00
$\text{Cl}_2(\text{aq}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$	+1.36

- (a) From the table above select the species which is the most powerful reducing agent. (1)
- (b) From the table above select
- (i) a species which, in acidic solution, will reduce $\text{VO}_2^+(\text{aq})$ to $\text{VO}^{2+}(\text{aq})$ but will **not** reduce $\text{VO}^{2+}(\text{aq})$ to $\text{V}^{3+}(\text{aq})$,
- (ii) a species which, in acidic solution, will oxidise $\text{VO}^{2+}(\text{aq})$ to $\text{VO}_2^+(\text{aq})$. (2)
- (c) The cell represented below was set up under standard conditions.
- $\text{Pt} | \text{Fe}^{2+}(\text{aq}), \text{Fe}^{3+}(\text{aq}) || \text{Tl}^{3+}(\text{aq}), \text{Tl}^+(\text{aq}) | \text{Pt}$ Cell e.m.f. = + 0.48 V
- (i) Deduce the standard electrode potential for the following half-reaction.
- $$\text{Tl}^{3+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Tl}^+(\text{aq})$$
- (ii) Write an equation for the spontaneous cell reaction. (3)

(Total 6 marks)

21. Concentrated nitric acid is a powerful oxidising agent. Concentrated nitric acid oxidises sulphur to sulphuric acid. Nitrogen dioxide and another product are also formed.

(i) Suggest a balanced equation for this reaction.

[1]

(ii) Deduce the change in oxidation number of nitrogen in this reaction.

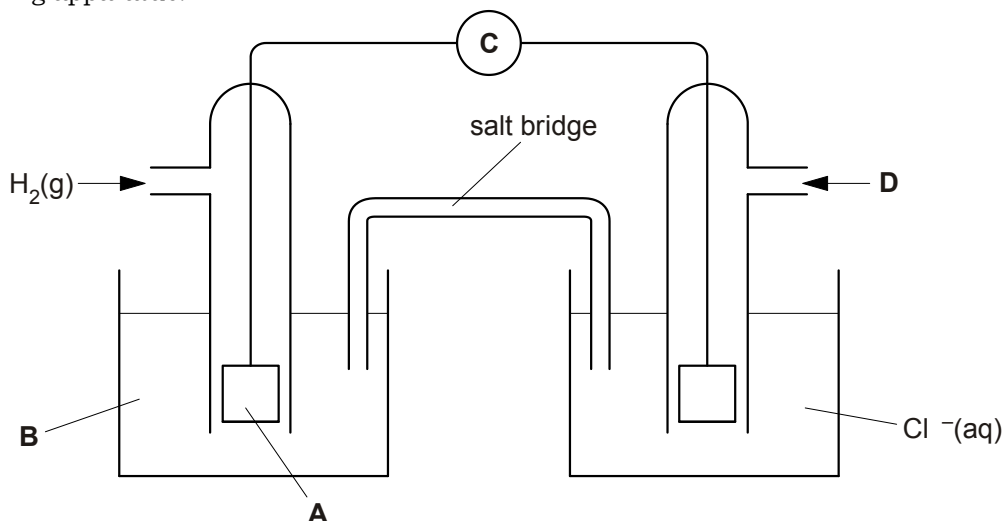
[1]

[Total 2 marks]

22. NO_2 reacts with oxygen and water to form nitric acid, HNO_3 . In the atmosphere, this contributes to acid rain. Construct a balanced equation for this formation of nitric acid and use oxidation numbers to show that this is a redox reaction.

[Total 2 marks]

23. The standard electrode potential of the $\frac{1}{2}\text{Cl}_2/\text{Cl}^-$ half-cell may be measured using the following apparatus.



- (a) Suggest suitable labels for **A**, **B**, **C** and **D**.

[2]

- (b) The half cell reactions involved are shown below.



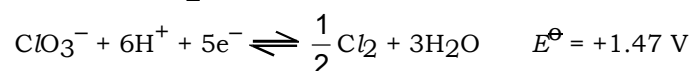
- (i) Use an arrow to show the direction of flow of electrons in the diagram of the apparatus. Explain your answer.

[2]

- (ii) The values of $E^\ominus$ are measured under standard conditions. What are the standard conditions?

[2]

- (c) The half cell reaction for $\text{ClO}_3^-/\frac{1}{2}\text{Cl}_2$ is shown below.

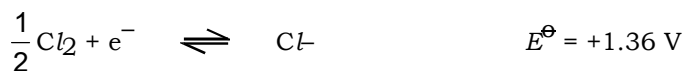


What does this tell you about the oxidising ability of ClO_3^- compared with Cl_2 ?
Explain your answer.

[2]

[Total 8 marks]

24. Chlorine gas may be prepared in the laboratory by reacting hydrochloric acid with potassium manganate(VII). The following standard electrode potentials relate to this reaction.

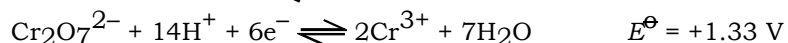


- (a) Define the term *standard electrode potential*. [3]
- (b) Determine the standard cell potential for a cell constructed from these two redox systems. [1]

[Total 4 marks]

25. Chromium is an important metallic element. Its compounds have a number of different oxidation states.

The following equations relate to half-cells involving iron and chromium ions.

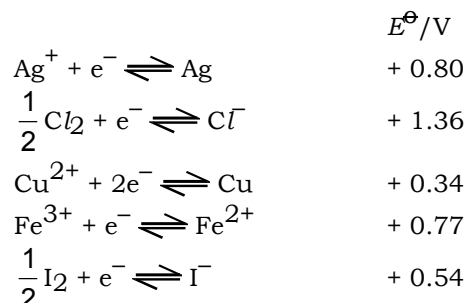


A cell was set up by combining these two half-cells.

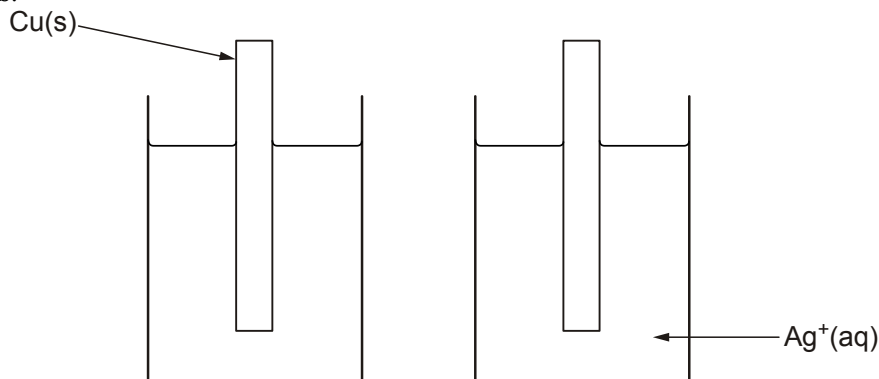
- (i) Derive a balanced equation for the reaction that would occur when the cell is in use.
Explain your reasoning in terms of oxidation and reduction. [3]
- (ii) Determine the emf of the cell under standard conditions. [1]

[Total 6 marks]

26. Some standard electrode potentials are shown below.



- (a) The diagram below shows an incomplete cell consisting of Cu/Cu^{2+} and Ag/Ag^+ half-cells.



- (i) Complete and label the diagram to show how the cell potential of this cell could be measured. [2]
- (ii) On the diagram, show the direction of **electron** flow in the circuit if a current was allowed. [1]
- (iii) Calculate the standard cell potential. [1]
- (iv) Write the overall cell reaction. [1]
- (b) Chlorine will oxidise Fe^{2+} to Fe^{3+} but iodine will not. Explain why, using the

electrode potential data.

[2]

[Total 7 marks]

27. Chlorine gas may be prepared in the laboratory by reacting hydrochloric acid with potassium manganate(VII). The following standard electrode potentials relate to this reaction.



- (a) Use the half-equations above to:

(i) Construct an ionic equation for the reaction between hydrochloric acid and potassium manganate(VII);

[2]

(ii) Determine the oxidation numbers of chlorine and manganese before and after the reaction has taken place;

[2]

(iii) State what is oxidised and what is reduced in this reaction.

[2]

- (b) If potassium manganate(VII) and very dilute hydrochloric acid are mixed, there is no visible reaction. Suggest why there is no visible reaction in this case.

[1]

[Total 7 marks]

28. Use the standard electrode potentials in the table below to answer the questions that follow.

I	$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Fe}(\text{s})$	$E^\ominus = -0.44 \text{ V}$
II	$\text{V}^{3+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{V}^{2+}(\text{aq})$	$E^\ominus = -0.26 \text{ V}$
III	$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$	$E^\ominus = 0.00 \text{ V}$
IV	$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}(\text{l})$	$E^\ominus = +0.40 \text{ V}$

An electrochemical cell was set up based on systems **I** and **II**.

- (i) Write half-equations to show what has been oxidised and what has been reduced in this cell.

2]

- (ii) Determine the cell potential of this cell.

[1]

[Total 3 marks]

29. Use the standard electrode potentials in the table below to answer the questions that follow.

I	$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Fe}(\text{s})$	$E^\ominus = -0.44 \text{ V}$
II	$\text{V}^{3+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{V}^{2+}(\text{aq})$	$E^\ominus = -0.26 \text{ V}$
III	$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$	$E^\ominus = 0.00 \text{ V}$
IV	$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}(\text{l})$	$E^\ominus = +0.40 \text{ V}$

An electrochemical fuel cell was set up based on systems **III** and **IV**.

- (i) Construct an equation for the spontaneous cell reaction. Show your working.

[2]

- (ii) Fuels cells based on systems such as **III** and **IV** are increasingly being used to generate energy.
Discuss **two** advantages and **two** disadvantages of using fuels cells for energy rather than using fossil fuels.

[4]

[Total 6 marks]

Marking Scheme

- | | | | | |
|----|-----|---|---|------|
| 1. | (a) | gains electrons (1) | 1 | |
| | | | | [1] |
| 2. | (a) | (i) Fe^{2+} | 1 | |
| | | (ii) F_2O | 1 | |
| | | (iii) Fe^{2+} | 1 | |
| | | Cl^- | 1 | |
| | | <i>Use list principle if more than two answers</i> | | |
| | (b) | (i) e.m.f. = $E(\text{rhs}) - E(\text{lhs})$ | 1 | |
| | | = $1.52 - 0.77 = 0.75$ | | |
| | | (0.75 scores first mark also) | 1 | |
| | | (ii) $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$ | 1 | |
| | | (iii) Decrease | 1 | |
| | | <i>(Increase is CE, no further marks)</i> | | |
| | | Equilibrium (or reaction) shifts to R | 1 | |
| | | <i>(or L if refers to half equation in table)</i> | | |
| | | <i>(or in favour of more Fe^{3+})</i> | | |
| | | <i>(or more Fe^{3+} formed)</i> | | |
| | | <i>(or more electrons formed)</i> | | |
| | | Electrode potential (for $\text{Fe}^{3+}/\text{Fe}^{2+}$) less positive (or decreases) | 1 | |
| | | | | [10] |
| 3. | (a) | Fe^{2+} or Fe(II) | 1 | |
| | (b) | (i) 6 or (VI) | 1 | |
| | | (ii) 3 or (III) | 1 | |
| | (c) | (i) 0.5 | 1 | |
| | | (ii) $2\text{Mn}^{2+} + 8\text{H}_2\text{O} + 5\text{S}_2\text{O}_8^{2-} \rightarrow 10\text{SO}_4^{2-} + 2\text{MnO}_4^- + 16\text{H}^+$ | | |
| | | Both SO_4^{2-} and MnO_4^- on right | 1 | |
| | | Balanced | 1 | |
| | | | | [13] |
| 4. | (a) | (i) 0.60 V | 1 | |
| | | (ii) $\text{H}_2\text{O} + \text{H}_2\text{SO}_3 \rightarrow \text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$ | 1 | |
| | (b) | (i) $2\text{IO}_3^- + 2\text{H}^+ 5\text{H}_2\text{O}_2 \rightarrow 5\text{O}_2 + \text{I}_2 + 6\text{H}_2\text{O}$ | 1 | |
| | | Species | | |
| | | Balanced | 1 | |
| | | (ii) The concentration of the ions change or are no longer standard or the e.m.f is determined when no current flows | 1 | |
| | | (iii) Unchanged | 1 | |
| | | (iv) Increased | 1 | |
| | | Equilibrium IO_3^-/I_2 displaced to the right | 1 | |
| | | Electrons more readily accepted or more reduction occurs or electrode becomes more positive (Q o L) | 1 | |
| | (c) | VO_2^+ | 1 | |
| | | 5 or V | 1 | |
| | | $\text{V}^{2+} + 2\text{H}_2\text{O} \rightarrow \text{VO}_2^+ + 4\text{H}^+ 3\text{e}^-$ | 1 | |
| | | | | [12] |
| 5. | (a) | Oxidising agents take/remove/accept/gain electrons (1) | | |

- Not 'electron pair' 1
- (b) (i) $\text{Cl}_2(\text{g})$ at 100 kPa / 1 bar / 1atm (1)
 $[\text{Cl}^-] = 1 \text{ molar} / 1\text{M}$ (1)
 Allow 1M HCl
 Temperature = 298K / 25°C (1)
 Do not use list principle for other incorrect species
- (ii) Lower $[\text{Cl}^-]$ or reduce temperature
 Increase pressure or concentration of Cl_2 (1)
 CE if change incorrect
 Equilibrium displace to right (1)
 or if reduced temperature given, reaction exothermic
 Allow a correct explanation when no change given 5
- (c) (i) Cl^- (1)
- (ii) Fe^{3+} ; (1) NO_3^- (1)
 Penalise by list principle
 Note: H^+ is incorrect
- (iii) V^{2+} , (1) Fe^{2+} (1)
 Penalise by list principle 5
- [11]
6. (a) (Standard) hydrogen (electrode) (1) 1
- (b) (i) To allow transfer of electrons / provide a reaction surface (1)
 (ii) 298 K (1)
 Both $\text{Fe}^{3+}(\text{aq})$ and $\text{Fe}^{2+}(\text{aq})$ have a concentration of 1 mol dm^{-3} (1) (QoL)
 OR $[\text{H}^+] = 1 \text{ mol dm}^{-3}$
NOT zero current or 100 kPa
- (c) +1.34 V (1) 3
 $2 \text{MnO}_4^- + 5 \text{H}_2\text{SO}_3 \rightarrow 2 \text{Mn}^{2+} + 5 \text{SO}_4^{2-} + 3 \text{H}_2\text{O} + 4 \text{H}^+$
 Correct species / order (1)
 Balanced and cancelled (1)
Allow one for $2 \text{MnO}_4^- + 5 \text{H}_2\text{SO}_3 \rightarrow 2 \text{Mn}^{2+} + 5 \text{SO}_4^{2-}$
- (d) (i) $\text{Ce}^{4+}(\text{aq})$ (1)
 (ii) $\text{VO}_2^+(\text{aq})$ (1); Cl_2 (1)
Penalise additional answers to zero
- (e) $\text{Pt} | \text{Fe}^{2+}(\text{aq}), \text{Fe}^{3+}(\text{aq}) || \text{Ce}^{4+}(\text{aq}), \text{Ce}^{3+}(\text{aq}) | \text{Pt}$
 Correct species (1)
 Correct order (1)
Deduct one mark for each error
- 2
- [12]
7. (a) Cell e.m.f.: 1.93 (v) CE if negative value given (1)
 Half equation: $\text{Mg} \rightarrow \text{Mg}^{2+} + 2 \text{e}^-$ (1)
 or $\rightleftharpoons$
Ignore state symbols
Mark on after an AE
- (b) Change in e.m.f.: increases (1)
Mark on even if incorrect

Explanation: Equilibrium displaced to Mg^{2+} or to the left **(1)**
cell reaction or overall reaction goes to the right
Electrode is more negative or E decreases
or gives more electron
or forms more Mg^{2+} ions
Mark separately

3

- (c) Cell e.m.f. : -0.84 (V) **(1)**

Explanation: Fe is giving electrons **or** forming Fe^{2+}
or reaction goes in the reverse direction **(1)**
Mark on after AE

2

**N.B. In (a) and (c) mark on if no value given,
 but CE in both (a) and (c) if e.m.f. = 0**

[7]

8. (a) (i) (standard) hydrogen (electrode) / hydrogen half cell **not**
 hydrogen cell **(1)**
 reference electrode / electrode to which others are compared **(1)**
 (ii) 0.00(V) / 0 / zero **(1)**
- (b) (i) emf = -0.14 - (-0.25)
 = + 0.11 V / allow 0.11V **not** - 0.11V **(1)**
 (ii) electrode D / Sn^{2+} / Sn / tin / right hand electrode **(1)**
 (iii) $\text{Ni} + \text{Sn}^{2+} \rightarrow \text{Ni}^{2+} + \text{Sn}$ (ignore state symbols) **(1)**
- (c) (i) e.m.f = -0.44 - (-0.14) = -0.30 (V) / emf for cell is - ve
 comparison of standard electrode potentials **(1)**
 +ve e.m.f for feasible reaction / tin is a weaker reducing agent
 therefore would not occur **(1)**

2

- (ii) manganese will decrease in size / disappear / eaten away / dissolves /
 solution turns (pale) pink **(1)**
 effervescence / bubbles (of colourless gas) / fizzing **not** gas
 given off **(1)**
 reaction likely to occur is $\text{Mn} + (2)\text{H}^+ \rightarrow \text{Mn}^{2+} + \text{H}_2$ **(1)**
or the same ideas expressed in words
 +ve e.m.f. / +1.18 V / Mn is strong reducing agent / has
 large - ve $E^{\ominus}$ **(1)**
(not just Mn is more reactive)

4

[12]

9. (a) oxidising agent accepts electrons **(1)**
- (b) Stronger oxidising agent H^+ **(1)**

$$\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2(\text{g})$$

$$\text{D}^+(\text{aq}) + \text{e}^- \rightarrow \frac{1}{2}\text{D}_2(\text{g})$$

Equation
$$\text{H}^+(\text{aq}) + \frac{1}{2}\text{D}_2(\text{g}) \rightarrow \text{D}^+(\text{aq}) + \frac{1}{2}\text{H}_2(\text{g})$$
 (1)
e.m.f.

$$-(-0.004) = +0.004 \text{ V}$$
 (1)
- (c) *Equation*
$$\text{AgF}(\text{s}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{F}^-(\text{aq})$$
 (1)
e.m.f.
$$= +0.78 - 0.22 = +0.56 \text{ V}$$
 (1)
- (d) (i) Silver lies above hydrogen in the electrochemical series

1

0.000

3

2

- Or $\text{Ag}^+(\text{aq})$ is a stronger oxidising agent than $\text{H}^+(\text{aq})$ (1)
 (ii) *Hydrogen halide* HI (1)
Equation $\text{H}^+(\text{aq}) + \text{I}^-(\text{aq}) + \text{Ag}(\text{s}) \rightarrow \text{AgI}(\text{s}) + \frac{1}{2} \text{H}_2(\text{g})$ (1)
e.m.f. $40.00 - (-0.15) = +0.15 \text{ V}$ (1) 4
[10]

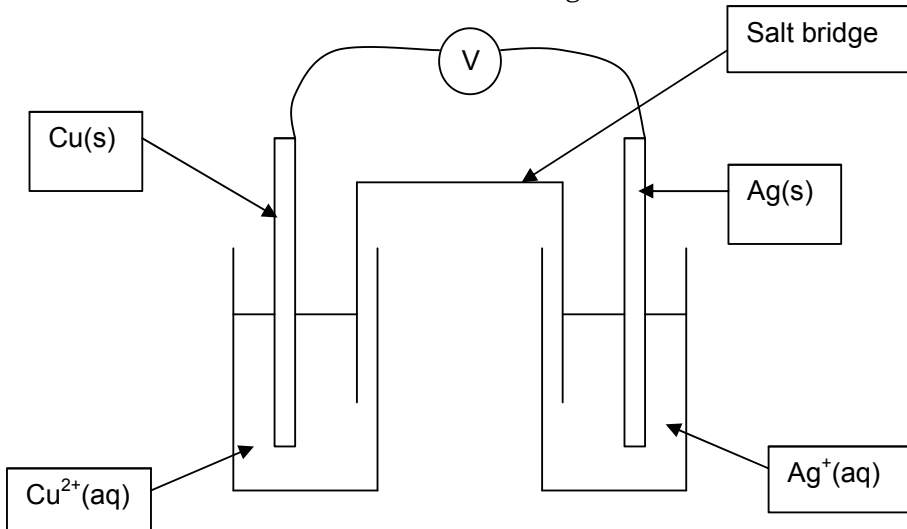
10. (a) *Reducing agent* Br_2 (1)
Half equation $\text{Br}_2 + 6\text{H}_2\text{O} \rightarrow 2\text{BrO}_3^- + 12\text{H}^+ + 10\text{e}^-$ (1) 2
 (b) (i) *Temperature* 298 K or 25°C (1)
Concentration 1.00 M or 1.00 mol dm^{-3} (1)
 (ii) *Secondary standard electrode* calomel
 or Ag/AgCl (1)
Reason Easier to use
 or hydrogen electrode (1)
 difficult to use 4
 (c) (i) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (1)
 (ii) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ (1)
 (iii) $E^\ominus$ for $[\text{Co}(\text{CN})_6]^{3-}/[\text{Co}(\text{CN})_6]^{4-}$ is negative with respect to H^+/H_2 (1)
 Electrons flow to H^+/H_2 and H^+ reduced (1) 4
[10]

11. (a) $\text{Fe} + \text{Cu}^{2+} \rightarrow \text{Cu} + \text{Fe}^{2+}$ (1)
 allow use of soluble Cu salt eg CuSO_4
 ignore state symbols 1
 (b) (i) $\text{Fe}(\text{s}) \mid \text{Fe}^{2+}(\text{aq}) \parallel \text{Cu}^{2+}(\text{aq}) \mid \text{Cu}(\text{s})$
 junctions correct (1)
 ignore state symbols
 allow alternative symbols for salt bridge
 allow if junctions are correct but order is wrong providing
 metals on each side of salt bridge are the same ie $\text{Fe} \mid \text{Fe}^{2+}$ not $\text{Fe} \mid \text{Cu}^{2+}$
 order of species correct (1)
 do not give this mark if cell reversed 2
 (ii) e.m.f. = $+0.34 - (-0.44)$
 = $+0.78 \text{ V}$ (1)
 must have + sign
 allow -0.78 V if reverse cell given in (i) 1
 (c) (i) e.m.f for cell must be positive for reaction to occur / be feasible / (1)
 spontaneous
 $\text{Cu}(\text{s}) + 2\text{H}^+ \rightarrow \text{products}$
 e.m.f = -0.34 therefore won't happen /
 sensible comparison of the magnitude of $E^\ominus$ for the electrodes
 eg 'Cu electrode more positive than hydrogen electrode hence won't work' (1)
 $\text{Cu}(\text{s}) + \text{NO}_3^- + 4\text{H}^+ \rightarrow \text{products}$
 e.m.f = $+0.96 - 0.34 = +0.62$ hence can occur /
 similar sensible comparison (1) 3
 (ii) $3\text{Cu} + 2\text{NO}_3^- + 8\text{H}^+ \rightarrow 3\text{Cu}^{2+} + 2\text{NO} + 4\text{H}_2\text{O}$
 species (1)
 balanced – this mark dependent on first mark (1) 2
 (d) (i) $2\text{Fe} + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Fe}^{2+} + 4\text{OH}^-$ or $2\text{Fe}(\text{OH})_2$ (1)
 ignore state symbols 1
 (ii) anode (only give this mark if explanation attempted) (1)
 Fe loses e^- (therefore negative pole) / oxidation occurs (1)
 this mark dependent on anode for first mark 2

- (iii) e.m.f. = + 0.06 V or reference to $E^\ominus$ for electrodes (1)
 reasoned argument (1)
 eg positive hence should occur / difference so small that reaction unlikely 2
[14]
12. (a) Name hydrogen electrode (1)
 Conditions 1 M H^+ (aq) or 1 M HCl(aq) or 0.5 M H_2SO_4 (aq) (1)
 298 K (1)
 Hydrogen gas at 1 bar or 100 kPa (1) 4
- (b) (i) $E^\ominus$ value 1.21 v (1)
 Equation $S_2O_8^{2-} + 2Ag \rightarrow 2SO_4^{2-} + 2Ag^+$ [2]
 (ii) Change, if any, in electrode potential Less positive or decrease (1)
 Explanation Equilibrium displaced to the left (1)
 More electrons released (1) 6
[10]
13. (i) Oxidising agents
 Cu^{2+} (aq) (1), H_2O_2 (aq) (1), Cl_2 (aq) (1) (3)
 if $Cr_2O_7^{2-}$ (aq) is included, deduct one mark
 (ii) $2I^- \rightarrow I_2$ I(-1) $\rightarrow$ I(0) (1)
 $Mn^{2+} \rightarrow MnO_4^-$ Mn(II) $\rightarrow$ Mn(VII) (1)
 $Cr_2O_7^{2-} \rightarrow 2CrO_4^{2-}$ no change (penalised above)
 $Cl_2 \rightarrow ClO^-$ Cl(0) $\rightarrow$ Cl(1) (1)
 (iii) Cu^{2+} (aq) + I^- (aq) + $e^- \rightarrow Cu^+$ (s) (1)
 I^- (aq) $\rightarrow \frac{1}{2} I_2$ (aq) + e^- (not reverse) (1)
 H_2O_2 (aq) + $2H^+$ (aq) + $2e^- \rightarrow 2H_2O$ (1) (1)
 Mn^{2+} (aq) + $4H_2O$ (1) $\rightarrow MnO_4^-$ (aq) + $8H^+$ (aq) + $5e^-$ (not reverse) (1)
 $\frac{1}{2} Cl_2$ (aq) + $e^- \rightarrow Cl^-$ (aq) (1)
 $\frac{1}{2} Cl_2$ (aq) + $2OH^-$ (aq) $\rightarrow ClO^-$ (aq) + H_2O (l) + e^- (not reverse) (1)
 Penalise missing or incorrect state symbols once only (12 MAX)
 11
[11]
14. (i) $SO_4^{2-} \rightarrow H_2S$: S from +6 to -2 (1)
 $I^- \rightarrow I_2$: I from -1 to 0 (1) 2
 (ii) $10H^+ + SO_4^{2-} + 8I^- \rightarrow 4I_2 + H_2S + 4H_2O$ (1) 1
[3]
15. Oxidation state of nitrogen goes from +5 to +4 (1)
 Oxidation state of oxygen goes from -2 to 0 (1);
 Correct linking of changes of oxidation state with
 reduction **and** with oxidation (1) 3
*If oxidation state of barium given is incorrect max 1 for the
 oxidation numbers.*
Allow ecf from wrong oxidation states for the correct linking
 mark
Both oxidation **and** reduction needed
[3]
16. (a) Emf/voltage/potential difference (of electrochemical cell) 1
 comprising a (Cu/Cu^{2+}) half cell combined with a standard

- hydrogen electrode 1
 1 atm, 1 mol.dm⁻³, 298K (all 3 needed but can transfer mark if stated in (b)) 1
- (b) Salt bridge and voltmeter 1
 Platinum electrode dipping into 1 mol dm⁻³ H⁺ 1
 Hydrogen gas feed 1
 (Accept a suitable alternative standard electrode)
- [6]**
17. (a) VO₂⁺ 1
 (b) (i) B and D 1
 (ii)
-
- Allow ecf from (b) (i)
 Solutions can be reversed. 4
 (iii) 298 K / 25 °C temperature
 all solutions 1 mol dm⁻³
 Both needed for 1 mark. Ignore any reference to pressure 1
- [7]**
18. (a) Hydrogen ion concentration: 1.00 mol dm⁻³ (1)
 Hydrogen gas pressure: 100 kPa (1) 2
 (b) Explanation of change: Equilibrium displaced to left (1)
 to reduce constraint (1)
 Change in electrode potential: Becomes negative or decreases (1)
 allow more negative 3
 (c) (i) 0.43V (1)
 (ii) Half-equation: 2Br⁻ → Br₂ + 2e⁻ (1)
 Overall equation: 2BrO₃⁻ + 10Br⁻ + 12H⁺ → 6Br₂ + 6H₂O (2) 4
 or BeO₃⁻ + 5Br⁻ + 6H⁺ → 3Br₂ + 3H₂O
 species (1)
 balanced (1)
- [9]**
19. (a) Platinum electrode (1)
 Hydrogen gas (1)
 H⁺(aq) or HCl or H₂SO₄ (1)
 298 k (1)
 100 kPa or 1 atm or 1 bar (1)
 1 M [H⁺] (1) 6
 (b) (i) 1.61 V (1)
 (ii) Positive (1) 2
 (c) (i) 1.36 V (1)
 Left to right (1)

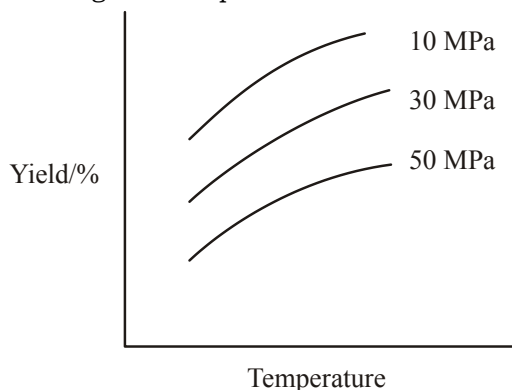
- Allow on diagram 2
- (d) $\text{Ce}^{4+} + \text{e}^- \rightarrow \text{Ce}^{3+}$ (1)
 $\text{V}^{3+} + \text{H}_2\text{O} \rightarrow \text{VO}^{2+} + 2\text{H}^+ + \text{e}^-$
or reversed
 or $\text{V}^{3+} + 2\text{H}_2\text{O} \rightarrow \text{VO}_2^+ + 4\text{H}^+ + 2\text{e}^-$ (1)
or reversed
 $\text{Ce}^{4+} + \text{V}^{3+} + \text{H}_2\text{O} \rightarrow \text{VO}^{2+} + 2\text{H}^+ + \text{Ce}^{3+}$
 or $2\text{Ce}^{4+} + \text{V}^{3+} + 2\text{H}_2\text{O} \rightarrow \text{VO}_2^+ + 4\text{H}^+ + 2\text{Ce}^{3+}$ (1) 3
 (e) VO^{2+} (1) VO_2^+ (1) 2
- [15]**
20. (a) most powerful reducing agent: Zn; 1
 (b) (i) reducing species: Fe^{2+} 1
 (ii) oxidising species: Cl_2 ; 1
 (c) (i) standard electrode potential 1.25 V; 1
 (ii) equation: $\text{Ti}^{3+} + 2\text{Fe}^{2+} \rightarrow 2\text{Fe}^{3+} + \text{Ti}^+$ balanced; 1
 correct direction; 1
- [6]**
21. (i) $6\text{HNO}_3 + \text{S} \rightarrow \text{H}_2\text{SO}_4 + 6\text{NO}_2 + 2\text{H}_2\text{O}$ /
 $4\text{HNO}_3 + \text{S} \rightarrow \text{H}_2\text{SO}_4 + 4\text{NO}_2 + \text{H}_2$ /
 $2\text{HNO}_3 + \text{S} \rightarrow \text{H}_2\text{SO}_4 + \text{NO}_2 + \frac{1}{2}\text{N}_2$ ✓ 1
 (ii) from (+)5 to (+)4 ✓ 1
- [2]**
22. $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$ (1)
 N from +4 to +5
 O from 0 to -2 (1) Could be below equation 2
- [2]**
23. (a) A = Platinum(electrode)
 B = $\text{H}^+(\text{aq})$ / $\text{HCl}(\text{aq})$ / other suitable acid
 C = Voltmeter / galvanometer
 D = $\text{Cl}_2(\text{g})$
 State symbols needed for B and D
 All correct = 2, 3 correct = 1 2
 (b) (i) Arrow marked on or close to wire via voltmeter pointing from
 hydrogen half cell to chlorine half cell 1
 Electrons flow to half cell with more +ve standard electrode
 potential 1
 (ii) Pressure = 1 Atm / 100 kPa
 Temp = 298 K / 25°C
 Concentration = 1 mol dm⁻³
 All 3 correct = 2 marks 2 correct = 1 mark 2
 (c) The standard electrode potential for $\text{ClO}_3^- / \frac{1}{2}\text{Cl}_2$ is more positive
 than that of $\frac{1}{2}\text{Cl}_2 / \text{Cl}^-$ 1
 ClO_3^- has a greater tendency to gain electrons than $\text{Cl}_2 / \text{ClO}_3^-$ is a
 better oxidising agent than Cl_2 1
 Alternative:
 Because $E^\ominus$ is positive, the reaction will go from left to right
 therefore ClO_3^- is reduced so it must be a better oxidising
 agent than chlorine.

- 24.** (a) Emf of a cell / voltage / potential difference / cell potential 1
 Comprising half cell combined with standard hydrogen electrode 1
 Conc = 1 mol.dm⁻³; Pressure (of H₂) = 1 atm; Temp = 298K 1
 (all of above = 1 mark)
 (b) +0.16 V (unit required) 1
- [8]**
- 25.** (a) $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{Fe}^{2+} \rightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$ 1
 $\text{Cr}_2\text{O}_7^{2-} / \text{Cr}^{3+}$ has more positive electrode potential 1
 Therefore $\text{Cr}_2\text{O}_7^{2-}$ is the stronger oxidising agent which
 oxidises Fe^{2+} to Fe^{3+} (ora) 1
 (b) Emf = (+) 0.56 V 1
- [4]**
- 26.** (a) (i) 1
 Diagram shows: 1
 Voltmeter + salt bridge + complete circuit 1
 Solution labelled Cu^{2+} and electrode labelled Ag 1
- 
- (ii) Direction from Cu(s) to Ag(s) (must be in / close to wire) 1
 (iii) $0.80 - 0.34 = 0.46 \text{ V}$ 1
 (iv) $\text{Cu} + 2\text{Ag}^+ \rightarrow \text{Cu}^{2+} + 2\text{Ag}$ 1
- (b) Standard Electrode Potential for chlorine is more positive than Fe^{3+} therefore it is a better oxidising agent than Fe^{3+} (do not accept $E^\ominus$ is larger or smaller) 1
 Standard Electrode Potential for iodine is less positive than Fe^{3+} therefore it is a poorer oxidising agent than Fe^{3+} 1
 (Accept release of electrons/equilibrium arguments)
- [7]**
- 27.** (a) (i) $2\text{MnO}_4^- + 10\text{Cl}^- + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 5\text{Cl}_2 + 8\text{H}_2\text{O}$ 1
 correct species on both sides of equation equation balanced 1
 (ignore electrons for first mark, penalise for balance)
 (ii) Chlorine $-1 \rightarrow 0$ 1
 Manganese $+7 \rightarrow +2$ 1
 Link to (i) and allow ecf

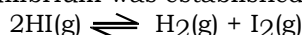
- (iii) Chloride ion oxidised (not chlorine) 1
 Manganate(VII) ion reduced (not manganese) 1
 (b) 0.16 V too small/rate too slow/insufficient activation energy/not standard conditions 1
[7]
- 28.** (i) oxidation: $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$ (1)
 reduction: $\text{V}^{3+} + \text{e}^- \rightarrow \text{V}^{2+}$ (1) 2
 (ii) $E_{\text{cell}} = 0.18 \text{ V}$ (1) 1
[3]
- 29.** (i) system **III** $\times 2$ and reversed + system **IV** (1)
 $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}/$
 $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$ (1) 2
 (ii) advantages:
 only H_2O formed/ non-polluting
 greater efficiency (1)
 disadvantages:
 H_2 difficult to store (1)
 H_2 difficult to manufactured initially /
 limited life cycle of H_2 adsorber/absorber (1) 4
[6]

7.0 EQUILIBRIA

1. (a) The diagram below shows the effect of temperature and pressure on the equilibrium yield of the product in a gaseous equilibrium.

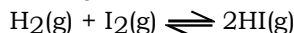


- (i) Use the diagram to deduce whether the forward reaction involves an increase or a decrease in the number of moles of gas. Explain your answer.
 (ii) Use the diagram to deduce whether the forward reaction is exothermic or endothermic. Explain your answer.
- (b) When a 0.218 mol sample of hydrogen iodide was heated in a flask of volume $V \text{ dm}^3$, the following equilibrium was established at 700 K.

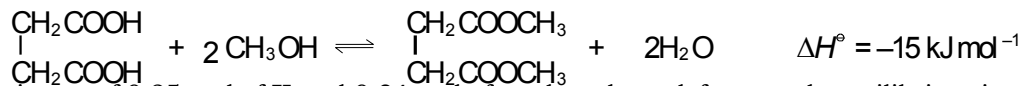


The equilibrium mixture was found to contain 0.023 mol of hydrogen.

- (i) Calculate the number of moles of iodine and the number of moles of hydrogen iodide in the equilibrium mixture.
 (ii) Write an expression for K_C for the equilibrium.
 (iii) State why the volume of the flask need not be known when calculating a value for K_C .
 (iv) Calculate the value of K_C at 700 K.
 (v) Calculate the value of K_C at 700 K for the equilibrium



2. Acid **X** reacts with methanol to form ester **Y** according to the following equation.

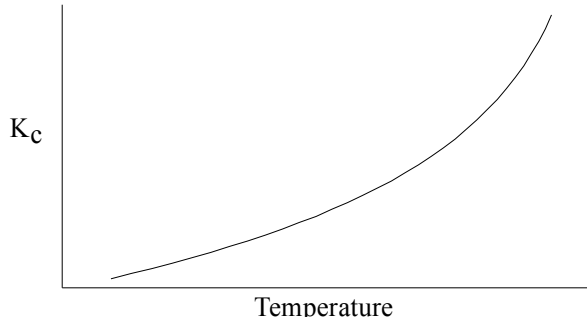


A mixture of 0.25 mol of **X** and 0.34 mol of methanol was left to reach equilibrium in the presence of a small amount of concentrated sulphuric acid. The equilibrium mixture thus formed contained 0.13 mol of **Y** in a total volume of $V \text{ dm}^3$.

- (a) Using **X** to represent the acid and **Y** to represent the ester, write an expression for the equilibrium constant, K_C , for this reaction. (1)
- (b) Calculate the number of moles of **X**, the number of moles of methanol and the number of moles of water in the equilibrium mixture. (3)
- (c) State why the volume V need not be known in calculating the value of K_C for the reaction. (1)
- (d) Calculate the value of K_C for this reaction and deduce its units. (3)
- (e) State the effect, if any, of increasing the temperature on the value of K_C (1)

(Total 9 marks)

3. As a first step in the manufacture of nitric acid it has been suggested that nitrogen monoxide, NO, can be formed from nitrogen and oxygen in a reversible reaction.
- (a) Write an equation for this reaction and deduce an expression for the equilibrium constant, K_C
- (b) The sketch graph below shows how the value of K_C for this reaction changes with temperature.

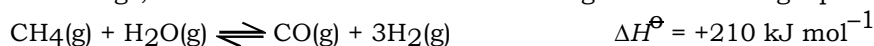


Use this graph to deduce whether the reaction is exothermic or endothermic. Explain your answer.

- (c) The value of K_C for this reaction is 1×10^{-5} at 1500 K. Explain the significance of this value for an industrial chemist interested in manufacturing nitrogen monoxide by the direct combination of the elements.
- (d) When cooled, nitrogen monoxide reacts with oxygen to form gaseous nitrogen dioxide, NO_2 , in a reversible reaction.
- (i) Write an equation for this reaction.
- (ii) State how an increase in pressure would change the position of the equilibrium and the value of the equilibrium constant for this reaction.

(Total 9 marks)

4. The manufacture of methanol can be achieved in two stages. In the first stage, methane and steam react according to the following equation.



Discuss, with reasons, the effects of increasing separately the temperature and the pressure on the yield of the products and on the rate of this reaction.

(6)
(Total 6 marks)

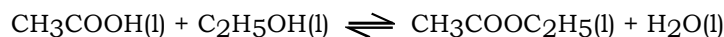
5. The reaction between hydrogen and iodine can be represented by the following equation:



- (a) Write a K_C expression for the decomposition of hydrogen iodide. At a given temperature, the value of K_C for this reaction is 20. What will be the value of K_C for the reaction between hydrogen and iodine at this temperature?
- (b) The pressure of an equilibrium mixture of hydrogen iodide, hydrogen and iodine was increased. State what, if anything, would happen to:
- (i) The rates of both forward and reverse reactions;
- (ii) The position of equilibrium;
- (iii) The value of the equilibrium constant.

(Total 6 marks)

6. (a) A flask containing a mixture of 0.200 mol of ethanoic acid and 0.110 mol of ethanol was maintained at 25 °C until the following equilibrium had been established.



The ethanoic acid present at equilibrium required 72.5 cm^3 of a 1.50 mol dm^{-3} solution of sodium hydroxide for complete reaction.

- Calculate the value of the equilibrium constant, K_c , for this reaction at 25°C .
- The enthalpy change for this reaction is quite small. By reference to the number and type of bonds broken and made, explain how this might have been predicted.

(9)

(Total 9 marks)

7. When ammonia gas is heated, a homogeneous, dynamic equilibrium is established between ammonia and its constituent elements. This decomposition is endothermic.

- (a) Explain the terms *homogeneous*, *dynamic* and *equilibrium*.

Write an equation for this decomposition and derive an expression for the equilibrium constant, K_c

(5)

- (b) State and explain the conditions under which a high equilibrium concentration of hydrogen would be obtained.

(4)

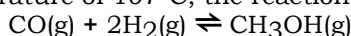
- (c) The decomposition of ammonia might in the future be used as an industrial method for the manufacture of hydrogen.

Explain why an industrial chemist might decide to use conditions different from those you have given in part (b) if large quantities of hydrogen were to be produced by this decomposition. Discuss the effect that using a catalyst would have on the equilibrium yield and on the amount of hydrogen which could be produced in a given time.

(6)

(Total 15 marks)

8. At a temperature of 107°C , the reaction



reaches equilibrium under a pressure of 1.59 MPa with 0.122 mol of carbon monoxide and 0.298 mol of hydrogen present at equilibrium in a vessel of volume 1.04 dm^3 . Use these data to answer the questions that follow.

- (a) Assuming ideal gas behaviour, determine the total number of moles of gas present. Hence calculate the number of moles of methanol in the equilibrium mixture.

(3)

- (b) Calculate the value of the equilibrium constant, K_c , for this reaction and state its units.

(3)

(Total 6 marks)

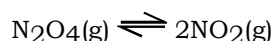
9. (a) The expression for an equilibrium constant, K_c , for a homogeneous equilibrium reaction is given below.

$$K_c = \frac{[A]^2[B]}{[C][D]^3}$$

- Write an equation for the forward reaction.
- Deduce the units of K_c
- State what can be deduced from the fact that the value of K_c is larger when the equilibrium is established at a lower temperature.

(3)

- (b) A 36.8 g sample of N_2O_4 was heated in a closed flask of volume 16.0 dm^3 . An equilibrium was established at a constant temperature according to the following equation.



The equilibrium mixture was found to contain 0.180 mol of N_2O_4

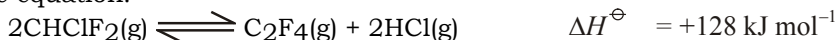
- Calculate the number of moles of N_2O_4 in the 36.8 g sample.
- Calculate the number of moles of NO_2 in the equilibrium mixture.

- (iii) Write an expression for K_C and calculate its value under these conditions.
- (iv) Another 36.8 g sample of N_2O_4 was heated to the same temperature as in the original experiment, but in a larger flask. State the effect, if any, of this change on the position of equilibrium and on the value of K_C compared with the original experiment.

(9)

(Total 12 marks)

10. Tetrafluoroethene, C_2F_4 , is obtained from chlorodifluoromethane, $CHClF_2$, according to the equation:



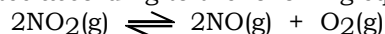
- (a) A 1.0 mol sample of $CHClF_2$ is placed in a container of volume 18.5 dm^3 and heated.
When equilibrium is reached, the mixture contains 0.20 mol of $CHClF_2$
 - (i) Calculate the number of moles of C_2F_4 and the number of moles of HCl present at equilibrium.
 - (ii) Write an expression for K_C for the equilibrium.
 - (iii) Calculate a value for K_C and give its units.
- (b)
 - (i) State how the temperature should be changed at constant pressure to increase the equilibrium yield of C_2F_4
 - (ii) State how the total pressure should be changed at constant temperature to increase the equilibrium yield of C_2F_4

(6)

(2)

(Total 8 marks)

11. Nitrogen dioxide dissociates according to the following equation.



When 21.3 g of nitrogen dioxide were heated to a constant temperature, T , in a flask of volume 11.5 dm^3 , an equilibrium mixture was formed which contained 7.04 g of oxygen.

- (a)
 - (i) Calculate the number of moles of oxygen present in this equilibrium mixture and deduce the number of moles of nitrogen monoxide also present in this equilibrium mixture.
 - (ii) Calculate the number of moles in the original 21.3 g of nitrogen dioxide and hence calculate the number of moles of nitrogen dioxide present in this equilibrium mixture.
- (b) Write an expression for the equilibrium constant, K_C , for this reaction. Calculate the value of this constant at temperature T and give its units.
- (c) The total number of moles of gas in the flask is 0.683. Use the ideal gas equation to determine the temperature T at which the total pressure in the flask is $3.30 \times 10^5 \text{ Pa}$.
(The gas constant $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)
- (d) State the effect on the equilibrium yield of oxygen and on the value of K_C when the same mass of nitrogen dioxide is heated to the same temperature T , but in a different flask of greater volume.

(4)

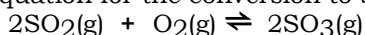
(4)

(3)

(2)

(Total 13 marks)

12. The manufacture of sulphuric acid is carried out on a large scale in most industrialised countries because it is needed by many other industries. In most countries the raw material is sulphur. The process consists of three main stages - the burning of sulphur in air, the conversion of sulphur dioxide to sulphur trioxide and the formation of sulphuric acid. The equation for the conversion to sulphur trioxide is



- (a) For the conversion to sulphur trioxide, the effect of temperature on the percentage yield at equilibrium is shown in **Figure 1**.

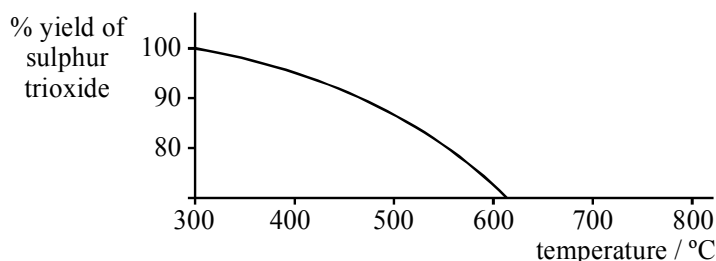
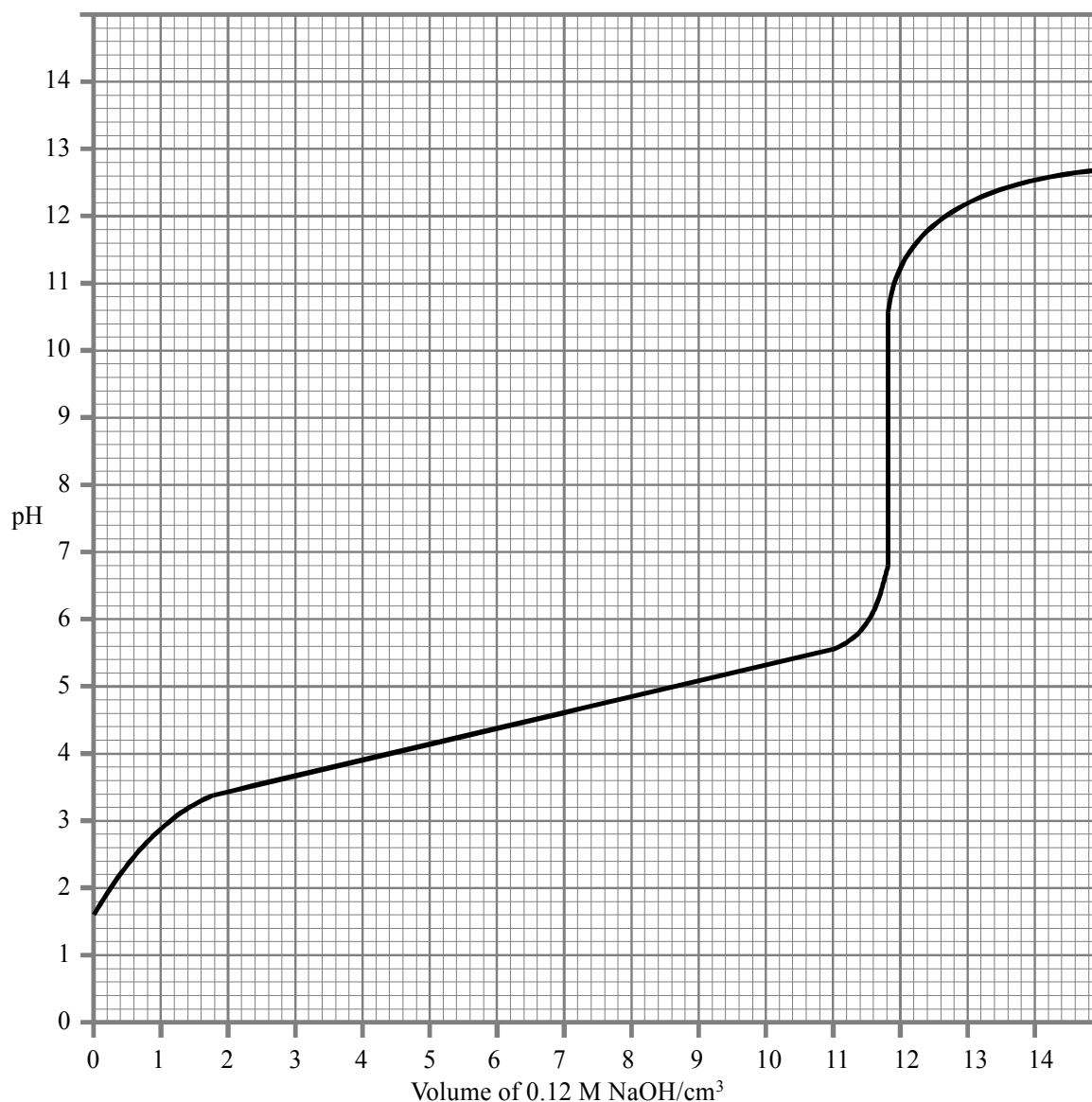


Figure 1

- (i) State whether this reaction is exothermic or endothermic and justify your answer by reference to **Figure 1**. (2)
- (ii) Give the name of the catalyst used in this reaction. State the effect of the catalyst on the percentage yield at equilibrium of sulphur trioxide and on the rates of both forward and reverse reactions. (4)
- (iii) A typical operating temperature is 450 °C. State **one** advantage and **one** disadvantage of operating the process at a temperature 50 °C lower. (2)
- (b) The operating pressure for the conversion of sulphur dioxide to sulphur trioxide is slightly higher than atmospheric pressure.
- (i) State the effect on the rate of the reaction of using a much higher pressure. (1)
- (ii) State the effect of using this much higher pressure on the percentage yield at equilibrium of sulphur trioxide and give a reason for your answer. (2)
- (Total 11 marks)**
13. (a) (i) Define the term *Bronsted-Lowry acid*. (1)
- (ii) What is meant by the term *strong* when describing an acid? (1)
- (iii) Give the value of the ionic product of water, K_w , measured at 298K, and state its units. (2)
- (b) At 298 K, 25.0 cm³ of a solution of a strong acid contained 1.50×10^{-3} mol of hydrogen ions. (4)
- (i) Calculate the hydrogen ion concentration in this solution and hence its pH. (1)
- (ii) Calculate the pH of the solution formed after the addition of 50.0 cm³ of 0.150 M NaOH to the original 25.0 cm³ of acid. (3)
- (c) A solution of a strong acid was found to have a pH of 0.5 (8)
- (i) Calculate the hydrogen ion concentration in this solution. (1)
- (ii) Calculate the volume of water which must be added to 25.0 cm³ of this solution to increase its pH from 0.5 to 0.7 (5)
- (Total 17 marks)**
14. (a) The pH of a 0.15 M solution of a weak acid, HA, is 2.82 at 300 K. (1)
- (i) Write an expression for the acid dissociation constant, K_a , of HA, and determine the value of K_a for this acid at 300 K, stating its units. (2)
- (ii) The dissociation of HA into its ions in aqueous solution is an endothermic process. How would its pH change if the temperature were increased? Explain your answer. (5)
- (b) Solution **A** contains n moles of a different weak acid, HX. The addition of some sodium hydroxide to **A** neutralises one third of the HX present to produce Solution **B**. (8)
- (i) In terms of the amount, n , how many moles of HX are present in Solution **B**? (1)
- (ii) Determine the ratio $\frac{[HX]}{[X^-]}$ in Solution **B**. (1)
- (iii) Solution **B** has a hydrogen ion concentration of 4.2×10^{-4} mol dm⁻³. Use this (6)

information and your answer to part (b)(ii) to determine the value of the acid dissociation constant of HX.

- (5)
- (c) Why is methyl orange **not** suitable as an indicator for the titration of HX with sodium hydroxide? (2)
- (d) Solution **B** can act as a buffer. Explain what this means and write an equation that shows how Solution **B** acts as a buffer if a little hydrochloric acid is added. (3)
- (Total 18 marks)**
15. (a) Explain the terms *acid* and *conjugate base* according to the Brønsted-Lowry theory. (2)
- (b) For each of the following reactions, give the formula of the acid and of its conjugate base.
- (i) $\text{NH}_3 + \text{HBr} \rightarrow \text{NH}_4^+ + \text{Br}^-$ (1)
- (ii) $\text{H}_2\text{SO}_4 + \text{HNO}_3 \rightarrow \text{HSO}_4^- + \text{H}_2\text{NO}_3^+$ (1)
- (c) (i) Write an equation to represent the dissociation of water. (1)
- (ii) Give the expression for the equilibrium constant, K_c , for the reaction in (c)(i) and use this to derive the expression for the ionic product of water, K_w . (3)
- (iii) The ionic product of water is $2.92 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at 313K. Calculate the pH of water at this temperature. (4)
- (iv) Given that the pH of water is 7.00 at 298 K, state whether the dissociation of water is endothermic or exothermic. Give a reason for your answer. (2)
- (Total 14 marks)**
16. (a) Give the Brønsted-Lowry definition of a base. State the essential feature of an acid-base reaction in aqueous solution, writing an ionic equation to illustrate your answer. (3)
- (b) Explain what is meant by the term *weak* when applied to acids and bases. (1)
- (c) In aqueous solution, the weak acid propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}(\text{aq})$, produces propanoate ions $\text{CH}_3\text{CH}_2\text{COO}^-(\text{aq})$. Write an expression for the acid dissociation constant, K_a , of propanoic acid and state its units. (2)
- (d) (i) Explain what is meant by the term *buffer solution*. (4)
- (ii) Identify two components that could be used to make a buffer solution.
- (iii) Give an example of the use of a buffer solution.
- (Total 10 marks)**
17. (a) The graph below shows how the pH changes when 0.12 M NaOH is added to 25.0 cm^3 of a solution of a weak monoprotic acid, HA.



- (i) Use the graph to calculate the initial concentration of the acid HA.
- (ii) Write an expression for the dissociation constant, K_a , of the weak acid HA.
- (iii) Determine the volume of sodium hydroxide added when $[HA] = [A^-]$ and use the graph to determine the pH at this point.
- (iv) Use your answers to part (a)(ii) and part (a)(iii) to determine the value of K_a for the acid HA.

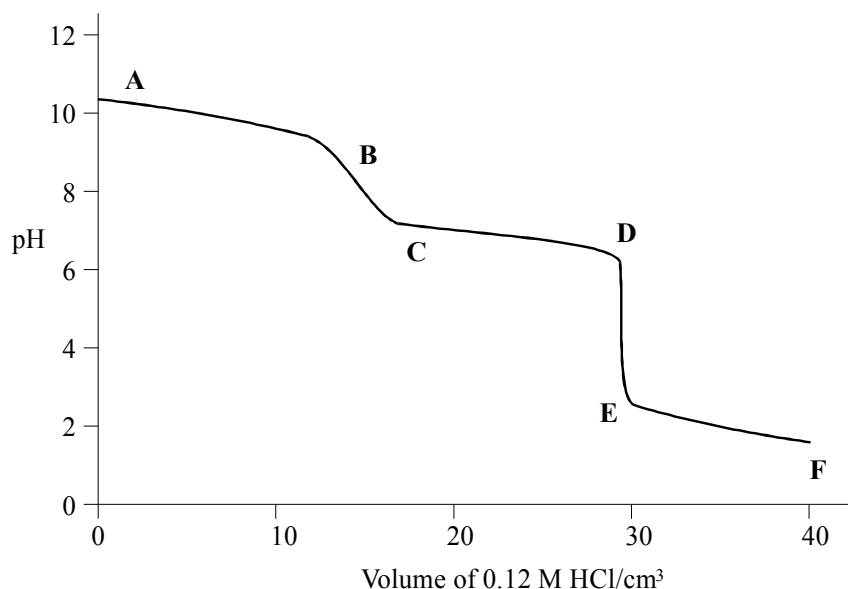
(9)

- (b) A buffer solution is formed, when approximately half of the original amount of the acid HA(aq) has been neutralised by the base NaOH(aq). Explain how this buffer solution is able to resist change in pH when
 - (i) a small amount of NaOH(aq) is added,
 - (ii) a small amount of HCl(aq) is added.

(4)

(Total 13 marks)

18. The graph below shows how the pH changes as 0.12 M HCl(aq) is added to 25.0 cm³ of a solution of sodium carbonate. There are two end-points. The second end-point is at 30.0 cm³.



- (a) Write equations for the reactions which occur in the solution between point A and point B on the graph and between point C and point D on the graph. (2)
- (b) Estimate the minimum volume of hydrochloric acid needed in this experiment for carbon dioxide to be produced from a well-stirred solution of sodium carbonate. (1)
- (c) Name an indicator which can be used to determine the end-point occurring between points D and E. Explain why this indicator does not change colour between points B and C. (2)
- (d) Use the end-point occurring between points D and E to calculate the concentration of sodium carbonate in the given solution. (3)
- (e) If the original solution had contained, in addition to sodium carbonate, an equal molar concentration of sodium hydrogen carbonate, at what volumes of hydrochloric acid would the two end-points have been detected? (2)

(Total 10 marks)

19. (a) Define the term *Brønsted-Lowry acid*. (1)
- (b) Write an equation for the reaction between gaseous hydrogen chloride and water. State the role of water in this reaction, using the Brønsted-Lowry definition. (2)
- (c) Write an equation for the reaction between gaseous ammonia and water. State the role of water in this reaction, using the Brønsted-Lowry definition. (2)
- (d) The ion H_2NO_3^+ is formed in the first stage of a reaction between concentrated nitric acid and an excess of concentrated sulphuric acid. In this first stage the two acids react in a 1:1 molar ratio. In the second stage, the H_2NO_3^+ ion decomposes to form the nitronium ion, NO_2^+ . Write equations for these two reactions and state the role of nitric acid in the first reaction. (3)
- (e) (i) Explain the term *weak acid*.
 (ii) Write an expression for the acid dissociation constant, K_a , of HA, a weak monoprotic acid.
 (iii) The value of the acid dissociation constant for the monoprotic acid HX is 144 mol dm^{-3} . What does this suggest about the concentration of undissociated HX in dilute aqueous solution?

- (iv) State whether HX should be classified as a strong acid or a weak acid. Justify your answer.

(5)

(Total 13 marks)

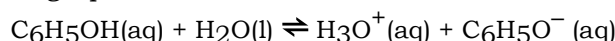
20. For acid HA, $K_a = 2.00 \times 10^{-4} \text{ mol dm}^{-3}$.

- (a) Write an equation for the reaction of HA with NaOH. (1)
- (b) A solution was formed by adding 15 cm^3 of 0.34 M NaOH to 25 cm^3 of 0.45 M HA.
- (i) Calculate the number of moles of $\text{A}^-(\text{aq})$ and $\text{HA}(\text{aq})$ in this solution. (You should neglect the small number of moles of $\text{A}^-(\text{aq})$ formed by ionisation of the remaining $\text{HA}(\text{aq})$.)
- (ii) Calculate the concentration of $\text{A}^-(\text{aq})$ and $\text{HA}(\text{aq})$ in this solution.
- (iii) Using an expression for K_a , calculate the pH of this solution.

(9)

(Total 10 marks)

21. (a) Phenol is a *weak acid*. The dissociation of phenol in aqueous solution is represented by the following equation:



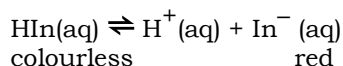
What is meant by the term *weak acid*?

(1)

- (b) (i) Write an expression for the acid dissociation constant, K_a , for phenol. (1)
- (ii) Write an expression linking K_a with $\text{p}K_a$. (1)
- (iii) The value of the acid dissociation constant, K_a , for phenol is $1 \times 10^{-10} \text{ mol dm}^{-3}$. Calculate the $\text{p}K_a$ value of phenol. (1)

- (iv) Ethanoic acid is a stronger acid than phenol. State whether the $\text{p}K_a$ value for ethanoic acid will be greater or smaller than that of phenol. (1)

- (c) The indicator phenolphthalein is a weak acid which can be represented by the formula HIn. It dissociates in solution and has a $\text{p}K_a$ value of 9.3.



- (i) Suggest and explain, with reference to the $\text{p}K_a$ value, the pH range of phenolphthalein. (2)
- (ii) State the colour change that would be observed at the end point in an acid-base titration using phenolphthalein if sodium hydroxide solution were being added from the burette. Explain, in terms of the species present, why this colour is formed. (2)
- (iii) State why phenolphthalein is unsuitable for a titration between a strong acid and a weak base. (1)

(1)

(Total 10 marks)

22. (a) (i) Define the term *pH*.
- (ii) Write an expression for the dissociation constant K_a for the weak acid HX.
- (iii) For HX, $K_a = 4.25 \times 10^{-5} \text{ mol dm}^{-3}$. Calculate the pH of a 0.45 M solution of this acid. (6)
- (b) In a 0.25 M solution, a different acid HY is 95% dissociated.
- (i) Calculate the pH of this solution.
- (ii) Calculate the value of K_a for the acid HY. (6)

(6)

(Total 12 marks)

23. (a) Define pK_a (1)
- (b) Calculate the pH of a 0.52M aqueous solution of the weak monoprotic (monobasic) acid HX ($pK_a = 3.72$). (4)
- (c) Write an expression for the acid dissociation constant K_a for HX. Use this to show that the pH of any sample of HX is 3.72 when half of the acid has been neutralised by a solution of sodium hydroxide. (3)
- (d) Explain why indicators cannot be used to determine the end-point of a titration between a weak acid and a weak base. (2)

(Total 10 marks)

24. The value of the acid dissociation constant, K_a , for ethanoic acid is $1.74 \times 10^{-5} \text{ mol dm}^{-3}$ at 298 K.

- (a) (i) Write an expression for K_a for ethanoic acid.
- (ii) Calculate the pH at 298 K of a $0.220 \text{ mol dm}^{-3}$ solution of ethanoic acid. (5)
- (b) A sample of the $0.220 \text{ mol dm}^{-3}$ solution of ethanoic acid was titrated against sodium hydroxide solution.
- (i) Calculate the volume of a $0.150 \text{ mol dm}^{-3}$ solution of sodium hydroxide required to neutralise 25.0 cm^3 of the ethanoic acid solution.
- (ii) From the list below, select the best indicator for this titration and explain your choice.

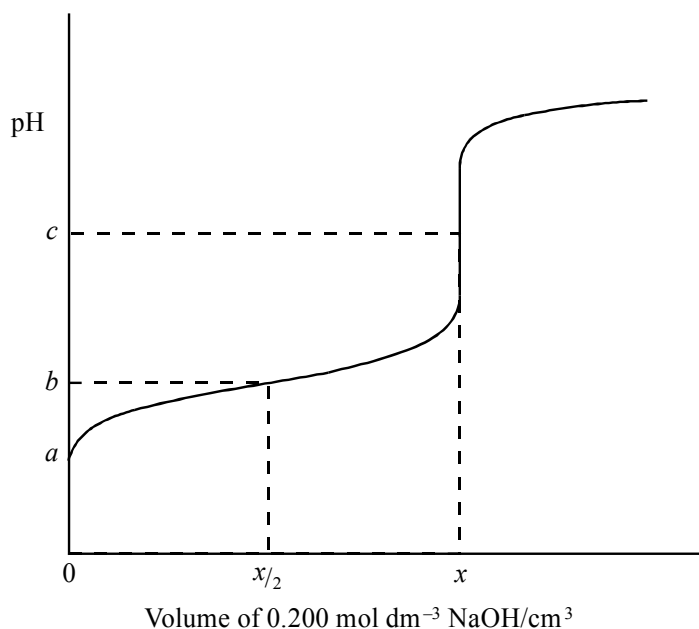
Name of indicator	pH range
bromophenol blue	3.0 – 4.6
methyl red	4.2 – 6.3
bromothymol blue	6.0 – 7.6
thymol blue	8.0 – 9.6

(5)

- (c) A buffer solution is formed when 2.00 g of sodium hydroxide are added to 1.00 dm^3 of a $0.220 \text{ mol dm}^{-3}$ solution of ethanoic acid. Calculate the pH at 298 K of this buffer solution.

(6)**(Total 16 marks)**

25. The sketch below shows the change in pH when a $0.200 \text{ mol dm}^{-3}$ solution of sodium hydroxide is added from a burette to 25.0 cm^3 of a $0.150 \text{ mol dm}^{-3}$ solution of the weak acid HA at 25°C .



- (a) The volume of sodium hydroxide solution added at the equivalence point is $x \text{ cm}^3$. Calculate the value of x . (2)
- (b) (i) Define the term pH.
(ii) The pH at the equivalence point is c . Suggest a value for c .
(iii) Identify a suitable indicator for detecting the equivalence point of the titration. (3)
- (c) The value of K_c for the weak acid HA at 25°C is $2.75 \times 10^{-5} \text{ mol dm}^{-3}$.
(i) Explain the term *weak* as applied to the acid HA.
(ii) Write an expression for K_a for the acid HA.
(iii) Calculate the pH of the $0.150 \text{ mol dm}^{-3}$ solution of acid HA before any sodium hydroxide is added, i.e. the pH at point a . (5)
- (d) Calculate the pH of the solution formed when $\frac{x}{2} \text{ cm}^3$ of the $0.200 \text{ mol dm}^{-3}$ solution of sodium hydroxide are added to 25.0 cm^3 of the $0.150 \text{ mol dm}^{-3}$ solution of HA, i.e. the pH at point b . (3)
- (Total 13 marks)**
26. (a) At 50°C , the ionic product of water, K_w , has the value $5.48 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$.
(i) Define the term K_w
(ii) Define the term pH
(iii) Calculate the pH of pure water at 50°C . Explain why pure water at 50°C is still neutral even though its pH is not 7. (5)
- (b) At 25°C , K_w has the value $1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. Calculate the pH at 25°C of
(i) a $0.150 \text{ mol dm}^{-3}$ solution of sodium hydroxide,
(ii) the solution formed when 35.0 cm^3 of this solution of sodium hydroxide is mixed with 40.0 cm^3 of a $0.120 \text{ mol dm}^{-3}$ solution of hydrochloric acid. (8)
- (c) In a $0.150 \text{ mol dm}^{-3}$ solution of a weak acid HX at 25°C , 1.80% of the acid molecules are dissociated into ions.
(i) Write an expression for K_a for the acid HX.
(ii) Calculate the value of K_a for the acid HX at this temperature and state its units. (5)

(Total 18 marks)

27. (a) By reference to the forces between molecules, explain why ammonia is very soluble in water. (2)
- (b) Aqueous solutions of ammonia have a pH greater than 7.
 (i) Write an equation for the reaction of ammonia with water.
 (ii) Explain why the pH of a solution containing 1.0 mol dm^{-3} of ammonia is less than 14 at 298 K. (3)
- (c) An ammonium ion in aqueous solution can behave as a Brønsted–Lowry acid. State what is meant by the term *Brønsted–Lowry acid*. (1)
- (d) State what is meant by the term *buffer solution*. Identify a reagent which could be added to a solution of ammonia in order to form a buffer solution. (3)
- (e) An acidic buffer solution is obtained when sodium ethanoate is dissolved in aqueous ethanoic acid.
 (i) Calculate the pH of the buffer solution formed at 298 K when 0.125 mol of sodium ethanoate is dissolved in 250 cm^3 of a 1.00 mol dm^{-3} solution of ethanoic acid.
 The acid dissociation constant, K_a , for ethanoic acid is $1.70 \times 10^{-5} \text{ mol dm}^{-3}$ at 298 K.
 (ii) Write an ionic equation for the reaction which occurs when a small volume of dilute hydrochloric acid is added to this buffer solution. (5)

(Total 14 marks)

28. The value of the acid dissociation constant, K_a , for the weak acid HA, at 298 K, is $1.45 \times 10^{-4} \text{ mol dm}^{-3}$.
 (a) Write an expression for the term K_a for the weak acid HA. (1)
- (b) Calculate the pH of a $0.250 \text{ mol dm}^{-3}$ solution of HA at 298 K. (4)
- (c) A mixture of the acid HA and the sodium salt of this acid, NaA, can be used to prepare a buffer solution.
 (i) State and explain the effect on the pH of this buffer solution when a small amount of hydrochloric acid is added.
 (ii) The concentration of HA in a buffer solution is $0.250 \text{ mol dm}^{-3}$. Calculate the concentration of A^- in this buffer solution when the pH is 3.59 (6)

(Total 11 marks)

29. (a) The pH of a $0.120 \text{ mol dm}^{-3}$ solution of the weak monoprotic acid, HX, is 2.56 at 298 K.
 (i) Write an expression for the term *pH*.
 (ii) Write an expression for the dissociation constant, K_a , for the weak acid HX and calculate its value at 298 K. (5)
- (b) (i) Write an expression for the ionic product of water, K_w , and give its value at 298 K.
 (ii) Hence, calculate the pH of a $0.0450 \text{ mol dm}^{-3}$ solution of sodium hydroxide at 298 K. (4)
- (c) A titration curve is plotted showing the change in pH as a $0.0450 \text{ mol dm}^{-3}$ solution of sodium hydroxide is added to 25.0 cm^3 of a solution of ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$. The titration curve obtained has two equivalence points (end points).
 (i) Write an equation for the reaction which is completed at the **first** equivalence

point.

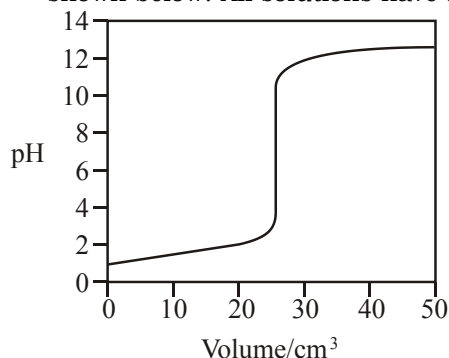
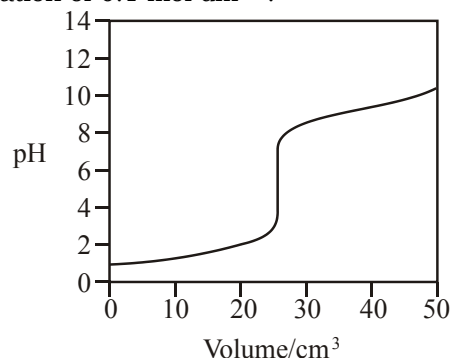
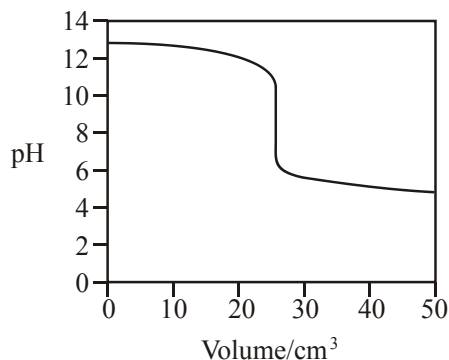
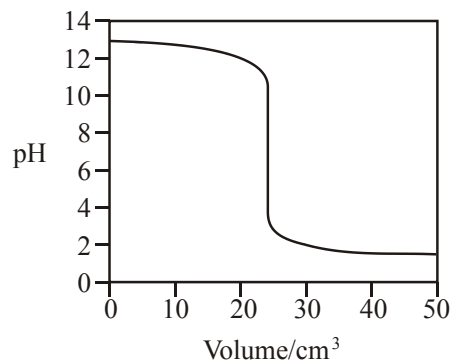
- (ii) When the **second** equivalence point is reached, a total of 41.6 cm^3 of $0.0450 \text{ mol dm}^{-3}$ sodium hydroxide has been added.
Calculate the concentration of the ethanedioic acid solution.

(4)

(Total 13 marks)

30. (a) Titration curves labelled **A**, **B**, **C** and **D** for combinations of different acids and bases are

shown below. All solutions have a concentration of 0.1 mol dm^{-3} .

**A****B****C****D**

- (i) Select from **A**, **B**, **C** and **D** the curve produced by the addition of
Ammonia to 25 cm^3 of hydrochloric acid
Ethanoic acid to 25 cm^3 of sodium hydroxide
Sodium hydroxide to 25 cm^3 of hydrochloric acid
(ii) A table of acid-base indicators and the pH ranges over which they change colour is shown below.

Indicator	pH range
Thymol blue	1.2 – 2.8
Bromophenol blue	3.0 – 4.6
Methyl red	4.2 – 6.3
Cresolphthalein	8.2 – 9.8
Thymolphthalein	9.3 – 10.5

Select from the table an indicator which could be used in the titration which produces curve **A** but not in the titration which produces curve **B**.

(4)

- (b) (i) Write an expression for the term *pH*.
(ii) A solution of potassium hydroxide has a pH of 11.90 at 25°C . Calculate the concentration of potassium hydroxide in the solution.

- (c) The acid dissociation constant, K_a , for propanoic acid has the value of $1.35 \times 10^{-5} \text{ mol dm}^{-3}$ at 25°C . (4)

$$K_a = \frac{[\text{H}^+][\text{CH}_3\text{CH}_2\text{COO}^-]}{[\text{CH}_3\text{CH}_2\text{COOH}]}$$

In each of the calculations below, give your answer to 2 decimal places.

- Calculate the pH of a $0.117 \text{ mol dm}^{-3}$ aqueous solution of propanoic acid.
- Calculate the pH of a mixture formed by adding 25 cm^3 of a $0.117 \text{ mol dm}^{-3}$ aqueous solution of sodium propanoate to 25 cm^3 of a $0.117 \text{ mol dm}^{-3}$ aqueous solution of propanoic acid.

(5)

(Total 13 marks)

31. (a) A sample of hydrochloric acid has a pH of 2.34
Write an expression for pH and calculate the concentration of this acid. (2)
- (b) A $0.150 \text{ mol dm}^{-3}$ solution of a weak acid, HX, also has a pH of 2.34
 - Write an expression for the acid dissociation constant, K_a , for the acid HX.
 - Calculate the value of K_a for this acid and state its units.
 - Calculate the value of $\text{p}K_a$ for the acid HX. Give your answer to two decimal places.

(5)

- (c) A 30.0 cm^3 sample of a $0.480 \text{ mol dm}^{-3}$ solution of potassium hydroxide was partially neutralised by the addition of 18.0 cm^3 of a $0.350 \text{ mol dm}^{-3}$ solution of sulphuric acid.
 - Calculate the initial number of moles of potassium hydroxide.
 - Calculate the number of moles of sulphuric acid added.
 - Calculate the number of moles of potassium hydroxide remaining in excess in the solution formed.
 - Calculate the concentration of hydroxide ions in the solution formed.
 - Hence calculate the pH of the solution formed. Give your answer to two decimal places.

(6)

(Total 13 marks)

32. In this question, give all pH values to 2 decimal places.
- (a)
 - Write expressions for the ionic product of water, K_w , and for pH.
 - At 318 K , the value of K_w is $4.02 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ and hence the pH of pure water is 6.70
State why pure water is not acidic at 318 K .
 - Calculate the number of moles of sodium hydroxide in 2.00 cm^3 of $0.500 \text{ mol dm}^{-3}$ aqueous sodium hydroxide.
 - Use the value of K_w given above and your answer to part (a)(iii) to calculate the pH of the solution formed when 2.00 cm^3 of $0.500 \text{ mol dm}^{-3}$ aqueous sodium hydroxide are added to 998 cm^3 of pure water at 318 K . (6)
- (b) At 298 K , the acid dissociation constant, K_a , for propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$, has the value $1.35 \times 10^{-5} \text{ mol dm}^{-3}$.
 - Write an expression for K_a for propanoic acid.
 - Calculate the pH of $0.125 \text{ mol dm}^{-3}$ aqueous propanoic acid at 298 K . (4)
- (c) Sodium hydroxide reacts with propanoic acid as shown in the following equation.

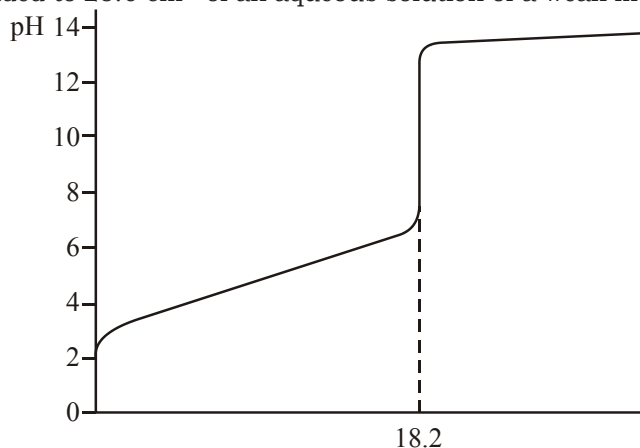
$$\text{NaOH} + \text{CH}_3\text{CH}_2\text{COOH} \rightarrow \text{CH}_3\text{CH}_2\text{COONa} + \text{H}_2\text{O}$$
A buffer solution is formed when sodium hydroxide is added to an excess of aqueous propanoic acid.

- (i) Calculate the number of moles of propanoic acid in 50.0 cm^3 of $0.125 \text{ mol dm}^{-3}$ aqueous propanoic acid.
- (ii) Use your answers to part (a)(iii) and part (c)(i) to calculate the number of moles of propanoic acid in the buffer solution formed when 2.00 cm^3 of $0.500 \text{ mol dm}^{-3}$ aqueous sodium hydroxide are added to 50.0 cm^3 of $0.125 \text{ mol dm}^{-3}$ aqueous propanoic acid.
- (iii) Hence calculate the pH of this buffer solution at 298 K.

(6)

(Total 16 marks)

33. The pH curve shown below was obtained when a $0.150 \text{ mol dm}^{-3}$ solution of sodium hydroxide was added to 25.0 cm^3 of an aqueous solution of a weak monoprotic acid, HA.



- (a) Use the information given to calculate the concentration of the acid. (2)
- (b)
 - (i) Write an expression for the acid dissociation constant, K_a , for HA.
 - (ii) Write an expression for $\text{p}K_a$
 - (iii) Using your answers to parts (b)(i) and (b)(ii), show that when sufficient sodium hydroxide has been added to neutralise half of the acid, $\text{pH of the solution} = \text{p}K_a$ for the acid HA (4)
- (c) Explain why dilution with a small volume of water does not affect the pH of a buffer solution. (2)
- (d)
 - (i) Calculate the change in pH when $0.250 \text{ mol dm}^{-3}$ hydrochloric acid is diluted with water to produce $0.150 \text{ mol dm}^{-3}$ hydrochloric acid.
 - (ii) Calculate the volume of water which must be added to 30.0 cm^3 of $0.250 \text{ mol dm}^{-3}$ hydrochloric acid in order to reduce its concentration to $0.150 \text{ mol dm}^{-3}$. (4)

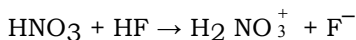
(Total 12 marks)

34. (a) Give the Brønsted–Lowry definition of an acid. (1)
- (b)
 - (i) Explain the term *weak* when applied to an acid or a base.
 - (ii) Give an example of a weak base and write an equation involving this weak base to illustrate the explanation you gave in part (i) above. (3)
- (c) In aqueous solution, the weak acid methanoic acid, HCOOH , produces aqueous methanoate ions, $\text{HCOO}^-(\text{aq})$.
- (i) Write an equation, including state symbols, for the formation of methanoate ions and H_3O^+ ions in an aqueous solution of methanoic acid.

- (ii) Identify one substance that acts as a Brønsted–Lowry base in the forward direction, and another in the reverse direction, of the equation you have written in part (i) above.
- (iii) Write an expression for the acid dissociation constant, K_a , of methanoic acid. (4)
- (d) (i) A buffer solution can be defined as a solution which resists change in pH under three different circumstances. Identify these circumstances.
- (ii) What would you add to methanoic acid in order to make a buffer solution?
- (iii) Apply your knowledge of equilibrium behaviour to the equation you wrote in your answer to part (c)(i) to suggest how this buffer solution is able to resist an increase in pH. (7)
- (Total 15 marks)**
35. (a) Write an equation, including state symbols, for the reaction of gaseous hydrogen chloride with water. (1)
- (b) Calculate the pH of a 1.26M solution of HCl. (2)
- (c) Suggest a value for the pH of a 1.26M solution of sodium chloride. Explain your answer. (2)
- (d) Barium hydroxide, $\text{Ba}(\text{OH})_2$, dissociates completely in water according to the equation
- $$\text{Ba}(\text{OH})_2(\text{s}) \rightarrow \text{Ba}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq})$$
- 45.0cm³ of 1.37M barium hydroxide were added to 95.0cm³ of 1.26M hydrochloric acid.
- (i) Calculate the number of moles of H^{+} ions in 95.0 cm³ of 1.26M hydrochloric acid.
- (ii) Calculate the number of moles of OH^{-} ions in 45.0 cm³ of 1.37 M barium hydroxide.
- (iii) Calculate the pH at 25 °C of the solution formed after mixing the acid and the base. (8)
- (Total 13 marks)**
36. HA is a weak monoprotic acid which can be used to make buffer solutions. It dissociates in water according to the equation
- $$\text{HA}(\text{aq}) \rightleftharpoons \text{H}^{+}(\text{aq}) + \text{A}^{-}(\text{aq})$$
- (a) Explain what is meant by the term *weak* when applied to an acid. (1)
- (b) A solution containing equal concentrations of undissociated HA and the anion A^{-} acts as a buffer solution. This solution is able to resist changes in pH. Use the equation above and your knowledge of equilibrium behaviour to suggest how the buffer solution is able to resist:
- (i) A decrease in pH when a small amount of strong acid is added;
- (ii) An increase in pH when a small amount of strong base is added. (4)
- (c) (i) Write an expression for the acid dissociation constant, K_a , of the acid HA
- (ii) Rearrange your expression for K_a to give an equation for the hydrogen ion concentration in the acid HA. Use this equation to suggest how the buffer solution is able to resist changes in pH on dilution. (4)
- (Total 9 marks)**
37. (a) An acid HA has $\text{p}K_a = 4.20$
- (i) Define the term $\text{p}K_a$
- (ii) Calculate the value of the dissociation constant, K_a , for the acid HA and state its units.
- (iii) Calculate the pH of a 0.830 M solution of the acid HA. (7)

- (b) A different acid, HX, has $K_a = 5.25 \times 10^{-5} \text{ mol dm}^{-3}$. A solution was formed by mixing 10.5 cm^3 of 0.800 M NaOH with 25.0 cm^3 of 0.920 M HX .
- Calculate the number of moles of X^- ions present in the solution formed.
(Ignore any X^- ions formed by dissociation of the excess acid HX)
 - Calculate the number of moles of HX which remain unreacted.
 - Calculate the concentrations of both X^- and HX and use these to determine the pH of the solution formed.
- (9)
- (c) State qualitatively how the pH of the solution formed in part (b) changes when a small volume of dilute hydrochloric acid is added. Use appropriate equations to explain your answer.
- (3)
- (Total 19 marks)
38. (a) State what is meant by the term *monoprotic acid* and give one example
- (2)
- (b) (i) Define pH.
(ii) What is the hydrogen ion concentration in a solution which has $\text{pH} = -0.20$?
- (2)
- (c) Calculate the pH of the solution formed when 35 cm^3 of 0.12 M NaOH are added to 25 cm^3 of 0.15 M HCl at 25°C .
- (7)
- (Total 11 marks)
39. All solutions in parts (a) to (d) below are maintained at 25°C .
- (a) Write equations to show the reaction of HCl(g) and KOH(s) with water.
- (2)
- (b) Write an expression to define the ionic product, K_w , of water.
- (1)
- (c) Calculate the pH of a 0.0160 M KOH solution and estimate the pH of a 0.100 M KCl solution. Give your reasoning.
(At 25°C , the value of K_w is $1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$)
- (7)
- (d) A mass, m , of solid KOH is added to 755 cm^3 of 0.0120 M HCl . The pH after this addition is 11.60 , measured at 25°C . The volume of the resulting solution is still 755 cm^3 .
- Calculate the number of moles of OH^- ions needed to neutralise exactly the H^+ ions present in the 755 cm^3 of 0.0120 M HCl .
 - Calculate the number of moles of OH^- ions in excess when the pH is 11.60
 - Use these results to calculate the total number of moles of KOH added.
 - Hence deduce the value of m .
- (8)
- (Total 18 marks)
40. (a) The hydrogen halides all react with water to form acids. Hydrogen fluoride forms a weak acid while the others all form strong acids. Write equations to show the reactions that occur when hydrogen fluoride and hydrogen chloride are dissolved in water.
- (2)
- (b) (i) Define the term pH
- (1)
- (ii) Calculate the pH of an aqueous solution of hydrochloric acid containing $0.050 \text{ mol dm}^{-3}$.
- (1)
- (c) (i) Write an expression for the dissociation constant, K_a , for hydrofluoric acid.
- (1)
- (ii) Calculate the pH of an aqueous solution of hydrofluoric acid of concentration $0.050 \text{ mol dm}^{-3}$ at 298 K , given that $K_a = 5.6 \times 10^{-4} \text{ mol dm}^{-3}$ at 298 K .

- (d) When hydrogen fluoride is dissolved in pure nitric acid, a reaction takes place that can be represented by the equation:

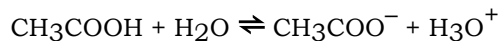


State, with a reason, which reactant acts as a Brønsted-Lowry acid in this reaction and give the formula of its conjugate base.

(3)

(Total 11 marks)

41. (a) When dissolved in water, ethanoic acid acts as a weak Brønsted-Lowry acid.



Explain the terms:

(1)

- (b) (i) Write an expression for the acid dissociation constant, K_a , for ethanoic acid.

(1)

- (ii) Calculate the $\text{p}K_a$ value of aqueous ethanoic acid.

$$K_a(\text{ethanoic acid}) = 1.70 \times 10^{-5} \text{ mol dm}^{-3} \text{ at } 25^\circ\text{C}$$

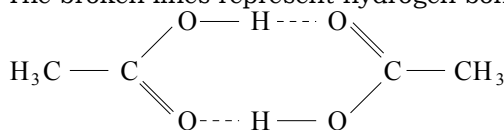
(1)

- (c) The pH ranges over which two indicators used in acid-base titrations change colour are given in table below.

Indicator	pH range
methyl orange	3.1 – 4.4
phenolphthalein	8.3 – 10.0

In a titration aqueous sodium hydroxide is run into a conical flask containing aqueous ethanoic acid.

- (i) State which indicator should be used and explain your answer. (3)
- (ii) State the colour change seen in the conical flask at the end point. (1)
- (iii) Write an equation for the reaction between ethanoic acid and sodium hydroxide in aqueous solution. (1)
- (d) You are supplied with aqueous sodium hydroxide and aqueous ethanoic acid, each solution having a concentration of 0.10 mol dm^{-3} . State briefly how you would prepare a buffer solution with a pH equal to the $\text{p}K_a$ of ethanoic acid. (2)
- (e) In the gaseous state, some ethanoic acid molecules are dimerised as shown in diagram below. The broken lines represent hydrogen bonds.



Explain how hydrogen bonds are formed between ethanoic acid molecules.

(3)

(Total 14 marks)

42. (a) State what is meant by the term *weak acid* and give one example. (2)
- (b) Write an expression for the dissociation constant, K_a , of the weak acid HA and state the units of K_a . (2)
- (c) When water is cooled, the pH increases but the water remains neutral. (2)
- (i) Explain why the pH increases.
- (ii) Explain why water remains neutral.
- (d) State the characteristic property of a buffer solution. (4)

(2)
(Total 10 marks)

43. Solution **S** is 0.16 M hydrochloric acid, HCl, a strong acid.
 Solution **W** is 0.16 M HX, a weak monoprotic acid. It has a pH of 2.74.
 Solution **Z** is 0.12 M barium hydroxide, Ba(OH)₂, a strong base.

- (a) (i) Explain the terms *weak* and *strong* as applied to acids or bases.
 (ii) Determine a value of the acid dissociation constant of the weak acid HX using the expression

$$K_a \approx \frac{[H^+]^2}{c}$$

where c is the original concentration of HX. Explain why it is reasonable to use this approximation.

(6)

- (b) Show details of **all** calculations in answering this part of the question.

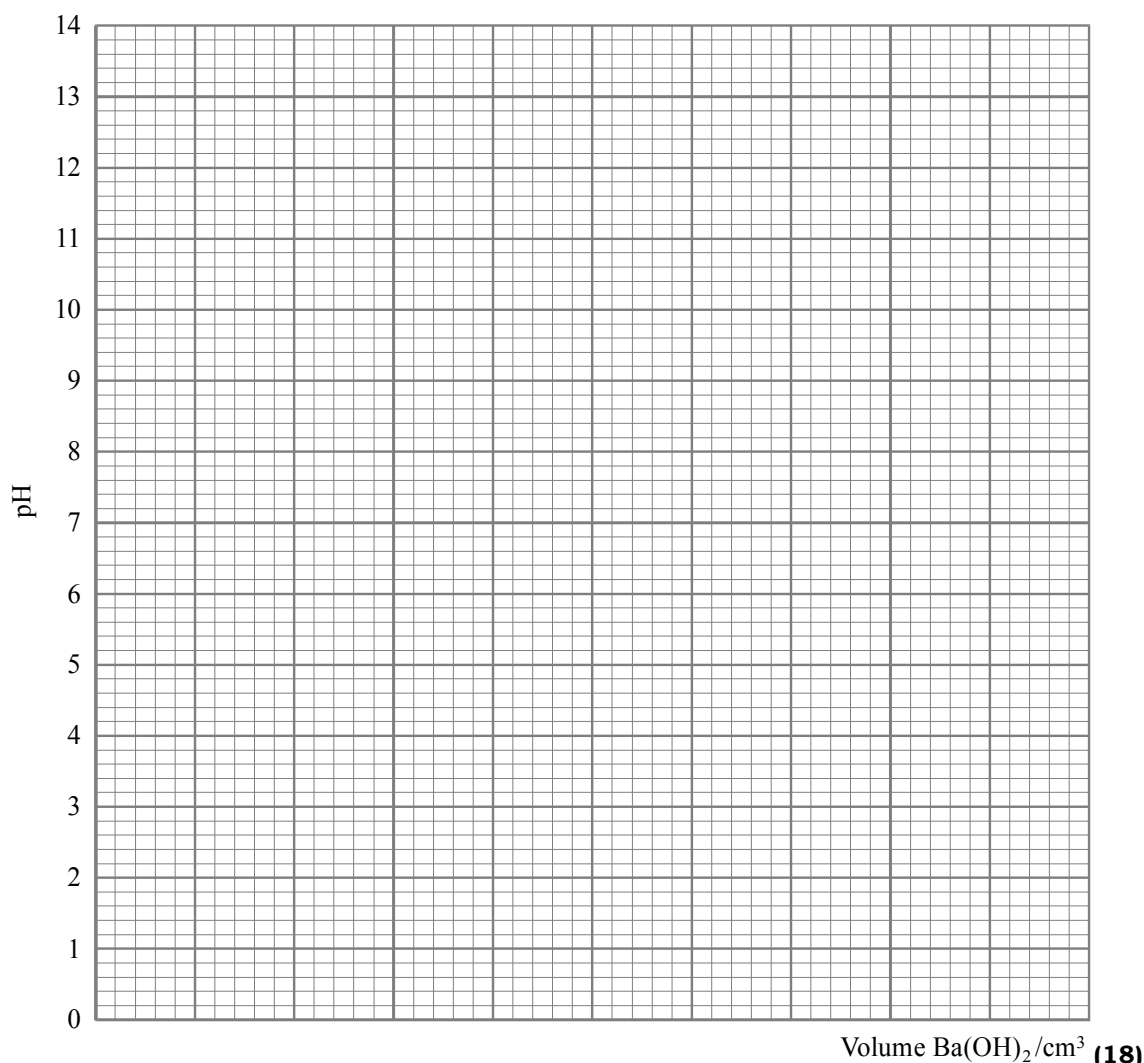
At a temperature of 25°C, 18 cm³ of solution **S** are titrated with solution **Z**. The titration is repeated using 18 cm³ of solution **W**.

- (i) Determine the equivalence volume (end-point) for these titrations and enter your result into the appropriate space in the incomplete table below. Enter also the half-equivalence volume and the double-equivalence volume in the appropriate space.
 (ii) Calculate the missing pH values for each of the two titration solutions and enter these into the table.

(You should assume that, at half-equivalence, [HX] = [X⁻] for the weak acid HX, and that, for both acids at double-equivalence, only the alkali that is in excess contributes to the pH of the resulting solution.)

	Start	Half equivalence	Equivalence	Double Equivalence
Volume/cm ³ Ba(OH) ₂ solution added	0.0			
pH for titration of S	0.80		7	
pH for titration of W	2.74		8.5	

- (iii) Plot these results on the graph below and use the points you have plotted to sketch the complete titration curves for solution **S** and solution **W** titrated with solution **Z**.



- (c) (i) Explain what is meant by the term *buffer solution*. Suggest how solution **W**, when half-neutralised, can behave as a buffer solution.
- (ii) State the difference between *acidic* and *basic* buffers. To which of these two types of buffer does a half-neutralised solution of **W** belong? What might you use to make a buffer solution of the other type?

(6)

(Total 30 marks)

44. This question concerns the weak acid, ethanoic acid, for which the acid dissociation constant, K_a , has a value of $1.74 \times 10^{-5} \text{ mol dm}^{-3}$ at 25°C .

$$K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$$

In each of the calculations below, give your answer to 2 decimal places.

- (a) Write an expression for the term *pH*. Calculate the pH of a $0.150 \text{ mol dm}^{-3}$ solution of ethanoic acid. (4)
- (b) A buffer solution is prepared by mixing a solution of ethanoic acid with a solution of sodium ethanoate.
- (i) Explain what is meant by the term *buffer solution*.
- (ii) Write an equation for the reaction which occurs when a small amount of hydrochloric acid is added to this buffer solution. (3)
- (c) In a buffer solution, the concentration of ethanoic acid is $0.150 \text{ mol dm}^{-3}$ and the

concentration of sodium ethanoate is $0.100 \text{ mol dm}^{-3}$.

- (i) Calculate the pH of this buffer solution.
- (ii) A 10.0 cm^3 portion of 1.00 mol dm^{-3} hydrochloric acid is added to 1000 cm^3 of this buffer solution.
Calculate the number of moles of ethanoic acid and the number of moles of sodium ethanoate in the solution after addition of the hydrochloric acid.
Hence, find the pH of this new solution.

(8)

(Total 15 marks)

45. A student carried out an investigation with aqueous solutions of nitric acid, sodium hydroxide, ethanoic acid and water.
Nitric acid, HNO_3 , is a strong Brønsted-Lowry acid.

- (a) Explain what is meant by a *strong acid* and a *Brønsted-Lowry acid*. [2]
- (b) What is the conjugate base formed from HNO_3 ? [1]
- (c) The student diluted $0.015 \text{ mol dm}^{-3}$ nitric acid with an equal volume of water and measured the pH of the diluted acid at 25°C .
(i) Calculate the pH of $0.015 \text{ mol dm}^{-3}$ nitric acid.

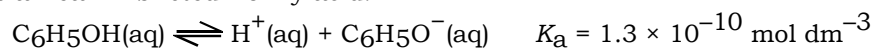
[2]

- (ii) Calculate the pH of the diluted acid.

[1]

[Total 6 marks]

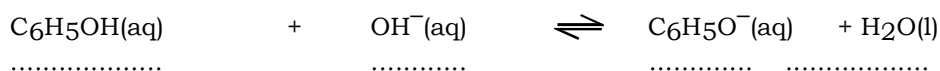
46. Phenol, $\text{C}_6\text{H}_5\text{OH}$, is a powerful disinfectant and antiseptic.
Phenol is a weak Brønsted-Lowry acid.



- (a) Define the following terms:

- (i) A Brønsted-Lowry acid, [1]
- (ii) A *weak* acid. [1]

- (b) When phenol is mixed with aqueous sodium hydroxide, an acid-base reaction takes place.



In the spaces above,

- label one **conjugate acid-base pair** as acid 1 and base 1,
- label the other **conjugate acid-base pair** as acid 2 and base 2.

[2]

- (c) A solution of phenol in water has a concentration of 38 g dm^{-3} .
The acid dissociation constant, K_a , of phenol is $1.3 \times 10^{-10} \text{ mol dm}^{-3}$.
(i) Write an expression for the acid dissociation constant, K_a , of phenol.

[1]

- (ii) Calculate the pH of this solution.

[5]

- (d) As part of an investigation, a student needed to prepare a buffer solution with a pH value of 8.71. From the K_a value of phenol, the student thought that a mixture of phenol and sodium phenoxide could be used to prepare this buffer solution.

The student decided to use a $0.200 \text{ mol dm}^{-3}$ solution of phenol, mixed with an equal volume of sodium phenoxide.

Use your knowledge of buffer solutions to determine the concentration of sodium phenoxide solution that the student would need to mix with the $0.200 \text{ mol dm}^{-3}$ phenol solution.

[Total 13 marks]

47. A student prepared two solutions.

- Solution **A** was made by mixing together 25 cm^3 $0.010 \text{ mol dm}^{-3}$ aqueous sodium hydroxide with 50 cm^3 $0.010 \text{ mol dm}^{-3}$ ethanoic acid, CH_3COOH . Solution **A** is a buffer solution.
- Solution **B** was made by mixing together 25 cm^3 $0.020 \text{ mol dm}^{-3}$ aqueous sodium hydroxide with 50 cm^3 $0.010 \text{ mol dm}^{-3}$ ethanoic acid, CH_3COOH . Solution **B** is **not** a buffer solution.

(i) What is meant by a *buffer solution*?

[1]

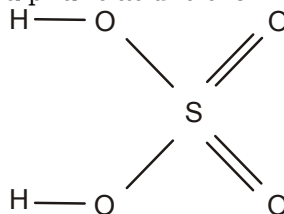
(ii) Explain why Solution **A** is a buffer solution whereas Solution **B** is **not**.

[4]

[Total 5 marks]

48. In order to obtain full marks in this question, you must show **all** your working clearly. In its reactions, sulphuric acid, H_2SO_4 , can behave as an acid, an oxidising agent and as a dehydrating agent.

The displayed formula of pure sulphuric acid is shown below.



Dilute sulphuric acid takes part in the typical acid reactions, reacting with metals, carbonates and bases.

Write balanced equations for the reaction of sulphuric acid with
a metal,
a carbonate
a base.

[Total 3 marks]

49. Compound **B** is an organic base. A student analysed this base by the procedure below. He first prepared a solution of **B** by dissolving 4.32 g of **B** in water and making the solution up to 250 cm^3 . The student then carried out a titration in which 25.00 cm^3 of this solution of **B** were neutralised by exactly 23.20 cm^3 of $0.200 \text{ mol dm}^{-3}$ HCl . 1 mole of **B** reacts with 1 mole of HCl .

Use this information to calculate the molar mass of base **B** and suggest its identity.

[Total 6 marks]

50. In sewage plants, biological activity can be reduced by increasing the pH of the water. This is achieved by adding small amounts of solid calcium hydroxide, $\text{Ca}(\text{OH})_2$, to the sewage water.

In all parts of this question, assume that measurements have been made at 25°C .

(a) The pH of aqueous solutions is determined by K_{W} .

K_{W} has a value of $1.0 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at 25°C .

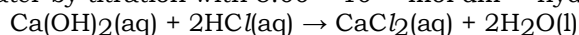
(i) What name is given to K_{W} ?

[1]

(ii) Write the expression for K_{W} .

[1]

(b) A chemist checked the concentration of aqueous calcium hydroxide, $\text{Ca}(\text{OH})_2$, in the sewage water by titration with $5.00 \times 10^{-3} \text{ mol dm}^{-3}$ hydrochloric acid.



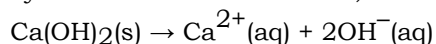
The chemist titrated 25.0 cm^3 of the sewage water with 21.35 cm^3 of HCl to reach the endpoint of the titration.

Calculate the concentration, in mol dm^{-3} , of the calcium hydroxide in the sewage water.

[3]

(c) The chemist analysed a sample of water from another part of the sewage works and he found that the calcium hydroxide concentration was $2.7 \times 10^{-3} \text{ mol dm}^{-3}$.

When solid calcium hydroxide dissolves in water, its ions completely dissociate.



Calculate the pH of this sample.

[3]

- (d) After further treatment, the water could be used for drinking. In the drinking water produced, the OH^{-} concentration was 100 times greater than the H^{+} concentration. What was the pH of this drinking water?

[1]

[Total 9 marks]

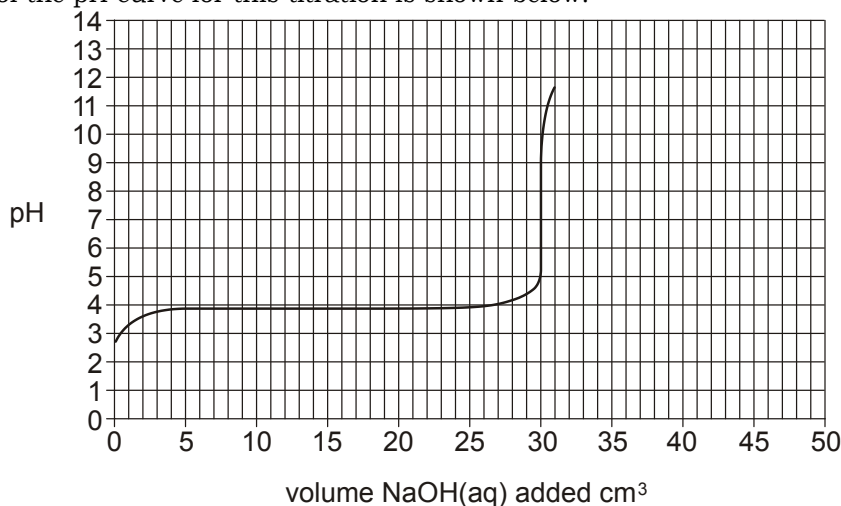
51. A student titrated the $1.50 \times 10^{-2} \text{ mol dm}^{-3}$ methanoic acid with aqueous sodium hydroxide.

A 25.00 cm^3 sample of the $\text{HCOOH}(\text{aq})$ was placed in a conical flask and the $\text{NaOH}(\text{aq})$ was added from a burette until the pH no longer changed.

- (i) Write a balanced equation for the reaction between $\text{HCOOH}(\text{aq})$ and $\text{NaOH}(\text{aq})$.

[1]

- (ii) Part of the pH curve for this titration is shown below.



Calculate the concentration, in mol dm^{-3} , of the aqueous sodium hydroxide.

[3]

- (iii) Calculate the pH of the aqueous sodium hydroxide.

$$K_{\text{W}} = 1.00 \times 10^{-14} \text{ mol dm}^{-3}$$

[2]

- (iv) The pH ranges in which colour changes for three acid-base indicators are shown below.

indicator	pH range
metacresol purple	7.4 – 9.0
2,4,6-trinitrotoluene	11.5 – 13.0
ethyl orange	3.4 – 4.8

Explain which of the three indicators is suitable for this titration.

[2]

[Total 8 marks]

Marking Scheme

1. (a) (i) Increase (if wrong no further marks in part (i)) 1
 higher P gives lower yield or moves to left 1
 Eqm shifts to reduce P or eqm favours side with fewer moles 1
 (ii) Endothermic if wrong no further marks in part (ii) 1
 increase T increases yield or moves to right 1
 Eqm shifts to reduce T or eqm favours endothermic direction 1
- (b) (i) Moles of iodine = 0.023 1
 Moles of HI = 0.172 1
 If $\times 2$ missed, max 1 in part (iv)
If wrong no marks in (i)
- (ii) $K_C = \frac{[H_2][I_2]}{[HI]^2}$ must be square brackets (penalise once in paper) 1
 – if round, penalise but mark on in (iv)
 if K_C wrong, no marks in (iv) either but mark on from a minor slip in formula
- (iii) V cancels in K_C expression
 or no moles same on top and bottom of expression
 or total moles reactants = moles products,
 i.e. total no of moles does not change 1
- (iv) $K_C = \frac{(0.023)^2}{(0.172)^2}$ 1
 = 0.0179 or 1.79×10^{-2}
 Conseq on (i) 1
Allow 0.018 or 1.8×10^{-2}
- (v) $K_C = 55.9$ or 56 1
Conseq i.e. (answer to (iv))⁻¹

[13]

2. (a) $K_C = \frac{[Y][H_2O]^2}{[X][CH_3OH]^2}$ (1)
if K_C expression wrong lose units mark in (e) also must be []
- (b) Moles of **X**: $0.25 - 0.13 = 0.12$ (1)
 Moles of methanol: $0.34 - 0.26 = 0.08$ (1)
 Moles of water: 0.26 (1) 3
- (c) Equal no. of moles on each side of equation (1)
OR V cancels out (provided not incorrectly qualified) 1
- (d) Calculation: $K_C = \frac{\left(\frac{0.13}{V}\right)\left(\frac{0.26}{V}\right)^2}{\left(\frac{0.12}{V}\right)\left(\frac{0.08}{V}\right)^2}$ (1)
 = 11(.4) (1)
Can score all 3 conseq on (b) and (c)
If different values from (c) used allow units only
(conseq on correct K_C)

Units of K_c : none (1)

but lose this mark if K_c is wrong even if none given

(e) decrease (1)

3
1

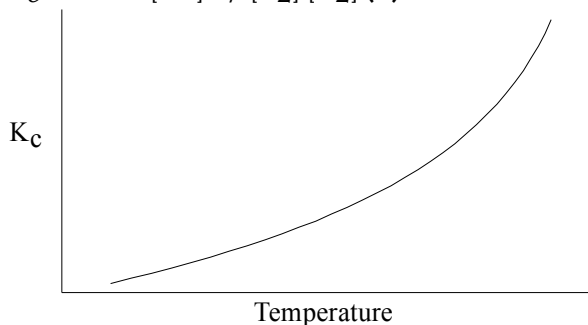
[9]

3. (a) Equation $N_2 + O_2 \rightleftharpoons 2NO$ (1)

K_c $[NO]^2 / [N_2] [O_2]$ (1)

2

(b)



As temperature increases K_c increases (or yield increases) (1)

Hence reaction endothermic (1)

2

(c) The product yield is very small (1)

Yield does not justify cost of producing high temp (1)

2

(d) (i) $2NO + O_2 \rightleftharpoons 2NO_2$ (1)

(ii) Change in equilibrium position Displaced to the right (1)

Change in equilibrium constant No change (1)

3

[9]

4. (must state correct effect on yield or rate to score the reason mark)

T effect: higher temp: yield greater or shifts equilibrium to right; 1
effect: higher temp: rate increased; 1
reason: endothermic

OR

more particles have $E > E_a$

1

OR

more successful/productive collisions;

1

P effect: higher pressure: yield less or shifts equilibrium to left; 1

effect: higher pressure: rate increased;

reason: increase in gas moles L to R

OR

greater collision frequency;

1

(Q of L mark)

[6]

5. (a) $K_c = \frac{[H_2][I_2]}{[HI]^2}$ (1)

0.05 or 1/20 (1)

2

(b) (i) forward rate increases (1)

reverse rate increases (1)

allow 1 mark for just 'increased'

allow 2 marks for 'both increased'

2

(ii) no change (1)

1

(iii) no change (1)

1

[6]

6. (a) (i) Moles NaOH = $mv/1000 = 1.50 \times 72.5/1000 = 0.108$ to 0.11 (1)

Moles of ethanoic acid at equilibrium = moles sodium hydroxide (1)

Moles ester = moles water (=moles acid reacted) (1)

$$= 0.200 - 0.108 = 0.090 \text{ to } 0.92 \text{ (1)}$$

$$\text{Moles ethanol} = 0.110 - 0.091 = 0.018 \text{ to } 0.020 \text{ (1)}$$

$$K_C = [\text{Ester}] [\text{Water}] / [\text{Acid}] [\text{Alcohol}] \text{ (1)}$$

Allow if used correctly

$$= (0.091)^2 / 0.109 \times 0.019 = 3.7 \text{ to } 4.9 \text{ (1)}$$

Ignore units

NB Allow the answer 4 one mark as correct knowledge

- (ii) Similar (types) of bond broken and made (1)
Same number of the bonds broken and made (1)
 any number if equal 2
 NB If a list given then the total number of each type of bond broken and made must be the same

[9]

7. (a) Homogeneous; All reactants in the same phase or state (1)
 Dynamic; Continuous or 'on-going' (1)
 Equilibrium: Concentrations of reactants and products constant
 or rates of forward and backward reactions equal (1)
 Equation; $2\text{NH}_3 \rightleftharpoons \text{N}_2 + 3\text{H}_2$ (Must be decomposition) (1)
 K_c ; $[\text{N}_2][\text{H}_2]^3 / [\text{NH}_3]^2$ (1)
 5
- (b) Conditions: decomposition favoured by high temp (1)
 since the reaction endothermic or logical
 statement with application of Le Chatelier's
 principle (1)
 decomposition favoured by low pressure (1)
 2 mole gas giving 4 moles gas or more gas moles
 on right (1)
 4
- (c) In practise low pressure means low production (1)
 low pressure means low rate (1)
 high temperature means high rate (1)
 high temperature expensive (1)
 equilibrium yield unaffected (1)
 rates of forward and backwards reactions
 increased by an equal amount (1)
 more hydrogen produced in a given time (1)
 Max 6

[15]

8. (a) Total moles $n = pV/RT$ (1)
 $= 1.59 \times 10^6 \times 1.04 \times 10^{-3} / (8.31 \times 380) = 0.524 \text{ moles (1)}$
 Moles of methanol $0.524 - 0.122 - 0.298 = 0.104 \text{ moles (1)}$ 3
- (b) $K_C = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2}$ (1)
 $= \frac{0.104}{0.122} \times \left(\frac{1.04}{0.298} \right)^2$
 $= 10.4$ (1)
 $\text{dm}^6 \text{ mol}^{-2}$ (1) 3

[6]

9. (a) (i) $\text{C} + 3\text{D} \longrightarrow 2\text{A} + \text{B}$ 1
 (ii) $\text{mol}^{-1} \text{dm}^3$ 1
 (iii) (forward reaction is) exothermic or more products formed 1

- (b) (i) for N_2O_4 Mr = 92.0 1
 $\text{Mol} = \frac{36.8}{92.0} = 0.400$ 1
 (ii) mol N_2O_4 reacted = $0.400 - 0.180 = 0.220$ 1
 mol NO_2 formed = 0.440 1
 (iii) $K_c = \frac{(\text{NO}_2)^2}{(\text{N}_2\text{O}_4)}$ 1
 $= \frac{(0.44/16)^2}{(0.18/16)}$ 1
 $= 0.067$ 1
 (iv) move to NO_2 / to right / forwards 1
 none 1
- [12]**
10. (a) (i) moles of $\text{C}_2\text{F}_2 = 0.40$ mark independently from HCl 1
 moles of $\text{HCl} = 0.80$ **not** consequential 1
 (ii) $K_c = \frac{[\text{C}_2\text{F}_4][\text{HCl}]^2}{[\text{CHClF}_2]^2}$
 wrong K_c means they can only 1
 score for units in (iii) consequ 1
 on their K_c
 (iii) $K_c = \frac{(0.40/18.5)(0.8/18.5)^2}{(0.20/18.5)^2}$ 1
 $= 0.35$ 1
 mol dm^{-3} 1
 (b) (i) increase 1
 (ii) decrease 1
- [8]**
11. (a) (i) Number of moles of O_2 at equilibrium: $\frac{7.04}{32} = 0.22$ (1) 1
 Number of moles of NO at equilibrium: 0.44 (1)
OR 2 × mol of oxygen
 (ii) Original number of moles of NO_2 : $\frac{21.3}{46} = 0.46(3)$ (1)
 Number of moles of NO_2 at equilibrium: $0.46(3) - 0.44 = 0.02(3)$ (1)
OR conseq on mol NO above
 3
- (b) Expression for K_c : $K_c = \frac{[\text{NO}]^2[\text{O}_2]}{[\text{NO}_2]}$ (1)
 $\text{Calculation: } K_c = \frac{\left(\frac{0.44}{11.5}\right)^2 \times \left(\frac{0.22}{11.5}\right)}{\left(\frac{0.023}{11.5}\right)^2} = 7.0(0) \text{ mol dm}^{-3}$
 (1) (1) (1)
If mol $\text{NO}_2 = 0.02$; $K_c = 9.26$ (9.3)
or conseq on values from (a)
If vol missed, score only K_c and units
If K_c wrong: max 2 for correct use of vol and conseq units
If K_c wrong and no vol: max 1 for conseq units
 3
- (c) $pV = nRT$ (1)

$$T = \frac{pV}{nR} = \frac{(3.30 \times 10^5) \times (11.5 \times 10^{-3})}{0.683 \times 8.31} (1)$$

(1) for using 11.5×10^{-3} as V

$$T = 669 \text{ K } (1)$$

(d) *Yield of oxygen*: increased **(1)**

Value of K_c : no effect **(1)**

4

2

[13]

12. (a) (i) exothermic **(1)**

yield decreases as temperature increases **(1)**

(this mark dependent on first mark)

2

(ii) vanadium (V) oxide / vanadium pentoxide / V_2O_5 **(1)**

yield of sulphur trioxide unchanged **(1)**

not just reference to position of equilibrium

(both / forward and reverse reaction) rates increase **(1)**

equally **(1)**

4

(iii) advantage – higher yield of (sulphur trioxide) / equilibrium

moves to the right **(1)**

not “saves energy costs”

disadvantage – slower rate of reaction / lower throughput **(1)**

2

(b) (i) rate increased **(1)**

1

(ii) % yield increased / more SO_3 / equilibrium shifts to the right **(1)**

fewer (gas) moles on right hand side of equation increased

pressure favours **(1)**

smaller number of moles / molecules / volume

2

[11]

13. (a) (i) A proton donor **(1)**

(ii) Fully ionised or fully dissociated **(1)**

(iii) 1.0×10^{-14} **(1)**

$\text{mol}^2 \text{dm}^{-6}$ **(1)**

4

(b) (i) $1.50 \times 10^{-3} / 25 \times 1000$ **(1)**

$= 0.06 \text{ mol dm}^{-3}$ **(1)**

1.2 **(1)**

(ii) Mol OH^- added $= 50 \times 0.150 / 1000 = 7.5 \times 10^{-3}$ **(1)**

Mol H^+ used $= 1.5 \times 10^{-3}$

(1)

Mol OH^- excess $= 6.0 \times 10^{-3}$

(1)

$[\text{OH}^-] = 6.0 \times 10^{-3} / 75 \times 1000 = 8.0 \times 10^{-2}$ **(1)**

$[\text{H}^+] = 10^{-14} / 8.0 \times 10^{-2} = 1.25 \times 10^{-13}$ **(1)**

pH $= 12.9$ **(1)**

8

(c) (i) 0.3 mol dm^{-3} **(1)**

(ii) $[\text{H}^+]$ at pH = 0.7 is 0.2 mol dm^{-3} **(1)**

$m_1 v_1 = m_2 v_2$ therefore $0.3 \times 25 = 0.2 \times v$ **(1)**

Hence $v = 37.5$ **(1)**

Water added $= 37.5 - 25 = 12.5$ **(1)**

5

[17]

14. (a) (i) $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} (1)$

$$K_a = \frac{[\text{H}^+]^2}{[\text{HA}]} (1)$$

$$[\text{H}^+] = 10^{-2.82} = 1.514 \times 10^{-3} \text{ mol dm}^{-3} (1)$$

$$K_a = \frac{(1.514 \times 10^{-3})^2}{0.15} = 1.53 \times 10^{-5} \quad (1) \text{ mol dm}^{-3} \quad (1) [2]$$

- (ii) Decreases (1)
Equilibrium shifts to right (endothermic process) (1)
hence $[H^+] \uparrow$ as $T \uparrow$ (1)
pH gets smaller 8
- (b) (i) $\frac{2}{3}n$ (1)
- (ii) $\left(\frac{2}{3}n\right) \div \left(\frac{1}{3}n\right) = 2$ (1)
- (iii) $K_a = \frac{[H^+][X^-]}{[HX]} = \frac{[H^+]}{2}$ (1)
 $= 2.1 \times 10^{-4} \text{ mol dm}^{-3}$ (1) 5
- (c) Weak acid/strong base thus pH at equivalence > 7 (1)
methyl orange has colour change at pH < 7 (1) 2
- (d) Buffer can resist change in pH (1)
on addition of small amounts of H^+ (or OH^-) (1)
 $H^+ (aq) + X^- (aq) \rightleftharpoons HX (aq)$ (1) 3

[18]

15. (a) proton donor (1)
substance formed when acid has lost proton / substance that becomes an acid
by gaining a proton (**not** just proton acceptor) (1) 2
- (b) (i) acid: HBr base: Br^-
(ii) acid: H_2SO_4 base: HSO_4^-
allow 1 in (b) if both acids / bases are correct (ie give 1 for a correct vertical pair) 2
- (c) (i) $H_2O \rightleftharpoons H^+ + OH^-$ / $2H_2O \rightleftharpoons H_3O^+ + OH^-$ (1)
(accept other types of arrow) 1
- (ii) $K_c = \frac{[H^+][OH^-]}{[H_2O]}$ / expression based on H_3O^+ equation (1)
[H_2O] is (effectively) constant / concentration of H_2O is large / equilibrium in (i) is to left (1)
 $(K_c \times [H_2O]) = K_w = [H^+][OH^-]$ (1) 3
- (iii) $[H^+] = [OH^-] / [H^+]^2 = 2.92 \times 10^{-14}$ (1)
 $[H^+] = \sqrt{2.92 \times 10^{-14}} = 1.71 \times 10^{-7}$ (1)
 $pH = -\log_{10}(1.71 \times 10^{-7})$ (1)
 $= 6.77$ (1) 4
- (iv) endothermic **and** attempt at reason (1)
more dissociation / ionization / H^+ ions at higher temperature (1)
if (iii) not completed, allow endothermic with sensible reason
for 1 mark if answer to (iii) is $pH > 7$, allow 1 mark for exothermic with attempt at reason 2

[14]

16. (a) Definition of a base Proton acceptor (1)
Essential feature Transfer of protons (1)
Equation $H^+ + OH^- \rightleftharpoons H_2O$
OR $H^+ + B \rightleftharpoons BH^+$ (1) 3

- (b) only partially dissociated in solution **(1)** 1
- (c) $K_a = \frac{[H^+(aq)][CH_3CH_2COO^-(aq)]}{[CH_3CH_2COOH(aq)]}$ **(1)**
- mol dm⁻³ **(1)** 2
- (d) (i) resists change in pH **(1)**
on addition of small amounts of strong acid or base **(1)**
- (ii) correct weak acid/co-base
or correct weak base/co-acid **(1)**
- (iii) any suitable use **(1)** 4
- [10]**
17. (a) (i) $0.12 \times 11.8 = M \times 25$ **(1)**
thus molarity = 0.057 **(1)**
- (ii) $K_a = \frac{[H^+][A^-]}{[HA]}$ **(1)**
- (iii) *Volume of NaOH(aq) added* 11.8/2 = 5.9 cm³ **(1)**
pH 4.3 to 4.35 **(1)**
- (iv) As $[HA] = [A^-]$ **(1)** $K_a = [H^+]$ **(1)**
 $pH = -\log_{10} [H^+]$ **(1)** hence $K_a = 10^{-4.3}$
 $= 5.0 \times 10^{-5}$ **(1)**
Note:- Mark K_a consequentially to pH in a(iii) 9
- (b) (i) The added OH⁻ reacts with HA or H⁺ **(1)**
The equilibrium, $HA \rightleftharpoons H^+ + A^-$, displacement to right
or HA ionises **(1)**
- (ii) The added H⁺ reacts with A⁻ **(1)**
The equilibrium, $HA \rightleftharpoons H^+ + A^-$, displaced to left **(1)** 4
- [13]**
18. (a) $CO_3^{2-} + H^+ \rightarrow HCO_3^-$ or $Na_2CO_3 + HCl \rightarrow NaHCO_3 + NaCl$ **(1)**
 $HCO_3^- + H^+ \rightarrow H_2O + CO_2$ or $NaHCO_3 + HCl \rightarrow NaCl + CO_2 + H_2O$ **(1)** or H_2CO_3 2
- (b) 15 cm³ 1
- (c) *Indicator* Methyl orange (allow other correct indicators) **(1)**
Explanation Methyl red changes colour over pH
range 3.2 – 4.4 (allow between 3 and 7) **(1)** 2
- (d) $CO_3^{2-} + 2H^+ \rightarrow H_2O + CO_2$
Mol H⁺ = $30 \times 0.12/1000 = 3.6 \times 10^{-3}$ **(1)**
Mol $CO_3^{2-} = 1.8 \times 10^{-3} = M \times 25/1000$ **(1)**
 $= 0.072 M$ **(1)** 3
- (e) *Volume of HCl(aq) added for first end-point* 15 **(1)**
Volume of HCl(aq) added for second end-point 45 **(1)** 2
- [10]**
19. (a) A proton donor 1
- (b) *Equation* $HCl(g) + H_2O(l) \rightleftharpoons H_3O^+(aq) + Cl^-(aq)$ **(1)**
Role of water base or proton acceptor **(1)** 2
- (c) *Equation* $NH_3(g) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$ **(1)**
Role of water acid or proton donor **(1)** 2
- (d) *Equation for formation* $H_2SO_4 + HNO_3 \rightleftharpoons HSO_4^- + H_2NO_3^+$ **(1)**

	Role of nitric acid	base or proton acceptor (1)	
	$\text{H}_2\text{NO}_3^+ \rightarrow \text{NO}_2^+ + \text{H}_2\text{O}$ (1)		3
(e)	(i) only partially dissociated in aqueous solution (1)		
	(ii) $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$ (1)		
	(iii) not very big (1)		
	(iv) strong (1)		
	Although HX is not fully dissociated, the relative concentration of undissociated HX is very small (1)		5
			[13]
20.	(a) $\text{NaOH} + \text{HA} \rightarrow \text{NaA} + \text{H}_2\text{O}$ (1)		
	or $\text{HA} + \text{OH}^- \rightarrow \text{A}^- + \text{H}_2\text{O}$		1
	(b) (i) Moles A^- = moles NaOH added (1)		
	$= 15 \times 0.34 \times 10^{-3} = 5.10 \times 10^{-3}$ (1)		
	Initial moles HA = $25 \times 0.45 \times 10^{-3} = 0.01125$ (1)		
	Allow 0.0110 – 0.0113		
	Moles NaOH added = 0.00510		
	Moles HA remaining = 6.15×10^{-3} (1)		
	Allow (5.90 – 6.20) $\times 10^{-3}$		
	Mark conseq		
	(ii) 5.10×10^{-3} moles A^- in (15 + 25) cm^3		
	Hence $[\text{A}^-] = 5.10 \times 10^{-3} \times 1000/40 = 0.1275$ (1)		
	Allow 0.127 – 0.128 and 0.13		
	6.15×10^{-3} moles HA in 40 cm^3		
	Hence $[\text{HA}] = 6.15 \times 10^{-3} \times 1000/40 = 0.1538$ (1)		
	Allow 0.147 – 0.155 and 0.15		
	Allow marks in (ii) conseq to answers in (i)		
	(iii) $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} = 2.00 \times 10^{-4}$		
	$[\text{H}^+] = 2.00 \times 10^{-4} \times 0.1538 / 0.1275$ (1)		
	$= 2.41 \times 10^{-4}$ (1)		
	Allow (2.29 – 2.44) $\times 10^{-4}$		
	pH = 3.62 (1)		
	Allow 3.61 – 3.64 and 3.6		
	Mark conseq to answers in (ii)		9
			[10]
21.	(a) only partially ionized / partially dissociated / not fully ionised (1)		
	not 'not ionised at all'		1
	(b) (i) $K_a = \frac{[\text{C}_6\text{H}_5\text{O}^-][\text{H}_3\text{O}^+]}{[\text{C}_6\text{H}_5\text{OH}]}$ (1)		
	accept $[\text{H}^+]$		
	do not accept with $[\text{H}_2\text{O}]$ included		
	must include charges		1
	(ii) $\text{p}K_a = -\log K_a$ (1)		
	allow $-\log(K_a)$ do not allow $-\log[K_a]$		1
	(iii) $\text{p}K_a = 10$ (ignore units) (1)		1
	(iv) lower / smaller number (1)		1
	(c) (i) at end point pH = $\text{p}K_a = 9.3$ (1)		
	colour change detectable over range of 2 pH units therefore range = 8.3 – 10.3		
	(1)		
	(allow 8 – 10)		2

- (ii) (colourless to) pink / red **(1)**
 $[\text{In}^-] \geq [\text{HIn}]$ / $[\text{In}^-]$ increases **(1)**
not just equilibrium shifts to right 2
- (iii) equivalence point / end point of titration below pH 7 more acidic / lower than phenolphthalein range / is about pH 4 **(1)**
not just the pH range is wrong 1
- [10]**
- 22.** (a) (i) $\text{pH} = -\log_{10} [\text{H}^+]$ Note; (aq) not required; Not $-\ln [\text{H}^+]$ **(1)**
- (ii) $K_a = [\text{H}^+] [\text{X}^-] / [\text{HX}]$ Note; (aq) not required **(1)**
 Allow $[\text{A}^-]$ and $[\text{HA}]$
 Do NOT allow $[\text{H}^+]^2 / [\text{HX}]$
- (iii) $K_a = 4.25 \times 10^{-5} = [\text{H}^+]^2 / [\text{HX}]$ **(1)**
 $[\text{H}^+] = \sqrt{0.45 \times 4.25 \times 10^{-5}}$ **(1)**
 $= 4.37 \times 10^{-3}$ **(1)**
not a conseq mark
 $\text{pH} = \underline{2.36}$ Mark conseq to $[\text{H}^+]$ above **(1)**
 or $\text{pH} = \frac{1}{2} \text{p}K_a - \frac{1}{2} \log_{10} [\text{HX}] = \frac{1}{2} \times 4.37 + \frac{1}{2} \times 0.346 = \underline{2.36}$
(1) **(2)** **(1)**
- Note pH = 2.4 scores max 3 6
- (b) (i) $[\text{H}^+] = 0.25 \times 0.95 = 0.2375$ **(1)**
 Allow 0.237 – 0.238 and 0.24
 $\text{pH} = 0.62$ **(1)**
 Allow 0.62 – 0.63
 Only allow pH mark if $[\text{H}^+]$ is correct
- (ii) $[\text{H}^+] = [\text{Y}^-] = 0.2375$ (or a value from b(i)) **(1)**
 $[\text{HY}] = 0.05 \times 0.25 = 0.0125$ **(1)**
 Allow 0.012 – 0.013
 $K_a = [\text{H}^+] [\text{Y}^-] / [\text{HY}]$
 $= (0.2375)^2 / 0.0125$ **(1)**
 $K_a = 4.51$ **(1)**
 Allow 4.3 – 4.8
 Ignore units
 CE if $[\text{HY}]$ is incorrect 6
- [12]**
- 23.** (a) $\text{p}K_a + -\log_{10} K_a$ 1
- (b) $K_a = 1.90 \times 10^{-4}$ **(1)**
 $K_a = [\text{H}^+]^2 / 0.52$ or $[\text{H}^+] = [\text{X}^-]$ **(1)**
 $[\text{H}^+] = \sqrt{(1.90 \times 10^{-4} \times 0.52)} = 9.94 \times 10^{-3}$ **(1)**
 $\text{pH} = -\log_{10} [\text{H}^+] = 2.00$ **(1)**
 or $\text{pH} = \frac{1}{2} \text{p}K_a - \frac{1}{2} \log [\text{HX}]$
 $= 1.86 - (-0.142) = 2.00$ 4
- (c) $K_a [\text{H}^+] [\text{X}^-] / [\text{HX}]$ **(1)**
 $[\text{HX}] = [\text{X}^-]$ at half neutralisation **(1)**
 Hence $K_a = [\text{H}^+]$ and $\text{p}K_a = \text{pH}$ **(1)** 3
- (d) There is no rapid/sharp/steep change in pH during a weak acid - weak base titration **(1)**
 Indicator need a sharp pH rise to change colour quickly **(1)** 2

[10]

24. (a) (i) $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$ (1)

(ii) (1) $K_a = \frac{[H^+]^2}{[CH_3COOH]}$ (1)

(2) $[H^+] = \sqrt{1.74 \times 10^{-5} \times 0.220} = 1.96 \times 10^{-3}$ (1)

(3) $pH = -\log_{10}[H^+]$ (1)

can score independently

(4) $pH = 2.71$ (1)

2 d.p. essential

If forget $\sqrt{\quad}$ can score (1) and (3) for pH = 5.42

5

(b) (i) moles acid = $\frac{25}{1000} \times 0.220$ (1) = 5.50×10^{-3}

$$= \frac{x}{10^3} \times 0.150$$

thus $x = 25 \times \frac{0.220}{0.150}$ **or** $5.50 \times 10^{-3} \times \frac{1000}{0.150}$

= 36.7 (or 37) cm^3 (or 36.6) (1)

NOT 36 NOR 37.0

units must match

(ii) *Indicator:* thymol blue (1)

Explanation: weak acid – strong base (1)
equivalent at pH >7 (1)
or high pH

5

(c) (1) mol NaOH added = $\frac{2.0}{40.0} = 0.050$ (1)

If wrong M_r : CE therefore lose marks (1) and (2) then mark on consequentially → max 4

(2) mol CH_3COOH left = $0.220 - 0.050 = 0.170$ (1)

(3) mol CH_3COO^- formed = 0.050 (1)

(4) $[H^+] = K_a \frac{[\text{acid}]}{[\text{salt}]}$ OR $pH = pK_a + \log \left(\frac{[A^-]}{[HA]} \right)$ etc (1)

If expression wrong no marks for 4 / 5 / 6

can score (1) to (4) in (5)

(5) $[H^+] = 1.74 \times 10^{-5} \times \frac{(0.170)}{(0.05)}$ OR $pH = 4.76 + \log \left(\frac{0.05}{0.17} \right)$ (1)

(6) $pH = 4.23$ (1)

Correct answer gets ✓✓✓✓✓✓

Mark (5) is for use of correct values of (acid moles) and (salt moles)

if one wrong allow pH conseq

if both wrong, no further marks

e.g. if candidate forgets substitution in (2)

he loses (2) and (5) but can score (1) (3) (4) (6) = max 4

for pH = 4.12 if $\frac{[\text{acid}]}{[\text{salt}]}$ upside down; answer 5.29 scores 3

for (1) (2) (3)

6

[16]

25. (a) moles HA = $\frac{25}{10^3} \times 0.150 = 3.75 \times 10^{-3}$ (1)
 vol NaOH = $\frac{3.75 \times 10^{-3}}{0.20} = 1.875 \times 10^{-2} \text{ dm}^3$ (1)
or 18.75 cm³

2

- (b) (i) pH = $-\log_{10} [\text{H}^+]$ (1)
 (ii) Value above 7 but below 11 (1)
 (iii) phenol red / thymol blue / phenolphthalein / thymolphthalein
i.e. indicator with $7 < \text{pK}_{\text{in}} < 11$

3

- (c) (i) Only slightly dissociated (1)
NOT “not fully dissociated / ionised”

(ii) $K_{\text{a}} = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$ (1)

NOT $\frac{[\text{H}^+]^2}{[\text{HA}]}$

- (iii) For weak acid alone:

$K_{\text{a}} = \frac{[\text{H}^+]^2}{[\text{HA}]}$ (1)

$\square [\text{H}^+] = \sqrt{(2.75 \times 10^{-5}) \times 0.15}$

$= 2.03 \times 10^{-3}$ (1)

$\square \text{pH} = 2.69$ (1)

pH should be given to 2 decimal places
penalise answer to 1 d.p. once in question

5

- (d) moles OH⁻ added = 1.875×10^{-3} = moles A⁻ = moles HA left (1)
or [A⁻] = [HA]
 $\rightarrow K_{\text{a}} = [\text{H}^+]$ **or** pH = pK_a (1)
 $\rightarrow \text{pH} = 4.56$ (1)

3

[13]

26. Penalise pH given to 1 dp first time it would have scored only

- (a) (i) $K_{\text{w}} = [\text{H}^+][\text{OH}^-]$ (1)
 (ii) pH = $-\log [\text{H}^+]$ (1)
or in words or below unless contradiction

(iii) Calculation: $[\text{H}^+] = \sqrt{5.48 \times 10^{-14}}$ (1)

$= 2.34 \times 10^{-7}$

thus pH = 6.63 or 6.64 (1)

Explanation: pure water $[\text{H}^+] = [\text{OH}^-]$ (1)

5

- (b) (i) $[\text{OH}^-] = 0.150$ $[\text{H}^+] = 10^{-14}/0.15 = 6.66 \times 10^{-14}$
 or pOH = 0.82
 pH = 13.18 (1)
 or pH = 13.17

(ii) moles OH⁻ = $(35 \times 10^{-3}) \times 0.150 = 5.25 \times 10^{-3}$ (1)^a

moles H⁺ = $(40 \times 10^{-3}) \times 0.120 = 4.8(0) \times 10^{-3}$ (1)^b

$$\text{excess moles of OH}^- = 4.5 \times 10^{-4} \text{ (1)}^c$$

$$[\text{OH}^-] = (4.5(0) \times 10^{-4}) \times 1000/75^d \text{ (1)}^e$$

$$[\text{H}^+] = \frac{10^{-14}}{6.00 \times 10^{-3}} = 1.66 \times 10^{-12} \text{ or pOH} = 2.22$$

$$\text{pH} = 11.78 \text{ (1)}^f$$

or 11.77

$$(c) \quad (i) \quad K_a = \frac{[\text{H}^+][\text{X}^-]}{[\text{HX}]} \text{ (1)}$$

$$(ii) \quad [\text{H}^+] = 1.80 \times 10^{-2} \times 0.150 = 2.70 \times 10^{-3} \text{ (1)}$$

$$K_a = \frac{[\text{H}^+]^2}{[\text{HX}]} \text{ (1)} = \frac{(2.70 \times 10^{-3})^2}{0.150} = 4.86 \times 10^{-5} \text{ (1) mol dm}^{-3} \text{ (1)}$$

$$\text{or } \frac{(2.70 \times 10^{-3})^2}{0.1473} = 4.95 \times 10^{-5}$$

Notes

- (a) If K_w includes H_2O allow 6.63 if seen otherwise no marks likely
- (b) (ii) If no vol, max 4 for a, b, c, f answer = 10.65
 If wrong volume max 5 for a, b, c, e, f
 If no subtraction max 3 for a, b, d
 If missing 1000 max 5 for a, b, c, d, f answer = 8.78
 If uses excess as acid, max 4 for a, b, d, f answer = 2.22
 If uses excess as acid and no volume, max 2 for a, b answer = 3.35
- (c) If wrong K_a in (i) max 2 in part (ii) for $[\text{H}^+]$ (1) and conseq units (1)
but mark on fully from minor errors
eg no [] or charges missing

[18]

27. (a) Hydrogen bonding (1)
 between H_2O and NH_3 (1) 2
- (b) (i) $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ (1)
 (ii) Ammonia is weak base (1)
NOT partially ionised
 Equilibrium to left or incomplete reaction (1) 3
- (c) A proton donor (1) 1
- (d) Buffer solution: A solution which resists change in pH (1)
 when small amounts of acid or base added or on dilution (1)
 Reagent: NH_4Cl (1) 3
Allow a correct strong acid
- (e) (i) $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} \text{ (1)}$
 $= \frac{[\text{H}^+][0.125 \times 4]}{1.00} \text{ (1)}$
 $[\text{H}^+] = 1.70 \times 10^{-5} / 0.125 \times 4 = 3.40 \times 10^{-5} \text{ (1)}$
 $\text{pH} = -\log_{10} [\text{H}^+] = 4.47 \text{ (1)}$
Allow pH conseq to $[\text{H}^+]$ if 2 place decimals given
- (ii) $\text{H}^+ + \text{CH}_3\text{COO}^- \rightarrow \text{CH}_3\text{COOH}$ (1) 5

[14]

28. (a) $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$ 1

(All three sets of square brackets needed, penalise missing brackets or missing charge once in the question)

(Don't penalise extra $[H^+]^2/[HA]$)

(b) $K_a = \frac{[H^+]^2}{[HA]}$ or $[H^+] = [A^-]$ 1

$$[H^+] = \sqrt{(1.45 \times 10^{-4}) \times 0.25}$$

$$= 6.02 \times 10^{-3}$$
 1

$$pH = 2.22$$
 1

(must be to 2dp)

(allow 4th mark consequential on their $[H^+]$)

1

(c) (i) pH (almost) unchanged 1
(Must be correct to score explanation)

H^+ removed by A^- forming HA
or acid reacts with salt
or more HA formed

1

(ii) $[H^+] = 10^{-3.59} = 2.57 \times 10^{-4}$ or 2.6×10^{-4} 1

$$[A^-] = \frac{K_a[HA]}{[H^+]}$$
 1

$$= \frac{(1.45 \times 10^{-4}) \times 0.25}{2.57 \times 10^{-4}}$$
 1

$$= 0.141 \text{ (mol dm}^{-3}\text{)}$$
 1

(Allow 0.139 to 0.141 and allow 0.14)

(If not used 3.59, to find $[H^+]$ can only score M2 for working)

(If 3.59 used but $[H^+]$ is wrong, can score M2 for correct method and conseq M4)

If wrong method and wrong expression, can only score M1)

(ii) Alternative scheme for first three marks of part (c)(ii)

$$pH = pK_a - \log \frac{[HA]}{[A^-]}$$
 1

$$pK_a = 3.84$$
 1

$$3.59 = 3.84 - \log \frac{0.250}{[A^-]}$$
 1

[11]

29. (a) (i) $pH = -\log [H^+]$ (1)

(ii) Expression for K_a : $K_a = \frac{[H^+][X^-]}{[HX]}$ (1)

Calculation: $pH = 2.56$ thus $[H^+] = 2.75 \times 10^{-3}$ (1)

$$K_a = \frac{[H^+]^2}{[HX]} = \frac{(2.75 \times 10^{-3})^2}{0.12} = 6.32 \times 10^{-5} \text{ (1) (mol dm}^{-3}\text{)}$$

or $[H^+] = [X^-]$ (1) 5

depending on approximate made, values of $K_a = 10^{-5} \times$

using $[HX] = 0.12$ 6.30 – 6.32

using $[HX] = 0.12 - 2.75 \dots$ 6.45 – 6.47

using 2.8 and $[HX] = 0.12$ 6.53

using 2.8 and $[HX] = 0.12 - 2.8$ 6.69

upside down K_a

(b) (i) Expression for K_w : $K_w = [H^+][OH^-]$ (1)

Value of K_w : $(1.0 \times 10^{-14} \text{ (mol}^2 \text{ dm}^{-6}) \text{ (1)}$
ignore units

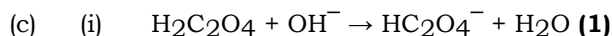
$$(ii) \quad [H^+] = \frac{1.0 \times 10^{-14}}{0.045} = 2.22 \times 10^{-13}$$

or $pOH = 1.35 \text{ (1)}$

thus $pH = 12.65 \text{ (1)}$

must be 2dp in final answer

4



(ii) $\text{mol } OH^- = (41.6 \times 10^{-3}) \times 0.0450 \text{ (1)} = 1.87 \times 10^{-3}$

$\text{mol } H_2C_2O_4 = 9.36 \times 10^{-4} \text{ (1)}$

$[H_2C_2O_4] = 9.36 \times 10^{-4} \times 10^3 / 25$
 $= 0.0374 \text{ (1)}$

4

if moles of $H_2C_2O_4$ not equal to half moles of OH^- , no further marks gained

*if $\text{mol } OH^- = 1.9 \times 10^{-3}$; hence $\text{mol } H_2C_2O_4 = 9.5 \times 10^{-4}$;
 $[H_2C_2O_4] = 0.038$*

[13]

30. (a) (i) B; 1
 C; 1
 A; 1
 (ii) cresolphthalein
 OR
 thymolphthalein; 1
- (b) (i) $-\log[H^+]$; 1
 (ii) $[H^+] = 1.259 \times 10^{-12}$ (or 1.26 or 1.3) OR $pOH = 14 - pH$; 1
 $[OH^-] = \frac{10^{-14}}{1.258 \times 10^{-12}}$ OR $= 2.10$; 1
 $= 7.9(4) \times 10^{-3}$; 1
(if $[H^+]$ is wrong allow 1 for $[OH^-] = K_w/[H^+]$ or as numbers)
- (c) (i) $K_a = [H^+]^2 / [CH_3CH_2COOH]$
 OR
 $[H^+]^2 / [HA]$
 OR
 $[H^+] = [A^-] \text{ etc.};$ 1
 $[H^+] = \sqrt{1.35 \times 10^{-5} \times 0.117}$ or expression without numbers; 1
 $= 1.257 \times 10^{-3}$
 $pH = 2.90$; 1
 (iii) $K_a = [H^+]$
 OR
 $pK_a = pH$; 1
 $pH = 4.87$; 1
(penalise 1dp once)

[13]

31. (a) $-\log [H^+]$ 1
 4.57×10^{-3} 1
allow 4.6×10^{-3}

ecf if [] wrong and already penalised

ignore units

(b) (i) $K_a = \frac{[H^+][X^-]}{[HX]}$ allow HA etc.

not $\frac{[H^+]^2}{[HX]}$ but mark on

If expression wrong allow conseq units in (ii) but no other marks in (ii) 1

(ii) $\frac{[H^+]^2}{[HX]} = \frac{(4.57 \times 10^{-3})^2}{[0.150]}$ 1

If use 4.6×10^{-3}

$K_a = 1.4(1) \times 10^{-4}$ and $pK_a = 3.85$ 1

$= 1.39 \times 10^{-4}$

allow $1.39 - 1.41 \times 10^{-4}$

mol dm^{-3}

(iii) $pK_a = 3.86$ 1

Penalise dp of final answer < or > 2 in pH once in paper

(c) (i) $\frac{30}{1000} \times 0.480 = 0.0144$ or $1.4(4) \times 10^{-2}$ 1

Mark is for answer (M1)

(ii) $\frac{18}{1000} \times 0.350 = 0.0063$ or 6.3×10^{-3} 1

Mark is for answer (M2)

(iii) $0.0144 - 2(0.0063) = 1.80 \times 10^{-3}$ 1

M3 is for (i) - 2(ii)

If x 2 missed, CE i.e. lose M3 and the next mark gained

(iv) $1.80 \times 10^{-3} \times \frac{1000}{48} = 0.0375$ (0.038) 1

M4 is for answer

If vol is not 48×10^{-3} (unless AE) lose M4 and next mark gained

If multiply by 48 - this is AE - i.e. lose only M4

If multiply by 48×10^{-3} this is AE - i.e. lose only M4

(v) $10^{-14} / 0.0375$ ($10^{-14} / 0.038$) 1

M5 for $K_w / [OH^-]$

$(= 2.66 \times 10^{-13})$ ($= 2.63 \times 10^{-13}$) 1

or pOH

or pOH = 1.426 (or pOH = 1.420)

If no attempt to use K_w or pOH lose both M5 and M6

pH = 12.57 (12.58) M6 1

Allow M6 conseq on AE in M5 if method OK

[13]

32. (a) (i) $[H^+][OH^-]$ 1
 $-\log [H^+]$ 1

(ii) $[H^+] = [OH^-]$ 1

(iii) $(2.0 \times 10^{-3}) \times 0.5 = 1.0 \times 10^{-3}$ 1

- (iv) $[H^+] = \frac{4.02 \times 10^{-14}}{1.0 \times 10^{-3}} \quad (= 4.02 \times 10^{-11})$ 1
- pH = 10.40 1
- (b) (i) $K_a = \frac{[H^+][CH_3CH_2COO^-]}{[CH_3CH_2COOH]}$ 1
- $= \frac{[H^+]}{[CH_3CH_2COOH]}$ 1
- $[H^+] = \sqrt{(1.35 \times 10^{-5}) \times 0.125} \quad (= 1.30 \times 10^{-3})$ 1
- pH = 2.89 1
- (c) (i) $(50.0 \times 10^{-3}) \times 0.125 = 6.25 \times 10^{-3}$ 1
- (ii) $(6.25 \times 10^{-3}) - (1.0 \times 10^{-3}) = 5.25 \times 10^{-3}$ 1
- (iii) mol salt formed = 1.0×10^{-3} 1
- $(H^+) = K_a \times \frac{[CH_3CH_2COOH]}{[CH_3CH_2COO^-]}$ 1
- $= (1.35 \times 10^{-5}) \times \frac{(5.25 \times 10^{-3})/V}{(1.0 \times 10^{-3})/V} (= 7.088 \times 10^{-5})$ 1
- pH = 4.15 1
- [16]**
33. (a) Concentration of acid: $m_1 v_1 = m_2 v_2$ hence $25 \times m_1 = 18.2 \times 0.150$
OR
moles NaOH = 2.73×10^{-3} ; 1
 $m_1 = 18.2 \times 0.150 / 25 = 0.109$; 1
- (b) (i) $K_a = [H^+][A^-] / [HA]$ not $K_a = [H^+]^2 / [HA]$; 1
- (ii) $pK_a = -\log K_a$; 1
- (iii) $[A^-] = [HA]$; 1
hence $K_a = [H^+][A^-] / [HA] = [H^+]$
and $-\log K_a = -\log [H^+]$; 1
- (c) ratio $[A^-] : [HA]$ remains constant; 1
hence as $[H^+] = K_a [HA] / [A^-]$; $[H^+]$ remains constant; 1
- (d) (i) pH of $0.250 \text{ mol dm}^{-3} \text{ HCl}$ = 0.60
and pH of $0.150 \text{ mol dm}^{-3} \text{ HCl}$ = 0.82; 1
pH change = 0.22; 1
- (ii) moles HCl = $30 \times 0.250 \times 10^{-3} = v \times 0.150 \times 10^{-3} = 7.50 \times 10^{-3}$
OR
 $v = 30 \times 0.250 \times 10^{-3} / 0.150 \times 10^{-3} = 50$; 1
water added = $50 - 30 = 20 \text{ cm}^3$; 1
- [12]**
34. (a) proton or H^+ donor (1) 1
- (b) (i) partially ionised or dissociated (1)
not fully
- (ii) NH_3 (1)
not NH_4OH – but allow in equation
not H_2O – but allow in equation if both weak acid and base stated
 $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$ (1) 3
- (c) (i) $HCOOH(aq) + H_2O(l) \rightleftharpoons HCOO^-(aq) + H_3O^+(aq)$ (1)

- (ii) allow $H_2O(aq)$
 H_2O or water **(1)**
 $HCOO^-$ or methanoate ion **(1)**
- (iii) $K_a = \frac{[H^+][HCOO^-]}{[HCOOH]}$ **(1)**
- (d) (i) allow $[OH_3^+]$ 4
 addition of small amounts of acid **(1)**
 addition of small amounts of base **(1)**
allow volumes, allow alkali, penalise missing small once only not weak dilution **(1)**
 (ii) sodium methanoate or sodium hydroxide **(1)**
allow salt of methanoic (or this) acid not just an ion (methanoate)
 (iii) OH^- added **(1)**
or base
 H^+ reacts with OH^- **(1)**
or forming water
 More $HCOOH$ dissociates to restore equilibrium **(1)**
allow equilibrium moves to right
 must only describe addition of base;
 if both base and acid addition given, MAX 1 ex 3 7
- [15]**
35. (a) $HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$ **(1)**
allow $H_2O(aq)$ 1
- (b) $pH = -\log[H^+]$ **(1)**
 $= -0.10 (\pm 0.01)$ **(1)** 2
- (c) $pH = 7$ **(1)**
 neutral solution or $[H^+] = [OH^-]$ **(1)** 2
- (d) (i) moles $H^+ = 1.26 \times \frac{95}{1000} = 0.1197$ moles **(1)**
range 0.120 ± 0.001
- (ii) moles $OH^- = 2 \times 1.37 \times \frac{45}{1000} = 0.1233$ moles **(1)**
range 0.123 ± 0.001
- (iii) $XS\ OH^- = 0.1233 - 0.1197 = 3.6 \times 10^{-3}$ moles **(1)**
range 0.001 to 0.005
 Volume = $95 + 45 = 140\ cm^3$ **(1)**
 $[OH^-] = 3.6 \times 10^{-3} \times \frac{1000}{140} = 0.0257\ M$ **(1)**
range 0.0071 to 0.0357
 $K_w = [H^+][OH^-] = 10^{-14}\ mol^2\ dm^{-6}$ **(1)**
- $[H^+] = \frac{K_w}{[OH^-]} = \frac{10^{-14}}{0.0257} = 3.89 \times 10^{-13}\ M$ **(1)**
range $(0.28\ to\ 1.35) \times 10^{-12}$
 $pH = 12.41$ **(1)**
must show 2 dp range 11.87 to 12.55 8
- [13]**
- If no $\times 2$ for $Ba(OH)_2$ then H^+ is in XS MAX 4 ex 6
 If no volume used then MAX 4 ex 6
 If no $\times 1000$ for molarity then MAX 4 ex 6
 Combinations of TWO of these MAX 2 ex 6

All THREE

ZERO

- 36.** (a) only partially dissociated in water 1
- (b) (i) Added H^+ reacts with A^- (1)
Equilibrium moves left
& HA forms, restoring $[H^+]$ and pH (1)
- (ii) Added OH^- reacts with H^+ forming H_2O (1)
Equilibrium moves right
& HA dissociates, restoring $[H^+]$ and pH (1) 4
- (c) (i) $K_a = \frac{[H^+][A^-]}{[HA]}$ 1
- (ii) Equation for $[H^+]$ $[H^+] = K_a \times \frac{[HA]}{[A^-]}$ (1)
[HA] and $[A^-]$ (1)
are altered to same extent
maintaining $[H^+]$ and pH (1) 3
- 37.** (a) (i) $pK_a = -\log_{10} K_a$ (1)
- (ii) 6.31×10^{-5} (1)
 mol dm^{-3} (1)
- (iii) $[H^+] = [A^-]$ or $K_a = [H^+]^2/[HA]$ (1)
Hence $6.31 \times 10^{-5} = [H^+]^2/0.830$
 $[H^+] = \sqrt{6.31 \times 10^{-5} \times 0.830} = 7.24 \times 10^{-3}$ (1)
 $pH = -\log_{10}[H^+]$ (1)
 $pH = 2.14$ (1)
Marked consequentially to a(ii) 7
- (b) (i) Mol NaOH = mol X^- (1)
 $mv/1000 = 0.800 \times 10.5/1000 = 8.4 \times 10^{-3}$ (1)
- (ii) Mol HX remaining = original mol HX – mol NaOH added (1)
 $(25 \times 0.92/1000) - 8.4 \times 10^{-3} = 0.0146$ (1)
- (iii) Concentration of X^- $8.4 \times 10^{-3} \times 1000/(25 + 10.5)$
 $= 0.237$ (1)
Concentration of HX $0.0146 \times 1000/35.5$
 $= 0.411$ (1)
 pH of solution $K_a = [H^+][X^-]/[HX]$ (1)
 $[H^+] = 5.25 \times 10^{-5} \times 0.411/0.237$ (1)
 $pH = 4.04$ (1)
Marked consequentially to b(iii) 9
- (c) Change in pH Very small fall or slight change (1)
Explanation $H^+ + X^- \rightarrow HX$ (1)
Equilibrium restored $HX \rightleftharpoons H^+ + X^-$ OR 3
- 38.** (a) Monoprotic acid An acid which gives only one proton (1)
Example HCl etc (1) 2
- (b) (i) $-\log_{10}[H^+]$ or in words (1)
(ii) 1.58 M (allow 1.6) (1) 2
- (c) Mol $H^+ = 25 \times 0.15 \times 10^{-3}$ (1) = 3.75×10^{-3}

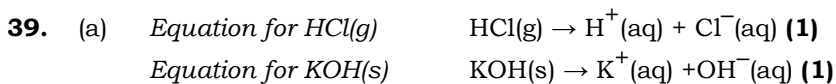
[9]

[19]

$$\begin{aligned}\text{Mol OH}^- &= 35 \times 0.12 \times 10^{-3} \text{ (1)} = 4.20 \times 10^{-3} \\ \text{Excess OH}^- &= 4.5 \times 10^{-4} \text{ (1)} \\ [\text{OH}^-] &= 4.5 \times 10^{-4} \times 1000/60 \text{ (1)} = 7.5 \times 10^{-3} \\ [\text{H}^+] &= 10^{-14}/7.5 \times 10^{-3} \text{ (1)} = 1.33 \times 10^{-12} \text{ (1)} \\ \text{pH} &= -\log_{10} 1.33 \times 10^{-12} \\ &= 11.9 \text{ (1)}\end{aligned}$$

NB Consequential marking if $[\text{OH}^-]$ not calculated to maximum of 5

7

[11]

2

(b) $K_w = [\text{H}^+][\text{OH}^-]$

1

(c) strong base, fully dissociated (1) or $[\text{OH}^-] = 0.016 \text{ M}$ (1)

$$[\text{H}^+] = \frac{K_w}{[\text{OH}^-]} \text{ (1)} = \frac{10^{-14}}{0.016} = 6.25 \times 10^{-13} \text{ M (1) (2)}$$

$$\text{pH} = -\log_{10}[\text{H}^+] \text{ (1)}$$

$$\text{pH} = 12.2 \text{ (1)}$$

neutral solution, $[\text{H}^+] = [\text{OH}^-]$ (1)

$$\text{pH} = 7 \text{ (1)}$$

7

(d) (i) 755 cm^3 of 0.012 M acid contain
 $\frac{0.012 \times 755}{1000} \text{ mol H}^+ = 9.06 \times 10^{-3} \text{ moles (1)}$

$$\text{moles OH}^- \text{ used for neutralisation} = 9.06 \times 10^{-3} \text{ (1)}$$

(ii) $\text{pH} = 11.6$ $[\text{H}^+] = 10^{-11.6} = 2.5 \times 10^{-12} \text{ M (1)}$

$$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} \text{ (1)} \quad [\text{OH}^-] = 3.98 \times 10^{-3} \text{ M (1) (2)}$$

$$\text{in } 755 \text{ cm}^3 \text{ there are } \frac{3.98 \times 10^{-3} \times 755}{1000}$$

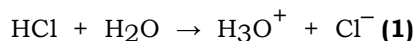
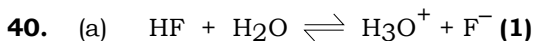
$$= 3.0 \times 10^{-3} \text{ mol (1)}$$

(iii) Total moles = $(9.06 + 3.0) \times 10^{-3} = 0.012 \text{ mol (1)}$

(iv) $M_r = 39 + 16 + 1 = 56$

$$m = 56 \times 0.012 = 0.68 \text{ g (1)}$$

8

[18]

2

for HF, must have reversible arrow

allow (aq) in HCl equation

(b) (i) $\text{pH} = -\log_{10}[\text{H}^+]$ or equivalent word definition (1)

1

allow $-\log[\text{H}_3\text{O}^+]$ or $-\log[\text{H}^+(\text{aq})]$

(ii) $[\text{H}^+] = 0.050 \text{ mol dm}^{-3}$

$$\text{pH} = 1.3(0) \text{ (1)}$$

1

if correct definition demonstrated in (ii), but word definition in (i) wrong, allow mark transfer from (ii) to (i)

(c) (i) $K_a = \frac{[\text{H}^+][\text{F}^-]}{[\text{HF}]} \text{ (1)}$

1

do **not** expression allow with $[\text{H}_2\text{O}]$; allow $[\text{H}_3\text{O}^+]$

allow consequential mark from wrong equation in (b) providing

- [H⁺] present
- (ii) $K_a = \frac{[H^+]^2}{[HF]}$ or $[H^+] = \sqrt{K_a[HF]}$ (1)
- $[H^+] = \sqrt{5.6 \times 10^{-4} \times 0.050}$
- = 0.0053 (1)
- pH = 2.3 / 2.28 (1) 3
- allow mark for correct pH from wrong [H⁺]
- (d) hydrogen fluoride **or** HF (1)
- donates a proton (to the nitric acid) (1)
- conjugate base F⁻ (this mark dependent on correct identification of acid) (1) 3
- [11]
41. (a) (i) proton donor (1) 1
- (ii) partially dissociated (into ions) (**not** weakly dissociated) (1) 1
- (b) (i) $K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$ (1) 1
- allow either [H₃O⁺] or [H⁺] in expression
- must include charges
- if [H₂O] included then no mark
- (ii) $pK_a = -\log(1.7 \times 10^{-5}) = 4.77 / 4.8 / 4.80$ (1) 1
- (c) (i) indicator: phenolphthalein (1)
- explanation: weak acid-strong base / pH change above 7.0 (1)
- link between pH change and indicator range (1) 3
- (ii) colourless to red / pink / purple (1) 1
- if methyl orange named as indicator - wrong but allow second explanation
- mark and colour change mark in (ii) (i.e. pink / red → yellow / orange)
- (iii) NaOH + CH₃COOH → CH₃COONa + H₂O (1)
- if charges shown they must be correct
- accept H⁺ + OH⁻ → H₂O 1
- (d) excess acid + NaOH / acid + NaOH = NaEt (1)
- in 2:1 ratio of volumes / 2:1 mol / 1:1 acid: salt (1) 2
- (e) (i) O-H / C=O bonds are polar (words or on diagram)
- due to different electronegativities of O and H (or O and C) (1)
- lone pairs of electrons on O atoms (1)
- attraction of δ⁺ H atom in O-H for δ⁻ O atom / lone pair in C=O
- between different molecules (1) Max
- 3
- [14]
42. (a) *Weak acid* An acid which only partially ionises (1)
- Example* Ethanoic, carbonic etc. (1) 2
- (b) *Expression* $K_a = \frac{[H^+][A^-]}{[HA]}$ (1)
- Units* mol dm⁻³ (or mol l⁻¹) (1) 2
- (c) (i) The dissociation of water is an endothermic process (1)
- Less dissociation on cooling (or equilibrium moves to water or
- K_w decreases) (1) less H⁺ (or [H⁺] lower) (1)
- (ii) Because [H⁺] = [OH⁻] (1) 4
- (d) Resists change in pH (1) on addition of small quantities of acid or base (1) 2
- [10]
43. (a) (i) Strong = fully dissociated)
- Weak = partially dissociated) in aqueous solution both needed (1) 1

$$(ii) \quad K_a \approx \frac{[H^+]^2}{[HX]} = \frac{(10^{-2.74})^2}{0.16} = 2.07 \times 10^{-5} \quad (1) \quad \text{mol dm}^{-3} \text{ (not M)} \quad (1) \quad 2$$

Explanation: HX barely dissociates OR $[H^+]$ very small (1)
so $[HX]_{eqm} = [HX] \text{ original}$ (1)

and $[H^+] = [X^-]$ in an aqueous solution of weak acid (1) 3

$$(b) \quad (i) \quad \text{moles } H^+ \text{ in } 18 \text{ cm}^3 = 18 \times 0.16 \times 10^{-3} = 2.88 \times 10^{-3} \quad (1)$$

$$\text{moles } OH^- \text{ in } V_{eqv} \text{ cm}^3 = \underline{2 \times} \quad (1) \quad (V_{eqv} \times 0.12 \times 10^{-3})$$

$$V_{eqv} = \frac{18 \times 0.16}{0.12 \times 2} = 12.0 \text{ cm}^3 \quad (1)$$

	Start	Half equivalence	Equivalence	Double equivalence
Volume/cm ³ Ba(OH) ₂ solution added	0.0	6.0	12.0	24.0
pH for titration of S	0.80	1.22	7	12.84
pH for titration of W	2.74	4.68	8.5	12.84

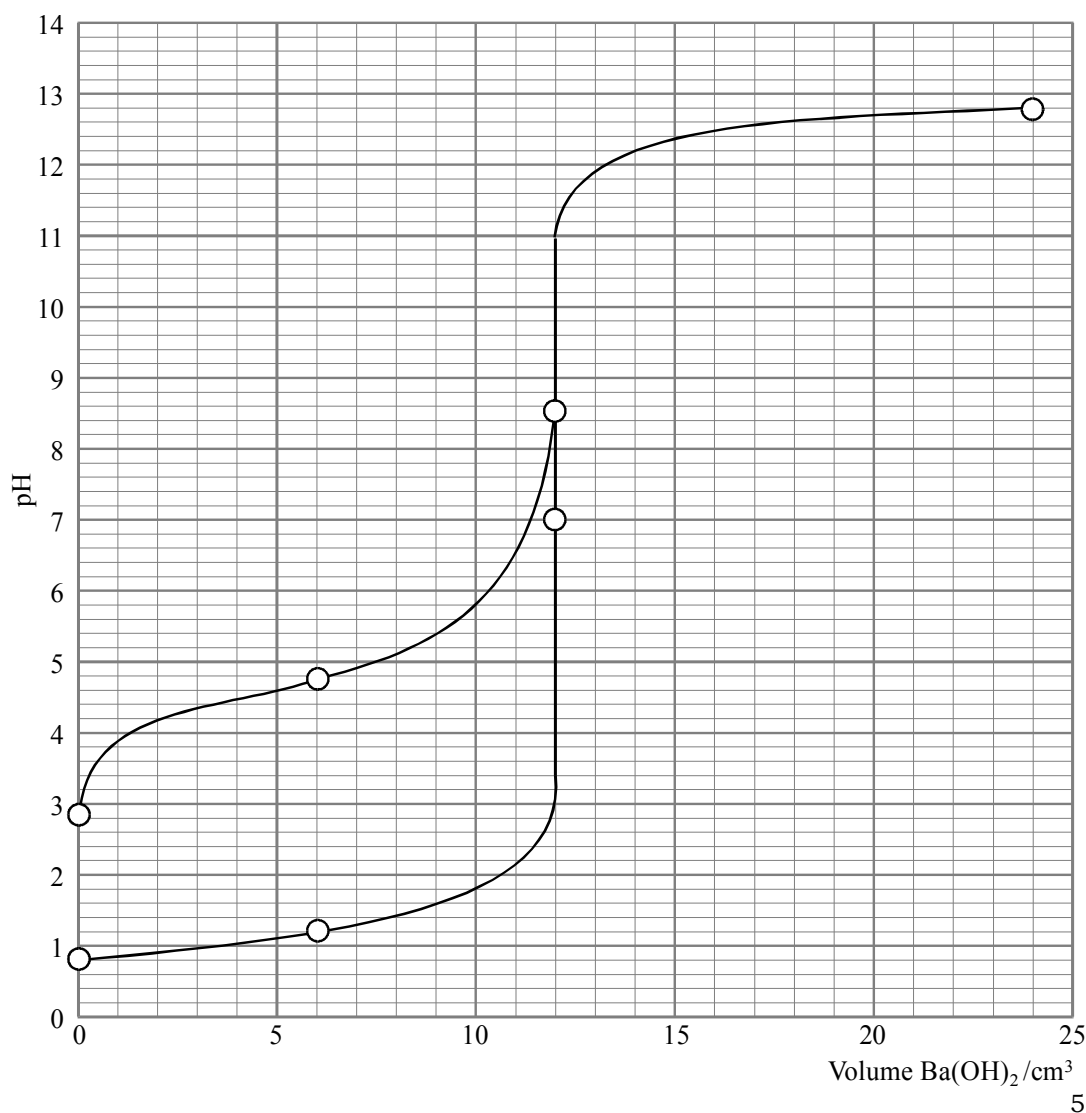
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(ii)	Strong acid S	Weak acid W
<i>Half-equivalence</i> Volume of base = 6 cm ³	9 cm ³ of strong acid S (1) in (18 + 6) cm ³ = 24 cm ³ (1) $[H^+] = 9 \times 0.16 / 24 = 0.06 \text{ M}$ (1) pH = 1.22 (1)	$[HX] = [X^-]$ (given) pH = pK _a <u>or</u> $[H^+] = K_a$ (1) = $-\log(2.07 \times 10^{-5})$ = 4.68 (1)
<i>2 × equivalence</i> Volume of base = 24 cm ³	12 cm ³ of base B in excess (1) in (18 + 24) cm ³ = 42 cm ³ (1) $[OH^-] = 12 \times 2 \times 0.12 / 42$ = 0.069 M (1) $[H^+] = K_w / [OH^-]$ (1) = $1.46 \times 10^{-13} \text{ M}$ (1) pH = 12.84 (1)	Same as for strong acid S pH = 12.84 (1)

[13 Max]

10

- (iii) *Graph:* Sensibly scaled volume axis (1)
4 points in text correctly plotted (–1 for each error) (2)
vertical portion at equivalence on graph for **S** (1)
single high pH curve for both **S** and **W** (1)
Weak acid (**W**) has marked pH rise at start (1)



- (c) (i) *Buffer Properties* resists change in pH (1)
on adding small amounts of acid or base (1)
W as a Buffer Plenty of X^- present to mop up H^+ (1)
Plenty of HX present to mop up OH^- (1)
OR equations showing same
e.g. $H^+ + X^- \rightarrow HX$, $OH^- + HX \rightarrow X^- + H_2O$

- (ii) *acid buffers* act at low pH, *basic buffers* act at high pH (1)
half-neutralised **W** is an *acid buffer* (1)
basic buffer: mix weak base with the salt of its co-acid (1)
OR correct specific example

6

[30]

44. (a) $pH = -\log[H^+]$ 1
 $K_a = \frac{[H^+]^2}{[CH_3COOH]}$ or $[H^+] = [A^-]$ 1
 $[H^+] = \sqrt{1.74 \times 10^{-5} \times 0.15}$ (or 1.62×10^{-3}) 1
 $pH = 2.79$ (penalise 1 dp or more than 2dp once in the qu) 1
(b) (i) Solution which resists change in pH / maintains pH 1
despite the addition of (small amounts of) acid/base (or dilution) 1
(ii) $CH_3COO^- + H^+ \rightarrow CH_3COOH$ 1

must show an equation full or ionic in which ethanoate ions are converted to ethanoic acid

- (c) (i) $[H^+] = \frac{K_a[CH_3COOH]}{[CH_3COO^-]}$ if rearrangement incorrect, no further marks 1
 $= 1.74 \times 10^{-5} \times \frac{0.15}{0.10}$ 1
 $(= 2.61 \times 10^{-5})$
 pH = 4.58 1
- (ii) M1 moles H^+ added = $10 \times 10^{-3} \times 1.0 = 0.01$ 1
 M2 moles ethanoic acid after addition = $0.15 + 0.01 = 0.16$ 1
 M3 moles ethanoate ions after addition = $0.10 - 0.01 = 0.09$ 1
 M4 $[H^+] = \frac{K_a[CH_3COOH]}{[CH_3COO^-]} = 1.74 \times 10^{-5} \times \frac{0.16/V}{0.09/V}$ 1
 $(= 3.09 \times 10^{-5})$
 M5 pH = 4.51 1

The essential part of this calculation is addition/subtraction of 0.01 moles to gain marks M2 and M3. If both of these are missing, only mark M1 is available. Thereafter treat each mark independently, except if the expression in M4 is wrong, in which case both M4 and M5 are lost.

[15]

alternative scheme for part (c)(i)

$$pH = pK_a - \log \frac{[CH_3COOH]}{[CH_3COO^-]} \quad 1$$

$$pK_a = 4.76 \quad 1$$

$$pH = (4.76 - \log \frac{0.15}{0.10}) = 4.58 \quad 1$$

alternative for penultimate mark of part (c)(ii)

$$pH = 4.76 - \log \frac{0.16}{0.09} \quad 1$$

45. (a) completely dissociates/ionised (1) 2
 proton donor (1)
- (b) NO_3^- (1) 1
- (c) (i) $pH = -\log[H^+] / -\log(0.015)$ (1) = 1.82 / 1.8 (1) (Not 2) 2
 (ii) $[H^+] = 0.0075 \text{ mol dm}^{-3}$
 $pH = -\log(0.0075) = 2.12 / 2.1$ (1) 1

[6]

46. (a) (i) H^+ / proton donor (1) 1
 (ii) partially dissociates/ionises (1) 1
- (b) $C_6H_5OH(aq) + OH^-(aq) \rightleftharpoons C_6H_5O^-(aq) + H_2O(l)$
 acid 1 base 2 (1) base 1 acid 2 (1)
 1 mark for each acid-base pair
- (c) (i) $K_a = \frac{[C_6H_5O^-][H^+]}{[C_6H_5OH]}$ (1) 1
- (ii) concentration = $38/94$ (1) = 0.40 mol dm^{-3} (1)
 (first mark for M_r of phenol – incorrect answer here will give ecf for remainder of question)
 $1.3 \times 10^{-10} \approx \frac{[H^+(aq)]^2}{0.40}$ (1)
 ('=' sign is acceptable)

$$[\text{H}^+] = \sqrt{\{(1.3 \times 10^{-10}) \times (0.40)\}} = 7.2 \times 10^{-6} \text{ mol dm}^{-3} \text{ (1)}$$

$$\text{pH} = -\log[\text{H}^+] = -\log 7.2 \times 10^{-6} = 5.14 \text{ (1)}$$

5

3 marks: $[\text{H}^+]$ (1); pH expression (1); calc of pH from $[\text{H}^+]$ (1)

Common errors:

Without square root, answer = 10.28 (1)(1)(0)

Use of 38 as molar concentration does not score 1st 2 marks.

This gives an answer of 4.15 for 3 marks (1)(1)(1)

(d) $[\text{H}^+(\text{aq})] = 1.99 \times 10^{-9} \text{ mol dm}^{-3} \text{ (1)}$

$$[\text{C}_6\text{H}_5\text{O}^-(\text{aq})] = K_a [\text{C}_6\text{H}_5\text{OH}(\text{aq})] / [\text{H}^+(\text{aq})] \text{ (1)}$$

$$[\text{C}_6\text{H}_5\text{O}^-(\text{aq})] = 0.013 \text{ mol dm}^{-3} \text{ (1)}$$

*Calculation should use half the original concentration of phenol to find the concentration of sodium phenoxide in the buffer. This should then be doubled back up again. Do **not** penalise an approach that uses the original concentration of phenol in the expression above.*

[13]
1

47.(i) a solution that minimises/resists/opposes pH changes (1)

(ii) The buffer must contain both CH_3COOH and CH_3COONa /

CH_3COO^- /weak acid and conjugate base(*) (1)

Solution **A** is a mixture of CH_3COOH (*) and CH_3COONa (*) /

/ has an excess of acid /is acidic (1)

Solution **B**, contains only CH_3COONa / only CH_3COO^-

/only the salt/ is neutral (1)

$\text{CH}_3\text{COOH}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{CH}_3\text{COONa}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ /

acid/alkali has been neutralised/

$\text{CH}_3\text{COOH}(\text{aq})$ and NaOH react together (1)

4

[5]

48. Correct equation for a metal (1)

Correct equation for a carbonate (1)

Correct equation for a base (1)

3

[3]

49. moles HCl in $23.2 \text{ cm}^3 = 0.200 \times 23.2/1000 = 4.64 \times 10^{-3} \text{ (1)}$

moles **B** in $25 \text{ cm}^3 = \text{moles HCl} = 4.64 \times 10^{-3} \text{ (1)}$

moles **B** in $250 \text{ cm}^3 = 4.64 \times 10^{-3} \times 10 = 4.64 \times 10^{-2} \text{ (1)}$

$4.64 \times 10^{-2} \text{ mol B}$ has a mass of 4.32 g

molar mass of **B** = $4.32/4.64 \times 10^{-2} = 93 \text{ g mol}^{-1} \text{ (1)}$

$93 - 16 = 77 \text{ (1)}$

Therefore **B** is phenylamine / $\text{C}_6\text{H}_5\text{NH}_2 \text{ (1)}$

6

There may be other valid structures that are amines. These can be credited provided that everything adds up to 93.

Answer could be a primary, secondary or tertiary amines.

[6]

50. (a) (i) Ionic product (1)

1

(ii) $K_w = [\text{H}^+(\text{aq})] [\text{OH}^-(\text{aq})] \text{ (1)}$ state symbols not needed

1

(b) moles of $\text{HCl} = \frac{5 \times 10^{-3} \times 21.35}{1000} = 1.067 \times 10^{-4} \text{ mol (1)}$

moles of $\text{Ca}(\text{OH})_2 = \frac{1.067 \times 10^{-4}}{2} = 5.34 \times 10^{-5} \text{ mol (1)}$

concentration of $\text{Ca}(\text{OH})_2 = 40 \times 5.34 \times 10^{-5}$

	$= 2.136 \times 10^{-3} \text{ mol dm}^{-3}$ (1)		
	2 marks for $4.27 \times 10^{-3} / 8.54 \times 10^{-3} \text{ mol dm}^{-3}$ (no factor of 4)	3	
(c)	$[\text{OH}^-] = 2 \times 2.7 \times 10^{-3} = 5.4 \times 10^{-3} \text{ mol dm}^{-3}$ (1)		
	$[\text{H}^+(\text{aq})] = \frac{K_w}{[\text{OH}^-(\text{aq})]} = \frac{1.0 \times 10^{-14}}{5.4 \times 10^{-3}} = 1.85 \times 10^{-12} \text{ mol dm}^{-3}$ (1)		
	$\text{pH} = -\log(1.85 \times 10^{-12}) = 11.73/11.7$ (1)	3	
	ecf is possible for pH mark providing that the $[\text{H}^+]$ value has been derived from $K_w/[\text{OH}^-]$		
	If pOH method is used, $\text{pOH} = 2.27$. would get 1 st mark, $\text{pH} = 14 - 2.27 = 11.73$ gets 2 nd mark.		
	Commonest mistake will be to not double OH^- and to use 2.7×10^{-3} This gives ecf answer of 11.43/11.4, worth 2 marks. $\text{pH} = 11.13$ from dividing by 2: worth 2 marks		
(d)	8 (1)	1	[9]
51.	(i) $\text{HCOOH} + \text{NaOH} \rightarrow \text{HCOONa} + \text{H}_2\text{O}$ (1) <i>state symbols not needed</i>	1	
	(ii) $n(\text{HCOOH}) = 0.0150 \times 25.00 / 1000 = 3.75 \times 10^{-4}$ (1) volume of $\text{NaOH}(\text{aq})$ that reacts is 30 cm^3 (1) so $[\text{NaOH}] = 3.75 \times 10^{-4} \times 1000 / 30 = 0.0125 \text{ mol dm}^{-3}$ (1)	3	
	(iii) $K_w = [\text{H}^+(\text{aq})][\text{OH}^-(\text{aq})]$ (1) $\text{pH} = -\log(1 \times 10^{-14} / 0.0125) = 12.10/12.1$ (1) (calc 12.09691001)	3	
	(iv) metacresol purple (1) pH range coincides with pH change during sharp rise OR pH 6-10 / coincides with equivalence point/end point (1)	2	[8]

8.0 REACTION KINETICS

1. (a) A fixed mass of marble is reacted with dilute hydrochloric acid at a constant temperature. Explain why the rate of the reaction is increased if the lumps of marble are reduced in size. (2)
- (b) The initial rate of the reaction between substances **A** and **B** was measured in a series of experiments and the following rate equation was deduced.

$$\text{rate} = k[\mathbf{A}][\mathbf{B}]^2$$

- (i) Complete the table of data below for the reaction between **A** and **B**.

Expt	Initial [A] /mol dm ⁻³	Initial [B] /mol dm ⁻³	Initial rate /mol dm ⁻³ s ⁻¹
1	0.020	0.020	1.2×10 ⁻⁴
2	0.040	0.040	
3		0.040	2.4×10 ⁻⁴
4	0.060	0.030	
5	0.040		7.2×10 ⁻⁴

- (ii) Using the data for Experiment 1, calculate a value for the rate constant, *k* and state its units. (7)

(Total 9 marks)

2. (a) The initial rate of reaction between ester **A** and aqueous sodium hydroxide was measured in a series of experiments at a constant temperature. The data obtained are shown below.

Experiment	Initial concentration of NaOH / mol dm ⁻³	Initial concentration of A / mol dm ⁻³	Initial rate / mol dm ⁻³ s ⁻¹
1	0.040	0.030	4.0 × 10 ⁻⁴
2	0.040	0.045	6.0 × 10 ⁻⁴
3	0.060	0.045	9.0 × 10 ⁻⁴
4	0.120	0.060	to be calculated

Use the data in the table to deduce the order of reaction with respect to **A** and the order of reaction with respect to NaOH. Hence calculate the initial rate of reaction in Experiment 4. (3)

- (b) In a further experiment at a different temperature, the initial rate of reaction was found to be 9.0 × 10⁻³ mol dm⁻³ s⁻¹ when the initial concentration of **A** was 0.020 mol dm⁻³ and the initial concentration of NaOH was 2.00 mol dm⁻³. (6)

Under these new conditions with the much higher concentration of sodium hydroxide, the reaction is first order with respect to **A** and appears to be zero order with respect to sodium hydroxide.

- (i) Write a rate equation for the reaction under these new conditions.
- (ii) Calculate a value for the rate constant under these new conditions and state its units.
- (iii) Suggest why the order of reaction with respect to sodium hydroxide appears to be zero under these new conditions. (6)

(Total 9 marks)

3. The rate of the reaction between substance **A** and substance **B** was studied in a series of experiments carried out at the same temperature. In each experiment the initial rate was measured using different concentrations of **A** and **B**. These results were used to deduce the order of reaction with respect to **A** and the order of reaction with respect to **B**.

- (a) What is meant by the term *order of reaction* with respect to **A**? (1)
- (b) When the concentrations of **A** and **B** were both doubled, the initial rate increased by a factor of 4. Deduce the **overall** order of the reaction. (1)
- (c) In another experiment, the concentration of **A** was increased by a factor of three and the concentration of **B** was halved. This caused the initial rate to increase by a factor of nine.
- (i) Deduce the order of reaction with respect to **A** and the order with respect to **B**.
- (ii) Using your answers from part (c)(i), write a rate equation for the reaction and suggest suitable units for the rate constant. (4)

(Total 6 marks)

4. The rate equation for a reaction between substances **A**, **B** and **C** is of the form:

$$\text{rate} = k[\text{A}]^x[\text{B}]^y[\text{C}]^z \text{ where } x + y + z = 4$$

The following data were obtained in a series of experiments at a constant temperature.

Experiment	Initial concentration of A /mol dm ⁻³	Initial concentration of B /mol dm ⁻³	Initial concentration of C /mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
1	0.10	0.20	0.20	8.0×10^{-5}
2	0.10	0.05	0.20	2.0×10^{-5}
3	0.05	0.10	0.20	2.0×10^{-5}
4	0.10	0.10	0.10	to be calculated

- (a) Use the data in the table to deduce the order of reaction with respect to **A** and the order of reaction with respect to **B**. Hence deduce the order of reaction with respect to **C**. (3)
- (b) Calculate the value of the rate constant, k , stating its units and also the value of the initial rate in experiment 4. (4)
- (c) How does the value of k change when the temperature of the reaction is increased? (1)

(Total 8 marks)

5. (a) A large excess of zinc was added to 100cm³ of 0.2 M hydrochloric acid. After the reaction had ended, 240 cm³ of hydrogen had been formed. In three further experiments, extra substances were added to the original mixture as shown in the table below. Fill in the table to show the total volume of hydrogen formed in each experiment and the qualitative effect of these additions on the initial rate of reaction compared to the original experiment.

Substances added to an excess of zinc and 100 cm ³ of 0.2 M hydrochloric acid	Volume of hydrogen/cm ³	Effect on initial rate of reaction
100cm ³ water		
10g zinc		
50 cm ³ 0.2 M hydrochloric acid		

- (b) The rate of reaction between compounds **A** and **B** was studied at a fixed temperature and some results obtained are shown in the table below. (6)

Experiment	Initial concentration of A /mol dm ⁻³	Initial concentration of B /mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
1	0.16	0.20	5.0×10^{-5}
2	0.24	0.20	7.5×10^{-5}
3	0.32	0.10	5.0×10^{-5}
4	0.12	0.15	to be calculated

Use the data in the table to deduce the order of reaction with respect to compound **A** and the order of reaction with respect to compound **B**. Hence calculate the initial rate of reaction in experiment 4.

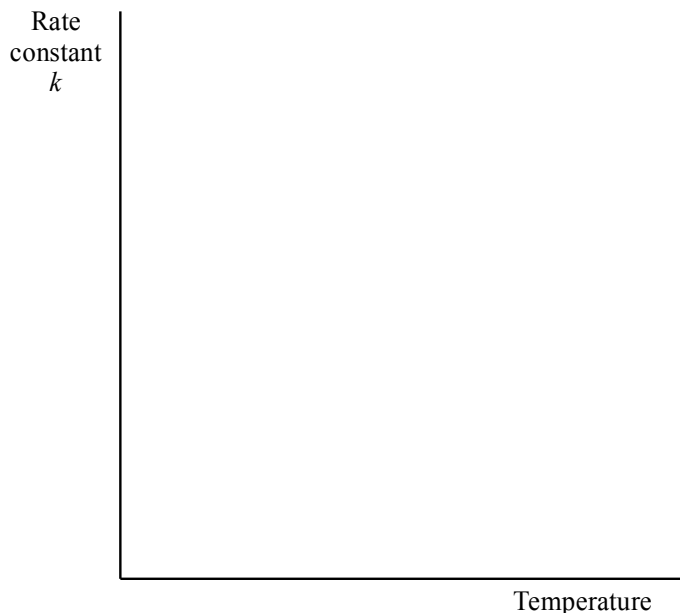
(4)

- (c) The rate equation for a reaction between substances **C** and **D** is:

$$\text{rate} = k[\text{C}]^2 [\text{D}]^2$$

The initial rate is found to be $7.5 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$ when the initial concentration of **C** is of 0.25 mol dm^{-3} and the initial concentration of **D** is 0.50 mol dm^{-3} .

- Calculate the value of the rate constant, k , at this temperature and deduce its units.
- On the axes below sketch a graph to show how the value of k varies as temperature is increased over a considerable range.



(4)

(Total 14 marks)

6. The initial rate of the reaction between the gases NO and H₂ was measured in a series of experiments at a constant temperature and the following rate equation was determined.

$$\text{rate} = k[\text{NO}]^2 [\text{H}_2]$$

- (a) Complete the table of data below for the reaction between NO and H₂

Experiment	Initial [NO] / mol dm ⁻³	Initial [H ₂] / mol dm ⁻³	Initial rate / mol dm ⁻³ s ⁻¹
1	3.0×10^{-3}	1.0×10^{-3}	1.8×10^{-5}
2	3.0×10^{-3}		7.2×10^{-5}

3	1.5×10^{-3}	1.0×10^{-3}	
4		0.50×10^{-3}	8.1×10^{-5}

(3)

- (b) Using the data from experiment 1, calculate a value for the rate constant, k , and state its units.

(3)

(Total 6 marks)

7. (a) The following data were obtained in a series of experiments on the rate of the reaction between compounds **A** and **B** at a constant temperature.

Experiment	Initial concentration of A /mol dm ⁻³	Initial concentration of B /mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
1	0.12	0.15	0.32×10^{-3}
2	0.36	0.15	2.88×10^{-3}
3	0.72	0.30	11.52×10^{-3}

- (i) Deduce the order of reaction with respect to **A**.
 (ii) Deduce the order of reaction with respect to **B**.

(2)

- (b) The following data were obtained in a series of experiments on the rate of the reaction between NO and O₂ at a constant temperature.

Experiment	Initial concentration of NO/mol dm ⁻³	Initial concentration of O ₂ /mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
4	5.0×10^{-2}	2.0×10^{-2}	6.5×10^{-4}
5	6.5×10^{-2}	3.4×10^{-2}	To be calculated

The rate equation for this reaction is

$$\text{rate} = k[\text{NO}]^2[\text{O}_2]$$

- (i) Use the data from experiment 4 to calculate a value for the rate constant, k , at this temperature, and state its units.
 (ii) Calculate a value for the initial rate in experiment 5.

(4)

(Total 6 marks)

8. (a) The initial rate of the reaction between substances **P** and **Q** was measured in a series of experiments and the following rate equation was deduced.

$$\text{rate} = k[\text{P}]^2[\text{Q}]$$

- (i) Complete the table of data below for the reaction between **P** and **Q**.

Experiment	Initial [P] / mol dm ⁻³	Initial [Q] / mol dm ⁻³	Initial rate / mol dm ⁻³ s ⁻¹
1	0.20	0.30	4.8×10^{-3}
2	0.10	0.10	
3	0.40		9.6×10^{-3}
4		0.60	19.2×10^{-3}

- (ii) Using the data from experiment 1, calculate a value for the rate constant, k , and deduce its units.

(6)

- (b) What change in the reaction conditions would cause the value of the rate constant to change?

(1)

(Total 7 marks)

9. (a) The following data were obtained in a series of experiments on the rate of the reaction between compounds **A** and **B** at a constant temperature.

Experiment	Initial concentration of A /mol dm ⁻³	Initial concentration of B /mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
1	0.15	0.24	0.45×10^{-5}
2	0.30	0.24	0.90×10^{-5}
3	0.60	0.48	7.20×10^{-5}

- (i) Show how the data in the table can be used to deduce that the reaction is first-order with respect to **A**.
 (ii) Deduce the order with respect to **B**.

(2)

- (b) The following data were obtained in a second series of experiments on the rate of the reaction between compounds **C** and **D** at a constant temperature.

Experiment	Initial concentration of A /mol dm ⁻³	Initial concentration of B /mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
4	0.75	1.50	9.30×10^{-5}
5	0.20	0.10	To be calculated

The rate equation for this reaction is

$$\text{rate} = k[\text{C}]^2[\text{D}]$$

- (i) Use the data from Experiment 4 to calculate a value for the rate constant, k , at this temperature. State the units of k .
 (ii) Calculate the value of the initial rate in Experiment 5.

(4)**(Total 6 marks)**

10. (a) The initial rate of the reaction between substances **A** and **B** was measured in a series of experiments and the following rate equation was deduced.

$$\text{rate} = k[\text{A}][\text{B}]$$

Expt	Initial [A] /mol dm ⁻³	Initial [B] /mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
1	0.30	0.30	1.5×10^{-2}
2		0.60	6.0×10^{-2}
3	0.45		9.0×10^{-2}
4	0.90	0.60	

- (i) Complete the table of data above.
 (ii) Using the data from experiment 1, calculate a value for the rate constant, k , and state its units.

(6)

- (b) Explain why the rate of the reaction between magnesium and dilute hydrochloric acid is increased much more by changing the magnesium from ribbon to powder than by doubling the concentration of the acid.

(2)**(Total 8 marks)**

11. (a) The following table shows the results of three experiments to investigate the rate of the reaction between compounds **J** and **K**. All three experiments were carried out at the same temperature.

	Experiment 1	Experiment 2	Experiment 3
Initial concentration of J /mol dm ⁻³	0.50×10^{-3}	1.00×10^{-3}	1.50×10^{-3}
Initial concentration of K /mol dm ⁻³	1.00×10^{-3}	1.00×10^{-3}	1.50×10^{-3}
Initial rate/mol dm ⁻³ s ⁻¹	0.15×10^{-3}	0.60×10^{-3}	1.35×10^{-3}

Use the data in the table to deduce the order of reaction with respect to **J** and the order of reaction with respect to **K**. Hence write an overall rate equation for the reaction.

(3)

- (b) In a reaction between compounds **L** and **M**, the order of reaction with respect to **L** is two and the order of reaction with respect to **M** is one. Given that the initial rate of reaction is 4.00×10^{-4} mol dm⁻³ s⁻¹ when the initial concentration of **L** is 2.00×10^{-2} mol dm⁻³ and the initial concentration of **M** is 5.00×10^{-2} mol dm⁻³, calculate the value of the rate constant for this reaction at this temperature and deduce its units.

(3)

- (c) The rate equation for the decomposition of a compound **N** has a rate constant with the unit s⁻¹. The rate constant is 4.31×10^3 s⁻¹ at 700 K and 1.78×10^4 s⁻¹ at a temperature *T*. Use this information to deduce the overall order of reaction and whether temperature *T* is greater or smaller than 700 K.

(3)

(Total 9 marks)

12. (a) A chemical reaction is first order with respect to compound **X** and second order with respect to compound **Y**.
- Write the rate equation for this reaction.
 - What is the overall order of this reaction?
 - By what factor will the rate increase if the concentrations of **X** and **Y** are **both** doubled?

(4)

- (b) The table below shows the initial concentrations of two compounds, **A** and **B**. And also the initial rate of the reaction that takes place between them at constant temperature.

Experiment	[A]/mol dm ⁻³	[B]/mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
1	0.2	0.2	3.5×10^{-4}
2	0.4	0.4	1.4×10^{-3}
3	0.8	0.4	5.6×10^{-3}

- Determine the overall order of the reaction between **A** and **B**. Explain how you reached your conclusion.
- Determine the order of reaction with respect to compound **B**. Explain how you reached your conclusion.
- Write the rate equation for the overall reaction.
- Calculate the value of the rate constant, stating its units.

(7)

(Total 11 marks)

13. The hydrolysis of methyl propanoate was studied in acidic conditions at 25°C and the rate equation was found to be

$$\text{rate} = k[\text{CH}_3\text{CH}_2\text{COOCH}_3][\text{H}^+]$$

- (a) Use the data below to calculate the value of the rate constant, *k*, at this temperature.

Deduce its units.

Initial rate of reaction / $\text{mol dm}^{-3} \text{ s}^{-1}$	Initial concentration of methyl propanoate / mol dm^{-3}	Initial concentration of hydrochloric acid / mol dm^{-3}
1.15×10^{-4}	0.150	0.555

(3)

- (b) The reaction in part (a) was repeated at the same temperature, but water was added so that the volume of the reaction mixture was doubled. Calculate the initial rate of reaction under these conditions.

(1)

(Total 4 marks)

14. (a) The following table shows the results of three experiments carried out at the same temperature to investigate the rate of the reaction between compounds **P** and **Q**.

	Experiment 1	Experiment 2	Experiment 3
Initial concentration of P /mol dm^{-3}	0.50	0.25	0.25
Initial concentration of Q /mol dm^{-3}	0.36	0.36	0.72
Initial rate/mol $\text{dm}^{-3} \text{ s}^{-1}$	7.6×10^{-3}	1.9×10^{-3}	3.8×10^{-3}

Use the data in the table to deduce the order with respect to **P** and the order with respect to **Q**.

(2)

- (b) In a reaction between **R** and **S**, the order of reaction with respect to **R** is one, the order of reaction with respect to **S** is two and the rate constant at temperature T_1 has a value of $4.2 \times 10^{-4} \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$.
- (i) Write a rate equation for the reaction. Calculate a value for the initial rate of reaction when the initial concentration of **R** is 0.16 mol dm^{-3} and that of **S** is 0.84 mol dm^{-3} .
- (ii) In a second experiment performed at a different temperature, T_2 , the initial rate of reaction is $8.1 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$ when the initial concentration of **R** is 0.76 mol dm^{-3} and that of **S** is 0.98 mol dm^{-3} . Calculate the value of the rate constant at temperature T_2 .
- (iii) Deduce which of T_1 and T_2 is the higher temperature.

(6)

(Total 8 marks)

15. (a) The initial rate of the reaction between compounds **A** and **B** was measured in a series of experiments at a fixed temperature. The following rate equation was deduced.

$$\text{rate} = k[\text{A}][\text{B}]^2$$

- (i) Complete the table of data below for the reaction between **A** and **B**.

Expt	Initial [A] /mol dm^{-3}	Initial [B] /mol dm^{-3}	Initial rate /mol $\text{dm}^{-3} \text{ s}^{-1}$
1	4.80×10^{-2}	6.60×10^{-2}	10.4×10^{-3}
2	4.80×10^{-2}	3.30×10^{-2}	

3		13.2×10^{-2}	5.20×10^{-3}
4	1.60×10^{-2}		10.4×10^{-3}

- (ii) Using the data for experiment 1, calculate a value for the rate constant, k , and state its units.

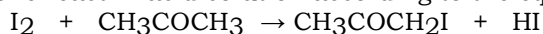
(6)

- (b) State how the value of the rate constant, k , would change, if at all, if the concentration of **A** were increased in a series of experiments.

(1)

(Total 7 marks)

16. Iodine and propanone react in acid solution according to the equation



The rate equation for the reaction is found to be

$$\text{rate} = k [\text{CH}_3\text{COCH}_3][\text{H}^+]$$

- (a) Deduce the order of reaction with respect to iodine and the overall order of reaction.

(2)

- (b) At the start of the experiment, the rate of reaction was found to be $2.00 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$ when the concentrations of the reactants were as shown below.

Reactant	Concentration / mol dm^{-3}
CH_3COCH_3	1.50
I_2	2.00×10^{-2}
H^+	3.00×10^{-2}

Use these data to calculate a value for the rate constant and deduce its units.

(3)

- (c) How can you tell that H^+ acts as a catalyst in this reaction?

(2)

(Total 7 marks)

17. The reaction between hydrogen and iodine can be represented by the following equation:



Kinetic data are available for both forward and reverse reactions. A series of experiments to investigate the kinetics of the forward reaction was carried out at a constant temperature. The results are shown in **Figure 1**.

Experiment	Initial concentration of H_2 / mol dm^{-3}	Initial concentration of I_2 / mol dm^{-3}	Initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	2.0×10^{-3}	3.0×10^{-3}	1.2×10^{-6}
2	6.0×10^{-3}	3.0×10^{-3}	3.6×10^{-6}
3	6.0×10^{-3}	6.0×10^{-3}	7.2×10^{-6}

Figure 1

- (a) Explain the meaning of each of the following terms:

- (i) rate of reaction;

(1)

- (ii) rate constant;

(1)

- (iii) overall order of reaction.

(2)

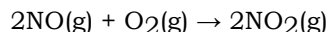
- (b) (i) Use the results in **Figure 1** to work out the order of the forward reaction with respect to both hydrogen and iodine, explaining your reasoning.

- (ii) Write the rate equation for the reaction between hydrogen and iodine. (4)
- (iii) Calculate the value of the rate constant for the forward reaction and give its units. (1)
- (c) The experiment was repeated with the following initial concentrations for reactants: (3)
- $[\text{H}_2] = 4.0 \times 10^{-3} \text{ mol dm}^{-3}$ $[\text{I}_2] = 5.0 \times 10^{-3} \text{ mol dm}^{-3}$
- Calculate the initial rate of the reaction. (1)
- (Total 13 marks)**

18. One cause of low-level smog is the reaction of ozone, O_3 , with ethene, C_2H_4 . The smog contains methanal, HCHO(g) . The equation for methanal production is shown below.
- $$\text{O}_3(\text{g}) + \text{C}_2\text{H}_4(\text{g}) \rightarrow 2\text{HCHO}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g})$$
- The rate of the reaction was investigated, using a series of different concentrations of either $\text{C}_2\text{H}_4(\text{g})$ or $\text{O}_3(\text{g})$, by measuring the initial rate of formation of HCHO(g) . The results are shown below.

experiment	$[\text{O}_3(\text{g})]$ / $10^{-7} \text{ mol dm}^{-3}$	$[\text{C}_2\text{H}_4(\text{g})]$ / $10^{-8} \text{ mol dm}^{-3}$	initial rate / $10^{-12} \text{ mol dm}^{-3} \text{ s}^{-1}$
1	0.5	1.0	1.0
2	2.0	1.0	4.0
3	4.0	2.0	16.0

- (i) Analyse and interpret the results to deduce the order of reaction of each reactant and the rate equation. Explain your reasoning. (5)
- (ii) Calculate the value of the rate constant and state the units. (3)
- (iii) Using the equation above, deduce the initial rate of **formation** of $\text{O}_2(\text{g})$ in experiment 1. Explain your reasoning. (1)
- (iv) The experiment was repeated at a higher temperature. How would the new conditions affect the rate of the reaction and the value of the rate constant? (1)
- [Total 10 marks]**
19. Nitrogen dioxide is one of the major pollutants in air, formed by reaction of nitrogen monoxide with oxygen.



- (a) What is meant by the *rate of reaction*? (1)
- (b) A series of experiments was carried out to investigate the kinetics of this reaction. The results are shown in the table below.

Experiment	$[\text{O}_2]$ / mol dm^{-3}	$[\text{NO}]$ / mol dm^{-3}	initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.00100	0.00100	7.10
2	0.00400	0.00100	28.4
3	0.00400	0.00300	256

- (i) For each reactant, deduce the order of reaction. Show your reasoning. [4]
- (ii) Deduce the rate equation for this reaction. [1]
- (iii) Calculate the rate constant, k , for this reaction. State the units for k . [2]
- [Total 8 marks]**

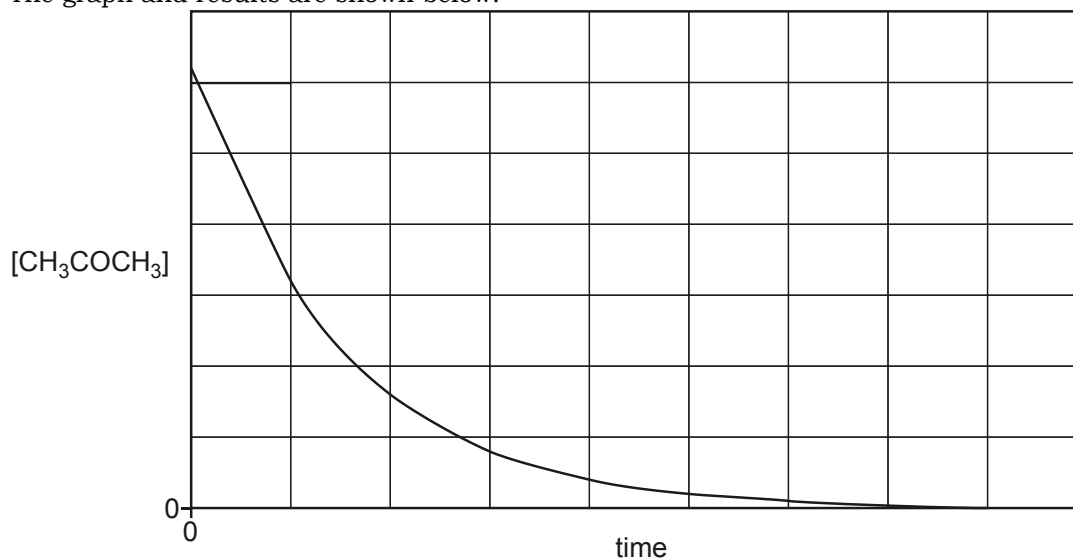
20. In this question, one mark is available for the quality of use and organisation of scientific terms.

Propanone reacts with iodine in the presence of dilute hydrochloric acid.

A student carried out an investigation into the kinetics of this reaction.

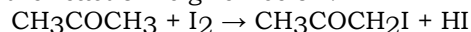
He measured how the concentration of propanone changes with time. He also investigated how different concentrations of iodine and hydrochloric acid affect the initial rate of the reaction.

The graph and results are shown below.



$[\text{CH}_3\text{COCH}_3]$ / mol dm^{-3}	$[\text{I}_2]$ / mol dm^{-3}	$[\text{H}^+]$ / mol dm^{-3}	initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1.5×10^{-3}	0.0300	0.0200	2.1×10^{-9}
1.5×10^{-3}	0.0300	0.0400	4.2×10^{-9}
1.5×10^{-3}	0.0600	0.0400	4.2×10^{-9}

The overall equation for the reaction is given below.



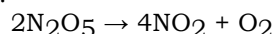
This is a multi-step reaction.

- What conclusions can be drawn about the kinetics of this reaction from the student's investigation? Justify your reasoning.
- Calculate the rate constant for this reaction, including units.
- Suggest the equations for a possible two-step mechanism for this reaction. Label the rate-determining step and explain your reasoning.

Quality of Written Communication [1]

[Total 14 marks]

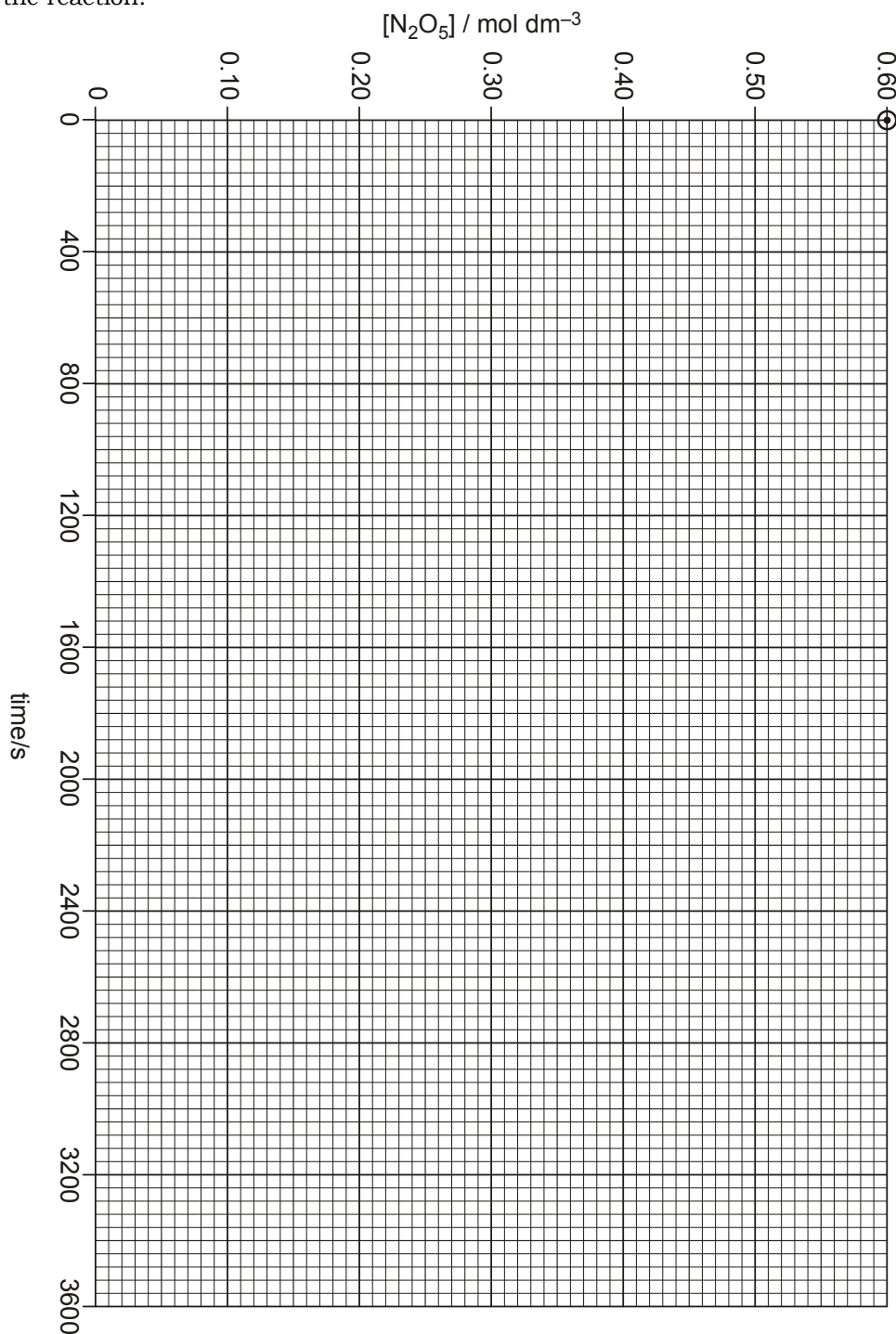
21. The decomposition of dinitrogen pentoxide, N_2O_5 , at 45°C was investigated. The reaction that takes place is shown below.



In an experiment, N_2O_5 with a concentration of 0.60 mol dm^{-3} was decomposed at 45°C .

At this temperature, the reaction has a constant half-life of 1200 s.

- (i) How can you tell that this reaction is first order with respect to N_2O_5 ? [1]
- (ii) Write down an expression for the rate equation of this decomposition. [1]
- (iii) Complete the graph below to show how the $[\text{N}_2\text{O}_5]$ changes over the first 3600 s of the reaction.

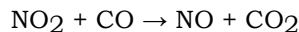


- (iv) The rate of this reaction can be determined from this graph. [2]
Show on the graph how the rate can be measured after 1200 s.
- (v) The rate can also be calculated from the rate equation. The rate constant for this reaction is $6.2 \times 10^{-4} \text{ s}^{-1}$. [1]
Calculate the initial rate of this reaction. State the units.

[2]

[Total 7 marks]

22. Nitrogen dioxide reacts with carbon monoxide emitted from car exhausts in the following reaction.



The rate equation for this reaction is $\text{rate} = k[\text{NO}_2]^2$.

This is a multi-step reaction. The first step is the rate-determining step.

- What is meant by the *rate-determining step*?
- Suggest a two-step reaction mechanism for this reaction that is consistent with the kinetic data and the overall reaction.

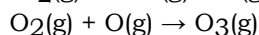
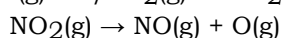
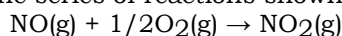
[1]

[2]

[Total 3 marks]

23. Nitrogen monoxide, NO, is involved in formation of ozone at low levels and the breakdown of ozone at high levels.

- In the lower atmosphere, NO is produced by combustion in car engines. Ozone is then formed following the series of reactions shown below.

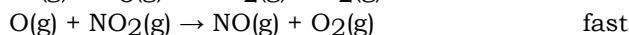
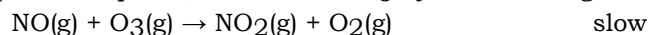


Write the overall equation for this reaction sequence.

Identify the catalyst and justify your answer.

[3]

- In the upper atmosphere, NO removes O₃ by the following reaction mechanism.



Suggest the rate equation for this process. Explain your reasoning.

[2]

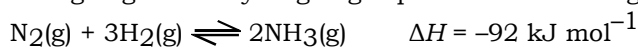
[Total 5 marks]

24. Some ammonia plants are run at 200–300 atm and 500 °C, with an iron catalyst.

- The hydrogen for the plants is obtained by reacting methane with steam. Construct a possible equation for this reaction.

[1]

- Nitrogen gas and hydrogen gas produce ammonia gas as shown below.



- Write the expression for K_c for this equilibrium.

[1]

- At 500 °C, $K_c = 8.00 \times 10^{-2} \text{ dm}^6 \text{ mol}^{-2}$.

At equilibrium, the concentration of N₂ is 1.20 mol dm^{-3} and the concentration of H₂ is 2.00 mol dm^{-3} .

Calculate the equilibrium concentration of ammonia under these conditions.

[3]

- In this question one mark is available for the quality of the use and organisation of scientific terms.

Discuss the advantages and disadvantages of running this reaction

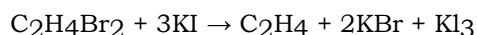
- at a pressure of 200–300 atm;
- at a temperature of 500°C;
- with an iron catalyst.

[6]

Quality of Written Communication [1]

[Total 12 marks]

25. 1, 2-Dibromoethane, C₂H₄Br₂, reacts with potassium iodide as shown in the equation below.

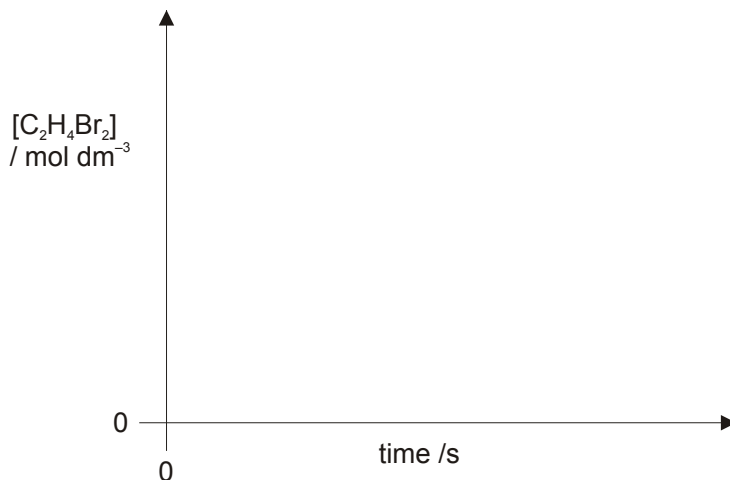


A series of experiments was carried out to investigate the kinetics of this reaction.

- In a first experiment the concentration of C₂H₄Br₂ was measured during the course of the reaction and a concentration-time graph was plotted. The reaction was

shown to be first order with respect to $\text{C}_2\text{H}_4\text{Br}_2$.

- (i) On the axes below, sketch a graph to show how $[\text{C}_2\text{H}_4\text{Br}_2]$ changed during the course of the reaction.

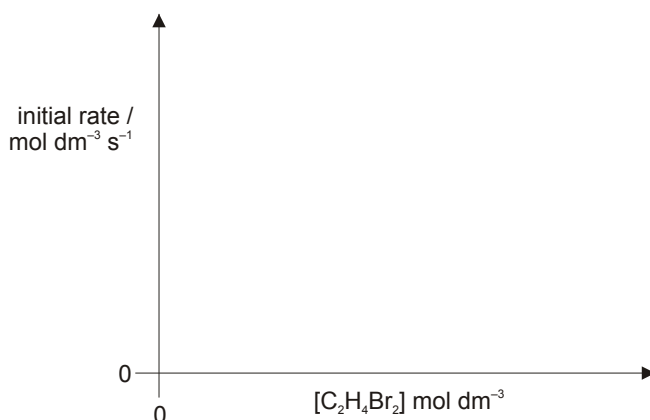


- (ii) Show on the graph how you would measure the **initial** rate of the reaction. [2]

- (iii) How would you use the graph to show that the reaction is first order with respect to $\text{C}_2\text{H}_4\text{Br}_2$? [1]

- (iv) The experiment was repeated using different initial concentrations of $\text{C}_2\text{H}_4\text{Br}_2$. [1]

Using the axes below, sketch a graph to show how the initial rate of the reaction changes with different concentrations of $\text{C}_2\text{H}_4\text{Br}_2$.



- (b) In a second experiment, the initial concentration of KI was varied and the initial rate was measured. The results are shown in the table below. [1]

experiment	$[\text{C}_2\text{H}_4\text{Br}_2]$ $/\text{mol dm}^{-3}$	$[\text{KI}]$ $/\text{mol dm}^{-3}$	initial rate $/\text{mol dm}^{-3} \text{s}^{-1}$
1	0.50	0.18	0.027
2	0.50	0.72	0.108

Deduce the order of reaction with respect to KI. Show your reasoning.

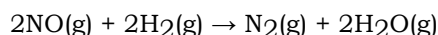
- (c) (i) Construct the rate equation for the reaction. [2]

- (ii) Calculate the rate constant, k , for this reaction. State the units for k . [1]

[3]

[Total 11 marks]

26. Nitrogen monoxide reacts with hydrogen at 500°C as in the equation below.



A series of experiments was carried out to investigate the kinetics of this reaction. The results are shown in the table below.

experiment	[NO] / mol dm ⁻³	[H ₂] / mol dm ⁻³	initial rate / mol dm ⁻³ s ⁻¹
1	0.10	0.20	2.6
2	0.10	0.50	6.5
3	0.30	0.50	58.5

In this question, one mark is available for the quality of spelling, punctuation and grammar.

- (i) For each reactant, deduce the order of reaction. Show your reasoning.

[4]

Quality of Written Communication [1]

- (ii) Deduce the rate equation for this reaction.

[1]

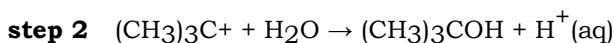
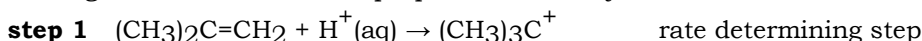
- (iii) Calculate the rate constant, k , for this reaction. State the units for k .

[3]

[Total 9 marks]

27. A student investigated the hydration of 2-methylpropene, $(\text{CH}_3)_2\text{C}=\text{CH}_2$, with dilute aqueous acid to form 2-methylpropan-2-ol, $(\text{CH}_3)_3\text{COH}$.

The following mechanism has been proposed for this hydration.



- (i) Step 1 is the rate-determining step for this hydration.
What is meant by the term *rate-determining step*?

[1]

- (ii) Write a balanced equation for the overall hydration reaction.

[1]

- (iii) Suggest the role of $\text{H}^+(\text{aq})$ in this mechanism. Explain your reason.

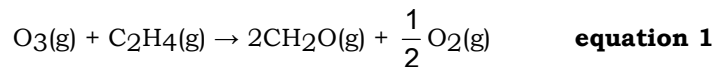
[2]

- (iv) Use the mechanism above to suggest the rate equation for this hydration.

[1]

[Total 5 marks]

28. One cause of low-level smog is the reaction of ozone, O_3 , with ethene. The smog contains methanal, $\text{CH}_2\text{O}(\text{g})$, and the equation for its production is shown below.



- (a) The rate of the reaction doubles when the initial concentration of either $\text{O}_3(\text{g})$ or $\text{C}_2\text{H}_4(\text{g})$ is doubled.

- (i) What is the order of reaction with respect to

[1]

- (ii) What is the overall order of the reaction?

[1]

- (iii) Write the rate equation for this reaction.

[1]

- (b) For an initial concentration of ozone of $0.50 \times 10^{-7} \text{ mol dm}^{-3}$ and one of ethene of $1.0 \times 10^{-8} \text{ mol dm}^{-3}$, the initial rate of methanal formation was $1.0 \times 10^{-12} \text{ mol dm}^{-3} \text{ s}^{-1}$.

- (i) How could the **initial** rate of methanal formation be measured from a concentration/time graph?

[2]

- (ii) Calculate the value of the rate constant and state the units.

[3]

- (iii) The initial rate of methanal formation is different from that of oxygen

formation in **equation 1**.
Explain why.

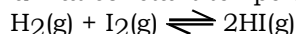
[1]

- (iv) The experiment was repeated but at a higher temperature. What would be the effect of this change on the rate and the rate constant of the reaction?

[2]

[Total 11 marks]

29. The preparation of hydrogen iodide, HI(g) , from hydrogen and iodine gases is a reversible reaction which reaches equilibrium at constant temperature.



- (a) Write the expression for K_c for this equilibrium.

[1]

- (b) A student mixed together 0.30 mol $\text{H}_2(\text{g})$ with 0.20 mol $\text{I}_2(\text{g})$ and the mixture was allowed to reach equilibrium. At equilibrium, 0.14 mol $\text{H}_2(\text{g})$ was present.

- (i) Complete the table below to show the amount of each component in the equilibrium mixture.

component	$\text{H}_2(\text{g})$	$\text{I}_2(\text{g})$	$\text{HI}(\text{g})$
initial amount / mol	0.30	0.20	0
equilibrium amount / mol			

[2]

- (ii) Calculate K_c to an appropriate number of significant figures. State the units, if any.

[3]

- (c) The student compressed the equilibrium mixture so that its volume was reduced. The temperature was kept constant.
Comment on the value of K_c **and** the composition of the equilibrium mixture under these new conditions.

[2]

[Total 8 marks]

30. When heated, phosphorus pentachloride, PCl_5 , dissociates.



A chemist placed a mixture of the three gases into a container. The initial concentration of each gas was the same: 0.30 mol dm^{-3} . The container was left until equilibrium had been reached.

Under these conditions, $K_c = 0.245 \text{ mol dm}^{-3}$.

- (a) Write an expression for K_c for this equilibrium.

[1]

- (b) Use the value of K_c for this equilibrium to deduce whether the concentration of each gas increases, decreases or stays the same as the mixture approaches equilibrium.
(i) Show your answer by placing a tick in the appropriate cells in the table below.

	initial concentration / mol dm^{-3}	greater than 0.30 mol dm^{-3}	less than 0.30 mol dm^{-3}	equal to 0.30 mol dm^{-3}
PCl_5	0.30			
PCl_3	0.30			
Cl_2	0.30			

[1]

- (ii) Explain your deduction.

[1]

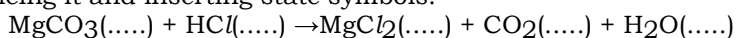
- (c) The chemist compressed the equilibrium mixture at constant temperature and allowed it to reach equilibrium under these new conditions.

- (i) Explain what happens to the value of K_C . [1]
- (ii) Explain what happened to the composition of the equilibrium mixture. [2]
- (d) The chemist heated the equilibrium mixture and the equilibrium moved to the left.
- (i) Explain what happens to the value of K_C . [1]
- (ii) Explain what additional information this observation reveals about the reaction. [2]

[Total 9 marks]

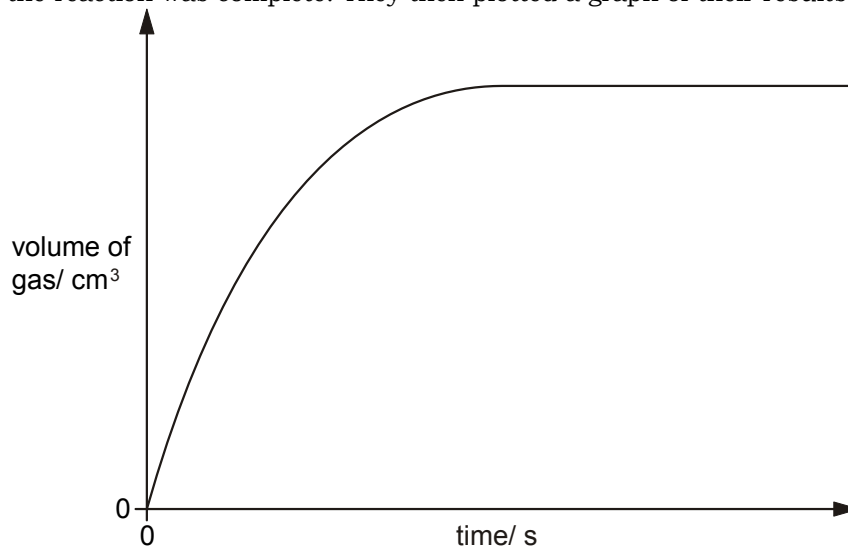
31. A group of students investigated the effect of concentration on the rate of a reaction. They used the reaction between magnesium carbonate and dilute hydrochloric acid and measured the rate at which the gas was collected.

- (a) An incomplete equation for this reaction is given below. Complete the equation by balancing it and inserting state symbols.



[2]

- (b) The students added dilute hydrochloric acid to some magnesium carbonate. The students collected the gas and measured the volume, at regular intervals, until after the reaction was complete. They then plotted a graph of their results.

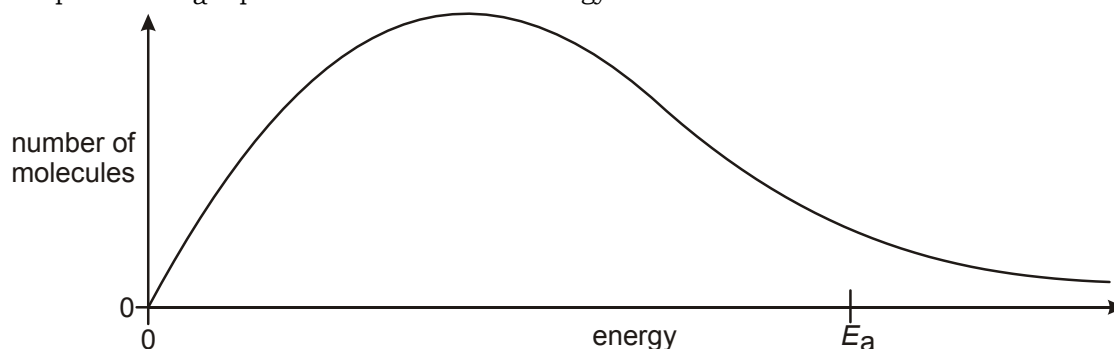


Use collision theory to explain the changes in the rate of the reaction as it proceeds.

[3]

[Total 5 marks]

- 32.** The diagram below shows the energy distribution of molecules at a particular temperature. E_a represents the activation energy of the reaction.



- (i) On the diagram, draw a second curve to represent the energy distribution of the same number of molecules at a higher temperature.

- (ii) Using your completed diagram, explain how an increase in temperature causes the rate of reaction to increase. [2]

[2]
[Total 4 marks]

33. The Boltzmann distribution can be used to show the effect of a change in temperature on the rate of a reaction.

- ☐ Draw a labelled Boltzmann distribution diagram.
- ☐ Explain the essential features of your diagram.
- ☐ Using your diagram, explain how an increase in temperature affects the rate of a reaction.

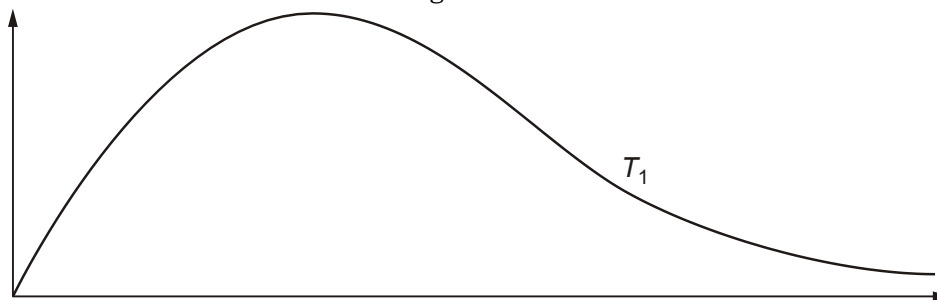
[Total 9 marks]

34. (a) A group of students were considering the factors that affect the rate of a chemical reaction. One of the students wrote the following.
The rate of a chemical reaction increases as the temperature is increased because there are more collisions.
Discuss to what extent this statement is true and describe any other factors that should be considered when looking at the effect of temperature on rate of reaction.

[5]

- (b) The students drew a Boltzmann distribution for a gas at temperature T_1 .

- (i) Label the axes on the students' diagram below.

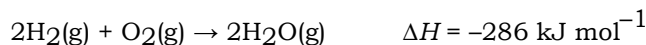


[2]

- (ii) Sketch, on the diagram, the distribution that would be obtained at a **lower** temperature, T_2 . Label this line T_2 .

[2]
[Total 9 marks]

35. The rate of the reaction between hydrogen and oxygen depends on the pressure as well as the temperature.



- (i) Describe and explain the effect of increasing the pressure on the **rate** of this reaction.
- (ii) A sudden rapid increase in the rate of a reaction causes an explosion to occur. Suggest why highly exothermic reactions such as this one are more likely to explode than other reactions.

[2]
[Total 4 marks]

36. In this question, one mark is available for the quality of written communication. What effect does a catalyst have on the rate of a reaction, and how does it achieve this effect?

[4]
Quality of Written Communication [1]
[Total 5 marks]

37. State le Chatelier's principle.

[Total 2 marks]

38. Esters are used as flavourings. They are made by a reversible reaction between a carboxylic acid and an alcohol.



- (a) Give **two** features of a reversible reaction, when a dynamic equilibrium has been set

up.

[2]

- (b) The production of esters is catalysed homogeneously by the presence of acids.

(i) What is meant by a catalyst?

[1]

(ii) Using the fact that acids are needed to catalyse this reaction, deduce the formula of the ion that acts as the catalyst.

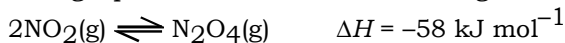
[1]

(iii) Catalysts do **not** affect the position of an equilibrium. Explain why not.

[1]

[Total 5 marks]

39. Nitrogen dioxide, NO_2 , is a brown gas whilst dinitrogen tetroxide, N_2O_4 , is a colourless gas. The following equilibrium between these two gases was set up.



Describe, and explain, what you would **see** after the following changes have been made and the system allowed to reach equilibrium again.

(i) The temperature is increased.

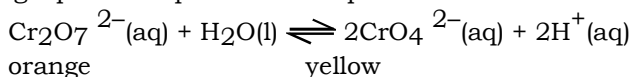
[3]

(ii) The pressure is increased.

[3]

[Total 6 marks]

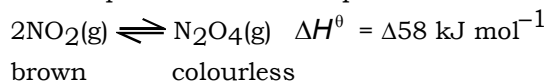
40. The following equation represents an equilibrium reaction.



Use le Chatelier's principle to describe and explain the colour change (if any) that might take place when dilute $\text{HCl}(\text{aq})$ is added to a solution containing $\text{K}_2\text{CrO}_4(\text{aq})$.

[Total 2 marks]

41. The following equation represents another equilibrium reaction.



Use le Chatelier's principle to describe and explain the colour change (if any) that might take place when

(i) a mixture of $\text{NO}_2(\text{g})$ and $\text{N}_2\text{O}_4(\text{g})$ is compressed at constant temperature,

[2]

(ii) a mixture of $\text{NO}_2(\text{g})$ and $\text{N}_2\text{O}_4(\text{g})$ is heated at constant pressure.

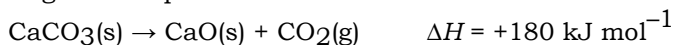
[2]

[Total 4 marks]

42. Limestone contains calcium carbonate, CaCO_3 .

Limestone is an important source of commercially important chemicals such as lime, calcium oxide, CaO .

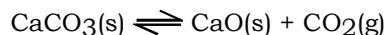
(a) If calcium carbonate is heated strongly in an open container, it decomposes according to the equation below.



Suggest **two** reasons why it is necessary to heat the calcium carbonate strongly to achieve decomposition.

[2]

(b) Another sample of calcium carbonate was placed in a closed container before being heated strongly. This allows an equilibrium to be set up. This equilibrium is shown below.



(i) What can you say about the rates of the forward and reverse reactions when the calcium carbonate starts to decompose?

[1]

(ii) What can you say about the rates of the forward and reverse reactions when the equilibrium has been established?

[1]

(iii) A valve allowed some of the carbon dioxide to escape.

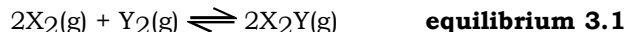
State and explain what happens to the composition of the mixture in the

container.

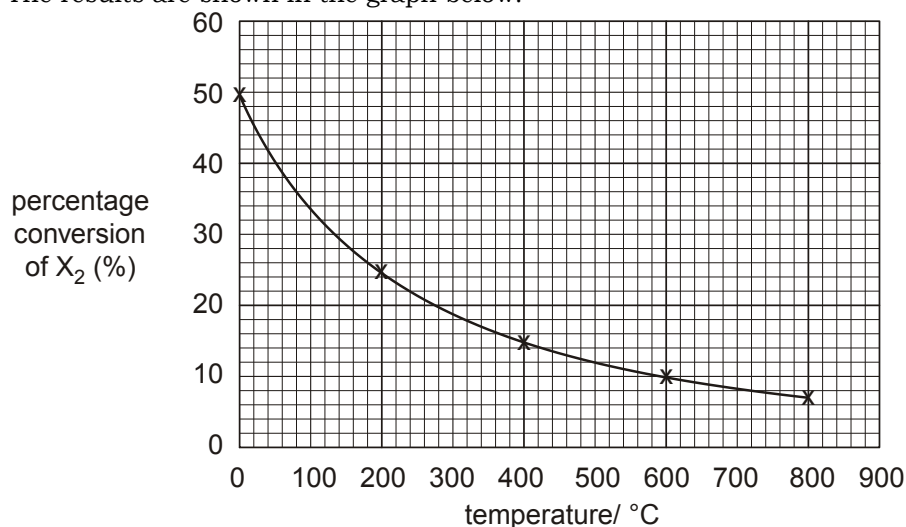
[3]

[Total 7 marks]

43. Many industrial processes, used to manufacture important chemicals, involve equilibrium reactions. Chemists use their understanding of rates of reaction and of yields at equilibrium to find the most economic conditions for the reactions. Chemists were investigating the production of a chemical, X_2Y , that could be formed from X_2 and Y_2 as shown in **equilibrium 3.1** below.



- (a) State le Chatelier's principle. [2]
- (b) State and explain the effect on **equilibrium 3.1** of a **decrease** in pressure on:
- the equilibrium position of the reaction, [2]
 - the rate of the reaction. [2]
- (c) The chemists measured the percentage conversion of X_2 at various temperatures. The results are shown in the graph below.

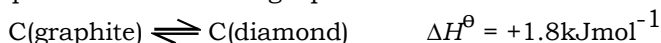


- Use the graph to predict the percentage conversion at 350 °C. [1]
 - The forward reaction in **equilibrium 3.1** is exothermic. Explain how the graph supports this statement. [2]
- (d) The chemists decided to use a catalyst in the process. State, and explain, the effect of using a catalyst on:
- the rate of conversion of X_2 and Y_2 into X_2Y , [2]
 - the percentage conversion at equilibrium of X_2 and Y_2 into X_2Y . [2]

[2]

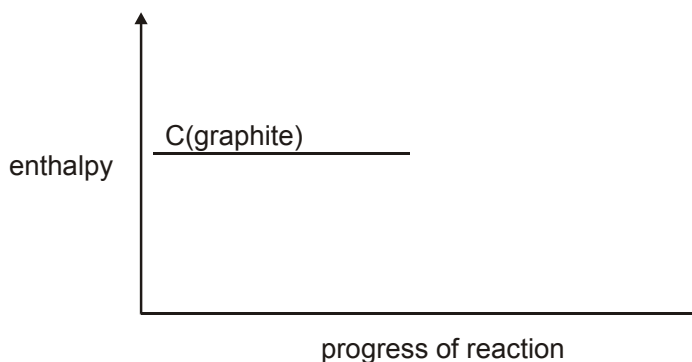
[Total 13 marks]

44. Diamond and graphite are two allotropes of carbon. Allotropes are different forms of the same element. Diamonds are made industrially from the much cheaper graphite. Under high pressure the following equilibrium exists.



substance	Density / gem^{-3}
graphite	2.25
diamond	3.51

- (a) Complete the following enthalpy profile diagram for the conversion of graphite into diamond, labelling ΔH^θ .



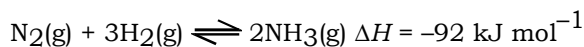
- (b) Suggest which allotrope of carbon is thermodynamically more stable. Give a reason for your choice. [2]
- (c) In which allotrope do the carbon atoms take up less space? Give a reason for your choice. [1]
- (d) State Le Chatelier's principle. [1]
- (e) Use Le Chatelier's principle and the information given at the start of the question to deduce the likely conditions of pressure and temperature needed to turn graphite into diamond. Explain your answer. [2]

[2]

[2]

[Total 10 marks]

- 45.** Ammonia, NH_3 , is made industrially by the Haber process. This is an equilibrium reaction.



Describe and explain why these conditions are a compromise between rate and equilibrium.

Marking Scheme

- 1.** (a) Increased surface area **(1)**
more collisions **(1)** 2
- (b) (i) Experiment 2 = 9.6×10^{-4} **(1)**
Experiment 3 = 0.010 **(1)**
Experiment 4 = 8.1×10^{-4} **(1)**
Experiment 5 = 0.035 **(1)**
- (ii) $k = \frac{1.2 \times 10^{-4}}{(0.020)(0.020)^2}$ **(1)** = 15 **(1)** mol⁻² dm⁶ s⁻¹ **(1)** 7
- [9]**
- 2.** (a) order wrt A = 1; 1
order wrt NaOH = 1; 1
Initial rate in Exp 4 = 2.4×10^{-3} ; 1
- (b) (i) r(ate) = k[A]
OR
r(ate) = k[A][NaOH]⁰; 1
(penalise missing [] but mark on)
(penalise missing [] once per paper)
(if wrong order, allow only units mark conseq on their rate eqs)
(penalise k_a or k_w etc)
- (ii) $k = \frac{9.0 \times 10^{-3}}{0.02}$; 1
= 0.45; 1
s⁻¹; 1
- (iii) (large) excess of OH⁻ or [OH⁻] is large/high; 1
[OH⁻] is (effectively) constant
OR
[A] is the limiting factor *(Q of L mark)* 1
- [9]**
- 3.** (a) Power (or index or shown as *x* in []^{*x*}) of concentration term
(in rate equation) **(1)** 1
- (b) 2 **(1)** 1
- (c) (i) Order with respect to **A**: 2 **(1)**
Order with respect to **B**: 0 **(1)**
- (ii) Rate equation: (rate =) k [A]² **(1)**
Allow conseq on c(i)
Units for rate constant: mol⁻¹ dm³ s⁻¹ **(1)** 4
conseq on rate equation
- [6]**
- 4.** (a) Order with respect to **A** 1 **(1)**
Order with respect to **B** 1 **(1)**
Order with respect to **C** 2 **(1)**
- 3**

(b) Value of k $K = \frac{8.0 \times 10^{-5}}{(0.1)(0.2)(0.2)^2} = 0.1$
 (1) (1)
 Units of k $\text{mol}^{-3} \text{dm}^9 \text{s}^{-1}$ (1)
 Initial rate $1.0 \times 10^{-5} (\text{mol dm}^{-3} \text{s}^{-1})$
 (1)

(c) increases (1)

4
1

[8]

5. (a)

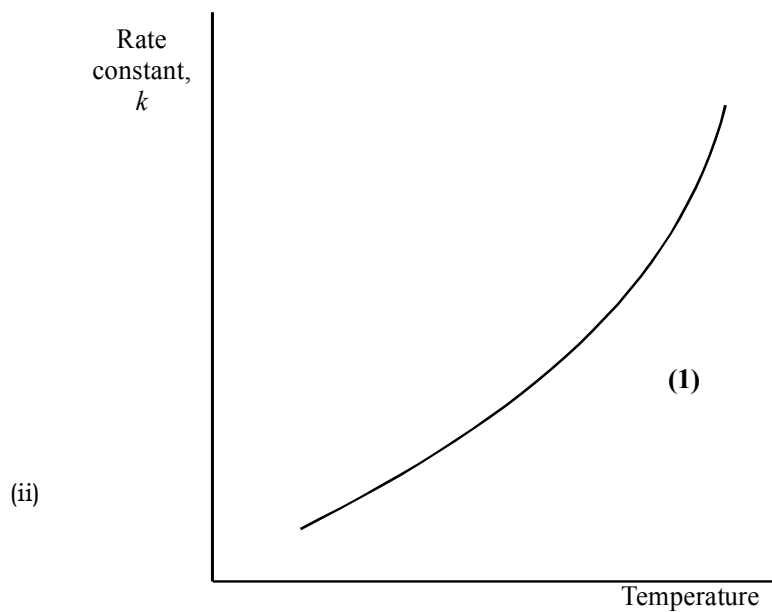
Substances added to an excess of zinc and 100 cm ³ of 0.2 M hydrochloric acid	Volume of hydrogen/cm ³	Effect on initial rate of reaction
100cm ³ water	240 (1)	decreased (1)
10g zinc	240 (1)	no change (1)
50 cm ³ 0.2 M hydrochloric acid	360 (1)	no change (1)

6

(b) Order with respect to A 1 (1)
 Order with respect to B 1 (1)
 Initial rate $2.8 \times 10^{-5} (\text{mol dm}^{-3} \text{s}^{-1})$ (1)
 either via $k = 1.56 \times 10^{-3}$ (1)
 or via table eg expts 2 & 4: $\text{rate} \times \frac{1}{2} \times \frac{3}{4} = \times \frac{3}{8}$ (1)

4

(c) (i) Calculation $k = \frac{7.5 \times 10^{-3}}{(0.25)^2 (0.50)^2}$ (1) = 0.48 (1)
 Units $\frac{\text{mol dm}^{-3} \text{s}^{-1}}{(\text{mol dm}^{-3})^2 (\text{mol dm}^{-3})^2} = \text{mol}^{-3} \text{dm}^9 \text{s}^{-1}$ (1)



4

[14]

6. (a) exp2 4.0×10^{-3}
 exp3 0.45×10^{-5}

1
1

	exp4	9.0×10^{-3}	1	
(b)		$\frac{1.8 \times 10^{-5}}{(3.0 \times 10^{-3})^2 (1.0 \times 10^{-3})}$	1	
		2000	1	
		$\text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$	1	[6]
7.	(a)	(i) 2 (1) (ii) 0 (1)	2	
	(b)	(i) Value of k: $k = \frac{\text{rate}}{[\text{NO}]^2 [\text{O}_2]} = \frac{6.5 \times 10^{-4}}{(5.012 \times 10^{-2})^2 (2.0 \times 10^{-2})} = 13$ Units of k: $\text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$ (1) (ii) rate = $13 (6.5 \times 10^{-2})^2 (3.4 \times 10^{-2})$ $= 1.9 \times 10^{-3} (\text{mol dm}^{-3} \text{s}^{-1})$ (1) If k wrong, the mark in (ii) may be gained conseq for their k $\times$ 1.437×10^{-4}	4	
				[6]
8.	(a)	(i) Experiment 2: $0.4(0) \times 10^{-3}$ (1) Experiment 3: 0.15 (1) Experiment 4: 0.28 (1) (ii) $k = \frac{4.8 \times 10^{-3}}{(0.20)^2 \times (0.30)} = 0.4(0) \text{ mol}^{-2} \text{dm}^6 \text{s}^{-1}$ (1) (1) (1)	6	
	(b)	(change in) <u>temperature</u> (1)	1	
				[7]
9.	(a)	(i) (Experiment 1 & 2) [A] doubled, ([B] constant,) <u>rate</u> doubled (1) stated or shown numerically (ii) 2 (1) or shown as ... $[\text{B}]^2$	2	
	(b)	(i) $k = \frac{9.30 \times 10^{-5}}{(0.75)^2 \times (1.50)} = 1.1(0) \times 10^{-4}$ (1) (1) units of k: $\text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$ (1) (ii) rate = $(1.10 \times 10^{-4}) \times (0.20)^2 \times (0.10)$ $= 4.4(1) \times 10^{-7} (\text{mol dm}^{-3} \text{s}^{-1})$ (1) for the answer Ignore units Conseq on (i) Upside down expression for k scores zero in (i) for 9073 but rate = $9073 \times (0.2)^2 \times (0.1) = 36(3)$ conseq scores (1) in (ii)	4	
				[6]

10. (a) (i)

Expt	Initial [A]/mol dm ⁻³	Initial [B]/mol dm ⁻³	Initial rate/mol dm ⁻³ s ⁻¹
1	0.30	0.30	1.5×10^{-2}
2	0.60 (1) (0.58 to 0.63)	0.60	6.0×10^{-2}
3	0.45	1.20 (1) (1.17 to 1.25)	9.0×10^{-2}
4	0.90	0.60	9.0×10^{-2} (1) (8.6 to 9.2 $\times 10^{-2}$)

(ii) $K = \frac{\text{rate}}{[A][B]} = \frac{1.5 \times 10^{-2}}{0.3 \times 0.3} \text{ (1)} = 0.16\dot{6} \text{ (1) (or 0.17 or 0.1\dot{6})}$
 units: $\text{mol}^{-1} \text{dm}^3 \text{s}^{-1} \text{ (1)}$

6

(b) surface area more (than doubled) **(1)**
 many more collisions **(1)**

2

[8]11. (a) 2 **(1)**
0 **(1)**

$$\text{rate} = k[J]^2 \text{ (1)}$$

3

(b) $k = \frac{4 \times 10^{-4}}{(2 \times 10^{-2})^2 (5 \times 10^{-2})} \text{ (1)} = 20 \text{ (1)}$
 $\text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$

3

(c) $\text{rate} = k []^n \quad \square []^n = \frac{\text{rate}}{k}$

units: $\frac{\text{mol dm}^{-3} \text{s}^{-1}}{\text{s}^{-1}} = \text{mol dm}^{-3}$ therefore $n = 1 \text{ (2)}$

greater/increase **(1)**

3

[9]12. (a) (i) $\text{Rate} = k[X][Y]^2 \text{ (2)}$ (ii) 3 **(1)**(iii) 8 **(1)**

4

(b) (i) Overall order of reaction 2 **(1)**

Explanation doubling both [A] and [B]
 leads to Rate $\times 4 \text{ (1)}$

(ii) Order with respect to B 0 **(1)**

Explanation doubling [A] at constant [B]
 also leads to rate $\times 4 \text{ (1)}$

(iii) $\text{Rate} = k[A]^2 \text{ (1)}$

(iv) $k = \frac{\text{Rate}}{[A]^2} = \frac{3.5 \times 10^{-4}}{(0.2)^2} \text{ (1)}$

$$= 8.75 \times 10^{-3} \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1} \quad (1) \quad 7$$

[11]

13. (a) $k = \text{rate} / [\text{CH}_3\text{CH}_2\text{COOCH}_3][\text{H}^+]$ or
 $= \frac{1.15 \times 10^{-4}}{(0.150)(0.555)}$ 1
 $= 1.38 \times 10^{-3} \text{ to } 1.4 \times 10^{-3}$ 1
 $\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$ 1
 (b) $\text{ans} = \text{rate constant} \times (\frac{1}{2} \times 0.150) \times (\frac{1}{2} \times 0.555)$ 1
ignore units
 $= \text{rate constant} \times 0.0208$
 $2.88 \times 10^{-5} \text{ (} 1.38 \times 10^{-3} \text{ gives } 2.87 \times 10^{-5} \text{)}$
Allow } 2.87 - 2.91 \times 10^{-5} \text{ (} 1.4 \times 10^{-3} \text{ gives } 2.91 \times 10^{-5} \text{)}

[4]

14. (a) order with respect to **P** is 2 1
 order with respect to **Q** is 1 1
 (b) (i) $\text{rate} = k[\text{R}][\text{S}]^2$ (if wrong expression, no further marks) 1
 $\text{rate} = (4.2 \times 10^{-4}) \times 0.16 \times 0.84^2$ 1
 $= 4.7 \times 10^{-5} \text{ (mol dm}^{-3} \text{ s}^{-1} \text{) ignore units even if wrong}$ 1
 (ii) $k = \frac{\text{rate}}{[\text{R}][\text{S}]^2} = \frac{8.1 \times 10^{-5}}{0.76 \times 0.98^2}$ 1
 $= 1.1 \times 10^{-4} *$ 1
 (iii) T_1 1
 If calculated value for $k > 4.2 \times 10^{-4}$, then answer to (iii) is T_2

[8]

15. (a) (i) Experiment 2 2.60×10^{-3} 1
 Experiment 3 0.60×10^{-2} 1
 Experiment 4 11.4×10^{-2} 1
 (ii) $k = \frac{10.4 \times 10^{-3}}{(4.80 \times 10^{-2})(6.60 \times 10^{-2})^2}$ 1
 $= 49.7$ 1
(Allow 49.8 and 50)
 $\text{mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$ 1
 (b) No change 1

[7]

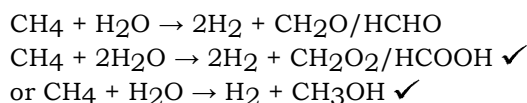
16. (a) Order with respect to iodine: 0 (1)
 Overall order: 2 (1) 2
 (b) Rate constant: $k = \frac{2 \times 10^{-5}}{(1.5) \times (3 \times 10^{-2})} = 4.4(4) \times 10^{-4} \text{ (1)}$
 Units: $\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} \text{ (1)}$ 3
 (c) Appears in rate equation (1)
OR implied by mention of concentration or order
 does not appear in (stoichiometric / overall) equation (1) 2

[7]

17. (a) (i) change in concentration / amount / mass / volume / (partial) pressure
 per unit time / per s (1) 1

- (ii) constant of (proportionality) in the rate equation **(1)**
allow reference to k in rate equation 1
- (iii) sum of
dependent on 2nd mark or reference to 'orders' **(1)**
powers in rate equation **(1)**
allow reference to annotated rate equation 2
- (b) (i) $(\text{H}_2) - 1$ **(1)**
eg rate trebles as $[\text{H}_2]$ trebles **not** rate doubles as $[\text{H}_2]$ doubles **(1)**
 $(\text{I}_2) - 1$ **(1)**
eg rate doubles as $[\text{I}_2]$ doubles **(1)**
mark these points independently
allow 1 mark for orders if not clear that both are first order 4
- (ii) rate = $k[\text{H}_2][\text{I}_2]$ **(1)**
must be $[\text{H}_2]$ not $[\text{H}]$
allow e.c.f from (b)(i)
allow mark for correct answer for (b)(ii), even if (b)(i) wrong 1
- (iii) rearrange correctly **(1)**
correct substitution and correct consequential answer **(1)**
correct units ($\text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$) **(1)**
allow e.c.f. from (b)(ii)
allow e.c.f. for incorrect arrangement 3
- (c) 4×10^{-6} **(1)**
allow e.c.f. from (b)(ii) and (b)(iii)
could be worked by ratio from values in table or from rate equation 1
- [13]**
18. (i) O_3 : Exp 2 has 4 times $[\text{H}_2]$ as Exp 1
and rate increases by 4 **(1)**,
so order = 1 with respect to O_3 **(1)**
 C_2H_4 : Exp 3 has $2 \times [\text{C}_2\text{H}_4]$ and $2 \times [\text{O}_3]$ as Exp 2;
and rate has increased by 4 **(1)**,
so order = 1 with respect to C_2H_4 **(1)**
rate = $k[\text{O}_3][\text{C}_2\text{H}_4]$ **(1)** 5
- (ii) use of $k = \text{rate} / [\text{O}_3][\text{C}_2\text{H}_4] = 1.0 \times 10^{-12} / (0.5 \times 10^{-7} \times 1.0 \times 10^{-8})$
to obtain a calculated value **(1)**
 $k = 2 \times 10^3$ **(1)**
units: $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$ **(1)** 3
- (iii) rate = $1.0 \times 10^{-12} / 4 = 2.5 \times 10^{-13} (\text{mol dm}^{-3} \text{s}^{-1})$ **(1)** 1
- (iv) rate increases and k increases **(1)** 1
- [10]**
19. (a) (change in) concentration/mass/volume with time 1
- (b) (i) O_2 :
Exp 2 has $4 \times [\text{O}_2]$ as Exp. 1: rate increases by 4 **(1)**,
so order = 1 with respect to O_2 **(1)**
 NO :
Exp 3 has $3 \times [\text{NO}]$ as Exp. 3: rate has increases by 9 **(1)**,
so order = 2 with respect to NO **(1)** 4
- (ii) rate = $k[\text{O}_2][\text{NO}]^2$ **(1)** 1
- (iii) $k = \frac{\text{rate}}{[\text{O}_2][\text{NO}]^2} = \frac{7.10}{0.0010 \times 0.0010^2} = 7.10 \times 10^9$ **(1)**
units: $\text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$ **(1)** 2
- [8]**
20. From graph, constant half-life **(1)**

	Therefore 1 st order w.r.t. [CH ₃ COCH ₃] (1)	2	
	From table, rate doubles when [H ⁺] doubles (1)		
	Therefore 1 st order w.r.t. [H ⁺] (1)	2	
	From table, rate stays same when [I ₂] doubles (1)		
	Therefore zero order w.r.t. [I ₂] (1)		
	Order with no justification does not score.	2	
	rate = k[H ⁺][CH ₃ COCH ₃] (1)		
	(from all three pieces of evidence)		
	$k = \frac{\text{rate}}{[\text{H}^+][\text{CH}_3\text{COCH}_3]} = \frac{2.1 \times 10^{-9}}{0.02 \times 1.5 \times 10^{-3}}$ (1)		
	= 7.0 × 10 ⁻⁵ (1) dm ³ mol ⁻¹ s ⁻¹ (1)	4	
	accept 7 × 10 ⁻⁵		
	rate determining step involves species in rate equation (1)		
	two steps that add up to give the overall equation (1)		
	The left hand side of a step that contains the species in rate-determining step (1)		
	i.e., for marking points 2 and 3:		
	CH ₃ COCH ₃ + H ⁺ → [CH ₃ COHCH ₃] ⁺		
	[CH ₃ COHCH ₃] ⁺ + I ₂ → CH ₃ COCH ₂ I + HI + H ⁺	3	
	organises relevant information clearly and coherently, using specialist vocabulary where appropriate		
	Use of the following four words/phrases: constant, half-life, order, doubles/x2 (1)	1	
			[14]
21.	(i) constant half-life (1)	1	
	(ii) rate = k[N ₂ O ₅] (1)	1	
	Common error will be to use '2' from equation.		
	(iii) curve downwards getting less steep (1)		
	curve goes through 1200,0.30; 2400,0.15; 3600,0.075 (1)	2	
	(iv) tangent shown on graph at t = 1200 s (1)	1	
	(v) 3.7(2) × 10 ⁻⁴ (1) mol dm ⁻³ s ⁻¹ (1)		
	ecf possible from (ii) using [N ₂ O ₅] ^x		
	(2 nd order answer: 2.2(3) × 10 ⁻⁴)	2	
			[7]
22.	(i) The slowest step (1)	1	
	(ii) 2NO ₂ → NO + NO ₃ (1)		
	NO ₃ + CO → NO ₂ + CO ₂ (1)	2	
	(or similar stage involving intermediates)		
			[3]
23.	(i) 1½O ₂ (g) → O ₃ (g)/		
	O ₂ (g) + ½O ₂ (g) → O ₃ (g) (1)		
	NO is a catalyst (1) as it is (used up in step 1 and) regenerated in step 2/ not used up in the overall reaction(1)		
	allow 1 mark for 'O/NO ₂ with explanation of regeneration.'	3	
	(ii) Rate = k[NO] [O ₃] (1)		
	Species in rate equation match those reactants in the slow step / rate determining step (1)	2	
			[5]
24.	(a) CH ₄ + H ₂ O → 3H ₂ + CO	1	
	CH ₄ + 2H ₂ O → 4H ₂ + CO ₂		



(b) (i) $k_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} \checkmark$ 1

$$[\text{NH}_3]^2 = (K_c \times [\text{N}_2] \times [\text{H}_2]^3) \checkmark$$

$$= 0.768 \checkmark$$

(ii) $[\text{NH}_3] = \sqrt{0.78} = 0.876/0.88 \text{ (mol dm}^{-3}\text{)} \checkmark$ 3

If no powers, then rearrangement mark only.

(c) **High pressure:**

adv: Fewer moles on r.h.s. $\rightarrow$ equilibrium moves to right $\checkmark$

Greater pressure $\rightarrow$ faster rate/more frequent collisions $\checkmark$

dis: Safety issues from (high) pressure

Expense of (high) pressure $\checkmark$ 3

High temperature:

adv: more collisions exceed activation energy/

more successful collisions/more energetic

collisions/molecules have more energy $\checkmark$

dis: Equilibrium moves to left/reverse direction because

(forward) reaction is exothermic $\checkmark$ 2

Catalyst:

lowers activation energy/

allows reaction to take place at a lower temperature $\checkmark$ 1

QWC: Uses 2 words following list in the correct context: 1

exothermic/endothermic, activation energy, collisions,

equilibrium/Le Chatelier

[12]

25. (a) rate of forward reaction = rate of reverse reaction (1)
concentrations of reactants and products are constant but they are constantly interchanging (1) 2

(b) (i) $K_c = [\text{CH}_3\text{OH}] / [\text{CO}] [\text{H}_2]^2$ (1) 1

(ii) use of $K_c = [\text{CH}_3\text{OH}] / [\text{CO}] [\text{H}_2]^2$ and moles to obtain a calculated value (1)

convert moles to concentration by $\div 2$: $[\text{CO}] = 3.10 \times 10^{-3} \text{ mol dm}^{-3}$;

$[\text{H}_2] = 2.60 \times 10^{-5} \text{ mol dm}^{-3}$; $[\text{CH}_3\text{OH}] = 2.40 \times 10^{-2} \text{ mol dm}^{-3}$ (1)

$K_c = [2.60 \times 10^{-5}] / [3.10 \times 10^{-3}] [2.40 \times 10^{-2}]^2 = 14.6 / 14.56$ (1)

If moles not converted to concentration, calculated K_c value = 3.64

(scores 1st and 3rd marks)

units: $\text{dm}^6 \text{ mol}^{-2}$ (1) 4

(c) (i) fewer moles of gas on right hand side (1) 1

(ii) None (1) 1

(d) (i) moved to left hand side/reactants increase/less products (1) 1

(ii) ΔH negative because high temperature favours the endothermic direction (1) 1

(e) (i) $\text{CH}_3\text{OH} + 1\frac{1}{2} \text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$ (1) 1

(ii) adds oxygen/oxygenated (1) 1

[13]

26. (a) (i) Curve downwards starting at $t = 0$ $\checkmark$ 2

with slope gradually levelling off with no increase $\checkmark$

(don't worry about hitting the x axis)

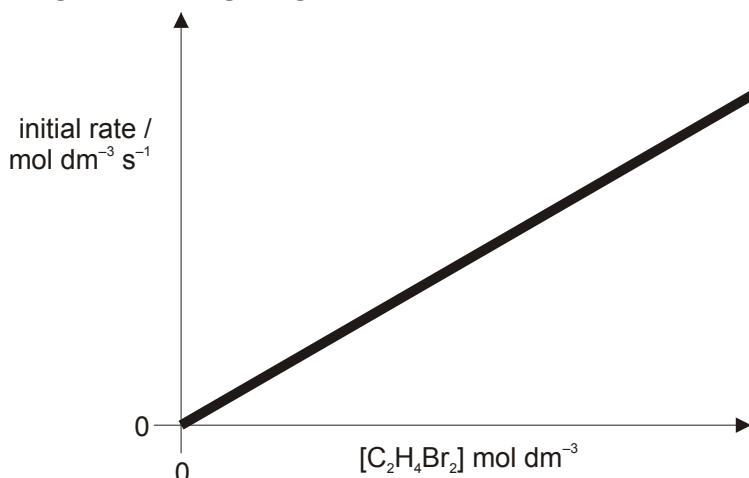
(ii) Tangent shown at start $\checkmark$ 1

(iii) Half-life is constant 1

OR: draw tangents and then plot a 2nd graph of tangent or rate against concentration, which is a straight line

- through the origin. ✓
(iv) Straight line through origin ✓

1



- (b) 4 times [KI], rate increases by 4 ✓,
so order = 1 with respect to KI ✓ independent marks

2

- (c) (i) $\text{rate}/r = k[\text{C}_2\text{H}_4\text{Br}_2][\text{KI}]$ ✓ or ecf from (b)

1

(ii) $k = \frac{\text{rate}}{[\text{C}_2\text{H}_4\text{Br}_2][\text{KI}]} = \frac{0.027}{0.50 \times 0.18}$ ✓
 $= 0.3(0)$ ✓ units: $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$ ✓
 units dependent on rate equation in (i).
 Mark independently.

3

[11]

27. (i) H_2 : Exp 2 has 2.5 times $[\text{H}_2]$ as Exp 1
and rate increases by 2.5 (1),
so order = 1 with respect to H_2 (1)

2

NO: Exp 3 has 3 x $[\text{NO}]$ as Exp 2;

and rate has increased by $9 = 3^2$ (1),
so order = 2 with respect to NO (1)

2

QWC At least two complete sentences where the meaning is clear.

1

(ii) $\text{rate} = k[\text{NO}]^2 [\text{H}_2]$ (1)

1

(iii) $k = \frac{\text{rate}}{[\text{NO}]^2 [\text{H}_2]} = \frac{2.6}{0.10^2 \times 0.20}$ (1)

$= 1300$ (1) units: $\text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$ (1)

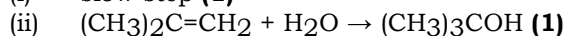
allow 1 mark for 7.69×10^{-4} or 1.3×10^x (x not 3)

3

[9]

28. (i) slow step (1)

1



1

- (iii) H^+ is a catalyst (1)

H^+ used in first step **and** formed in second step/
regenerated/ not used up (1)

2

(iv) $\text{rate} = k [(\text{CH}_3)_2\text{C}=\text{CH}_2] [\text{H}^+]$ (1)

common error will be use of H_2O instead of H^+

1

[5]

29. (a) (i) O_3 : 1
and C_2H_4 (1)

1

- (ii) 2 (1)

1

- (iii) rate = $k[\text{O}_3][\text{C}_2\text{H}_4]$ (1) 1
- (b) (i) measure gradient/tangent (1)
at $t = 0$ /start of reaction (1) 2
- (ii) $k = \frac{\text{rate}}{[\text{O}_2][\text{C}_2\text{H}_4]}$ (1)
 $k = \frac{1.0 \times 10^{-12}}{0.5 \times 10^{-7} \times 1.0 \times 10^{-8}} = 2 \times 10^3$ (1) $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$ (1) 3
- (iii) 2 mol CH_2O forms for every 0.5 mol O_2 /
stoichiometry of $\text{CH}_2\text{O} : \text{O}_2$ is **not** 1:1 (1) 1
- (iv) rate increases (1)
 k increases (1) 2
- [11]**
30. (a) $K_C = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$ (1) 1
- (b) (i)

H_2	I_2	HI
0.30	0.20	0
0.14	0.04	0.32
(1)		(1)

 2
- (ii) $K_C = \frac{0.32^2}{0.14 \times 0.04} = 18.28571429$ (1)
 $= 18$ (to 2 sig figs) (1)
no units (1)
(or ecf based on answers to (i) and/or (a)) 3
- (c) K_C is constant (1)
Composition of mixture is the same (1) 2
- [8]**
31. (a) $K_C = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]}$ (1) 1
- (b) (i) $\text{PCl}_5 > 0.3 \text{ mol dm}^{-3}$; PCl_3 and $\text{Cl}_2 < 0.3 \text{ mol dm}^{-3}$ (1) 1
- (ii) At start, system is out of equilibrium with too much PCl_3
and Cl_2 and not enough PCl_5 /
 $\frac{0.3 \times 0.3}{0.3} = 0.3$ is greater than $K_C = 0.245 \text{ mol dm}^{-3}$ (1) 1
- (c) (i) K_C does not change as temperature is the same (1) 1
- (ii) Fewer moles on left hand side (1)
system moves to the left to compensate for increase in
pressure by producing less molecules (1) 2
- (d) (i) K_C decreases (as more reactants than products)(1) 1
- (ii) Forward reaction is exothermic/
reverse reaction is endothermic (1)
equilibrium $\rightarrow$ left to oppose increase in energy/
because K_C decreases (1) 2
- [9]**
31. (a) (i) $\text{MgCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
balancing ✓
state symbols ✓ 2
- (b) (as the reaction proceeds) the concentration decreases ✓
(rate) of collision decreases ✓
reaction stops when all of one reagent is used up ✓ 3
- [5]**

32. (i) curve displaced to the right (1)
maximum is lower (1) 2
(ii) area under curve exceeding E_a = number of molecules that can react (1)
at higher temperature, area under curve $> E_a$ is greater so more can react (1) 2 [4]
33. sketch distribution to show axes labelled number/ fraction of molecules/atoms and energy (1)
shape starting at origin, maximum, approaching but not crossing x-axis (1) for **both** graphs
explanation of distribution 2 from
no molecules with no energy
few very energetic molecules
most have average amounts of energy
area under curve is the number of molecules (2)
distribution at higher temperature shown on diagram
hump lower than original (1)
and to RHS of original (1)
 E_a marked (1)
rate increases with an increase in temperature (1) 9
since more molecules have energy $> E_a$ (1) [9]
34. (a) the statement is true because there are more collisions (as temperature increases) ✓
increase in temperature increases the velocity/ energy of particles ✓
rate increases (with increase in temperature) more than can be explained by this/
but not all collisions are successful ✓
to be successful collisions must exceed E_a ✓
if temperature increased higher proportion of collisions exceed E_a ✓ 5
(b) (i) y axis: fraction/ number of particles/ molecules/ atoms ✓ 2
x axis: energy/ velocity ✓
(ii) line labelled T_2 with higher maximum ✓
maximum to LHS of original line ✓
(must start at 0.0, be below original curve at higher energies,
cut the other curve only once and not cross the □ axis 2 [9]
35. (i) more particles in a given volume (1)
so more frequent collisions (1)
(ii) the reaction gets hotter as it continues because it is exothermic (1)
this makes the reaction faster (1) 4]
36. (adding a catalyst):
☐ speeds up a reaction
☐ provides an alternative route *or* forms an intermediate of some sort
☐ of lower E_{act} (can be read into a label on a Boltzmann distribution)
☐ so more molecules have $E > E_{act}$ *or* more collisions are successful
☐ weakens bonds in the reactants
[any 4 points.] ✓✓✓✓ [5]
37. when the conditions on a system in equilibrium are changed (1)
the equilibrium moves to minimise the effects of the change/
counteract/ resist/ oppose the change (1) [2]
38. (a) any two from
rate of forward reaction = **rate** reverse reaction ✓
macroscopic properties remain constant/ concentrations remain constant ✓
closed system needed ✓ 2

- (i) a substance that alters the rate of a reaction without being used up /
a substance that lowers the activation energy (for a reaction) by providing an alternative route ✓ 1
- (ii) H^+ ✓ 1
- (iii) they alter the rate of the forward and the reverse reaction **by the same amount** ✓ 1 [5]
39. (i) becomes brown/ darker/ colour more intense (1)
moves towards LHS/ towards NO_2 (1)
forward reaction is exothermic/ **reverse** reaction is endothermic (1) 3
- (ii) becomes less brown/ pale/ colourless (1)
moves towards RHS/ towards N_2O_4 (1)
fewer moles on RHS (1) 3 [6]
40. (from yellow) to orange ✓
increasing $[H^+]$ or more acid/HCl
moves equilibrium/reaction to the left or produces more $Cr_2O_7^{2-}$ ✓ [2]
41. (i) turns lighter brown/colourless ✓
(equilibrium/reaction moves to the right):
fewer molecules/particles/ moles on right or 2 moles $\rightarrow$ 1 mole ✓ 2
- (ii) turns darker (brown) ✓
(equilibrium/reaction moves to the left): $L \rightarrow R$ /forward rxn is exothermic. ✓ 2
- [in (i) and (ii) mark the observation first, and then the reason. Each mark is unconditional on the other.] [in (ii), if neither mark is scored and you are convinced that the only error is mixing up endo/exo-thermic, you may award [1] mark]
- [4]
42. (a) to overcome activation energy (1)
reaction is endothermic (1)
to break bonds – if type of bonds stated must be ionic or covalent (1)
A2 answer based on polarisation of carbonate by Ca^{2+} is acceptable 2
- (b) (i) rate forward > rate backward (1) 1
- (ii) rate forward = rate backward (1) 1
- (iii) equilibrium moves to RHS (1)
use of le Chatelier (1)
more CaO /product / less $CaCO_3$ / reactant present (1) 3 [7]
43. (a) when the conditions on a reaction in **equilibrium** are **changed/ disturbed** ✓
the (equilibrium) moves in the direction to minimise the effects of the change ✓ 2
- (b) (i) equilibrium moves to the LHS/ more X_2 and Y_2 are produced ✓
more moles (of gas)/ particles on LHS ✓ 2
- (ii) rate becomes less as there are less particles in a unit volume/ concentration less/ more space between particles ✓
therefore there are less (frequent) collisions ✓ 2
- (c) (i) 16–17 □ ✓ 1
- (ii) as the temperature increases the conversion decreases ✓
(equilibrium) has moved to LHS/ has moved in endothermic direction ✓ 2
- (d) (i) increases ✓
because more collisions exceed (lowered) E_a / because the catalyst provides an alternative route with a lower activation energy ✓ 2
- (ii) no change ✓
forwards and reverse rates increased by **same** amount ✓ 2 [13]
44. (a) diamond line drawn higher than graphite line (1)

activation energy profile linking the two and going higher than both lines (1)

- (b) graphite as it is lower in energy (1)
- (c) diamond as it is more dense (1)
- (d) when the conditions of a reaction at equilibrium are changed (1)
the reaction/position of equilibrium moves to counteract the change (1)
- (e) pressure should be high (1)
so particles will rearrange to diamond as it occupies less space (1)
temperature should be high (1)
so reaction moves in endothermic direction (from graphite to diamond) (1)

[10]

45.

rate

(increased) pressure increases rate because molecules are closer together/ more concentrated ✓

(increased) temperature increases rate because molecules are moving faster/ have more energy ✓

equilibrium

increased pressure pushes equilibrium to RHS ✓ because fewer (gas) moles/ molecules on RHS ✓

increased temperature pushes equilibrium to LHS ✓ because (forward) reaction is exothermic ✓

compromise

if temperature is **too** high, low yield ✓

if temperature is **too** low, slow rate ✓

if pressure is too high, increased costs/ safety issues ✓

[9]

9.0 INORGANIC CHEMISTRY

1. Write balanced equations to show the reaction of water with:
- (i) sodium (1)
- (ii) sodium oxide. (1)

(Total 1 marks)

2. Write equations for the reactions of phosphorus(V) oxide and sulphur dioxide with water. In each case predict the approximate pH of a 1M aqueous solution of the product.

(Total 4 marks)

3. Apart from argon, the Period 3 elements all form oxides.

- (i) Write an equation to represent the reaction of aluminium with oxygen. (1)
- (ii) The reaction in (b)(i) occurs very readily. Suggest why aluminium saucepans can be safely heated on a gas cooker. (1)
- (iii) Complete the following table to show information about Period 3 oxides.

	Sodium oxide	Silicon dioxide	Phosphorus(V) oxide	Sulphur dioxide
Physical state at room temperature				
Type of bonding present				

(4)

- (iv) Write an equation for the reaction of sodium oxide with water. (1)
- (v) Write an equation for the reaction of sulphur dioxide with water. (1)
- (vi) Samples of silicon dioxide and phosphorus(V) oxide are added to separate samples of pure water and each mixture is stirred. State the approximate pH value of each sample after the stirring. (2)

(Total 10 marks)

4. (a) Explain, with reference to the bonding in sodium oxide, why this compound reacts with water to form a solution with a pH of 14. (3)
- (b) What general type of oxide forms acidic solutions in water? Give the formula of one such oxide. (2)

(Total 5 marks)

5. The following two-stage method was used to analyse a mixture containing the solids magnesium, magnesium oxide and sodium chloride.

Stage 1

A weighed sample of the mixture was treated with an excess of dilute hydrochloric acid. The sodium chloride dissolved in the acid. The magnesium oxide reacted to form a solution of magnesium chloride. The magnesium also reacted to form hydrogen gas and a solution of magnesium chloride. The hydrogen produced was collected.

- (a) Write equations for the two reactions involving hydrochloric acid.
- (b) State how you would collect the hydrogen. State the measurements that you would make in order to calculate the number of moles of hydrogen produced. Explain how

your results could be used to determine the number of moles of magnesium metal in the sample.

(8)

Stage 2

Sodium hydroxide solution was added to the solution formed in **Stage 1** until no further precipitation of magnesium hydroxide occurred. This precipitate was filtered off, collected, dried and heated strongly until it had decomposed completely into magnesium oxide. The oxide was weighed.

- (c) Write equations for the formation of magnesium hydroxide and for its decomposition into magnesium oxide.
- (d) When a 6.25 g sample of the mixture of the three solids was analysed as described above, the following results were obtained.
 Hydrogen obtained in **Stage 1**: 0.0528 mol
 Mass of magnesium oxide obtained in **Stage 2**: 6.41 g
 Use these results to calculate the number of moles of original magnesium oxide in 100 g of the mixture.

(7)

(Total 15 marks)

6. Describe the trend in pH of the solutions formed when the oxides of the Period 3 elements, sodium to sulphur, are added separately to water. Explain this trend by reference to the structure and bonding in the oxides and by writing equations for the reactions with water.

(19)

(Total 19 marks)

7. (a) Write equations to show what happens when the following oxides are added to water and predict approximate values for the pH of the resulting solutions.

- (i) sodium oxide
 (ii) sulphur dioxide

(4)

- (b) What is the relationship between bond type in the oxides of the Period 3 elements and the pH of the solutions which result from addition of the oxides to water?

(2)

(Total 6 marks)

8. In the question below, Z is one of the Period 3 elements Na, Mg, Al, Si or P.

The oxide of element **Z** is a crystalline solid with a very high melting point. This oxide is classified as an acidic oxide but it is not soluble in water.

- (i) Deduce the type of crystal shown by the oxide of element **Z**.
 (ii) Identify element **Z**.
 (iii) Write an equation for a reaction which illustrates the acidic nature of the oxide of element **Z**.

(4)

(Total 4 marks)

9. (a) The Period 3 elements, Na, Mg, Al, Si, P and S, all form oxides when the elements are burned in an excess of oxygen.

- (i) Give the formula of an oxide of **one** of these elements in which the element is not in its highest oxidation state. Give the oxidation state of the element in this oxide.
 (ii) Write an equation for the reaction in which phosphorus(V) oxide is formed from phosphorus and oxygen.

(3)

- (b) The melting points of some of the oxides formed by Period 3 elements are given in a random order below.

Oxide	A	B	C	D	E
$T_m / ^\circ\text{C}$	2852	-73	1610	1275	300

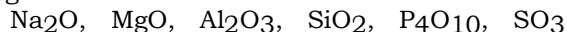
- (i) Using the letters **A** to **E**, give **two** oxides which have simple molecular structures.
 Explain your answer.
- (ii) Give a simple chemical test which could be used to show which of the oxides

in the table is sodium oxide. State the observation you would make.

(6)

(Total 9 marks)

10. Consider the following oxides.



- (a) Identify one of the oxides from the above which
- can form a solution with a pH less than 3
 - can form a solution with a pH greater than 12
- (b) Write an equation for the reaction between
- MgO and HNO_3
 - SiO_2 and NaOH
 - Na_2O and H_3PO_4
- (c) Explain, in terms of their type of structure and bonding, why P_4O_{10} can be vaporised by gentle heat but SiO_2 cannot.

(4)

(Total 9 marks)

11. (a) (i) Write an equation for the reaction of sodium with cold water.
- (ii) By referring to the bonding and structure of silicon, suggest why silicon does not react with cold water.
- (b) The oxides of elements of Period 3 react to form salts with aqueous sodium hydroxide **or** with aqueous sulphuric acid **or** with both of these reagents. Give the formula of an oxide of an element of Period 3 that reacts with:
- aqueous sodium hydroxide **only**;
 - aqueous sulphuric acid **only**;
 - aqueous sodium hydroxide **and** aqueous sulphuric acid.

(1)

(1)

(1)

(Total 6 marks)

12. (a) **P** and **Q** are oxides of Period 3 elements.
- Oxide **P** is a solid with a high melting point. It does not conduct electricity when solid but does conduct when molten or when dissolved in water. Oxide **P** reacts with water forming a solution with a high pH.
- Oxide **Q** is a colourless gas at room temperature. It dissolves in water to give a solution with a low pH.
- Identify **P**. State the type of bonding present in **P** and explain its electrical conductivity. Write an equation for the reaction of **P** with water.
 - Identify **Q**. State the type of bonding present in **Q** and explain why it is a gas at room temperature. Write an equation for the reaction of **Q** with water.
- (b) **R** is a hydroxide of a Period 3 element. It is insoluble in water but dissolves in both aqueous sodium hydroxide and aqueous sulphuric acid.
- Give the name used to describe this behaviour of the hydroxide.
 - Write equations for the reactions occurring.
 - Suggest why **R** is insoluble in water.

(9)

(6)

(Total 15 marks)

13. In their reactions, calcium and strontium each lose electrons to form ions with a 2+ charge. The first and second ionisation energies of calcium and strontium are shown below.

	1st ionisation energy / kJ mol^{-1}	2nd ionisation energy / kJ mol^{-1}
calcium	590	1145
strontium	550	1064

- [3]**
[Total 6 marks]

14. This question refers to the elements in the first four periods of the Periodic Table.

																H																	He
Li	Be															B	C	N	O	F	Ne												
Na	Mg															Al	Si	P	S	Cl	Ar												
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr																

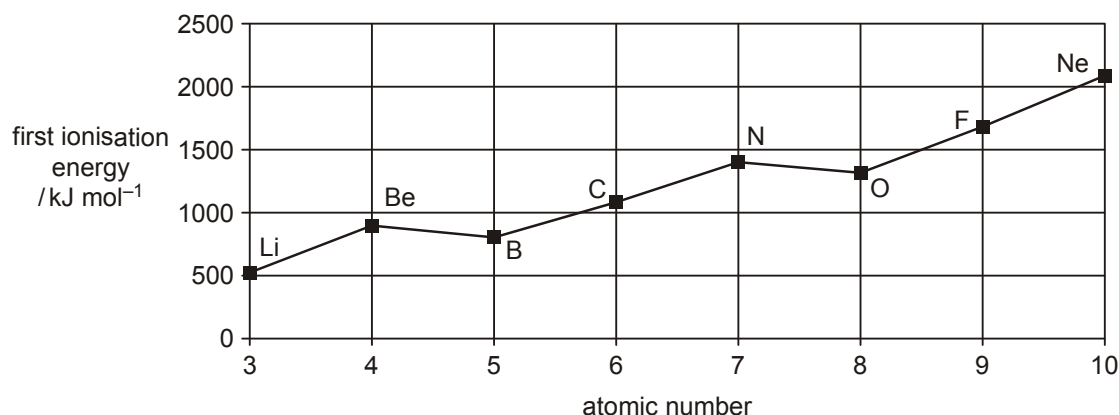
Ionisation energies provide information about the model for the electron structure of elements.

- (i) Explain why first ionisation energies show a general increase across Period 3, Na–Ar. [3]
- (ii) Write an equation, including state symbols, to represent the third ionisation energy of sodium. [1]
- (iii) Element **X** is in Period 3 of the Periodic Table, Na–Ar.
The first six ionisation energies of an element **X** are shown below.

ionisation number	1st	2nd	3rd	4th	5th	6th
ionisation energy /kJ mol⁻¹	789	1577	3232	4 556	16091	19 785

Predict, with reasons, the identity of element **X**.

- 15.** The diagram below shows the variation in the first ionisation energies of elements across Period 2 of the Periodic Table.



- (i) Define the term *first ionisation energy*. [3]
- (ii) Explain why the first ionisation energies show a **general** increase across Period 2. [2]
- (iii) Explain why the first ionisation energy of B is **less** than that of Be.

[2]
[Total 7 marks]

- (ii) The element that forms a 3+ ion with the same electronic configuration as Ne. [1]
- (iii) The element that has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^3$. [1]
- (iv) An element that forms a compound with hydrogen with tetrahedral molecules. [1]
- (v) An element that forms a compound with hydrogen with pyramidal molecules. [1]
- (vi) The element that forms a chloride $XC\ell^2$ with a molar mass of 95.3 g mol^{-1} . [1]
- (vii) The element with the largest atomic radius. [1]
- (viii) The element in Period 3 with the highest boiling point. [1]
- [Total 8 marks]**

Marking Scheme

1. (i) $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ (1)
 (ii) $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$ (1) [2]
2. Phosphorus (V) oxide $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$ (1)
 Approximate pH O (1)
 allow -1 to 0.5
 Sulphur dioxide $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$ (1)
 Approximate pH 3 (1)
 allow 1 to 4 [4]
3. (i) $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ (1) 1
 (ii) aluminium is protected by an **oxide** layer (1) 1
- | | | | | |
|------------------------------------|--------------|-----------------|---------------------|-----------------|
| | Sodium oxide | Silicon dioxide | Phosphorus(V) oxide | Sulphur dioxide |
| Physical state at room temperature | solid | solid | solid | gas |
- (iii) allow abbreviations (s) and (g)
- | | | | | |
|-------------------------|-------|----------|----------|----------|
| Type of bonding present | ionic | covalent | covalent | covalent |
|-------------------------|-------|----------|----------|----------|
- must give ionic/covalent but ignore additional information about structure
 if 8 correct, give 4 marks
 if 6 or 7, give 3
 if 4 or 5, give 2
 if 2 or 3, give 1 4
- (iv) $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$ (1)
 accept ionic charges (if correct) for Na compounds 1
- (v) $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$ (1)
 do **not** accept SH_2O_3
 accept ions on RHS 1
- (vi) silicon dioxide 7 (1)
 phosphorus(v) oxide 0 – 3 (1)
 (must give values) 2
- [10]
4. (a) ionic (1)
 O^{2-} ion reacts with water (1)
 forming OH^- (or NaOH) (1) 3
- (b) General type covalent (1) (or non-metal or molecular)
 Formula SO_2 (1) etc 2
- [5]
5. (a) $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$ 1
 $\text{MgO} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{O}$ 1
 Allow ionic equations

	(b)	Hydrogen collection	1	
		Using a gas syringe or measuring cylinder/ graduated vessel over water		
		<i>Allow if shown in a diagram</i>		
		Measurements		
		(i) P	1	
		(ii) T	1	
		(iii) V	1	
		Use ideal gas equation to calculate mol hydrogen or mass/Mr	1	
		Mol H ₂ = mol Mg (Mark consequentially to equation)	1	
	(c)	MgCl ₂ + 2NaOH → Mg(OH) ₂ + 2NaCl	1	
		Species	1	
		Balanced	1	
		<i>Allow an ionic equation</i>		
		Mg(OH) ₂ → MgO + H ₂ O	1	
6.	(a)	(i) Equation Na ₂ O + H ₂ O → 2NaOH (1)		
		pH 11-14 (1)		
		(ii) Equation SO ₂ + H ₂ O → H ₂ SO ₃ (1)		
		pH 2-5 (1)	4	
		(b) Covalent oxides → acidic solutions (1)		
		ionic oxides → alkaline solutions (1)	2	
7.		(i) Macromolecular/giant covalent/giant molecular	1	
		(ii) Silicon/Si	1	
		(iii) e.g. CaO + SiO ₂ CaSiO ₃	1	
		Base	1	
		Balanced	1	
8.	(a)	(i) SO ₂	1	
		+4	1	
		(ii) 4P + 5O ₂ → 2P ₂ O ₅	1	
		or P ₄ + 5O ₂ → P ₄ O ₁₀		
		(b) (i) B	1	
		E	1	
		They have low melting points		
		or there are weak van der Waals forces between molecules	1	
		(ii) Add water or heat in a flame	1	
		Test pH check flame colour	1	
9.	(a)	(i) can form a solution with pH less than 3: P ₄ O ₁₀ or SO ₃ (1)		
		(ii) can form a solution with with a pH greater than 12: Na ₂ O (1)	2	
		<i>penalise any wrong answer to zero</i>		
		(b) (i) MgO + 2HNO ₃ → Mg(NO ₃) ₂ + H ₂ O or an ionic equation (1)		
		i.e. MgO + 2H ⁺ → Mg ²⁺ + H ₂ O		
		<u>not</u> O ²⁻ + 2H ⁺ → H ₂ O		
		(ii) 2NaOH + SiO ₂ → Na ₂ SiO ₃ + H ₂ O or ionic equation (1)		
		i.e. SiO ₂ + 2OH ⁻ → 2Na ⁺ + H ₂ O		
		(iii) 3Na ₂ O + 2H ₃ PO ₄ → 2Na ₃ PO ₄ + 3H ₂ O etc or ionic equation (1)	3	
		i.e. Na ₂ O + 2H ⁺ → 2Na ⁺ + H ₂ O		
10.	(a)	(i) NaOH may be shown as ions. Balanced using H ₂ or ½ H ₂		
		2Na + 2H ₂ O → 2NaOH + H ₂	1	
		(ii) silicon forms a giant covalent / atomic lattice / has a		
		macromolecular structure / has diamond structure (1)		
		contains many covalent bonds / forms 4 bonds per		

[6]

[4]

[9]

- atom / lattice is strong / the bonding is strong / bonds are strong / silicon is non-polar (1) 2
condone 'bond is strong'
- (b) (i) SiO_2 / P_2O_5 / P_4O_{10} / P_2O_3 / P_4O_6 / SO_2 / SO_3 / Cl_2O / ClO_2 / Cl_2O_6 / Cl_2O_7 1
(ii) Na_2O / Na_2O_2 / MgO 1
(iii) Al_2O_3 must give formulae 1
- [6]**
11. (a) (i) Deductions:
Ionic (1)
Ions not free to move in the solid state (1)
Ions free to move when molten or in aqueous solution (1)
Identity of **P**: Na_2O or sodium oxide (1)
N.B. If a formula given this must be correct
Equation: $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2 \text{NaOH}$ (1) 5
- (ii) Deductions:
Covalent
Intermolecular forces are weak or van der Waals forces, or dipole-dipole
N.B. Any answer including a reference to hydrogen bonding is incorrect
Identity of **Q**: SO_2 or sulphur dioxide (1)
Equation: $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$ (1)
NB Allow max one for SO_3 4
- (b) (i) Amphoteric (1)
(ii) Equation with NaOH
 $\text{Al}(\text{OH})_3 + \text{NaOH} \rightarrow \text{NaAl}(\text{OH})_4$
OR $\text{Al}(\text{OH})_3(\text{H}_2\text{O})_3 + \text{OH}^- \rightarrow [\text{Al}(\text{OH})_4(\text{H}_2\text{O})_2]^- + \text{H}_2\text{O}$
OR $\text{Al}(\text{OH})_3 + \text{OH}^- \rightarrow [\text{Al}(\text{OH})_4]^-$
R identified as $\text{Al}(\text{OH})_3$ or $\text{Al}(\text{OH})_3(\text{H}_2\text{O})_3$ (1)
A balanced equation (1)
N.B. Allow equation with six co-ordinate Aluminium and up to six OH^- ligands
N.B. Allow equation mark if $\text{M}(\text{OH})_3$ given in a balanced equation
Equation with H_2SO_4
 $2\text{Al}(\text{OH})_3 + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 6\text{H}_2\text{O}$
OR $\text{Al}(\text{OH})_3(\text{H}_2\text{O})_3 + \text{H}^+ \rightarrow [\text{Al}(\text{OH})_2(\text{H}_2\text{O})_4]^+ + \text{H}_2\text{O}$
NB Allow equations with six co-ordinate Aluminium and up to six H_2O ligands
NB Allow equation mark if $\text{H}(\text{OH})_3$ given in a balanced equation
Correct Al species as product (1)
A balanced equation (1)
- (iii) Large lattice energy
or strong covalent bonds
or ΔH_{soln} is very positive
or sum of hydration energies less than covalent bond energies (1) 6
- [15]**
- (c) P_4O_{10} is a molecular (structure) or simple covalent (1)
Weak intermolecular forces or van der Waals forces (between molecules) (1)
 SiO_2 is a macromolecule / giant covalent / giant molecule (1)
Not giant lattice
(Strong) covalent bonds (between atoms) must be broken (1) 4

[9]

- (d) **Allow 2 significant figures in these calculations and ignore additional figures**

EITHER

Mol MgO obtained stage 2 = mass MgO / MrMgO 1

= 6.41 / 40.(3) = 0.159 Allow 0.16 1

Allow method mark if formula of magnesium oxide or M_r incorrect

Moles of Mg = moles of H_2 henceMol original MgO = mol MgO from stage 2 - mol H_2 1

= 0.159 - 0.0528 = 0.106 Allow 0.11 1

Mark consequentially to moles of magnesium oxide determined above

OR

Mass MgO formed from Mg = 0.0528 $\times M_r$ MgO {or 40.(3)} (1)

= 2.13 g Allow 2.1 (1)

Allow method mark if formula of magnesium oxide or M_r incorrect

Mass original MgO = total mass MgO - mass formed from Mg (1)

= 6.41 - 2.13 = 4.28 g Allow 4.3 (1)

Mark consequentially mass of magnesium oxide determined above

[15]

12.

the trend is a decrease in pH (or from alkaline to acid) (1)(can be implied from separate values)

 $Na_2O + H_2O \rightarrow 2NaOH$ product (1) equation (1)
(allow $Na^+ OH^-$, ignore state symbols)
 Na_2O is ionic lattice (1)

(if lattice is not mentioned lose mark only once ie allow ionic for MgO , Al_2O_3)

 $MgO + H_2O \rightarrow Mg(OH)_2$ product (1) equation (1)
(allow $Mg^{2+} + 2OH^-$)
 MgO is ionic lattice (1)

 MgO sparingly soluble (1)

 Al_2O_3 is ionic lattice or covalent macromolecular (1)(if covalent

not mentioned lose mark only once)

insoluble in water or no reaction (1)(if formula wrong lose one mark)

 SiO_2 is covalent macromolecular (1)(if covalent not mentioned lose

mark only once) insoluble in water or no reaction (1)(formula wrong lose 1 mark)

 $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$ product (1) equation (1)(allow P_2O_5 , P_4O_6 , P_2O_3)

 H_3PO_4 is a strong acid or very acidic (1)

P_4O_{10} is covalent molecular (1)(if covalent or molecular not mentioned lose mark once only)

 $SO_2 + H_2O \rightarrow H_2SO_3$ Product (1) equation (1) or $SO_3 + H_2O \rightarrow H_2SO_4$

Product (1) equation (1)

 H_2SO_3 is a weak acid (1) H_2SO_4 is a strong acid or very acidic (1)

 SO_2 is covalent molecular (1) SO_3 is covalent molecular (1)

(Choose the best of the above two answers if both given)

max

19

[19]

13. (i) $Ca^+(g) \rightarrow Ca^{2+}(g) + e^-$
 Equation with correct charges and 1 electron lost ✓
 state symbols ✓
 '-' not required on 'e'
 (ii) same number of protons or same nuclear charge attracting

2

- less electrons/
electron removed from an ion/
less electron-electron repulsion (**not** less shielding)/
ion is smaller✓ 1
- (iii) atomic radii of Sr > atomic radii of Ca/
Sr has electrons in shell further from nucleus than Ca/
Sr has electrons in a higher energy level/
Sr has more shells ✓
Therefore less attraction ✓
Sr has **more** shielding than Ca ✓
(**'more'** is essential) 3
increased nuclear charge is outweighed / despite increased nuclear chargeby at least one of the factors above ✓
- [6]
14. (i) atomic radii decrease /similar shielding /electrons added to same shell (1)
number of protons in the nucleus increases (1)
nuclear attraction increases (1) 3
- (ii) $\text{Na}^{2+}(\text{g}) \rightarrow \text{Na}^{3+}(\text{g}) + \text{e}^-$: equation and state symbols (1) 1
- (iii) large jump (in energy) between the 4th and 5th ionisation energies (1)
four electrons in outer shell so element is Si (1) 2
- [6]
15. (i) Energy change when each atom in 1 mole ✓
of gaseous atoms ✓
loses an electron ✓ (to form 1 mole of gaseous 1M^+ ions). 3
- (ii) increasing nuclear charge/number of protons ✓
electrons experience greater attraction or *pull* / atomic
radius decreases / electrons added to same shell /same or
similar shielding ✓ 2
- (iii) In B, electron being removed is at a higher energy /
In Be, electron being removed is at a lower energy ✓
An s electron is lost in Be AND a p electron is lost in B ✓ 2
- [7]
16. (i) They have different numbers of protons/
Ba has one more proton/Ba has 56 p^+ ; Cs has 55 p^+ ✓ 1
(ignore electrons: any mention of 'neutrons' is wrong)
- (ii) s ✓ 1
- (iii) Cs to Ba: nuclear charge increases/more protons ✓
electrons are in: the same shell/sub-shell/orbital
/similar shielding/same shielding ✓ 3
attraction increases/pull increases ✓ORA
- (iv) smaller ✓ 2
shell has been lost/less shielding/less electron
repulsion/proton : electron ratio larger ✓
mark separately
- [7]
17. (a) 3
- | element | structure | bonding | |
|---------|-----------|----------|----------------------------|
| Mg | giant | metallic | ✓ ☐ |
| Si | giant | covalent | ✓ ☐ |
| S | simple | covalent | ✓ ☐ |
- 1 mark for each correct row
- (b) Si has strong forces between atoms/
covalent bonds are broken ✓
P has weak forces between molecules/
intermolecular forces/van der Waals' forces are broken ✓ 2

- (c) From Na $\rightarrow$ Al, no of **delocalised** electrons increases ✓
 charge on positive ion increases/
 ionic size decreases/
 charge density increases ✓
 attraction between + ions and electrons increases/
 metallic bonding gets stronger ✓

2

[7]

18. (i) O ✓
 (ii) Al ✓
 (iii) P ✓
 (iv) C/Si ✓
 (v) N/P ✓
 (vi) Mg ✓
 (vii) Na ✓
 (viii) Si ✓

1

1

1

1

1

1

1

1

[8]

10.1 INTRODUCTORY TOPICS

1. The table below gives the names and structures of three isomeric alkenes.

Name	Structure
but-1-ene	$\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$
but-2-ene	$\text{CH}_3\text{CH}=\text{CHCH}_3$
methylpropene	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{C}=\text{CH}_2 \end{array}$

Give the molecular formula and the empirical formula of but-2-ene.

(2)

2. (a) Define the term *structural isomerism*.

(2)

- (b) Draw displayed formulae for the two structural isomers of C_4H_{10} .

(2)

- (c) A hydrocarbon, **W**, contains 92.3% carbon by mass. The relative molecular mass of **W** is 78.0

- (i) Calculate the empirical formula of **W**.
 (ii) Calculate the molecular formula of **W**.

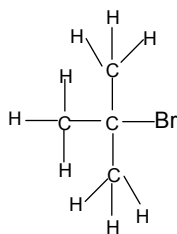
(4)

- (d) Give the name and draw the graphical (displayed) formula of an alkene that is an isomer of but-1-ene **and** that has a different carbon skeleton.

(2)

(Total 10 marks)

3. There are four structural isomers of molecular formula $\text{C}_4\text{H}_9\text{Br}$. The structural formulae of two of these isomers are given below.



Isomer 1



Isomer 2

- (i) Draw the structural formulae of the remaining two isomers.
 (ii) Name isomer 1.

(3)

4. Analysis of T showed that it has the following composition by mass: Carbon 45.86 %, hydrogen 8.92 % and chlorine 45.22 %.

- (i) Calculate the empirical formula of substance T.

(2)

- (ii) The relative molecular mass of T is 78.5. Suggest two displayed formulae which could represent T.

(2)

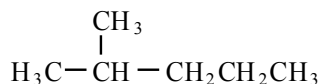
(Total 4 marks)

5. The alkanes form an homologous series of hydrocarbons. The first four straight-chain alkanes are shown below.

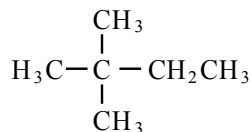
methane	CH ₄
ethane	CH ₃ CH ₃
propane	CH ₃ CH ₂ CH ₃
butane	CH ₃ CH ₂ CH ₂ CH ₃

- (a) (i) State what is meant by the term *hydrocarbon*.
 (ii) Give the general formula for the alkanes.
 (iii) Give the molecular formula for hexane, the sixth member of the series. (3)
- (b) Each homologous series has its own general formula. State **two** other characteristics of a homologous series. (2)
- (c) Branched-chain structural isomers are possible for alkanes which have more than three carbon atoms.
 (i) State what is meant by the term *structural isomers*.
 (ii) Name the **two** isomers of hexane shown below.

Isomer 1



Isomer 2

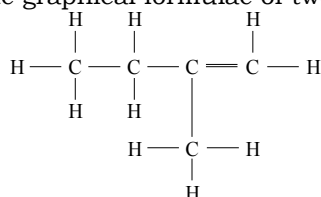


- (iii) Give the structures of **two** other branched-chain isomers of hexane.

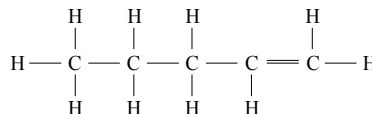
(6)

(Total 11 marks)

6. (a) Four members of the homologous series of alkenes are ethene, propene, but-1-ene and pent-1-ene.
 (i) Give **one** structural feature of the compounds that makes them members of the homologous series of alkenes. (1)
 (ii) State the trend in the boiling points from ethene to pent-1-ene. (1)
- (b) There are five structural isomers of the molecular formula C₅H₁₀ which are alkenes. The graphical formulae of two of these isomers are given.



Isomer 1



Isomer 2

Draw the displayed formulae of **two** of the remaining alkene structural isomers.

(2)

- (c) Draw the displayed formulae of **three** alkenes which have the formula C_4H_8 . (3)

(Total 7 marks)

7. (a) Give the structural formula of 2-bromo-3-methylbutane. (1)

- (b) There are nine structural isomers of molecular formula $C_4H_8Br_2$, three of which have branched carbon chains. Give the names and draw the displayed formulae for **any two** of the branched chain isomers of $C_4H_8Br_2$. (4)

- (c) There are eight structural isomers with the molecular formula $C_5H_{11}Br$. Four of these are classed as primary, three as secondary and one as tertiary.

Draw the displayed formula of the tertiary compound with molecular formula $C_5H_{11}Br$. (1)

(Total 6 marks)

8. (a) Draw the displayed formula for pent-2-ene. (1)

- (b) Distinguish clearly between the terms positional, chain and functional isomerism. (4)

- (c) Draw the graphical formula of a positional isomer, a chain isomer and a functional isomer of pent-2-ene and name them. (6)

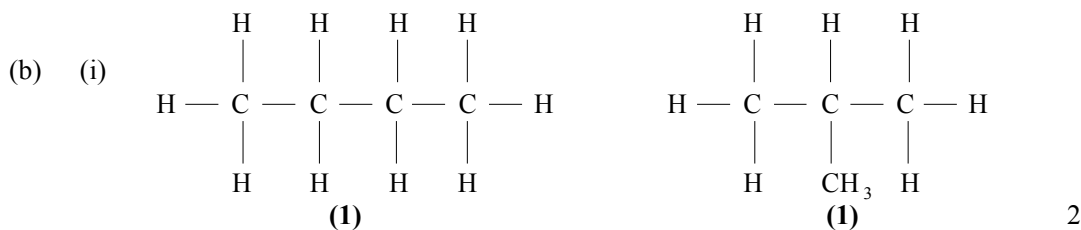
Marking Scheme

1. (a) *Molecular formula:* C₄H₈ (1)

Empirical formula: CH₂ (1) [2]

2. (a) same molecular formula / same number of each type of atom (1)

different arrangements of atoms (in the molecule) (1)
(**not** just same structural formula) 2



- (c) (i) %H = 100 - 92.3 = 7.7 (1)

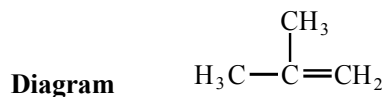
C 92.3/12 = 7.7

H 7.7/1 = 7.7 (1)

= 1:1 ratio so empirical formula = CH (1)

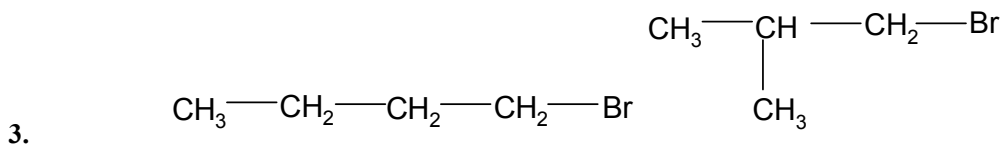
- (ii) C₆H₆

- (d) **Name** methylpropene (1)



4

2
[10]



Isomer 3 (1)

Isomer 4 (1)

- (ii) 2-bromomethylpropane (1) [3]

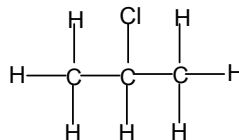
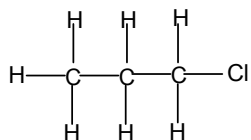
4. i) C: 45.86/12 = 3.82 H: 8.92/1 = 8.92 Cl: 45.22/35.5 = 1.27

1.27 = 3 /1.27 = 7 /1.27 = 1



(2)

ii) $C_3H_7Cl = 78.5$ so empirical formula = molecular formula



(2)

[4]

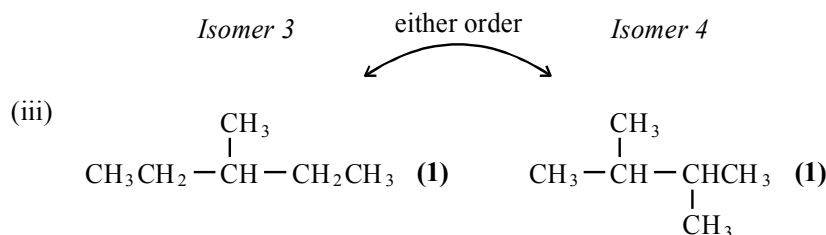
5. (a) (i) Molecule/compound/consists/composed/made up of hydrogen and carbon only (1)
 (ii) C_nH_{2n+2} (1)
 (iii) C_6H_{14} only (1)
 (b) Chemically similar / react in same way / same chemistry
 Differ by CH_2
 gradation in physical properties OR specified trend e.g. b.p.
 same functional group

Any 2, 2 marks 1 + 1

Not same molecular formula

2

- (c) (i) Same molecular formula (1)
 NOT same Mr
 different structural formula / structures (1) (or atoms arranged in different way)
 NOT different spatial arrangement
 Only credit M2 if M1 correct
 (ii) 2-methylpentane (1)
 2,2-dimethylbutane (1)



OR correct condensed / structural formula

Penalise "sticks" once

Penalise absence of vertical bonds once

penalise badly drawn bonds once

6

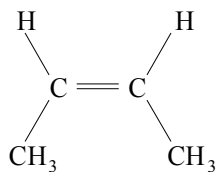
6. (a) (i) the same general formula or C_nH_{2n} / the same functional group / a $C=C$ / a double bond / differ from their immediate neighbour by CH_2 [11]
 1

allow 'all straight chain alkenes'

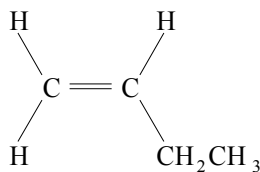
- (ii) increases
 not just 'pent-1-ene highest'; allow 'ethene lowest, pent-1-ene highest' 1
 (b) appropriate structure for pent-2-ene (1)
 appropriate structure for 2-methylbut-2-ene (1)
 appropriate structure for 3-methylbut-1-ene (1) max 2

accept structures with CH_3- $-\text{C}_2\text{H}_5$ and $-\text{CH}_2-$

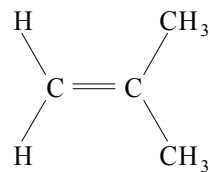
(c)



(1)



(1)



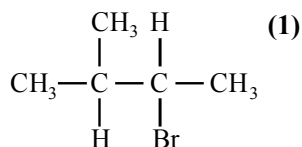
(1)

allow C_2H_5

3

[7]

7. (a)



1

(b)

any two pairs of marks

1,1-dibromo-(2-)methylpropane (1)

graphical formula to suit $(\text{CH}_3)_2\text{CHCHBr}_2$ (1)

1,2-dibromo-(2-)methylpropane (1)

graphical formula to suit $(\text{CH}_3)_2\text{C}(\text{Br})\text{CH}_2\text{Br}$ (1)

1,3-dibromo-(2-)methylpropane (1)

graphical formula to suit $\text{BrCH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{Br}$ (1)

allow unambiguous names

mark name and structure independently

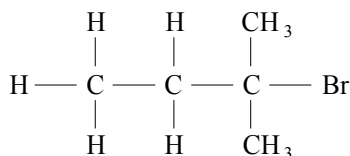
accept order of bromo / methyl reversed

penalise once for each of

numbering from wrong end and di in dibromo omitted

max 4

(c)



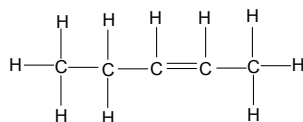
accept $\text{CH}_3-\text{CH}_2\text{-----}$ (1)

1

[6]

8.

(a)



1

(b)

same molecular formula (1) but

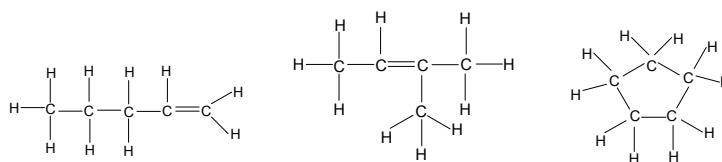
(positional) functional group in different position (1)

(chain) different carbon skeleton/arrangement of carbon atoms (1)

(functional) different functional group (1)

4

(c)



positional (1)

chain (1)

functional (1)

pent-1-ene (1)

methylbut-2-ene (1)

cyclopentane (1)

6

formula, type of isomerism and name all correct = 2 marks

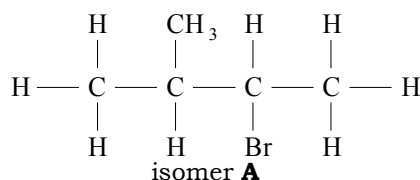
any two of formula, type of isomerism and name = 1 mark

allow correct 3- or 4- membered ring with correct branches for functional

[11]

10.2 HYDROCARBONS

1. (a) (i) Name the process used to separate petroleum into fractions.
 (ii) Give the molecular formula for an alkane with nine carbon atoms.
 (iii) Write an equation for the complete combustion of the alkane $C_{11}H_{24}$
 (iv) Write an equation for the incomplete combustion of $C_{11}H_{24}$ to produce carbon and water only. (4)
- (b) Alkenes can be produced by cracking the naphtha fraction obtained from petroleum.
 (i) Write an equation for the thermal cracking of one molecule of $C_{10}H_{22}$ to give one molecule of propene and one molecule of an alkane only.
 (ii) Draw the structure of the chain isomer of but-1-ene. (2)
- (c) The alkanes and the alkenes are examples of homologous series of compounds. One feature of an homologous series is the gradual change in physical properties as the relative molecular mass increases. State **two** other general features of an homologous series of compounds. (2)
- (Total 8 marks)**
2. The fractions obtained from petroleum contain saturated hydrocarbons that belong to the homologous series of alkanes.
- (a) Any homologous series can be represented by a general formula.
 (i) State **two** other characteristics of homologous series.
 (ii) Name the process which is used to obtain the fractions from petroleum.
 (iii) State what is meant by the term *saturated*, as applied to hydrocarbons. (4)
- (b) Decane has the molecular formula $C_{10}H_{22}$
 (i) State what is meant by the term *molecular formula*.
 (ii) Give the molecular formula of the alkane which contains 14 carbon atoms.
 (iii) Write an equation for the incomplete combustion of decane, $C_{10}H_{22}$, to produce carbon and water only. (3)
- (c) When petrol is burned in an internal combustion engine, some nitrogen monoxide, NO, is formed. This pollutant is removed from the exhaust gases by means of a reaction in a catalytic converter.
 (i) Write an equation for the reaction between nitrogen and oxygen to form nitrogen monoxide.
 (ii) Identify a catalyst used in a catalytic converter.
 (iii) Write an equation to show how nitrogen monoxide is removed from the exhaust gases as they pass through a catalytic converter. (3)
- (Total 10 marks)**
3. (a) Compound **A** ($M_r = 215.8$) contains 22.24% carbon, 3.71% hydrogen and 74.05% bromine by mass. Show that the molecular formula of **A** is $C_4H_8Br_2$. (3)
- (b) There are nine structural isomers of molecular formula $C_4H_8Br_2$, three of which have branched carbon chains. Give the names and draw the displayed formulae for **any two** of the branched chain isomers of $C_4H_8Br_2$. (4)
- (Total 7 marks)**
4. There are eight structural isomers with the molecular formula $C_5H_{11}Br$. Four of these are classed as primary, three as secondary and one as tertiary. The graphical formula of one of the secondary compounds, isomer **A**, is shown below.



- (a) Give the name of isomer **A**. (2)
- (b) Explain what is meant by the term *structural isomers*. (2)

(Total 4 marks)

5. The table below gives some of the names and structures of isomers having the molecular formula $\text{C}_4\text{H}_9\text{Br}$

Structure	Name
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$	
$ \begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C} - \text{C} - \text{CH}_3 \\ \\ \text{Br} \end{array} $	2-bromo-2-methylpropane
	1-bromo-2-methylpropane
$ \begin{array}{c} \text{CH}_3\text{CH}_2 - \text{CH} - \text{CH}_3 \\ \\ \text{Br} \end{array} $	2-bromobutane

Complete the table.

(Total 2 marks)

6. Petroleum is separated into fractions by fractional distillation. The petrol fraction (C_4 to C_{12}) is burned in internal combustion engines and the naphtha fraction (C_7 to C_{14}) is cracked.
- (a) Petroleum is separated into fractions when it is heated and the vapour mixture is passed into a fractionating column. (3)
- Explain what is meant by the term *fraction* as applied to fractional distillation.
 - State a property of the molecules in petroleum which allows the mixture to be separated into fractions.
 - Describe the temperature gradient in the column.
- (b) The fractions from petroleum contain alkane hydrocarbons. (2)
- Write an equation for the incomplete combustion of the alkane C_8H_{18} to produce carbon monoxide and water only.
 - One isomer of C_8H_{18} is 2,2,3-trimethylpentane. Draw the structure of this isomer. (1)
- (c) State **one** economic reason for the cracking of petroleum fractions. (2)
- (d) Identify a catalyst used in catalytic cracking. (1)
- (e) Identify the different type of hydrocarbon produced in a high percentage by the (1)

thermal cracking of alkanes.

(1)

(Total 8 marks)

7. Many hydrocarbon compounds burn readily in air.

- (a) Write an equation to show the complete combustion of $C_{15}H_{32}$
 (b) One of the gaseous products of the incomplete combustion of methane in gas fires is known to be poisonous. Identify this product and write an equation for the reaction in which it is formed from methane.

(Total 4 marks)

8. (a) Crude oil is separated into fractions by fractional distillation. Outline how different fractions are obtained by this process.

(3)

- (b) The table below gives details of the supply of, and demand for, some crude oil fractions.

Fractions	Approximate %	
	Typical supply from crude oil	Global demand
Gases	2	4
Petrol and naphtha	16	27
Kerosine	13	8
Gas oil	19	23
Fuel oil and bitumen	50	38

- (i) Use the data given above to explain why catalytic cracking of crude oil fractions is commercially important.
 (ii) Give the two main types of product obtained by catalytic cracking.

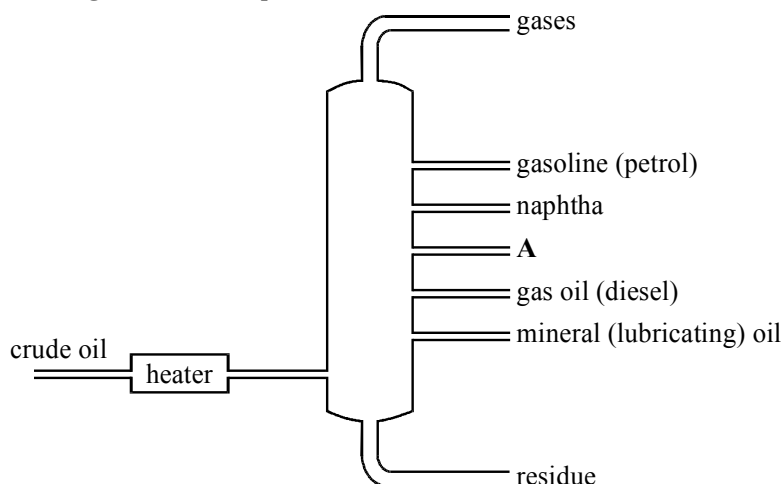
(4)

- (c) Name a catalyst used in catalytic cracking and outline the industrial conditions used in the process.

(3)

(Total 10 marks)

9. (a) The diagram below represents the industrial fractional distillation of crude oil.



- (i) Identify fraction **A**.
 (ii) What property of the fractions allows them to be separated in the column?

(2)

- (b) A gas oil fraction from the distillation of crude oil contains hydrocarbons in the C_{15} to C_{19} range. These hydrocarbons can be cracked by strong heating.

- (i) Write the molecular formula for the alkane with 19 carbon atoms.

- (ii) Write an equation for one possible cracking reaction of the alkane $C_{16}H_{34}$ when the products include ethene and propene in the molar ratio 2:1 and only one other compound.

(3)

(Total 5 marks)

10. Name the alkene $CH_3CH_2CH=CH_2$

(Total 1 mark)

11. The table below shows the fractions obtained from crude oil:

Name of fraction	Number of carbon atoms	Use
Gases	1-4	bottled fuels for camping
Petrol (gasoline)	4-12	fuel for cars
Naphtha	7-14	
	11-15	jet fuel
Gas oil (diesel)	15-19	central heating fuel
Mineral oil	20-30	lubrication
Fuel oil	30-40	
Wax	41-50	Candles
Bitumen	over 50	road surfacing

- (a) Complete the table above by writing in the empty boxes
- (b) Give **one** structural formula in each case for the following components of crude oil:
- (i) the isomer of C_5H_{12} with the lowest boiling point
- (ii) a saturated compound with molecular formula C_5H_{10}
- (c) Give a molecular formula for each of the following components of crude oil:
- (i) the alkane which, on cracking, forms, as the only products, two moles of ethene and one mole of butane per mole of alkane.
- (ii) the straight chain alkane found in the petrol fraction which contains the lowest percentage by mass of hydrogen. Refer to the table at the start of the question to deduce your answer.

(3)

(2)

(2)

(Total 7 marks)

12. (a) Central heating fuel is obtained by distillation of crude oil. An alkane present in central heating fuel contains 16 carbon atoms per molecule. When a molecule of this alkane is cracked, propene is formed.
- (i) Name the crude oil fraction used as central heating fuel.
- (ii) Write an equation for the cracking of the 16-carbon alkane to form octane, propene and ethene as the only products.
- (iii) Give one important commercial use of propene.
- (b) A catalytic converter in the exhaust system of a car contains a ceramic honeycomb covered with a thin coating of the catalyst. When hot gases containing nitrogen monoxide and unburnt octane are passed over the catalyst, they react to form nitrogen, carbon dioxide and water.
- (i) Explain why the catalyst is coated on a honeycomb.
- (ii) Write an equation for the reaction of octane with nitrogen monoxide to form nitrogen, carbon dioxide and water.

(4)

(4)

(Total 8 marks)

13. (a) Gas oil (diesel), kerosine (paraffin), mineral oil (lubricating oil) and petrol (gasoline) are four of the five fractions obtained by the fractional distillation of crude oil within the temperature range 40–400 °C. Identify the missing fraction and state the order in which the five fractions are removed as the fractionating column is ascended. Give **two** reasons why the fractions collect at different levels in the fractionating column. (4)
- (b) Thermal cracking of large hydrocarbon molecules is used to produce alkenes. State the type of mechanism involved in this process. Write an equation for the thermal cracking of $C_{21}H_{44}$ in which ethene and propene are produced in a 3:2 molar ratio together with one other product. (3)
- (c) Write equations, where appropriate, to illustrate your answers to the questions below. (10)
- (i) Explain why it is desirable that none of the sulphur-containing impurities naturally found in crude oil are present in petroleum fractions.
- (ii) The pollutant gas NO is found in the exhaust gases from petrol engines. Explain why NO is formed in petrol engines but is not readily formed when petrol burns in the open air.
- (iii) The pollutant gas CO is also found in the exhaust gases from petrol engines. Explain how CO and NO are removed from the exhaust gases and why the removal of each of them is desirable. (Total 17 marks)
14. (a) Petrol engines in cars produce a number of pollutants which can be removed by catalytic converters. Discuss this statement, indicating what the pollutants are, how they arise and how they are removed as efficiently as possible in a catalytic converter. Write equations for any reactions you discuss. (11)
- (b) Petrol is obtained, not only by fractional distillation of crude oil, but also by cracking of hydrocarbons from heavy fractions. State why hydrocarbons from heavy fractions are cracked and explain why these hydrocarbons are less easy to ignite than those in petrol. (4)
- (Total 15 marks)
15. Ethene and other important hydrocarbons can be produced industrially from decane, $C_{10}H_{22}$. Name the process involved. Write **two** equations for reactions in which ethene is formed from decane by this process. Explain the economic importance of the process. (Total 7 marks)
16. The petroleum and petrochemicals industries use a small number of processes to produce a large number of final products. Three of these processes are fractional distillation, vacuum distillation and cracking.
- (a) Describe briefly how fractional distillation can be used to convert crude oil into a small number of fractions. (4)
- (b) The residue from the fractional distillation process is usually vacuum distilled. This process allows the compounds in the residue to boil at temperatures much lower than their normal boiling points. State whether the sizes of the molecules of the compounds in the residue are smaller or larger than those in the other fractions and suggest why the process of vacuum distillation is used. (2)
- (c) Cracking is carried out by heating long-chain hydrocarbons, often in the presence of steam and a catalyst. Steam cracking is used to produce a mixture of alkanes and alkenes in reactions such as the following:

$$C_9H_{20} \rightarrow C_6H_{14} + C_3H_6$$
- (i) Give a name for each compound produced in this reaction. (2)
- (ii) Nonane, C_9H_{20} , can be cracked to give other products. One molecule of nonane can give two molecules of ethene, one molecule of ethane and one molecule of another substance. What is the molecular formula of the other

substance?

(1)

- (d) Halogenoalkanes have been used for a long time as anaesthetics, for example trichloromethane and 2-bromo-2-chloro-1,1,1-trifluoroethane(halothane). Give the formulae of both trichloromethane and 2-bromo-2-chloro-1,1,1-trifluoroethane.

(2)

(Total 11 marks)

17. (a) Crude oil is separated into fractions by fractional distillation.

(i) Explain the term *fraction*.

(ii) Outline how different fractions are obtained by this process.

(4)

- (b) Gas oil (diesel), gasoline (petrol), kerosene (paraffin) and naphtha are fractions obtained from crude oil.

Write the fractions gas oil, gasoline and kerosene in order of increasing boiling temperature and give one specific use for each of the fuels, gas oil and kerosene.

(3)

- (c) The hydrocarbon $C_{17}H_{36}$ is found in the gas oil fraction obtained by fractional distillation of crude oil.

(i) Name a fraction which is obtained higher up the distillation column than gas oil and give one reason why it is obtained higher up the column.

(ii) Write an equation for the complete combustion of the hydrocarbon $C_{17}H_{36}$

(4)

- (d) Write an equation for the complete combustion of the alkane which requires 11 moles of oxygen per mole of alkane for complete combustion.

(2)

(Total 13 marks)

18. (a) The table below gives details of the supply of, and demand for, some crude oil fractions.

Fractions	Approximate %	
	Typical supply from crude oil	Global demand
Gases	2	4
Petrol and naphtha	16	27
Kerosine	13	8
Gas oil	19	23
Fuel oil and bitumen	50	38

(i) Use the data given above to explain why catalytic cracking of crude oil fractions is commercially important.

(ii) Give the two main types of product obtained by catalytic cracking.

(4)

- (b) Name a catalyst used in catalytic cracking and outline the industrial conditions used in the process.

(3)

- (c) The naphtha fraction is thermally cracked to produce a mixture containing methane, ethene, butene and several other molecules.

(i) Suggest how the mixture produced by cracking is separated into individual compounds.

(ii) Write an equation for the cracking of an alkane which has nine carbon atoms per molecule into methane, ethene and butene.

(3)

- (d) Write an equation for the cracking of $C_{17}H_{36}$ to form octane, propene and ethene

only.

(1)

- (e) Give a molecular formula of the alkane which, on cracking, forms, as the only products, two moles of ethene and one mole of butane per mole of alkane.

(1)

(Total 12 marks)

19. (a) Petrol engines in cars produce a number of pollutants which can be removed by catalytic converters.
Discuss this statement, indicating what the pollutants are, how they arise and how they are removed as efficiently as possible in a catalytic converter. Write equations for any reactions you discuss.

(12)

20. (a) Most of the ethene used by industry is produced when ethane is heated to 900°C in the absence of air. Write an equation for this reaction.

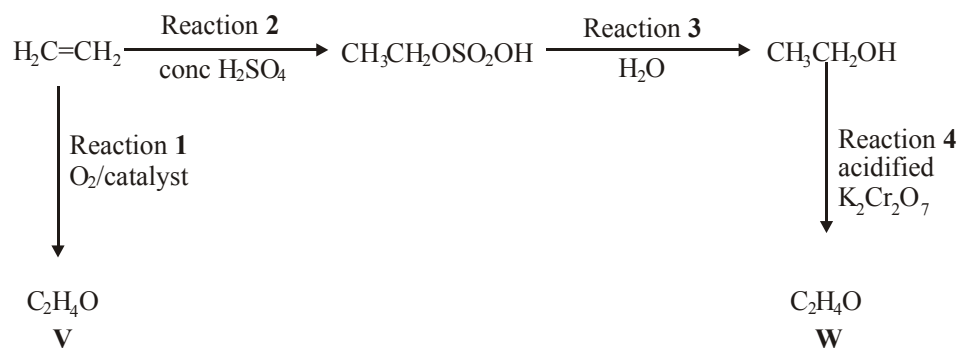
(1)

- (b) Name the type of polymerisation which occurs when ethene is converted into poly(ethene).

(1)

(Total 2 marks)

21. Consider the following reaction scheme, which leads to the formation of two compounds **V** and **W**.

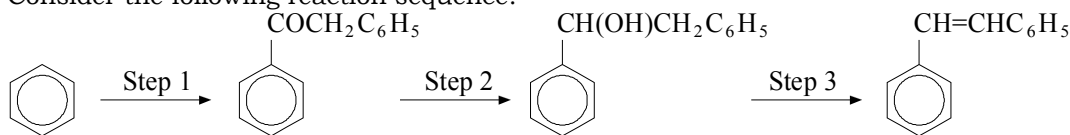


Name and outline a mechanism for Reaction 2.

(5)

(Total 5 marks)

22. Consider the following reaction sequence:



Name the type of isomerism shown by the final product, 1,2-diphenylethene, and explain why this compound can exist in two stereoisomeric forms.

(2)

(Total 2 marks)

23. But-2-ene can exist in two isomeric forms. Give the structures of these two isomers and name the type of isomerism.

(3)

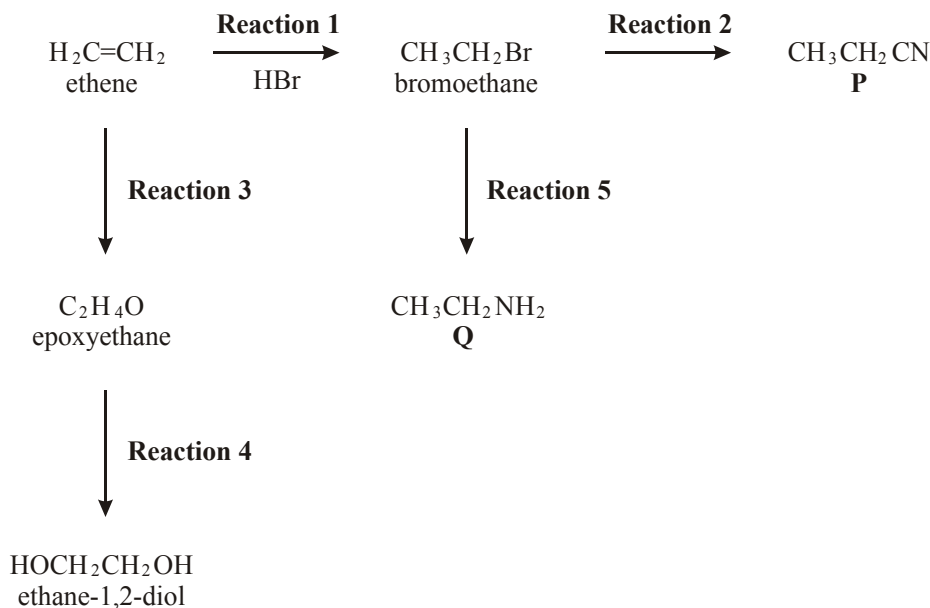
(Total 3 marks)

24. (i) Write an overall equation for the reaction of cyclohexene with hydrogen.
(ii) By stating a reagent and an observation, give a chemical test which would show that this hydrogenation reaction has gone to completion.

(3)

(Total 3 marks)

25. Consider the following scheme of reactions.



In **Reaction 1**, ethene undergoes electrophilic addition with hydrogen bromide.

- State what is meant by the term *electrophile*.
- Outline a mechanism for this reaction.

(5)

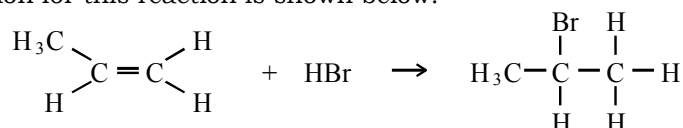
(Total 5 marks)

26. (a) Draw and name the geometrical isomers of pent-2-ene. (2)
- (b) Pent-1-ene reacts with hydrogen bromide to produce 2-bromopentane as the major product.
- Outline the mechanism for this reaction.
 - Identify the minor product formed in this reaction.
 - Explain why 2-bromopentane is the major product of this reaction.

(7)

(Total 9 marks)

27. Propene reacts with hydrogen bromide by an electrophilic addition mechanism forming 2-bromopropane as the major product. The equation for this reaction is shown below.

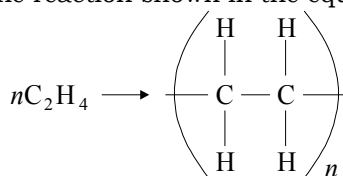


- Outline the mechanism for this reaction, showing the structure of the intermediate carbocation formed.
- Give the structure of the alternative carbocation which could be formed in the reaction between propene and hydrogen bromide.

(5)

(Total 12 marks)

28. (a) How do the physical properties of the alkene homologous series change as the chain length increases? (2)
- (b) For the reaction shown in the equation:



Give the name of this process:

(2)

- (c) Most ethanol used in the chemical industry is manufactured by reacting together water and ethene.

(i) Write a balanced equation for this reaction.

(1)

(ii) Outline the reaction conditions for this reaction.

(3)

(iii) Classify this reaction, indicating the type of initial attack on the ethene.

(2)

(Total 10 marks)

29. Compounds with double bonds between carbon atoms can exhibit geometrical isomerism.

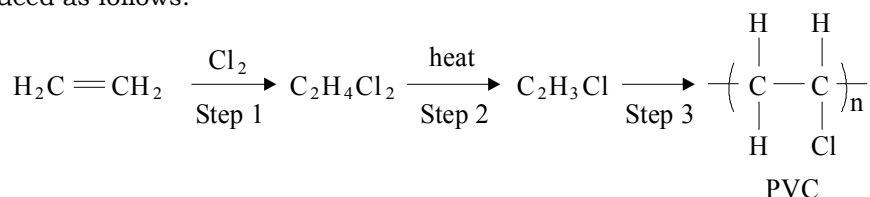
(i) Draw structures for the two geometrical isomers of 1,2-dichloroethene.

(ii) What feature of the double bond prevents isomer 1 from changing into isomer 2?

(3)

(Total 3 marks)

30. The polymer poly(chloroethene), commonly known as poly(vinyl chloride) or PVC, can be produced as follows:



- (a) Using your knowledge of the reaction between bromine and ethene, name and outline a mechanism for Step 1.

(5)

- (b) Write an equation for Step 2 showing clearly the structure of the organic product.

(1)

- (c) Plasticisers are often incorporated into polymers such as PVC. Name a type of compound used as a plasticiser.

(1)

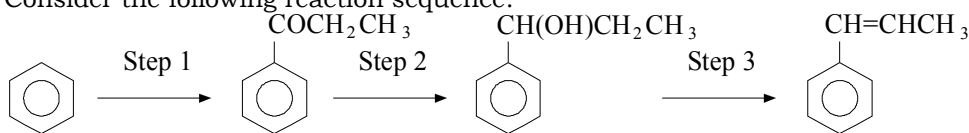
- (d) (i) Draw the structure of the organic product of the reaction of $\text{C}_2\text{H}_4\text{Cl}_2$ with an excess of warm aqueous sodium hydroxide.

(ii) Suggest why $\text{C}_2\text{H}_3\text{Cl}$, the organic product of Step 2, does not react with warm aqueous sodium hydroxide.

(3)

(Total 10 marks)

31. Consider the following reaction sequence:



- (a) Explain why the final product, 1-phenylpropene, is formed as a mixture of two isomers.

(2)

- (b) When 1-phenylpropene is treated with hydrogen bromide, two compounds are formed which are structural isomers.

(i) Give the structures of the two isomers.

(ii) Name the type of mechanism involved.

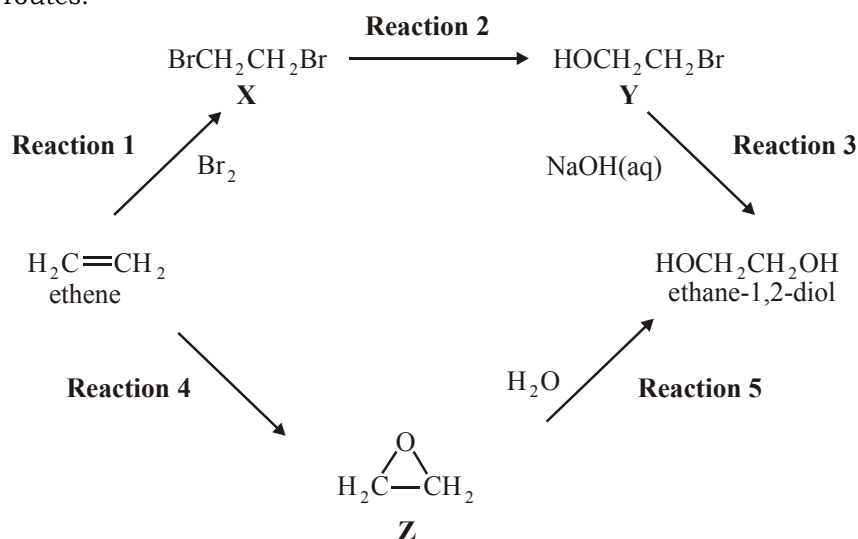
(iii) By reference to the structures of the two carbonium ion intermediates formed, suggest why the two isomers are obtained in unequal amounts.

(6)

(Total 8 marks)

32. Four members of the homologous series of alkenes are ethene, propene, but-1-ene and pent-1-ene.
- Give **one** structural feature of the compounds that makes them members of the homologous series of alkenes. (1)
 - State the trend in the boiling points from ethene to pent-1-ene. (1)
 - Give the name and draw the displayed formula of an alkene that is an isomer of but-1-ene **and** that has a different carbon skeleton. (2)
- (Total 4 marks)
33. Ethanol is produced industrially from ethene. Name the catalyst and state the conditions used in the industrial process and write an equation for the reaction. (4)
- (Total 4 marks)

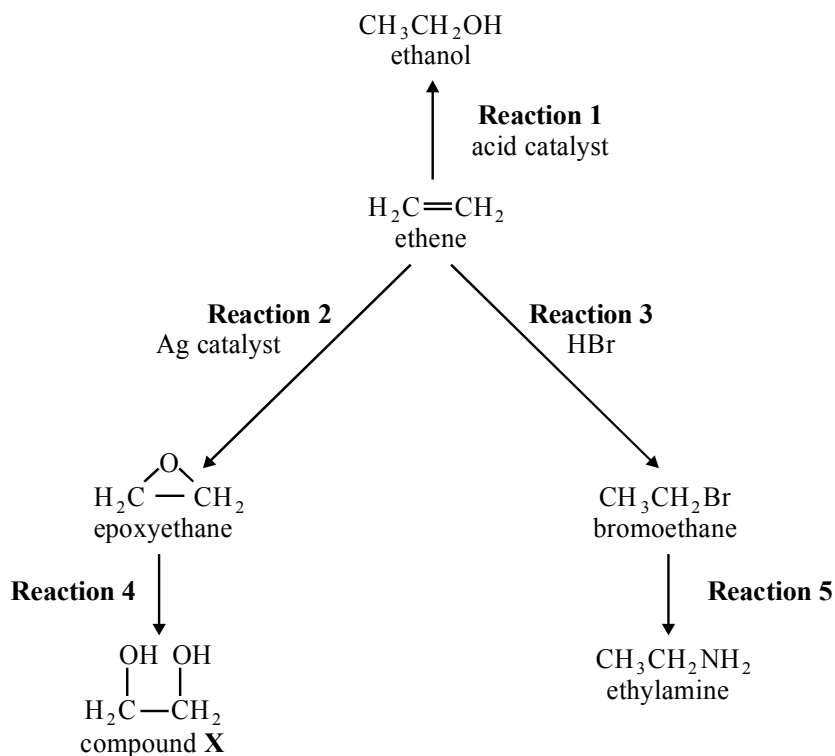
34. Consider the following scheme of reactions for making ethane-1,2-diol from ethene by two different routes.



Name compound **X** and name a mechanism for **Reaction 1**. Explain why ethene is able to react with bromine in this reaction, given that bromine molecules are non-polar.

- (4)
- (Total 4 marks)
35. Addition reactions to alkenes can result in the formation of isomeric compounds. Choose an alkene with molecular formula C_4H_8 which reacts with HBr to form two structural isomers. Give the structures of these two isomers and name the type of structural isomerism shown. Outline a mechanism for the formation of the major product. (7)
- (Total 7 marks)

36. Ethene can be converted into a variety of useful products as illustrated below.



- (a) Give a reagent for **Reaction 1** (1)
- (b) Outline a mechanism for **Reaction 3**. (4)

(Total 5 marks)

37. Describe the arrangement of bonds around the carbon atoms in ethane and in ethene, and state the bond angle in each compound. Compare the relative lengths of the carbon-carbon bonds in the two molecules and explain why they are different. (6)

(Total 6 marks)

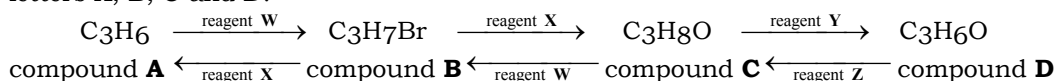
38. There are several non-cyclic structural isomers with the molecular formula C_6H_{12} . (a) One of these isomers, 2-methylpent-2-ene, $(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_3$, reacts with hydrogen bromide. Name the major product and account for its formation by reference to the mechanism of the reaction. (7)
- (b) Identify one linear alkene of formula C_6H_{12} which can exist as a pair of stereoisomers. State the type of stereoisomerism shown, name the alkene and draw the structures of the two isomers. (4)

(Total 11 marks)

39. When 2-methylpent-2-ene reacts with hydrogen bromide two products are formed, one major and one minor, each with molecular formula $\text{C}_6\text{H}_{13}\text{Br}$. Name these two products and account for their formation by reference to the mechanism of the reaction. (Total 15 marks)

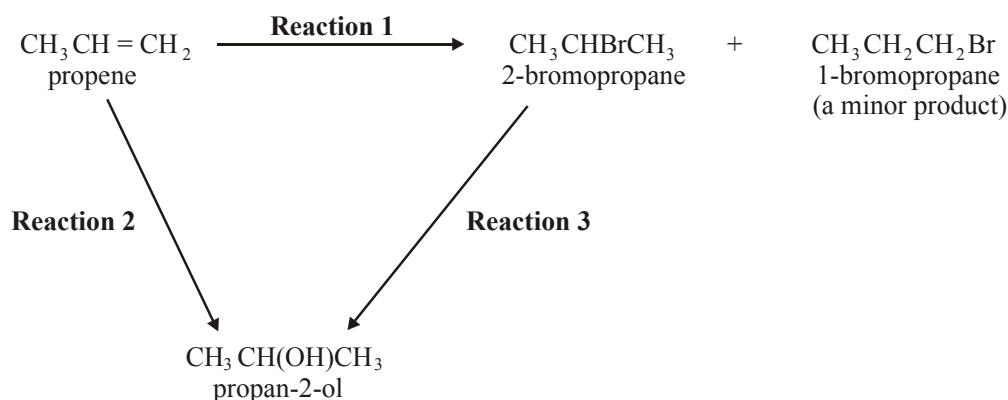
(Total 15 marks)

40. This question is concerned with the six reactions shown in the following scheme. The four compounds involved are represented by their molecular formulae and labelled using the letters **A**, **B**, **C** and **D**.



- (a) State the names of the three homologous series to which compounds **A**, **B** and **C** belong. (3)
- (b) Give the name of the type of reaction occurring in each of the following conversions:

- (i) compound **A** to compound **B**; (1)
- (ii) compound **B** to compound **C**; (1)
- (c) Give the name **or** formula of reagent **X** and state the different conditions under which it would be used in the conversions of compound **B** to compound **C**, and of compound **B** to compound **A**. (3)
- (f) Write an equation for each of the following reactions:
- (i) the conversion of compound **B** to compound **A**; (1)
- (ii) the conversion of compound **B** to compound **C**. (1)
- (g) Outline a possible mechanism for the conversion of compound **B** to compound **C**. (3)
- (Total 13 marks)**
41. (i) Name the alkene $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$
- (ii) Explain why $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$ does not show geometrical isomerism.
- (iii) Draw an isomer of $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$ which does show geometrical isomerism.
- (iv) Draw another isomer of $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$ which does not show geometrical isomerism.
- (Total 4 marks)**
42. The alkene $\text{CH}_3\text{CH}=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_3$ reacts with hydrogen bromide to form 3-bromo-3-methylpentane, $\text{CH}_3\text{CH}_2\text{CBr}(\text{CH}_3)\text{CH}_2\text{CH}_3$, as the major product.
- (a) Give the name of this alkene and state the type of stereoisomerism shown by this compound. (2)
- (b) Name and outline a mechanism for the reaction between the alkene and hydrogen bromide. (5)
- (c) Give the structure of the isomeric product also formed in the above reaction and explain why the two isomers are obtained in unequal amounts. (3)
- (Total 10 marks)**
43. Many naturally-occurring organic compounds can be converted into other useful products. Oleic acid can be obtained from vegetable oils. Oleic acid is an example of an unsaturated compound.
- $$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$$
- oleic acid
- (i) Deduce the molecular formula and the empirical formula of oleic acid.
- (ii) State what is meant by the term *unsaturated*.
- (iii) Identify a reagent for a simple chemical test to show that oleic acid is unsaturated. State what you would observe when oleic acid reacts with this reagent. (5)
- (Total 5 marks)**
44. Consider the following reaction scheme.



- (a) (i) Name the mechanism for **Reaction 1**.
 (ii) Explain why 1-bromopropane is only a minor product in **Reaction 1**. (3)
- (b) **Reaction 2** proceeds in two stages.
 Stage 1 $\text{CH}_3\text{CH}=\text{CH}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CH}_3\text{CH}(\text{OSO}_2\text{OH})\text{CH}_3$
 Stage 2 $\text{CH}_3\text{CH}(\text{OSO}_2\text{OH})\text{CH}_3 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH(OH)CH}_3 + \text{H}_2\text{SO}_4$
 (i) Name the class of alcohols to which propan-2-ol belongs.
 (ii) State the overall role of the sulphuric acid in **Reaction 2**. (6)

(Total 9 marks)

45. This question concerns the chemistry of ethene and compounds derived from it. Consider the following statements and then answer the questions below.
- ☐ Ethene may be polymerised to form poly(ethene).
 - ☐ Treatment of ethene with bromine gives a compound **C**.
 - ☐ Compound **C** may be converted in the laboratory into a compound **D**, which has a percentage composition by mass of: C 38.71%; H 9.68%; O 51.61%. The relative molecular mass, M_r , of **D** is 62.
- (a) (i) Explain what is meant by the term *polymerisation*. (2)
 (ii) Write an equation to represent the polymerisation of ethene. (1)
- (b) Give the name of compound **C**. (1)
- (c) Give the name of the mechanism of the reaction between ethene and bromine. Draw the mechanism for this reaction. (4)
- (d) (i) Use the analytical data provided at the start of the question to deduce the molecular formula of compound **D**. (3)
 (ii) Give the reagent(s) and condition(s) for the conversion of **C** into **D**. (2)
 (iii) Write an equation for the conversion of **C** into **D**. (2)

(Total 15 marks)

Marking Scheme

1. (a) (i) fractional distillation or fractionation 1
 (ii) C_9H_{20} only 1
 (iii) $C_{11}H_{24} + 17O_2 \rightarrow 11CO_2 + 12H_2O$ 1
 (iv) $C_{11}H_{24} + 6O_2 \rightarrow 11C + 12H_2O$ 1
- (b) (1) $C_{10}H_{22} \rightarrow C_3H_6 + C_7H_{16}$ 1
 (ii) correctly drawn structure of methylpropene 1
(insist on clearly drawn C-C and C=C bonds)
- (c) Any two from 2
☐ chemically similar or chemically the same or react in the same way
☐ same functional group
☐ same general formula
☐ differ by CH_2
(penalise same molecular formula or same empirical formula)
- [8]**
2. (a) (i) any two from:
 show a gradation/trend/gradual change in physical properties/
 a specified property
 differ by CH_2
 chemically similar or react in the same way
 have the same functional group 2
(penalise 'same molecular formula')
(penalise 'same empirical formula')
- (ii) fractional distillation or fractionation 1
 (iii) contains only single bonds or has no double bonds
 (credit 'every carbon is bonded to four other atoms' provided
 it does not contradict by suggesting that this will always be H) 1
- (b) (i) the molecular formula gives the actual number of atoms of each
element/type in a molecule/hydrocarbon/compound/formula 1
(penalise 'amount of atoms')
(penalise 'ratio of atoms')
- (ii) $C_{14}H_{30}$ only 1
*(penalise as a contradiction if correct answer is accompanied
 by other structural formulae)*
- (iii) $C_{10}H_{22} + 5\frac{1}{2}O_2 \rightarrow 10C + 11H_2O$ 1
(or double this equation)
- (c) (i) $\frac{1}{2}N_2 + \frac{1}{2}O_2 \rightarrow NO$ 1
(or double this equation)
- (ii) Platinum or palladium or rhodium 1
- (iii) $2CO + 2NO \rightarrow 2CO_2 + N_2$ or
 $2NO \rightarrow N_2 + O_2$ or
(ignore extra O_2 molecules provided the equation balances)
 $C + 2NO \rightarrow CO_2 + N_2$
(or half of each of these equations)
 $C_8H_{18} + 25NO \rightarrow 8CO_2 + 12\frac{1}{2}N_2 + 9H_2O$ 1
(or double this equation)
- [10]**
3. (a) $C \ 22.24/12 = 1.85 \quad H \ 3.71/1 = 3.71 \quad Br \ 74.05/79.9 = 0.927 \quad \mathbf{(1)}$
 ratio C:H:Br = 2:4:1 therefore $C_2H_4Br \quad \mathbf{(1)}$
 empirical mass = 107.9 therefore mol formula = $215.8/107.9 \times C_2H_4Br = C_4H_8Br_2$
(1)
 must use % to justify answer
or
 C $(22.24/100) \times 215.8 = 47.99$ i.e. $48/12 = 4$ carbon atoms **(1)**
 H $(3.71/100) \times 215.8 = 8.01$ i.e. $8/1 = 8$ hydrogen atoms **(1)**

- Br $(74.05/100) \times 215.8 = 159.8$ i.e. $159.8/79.9 = 2$ bromine atoms **(1)**
or
 C $(48/215.8) \times 100 = 22.24\%$ **(1)**
 H $(8/215.8) \times 100 = 3.71\%$ **(1)**
 Br $(159.8/215.8) \times 100 = 74.05\%$ **(1)** 3
- (b) **any two pairs of marks**
 1,1-dibromo-(2-)methylpropane **(1)**
 graphical formula to suit $(\text{CH}_3)_2\text{CHCHBr}_2$ **(1)**
 1,2-dibromo-(2-)methylpropane **(1)**
 graphical formula to suit $(\text{CH}_3)_2\text{C}(\text{Br})\text{CH}_2\text{Br}$ **(1)**
 1,3-dibromo-(2-)methylpropane **(1)**
 graphical formula to suit $\text{BrCH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{Br}$ **(1)**
 allow unambiguous names
 mark name and structure independently
 accept order of bromo / methyl reversed
 penalise once for each of
 numbering from wrong end and di in dibromo omitted max
 4
- [7]**
4. (a) 2-bromo-3-methylbutane
 correct spelling each of bromo, methyl and butane **(1)**
 for numbers – 2 & 3 either order **(1)** 2
- (b) compounds with the same molecular formula / compounds or molecules
 with the same number and type of atoms
not atoms or elements instead of compounds **(1)**
 different structural formulae / different arrangement of atoms / different
 structures / different graphical (displayed) formulae / functional groups in
 different places **(1)** 2
- [4]**
5. 1(-)bromobutane 1
 correct structure for 1-bromo-2-methylpropane 1
(C–C bonds must be clear where drawn)
- [2]**
6. (a) (i) compounds/mixtures/alkanes/hydrocarbons/molecules with a
 boiling point range/similar boiling point/similar number of
 carbon atoms/similar chain length; 1
(insist on “similar” rather than “same”)
(ignore references to size or M_r)
(penalise references to bond breaking/cracking as contradictions)
- (ii) molecules have different boiling points/intermolecular forces/sizes/chain
 lengths/ M_r ; 1
(ignore references to melting points)
(credit the idea that molecules condense at different temperatures)
- (iii) the column has a higher temperature at the base (*Q of L mark*)
 OR
 the column has a lower temperature at the top; 1
(the statement needs to be expressed in good English and show a clear understanding of the correct temperature difference) (penalise “negative OR positive temperature gradient” without qualification to what the candidate means, otherwise ignore) (ignore references to the boiling points of the molecules) (credit correct statements which use specific temperatures with a maximum temperature of 500 °C at the base)
- (b) (i) $\text{C}_8\text{H}_{18} + 8 \frac{1}{2} \text{O}_2 \rightarrow 8\text{CO} + 9\text{H}_2\text{O}$; 1
(or double this equation)
- (ii) correctly drawn structure of 2,2,3-trimethylpentane 1

(penalise the use of 'sticks' once on the paper, including the structures in the 2(a)(ii) and 2(c)(iii) mechanisms) (credit correctly condensed structures)

- (c) cracking produces/makes ethene/propene/alkenes/motor fuels/petrol
OR
cracking makes more useful products/high(er) value products
OR

cracking satisfies the high demand for small(er) products; 1
(ignore the idea that cracking makes or leads to plastics or polyethene) (high demand needs to be qualified)

- (d) zeolite
OR
aluminosilicate OR Al_2O_3 ; 1

- (e) alkene(s); 1
(credit "small or short chain alkenes")
(penalise "cycloalkenes")
(penalise additional types of compounds (e.g. branched alkanes) as a contradiction)
(do not credit examples or formulae, but ignore if these are correct and in addition to the word "alkene")

[8]

7. (a) $\text{C}_{15}\text{H}_{32} + 23 \text{O}_2 \rightarrow 15 \text{CO}_2 + 16 \text{H}_2\text{O}$

Products (1)

Balance (1)

If wrong reactant C.E

- (b) Identity of product: CO or carbon monoxide (1)

Equation: $\text{CH}_4 + \frac{3}{4} \text{O}_2 \rightarrow \text{CO} + 2 \text{H}_2\text{O}$ (1)

Any balanced equation using CH_4 , producing CO

Not could also make C + CO_2

[4]

8. (a) Crude oil is heated to vaporise it / **oil vaporised (1)**
(Vapour passed into fractionating) tower / column (1)

Top of tower cooler than bottom

or **negative temperature gradient (1)**

fractions separated by b.p

OR condensed at different temperatures OR levels

OR low boiling fractions at the top

OR at the top small molecules or light components (1)

max

3

- (b) (i) Identify shortfall in supply - e.g. petrol / small molecules (1)
Higher value products **OR more useful products (1)**
OR cracking produces more of material (problem solving)

- (ii) Motor fuels
Aromatic hydrocarbons
Branched alkanes / hydrocarbons
Cycloalkanes

Any two (2)

Ignore specific fractions, alkanes, shorter alkanes, penalise alkenes, and hydrogen

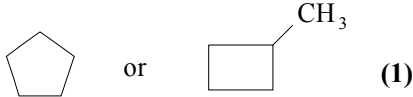
4

- (c) *Catalyst:* Zeolite / aluminosilicate (1)
Conditions: High temp OR around 450°C [300 – 600] $^\circ\text{C}$ **NOT heat / warm (1)**
Slight pressure [1 atm to 10 atm **OR 1 megaPa, 1000 kPa**] (1)
NOT high pressure

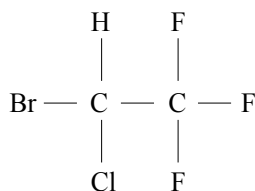
3

[10]

9. (a) (i) Kerosine or paraffin (1)
 (ii) Boiling point (1) 2
 (b) (i) C₁₉H₄₀ (1)
 (ii) C₁₆H₃₄ → 2C₂H₄ + C₃H₆ + C₉H₂₀
 or C₁₆H₃₄ → 4C₂H₄ + 2C₃H₆ + C₂H₆ (2) 4 [5]
10. but-1-ene (1) [1]
11. (a) petrochemicals (1)
 Kerosine or paraffin (1)
 Power stations or ships (1) 3
- (b) (i)
$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array} \quad (1)$$

 (ii) 2
 (1)
- (c) (i) C₈H₁₈ (1)
 (ii) C₁₂H₂₆ (1) 2 [7]
12. (a) (i) Gas oil or diesel (1)
 (ii) C₁₆H₃₄ (1) → C₈H₁₈ + 2C₃H₆ + C₂H₄ eqⁿ (1)
 (iii) To produce polymers (1) 4
- (b) (i) large surface area (1)
 faster reaction (1)
 (ii) C₈H₁₈ + 25NO → 8CO₂ + 9H₂O + 12½N₂ (2) 4 [8]
13. (a) Missing fraction = naphtha (*allow naphtha from list if not quoted separately*) (1) Order = mineral oil (lubricating oil), gas oil (diesel), kerosene (paraffin), naphtha, petrol (gasoline) (1)
Mark order consequential on M1 (if no missing fraction given, M2 = 0) Accept correct reversed order
 Negative temperature gradient on the column
or temperature of column decreases upwards (1)
 Larger molecules **or** heavier fractions condense at higher temperatures **or** lower down the column **or** reference to different boiling points (*ignore mp*) (1) 4
 (b) Type of mechanism = (free) radical / homolytic fission - **used in complete sentence phrase** (1)
 C₂₁H₄₄ → 3 C₂H₄ + 2 C₃H₆ + C₉H₂₀ **correct alkenes** (1)
Accept CH₂CH₂ & CH₂CHCH₃ all correct (1) 3
 (c) (i) Sulphur (containing impurities) burn to form **or forms** SO₂ **or** correct oxides of sulphur (*if oxide identified, must be correct*) (1)
OR equation: e.g. S + O₂ → SO₂ **or** H₂S + 1½O₂ → SO₂ + H₂O
 Leading to acid rain (*must have specified oxides of S or burning*)
or toxic product **or** respiratory problems (1)
 (ii) NO formed by reaction between N₂ and O₂ from the air (1)
OR N₂ + O₂ → 2NO
 High combustion temperature **or** spark in engine (1)
 provides E_A **or** sufficient heat / energy to break N≡N (1)

- (iii) Need to remove NO as forms acid rain **or** toxic product **or** causes respiratory problems **(1)**
 $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ **(1)**
 $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$ **(1)**
 Need to remove CO as it is poisonous **(1)**
 Catalytic converter **(1)**
 uses Pt / Rh / Pd / Ir (*wrong answer cancels a correct one*) **(1)**
 Provides active sites / reduces E_A **(1)**
 Forms $\text{N}_2 + \text{CO}_2$ **(1)**
 $2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ (*correct equation worth last 2 marks*) **(1)** max 10
- [17]**
14. (a) pollutants: CO **(1)**
 NO or NO_2 **(1)**
 unburned hydrocarbons **(1)**
 CO from incomplete combustion **(1)**
 eg $\text{C}_8\text{H}_{18} + 8 \frac{1}{2} \text{O}_2 \rightarrow 8\text{CO} + 9\text{H}_2\text{O}$ (eqⁿ 1)
 NO from $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$ **(1)**
 spark **(1)** max 7
- removal: reaction between NO_x and CO or C_xH_y
 to form harmless products **(1)**
 eqⁿ : $2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ **(2)**
 $\text{C}_8\text{H}_{18} + 25\text{NO} \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O} + 12\frac{1}{2}\text{N}_2$ **(2)**
 one of Pt/Rh/Pd catalyst **(1)** max 4
- (b) Demand for heavy fraction: low or for petrol: high **(1)**
 Supply of heavy fraction: high or of petrol: low **(1)**
 larger Mr are less volatile/have higher bp **(1)**
 due to stronger intermolecular forces **(1)** 4
- [15]**
15. Cracking **(1)**
 radical mechanism **(1)**
 Any two equations e.g. $\text{C}_{10}\text{H}_{22} \rightarrow \text{C}_2\text{H}_4 + \text{C}_8\text{H}_{18}$
 $\text{C}_{10}\text{H}_{22} \rightarrow 2\text{C}_2\text{H}_4 + \text{C}_6\text{H}_{14}$ **(2)**
- $\text{C}_{10}\text{H}_{22}$ or larger alkanes:
 low demand/high abundance/less useful **(1)**
 C_2H_4 or smaller alkanes: high demand/low abundance/more useful **(1)**
 Uses: ethene to make polymers/plastics/ethanol **(1)**
 octane or smaller alkanes - for petrol or fuels **(1)** 8
- [8]**
16. (a) heated / vaporised / boiled
 passed into column / tower
 condense at different heights / liquefy at different heights
 similar molecules (size, bp, mass) condense together / **(1)**
 small molecules at the top and big molecules at the bottom 4
- (b) larger **(1)**
 reduces decomposition **(1)** 2
- (c) (i) hexane or valid isomers **(1)**
 propene **(1)** 2
 (ii) C_3H_6 **(1)** 1
- (d) CHCl_3 **(1)**
 $\text{C}_2\text{HBrClF}_3$ or correct structural formula **(1)**



2

[11]

17. (a) (i) (group of molecules) with similar bp/chain length/size (1)
 (ii) Crude oil is heated to vaporise it / **oil vaporised** (1)
 (Vapour passed into fractionating) tower / column (1)
 Top of tower cooler than bottom
 or **negative temperature gradient** (1)
 fractions separated by b.p
OR condensed at different temperatures OR levels
OR low boiling fractions at the top
OR at the top small molecules or light components (1) max 3
 4
- (b) gasolene, kerosene, gas oil (1)
 lorries, trains (1)
 jet engines (1) 3
- (c) (i) Kerosine (paraffin) / naptha / gasoline (petrol) / LPG (1)
 smaller molecules or lower bp (1)
 (ii) $\text{C}_{17}\text{H}_{36} + 26\text{O}_2 \rightarrow \underbrace{17\text{CO}_2 + 18\text{H}_2\text{O}}_{\text{(1) (1)}}$ 4
- (d) $\text{C}_7\text{H}_{16} + 11\text{O}_2 \rightarrow 7\text{CO}_2 + 8\text{H}_2\text{O}$
 (1) eqⁿ (1)
 2 [13]
18. (a) (i) Identify shortfall in supply - e.g. petrol / small molecules (1)
 Higher value products **OR more useful products** (1)
OR cracking produces more of material (problem solving)
 (ii) Motor fuels
 Aromatic hydrocarbons
 Branched alkanes / hydrocarbons
 Cycloalkanes
Any two (2)
Ignore specific fractions, alkanes, shorter alkanes, penalise alkenes, and hydrogen 4
- (b) *Catalyst:* Zeolite / aluminosilicate (1)
Conditions: High temp OR around 450 °C [300 – 600] °C **NOT heat / warm** (1)
 Slight pressure [□ 1 atm to 10 atm **OR 1 megaPa, 1000 kPa**] (1)
NOT high pressure 3
- (c) (i) fractional distillation (1)
 (ii) $\text{C}_9\text{H}_{20} \rightarrow \text{CH}_4 + 2\text{C}_2\text{H}_4 + \text{C}_4\text{H}_8$ (1)
 (1) 3
- (d) $\text{C}_{17}\text{H}_{36} \rightarrow \text{C}_8\text{H}_{18} + \text{C}_3\text{H}_6 + 3\text{C}_2\text{H}_4$ } (1)
 or $2\text{C}_{17}\text{H}_{36} \rightarrow 2\text{C}_8\text{H}_{18} + 4\text{C}_3\text{H}_6 + 3\text{C}_2\text{H}_4$ } 1
- (e) C_8H_{18} (1) [12]
19. pollutants: CO (1)
 NO or NO₂ (1)

unburned hydrocarbons (1)

CO from incomplete combustion (1)

eg $\text{C}_8\text{H}_{18} + 8 \frac{1}{2} \text{O}_2 \rightarrow 8\text{CO} + 9\text{H}_2\text{O}$ (eqⁿ 1)

NO from $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$ (1)

spark (1)

max 7

removal: reaction between NO_x and CO or C_xH_y

to form harmless products (1)

eqⁿ: $2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ (2)

$\text{C}_8\text{H}_{18} + 25\text{NO} \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O} + 12\frac{1}{2}\text{N}_2$ (2)

one of Pt/Rh/Pd catalyst (1)

max 5

[12]

20. (a) $\text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_4 + \text{H}_2$

1

(b) Addition

1

(ignore self or chain as a preface to "addition")

(penalise additional)

[2]

21. Electrophilic addition

1

M1: curly arrow from C=C bond towards/alongside the side of H atom on $\text{H}_1\text{OSO}_2\text{OH}$

1

(penalise M1 if arrow to H_2SO_4 OR to formal charge on H of H_1O bond)

(ignore partial charges on H and O of H_2SO_4 , but penalise if these are incorrect on the H atom being attacked)

(credit M1 and M2 if correct curly arrow to H^+ provided the anion is present)

M2: curly arrow from H-O bond towards/alongside the side of the O atom on $\text{H-OSO}_2\text{OH}$

1

(credit the arrow even if there are partial or formal charges on H and O but the structure of H_2SO_4 is correct)

M3: correct structure of the carbocation

1

(penalise use of 'sticks' in this structure)

M4: curly arrow from lone pair on an individual oxygen atom of (correct formula for) hydrogensulphate ion towards/alongside C atom bearing the positive charge

1

(insist that the an ion has the correct formula with a lone pair of electrons and a negative charge)

[5]

22. Type of isomerism

geometrical or E-Z (1)

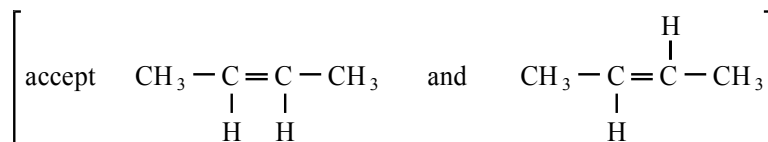
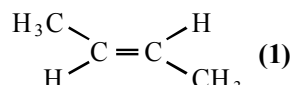
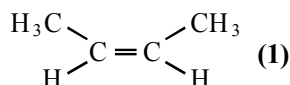
Explanation

restricted rotation or
double bond rigid (1)

2

[2]

23.



Credit 1 mark for a correct formula for but-2-ene

Credit 1 mark for any pair of cis / trans isomers

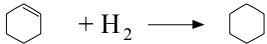
Geometric(al)

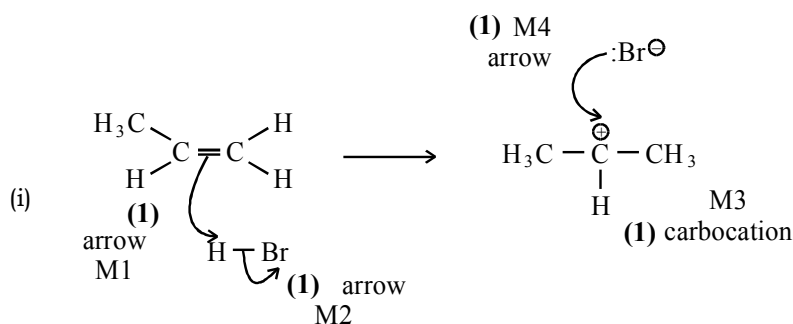
Or cis-trans

Or diastereoisomerism

NOT stereoisomerism

3

- [3]**
24. (i)  + H₂ → 
 or C₆H₁₀ + H₂ → C₆H₁₂
 (ii) Reagent(s) Br₂ or KMnO₄ (1)
 Observation(s) no charge (1)
- [3]**
25. (i) Electron pair/ lone pair acceptor OR seeking/bonds with an electron pair 1
(insist on reference to a pair of electrons)
 (ii) M1 curly arrow from middle of C=C bond of the alkene towards/ alongside the H atom of the H-Br; 1
(penalise arrows which go towards one of the carbon atoms)
(ignore a partial negative charge on the C=C)
 M2 curly arrow from H-Br bond to side of Br atom; 1
(penalise M2 if there are formal charges on HBr or if there are partial charges which are the wrong)
(penalise M2 if the single bond has two dots in addition to the line)
 M3 correct structure for carbocation; 1
(penalise M3 if the positive charge is placed on the end of a bond) *(penalise M3 if any alkene other than ethene is used - all other marks can score)*
 M4 curly arrow from lone pair on bromide ion to the positive carbon 1
 of carbocation, ensuring that bromide ion has a negative charge;
- [5]**
26. (a) M1 credit a correct structure for either geometrical isomer and its designation as either *cis* or *trans*. 1
 OR credit two correct geometrical isomer structures (ignore the names)
 OR credit two correct names for *cis* pent-2-ene and *trans* pent-2-ene (ignore the structures)
 M2 credit a second mark if all four parts of the required structures and names are correct. 1
(credit "linear" structures)
(insist on the alkyl groups being attached clearly by C-C bonds)
 (b) (i) M1 curly arrow from middle of C = C bond to H atom on H-Br 1
(penalise M1 if partial negative charge or formal positive charge on H)
(penalise M1 if pent-2-ene is used)
 M2 curly arrow from H-Br bond to side of Br atom 1
 M3 correct structure for correct secondary carbocation 1
 M4 curly arrow from lone pair on bromide ion to the positive carbon of carbocation, ensuring that bromide ion has a negative charge. 1
(with the exception of pent-2-ene, if the wrong alkene is used, only penalise the structure M3)
(penalise the use of two dots in addition to a covalent bond, once only)
 (ii) 1-bromopentane 1
 (iii) M1 2-bromopentane is formed *via* the secondary (or 2°) carbocation 1
 OR 1-bromopentane is formed *via* the primary (or 1°) carbocation
 M2 a secondary carbocation is more stable than a primary carbocation - 1
 award this mark only if the quality of language justifies the award.
(the argument must involve clear statements about carbocations)
- [9]**
- 27.

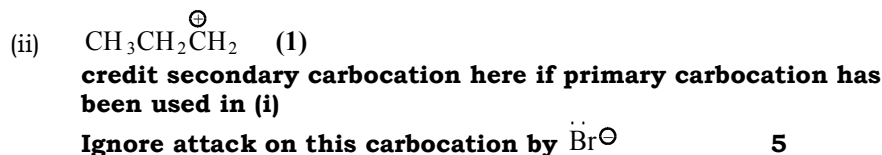


If wrong carbocation, lose structure mark

If wrong alkene, lose structure mark

Can still score $\frac{3}{4}$ i.e. penalise M3

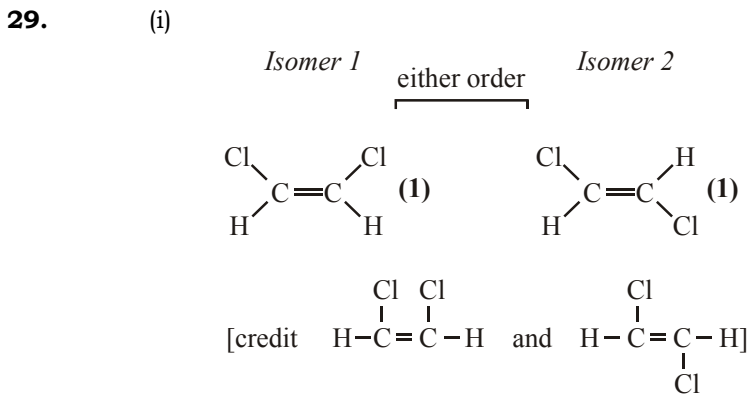
Penalise M2 if polarity included incorrectly
 no bond between H and Br
 bond is shown as $\overset{\cdot\cdot}{\text{H}}-\text{Br}$ or $\text{H}-\overset{\cdot\cdot}{\text{Br}}$



[5]

28. (a) melting point increases (1)
 boiling point increases (1)
or they are liquids, the higher members are solids (1)
 density increases (1)
 viscosity increases (1) max 2
- (b) addition (1) 2
 polymerisation (1)
- (c) (i) $\text{C}_2\text{H}_4 + \text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_5\text{OH}$ - must show the functional group (1) 1
 (ii) vapour phase / high temperature ($300 \pm 50^\circ\text{C}$) (1)
 high pressure $70\text{cl} \pm 20$ (1)
 if high T and high p , then only 1 mark, value for either gives 2nd mark
 strong acidic catalyst / H_3PO_4 (1) 3
 (iii) electrophilic (1)
 addition (1) 2

[10]

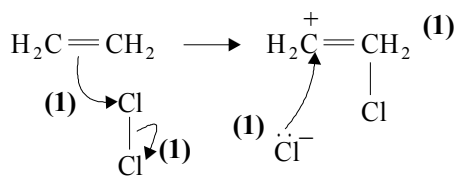


- (ii) restricted rotation OR no rotation OR cannot rotate (1)

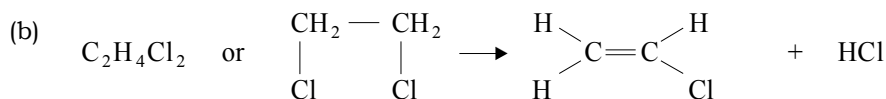
3

[3]

30. (a) electrophilic addition (1)



5



1

(1)

(c) ester or alkoxy alcohol (1)

1

(d) (i) $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$ (1)

(ii) high electron density of double bond (1)

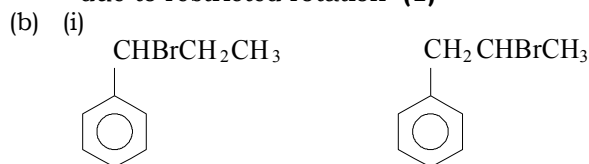
repels OH^- or nucleophile (1)

3

[10]

31. (a) geometrical or cis-trans isomers (1)
due to restricted rotation (1)

2



(1)

(1)

(ii) electrophilic addition (1)

(iii) $\text{C}_6\text{H}_5\text{CH}^+\text{CH}_2\text{CH}_3$ (1) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}^+\text{CH}_3$ (1)
both secondary but one is more stable (1)

6

[8]

32. (i) the same general formula or C_nH_{2n} / the same functional group / a $\text{C}=\text{C}$ / a double bond / differ from their immediate neighbour by CH_2

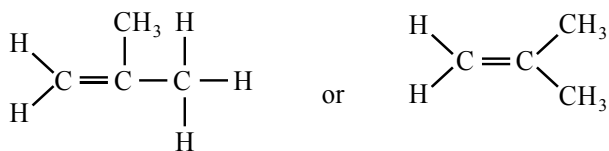
1

allow 'all straight chain alkenes'

(ii) increases

(iii) methylpropene / 2-methylpropene **not** 2-methylprop-2-ene (1)

1

ignore wrong punctuation

allow 1 mark for but-2-ene with its correct structure (1)

2

minimum structure is $\text{CH}_2=\text{CCH}_3$ or $\text{CH}_2=\text{C}(\text{CH}_3)\text{CH}_3$



[4]

33. Catalyst (c) phosphoric acid or (c) sulphuric acid (1)

Not dilute

accept correct formula

Conditions Temp = High or 200–500°C (1)

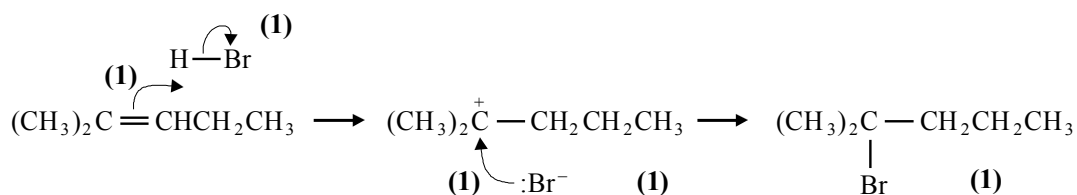
Temp = medium or moderate or 50–100°C

Pressure = High or 5–20 Mpa or 50–200 atmos

Pressure = High or 2–4 Mpa or 20–40 atmos

If. wrong, no catalyst given, allow phosphoric acid conditions

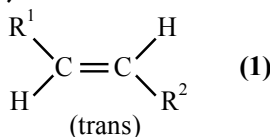
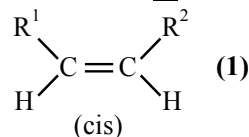
- Equation $\text{CH}_2 = \text{CH}_2 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$
 allow C_2H_4 allow $\text{C}_2\text{H}_5\text{OH}$
 not CH_2CH_2 4
34. M1 X is 1,2-dibromoethane only 1 [4]
 M2 electrophilic addition 1
 (both words needed)
 M3 the double bond is a centre of electron density 1
 OR electron-rich
 OR nucleophilic
 OR a source of an electron pair
 OR a pi cloud/bond of electrons
 M4 a dipole or polarity is induced/created/formed in the Br-Br bond/molecule - 1
 award this mark only if the quality of language justifies the award. [15]
35. An appropriate alkene; $\text{CH}_3\text{CH}_2\text{CHCH}_2$ or $(\text{CH}_3)_2\text{CCH}_2$ 1
 Isomer 1 1
 Isomer 2 1
 Position isomerism 1
 Mechanism
 electrophilic attack and electron shift to Br (Unless H^+ used) 1
 carbocation 1
 reaction with carbocation 1
 [Allow mechanism marks for the alkene $\text{CH}_3\text{CHCHCH}_3$]
 [Allow one mark if mechanism for minor product given] [7]
36. (a) Reaction 1 H_2O or steam (1)
 (b) 4
- M1 (1) M2 (1) M3 (1) M4 structure (1)
 Penalise M2 incorrect δ^+ / δ^-
 Penalise δ^- on alkene (M1)
 Penalise dots on bonds once
 Penalise M4 (structure) for use of wrong alkene
 Penalise M1 for use of Br_2 [5]
37. ethane tetrahedral (or 3d shown in diagram) (1) $109(\frac{1}{2})^\circ$ (1)
 ethene (trigonal) planar (1) 120° (1)
 bond lengths: C – C in ethane longest or }
 C = C in ethene shortest } (1)
 density
 in ethene : 2 pairs of e/greater electron / π bond (1)
 6
38. (a) [6]



2-bromo-2-methylpentane (1)

tertiary carbonium ion more stable than secondary (1)

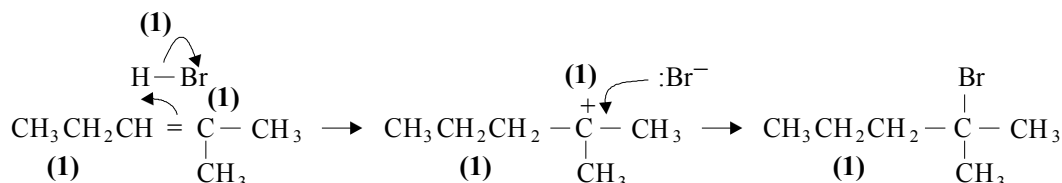
7

(b) geometrical or cis-trans isomerisation (1)hex-3-ene or hex-2-ene (1)

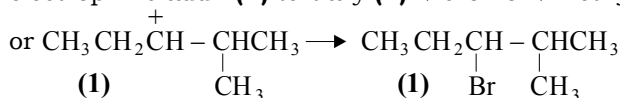
4

[11]

39.



electrophilic addn (1) tertiary (1) 2-bromo-2-methylpentane (1)



secondary (1) 3-bromo-2-methylpentane (1)

tertiary more stable than secondary (1)

[13]

40. (a) A – alkene (1)

B – halogenoalkane / bromoalkane / alkyl halide / haloalkane (1)

C – alcohol (ignore primary, secondary) (1)

3

(b) (i) addition ignore nucleophilic / electrophilic / free radical (1)

1

(ii) substitution **not** replacement / displacement (1)

1

(c) Sodium hydroxide / NaOH / KOH **not** just hydroxide (1)(B to C) aqueous **not** dilute (1)

(B to A) alcoholic (1)

mark alternatives as (d)

ignore references to concentration and temperature

3

(d) (i) $\text{CH}_3\text{CH}(\text{CH}_3)\text{Br} + \text{NaOH} \rightarrow \text{CH}_3\text{CH}=\text{CH}_2 + \text{NaBr} + \text{H}_2\text{O}$ (1)(ii) $\text{CH}_3\text{CH}(\text{CH}_3)\text{Br} + \text{NaOH} \rightarrow \text{CH}_3\text{CH}(\text{CH}_3)\text{OH} + \text{NaBr}$ (1)allow molecular formulae $\text{C}_3\text{H}_7\text{Br}$; $\text{C}_3\text{H}_8\text{O}$; C_3H_6 allow ionic versions (with OH^- , Br^-)

2

(e) arrow from O of OH^- to C joined to Br (1)

lone pair not needed

C-Br polarity shown by δ^+ δ^- **or**heterolytic fission of C-Br bond shown by arrow from bond between C and Br to Br **or**

intermediate with partial bonds and minus sign (1)

 Br^- as product (1)

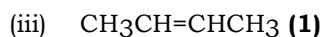
allow all 3 marks if 1-bromopropane identified as B

3

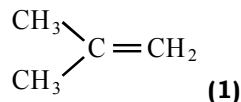
[13]

41. (i) but-1-ene (1)

(ii) two H on one carbon of double bond (1)



(iv)



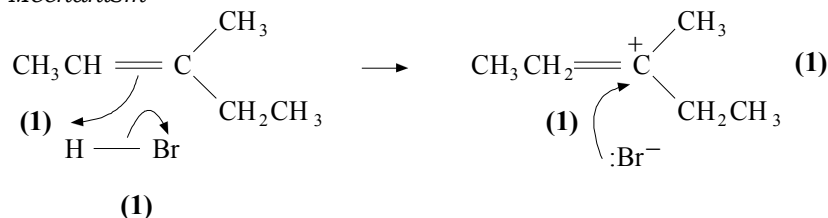
4

[4]

42. (a) Name of alkene 3-methylpent-2-ene (1)
 Type of stereoisomerism geometrical or cis-trans (1)

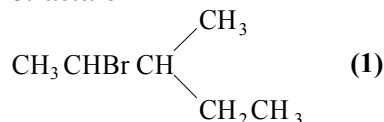
2

- (b) Name of mechanism electrophilic addition (1)
 Mechanism



5

- (c) Structure



Explanation $(\text{CH}_3\text{CH}_2)_2\text{C}^+\text{CH}_3$ or tertiary carbonium ion more stable (1)

than $\text{CH}_3\text{C}^+\text{HCH}(\text{CH}_3)\text{CH}_2\text{CH}_3$ or secondary carbonium ion (1)

3

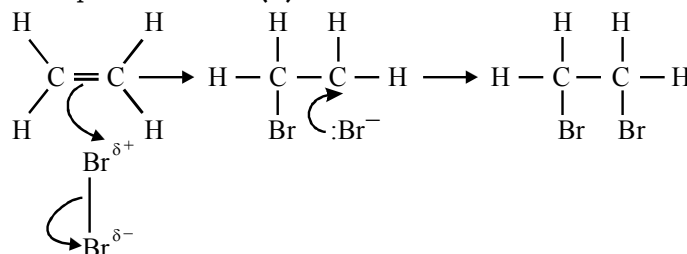
[10]

43. (i) $\text{C}_{18}\text{H}_{34}\text{O}_2$ Only; 1
 $\text{C}_9\text{H}_{17}\text{O}$ Only; 1
 (empirical formula is not consequential on molecular formula)
- (ii) (An unsaturated compound) contains (at least) one double bond
 OR
 Contains $\text{C}=\text{C}$; 1
 (must be a positive statement)
- (iii) M1: Bromine water
 OR
 $\text{Br}_2(\text{aq})$
 OR
 Bromine
 OR
 Br_2 ; 1
 (penalise "bromide water", but mark on)
- M1: decolourised or goes colourless
 OR
 from brown/red/orange/yellow to colourless; 1
 (Must be "colourless" not "clear" for M2)
 (chemical error if no reagent or wrong reagent, loses both marks) (credit KMnO_4 for M1, (purple) to colourless for M2 (if acidified) OR (purple) to brown/brown precipitate (if alkaline or unspecified) (No credit for hydrogen or iodine as reagents)

[5]

44. (a) (i) Electrophilic addition (Both words required) 1
- (ii) M1 the reaction to form 1-bromopropane goes via the primary carbocation OR 1° carbocation 1
OR via $\text{CH}_3\text{CH}_2\text{CH}_2^+$
M2 primary carbocations are less stable than secondary carbocations. 1
(Credit converse arguments for M1 and M2 i.e. the reaction to form 2-bromopropane goes via the secondary carbocation, M1, and secondary carbocations are more stable than primary carbocations, M2)
(Accept the use of "carbonium ions" as an alternative to carbocation)
- (b) (i) Secondary OR 2° 1
(ii) Catalyst ONLY 1
(Ignore homogeneous, heterogeneous)

[9]

45. (a) (i) the joining together of monomers / small molecules (1) to form long chains / large molecules (1) 2
(ii) $n\text{CH}_2 = \text{CH}_2 \rightarrow (-\text{CH}_2-\text{CH}_2-)_n$ (1)
allow $n\text{CH}_2\text{CH}_2$ **not** $n\text{C}_2\text{H}_4$ 1
- (b) 1,2-dibromoethane (1) 1
- (c) electrophilic addition (1)
- 
- words or diagrams to show attack by p electrons on Br atom
and either δ^+/δ^- on Br_2 or e^- shift on $\text{Br}-\text{Br}$ (1)
correct carbocation intermediate (allow triangular representation) (1)
attack by Br^- (onto +ve carbon) leading to correct product (1) 4
- (d) (i) $\text{C } 38.71/12 = 3.23$; $\text{H } 9.68/1 = 9.68$; $\text{O } 51.61/16 = 3.23$ (1)
ratio $\text{C}:\text{H}:\text{O} = 1:3:1$ / empirical formula = CH_3O (1)
empirical mass = 31 so molecular formula = $2 \times \text{CH}_3\text{O} = \text{C}_2\text{H}_6\text{O}_2$ (1) 3
(ii) reagent = NaOH / KOH (1)
conditions = aqueous solution (dependent on first mark) (1) 2
(iii) $\text{CH}_2\text{BrCH}_2\text{Br} + 2\text{NaOH} \rightarrow \text{CH}_2(\text{OH})\text{CH}_2\text{OH} + 2\text{NaBr}$
product = $\text{CH}_2(\text{OH})\text{CH}_2\text{OH}$ (condone missing brackets) (1)
correctly balanced (1) 2
if $\text{C}_2\text{H}_6\text{O}_2$ given, allow second mark only
for $\text{CH}_2\text{BrCH}_2\text{Br} + 2\text{H}_2\text{O} \rightarrow \text{CH}_2(\text{OH})\text{CH}_2(\text{OH}) + 2\text{HBr}$
allow 2 marks if reagent in (ii) is H_2O or aqueous solution

[15]

10.3 ORGANIC HALOGEN COMPOUNDS

1. Methylbenzene is converted into (chloromethyl)benzene in a free radical substitution reaction.
- $$\text{C}_6\text{H}_5\text{CH}_3 + \text{Cl}_2 \longrightarrow \text{C}_6\text{H}_5\text{CH}_2\text{Cl} + \text{HCl}$$
- Write an equation for the initiation step. (1)
 - Write equations for the **two** propagation steps. (2)
 - Give the formula of another possible organic product of the reaction. (1)
- (Total 4 marks)**
2. The reaction of bromine with ethane is similar to that of chlorine with ethane. Three steps in the bromination of ethane are shown below.
- Step 1 $\text{Br}_2 \longrightarrow 2\text{Br}^\bullet$
- Step 2 $\text{Br}^\bullet + \text{CH}_3\text{CH}_3 \longrightarrow \text{CH}_3\text{CH}_2^\bullet + \text{HBr}$
- Step 3 $\text{CH}_3\text{CH}_2^\bullet + \text{Br}_2 \longrightarrow \text{CH}_3\text{CH}_2\text{Br} + \text{Br}^\bullet$
- Name this type of mechanism.
 - Suggest an essential condition for this reaction.
 - Steps 2 and 3 are of the same type. Name this type of step.
 - In this mechanism, another type of step occurs in which free-radicals combine. Name this type of step. Write an equation to illustrate this step. (5)
 - Further substitution in the reaction of bromine with ethane produces a mixture of liquid organic compounds.
 - Name a technique which could be used to separate the different compounds in this mixture.
 - Write an equation for the reaction between bromine and ethane which produces hexabromoethane, C_2Br_6 , by this substitution reaction. (2)
- (Total 7 marks)**
3. (a) Dichloromethane, CH_2Cl_2 , is one of the products formed when chloromethane, CH_3Cl , reacts with chlorine.
- Name the type of mechanism involved in this reaction and write an equation for each of the steps named below.
 - Write an overall equation for the formation of dichloromethane from chloromethane. (5)
- (b) A compound contains 10.1% carbon and 89.9% chlorine by mass. Calculate the molecular formula of this compound, given that its relative molecular mass (M_r) is 237.0. (3)
- (c) Suggest the formulae of two bromine-containing organic compounds formed when dibromomethane, CH_2Br_2 , reacts with bromine. (2)
- (Total 10 marks)**
4. (a) When 3-bromo-2,3-dimethylpentane, $(\text{CH}_3)_2\text{CHCBr}(\text{CH}_3)\text{CH}_2\text{CH}_3$, reacts with aqueous potassium hydroxide, an alcohol is formed.
- Name the type of reaction taking place and give the role of the reagent.
 - Outline a mechanism for the reaction, showing clearly the structure of the alcohol formed. (5)
- (b) When 3-bromo-2,3-dimethylpentane reacts with ethanolic potassium hydroxide, three structurally isomeric alkenes are formed.
- Name the type of reaction taking place and give the role of the reagent.
 - One of the reaction products is 2,3-dimethylpent-2-ene. Give the structure of this alkene and outline a mechanism for its formation.

- (iii) Give the structures and names of the other two alkenes which are also formed.

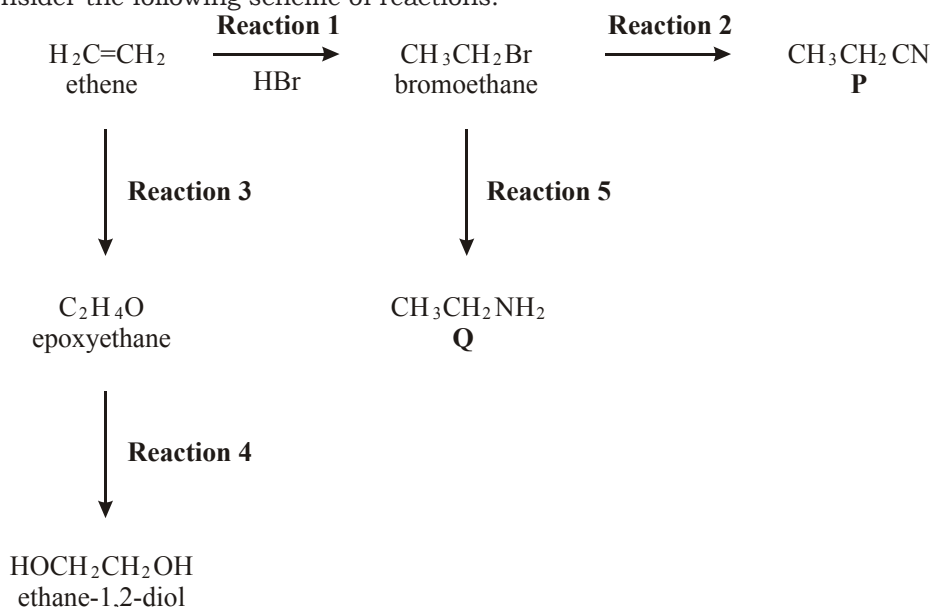
(10)
(Total 15 marks)

5. Bromoethane, $\text{CH}_3\text{CH}_2\text{Br}$, reacts with sodium hydroxide in an elimination reaction to form ethene.

- (i) Outline a mechanism for this elimination reaction.
(ii) Suggest **one** reason why this method for making ethene is not used in industry.

(4)
(Total 4 marks)

6. Consider the following scheme of reactions.



In **Reactions 2** and **5**, bromoethane undergoes nucleophilic substitution.

- (i) Identify a reagent for **Reaction 2**. Name the organic product, **P**.
(ii) Identify a reagent for **Reaction 5**. Name the organic product, **Q**.
(iii) Outline a mechanism for **Reaction 5**.

(8)
(Total 8 marks)

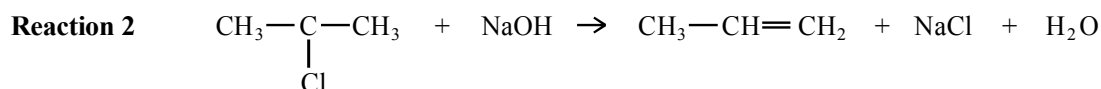
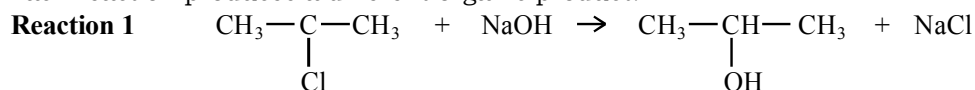
7. (a) Draw the structure of 2-bromo-3-methylbutane.

- (b) (i) Draw the structure of methylbut-2-ene.
(ii) Methylbut-2-ene is formed when 2-bromo-3-methylbutane is treated with ethanolic potassium hydroxide. Name and outline the mechanism for this reaction.

- (c) Name the isomer of methylbut-2-ene which is also formed when 2-bromo-3-methylbutane is treated with ethanolic potassium hydroxide.

(1)
(Total 7 marks)

8. When 2-chloropropane reacts with sodium hydroxide, two different reactions occur. Each reaction produces a different organic product.



- (i) Outline a mechanism for **Reaction 1** and state the role of the hydroxide ion in

- this reaction.
 (ii) Outline a mechanism for **Reaction 2** and state the role of the hydroxide ion in this reaction.

(7)

(Total 7 marks)

9. (a) A substitution reaction occurs when 2-bromopropane reacts with aqueous sodium hydroxide.

- (i) Draw the structure of the organic product of this reaction and give its name.
 (ii) Name and outline the mechanism for this reaction.

(5)

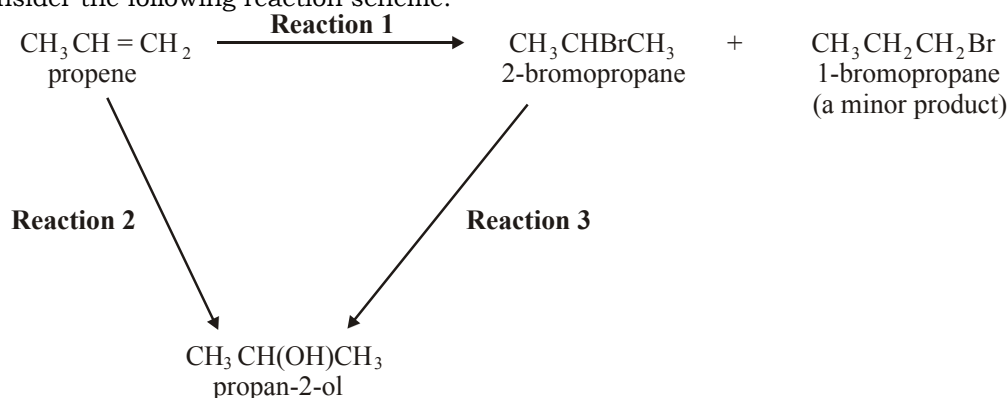
- (b) Under different conditions, 2-bromopropane reacts with sodium hydroxide to produce propene.

- (i) Name the mechanism for this reaction.
 (ii) State the role of sodium hydroxide in this reaction.

(2)

(Total 7 marks)

10. Consider the following reaction scheme.



- (a) Give a suitable reagent and state the essential conditions required for **Reaction 3**.
 (b) The reagent used for **Reaction 3** can also be used to convert 2-bromopropane into propene. State the different conditions needed for this reaction.

(2)

(1)

(Total 3 marks)

11. Compound **X**, $(\text{CH}_3)_2\text{CHCN}$, can be formed from a haloalkane, $\text{C}_3\text{H}_7\text{Br}$.

- (i) Name compound **X**.
 (ii) Give the reagent and conditions necessary to form **X** from $\text{C}_3\text{H}_7\text{Br}$.
 (iii) Name and outline the mechanism for this reaction, showing clearly the structure of $\text{C}_3\text{H}_7\text{Br}$.

(7)

(Total 7 marks)

12. (a) The equation below shows the reaction of 2-bromopropane with an excess of ammonia.



Name and outline the mechanism involved.

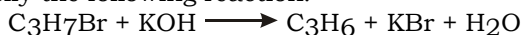
(5)

- (b) When 2-bromopropane is heated with ethanolic potassium hydroxide, an elimination reaction occurs. State the role of potassium hydroxide and outline a mechanism for this reaction.

(5)

(Total 10 marks)

13. Classify the following reaction.



(1)

(Total 1 mark)

14. (i) Give the structural formula and name of the organic product of the reaction

between $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ and potassium cyanide.

- (ii) Name and outline the mechanism involved in this reaction.
- (iii) Suggest why $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{F}$ reacts less rapidly than $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ with potassium cyanide.

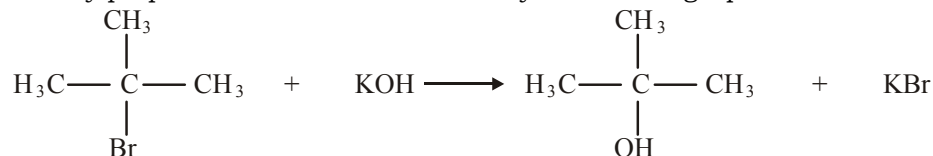
(6)

(Total 6 marks)

15. (a) Name and outline a mechanism for the reaction of 2-bromo-2-methylpropane with ethanolic potassium hydroxide to form the alkene 2-methylpropene, $(\text{CH}_3)_2\text{C}=\text{CH}_2$

(4)

- (b) When 2-bromo-2-methylpropane reacts with aqueous potassium hydroxide, 2-methylpropan-2-ol is formed as shown by the following equation.



State the role of the hydroxide ions in this reaction.

(1)

- (c) Write an equation for the reaction that occurs when $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ reacts with an excess of ammonia. Name the organic product of this reaction.

(3)

(Total 8 marks)

16. Alkenes can be made in the laboratory from halogenoalkanes by heating them with a suitable reagent. By a similar method, cyclohexene can be prepared from bromocyclohexane.

- (i) Suggest the name of a suitable reagent and state an essential condition, other than heat, for this reaction, starting from bromocyclohexane. (2)
- (ii) Using graphical/displayed formulae, write an equation for this reaction. (2)
- (iii) Give the name of this type of reaction. (1)

(1)

(Total 5 marks)

17. (a) (i) Write an equation for the reaction between 1-bromopropane and potassium cyanide.

(1)

- (ii) Give the name of the type of mechanism involved in the reaction in (b)(i).

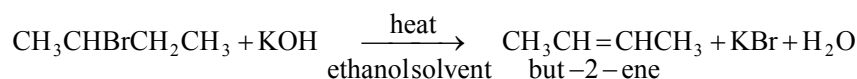
(1)

- (b) Give the formula of the attacking species in (b) and state what feature of its structure is responsible for its role.

(2)

(Total 4 marks)

18. Consider the following reaction in which an alkene is formed from a haloalkane.



- (a) Name the haloalkane used in this reaction.

(1)

- (b) Name and outline a mechanism for this reaction.

(4)

(Total 5 marks)

19. (a) The reaction of chloromethane with chlorine to form trichloromethane is a free radical substitution involving several propagation steps.

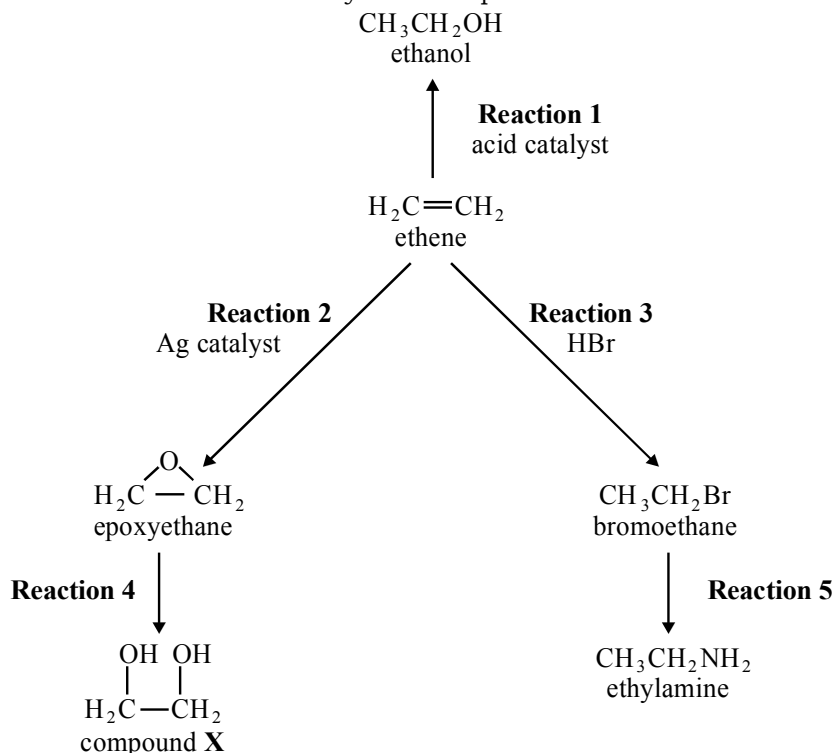
- (i) Write an overall equation for this reaction.
- (ii) What is meant by the term *propagation step*?
- (iii) Write an equation for a propagation step in which chloromethane reacts.
- (iv) Write an equation for a propagation step in which trichloromethane is formed.

(4)

- (b) Write an equation and outline a mechanism for the reaction of chloromethane with an **excess** of ammonia.

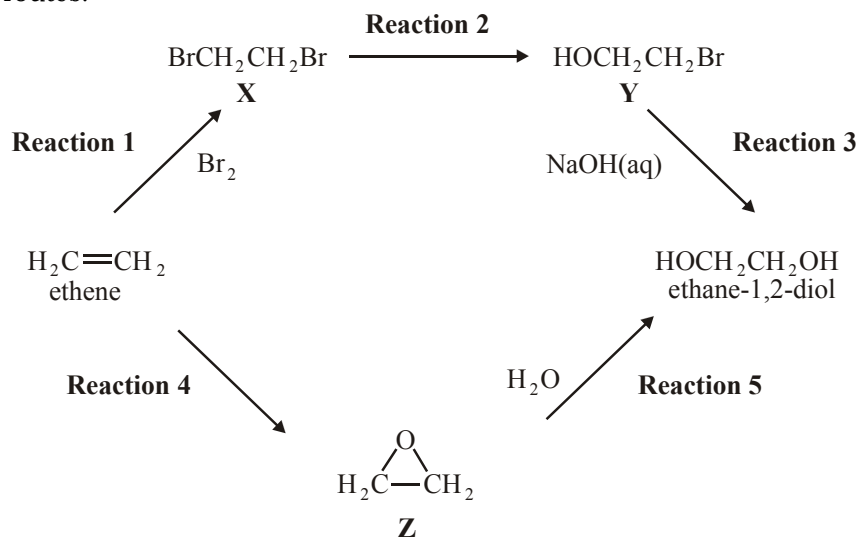
20. When a mixture of chlorine with an excess of methane is irradiated with ultraviolet light, a reaction occurs with chloromethane as the main organic product. Write an equation and a mechanism for the formation of chloromethane. (5)
(Total 9 marks)

21. Ethene can be converted into a variety of useful products as illustrated below. (5)
(Total 5 marks)



(b) Give a reagent for **reaction 5**.

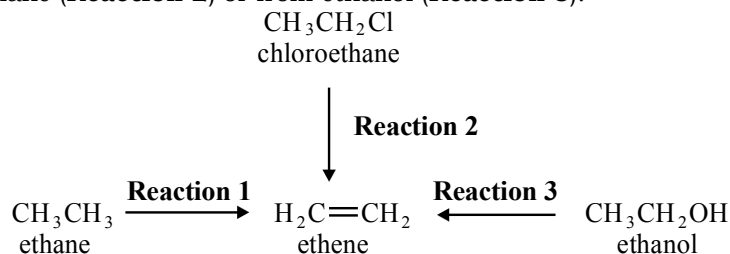
22. Consider the following scheme of reactions for making ethane-1,2-diol from ethene by two different routes. (1)
(Total 1 mark)



Name and outline a mechanism for **Reaction 3**. Explain why compound **Y** is susceptible to attack by hydroxide ions.

23. Ethene is an important starting point for the manufacture of plastics and pharmaceutical (4)
(Total 4 marks)

chemicals. Most of the ethene used by industry is produced by the thermal cracking of ethane obtained from North Sea gas (**Reaction 1**). It is also possible to make ethene either from chloroethane (**Reaction 2**) or from ethanol (**Reaction 3**).



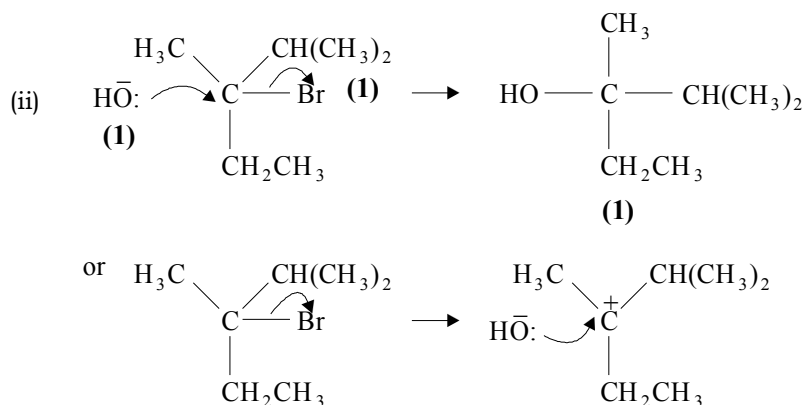
- (a) Give essential conditions and reagents for **Reaction 2**. (2)
- (b) Name and outline a mechanism for **Reaction 2**. Suggest a reason why chloroethane is **not** chosen by industry as a starting material to make ethene commercially. (5)
- (Total 7 marks)**
24. Reaction of 2-bromobutane with potassium hydroxide can produce two types of product depending on the solvent used. In aqueous solution, the formation of an alcohol, **E**, is more likely but in ethanolic solution the formation of alkenes is more likely.
- (a) For each type of product, name the type of reaction occurring and state the role of the potassium hydroxide. (4)
- (b) Name alcohol **E** and draw its structural formula. By reference to the structure of the halogenoalkane, explain why the initial step in the mechanism of the reaction producing the alcohol occurs. (5)
- (Total 9 marks)**
25. The structural formulae of compounds **A**, **B**, **C** and **D**, which are the four structural isomers of molecular formula $\text{C}_4\text{H}_9\text{Cl}$, are given below.
- | | | | |
|---|---|---|----------|
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$ | $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{CH}_3$ | $(\text{CH}_3)_2\text{CHCH}_2\text{Cl}$ | |
| $(\text{CH}_3)_2\text{C}(\text{Cl})\text{CH}_3$ | | | |
| A | B | C | D |
- (i) Give the name of compound **D**. (1)
- (ii) When refluxed with an alcoholic solution of KOH, compound **B** undergoes an elimination reaction. Two structurally isomeric products are formed. Draw the graphical formulae of these two structural isomers. (2)
- (iii) Draw the graphical formula for the product of the reaction between compound **C** and $\text{NaOH}(\text{aq})$. Give an outline of the mechanism involved in this reaction. (4)
- (Total 7 marks)**
26. In aqueous ethanolic alkali, 2-bromo-2-methylbutane undergoes either substitution or elimination reactions to produce an alcohol or a mixture of two alkenes, respectively. Give the structures and names of these three compounds. Account for the formation of the various products by reference to the mechanisms of the reactions involved. (14)
- (Total 14 marks)**

Marking Scheme

- 1.**
- (i) $\text{Cl}_2 \longrightarrow 2\text{Cl}\bullet$ / $\frac{1}{2} \text{Cl}_2 \longrightarrow \text{Cl}\bullet$ **(1)** 1
- (ii) $\text{Cl}\bullet + \text{C}_6\text{H}_5\text{CH}_3 \longrightarrow \text{C}_6\text{H}_5\text{CH}_2\bullet + \text{HCl}$ **(1)**
 $\text{C}_6\text{H}_5\text{CH}_2\bullet + \text{Cl}_2 \longrightarrow \text{Cl}\bullet + \text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ **(1)** 2
- (iii) $\text{C}_6\text{H}_5\text{CHCl}_2$ / $\text{C}_6\text{H}_5\text{CCl}_3$ / $\text{C}_6\text{H}_5\text{CHCHC}_6\text{H}_5$ /
 other correct possible answer **(1)** 1
- [4]**
- 2.**
- (a) (i) (free-)radical substitution 1
(both words required for the mark)
- (ii) uv light OR sunlight OR high temperature OR 150°C to 500°C 1
- (iii) Propagation 1
(ignore "chain", "first", "second" in front of the word propagation)
- (iv) Termination 1
 $\bullet\text{CH}_2\text{CH}_3 + \text{Br}\bullet \longrightarrow \text{CH}_3\text{CH}_2\text{Br}$
 OR $2\bullet\text{CH}_2\text{CH}_3 \longrightarrow \text{C}_4\text{H}_{10}$ 1
(penalise if radical dot is obviously on CH₃, but not otherwise)
(penalise C₂H₅·)
(credit $2\text{Br}\bullet \longrightarrow \text{Br}_2$)
(ignore "chain" in front of the word termination)
- (b) (i) Fractional distillation OR fractionation 1
(credit gas-liquid chromatography, GLC)
- (ii) $\text{CH}_3\text{CH}_3 + 6\text{Br}_2 \longrightarrow \text{C}_2\text{Br}_6 + 6\text{HBr}$ 1
(credit C₂H₆ for ethane)
- [7]**
- 3.**
- (a) (i) (free-) radical substitution 1
(both words required for the mark)
 initiation $\text{Cl}_2 \rightarrow 2\text{Cl}\bullet$ 1
(credit correct half arrows, but penalise double headed arrows)
 first propagation $\text{CH}_3\text{Cl} + \text{Cl}\bullet \rightarrow \bullet\text{CH}_2\text{Cl} + \text{HCl}$ 1
 second propagation $\bullet\text{CH}_2\text{Cl} + \text{Cl}_2 \rightarrow \text{CH}_2\text{Cl}_2 + \text{Cl}\bullet$ 1
(penalise the absence of dots on radicals once only)
(penalise radical dot on Cl of CH₂Cl once only)
- (ii) $\text{CH}_3\text{Cl} + \text{Cl}_2 \rightarrow \text{CH}_2\text{Cl}_2 + \text{HCl}$ 1
(penalise if any radicals appear in this equation)
- (b) M1: mol C = $10.1/12.0$ and mol Cl = $89.9/35.5$ 1
 M2: Ratio $0.842 : 2.53$ OR $1 : 3$ OR CCl_3 1
 M3: $237.0/\text{Mr of CCl}_3 = 237.0/118.5 = 2$ Therefore C_2Cl_6 1
(correct answer gains full credit)
 OR
 M1: $237.0 \times 10.1/100$ and $237 \times 89.9/100$ 1
 M2: Ratio $23.9/12.0 : 213/35.5$ OR $2 : 6$ 1
 M3: C_2Cl_6 1
(correct answer gains full credit)
- (c) any two from CHBr_3 or CBr_4 or $\text{C}_2\text{H}_2\text{Br}_4$ (or $\text{CHBr}_2\text{CHBr}_2$) or C_2Br_6
 (or CBr_3CBr_3) 2
(ignore HBr or H₂)
(ignore equations and ignore names when given in addition to formulae)
(penalise names alone)

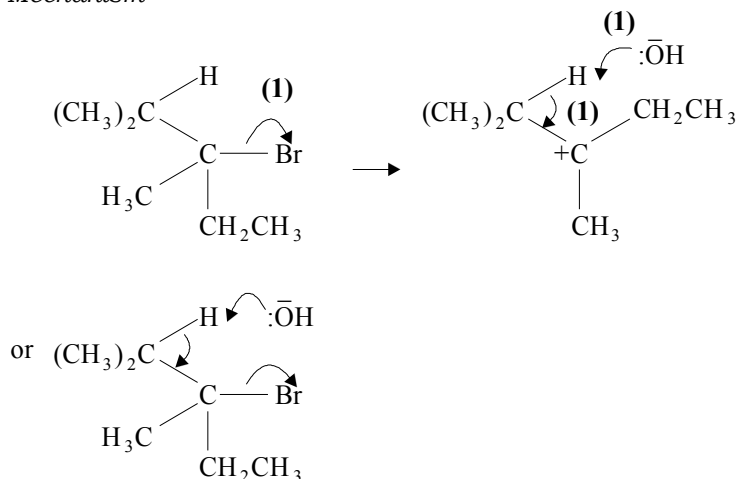
[10]

4. (a) (i) substitution or hydrolysis (1)
nucleophile (1)



5

- (b) (i) *Type of reaction* elimination (1)
Role of reagent base or proton acceptor (1)
(ii) *Structure of the alkene* $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_3$ (1)
Mechanism



- (iii) *Structure of second alkene* $(\text{CH}_3)_2\text{CHC}(\text{CH}_3)=\text{CHCH}_3$ (1)
Name 3,4-dimethylpent-2-ene (1)
Structure of third alkene $(\text{CH}_3)_2\text{CHC}(\text{C}_2\text{H}_5)=\text{CH}_2$ (1)
Name 2-ethyl-3-methylbut-1-ene (1)

10

[15]

5. (i) M1 curly arrow from lone pair of electrons on oxygen of hydroxide ion 1
(insist on a lone pair of electrons on the oxygen atom and a negative charge, but only credit this mark if the attack is to a correct H atom)

M2 curly arrow from the middle of the C-H bond to the middle of the C-C bond.

1

(only credit this mark if the arrow originates from the correct C-H bond and if an attempt has been made at M1)

M3 curly arrow from the middle of the C-Br bond towards/alongside the Br atom.

1

(credit M3 independently unless the bond breaking is contradicted by an additional arrow)

(penalise M3 curly arrow if the C-Br has a formal positive charge)

(ignore partial charges on the C-Br bond, but penalise if incorrect)

(credit full marks for an E1 mechanism, with M2 awarded for a correct curly arrow on the correct carbocation)

(award a maximum of two marks for an incorrect haloalkane)

(ignore products)

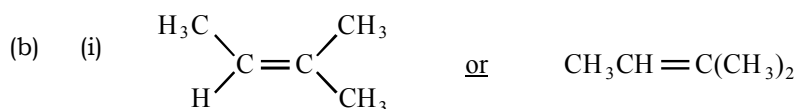
- (ii) Haloalkane/C₂H₅Br is made from ethane
 OR haloalkane is not (readily) available
 OR haloalkane is expensive
 OR it is (too) expensive/costly
 OR (reaction) yield is too low/poor
 OR it is too slow
 OR a valid reference to nucleophilic substitution/alcohol formation occurring as an alternative reaction. 1
(ignore references to temperature or to energy consumption)
(do not credit statements which refer to the idea that this route is not chosen, because industry chooses another route e.g. cracking)

[4]

6. (i) M1: potassium cyanide OR KCN OR sodium Cyanide OR NaCN; 1
(ignore conditions - dissolved in (aq) or (alc) or KOH(aq) all work) (penalise HCN)
 M2: propanenitrile; 1
(credit propan-1-nitrile OR propan nitrile, but not propanitrile)
- (ii) M1: ammonia OR NH₃; 1
(If formula is written, insist that it is correct)
(ignore conditions, but penalise acidic)
 M2: ethylamine; 1
(credit aminoethane)
- (iii) M1: curly arrow from lone pair on nitrogen of (correct formula for) ammonia towards/alongside C atom of C-Br; 1
(penalise M1 if formula of ammonia is wrong or has a negative charge or has no lone pair or arrow is from negative charge)
 M2: curly arrow from C-Br bond towards/alongside side Br atom; 1
(credit M2 independently)
(penalise M2 if formal positive charge on C atom of C-Br)
 M3: correct structure of the ethylammonium ion; 1
(credit the structure drawn out with all four bonds around the nitrogen atom OR written as C₂H₅NH₃⁺ OR CH₃CH₂NH₃⁺)
 M4: curly arrow from the middle of one of the H-N bonds towards the positive N atom; 1
(possible to credit M4 on an incorrect ethylammonium ion with no positive charge)
(ignore use of ammonia or bromide ion etc. to remove proton from ethylammonium ion)
(If the wrong haloalkane is used, award MAX. 3 marks for the mechanism) (If S_N 1 mechanism is used, give full credit in which M1 is for a curly arrow from the lone pair of the N atom of (correct formula for) ammonia towards/alongside the positive carbon atom of CH₃CH₂⁺)

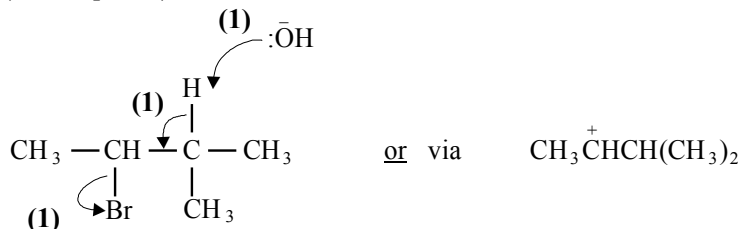
[8]

7. (a) $\text{CH}_3 - \underset{\text{Br}}{\text{CH}} - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_3$ or $\text{CH}_3\text{CHBrCH}(\text{CH}_3)_2$ 1
 (1)



(1)

(ii) (nucleophilic) elimination



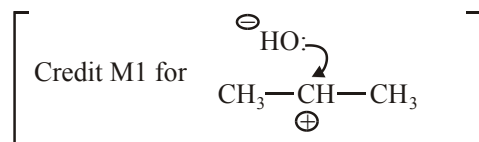
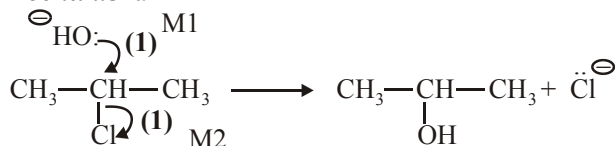
(c) 3-methylbut-1-ene (1)

5
1

[7]

8.

(i) Mechanism:



M1 and M2 independent

Curly arrows must be from a bond or a lone pair

Do not penalise sticks

Penalise M1 if NaOH precedes (penalise this once)Penalise incorrect $\square + \square -$ for M2

Penalise + on C atom for M2

Only allow M1 for incorrect haloalkane

Role of the hydroxide ion: nucleophile (1)

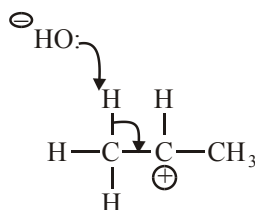
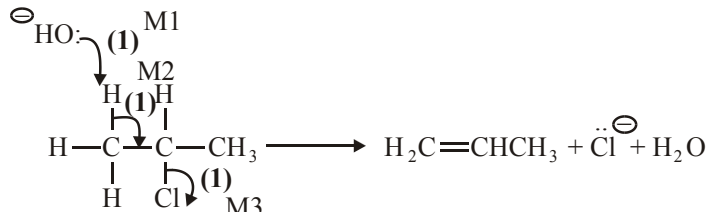
electron pair donor

lone pair donor

NOT nucleophilic substitution

(ii)

Mechanism:



Only allow M1 and M2 for incorrect haloalkane unless RE on

(i)+ charge on H on molecule, penalise M1
 M3 independent
 M2 must be to correct C-C
 M1 must be correct H atom
 Credit M1 and M2 via carbocation mechanism

No marks after any attack of C by OH⁻

Role of the hydroxide ion: base (1)
 proton acceptor
 accepts H⁺

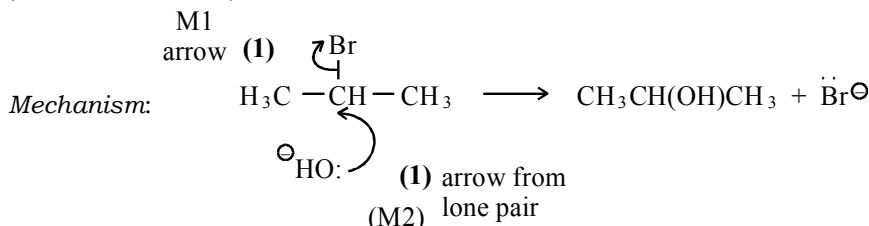
7

[7]

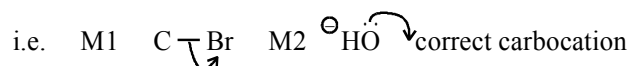
9. (a) (i) Structure: $\begin{array}{c} \text{OH} \\ | \\ \text{H}_3\text{C}-\text{CH}-\text{CH}_3 \end{array}$ (1) [insist on
C-OH bond]
No credit for propan-1-ol even when named correctly
Credit propane-2-ol

Name: propan-2-ol (1)
Not 2-hydroxypropane

- (ii) Name of mechanism: nucleophilic substitution (1) (both words)
(NOT S_N1 or S_N2)



penalise incorrect polarity on C-Br (M1)
Credit the arrows even if incorrect haloalkane
If S_N1, both marks possible



5

- (b) (i) elimination (1)
 (ii) base (1)
OR proton acceptor
NOT nucleophile (base)

2

[7]

10. (a) M1 NaOH OR KOH OR correct name 1
 M2 aqueous or solution in water (ignore heat, reflux etc.) 1
 (Penalise M1 for hydroxide ion alone, but mark on and credit M2)
 (Credit M2 ONLY for H₂O as reagent and heat / warm / T=50 to 100°C)
 (NaOH(aq) scores M1 and M2 provided it is not contradicted)
 (Penalise M2 if NaOH(aq) followed by concentrated or ethanol)
 (Penalise M1 and M2 if followed by acid)
- (b) Ethanolic OR alcoholic OR CH₃CH₂OH / CH₃OH solvent OR 1
 aqueous ethanol/alcohol
 OR higher temperature (must be comparative)
 (Ignore heat or heat under reflux)
 (Credit part (c) independently from part (b))
 (Penalise "ethanoic")

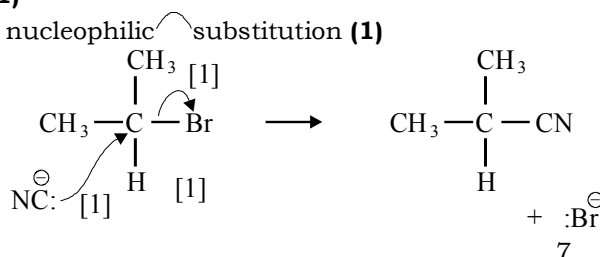
11. (i) 2-methylpropanenitrile (1)
 (ii) Reagent KCN (1)

[3]

Conditions alcoholic/aq (1)

(iii) Name of mechanism nucleophilic substitution (1)

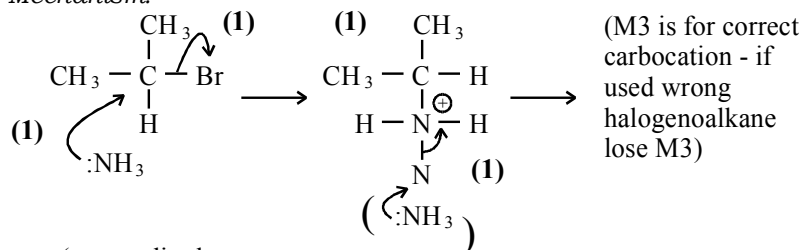
Mechanism



[7]

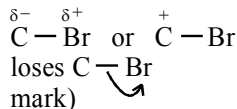
12. (a) Name of mechanism: nucleophilic substitution (1)

Mechanism:

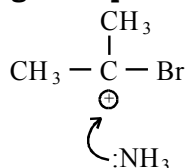


(M3 is for correct carbocation - if used wrong halogenoalkane lose M3)

(wrong dipole

Marks S_N1 using same points

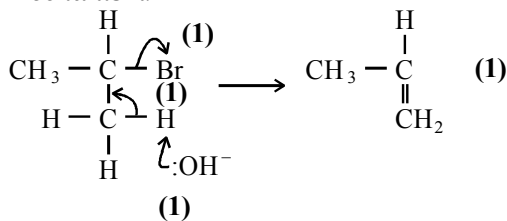
□ M2 requires



5

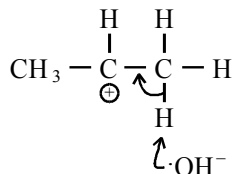
(b) Role of potassium hydroxide: Base (1)

Mechanism:



Mark E1 using same points

∴ M2/M3



5

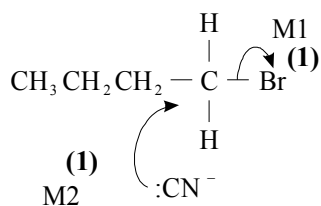
[10]

13. elimination (1)

1

[1]

14. (i) CH₃CH₂CH₂CH₂CN (1) Not C₄H₉CN
Pentanenitrile (1)
allow pentanonitrile, ignore numbers
- (ii) Nucleophilic substitution (1)



M1 formation of carbocation including C⁺ Br⁻

M2 attack by :CN⁻

allow mechanism showing 'Cl'

If '+' on C' lose M1

If K-CN shown lose M2

If δ^- C- δ^+ Br lose M1

- (iii) C-F bond is strong (er) (1)

Must imply C-F is stronger than C-Br bond

ignore references to electronegativity, bond polarity etc.

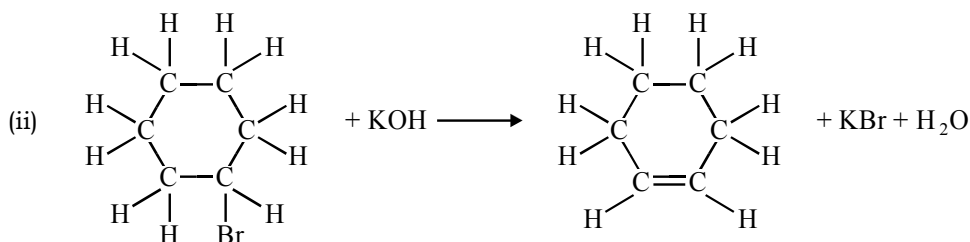
6

[6]

15. (a) (base) elimination 1
- (penalise other words before 'elimination' e.g. nucleophilic)
- M1: curly arrow from lone pair of electrons on oxygen of hydroxide ion 1
- (insist on a lone pair of electrons on the oxygen atom and a negative charge, but only credit this mark if the attack is to a correct H atom)
- M2: curly arrow from the middle of the C-H bond to the middle of the C-C bond 1
- (only credit this mark if the arrow originates from the correct C-H bond and if an attempt has been made at M1)
- M3: curly arrow from the middle of the C-Br bond towards/alongside the Br atom
- (credit M3 independently unless the bond breaking is contradicted by an additional arrow)
- (penalise curly arrow if the C-Br has a formal positive charge)
- (credit full marks for an E1 mechanism, with M2 awarded for a correct curly arrow on the correct carbocation)
- (award a maximum of two marks for either an incorrect haloalkane or an incorrect organic product)
- (maximum 2 marks for use of .sticks. for the haloalkane, unless RE from 2(b), when credit can be given)
- (b) nucleophile or electron pair donor 1
- (penalise 'base')
- (c) CH₃CH₂CH₂CH₂Br + 2NH₃ → CH₃CH₂CH₂CH₂NH₂ + NH₄Br 2
- (M1 correct product)
- (M2 balanced equation using 2NH₃ and leading to NH₄Br)
- (penalise M1 for use of C₄H₉NH₂ or for incorrect haloalkane, but allow consequent correct balancing of equation with 2 moles of ammonia)
- (1-)butylamine 1
- (credit 1-aminobutane and butyl-1-amine)
- (award QoL mark for correct spelling)

[8]

16. (i) potassium or sodium hydroxide (or other specific strong base) (1)
- allow formula; **not** just 'hydroxide'
- in ethanolic /alcoholic solution (independent of first mark) (1) 2
- allow 'in ethanol', **not** just 'ethanol'/ 'with ethanol'
- ignore references to heat / pressure**



structure of cyclohexene showing double bond **(1)**

minimum structure needed is a ring with CH₂ units implied and double bond:

do not allow KOH over arrow

equation correct overall ie balanced

allow ionic equation **(1)**

(iii) elimination / dehydrohalogenation

ignore nucleophilic / electrophilic etc.

2

1

17. (a) (i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br} + \text{KCN} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + \text{KBr}$ **(1)** 1 **[5]**

allow C₃H₇Br

allow C₄H₇N

(ii) nucleophilic substitution / SN2 **(1)** 1

(b) CN^- or NC^- **(1)**

lone pair of electrons on C atom **(1)** 2

18. (a) 2-bromobutane; 1

(b) Elimination; 1

(penalise "nucleophilic" OR "electrophilic" before the word "elimination")

M1: curly arrow from lone pair on oxygen of hydroxide ion to H atom on correct C-H adjacent to C-Br; 1

(penalise M1 if KOH shown as covalent with an arrow breaking the bond)

M2: curly arrow from single bond of adjacent C-H to adjacent single bond C-C; 1
(only credit M2 if M1 is being attempted to correct H atom)

M3: curly arrow from C-Br bond to side of Br atom; 1
(credit M3 independently unless arrows contradict)
(Credit possible repeat error from 2(c)(iii) for M3)
(If the wrong haloalkane is used OR but-1-ene is produced, award MAX. 2 marks for the mechanism)
(If E1 mechanism is used, give full credit in which M1 and M2 are for correct curly arrows on the correct carbocation)

19. (a) (i) $\text{CH}_3\text{Cl} + 2\text{Cl}_2 \rightarrow \text{CHCl}_3 + 2\text{HCl}$ **(1)**

(ii) step in which radicals are used and formed **(1)**

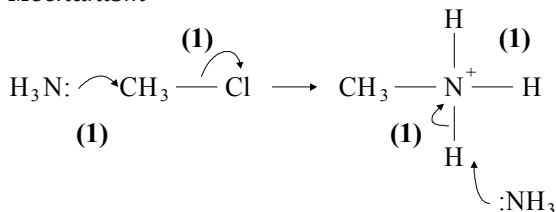
(iii) $\text{CH}_3\text{Cl} + \text{Cl}\bullet \rightarrow \text{CH}_2\text{Cl}\bullet + \text{HCl}$ **(1)**

(iv) $\bullet\text{CHCl}_2 + \text{Cl}_2 \rightarrow \text{CHCl}_3 + \text{Cl}\bullet$ 4

(b) Equation

$\text{CH}_3\text{Cl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{NH}_2 + \text{NH}_4\text{Cl}$ **(1)**

Mechanism



5

[9]

20. $\text{CH}_4 + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{HCl}$ (1)
 Initiation: $\text{Cl}_2 \longrightarrow 2\text{Cl}^\bullet$ (1)
 Propagation: $\text{CH}_4 + \text{Cl}^\bullet \longrightarrow \text{CH}_3^\bullet + \text{HCl}$ (1)
 $\text{CH}_3^\bullet + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{Cl}^\bullet$ (1)
 Termination: $\text{Cl}^\bullet + \text{Cl}^\bullet \longrightarrow \text{Cl}_2$ (1)
 or $\text{CH}_3^\bullet + \text{CH}_3^\bullet \longrightarrow \text{C}_2\text{H}_6$ 5
21. Reaction 5 NH_3 (1) 1 [5]
 For Reaction 4; credit $\text{dil H}_2\text{SO}_4$ OR $\text{H}_2\text{SO}_4(\text{aq})$ OR $\text{HCl}(\text{aq})$
 but NOT steam and NOT $\text{NaOH}(\text{aq})$
22. nucleophilic substitution 1 [1]
 (both words needed)
 Mechanism M1 curly arrow from lone pair on oxygen of hydroxide ion to C atom of C-Br 1
 Mechanism M2 curly arrow from C-Br bond to side of Br atom 1
 (a possible repeat error here from Question 4a)
 (award a maximum of one mark for the wrong haloalkane)
 (credit an $\text{S}_{\text{N}}1$ mechanism in which M1 will be a curly arrow from the lone pair on oxygen of the hydroxide ion to the correct positive carbon atom)
 Y is susceptible to attack by hydroxide ions for one of the following reasons 1
 o the C-Br bond is polar
 o the carbon atom is partially positive (or shown as such)
 o the carbon atom is electron deficient
23. (a) Reaction 2: NaOH OR KOH (1) M1 alcohol (ic) OR ethanol (ic)(1) M2 [4]
 ignore heat
 Condition mark linked to correct reagent but award M2 if OH^- or base or alkali mentioned
- (b) Mechanism: 2
- $\text{HO}^\ominus$ (1) M1

 $\text{H}-\text{C}-\text{C}-\text{H} \longrightarrow \text{H}_2\text{C}=\text{CH}_2 + \text{H}_2\text{O} + \text{Cl}^\ominus$
 H Cl
 Award M3 (C-Cl) independently
 M1 and M2 must be to / from correct places
 E1 mechanism possible in which M2
- Name: of mechanism = elimination (1)
 NOT dehydrohalogenation
 Ignore "base" OR "nucleophilic" before elimination
- Reason: Reaction 2 has (very) low yield (1) 5
 QoL OR chloroethane has to be made (from ethane)
 OR chloroethane is expensive
 OR chloroethane is not readily available
24. (a) Alcohol: Reaction = Substitution (/ hydrolysis) (1) [7]

Ignore reference to nucleophilic, but electrophilic give zero

Alcohol: Role = nucleophile (/ lone pair donor) **(1)**

Alkene: reaction = elimination **(1)**

Ignore ref to nucleophilic or electrophilic

Alkene: base (/ proton acceptor) **(1)**

4

(b) Alcohol = butan-2-ol **(1)**

Not 2-hydroxybutane or but-2-ol

Appropriate structure for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ **(1)**

Brackets not essential

$\text{S}_{\text{N}}2$ version

$\text{S}_{\text{N}}1$ version

$\delta^+ \quad \delta^-$
C – Br bond is polar

C–Br bond is polar **(1)**

Lone pair of OH^-

C–Br bond breaks **(1)**

Attacks the $\text{C}^{\delta+}$

forming carbocation / carbonium ion **(1)**

M1 can be scored from a diagram, M2 and M3 from written explanation only

5

25.

(i) 2-chloro(2-)-methylpropane / (2)methyl 2 chloropropane **(1)**

1

(ii) appropriate unambiguous formula for **either** but-1-ene **or** but-2-ene **(1)**

appropriate unambiguous formula for the remaining structural isomer
allow 1 mark if candidate draws cis and trans but-2-ene **(1)**

2

(iii) unambiguous structure for 2-methylpropan-1-ol – may be from mechanism **(1)**

curly arrow / attack by OH^- curly arrow from lone pair or charge only **(1)**

do **not** allow if Na –OH

curly arrow from bond to Cl / dipole shown on

C–Cl bond / intermediate showing 3 full and 2 partial bonds to C **(1)**

loss of Cl^-

NaCl **or** $\text{Na}^+:\text{Cl}^-$ **(1)**

$^-$ **not** allowed

4

if $\text{S}_{\text{N}}1$ mechanism given:

first mark as above - independent

second mark for correct carbocation formed including curly

arrow from C to Cl or $\text{C}^{\text{S}+} - \text{Cl}^{\text{S}-}$

third mark for hydroxide attack as above

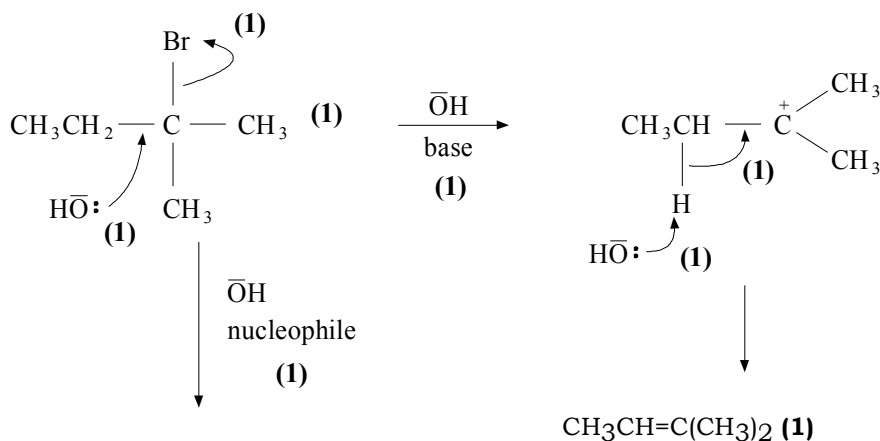
final mark not available (wrong mechanism)

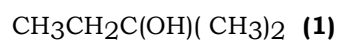
penalise missing proton once only

[9]

[7]

26.



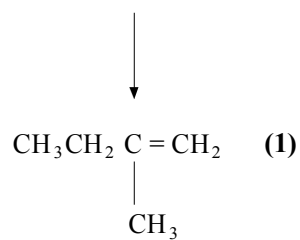
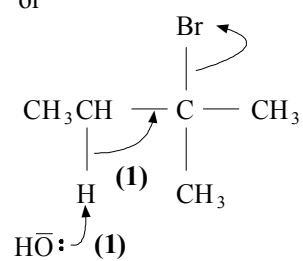


2-methylbutan-2-ol (1)

2-methylbut-2-ene (1)

also loss of H^+ from CH_3 (1)

or



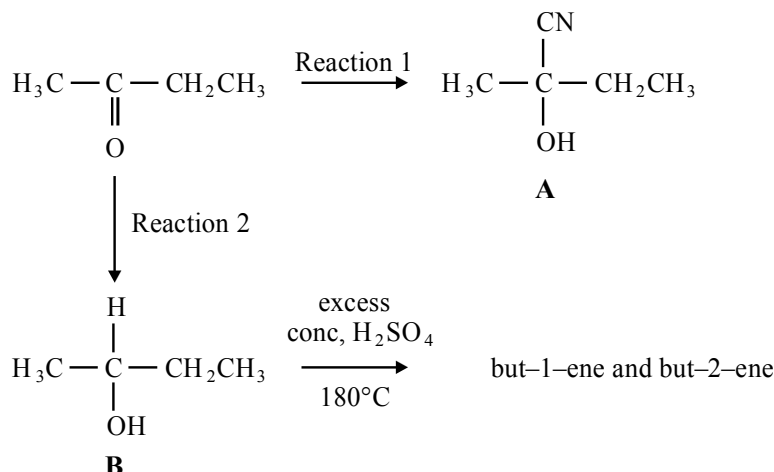
2-methylbut-1-ene (1)

14

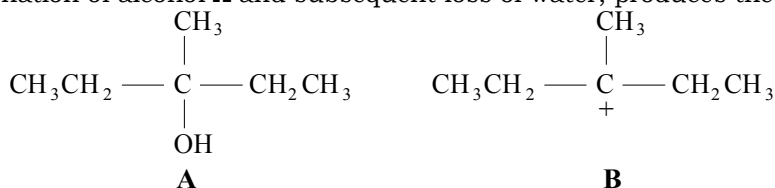
[14]

10.4 HYDROXY COMPOUNDS

1. Consider the reaction scheme below which starts from butanone.



- (a) When 5.0 g of butanone were used to carry out Reaction 1, the yield was 64% of the theoretical maximum. Calculate the mass of compound **A** formed. (4)
- (b) Draw the structure of the repeating unit in the polymer which can be formed from but-1-ene. (1)
- (Total 5 marks)
2. Ethanol is produced industrially from ethene. Name the catalyst and state the conditions used in the industrial process and write an equation for the reaction. (4)
- (Total 4 marks)
3. (a) Write an equation for the complete combustion of butan-2-ol in an excess of oxygen. (1)
- (b) But-1-ene and other products can be made by the dehydration of butan-2-ol.
- (ii) Explain why but-1-ene does not show geometrical isomerism.
- (iii) An alternative dehydration of butan-2-ol produces geometrical isomers. Draw the structure of one of these geometrical isomers and give its full name. (3)
- (Total 4 marks)
4. Protonation of alcohol **A** and subsequent loss of water, produces the intermediate **B**.



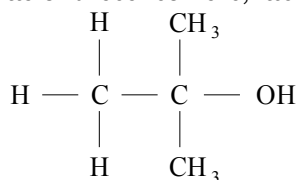
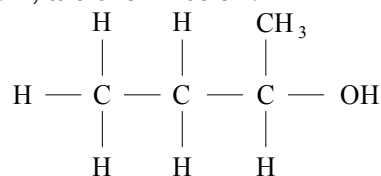
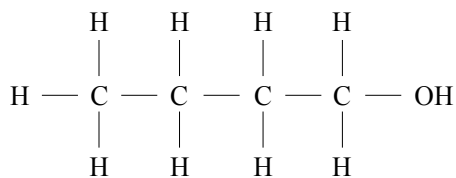
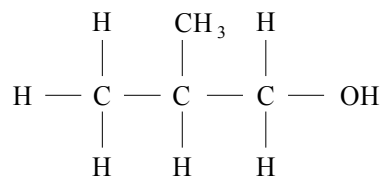
- (a) (i) Name alcohol **A**.
- (ii) What type of species is intermediate **B**?
- (iii) Draw the structures of the two alkenes which can be formed from species **B** by removal of a proton. Label as **C** the alkene which shows geometrical isomerism. (5)
- (b) The intermediate **B** is readily attacked by nucleophiles such as water. What is the essential feature of a nucleophile? (1)
- (c) (i) State what final colour you would see if alcohol **A** were warmed with acidified potassium dichromate(VI). Explain your answer.
- (ii) Draw two structural isomers of alcohol **A** which form branched chain ketones

when heated with acidified potassium dichromate(VI), but which could not form alkene **C** on dehydration.

(5)

(Total 11 marks)

5. There are four structurally isomeric alcohols of molecular formula $C_4H_{10}O$. Graphical formulae of these isomers, labelled **A**, **B**, **C** and **D**, are shown below.

**A****B****C****D**

- Identify the type of alcohol represented by **A** and by **B**. (2)
- Give the name of alcohol **A**. (1)
- Select one of the alcohols **A**, **B**, **C** or **D** which will, on oxidation, produce an aldehyde.
 - Give the structural formula of the aldehyde produced by this reaction. (1)
 - State the reagents and conditions required for the aldehyde to be the main product of the oxidation reaction. (3)
- All the alcohols **A**, **B**, **C** and **D** may be readily dehydrated.
 - Explain what is meant by the term *dehydration*. (1)
 - State the type of compound formed by dehydration of alcohols. (1)
 - Suggest suitable reagent(s) and condition(s) for the dehydration of alcohols. (2)
 - Select one of the alcohols **A**, **B**, **C** or **D** which, on dehydration, would give a single product. Draw the structural formula of this product. (1)
 - Select one of the alcohols **A**, **B**, **C** or **D** which, on dehydration, would give two products which are **structurally** isomeric. Draw structural formulae for these two structural isomers and explain why the formation of two structural isomers is possible in this case. (4)

(Total 16 marks)

6. Butan-1-ol can be oxidised by acidified potassium dichromate(VI) using two different methods.
- In the first method, butan-1-ol is added dropwise to acidified potassium dichromate(VI) and the product is distilled off immediately.
 - Using the symbol [O] for the oxidising agent, write an equation for this oxidation of butan-1-ol, showing clearly the structure of the product. State what colour change you would observe.
 - Butan-1-ol and butan-2-ol give different products on oxidation by this first method. By stating a reagent and the observation with each compound, give a simple test to distinguish between these two oxidation products.

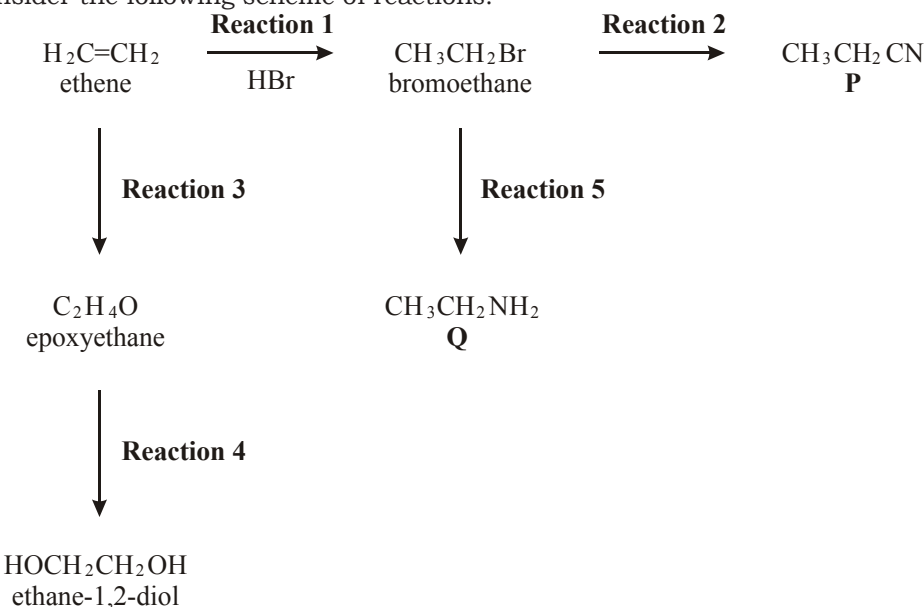
- (b) In a second method, the mixture of butan-1-ol and acidified potassium dichromate(VI) is heated under reflux. Identify the product which is obtained by this reaction. (6)

- (c) Give the structures and names of two branched chain alcohols which are both isomers of butan-1-ol. Only isomer 1 is oxidised when warmed with acidified potassium dichromate(VI). (1)

(4)

(Total 11 marks)

7. Consider the following scheme of reactions.



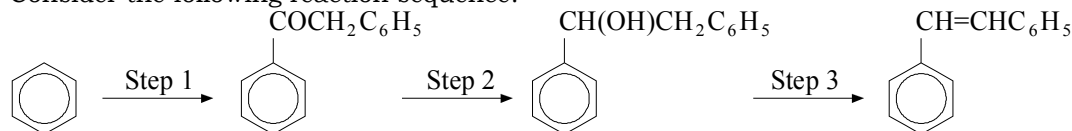
- (a) In **Reaction 1**, ethene undergoes electrophilic addition with hydrogen bromide.
 (i) State what is meant by the term *electrophile*.
 (ii) Outline a mechanism for this reaction. (5)

(8)

- (b) In **Reactions 2** and **5**, bromoethane undergoes nucleophilic substitution.
 (i) Identify a reagent for **Reaction 2**. Name the organic product, **P**.
 (ii) Identify a reagent for **Reaction 5**. Name the organic product, **Q**.
 (ii) Outline a mechanism for **Reaction 5**. (8)

(Total 13 marks)

8. Consider the following reaction sequence:

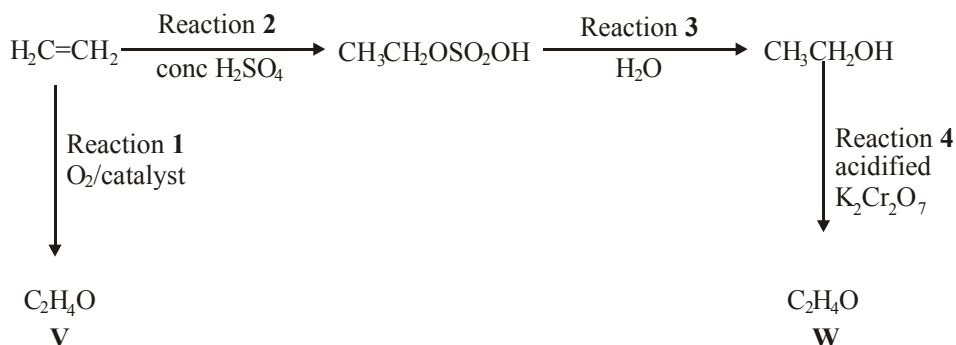


- (a) Suggest a suitable reagent for Step 3. Name the mechanism for this reaction. (2)
 (b) Name the type of isomerism shown by the final product, 1,2-diphenylethene, and explain why this compound can exist in two stereoisomeric forms. (2)

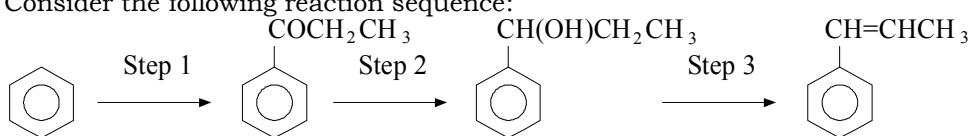
(2)

(Total 4 marks)

9. Consider the following reaction scheme, which leads to the formation of two compounds **V** and **W**.



- (a) In Reaction 4, compound **W** is distilled from the reaction mixture.
- Name compound **W** and draw its structure.
 - Name the type of reaction shown by Reaction 4.
- (3)
- (Total 3 marks)**
10. (a) Ethanol, $\text{C}_2\text{H}_5\text{OH}$, can be made from glucose, $\text{C}_6\text{H}_{12}\text{O}_6$
- Write an equation to represent this reaction.
 - Give the name of this process for making ethanol.
- (2)
- (b) Ethanol can be used as a fuel in the internal combustion engine of a motor car.
- Write an equation for the complete combustion of ethanol.
 - Identify a pollutant produced when ethanol is burned in a limited supply of air.
 - Nitrogen monoxide, NO , is a pollutant gas produced by motor cars. Write an equation to represent a reaction occurring in the catalytic converter which decreases the amount of this pollutant.
- (3)
- (c) Ethene can be formed by the dehydration of ethanol using concentrated sulphuric acid.
- Name a mechanism for this reaction.
- (1)
- (Total 6 marks)**
11. Consider the following reaction sequence:



- (a) For step 3, name the type of reaction taking place and suggest a suitable reagent or combination of reagents.
- (2)
- (Total 2 marks)**
12. (a) An alcohol containing carbon, hydrogen and oxygen only has 64.9% carbon and 13.5% hydrogen by mass. Using these data, show that the empirical formula of the alcohol is $\text{C}_4\text{H}_{10}\text{O}$.
- (3)
- (b) The structural formulae of two of the four possible alcohols of molecular formula $\text{C}_4\text{H}_{10}\text{O}$ are shown below.



Isomer 1

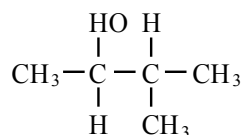
Isomer 2

- What type of alcohol is Isomer 1? Suggest a reason why this type of alcohol is not easily oxidised.
- Draw the structural formulae of the two remaining alcohols of molecular formula $\text{C}_4\text{H}_{10}\text{O}$

- (c) Isomer 2 was oxidised by adding it dropwise to acidified potassium dichromate(VI) solution and immediately distilling off the product. When this product was treated with Fehling's solution, a red precipitate was formed. (4)
- State the type of product distilled off during the oxidation by acidified potassium dichromate(VI) solution.
 - Write an equation for the oxidation by potassium dichromate(VI), showing clearly the structure of the organic product. Use [O] to represent the oxidising agent.
 - Name and draw a structure for the organic product formed by the reaction with Fehling's solution.
- (d) State **one** advantage and **one** disadvantage of the production of ethanol by the hydration of ethene compared to the fermentation of glucose. (5)

(2)
(Total 14 marks)

13. In the presence of reagent **X**, the alcohol shown below undergoes a reaction to form two isomeric alkenes.



- Name this alcohol.
- Give the name of the type of reaction involved in the formation of the two alkenes.
- Suggest the identity of reagent **X**.
- Give the structural formulae of the two isomeric alkenes.

(5)
(Total 5 marks)

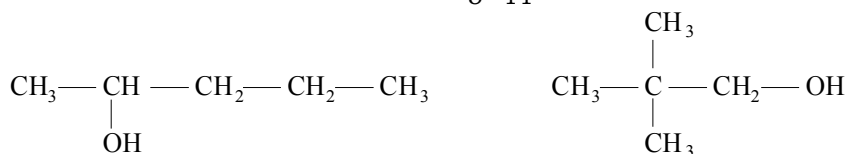
14. Many naturally-occurring organic compounds can be converted into other useful products.

Glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, can be fermented to make ethanol, which can then be dehydrated to make the unsaturated compound, ethene.

- Write an equation for the fermentation of glucose to form ethanol.
- Identify a catalyst for the dehydration of ethanol to form ethene. Write an equation for this reaction.

(3)
(Total 3 marks)

15. The structures of two alcohols of formula $\text{C}_5\text{H}_{11}\text{OH}$ are shown below.



- A third alcohol of formula $\text{C}_5\text{H}_{11}\text{OH}$ is tertiary. Draw the displayed formula and give the name of this alcohol. (2)
- Draw the displayed formulae of **two** alkenes formed when pentan-2-ol is heated with concentrated sulphuric acid. (2)
- State the type of reaction taking place in (a)(ii). (1)
- Explain why the type of reaction taking place in (a)(ii) does not occur when 2,2-dimethylpropan-1-ol is heated with concentrated sulphuric acid. (1)

(1)
(Total 6 marks)

16. When 3-bromo-3-methylpentane, $\text{CH}_3\text{CH}_2\text{CBr}(\text{CH}_3)\text{CH}_2\text{CH}_3$, is treated with aqueous ethanolic potassium hydroxide, a mixture of 3-methylpentan-3-ol, 3-methylpent-2-ene and an isomeric alkene is formed.
- Name the type of mechanism occurring in the formation of 3-methylpentan-3-ol from 3-bromo-3-methylpentane.
 - Why is it difficult to oxidise 3-methylpentan-3-ol? (2)
 - Name and outline a mechanism for the formation of 3-methylpent-2-ene in the above reaction. (4)
 - Give the structure and name of the isomeric alkene formed together with 3-methylpent-2-ene in the above reaction. (2)
- (Total 8 marks)**
17. When 3-bromo-2,3-dimethylpentane, $(\text{CH}_3)_2\text{CHCBr}(\text{CH}_3)\text{CH}_2\text{CH}_3$, reacts with aqueous potassium hydroxide, an alcohol is formed.
- Name the type of reaction taking place and give the role of the reagent.
 - Outline a mechanism for the reaction, showing clearly the structure of the alcohol formed.
 - Why is it difficult to oxidise this alcohol? (6)
- (Total 6 marks)**
18. Glucose can be used as a source of ethanol. Ethanol can be burned as a fuel or can be converted into ethene.
- $$\begin{array}{ccccc} \text{C}_6\text{H}_{12}\text{O}_6 & + & \text{CH}_3\text{CH}_2\text{OH} & \longrightarrow & \text{H}_2\text{C}=\text{CH}_2 \\ \text{glucose} & & \text{ethanol} & & \text{ethene} \end{array}$$
- Name the types of reaction illustrated by the two reactions above. (2)
 - State what must be added to an aqueous solution of glucose so that ethanol is formed.
 - Identify a suitable catalyst for the conversion of ethanol into ethene. (2)
 - State the class of alcohols to which ethanol belongs.
 - Give **one** advantage of using ethanol as a fuel compared with using a petroleum fraction. (2)
 - Most of the ethene used by industry is produced when ethane is heated to 900°C in the absence of air. Write an equation for this reaction. (1)
 - Name the type of polymerisation which occurs when ethene is converted into poly(ethene). (1)
- (Total 8 marks)**
19. (a) (i) Pentan-1-ol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$, was oxidised by adding it dropwise to acidified potassium dichromate(VI) and distilling off the organic product immediately. Write an equation for this reaction showing clearly the structural formula of the organic product. You may use [O] to represent the oxidising agent.
- (ii) Pentan-2-ol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$, was oxidised by heating it under reflux with acidified potassium dichromate(VI). Write an equation for this reaction showing clearly the structural formula of the organic product. You may use [O] to represent the oxidising agent. (4)
- (b) By stating the reagents, conditions and observations, show how you would distinguish between the two oxidation products formed in parts (a)(i) and (a)(ii). (4)
- (Total 8 marks)**
20. Some alcohols can be oxidised to form aldehydes, which can then be oxidised further to form carboxylic acids.
- Some alcohols can be oxidised to form ketones, which resist further oxidation.
- Other alcohols are resistant to oxidation.

- (a) Draw the structures of the **two** straight-chain isomeric alcohols with molecular formula, $C_4H_{10}O$ (2)
- (b) Draw the structures of the oxidation products obtained when the two alcohols from part (a) are oxidised separately by acidified potassium dichromate(VI). Write equations for any reactions which occur, using [O] to represent the oxidising agent. (6)
- (c) Draw the structure and give the name of the alcohol with molecular formula $C_4H_{10}O$ which is resistant to oxidation by acidified potassium dichromate(VI). (2)
- (Total 10 marks)**
- 21.** The three compounds $CH_3CH_2CH_2CH_2OH$, $(CH_3)_3COH$ and $CH_3CH_2CH_2CHO$ can be distinguished by use of the following reagents:
- potassium dichromate(VI) acidified with dilute sulphuric acid
 - Tollens' reagent
- (a) Identify which of these three organic compounds would reduce acidified potassium dichromate(VI). Give the structures of the organic products formed. Write a half-equation for the reduction of dichromate(VI) ions in acidic solution. (6)
- (b) Identify which one of these three organic compounds would reduce Tollens' reagent. Give the structure of the organic product formed. Write a half-equation for the reduction of Tollens' reagent. (3)
- (Total 9 marks)**
- 22.** Ethene can be converted into a variety of useful products as illustrated below.
- ```

graph TD
 Ethene["H2C=CH2
ethene"]
 Ethanol["CH3CH2OH
ethanol"]
 Epoxyethane["H2C-CH2
epoxyethane"]
 Bromoethane["CH3CH2Br
bromoethane"]
 CompoundX["H2C-CH2
OH OH
compound X"]
 Ethylamine["CH3CH2NH2
ethylamine"]

 Ethene -- "Reaction 1
acid catalyst" --> Ethanol
 Ethene -- "Reaction 2
Ag catalyst" --> Epoxyethane
 Ethene -- "Reaction 3
HBr" --> Bromoethane
 Epoxyethane -- "Reaction 4" --> CompoundX
 Bromoethane -- "Reaction 5" --> Ethylamine

```
- (a) Give a reagent for each of **Reactions 1** and **5**. (2)
- (b) Outline a mechanism for **Reaction 3**. (4)
- (c) Ethanol can be manufactured from ethene as shown in **Reaction 1** or by the fermentation of sugars. Outline the essential conditions and give an equation for the fermentation reaction. Compare the relative rates and the purity of the product obtained in each case by these two manufacturing processes. (5)
- (Total 11 marks)**
- 23.** Ethanoic acid can be formed from bromoethane in two steps. Give reagents and

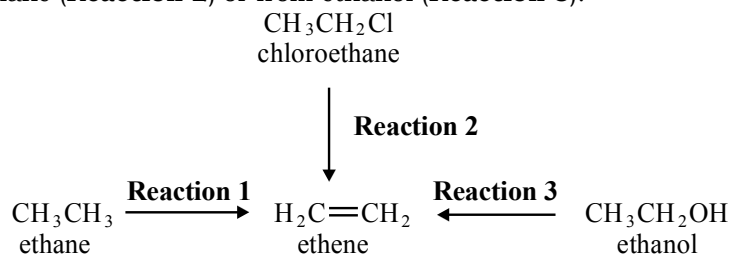
conditions for each step. Include in your answer the name of the mechanism for the first step and an equation for the second step.

(Total 9 marks)

24. (a) When a mixture of chlorine with an excess of methane is irradiated with ultraviolet light, a reaction occurs with chloromethane as the main organic product. Write an equation and a mechanism for the formation of chloromethane. (5)
- (b) Ethanol can be produced industrially either by the direct hydration of ethene or by fermentation. State the conditions for each method and compare the two processes by giving two advantages and one disadvantage of direct hydration. (9)

(Total 14 marks)

25. Ethene is an important starting point for the manufacture of plastics and pharmaceutical chemicals. Most of the ethene used by industry is produced by the thermal cracking of ethane obtained from North Sea gas (**Reaction 1**). It is also possible to make ethene either from chloroethane (**Reaction 2**) or from ethanol (**Reaction 3**).

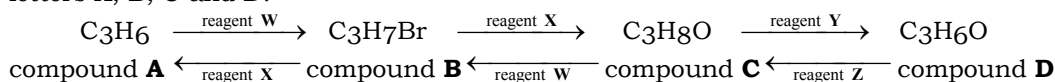


- (a) Give essential conditions and reagents for each of **Reactions 2** and **3**.
- (b) Name and outline a mechanism for **Reaction 2**. Suggest a reason why chloroethane is **not** chosen by industry as a starting material to make ethene commercially.
- (c) Name a mechanism for **Reaction 3**. Suggest why this route to ethene may become used more commonly in the future as supplies of North Sea gas begin to run out.

(4 + 5 + 2)

(Total 11 marks)

26. This question is concerned with the six reactions shown in the following scheme. The four compounds involved are represented by their molecular formulae and labelled using the letters **A**, **B**, **C** and **D**.



- (a) State the names of the three homologous series to which compounds **A**, **B** and **C** belong. (3)
- (b) Give the name of the type of reaction occurring in each of the following conversions: (3)
- compound **A** to compound **B**;
  - compound **B** to compound **C**;
  - compound **C** to compound **D**.
- (c) Give the name **or** formula of reagent **X** and state the different conditions under which it would be used in the conversions of compound **B** to compound **C**, and of compound **B** to compound **A**. (3)
- (d) Identify, by name **or** formula, the substances present in reagent **Y** and state an essential condition under which they would be used in the conversion of compound **C** to compound **D**. (3)
- (f) Write an equation for each of the following reactions: (2)
- the conversion of compound **B** to compound **A**;
  - the conversion of compound **B** to compound **C**.
- (g) Outline a possible mechanism for the conversion of compound **B** to compound **C**. (3)

(Total 17 marks)

## Marking Scheme

- 4



[9]

- 4

- 1

- 3

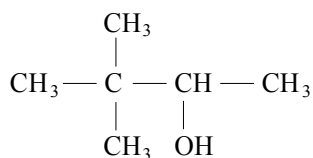
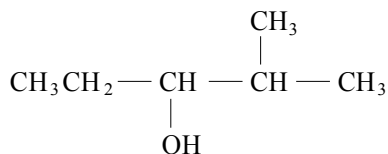
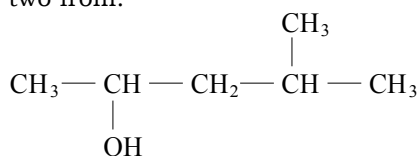
[4]

- C(1)

- 5

- (c) (i) orange **(1)**  
tertiary alcohol **(1)**  
not oxidised **(1)**

(ii) two from:



**(2x1)**

5

**[11]**

5. (a) **A** = tertiary alcohol/ 3° **(1)**  
**B** = secondary alcohol/ 2° **(1)** 2
- (b) (2)-methylpropan-2-ol **(1)** 1
- (c) (i) (selects isomer **C** and) gives suitable structure for butanal **(1)**  
**or**  
(selects isomer **D** and) gives suitable structure for 2-methylpropanal **(1)** 1
- (ii) (sodium / potassium) dichromate / manganate VII **(1)**  
acidic conditions /  $\text{H}^+$  (this mark dependent on first mark) **(1)**  
**not** HCl with  $\text{KMnO}_4$   
distill(ation) **not** reflux **(1)** 3
- (d) (i) removal of water **(1)** 1
- (ii) alkene / ether (**not** a named example) **(1)** 1
- (iii) concentrated sulphuric acid / concentrated or solid phosphoric acid  
(strong) heat / high temperature / 150 – 200 °C / reflux  
**or**  
suitable catalyst eg aluminium oxide / broken porcelain **(1)**  
strong heat / high temperature / 300+ °C **(1)**  
mark for conditions dependent on mark for reagents 2
- (iv) (selects isomer **C** and) gives suitable structure for but-1-ene  
(accept but-2-ene structure if primary to secondary carbocation rearrangement mentioned)  
**or**  
(selects isomer **A** or **D** and) gives suitable structure for 2-methylpropene **(1)**  
also accept structures for ethers 1
- (v) selects isomer **B** **(1)**  
suitable structure for but-1-ene (mark independently of isomer chosen) **(1)**  
suitable structure for but-2-ene (either cis or trans **not** both) **(1)**  
dehydration 'involves the removal of OH and H to make water, and H can be  
either from C<sub>1</sub> or from C<sub>3</sub> so two isomers formed **(1)** 4

**[16]**

6. (a) (i) *Equation*
- $$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + [\text{O}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2-\text{C} \begin{array}{l} \text{O} \\ \text{H} \end{array} + \text{H}_2\text{O} \quad (1)$$
- (ii) *Colour change* orange  $\longrightarrow$  green  
*Reagent* ammoniacal silver nitrate/Tollens (1)  
*Observation with oxidation product of butan-1-ol* silver  
 mirror (1)  
*Observation with oxidation product of butan-2-ol* no  
 reaction (1) 6
- (b) Butanoic acid (1) 1
- (c) *Structure of isomer 1* *Structure of isomer 2*
- $$\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2\text{OH} \\ | \\ \text{H} \end{array}$$

(1)

*Name of isomer 1* 2-methylpropan-1-ol (1)

*Name of isomer 2* 2-methylpropan-2-ol (1)

$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}_3\text{C}-\text{C}-\text{CH}_3 \\ | \\ \text{OH} \end{array}$$

(1)
- 4 [11]
7. (a) (i) Electron pair/ lone pair acceptor OR seeking/bonds with an electron pair 1  
*(insist on reference to a pair of electrons)*
- (ii) M1 curly arrow from middle of C=C bond of the alkene towards/ alongside the H atom of the H-Br; 1  
*(penalise arrows which go towards one of the carbon atoms)*  
*(ignore a partial negative charge on the C=C)*  
 M2 curly arrow from H-Br bond to side of Br atom; 1  
*(penalise M2 if there are formal charges on HBr or if there are partial charges which are the wrong)*  
*(penalise M2 if the single bond has two dots in addition to the line)*  
 M3 correct structure for carbocation; 1  
*(penalise M3 if the positive charge is placed on the end of a bond) (penalise M3 if any alkene other than ethene is used - all other marks can score)*  
 M4 curly arrow from lone pair on bromide ion to the positive carbon 1  
 of carbocation, ensuring that bromide ion has a negative charge;
- (b) (i) M1: potassium cyanide OR KCN OR sodium Cyanide OR NaCN; 1  
*(ignore conditions - dissolved in (aq) or (alc) or KOH(aq) all work) (penalise HCN)*  
 M2: propanenitrile; 1  
*(credit propan-1-nitrile OR propan nitrile, but not propanitrile)*
- (ii) M1: ammonia OR NH<sub>3</sub>; 1  
*(If formula is written, insist that it is correct)*  
*(ignore conditions, but penalise acidic)*  
 M2: ethylamine; 1  
*(credit aminoethane)*
- (iii) M1: curly arrow from lone pair on nitrogen of (correct formula for) ammonia towards/alongside C atom of C-Br; 1  
*(penalise M1 if formula of ammonia is wrong or has a negative charge or has no lone pair or arrow is from negative charge)*  
 M2: curly arrow from C-Br bond towards/alongside side Br atom; 1

- (credit M2 independently)  
(penalise M2 if formal positive charge on C atom of C-Br)
- M3: correct structure of the ethylammonium ion; 1  
(credit the structure drawn out with all four bonds around the nitrogen atom OR written as  $C_2H_5NH_3^+$  OR  $CH_3CH_2NH_3^+$ )
- M4: curly arrow from the middle of one of the H-N bonds towards the positive N atom; 1  
(possible to credit M4 on an incorrect ethylammonium ion with no positive charge)  
(ignore use of ammonia or bromide ion etc. to remove proton from ethylammonium ion)  
(If the wrong haloalkane is used, award MAX. 3 marks for the mechanism) (If SN 1 mechanism is used, give full credit in which M1 is for a curly arrow from the lone pair of the N atom of (correct formula for) ammonia towards/alongside the positive carbon atom of  $CH_3CH_2^+$ )
- 8.** (a) Reagents  $H_2SO_4$  or  $H_3PO_4$  or  $Al_2O_3$  (1) [17]  
Name of mechanism elimination (1)  
Mechanism  
(b) Type of isomerism geometrical or cis-trans (1)  
Explanation restricted rotation or double bond rigid (1) 2
- 9.** (a) (i)ethanal 1  
correct structure for ethanal 1  
(aldehyde functional group must be drawn out)  
(ii) oxidation or redox 1
- 10.** (a) (i)  $C_6H_{12}O_6 \longrightarrow 2C_2H_5OH + 2CO_2$  1  
(Or  $CH_3CH_2OH$ )  
(Ignore state symbols in the equation)  
(ii) Fermentation 1  
(b) (i)  $C_2H_5OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O$  1  
(Or  $C_2H_6O$  or  $CH_3CH_2OH$ )  
(ii) CO or carbon monoxide or C or carbon ONLY 1  
(iii)  $2CO + 2NO \longrightarrow 2CO_2 + N_2$   
OR  $2NO \longrightarrow N_2 + O_2$   
OR  $2NO + C \longrightarrow N_2 + CO_2$   
OR  $C_8H_{18} + 25NO \longrightarrow 8CO_2 + 12\frac{1}{2}N_2 + 9H_2O$  1  
(In equation 2, allow additional  $O_2$  on both sides of the equation)  
(c) Elimination 1  
(Penalise additional words such as "electrophilic")
- 11.** Step 3 [6]  
dehydration or elimination (1)  
 $H_2SO_4$  or  $H_3PO_4$  or  $Al_2O_3$  (1) 2
- 12.** (a) % O = 21.6 % (1) [2]  
If % O not calculated only M2 available

$$\begin{array}{ccc} \text{C} \frac{64.9}{12} & \text{H} \frac{13.5}{1} & \text{O} \frac{21.6}{16} \text{ (1)} \\ = 5.41 & = 13.5 & = 1.35 \end{array}$$

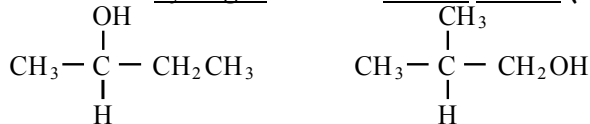
Ratio: 4 : 10 : 1 ( $\longrightarrow$  C<sub>4</sub>H<sub>10</sub>O) (1)

**If arithmetic error in any result lose M3**

**If percentage composition calculation done zero**

3

- (b) (i) *Type of alcohol:* Tertiary (1)  
*Reason:* No hydrogen atom on central carbon (1)



(ii)

(1)

(1)

*Isomer 3**Isomer 4*

**Penalise missing bonds / incorrect bonds once per paper**

4

- (c) (i) Aldehyde (1)  
**Ignore named aldehydes or their structures, penalise wrong named compound**
- (ii) CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH + [O]  $\longrightarrow$  CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CHO + H<sub>2</sub>O (1)  
 Balanced (1)  
**C<sub>4</sub>H<sub>10</sub>O is OK as a reactant**  
**[O] can be over arrow**  
**C<sub>3</sub>H<sub>7</sub>CHO not accepted for product, but C<sub>2</sub>H<sub>5</sub>CH<sub>2</sub>CHO is OK**  
**If use C<sub>3</sub> or C<sub>5</sub> compounds no marks in (ii) C.E of wrong alcohol**

- (iii) *Name* Butanoic acid (1)

*Structure:* CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH (1)

5

- (d) *Advantage:* Fast reaction OR pure product OR continuous process  
 OR cheap OR high yield, 100% alcohol (1)  
*Disadvantage:* High technology OR ethene from non-renewable source  
 OR expensive equipment not just costly (1)

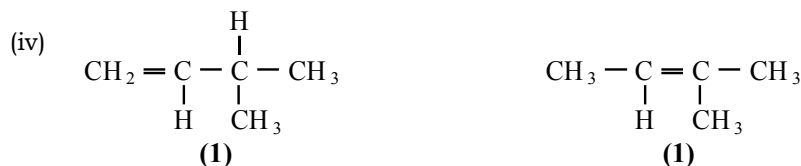
**Not answers based on fermentation**

2

[14]

13.

- (i) 3-methylbutan-2-ol (1)  
**No alternatives**
- (ii) elimination or dehydration (1)
- (iii) (c) H<sub>2</sub>SO<sub>4</sub> or (c) H<sub>3</sub>PO<sub>4</sub> – name or correct formula (1)

*Alkene 1**Alkene 2*

**Double bond must be shown**

**Accept any correct unambiguous structures**

**if but-1-ene and but-2-ene offered, allow M2**

5

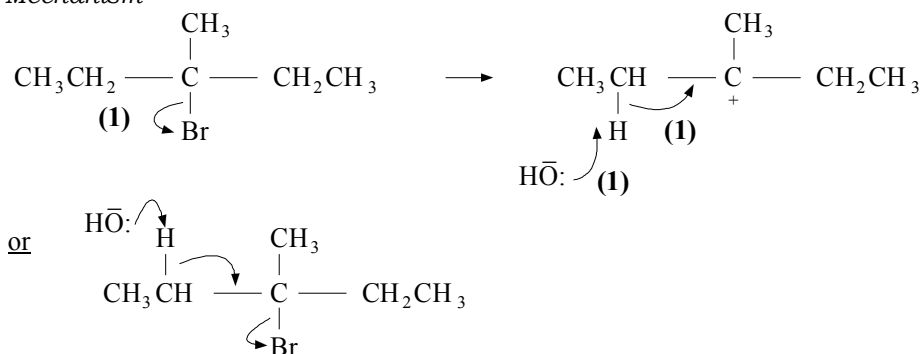
[5]

14. (i)  $C_6H_{12}O_6 \longrightarrow 2C_2H_5OH + 2CO_2$ ; 1  
(penalise  $C_2H_6O$  once only in this question)
- (ii) Concentrated  $H_2SO_4$  OR concentrated  $H_3PO_4$  OR  $Al_2O_3$ ; 1  
(penalise *aqueous* or *dilute* as a contradiction)
- $C_2H_5OH \longrightarrow C_2H_4 + H_2O$  OR  $C_2H_5OH \longrightarrow H_2C = CH_2 + H_2O$ ; 1  
(penalise  $CH_2 \cdot CH_2$  and  $CH_2-CH_2$  and  $CH_2:CH_2$  for ethene)
15. (a) (i) correct graphical formula for tertiary alcohol  
allow  $CH_3$  not  $C_2H_5$  (1)  
2-methylbutan-2-ol / 2-hydroxy-2-methylbutane / 2-methyl-2-hydroxybutane award name mark even if it follows  
incorrect formula (1) 2
- (ii) graphical formula of pent-1-ene (1)  
graphical formula of pent-2-ene (1) 2  
accept geometrical isomers of pent-2-ene if clearly shown to be different
- (iii) dehydration / elimination (1) 1
- (iv) no H atoms on C atom next to C-OH / three methyl groups on C (1) 1

[3]

16. (a) (i) nucleophilic substitution (1)  
(ii) tertiary alcohol or no H atom available (1) 2
- (b) *Name of mechanism* elimination (1)  
*Mechanism*

[4]



4

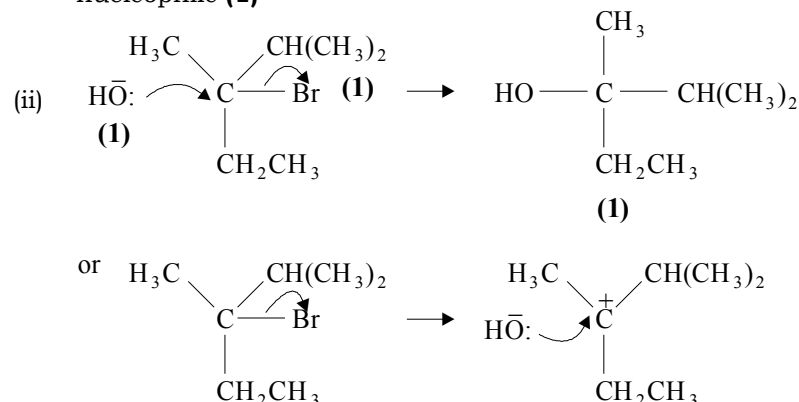
- (c) *Structure*  

$$\begin{array}{c}
 \text{CH}_3\text{CH}_2 - \text{C} = \text{CH}_2 \\
 | \\
 \text{CH}_2\text{CH}_3
 \end{array}$$
 (1)
- Name*  
2-ethylbut-1-ene (1)

2

17. (i) substitution or hydrolysis (1)  
nucleophile (1)

[8]



- (iii) tertiary or no C-H (1)

6



- 18.** (a) M1 fermentation 1 [6]  
 M2 dehydration or elimination 1  
 (b) (i) yeast OR zymase OR an enzyme 1  
 (ii) concentrated sulphuric or phosphoric acid 1  
*(penalise aqueous or dilute as a contradiction)*  
 (c) (i) primary or 1° 1  
 (ii) sugar or glucose or ethanol is renewable 1  
 OR ethanol does not contain sulphur-containing impurities  
 OR ethanol produces less pollution or is less smoky or less CO/C  
*(the objective is a positive statement about ethanol)*  
*(penalise the idea that ethanol is an infinite source or vague statements that ethanol has less impurities)* *(penalise the idea that ethanol produces no pollution)*  
 (d)  $\text{C}_2\text{H}_6 \longrightarrow \text{C}_2\text{H}_4 + \text{H}_2$  1  
 (e) Addition 1  
*(ignore self or chain as a preface to "addition")*  
*(penalise additional)*

[8]

- 19.** (a) (i)  $\text{CH}_3 \text{CH}_2 \text{CH}_2 \text{CH}_2 \text{CH}_2 \text{OH} + [\text{O}] \longrightarrow \text{CH}_3 \text{CH}_2 \text{CH}_2 \text{CH}_2 \text{CHO} + \text{H}_2\text{O}$  (1)  
*(equation balances (1))*

  
 allow  $\text{C}_4 \text{H}_9 \text{CH}_2 \text{OH}$  product must show aldehyde group  
 or  $\text{C}_5 \text{H}_{11} \text{OH}$  e.g.  $\text{C}_4 \text{H}_9 \text{CHO}$

**If ketone shown here, allow balance**

**If wrong alcohol used, allow balance**

- (ii)  $\text{CH}_3 \text{CH}_2 \text{CH}_2 \text{CH}(\text{OH})\text{CH}_3 + [\text{O}] \longrightarrow \text{CH}_3 \text{CH}_2 \text{CH}_2 \text{COCH}_3 + \text{H}_2\text{O}$  (1)  
*(equation balances (1))*

  
 allow  $\text{C}_3 \text{H}_7 \text{C}(\text{OH}) \text{CH}_3$  product must show CO in ketone  
 or  $\text{C}_5 \text{H}_{11} \text{OH}$  e.g.  $\text{C}_3 \text{H}_7 \text{COCH}_3$

If aldehyde shown here, allow balance

If wrong alcohol used, allow balance

4

- (b) Fehling's solution or Tollen's reagent or Potassium dichromate (1)

☐ Boil, heat, warm  $\longrightarrow$  ammoniacal silver nitrate

**Conditions tied to reagent**

orange or brown or red precipitate or solid etc  
 or silver mirror grey/black precipitate or solid  
 or (orange) turns green or blue  
**observation tied to reagent**

4

No precipitate

or no silver mirror

or no colour change (1)

or no reaction  $\longrightarrow$

Not "nothing"

If no reagent or wrong reagent quoted, mark as 'CE'

Any appropriate reagent - e.g. - Benedicts, but if  $\text{Cr}_2\text{O}_7^{2-}$  or  $[\text{Ag}(\text{CNH}_3)_2]^+$

or  $\text{MnO}_4^-$  given, do not allow reagent mark.

[8]

- 20.** (a) M1:  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ ; 1  
 M2:  $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ ; 1

(penalise incorrect alcohols in part (a), but mark consequentially in part (b) and in part (c), if relevant)  
(if three alcohols drawn, award MAX. 1 mark)

- (b) M1, M2 and M3: Correct structures for butanal, butanone and butanoic acid; 3  
(award these structure marks wherever the structures appear, but insist that the C=O is shown in each structure and additionally, the C-O in the carboxylic acid)
- M4: balanced equation for the reaction of butan-1-ol with [O] to produce butanal and water; 1
- M5: balanced equation for the reaction of butan-1-ol with [O] to produce butanoic acid and water
- OR
- balanced equation for the reaction of butanal with [O] to produce butanoic acid; 1
- M6: balanced equation for the reaction of butan-2-ol with [O] to produce butanone and water; 1
- (Credit condensed structures or molecular formulas in each equation, provided it is obvious to which reaction the equation refers) (Insist that whatever formula is used in each equation that it is a conventional representation of the compound; for example penalise  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COH}$  for butanal)
- (c) M1: Correct structure for 2-methylpropan-2-ol; 1
- M2: 2-methylpropan-2-ol
- OR
- methylpropan-2-ol; 1
- (penalise on every occasion in parts (a) and (c), structures for the alcohols that are presented with the alcohol functional group as C-H-O)

[10]

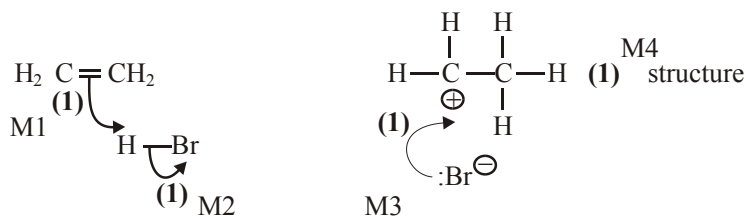
21. (a)  $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$  reduced by  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$  (1)  
oxidised to  $\text{CH}_3(\text{CH}_2)_2\text{CHO}$  (1)  
and  $\text{CH}_3(\text{CH}_2)_2\text{COOH}$  (1)  
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$  (1)  
oxidised to  $\text{CH}_3(\text{CH}_2)_2\text{COOH}$  (1)
- Equation:  $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$  (1) 6
- Note: Deduct one if all three compounds given as reducing agents.

- (b) Tollens' reduced by  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$  (1)  
oxidised to  $\text{CH}_3(\text{CH}_2)_2\text{COOH}$  (1)
- Equation  $[\text{Ag}(\text{NH}_3)_2]^+ + \text{e}^- \longrightarrow \text{Ag} + 2\text{NH}_3$  (1) 3

[9]

22. (a) Reaction 1  $\text{H}_2\text{O}$  or steam (1)  
Reaction 5  $\text{NH}_3$  (1) 2
- For Reaction 4; credit  $\text{dil H}_2\text{SO}_4$  OR  $\text{H}_2\text{SO}_4(\text{aq})$  OR  $\text{HCl}(\text{aq})$  but NOT steam and NOT  $\text{NaOH}(\text{aq})$

- (b) 4



Penalise M2 incorrect  $\delta+$  /  $\delta-$

Penalise  $\delta-$  on alkene (M1)

Penalise dots on bonds once

Penalise M4 (structure) for use of wrong alkene

Penalise M1 for use of Br<sub>2</sub>

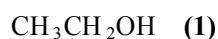
- (c) Water OR aqueous solution OR (aq) in equation (1) M1  
 Yeast OR enzyme/zymase OR  $T \leq 45^\circ\text{C}$  M2  
*but T not below  $20^\circ\text{C}$  and allow warm*  
*N.B. yeast and  $T=60^\circ$  ✗ con*  
*Ignore pH*  
*Ignore anaerobic / oxygen*  
*Ignore time*  
*Ignore pressure*  
 $\text{C}_6\text{H}_{12}\text{O}_6 \longrightarrow 2\text{C}_2\text{H}_5\text{OH}$  (or  $\text{CH}_3\text{CH}_2\text{OH}$ ) +  $2\text{CO}_2$  M3  
*Allow  $\text{C}_{12}\text{H}_{22}\text{O}_{11}$  if balanced equation*  
*M4 OR M5 needs the use of good English and correct chemistry to gain credit*  
 M4: The rate of fermentation is slower (1)  
 OR The rate of hydration is faster  
 QoL OR (The rate of) fermentation is slow and  
 (the rate of) hydration is fast  
*reference correctly to time rather than rate gains credit*  
 M5: The product of fermentation is less pure or lower purity  
 OR The product of hydration is more pure or higher purity  
 OR The product of fermentation is impure and that of hydration is pure  
 OR Specific reference to 10–15% versus 90–100%  
 OR correct reference to higher or lower yield 5

[11]

23.



NaOH (1) aqu or warm (1)  
 nucleophilic substitution (1)



$\text{K}_2\text{Cr}_2\text{O}_7$   $\text{H}_2\text{SO}_4$   
 Reflux (1)



[9]

24. (a)  $\text{CH}_4 + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{HCl} \quad (1)$   
 Initiation:  $\text{Cl}_2 \longrightarrow 2\text{Cl}^\bullet \quad (1)$   
 Propagation:  $\text{CH}_4 + \text{Cl}^\bullet \longrightarrow \text{CH}_3^\bullet + \text{HCl} \quad (1)$   
 $\text{CH}_3^\bullet + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{Cl}^\bullet \quad (1)$   
 Termination:  $\text{CH}_3^\bullet + \text{Cl}^\bullet \longrightarrow \text{CH}_3\text{Cl} \quad (1)$   
 or  $\text{CH}_3^\bullet + \text{CH}_3^\bullet \longrightarrow \text{C}_2\text{H}_6$

5

- (b) Hydration:  $\text{H}_3\text{PO}_4$  (1)  
 300°C (1)  
 65 atmos (1)  
 Fermentation: yeast (1)  
 35°C (1)  
 air free (1)  
 Advantages: Two from  
 faster/purer product/continuous process  
 (cheaper on manpower) (2)  
 Disadvantage: ethene is non-renewable resource (1) 9
- [14]**
25. (a) Reaction 2:  $\text{NaOH}$  OR  $\text{KOH}$  (1) M1 alcohol (ic) OR ethanol (ic)(1) M2  
 ignore heat  
 Condition mark linked to correct reagent but award M2 if  $\text{OH}^-$   
 or base or alkali mentioned  
Reaction 3: concentrated  $\text{H}_2\text{SO}_4$  OR  $\text{H}_3\text{PO}_4$  M1 (1) heat  
 (1) M2  
 OR 150°C - 200°C 4  
 Condition mark linked to correct reagent but award M2 if  
 $\text{H}_2\text{SO}_4$  or  $\text{H}_3\text{PO}_4$ , but not concentrated  
 Penalise reagent and condition if dilute  $\text{H}_2\text{SO}_4$  /  $\text{H}_3\text{PO}_4$
- (b) Mechanism:
- $\text{HO}^- \text{ (1) M1}$   
 $\text{H} \text{ (1) M2}$   
 $\text{H} \text{ (1) M2}$   
 $\text{H}-\text{C}-\text{C}-\text{H} \rightarrow \text{H}_2\text{C}=\text{CH}_2 + \text{H}_2\text{O} + \text{Cl}^-$   
 $\text{H} \text{ (1) M3}$   
 $\text{Cl}$
- Award M3 (C-Cl) independently  
 M1 and M2 must be to / from correct places  
 E1 mechanism possible in which M2
- $\text{H} \text{ (1) M2}$   
 $\text{H} \text{ (1) M2}$   
 $\text{H}-\text{C}-\text{C}-\text{H}$   
 $\text{H} \text{ (1) M2}$   
 $\text{H}$
- Name: of mechanism = elimination (1)  
 NOT dehydrohalogenation  
 Ignore "base" OR "nucleophilic" before elimination
- Reason: Reaction 2 has (very) low yield (1) 5  
 QoL OR chloroethane has to be made (from ethane)  
 OR chloroethane is expensive  
 OR chloroethane is not readily available
- (c) Name of mechanism = elimination (1)  
 NOT dehydration alone
- Reason: Ethanol could come from (fermentation of) renewable 2  
 QoL sugars / glucose / carbohydrates / sources (1)
- [11]**
26. (a) **A** – alkene (1)  
**B** – halogenoalkane / bromoalkane / alkyl halide / haloalkane (1)  
**C** – alcohol (ignore primary, secondary) (1) 3
- (b) (i) addition ignore nucleophilic / electrophilic / free radical (1) 1  
 (ii) substitution **not** replacement / displacement (1) 1  
 (iii) oxidation **not** reduction; **not** redox; allow dehydrogenation (1) 1
- (c) Sodium hydroxide /  $\text{NaOH}$  /  $\text{KOH}$  **not** just hydroxide (1)  
**(B to C)** aqueous **not** dilute (1)

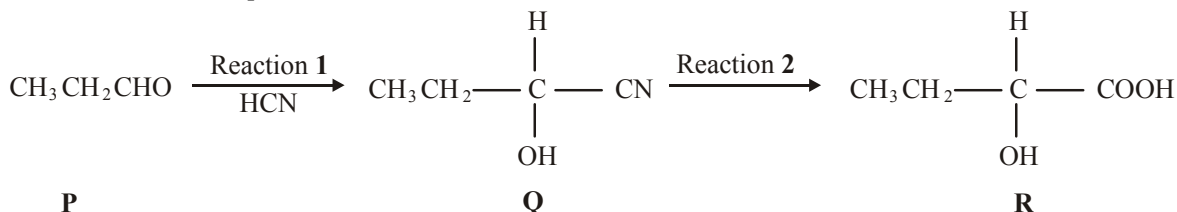
- (B to A)            alcoholic **(1)**  
 mark alternatives as (d)  
 ignore references to concentration and temperature 3
- (d) sodium (or potassium) dichromate /  $\text{Na}_2\text{Cr}_2\text{O}_7$  **or (1)**  
 named alkali **or** water **or** aqueous  
 sulphuric acid /  $\text{H}_2\text{SO}_4$   
 ignore dilute / concentrated **(1)**  
 allow  $\text{HCl}$ ,  $\text{H}_3\text{PO}_4$ ,  $\text{HNO}_3$   
 allow  $\text{KMnO}_4$  with  $\text{H}_2\text{SO}_4$  /  $\text{H}_3\text{PO}_4$  /  $\text{HNO}_3$  **not**  $\text{HCl}$   
 allow 1 mark for acidified dichromate or dichromate /  $\text{H}^+$   
 heat / reflux / boil / warm / temperature  $40^\circ\text{C}$  -  $100^\circ\text{C}$  **(1)**  
 this mark dependent on 'dichromate' or 'manganate' 3
- (e) (i)  $\text{CH}_3\text{CH}(\text{CH}_3)\text{Br} + \text{NaOH} \longrightarrow \text{CH}_3\text{CH}=\text{CH}_2 + \text{NaBr} + \text{H}_2\text{O}$  **(1)**  
 (ii)  $\text{CH}_3\text{CH}(\text{CH}_3)\text{Br} + \text{NaOH} \longrightarrow \text{CH}_3\text{CH}(\text{CH}_3)\text{OH} + \text{NaBr}$  **(1)**  
 allow molecular formulae  $\text{C}_3\text{H}_7\text{Br}$ ;  $\text{C}_3\text{H}_8\text{O}$ ;  $\text{C}_3\text{H}_6$   
 allow ionic versions (with  $\text{OH}^-$ ,  $\text{Br}^-$ ) 2
- (f) arrow from O of  $\text{OH}^-$  to C joined to Br **(1)**  
 lone pair not needed  
 C-Br polarity shown by  $\delta^+$   $\delta^-$  **or**  
 heterolytic fission of C-Br bond shown by arrow from bond between C  
 and Br to Br **or**  
 intermediate with partial bonds and minus sign **(1)**  
 $\text{Br}^-$  as product **(1)**  
 allow all 3 marks if 1-bromopropane identified as **B** 3

**[17]**

## 10.5 CARBONYL COMPOUNDS

## 10.6 CARBOXYLIC ACIDS &amp; DERIVATIVES

1. Consider the sequence of reactions below.



- (a) Name and outline a mechanism for Reaction 1. (5)
- (b) (i) Name compound **Q**  
 (ii) The molecular formula of **Q** is  $\text{C}_4\text{H}_7\text{NO}$ . Draw the structure of the isomer of **Q** which shows geometrical isomerism and is formed by the reaction of ammonia with an acyl chloride. (3)
- (c) Draw the structure of the main organic product formed in each case when **R** reacts separately with the following substances:  
 (i) methanol in the presence of a few drops of concentrated sulphuric acid;  
 (ii) acidified potassium dichromate(VI);  
 (iii) concentrated sulphuric acid in an elimination reaction. (3)

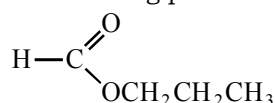
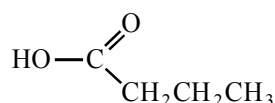
(3)  
(Total 11 marks)

2. Alcohol **X** has the structure  $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}(\text{CH}_3)_2$

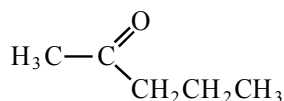
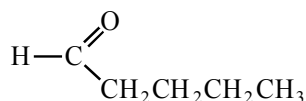
- (a) (i) Name alcohol **X**.  
 (ii) Name and outline the mechanism for the reaction occurring when alcohol **X** is converted into 2,3-dimethylbut-2-ene in the presence of a strong acid.  
 (iii) Give the structure of, and name an isomer of 2,3-dimethylbut-2-ene which is also formed in the reaction. Explain why two products are obtained. (10)
- (b) (i) Write an equation for the reaction between alcohol **X** and ethanoyl chloride. Name and outline a mechanism for this reaction, using  $\text{ROH}$  to represent the alcohol in the mechanism.  
 (ii) Give an alternative method for obtaining the organic reaction product in part (b) (i), starting from alcohol **X**, other than using ethanoic anhydride. State the type of reaction, the reagent(s) used and the reaction conditions. (9)

(9)  
(Total 19 marks)

3. (a) Consider the following pair of isomers.

**C****D**

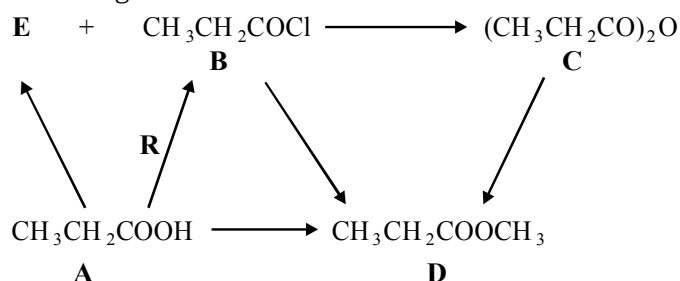
- (i) Name compound **C**.  
 (ii) Identify a reagent which could be used in a test-tube reaction to distinguish between **C** and **D**. In each case, state what you would observe. (4)
- (b) Consider the following pair of isomers.

**E****F**

- (i) Name compound **E**.  
 (ii) Identify a reagent which could be used in a test-tube reaction to distinguish between **E** and **F**. In each case, state what you would observe. (4)  
 (c) Draw the structure of the chain isomer of **F** which shows optical isomerism. (1)

**(Total 9 marks)**

4. This question is about compounds **A**, **B**, **C**, **D** and **E** and their interconversions. Some of these are shown in the diagram below.



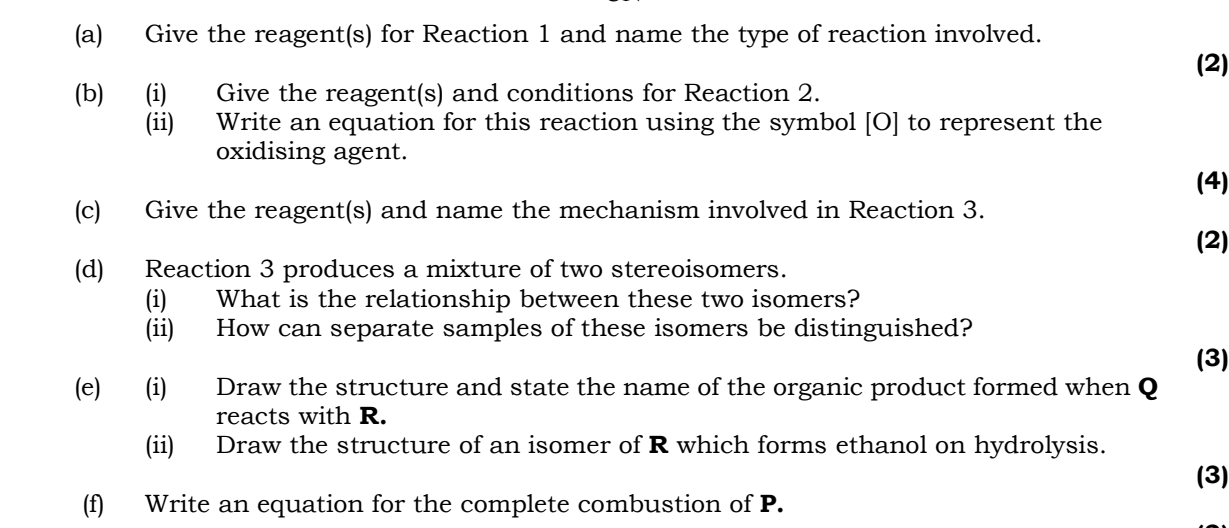
- (a) (i) Give the names of compounds **B** and **C**. (2)  
 (ii) Describe one observation that would be made when compound **B** reacts with water. (1)  
 (iii) Draw the graphical formula of the organic product formed when compound **B** reacts with ammonia. (1)  
 (iv) When compound **C** is heated in water, it is hydrolysed to a carboxylic acid. Write the equation for the reaction that occurs. (1)  
 (b) Compounds **A**, **B** and **C** will each react with the same alcohol to form compound **D**.  
 (i) Give the name of the alcohol and the name of compound **D**. (2)  
 (ii) State the appropriate reaction conditions for the formation of compound **D** from each of the compounds **A**, **B** and **C**. (4)

**(Total 11 marks)**

5. (a) (i) Write an equation for the reaction of butan-2-ol with ethanoic acid, showing clearly the structure of the organic product.  
 (ii) Name the type of organic compound formed in part (a)(i) and suggest a use for this compound.  
 (iii) Give a homogeneous catalyst for the reaction in part (a)(i) and state the meaning of the term *homogeneous*. (6)  
 (b) Write an equation for the complete combustion of butan-2-ol in an excess of oxygen. (1)  
 (c) But-1-ene and other products can be made by the dehydration of butan-2-ol.  
 (i) Outline a mechanism for the dehydration of butan-2-ol into but-1-ene.  
 (ii) Explain why but-1-ene does not show geometrical isomerism.  
 (iii) An alternative dehydration of butan-2-ol produces geometrical isomers. Draw the structure of one of these geometrical isomers and give its full name. (7)

**(Total 14 marks)**

6. Consider the following reaction scheme.



**7.** (a) The structures of two alcohols of formula  $\text{C}_5\text{H}_{11}\text{OH}$  are shown below.



- 303

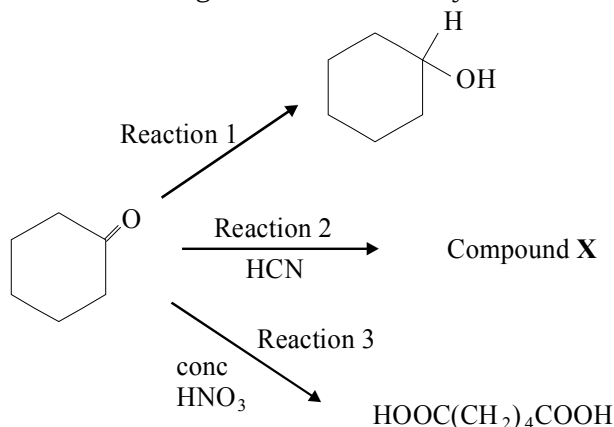


(iii) Write an equation for the reaction taking place in Step 3.

(2)

(Total 10 marks)

8. Consider the following three reactions of cyclohexanone,  $C_6H_{10}O$ .



(a) Give a suitable reagent for Reaction 1.

(1)

(b) Name the type of reaction and outline a mechanism for Reaction 2.

(5)

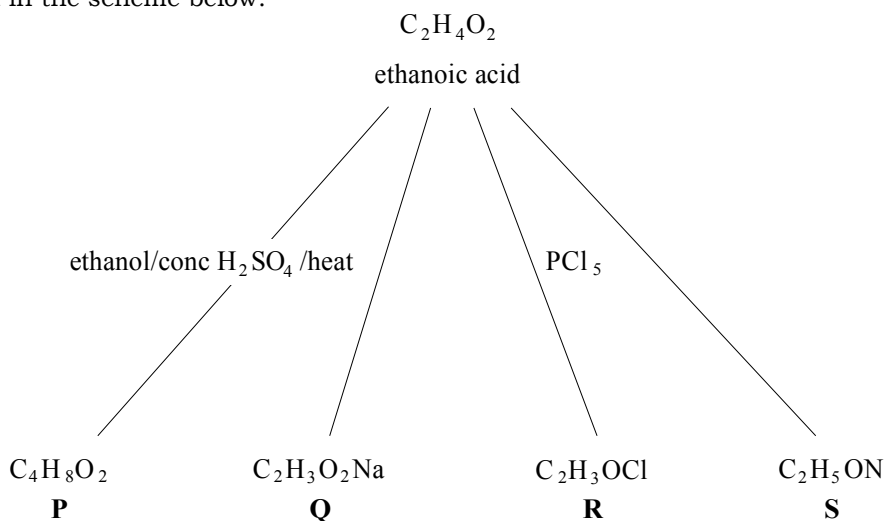
(c) (i) Name the organic product of Reaction 3.

(ii) Calculate the maximum mass of this organic product that could be formed if 2.40 g of cyclohexanone were allowed to react in Reaction 3.

(4)

(Total 10 marks)

9. The molecular formulae of some compounds that can be prepared from ethanoic acid are given in the scheme below.



(a) (i) Give the name and graphical (displayed) formula of **P**.

(2)

(ii) Give the name of the type of reaction which occurs when **P** is formed from ethanoic acid.

(1)

(b) Ethanoic acid can be obtained from **P**.

(i) Give the name of the reagent(s) and state the conditions required.

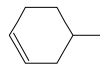
(2)

(ii) Write a balanced equation for the reaction

(1)

(c) (i) State the reagent and reaction conditions that could be used for converting ethanoic acid into **Q**.

(2)

- (ii) Give the name and graphical formula of the organic product of the reaction between anhydrous samples of **Q** and **R**. (2)
- (iii) State how the product formed in (c)(ii) could be converted into ethanoic acid and write an equation for the reaction. (2)
- (d) (i) Give the name and graphical formula of the amide **S**. (2)
- (ii) State the reagent(s) and reaction conditions that could be used for converting ethanoic acid into **S**. (2)
- (iii) Write a balanced equation for the reaction between **S** and aqueous hydrochloric acid. (1)
- (Total 17 marks)**
10. (a) Addition reactions to both alkenes and carbonyl compounds can result in the formation of isomeric compounds.
- (i) Choose an alkene with molecular formula  $C_4H_8$  which reacts with  $HBr$  to form two structural isomers. Give the structures of these two isomers and name the type of structural isomerism shown. Outline a mechanism for the formation of the major product.
- (iii) Using  $HCN$  and a suitable carbonyl compound with molecular formula  $C_3H_6O$ , outline a mechanism for an addition reaction in which two isomers are produced. Give the structures of the two isomers formed and state the type of isomerism shown. (14)
- (b) Explain why ethanoyl chloride reacts readily with nucleophiles. Write an equation for one nucleophilic addition-elimination reaction of ethanoyl chloride. (A mechanism is not required.) (4)
- (Total 18 marks)**
11. The two functional groups in compound **A**,   $CHO$ , behave independently.
- (a) State what would be observed if a few drops of compound **A** were added to Fehling's solution and heated. Give the structure of the organic reaction product. (2)
- (b) Using  $RCHO$  to represent compound **A**, write an equation for the reaction between  $RCHO$  and hydrogen cyanide. State the type of reaction taking place and outline a mechanism. (5)
- (c) State the type of isomerism shown by 2-hydroxypropanoic (*lactic*) acid,  $CH_3CH(OH)COOH$ , and point out the structural feature of the molecule which causes the existence of two isomers. With the aid of diagrams, show the structural relationship between the two isomers and state how these isomers can be distinguished. (5)
- (Total 12 marks)**
12. (a) Compound **A**,  $HCOOCH_2CH_2CH_3$ , is an ester. Name this ester and write an equation for its reaction with aqueous sodium hydroxide. (2)
- (d) A naturally-occurring triester, shown below, was heated under reflux with an excess of aqueous sodium hydroxide and the mixture produced was then distilled. One of the products distilled off and the other was left in the distillation flask.
- $$\begin{array}{c}
 CH_3(CH_2)_{16}COOCH_2 \\
 | \\
 CH_3(CH_2)_{16}COOCH \\
 | \\
 CH_3(CH_2)_{16}COOCH_2
 \end{array}$$
- (i) Draw the structure of the product distilled off and give its name. (2)
- Structure*

- (ii) Give the formula of the product left in the distillation flask and give a use for it.

(4)

(Total 6 marks)

13. (a) **P**, **Q** and **R** have the molecular formula  $C_6H_{12}$

All three are branched-chain molecules and none is cyclic.

**P** can represent a pair of optical isomers.

**Q** can represent a pair of geometrical isomers.

**R** can represent another pair of geometrical isomers different from **Q**.

Draw one possible structure for one of the isomers of each of **P**, **Q** and **R**.

(3)

- (b) Butanone reacts with reagent **S** to form compound **T** which exists as a racemic mixture. Dehydration of **T** forms **U**,  $C_5H_7N$ , which can represent a pair of geometrical isomers.

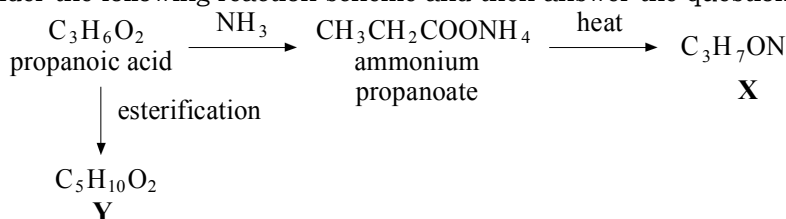
(i) State the meaning of the term *racemic mixture* and suggest why such a mixture is formed in this reaction.

(ii) Identify reagent **S**, and draw a structural formula for each of **T** and **U**.

(6)

(Total 9 marks)

14. Consider the following reaction scheme and then answer the questions below.



- (a) Draw the displayed formula of propanoic acid.

(1)

- (b) Propanoic acid may be converted into compound **Y** by an esterification reaction.

(i) Give the reagent(s) and condition(s) required for the formation of compound **Y** from propanoic acid.

(3)

(ii) Give the name of compound **Y**.

(1)

(iii) Write an equation for the esterification reaction.

(2)

- (c) Propyl ethanoate is an ester that is structurally isomeric with compound **Y**. When propyl ethanoate is heated with aqueous sodium hydroxide, two products are formed.

(i) Explain what is meant by the term *structural isomerism*.

(2)

(ii) Give the names or structures of the two products of this reaction.

(2)

(Total 11 marks)

15. There are several non-cyclic structural isomers with the molecular formula  $C_6H_{12}$ .

- (a) One of these isomers, 2-methylpent-2-ene,  $(CH_3)_2C=CHCH_2CH_3$ , reacts with hydrogen bromide. Name the major product and account for its formation by reference to the mechanism of the reaction.

(7)

- (b) Identify one linear alkene of formula  $C_6H_{12}$  which can exist as a pair of stereoisomers. State the type of stereoisomerism shown, name the alkene and draw the structures of the two isomers.

(4)

- (c) One of the branched, non-cyclic structural isomers of  $C_6H_{12}$  can exhibit a different type of stereoisomerism from that shown in part (b). State the type of stereoisomerism shown, name the alkene and draw the two structures.

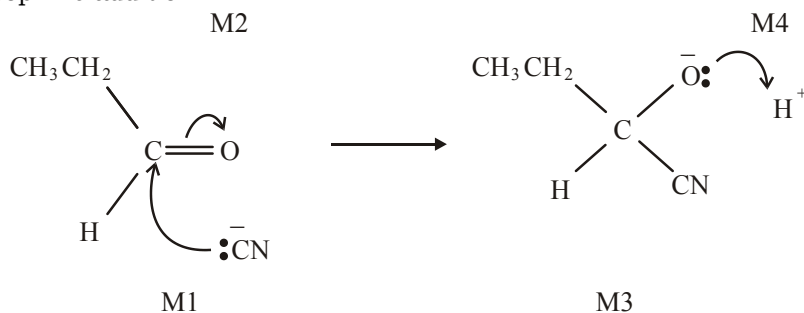
(4)

(Total 15 marks)

# Marking Scheme

1. (a) nucleophilic addition

1



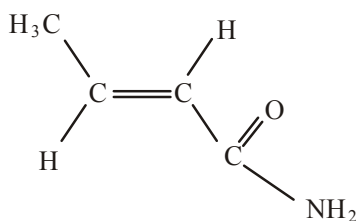
4

- (b) (i) 2-hydroxybutanenitrile

1

(ii)

2

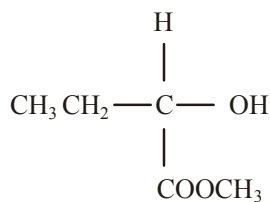


(allow 1 for amide even if not  $\text{C}_4\text{H}_7\text{NO}$ , i.e.  $\text{RCONH}_2$ )

(if not amide, allow one for any isomer of  $\text{C}_4\text{H}_7\text{NO}$  which shows geometric isomerism)

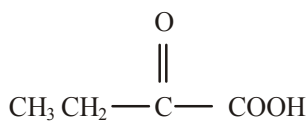
- (c) (i)

1



(ii)

1



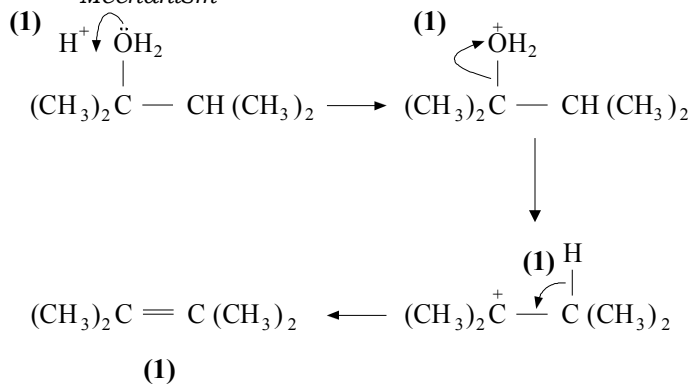
(iii)  $\text{CH}_3\text{CH}=\text{CHCOOH}$

1

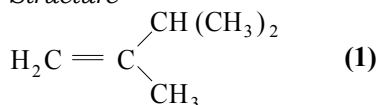
[11]

2. (a) (i) elimination (1)  
 (ii) Mechanism

2, 3 - dimethylbutan - 2 - ol (1)



(iii) Structure

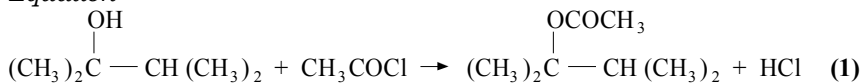


Name of isomer 2, 3 – dimethylbut – 1 – ene (1)

Explanation loss of  $\text{H}^+$  or  $\text{H}$  (1)  
 from end C also possible (1)

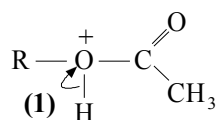
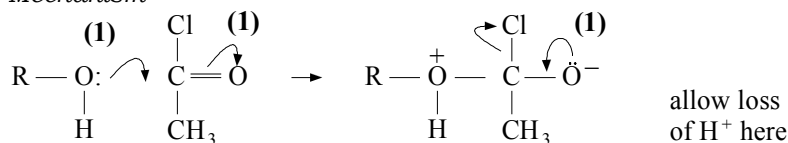
10

(b) (i) Equation



Name of mechanism addition – elimination (1)

Mechanism



(ii) Type of reaction esterification (1)

Reagent(s)  $\text{CH}_3\text{COOH}$  or ethanoic acid (1)

Conditions strong acid catalyst (1)  
or  $\text{H}_2\text{SO}_4$  or  $\text{HCl}$

9

[19]

3. (a) (i) propyl methanoate (1)

not propanyl

- ☐ A wrong reagent or no reagent scores zero
- ☐ An incomplete reagent such as silver nitrate for Tollens, or potassium dichromate loses the reagent mark, but can get both observation marks
- ☐ penalise observations which just say colour change occurs or only state starting colour

(ii) Reagent:  $\text{NaHCO}_3$  (1)Observation with **C**: no reaction (1)Observation with **D**: effervescence (1)

4

for **C** and **D** NOT Tollens

| Test                      | an identified (hydrogen) carbonate | acidified $\text{K}_2\text{Cr}_2\text{O}_7$ | acidified $\text{KMnO}_4$ | correct metal           | UI or stated indicator              | $\text{PCl}_5$ |
|---------------------------|------------------------------------|---------------------------------------------|---------------------------|-------------------------|-------------------------------------|----------------|
| Observation with <b>C</b> | no reaction                        | goes green                                  | goes colourless           | no reaction             | no change                           | no reaction    |
| observation with <b>D</b> | bubbles or $\text{CO}_2$           | no change                                   | no change                 | bubbles or $\text{H}_2$ | red or correct colour<br>pH 3 – 6.9 | (misty) fumes  |

(b) (i) Reagent: pentan-2-one (1)

or 2-pentanone

but not pent-2-one or pentyl

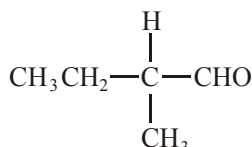
- (ii) *Reagent:* Tollen's or Fehling's (1)  
*Observation with E:* no reaction (1)  
*Observation with F:* silver mirror or red ppt (1)  
 for **E** and **F**

4

| Test                      | Tollens                         | Fehling's or Benedict's     | iodoform or I <sub>2</sub> /NaOH | acidified K <sub>2</sub> Cr <sub>2</sub> O <sub>7</sub> | Schiff's    |
|---------------------------|---------------------------------|-----------------------------|----------------------------------|---------------------------------------------------------|-------------|
| observation with <b>E</b> | no reaction                     | no reaction                 | yellow (ppt)                     | no change                                               | no reaction |
| observation with <b>F</b> | silver or mirror or grey or ppt | red or ppt not red solution | no reaction                      | goes green                                              | goes pink   |

(c) (1)

1



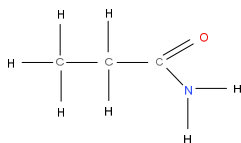
*must be aldehyde. Allow C<sub>2</sub>H<sub>5</sub> for CH<sub>3</sub>CH<sub>2</sub> otherwise this is the only answer*

4. (a) (i) **B:** propanoyl chloride (or consequentially on part (a) (ii)) (1)  
**C:** propanoic anhydride (or consequentially on part (a) (ii)) (1)  
 do **not** allow formulae  
 (ii) effervescence / misty fumes / steamy fumes / fumes /  
 solution becomes warm / fizzing **not** just gas (1)  
 (iii)

[9]

2

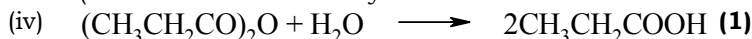
1



(1)

1

(the minimum necessary for the mark is C=O and C-N shown)



1

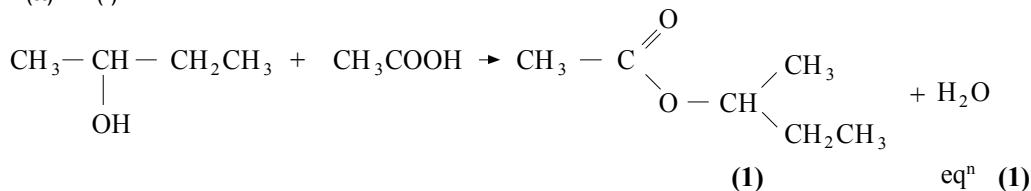
allow C<sub>2</sub>H<sub>5</sub>.....

- (b) (i) methanol (1)  
 methyl propanoate (or consequentially on part (a) (ii)) (1)  
 do **not** allow formulae  
 (ii) **A:** in presence of (concentrated) sulphuric acid  
 / H<sub>2</sub>SO<sub>4</sub> / strong acid / gaseous hydrogen chloride or HCl  
 allow dilute H<sub>2</sub>SO<sub>4</sub> (1)  
 heat / reflux (but only if first mark awarded) (1)  
 allow 1 mark for acidic conditions / H<sup>+</sup> and heat  
**B:** room temperature / in the cold / not heated / cooling **not** acid (1)  
**C:** heat / reflux **not** acid (1)

4

[11]

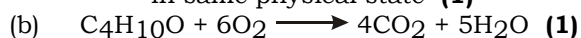
5. (a) (i)



- (ii) ester (1)  
 solvent, flavourings (1)

- (iii) conc  $\wedge$  H<sub>2</sub>SO<sub>4</sub> (1)  
 in same physical state (1)

6



1

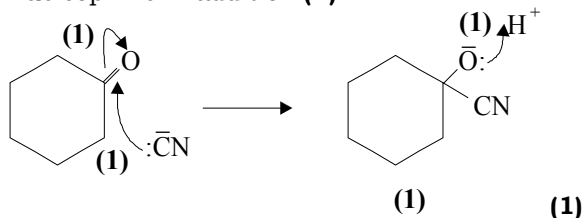
[14]

- [16]

- [10]**

8. (a)  $\text{NaBH}_4$  (1) 1

(b) nucleophilic addition (1)



5

(c) (i) hexanedioic acid (1)



$\text{Mr} = 98$  (1)

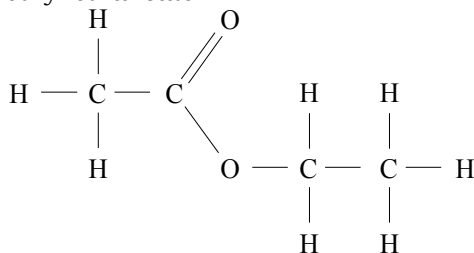
$\text{Mr} = 146$  (1)

$$2.40 \text{ g} \longrightarrow \frac{2.40}{98} \times 146 = 3.58 \text{ g (1)}$$

4

[10]

9. (a) (i) ethyl ethanoate



(1)

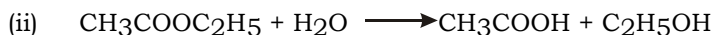
2

(ii) esterification / condensation / addition - elimination (1)

1

- (b) (i) aqueous / dilute sulphuric / hydrochloric acid (allow  $\text{HCl(aq)}$ ;  $\text{H}_2\text{SO}_4\text{(aq)}$  not water) (1)  
temp.  $< 100^\circ$  / warm / heat / reflux (this mark dependent on sensible reagent) (1)

2



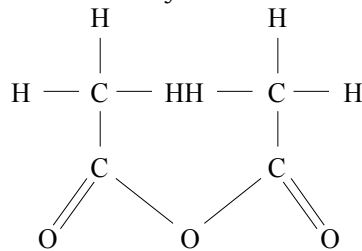
(allow  $\text{C}_4\text{H}_8\text{O}_2$ ,  $\text{C}_2\text{H}_4\text{O}_2$  but must have  $\text{C}_2\text{H}_5\text{OH}$ ) (1)

1

- (c) (i) sodium hydroxide / sodium carbonate / sodium hydrogen carbonate (allow formula) (1)  
room temperature / aqueous (2nd mark dependent on correct reagent) (1)

2

(ii) ethanoic anhydride

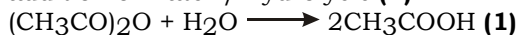


(1)

2

(methyl groups can be shown as  $-\text{CH}_3$  but the C-C bond must be drawn)

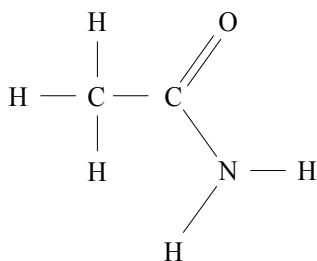
(iii) addition of water / hydrolysis (1)



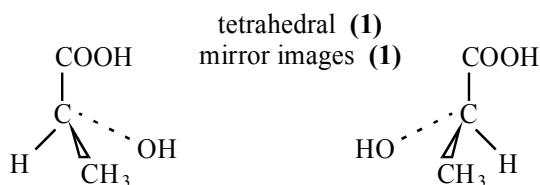
2

- (d) (i) ethanamide **not** ethylamide.





- (1) 2
- (ii) ammonia (not if dilute implied) / ammonium carbonate /  $\text{PCl}_5$  followed by  $\text{NH}_3$  (allow formulae) (1) 2
- heat or temperature  $< 100^\circ$  (1) 2
- (iii)  $\text{CH}_3\text{CONH}_2 + \text{HCl} + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{COOH} + \text{NH}_4\text{Cl}$  (1) 1
- [17]**
10. (a) (i) An appropriate alkene;  $\text{CH}_3\text{CH}_2\text{CHCH}_2$  or  $(\text{CH}_3)_2\text{CCH}_2$  1
- Isomer 1 1
- Isomer 2 1
- Position isomerism 1
- Mechanism
- electrophilic attack and electron shift to Br (Unless  $\text{H}^+$  used) 1
- carbocation 1
- reaction with carbocation 1
- [Allow mechanism marks for the alkene  $\text{CH}_3\text{CHCHCH}_3$ ]
- [Allow one mark if mechanism for minor product given]
- (ii) An appropriate carbonyl;  $\text{CH}_3\text{CH}_2\text{CHO}$  1
- Mechanism nucleophilic attack and electron shift to O 1
- anion intermediate 1
- reaction with anion 1
- [Allow mechanism marks for the carbonyl  $(\text{CH}_3)_2\text{CO}$ ]
- Isomer 1 1
- Isomer 2 1
- Optical isomerism 1
- NB Isomer structures must be tetrahedral
- NB Penalise "stick" structures once in part (a)
- (b) QoL Large charge on carbonyl carbon atom due to bonding to O and Cl 1
- Nucleophiles have electron pairs which can be donated 1
- Equation Species 1
- Balanced 1
- [18]**
11. (a) Observation orange or red-brown precipitate (1)
- Structure
- (1) 2
- (b) Equation  $\text{RCHO} + \text{HCN} \longrightarrow \text{RCH(OH)CN}$  (1)
- Type of reaction nucleophilic addition (1)
- Mechanism
- (1) 5
- (c) Type of isomerism optical (1)
- Structural feature atom with 4 different substituents or asymmetric or chiral (1)
- Isomer 1 Isomer 2



Method of distinguishing opposite rotation of polaroid light (1)

5

[12]

12. (a) propyl methanoate;

1

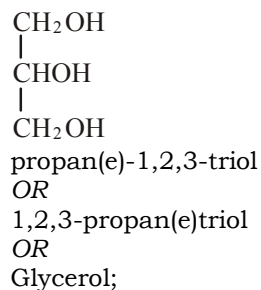


OR



1

- (b) (i)



1

- (ii)  $\text{CH}_3(\text{CH}_2)_{16}\text{COONa}$  or  $\text{C}_{17}\text{H}_{35}\text{COONa}$  or  $\text{C}_{18}\text{H}_{35}\text{O}_2\text{Na}$ ;

1

(ignore 3 in front of formula but not if indicating trimer)

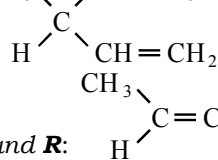
(not just anion and penalise Na shown as covalently bonded) soap -  
allow with detergent but not detergent alone;

1

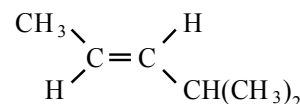
[6]

13. (a) Structure of **P**:  $\text{CH}_3-\text{CH}_2\text{CH}_3$  (1)

3



and

Structures of **Q** and **R**:NOT  $\text{C}_3\text{H}_7$ 

(1)

(1)

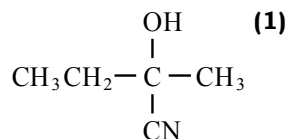
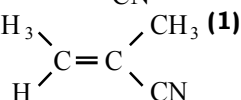
**Q and R in any order**

- (b) (i) Racemic mixture: equal mixture of optical isomers / enantiomers

**OR in explanation**Explanation: planar ( $>\text{C}=\text{O}$ ) (1)

attack from either side is equally likely (1)

- (ii) Reagent **S**:  $\text{HCN}$  or ( $\text{KCN}$  /  $\text{HCl}$  **or**  $\text{H}_2\text{SO}_4$ ) (1)

Compound **T**:Compound **U**:

6

[9]

4. (a) graphical structure for  $\text{CH}_3\text{CH}_2\text{COOH}$  (1)

allow  $\text{CH}_3\text{CH}_2-$  or  $\text{C}_2\text{H}_5-$ 

1

- (b) (i) ethanol /  $\text{CH}_3\text{CH}_2\text{OH}$  /  $\text{C}_2\text{H}_5\text{OH}$  **not** just alcohol (1)

(concentrated or dilute) sulphuric acid / HCl / strong acid / H<sup>+</sup>**not** just acid solutionheat / reflux / warm / temperature < 100 °C **(1)**

allow second and third marks if alcohol given

third mark is dependent on first and second marks

second mark is independent on first mark

3

(ii) ethyl propanoate **(1)**

1

(iii)  $\text{CH}_3\text{CH}_2\text{COOH} + \text{CH}_3\text{CH}_2\text{OH} \longrightarrow \text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3 + \text{H}_2\text{O}$  $\text{C}_3\text{H}_6\text{O}_2 + \text{C}_2\text{H}_6\text{O} \longrightarrow \text{C}_5\text{H}_{10}\text{O}_2 + \text{H}_2\text{O}$  (minimum for mark) **(1)** $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$  **or**  $\text{C}_2\text{H}_5\text{CO}_2\text{C}_2\text{H}_5$  **(1)**

2

(c) (i) compounds with the same molecular formula /

same numbers of same atoms **(1)**but different structural formulae / structures **or** **(1)**

atoms joined / bonded / linked in different orders / C skeletons

**not** atoms arranged differently

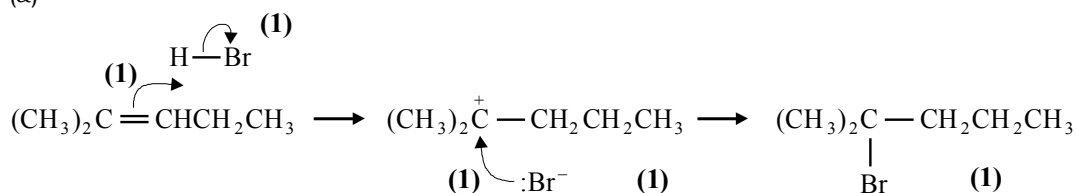
2

(ii) sodium ethanoate /  $\text{CH}_3\text{COONa}$  **(1)**propan-1-ol /  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$  / propanol / propan-2-ol **(1)**

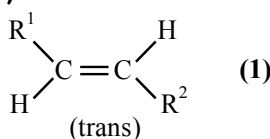
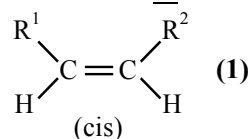
2

**[11]**

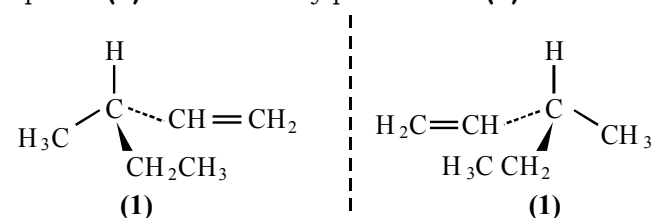
5. (a)

2-bromo-2-methylpentane **(1)**tertiary carbonium ion more stable than secondary **(1)**

7

(b) geometrical or cis-trans isomerisation **(1)**hex-3-ene or hex-2-ene **(1)**

4

(c) optical **(1)** 3-methylpent-1-ene **(1)**

4

**[15]**